END OF WELL REPORT

AZADEGAN OIL FIELD

END OF WELL REPORT

AZNS-040 (PAD #50), TYPE: IIF

Drilling Department

FEBRUARY 2024

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1. General Information

1.1 Field Introduction

Azadegan South Oil Field is situated in West Karoun, Khuzestan province on a large long-axis anticline with N-S trending with approximate dimensions about 20^{km} width and 60^{km} length near to Iran-Iraq border, It is located in The Northwest of The YADAVARAN Field and in The West of JUFEYR Field. The Average Monthly Temperature is 45$^{°c}$ in summer, and Highest Temperature is 52°c. The Lowest Temperature is -4.4$^{°c}$ in winter. Maximum Rainfall is 78^{mm} in February, Winter Season.

The trap area of the south Azadegan is limited to 43.2x17.2^{km}. The Azadegan Oil field was discovered in 1999, when Well AZN-1 was drilled vertically down to the Cretaceous Fahliyan Formation with total depth of 4,643^{m}. The results of this well, revealed the reservoir characteristics of the field. Following to the result of the well no. AZN-1, another 5 exploration wells was drilled in different parts of this huge field down to Fahliyan Formation. In the later stage the well No.10 assumed as exploration well for evaluation of deeper formations. This well was drilled in 2008, down to the depth of 5340^{m} (Neyriz Fm.), mainly to evaluate the Jurassic formations.

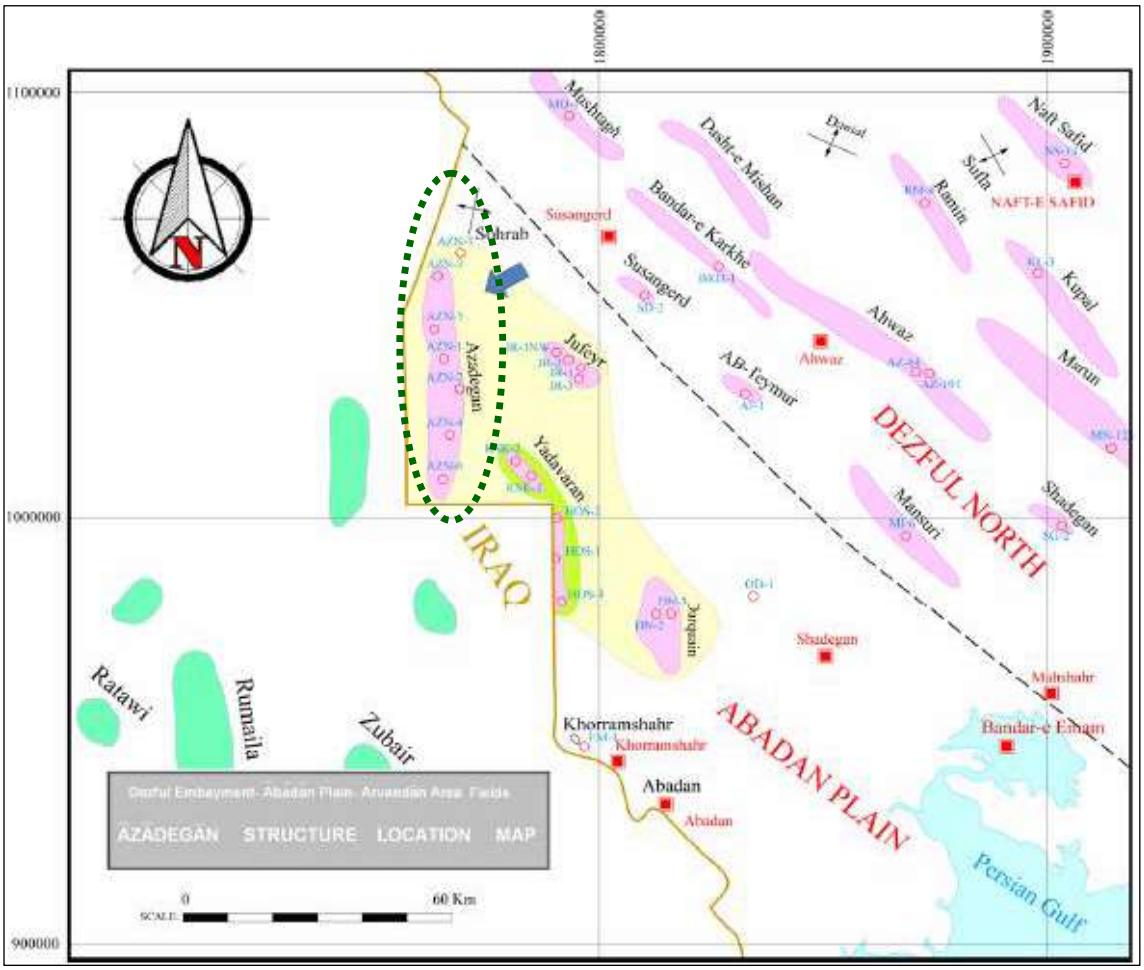


Figure 1-1: AZNS Oilfield Location Map

1.2 AZNS-040 Well Location Map

This well is located on Pad-50 near wells AZNS-036 and AZNS-017S.T-1.
The pad-50 is located on center of South Azadegan Oil field in Block-II.

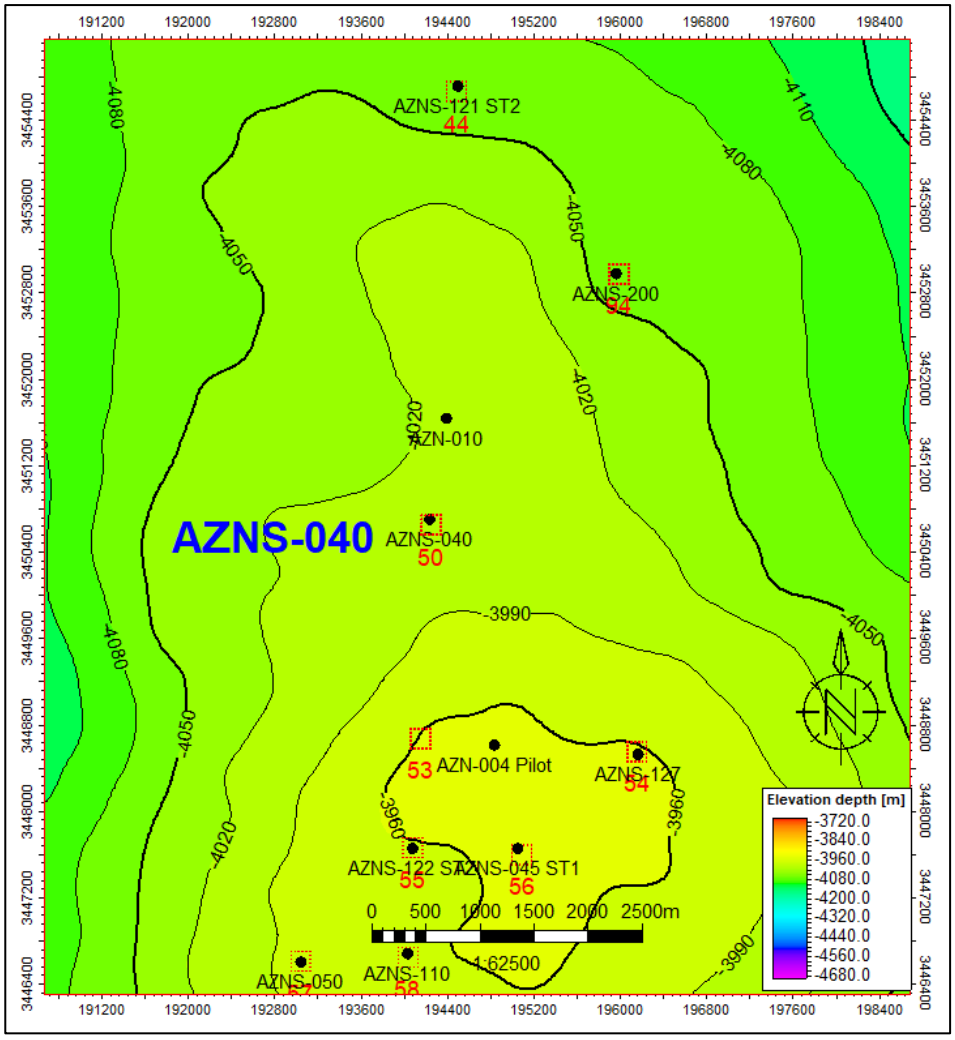


Figure 1-2: AZNS-040 Well Location Map

2. Well History

2.1 Well Informaion

Rig name	208	**Owner**	PEDEC/PPL
Rig type	Land Rig 2000HP	**Contractor**	MSA/OEOC
Well Name	AZNS-40	**Management Contractor**	MSA
Well Type	Vertical IIF	**Field**	South Azadegan
Completion Type	4-1/2" * 3-1/2", 10K	**RTE**	13.45
Planned TD	4200^{m}	**GLE**	3.95
Actual TD	4200^{m}		
Surface Coordinate		E: 194,235^{m} - N: 3,450,715^{m}	
Target Coordinate		E: 194,235^{m} - N: 3,450,715^{m}	
Drilling & Suspension		**Completion**	
Spud Date	07/13/2023	**Spud Date**	11/22/2023
Released Date	12/14/2023	**Released Date**	12/14/2023
End of Drilling	09/23/2023	**End of Drilling**	-
Rig Days Actual/Plan	155 / 102	**Rig Days Actual/Plan**	-

Size (in)	Shoe Depth (mMD)	Specification
20	152.5	K-55, 117^{ppf}, Big Omega
13-3/8	1493	N-80, 68^{ppf}, BTC
9-5/8	3933	TP110SS, 53.5^{ppf}, TPCQ
7	3812.2 – 4189.5	TP110SS, 32^{ppf}, TPCQ
SN Packer	3809.6 – 3812.2	L-80, 29^{ppf}

Table 2-1: AZNS-040 Re-Entry Well Information

2.2 AZNS-040 Well History

The AZNS-040 well was drilled from 26-Mar-2014 to 10-May-2014 by CNPCI to total depth 1493^{m} and suspended. The 26" hole section was drilled to 153^{m} with Mw 65.5 to 71.8^{pcf} and cased by K-55, 117^{ppf} Big Omega 20" casing. The 17-1/2" hole section was drilled f/153 to 1493^{m} with Mw71.8 to 85.5^{pcf}, totally used 5 bits during drilling this section (1 used R.bit, 3 used PDC bit & 1 new PDC bit), this hole section was cased by 13-3/8" N-80, 68^{ppf} BTC casing to 1492^{m} and suspended.

2.3 Breif Summary of Re-Entry Operation

The AZNS-40 well Re-Entry was drilled by rig OEOC-208 from 13-July-2023 to 14-Dec-2023 to total depth 4200^{m} as described in the following:

12-1/4" Hole Section Summary

Date	16 – July - 2023
Drilling Depth	3934^{m}
Casing Size / Shoe	9 5/8" 53.5#, TP110S, TPCQ – 3934^{m}
Formation	Fahliyan
FIT Depth / Pressure Test	1482^{m} / 480^{psi}
Mud Weight	73 – 85^{pcf}

Table 2-2: AZNS-040 12 – 1/4" Hole Section Summary

This section drilling started at 16-July-2023 with 12-1/4" bit from 1493^{m} to 1510^{m} & POOH to surface due to change BHA with new PDC bit, Commence Drlg f/1510 to 1556.5^{m}, suddenly pressure dropped 390^{psi} due to twist-off string from box of 8-1/2" DC & fish left in the hole, POOH to surface with 1-2 BPH loss (fish left in hole: 12-1/4" PDC bit + 8-1/2" B.Sub + 8.05^{m} 8-1/2" DC, fish length: 9.45^{m}), recover the fish with 20^{klbs} over-pull by overshot successfully, POOH to surface & L/D BHA, RIH with new 12-1/4" PDC bit & continue Drlg to 1930^{m} with 4-8 BPH loss while pumping Hi-Vis Pill periodically, stop Drlg & POOH as condition trip to 1492^{m} with good condition, run totco survey (Inc: 0.75 Deg, 2-3 BPH loss), RIH f/1492^{m} to BTM & commence Drlg from 1930^{m} while increasing Mw f/73 to 74^{pcf}, stop Drlg @ 1960^{m} due to observe 30% Chert in geologist sample & POOH to surface with 2-3 BPH loss to change BHA, P/U new 12-1/4" B.Bit & RIH to BTM, commence Drlg f/1960 to 2010^{m}, B/R & R/D each drilled Std, continue Drlg to 2115^{m} with 3-8 BPH loss while pumping Hi-Vis Pill periodically, suddenly lost 60^{klbs} SW & 500^{psi} dropped due to twist of string f/mandrel

of 8" D.Jar (fish left in hole: 12-1/4" B.Bit + B.Sub + 2 Jts 8-1/2" DC + 12-1/4" S.Stab + 10 Jts 8-1/2" DC + 9.8^{m} of 8" D.Jar. LOF: 119.5^{m}), POOH to surface & try to catch fish w/overshot & recover the fish successfully by 30^{kllbs} over-pull, RIH w/12-1/4" B.Bit, try to make Circ. but string was plug, try to unplug string several times without succeed, POOH to surface, string plugged f/first DC above B.Sub by cutting & o-ring of D.Jar, unplug same, RIH back to BTM, commence Drlg f/2115^{m} w/3-7 BPHL. Continue Drlg to 2173^{m} w/2-6 BPHL. CBU & POOH to 2169^{m}, run totco survey at 2169^{m} (Inc: 0.75 deg, 2-3 BPHL.), RIH to BTM & continue Drlg to 2191^{m} w/2-6 BPHL, due to change bit start POOH to surface, P/U 12-1/4" PDC bit & RIH w/Drlg BHA to BTM, commence Drlg w/low parameters to 2197^{m} while add KCl Solution in active mud Sys. f/2% to 3% w/2-3 BPHL, suddenly loss 30^{klbs} SW & pressure drop due to twist off string f/pin of 8-1/2" DC (fish left in hole: 12-1/4" PDC bit + 8-1/2" B.Sub + 2 Jts 8-1/2" DC + 12-1/4" S.Stab + 6 Jts 8-1/2" DC, LOF: 73^{m}), POOH to surface & RIH w/overshot, recover the fish w/20^{klbs} over-pull successfully, POOH to surface & L/D same, P/U new PDC bit & RIH to BTM, commence Drlg to 2491^{m} while increasing Mw f/75 to 77^{pcf} stepwise w/2-8 BPHL, Cir. out 20^{bbl} hi-vis pill, POOH to 2160^{m} as wiper trip w/good condition, RIH back to 2480^{m}, run totco survey (Inc: 1 deg, 1-3 BPHL.), RIH to BTM & continue Drlg to 2850^{m} w/high TQ & TQ fluctuation in some intervals, 6-10BPHL, continue Drlg to 2854^{m} w/8-10 BPHL, in order to wiper trip B/R last stand f/BTM to 2830^{m}, due to carelessness of night driller & apply 170^{klbs} abnormal over-pull without permission, suddenly lost 120^{klbs} SW & left in hole & burst TDS hydraulic hose because of jumping TDS, M/U Circ. swage on string establish Circ. w/zero pressure, POOH to surface & observed string was twist off f/above shoulder of pin of 5" DPs, (fish left in hole: all BHA + 1854^{m} 5" DPs "G", LOF: 2049^{m}), RIH w/overshot to TOF, try to catch fish & recover w/180^{klbs} excess weight, due to abnormal drag pump 40^{bbl} brine & 30^{bbl} hi-vis pill & displace brine around BHA, pump out w/overshot & recover whole fish successfully w/6-8 BPHL & L/D same, P/U PDC bit & RIH to 2852^{m} w/good condition, took weight & stall string at several depth w/3-5 BPHL, POOH to 2842^{m} & run totco survey (Inc: 1 deg, 1-3 BPHL.), run back to BTM & commence Drlg f/2854^{m} to 2900^{m}, B/R & R/D each stand w/3-6 BPHL, continue Drlg to 2919^{m} w/low progress, due to low ROP stop Drlg & POOH to surface to check the bit w/5-7 BPHL, P/U 12-1/4" TCI bit & 9-5/8" Junk Sub & start RIH to BTM, Circ. out 30^{bbl} hi-vis pill, work on junk several times, resume Drlg f/2919^{m} to 2923^{m} w/low parameter due to bit bouncing & 5-7 BPHL, while increasing Mw f/79.5 to 80^{pcf}, due to change BHA start POOH to surface w/3-5 BPHL & L/D same, P/U new PDC bit, commence RIH to 2918.5^{m}, wash & R/D to 2923^{m} while pumping 40^{bbl} hi-vis pill w/2-4 BPHL, commence Drlg to 2944^{m} w/3-6 BPHL while pumping 25^{bbl} hi-vis LCM pill periodically & increasing Mw f/80 to 80.5^{pcf}, continue Drlg to 2945^{m} w/low parameters, due to low progress commence POOH from BTM w/8-10 BPHL, stuck string at 2503^{m}, got free by apply weight downward, continue POOH to surface w/4-6 BPHL & L/D same, P/U 12-1/4" TCI bit & 2 No. of 9-5/8" Junk Sub, start RIH to BTM & Circ. out 30^{bbl} hi-vis pill w/2-4 BPHL, work on junk several times, resume Drlg to 2961.5^{m} w/2-7 BPHL, due to low ROP stop Drlg & CBU, work on junk & start POOH to surface w/2-5 BPHL & L/D bit & Junk Sub, recover 22 gr junk, P/U PDC bit & RIH Drlg BHA to BTM, commence Drlg w/low parameters to 3069^{m} & 5-8 BPHL while pumping 70^{bbl} low-vis/hi-vis pill, continue Drlg to 3275^{m} w/3-8 BPHL, while pumping hi-vis LCM pill periodically, B/R & R/D each drilled Std, B/R f/BTM to 3245^{m}, POOH to 3180^{m} as condition trip w/good condition, f/3180^{m} to 3100^{m}

hard B/R & observed over-pull at 3152[m] & 3125[m], POOH to 2960[m] w/good condition, RIH back to BTM & continue Drlg to 3350[m] w/4-10 BPHL while increasing Mw f/80.5 to 82[pcf] & add KCl 0.5% to 5%, continue Drlg to 3497[m] w/5-10 BPHL while increasing Mw f/82 to 83[pcf] while pumping hi-vis LCM pill periodically, continue Drlg f/3497[m] to 3552[m] w/5-10 BPHL, suddenly lost 40[klbs] SW, start POOH to surface, string twist off observed f/shock sub of 8" D.Jar (fish left in hole: 12-1/4" PDC bit + B.Sub + 2 Jts 8-1/2" DC + 12-1/4" S.Stab + 10 Jts 8-1/2" DC + 8" Flex Jt + 23[cm] of shock sub, LOF: 87.36[m]), RIH w/overshot, try to catch fish & recover all components of fish successfully w/40[klbs] excess weight, RIH back w/same PDC bit to 1000[m] w/4-8 BPHL, due to bad weather stop operation while slow Circ. for 4.5[hrs], continue RIH to BTM w/good condition, 4-8 BPHL, run totco survey at 3546[m] (Inc: 1 deg, 4-8 BPHL), commence Drlg f/BTM to 3656[m] w/low parameters & hi TQ while increasing Mw f/83 to 84[pcf], continue Drlg to 3910[m] w/HI TQ & 5-10 BPHL while increasing Mw f/84 to 85[pcf], continue Drlg to 3934[m] (CSG point) w/controlled parameters & 5-10 BPHL, in order to wiper trip commence POOH f/BTM, string took weight at 3854[m] w/maximum 30[klbs], work on stuck downward max 50[klbs], string got free, B/R & R/D several times. POOH w/good condition to surface w/5-8 BPHL, perform Full Suite Log f/3934[m] to 1492[m] & GR Log f/300[m] to surface, perform LDT/CNT/NGRS Log f/BTM to 1492[m], run 9-5/8" CSG (53.5#, TP110S, TPCQ) to 3934[m], install related wellhead & Acc. & perform pressure test.

8-1/2” Hole Section Summary

Date	10 – September - 2023
Drilling Depth	4200[m]
Liner Size / Lap, Shoe	7" 32#, TP110S, TPCQ / 3812[m] – 4189.5[m]
Formation	Fahliyan
Mud Weight	100 – 104[pcf]

Table 2-3: AZNS-040 8 – 1/2" Hole Section Summary

This section was drilled at 10-Sep-2023 with 8-1/2" TCI bit f/3934[m] to 3939[m] w/low parameters, start POOH to surface, P/U 8-1/2" B.bit & 9-5/8" scraper & Junk Sub, RIH to BTM & work on junk while scrap f/3885[m] to 3932[m] several times, commence POOH to surface & L/D same, perform CCL/GR/SBT/VDL wireline logging f/3928[m] to 1385[m], P/U & M/U new PDC bit & RIH w/Drlg BHA to 3923[m], perform FIT by apply 2100[psi] stepwise for 15min. displace the well w/limestone mud 100[pcf], RIH to BTM & commence Drlg to 4070[m] w/low parameters while increasing Mw f/100 to 104[pcf] stepwise due to TQ & gas to 92%, B/R & R/D each drilled Std, 1-3 BPHL, continue Drlg to 4200[m] (TD) w/1-4 BPHL, commence POOH to surface w/1-3 BPHL, perform Full Suite wireline logging as run #1 f/TD to 3933[m] (CSG Shoe) successfully, LDT/CNT/NGS as run #2 to 3958[m] w/no succeed due to stuck wireline logging tools,

run new LDT/CNT/NGS logging tools as run #3 f/TD to 3933^{m} successfully, perform LDT/NGS wireline logging as run #4 f/TD to 3900^{m} successfully, run 7" liner w/PKR (32#, TP110S, TPCQ) f/3812^{m} to 4189.5^{m}, scrap to L.lap & perform pressure test, dry test was not OK, RIH & set SN packer on 7" L.lap & perform dry test was OK.

First Liner Lap Dry Test Report

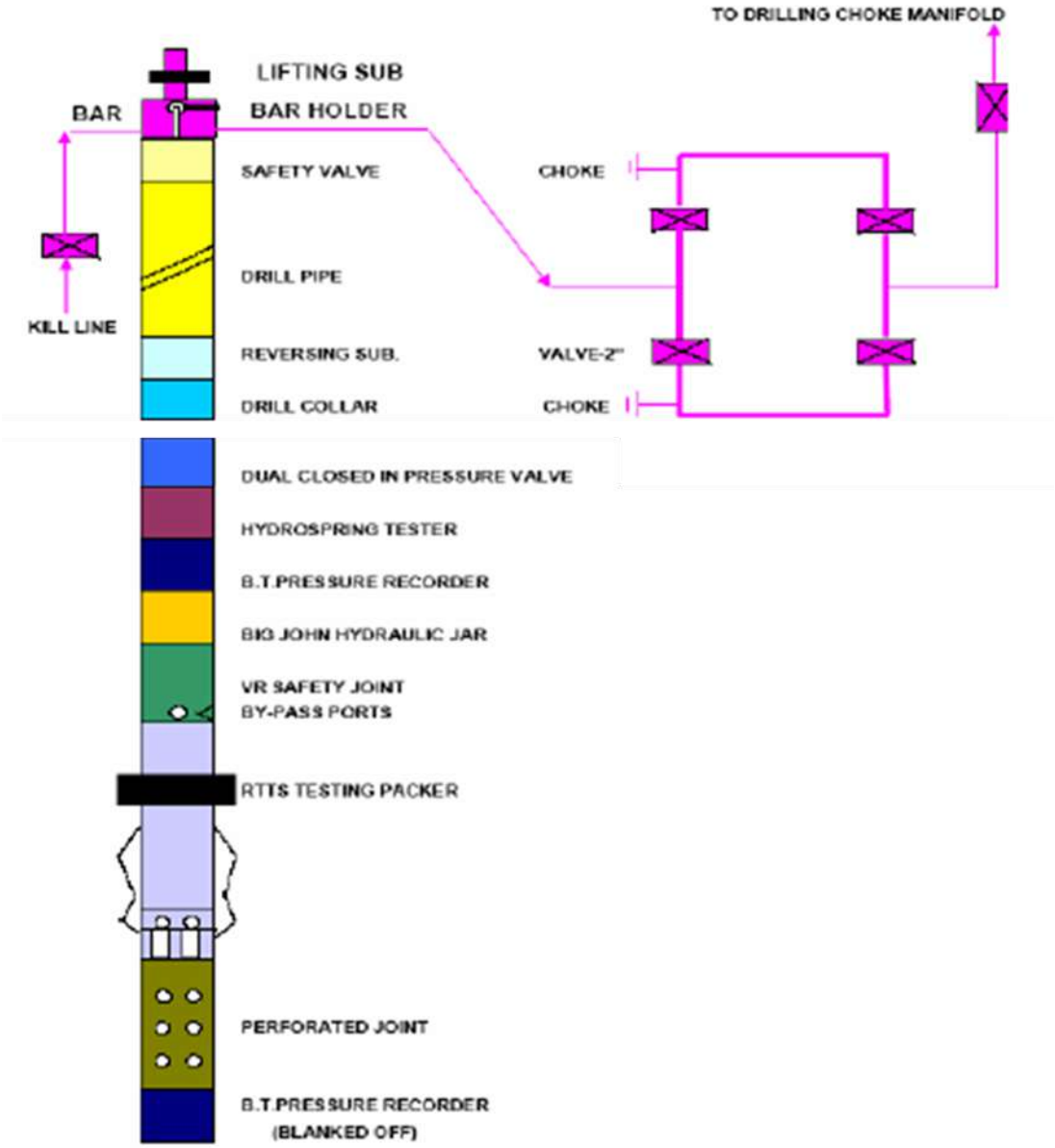
7” Liner Data
32 ppf Weight
110ss Grade
5.969” Drift
TP-CQ premium connection

Table 2-4: AZNS-040 7" Liner Information

Exposed Formation Characteristics	
Datum Depth: 4000 m TVD	
Datum Pressure: 9300 psi (0.29 psi/ft)	
Cushion Type	Fresh Water & S.KCl.Poly.Lst Mud
Cushion Density	62.4 & 104
Packer Depth	3792^{m}
Datum Depth(4000^{m}) Pressure	9300^{psi}
Packer Depth Pressure	8807^{psi}
7” Liner Lap Pressure	8855^{psi}
Drive Force	2000^{psi}
Cushion Depth	500^{m} Water & 2573^{m} Mud

Table 2-5: AZNS-040 First Dry Test Information

DRILL STEM TEST STRING



TIME	EVENTS
27-Sep-23	
10:00	Connect Memory Gauges battery
13:10	RIH DST tools
28-Sep-23	
21:06	Set packer
21:08	Hydro Spring opened (Open the well)
29-Sep-23	
00:25	DCIP closed (Close the well)
06:25	Drop the Bar and start circulating
07:39	Unset Packer
08:48	POOH DST String
21:59	String at Surface
30-Sep-23	
00:35	Disconnect Memory Gauges
	End of job

EVENTS TABLE

Calendar (yyyy/mm/dd hh:mm:ss)	Elapsed time (hours)	Pressure (psia)	Temperature (degF)	Comment
2023/09/27 10:00:01	0.00028	14.869	75.503	Connect Memory Gauge
2023/09/27 13:16:32	3.27556	19.626	97.632	Start RIH DST String
2023/09/28 21:06:02	35.10056	8957.173	241.186	Set Packer
2023/09/28 21:08:12	35.13667	8981.000	241.186	Open the Well (Open Hydrospring)
2023/09/28 21:08:22	35.13944	6820.022	241.125	Hydrospring at Open Position
2023/09/29 00:25:02	38.41722	6779.555	241.593	Close the Well fromDCIP
2023/09/29 07:38:52	45.64778	7126.281	242.268	Unset Packer
2023/09/29 08:48:32	46.80889	9056.177	242.312	Start POOH DST String
2023/09/29 21:59:02	59.98389	16.397	105.264	DST String at Surface

Figure 2-1: AZNS-040 First Dry Test Event

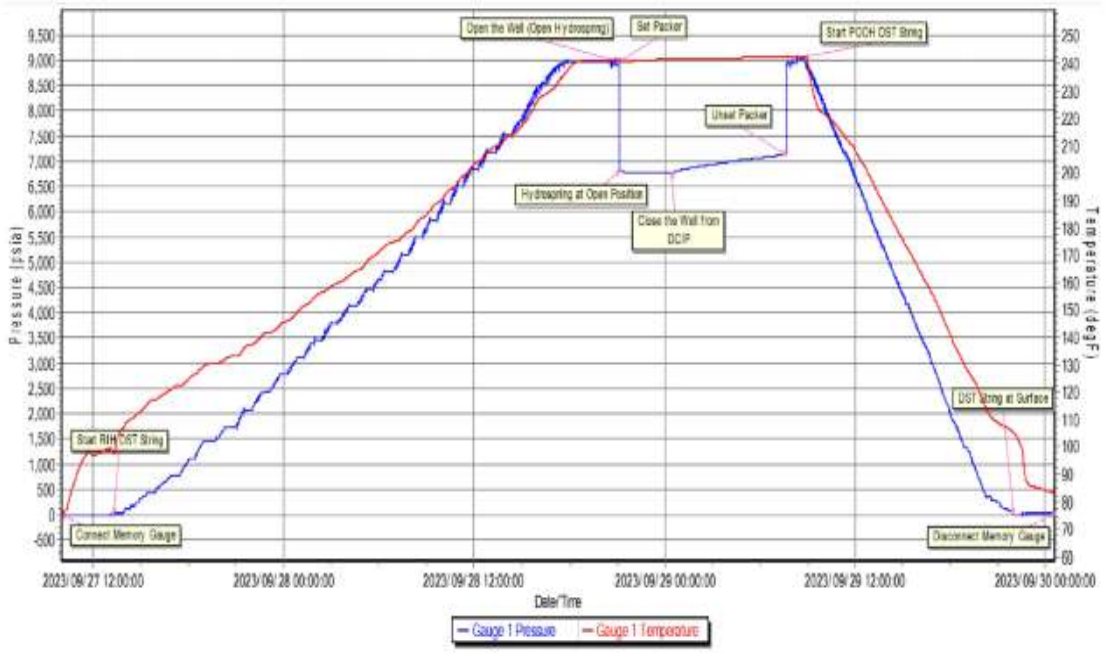


Figure 2-2: AZNS-040 First Total Job Chart

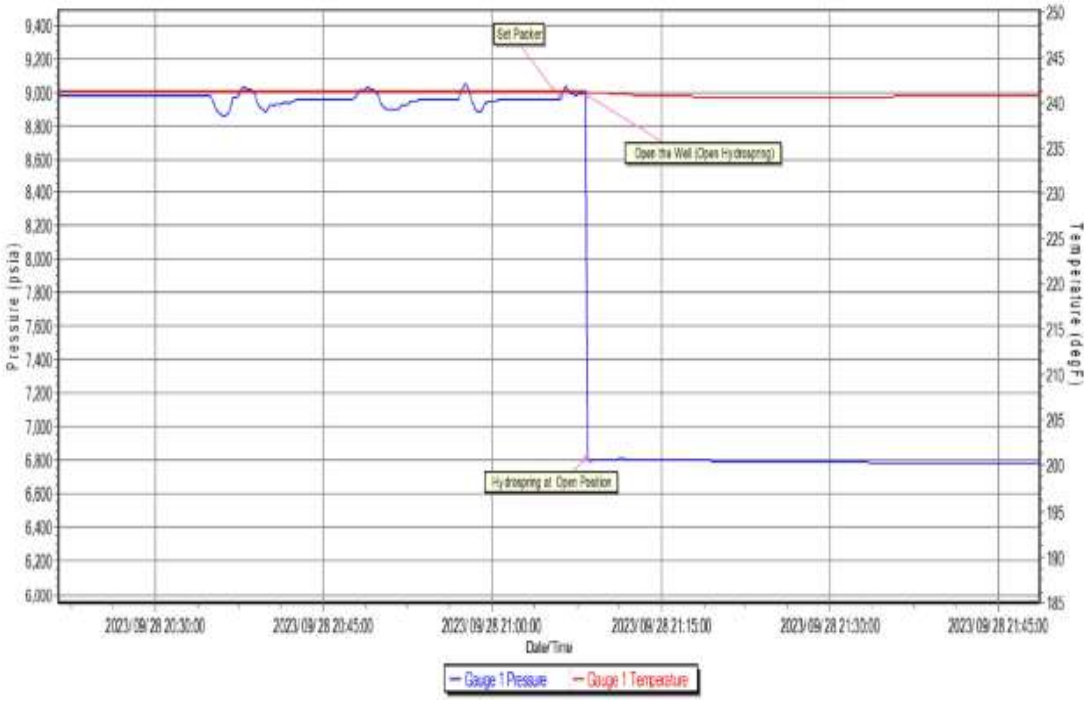


Figure 2-3: AZNS-040 Set Packer & Open the Well Chart

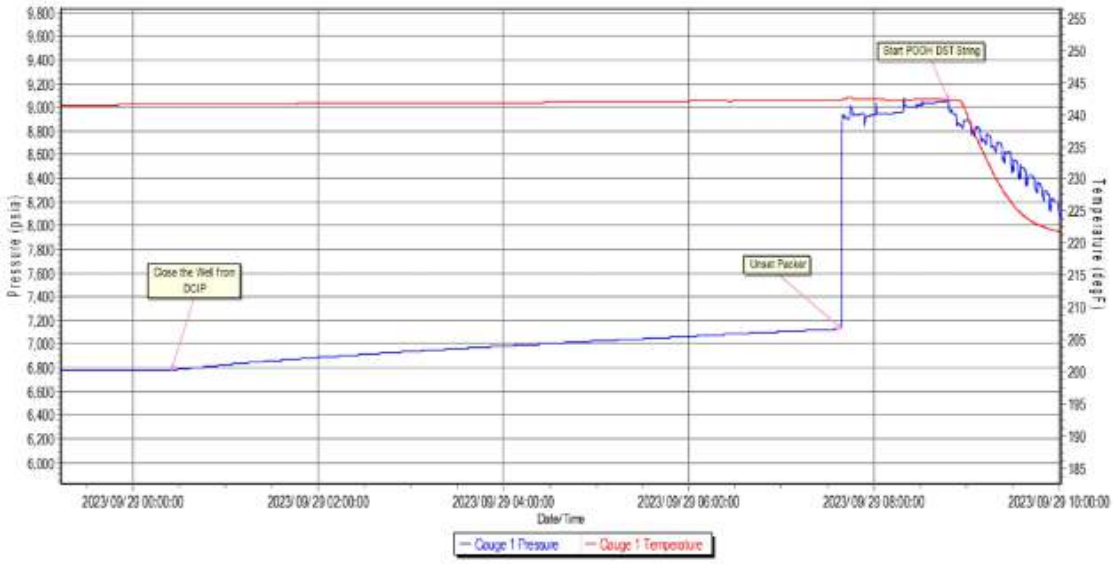


Figure 2-4: AZNS-040 Close the Well – Bottom BT

Row	Description	Time	Bottom Bt.	Middle Bt.
1	Initial Mud Static Pr. (psi)		8980	8964
2	Open Well Pr. (psi)	1402/07/06 21:09	6800	6782
		1402/07/07 00:17	6779	6765
3	Close Well Pr. (psi)	1402/07/07 00:21	6779	6762
		1402/07/07 07:38	7126	7112
4	Final Hydrostatic Pr. (psi)		8940	8924

Table 2-6: AZNS-040 First Downhole Test Result Data

Second Liner Lap Dry Test Report

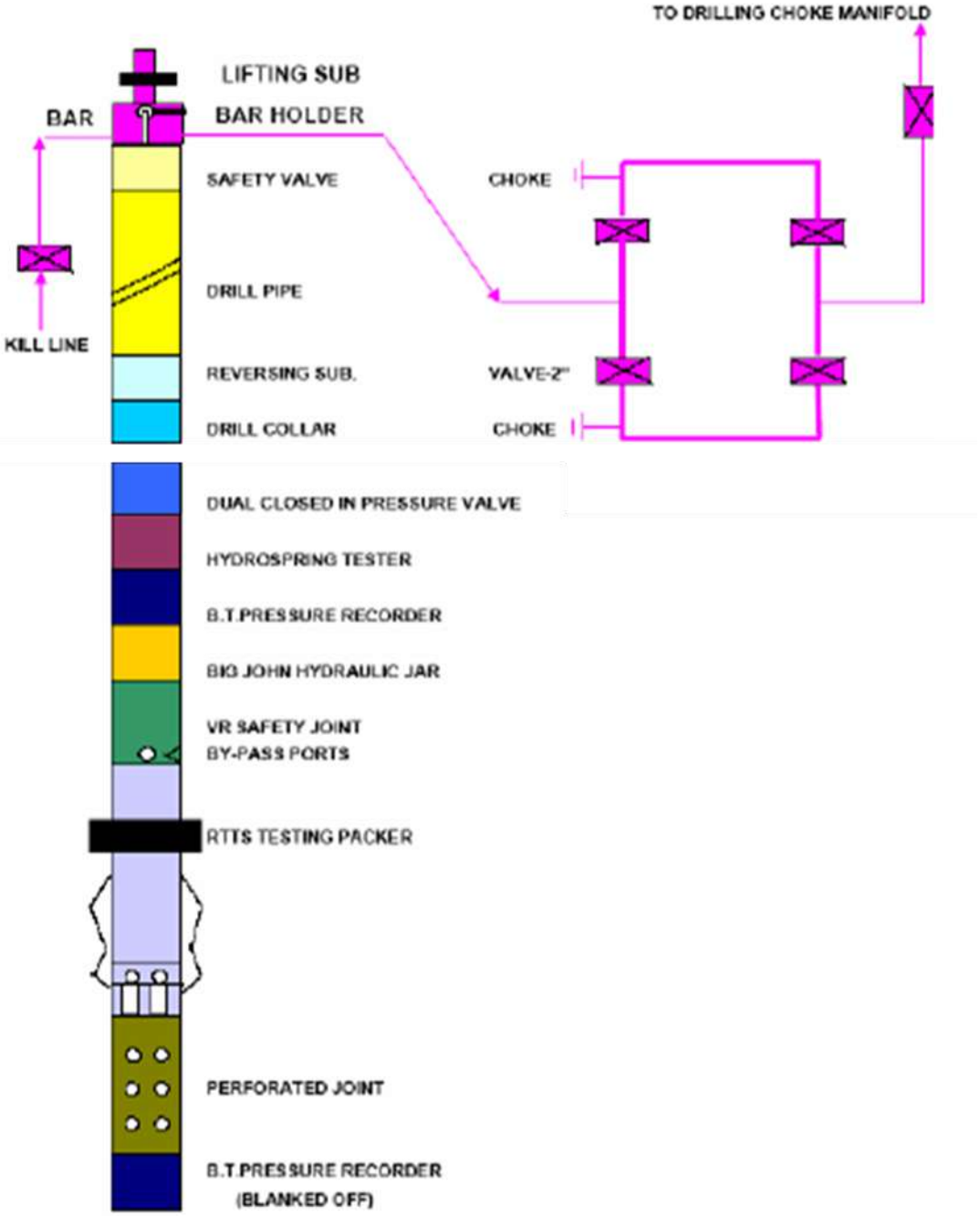
7” Liner Data
32 ppf Weight
110ss Grade
5.969” Drift
TP-CQ premium connection

Table 2-7: AZNS-040 7" Liner Information

Exposed Formation Characteristics	
Datum Depth: 4000^{m} TVD	
Datum Pressure: 9300^{psi} (0.29 psi/ft)	
Cushion Type	Fresh Water & S.KCl.Poly.Lst Mud
Cushion Density	62.4 & 104
Packer Depth	3792^{m}
Datum Depth(4000^{m}) Pressure	9300^{psi}
Packer Depth Pressure	8807^{psi}
7” Liner Lap Pressure	8855^{psi}
Drive Force	2000^{psi}
Cushion Depth	500^{m} Water & 2573^{m} Mud

Table 2-8: AZNS-040 Second Dry Test Information

DRILL STEM TEST STRING



TIME	EVENTS
03-Oct-23	
13:50	Connect Memory Gauges battery
22:52	RIH DST tools
04-Oct-23	
00:00	Continue RIH
05-Oct-23	
04:47	Set packer
04:49	Hydro Spring opened (Open the well)
07:55	DCIP closed (Close the well)
13:55	Drop the Bar and start circulating
14:35	Unset Packer
15:37	POOH DST String
06-Oct-23	
09:21	Disconnect Memory Gauges
	End of job

EVENTS TABLE

Calendar (yyyy/mm/dd hh:mm:ss)	Elapsed time (hours)	Pressure (psia)	Temperature (degF)	Comment
2023/10/03 13:50:01	0.00028	14.734	80.844	Connect Memory Gauge
2023/10/03 22:52:32	9.04222	15.544	93.895	Start RIH
2023/10/05 04:47:32	38.95889	8977.910	244.345	Set Packer
2023/10/05 04:49:12	38.98667	8984.226	244.346	Open the Well
2023/10/05 04:49:32	38.99222	6900.679	244.246	Hydrospring Opened
2023/10/05 07:55:02	42.08389	6875.082	244.191	Close the Well from DCIP
2023/10/05 14:35:22	48.75611	6993.380	244.664	Unset Packer
2023/10/05 15:37:52	49.79778	9037.895	245.384	Start POOH DST String
2023/10/06 08:54:02	67.06722	15.489	98.303	DST at Surface

Figure 2-5: AZNS-040 Second Dry Test Event

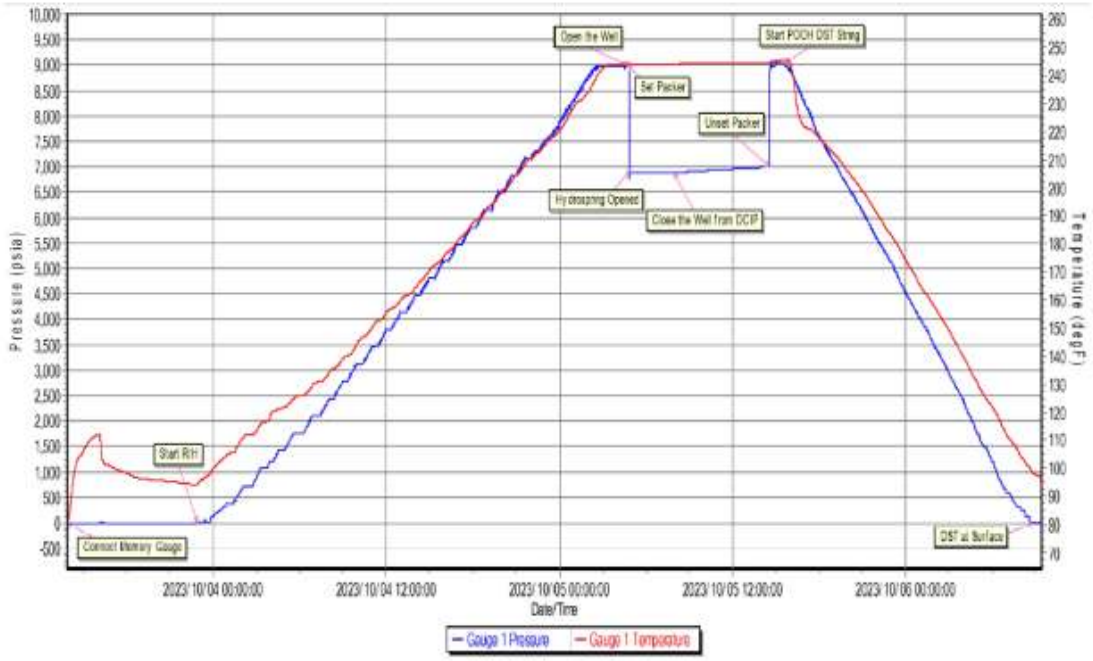


Figure 2-6: AZNS-040 Second Total Job Chart

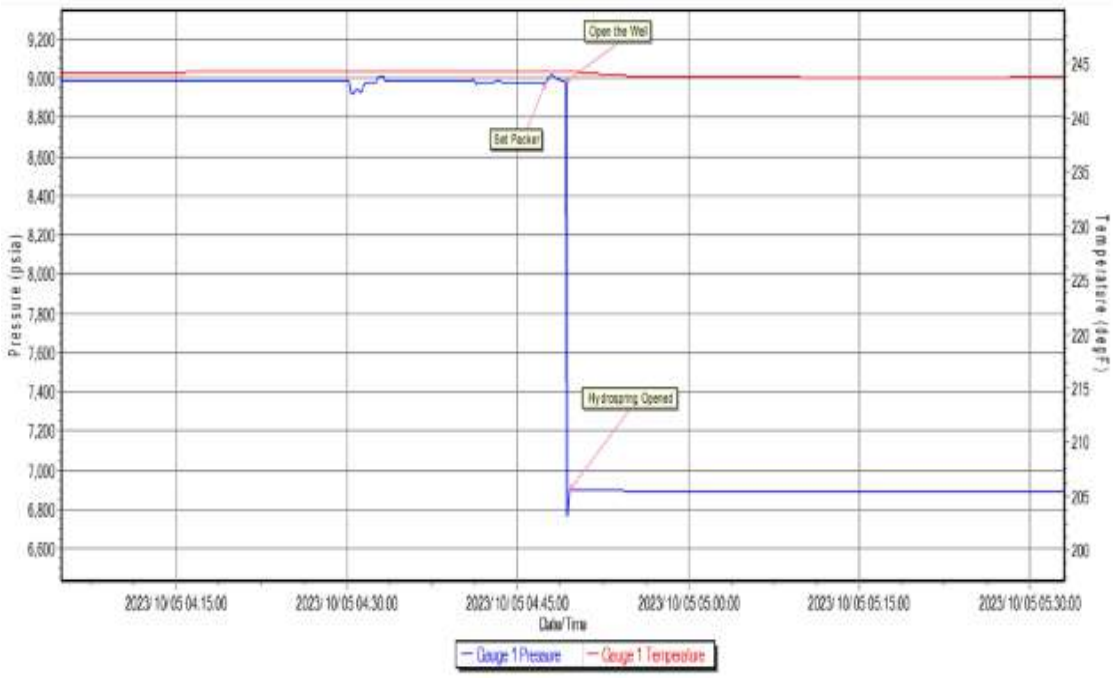


Figure 2-7: AZNS-040 Set Packer & Open the Well Chart

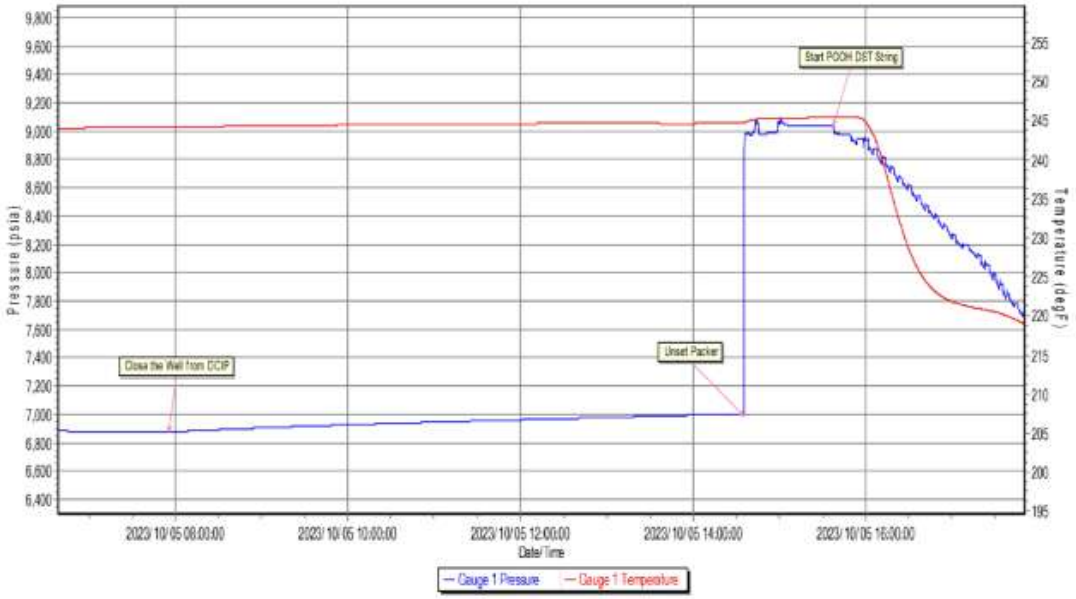


Figure 2-8: AZNS-040 Close the Well – Bottom BT

Row	Description	Time	Bottom Bt.	Middle Bt.
1	Initial Mud Static Pr. (psi)		8982	8968
2	Open Well Pr. (psi)	1402/07/13 04:50	6896	6880
		1402/07/13 07:50	6877	6862
3	Close Well Pr. (psi)	1402/07/13 07:51	6875	6859
		1402/07/13 14:35	6993	6961
4	Final Hydrostatic Pr. (psi)		8978	8958

Table 2-9: AZNS-040 Second Downhole Test Result Data

Completion Summary

RIH wireline w/4.5" gun & CCL tools as dummy run to 4140^{m} while correlate depth, POOH to surface & R/D same, perform perforation operation in #7 runs (perforation interval f/4039^{m} to 4112^{m}), RIH w/ 5-7/8" M.M J.mill & clean-out BHA to 4154^{m}, POOH to surface, perform CCL/GR logs f/4130^{m} to 3900^{m}, as client request log repeat in same interval, POOH & R/D wireline logging equipment, RIH w/ 5-7/8" M.M J.mill & clean-out BHA to 4145^{m}, CBU w/ 50 GPM while decreasing Mw f/104 to 102.5^{pcf}, POOH to surface, rig was waiting due to shortage of completion tubing from 13-Oct to 25-Nov, decide to open BOP rams & POOH 7" scraper while L/D tubulars, due to inconsistency driller start to POOH string to 530^{klbs} over-pull while top rams still closed, suddenly 5" DP Jt cut 2^{m} below RT & string drop into the well, (fish left in hole: 3 Stds 4-3/4" DC + 2 Stds 3-1/2" HWDP + 4-3/4" D.pin + 7" Scraper + 4-3/4" B.Sub + 1 Std 3-1/2" HWDP + 10 Stds 3-1/2" DPs + XOS 3-1/2" * 4-1/2" + 43 Stds & 1 Jt 5" DPs, LOF: 1700^{m}), P/U overshot & fishing BHA to TOF at 2454^{m}, engage with fish & commence POOH f/5415^{m} to 3800^{m}, observe bubble, close top rams & establish Circ. thru C/M to uniform completion fluid, observe oil on shakers, start killing well from 27-Nov to 29-Nov, commence POOH overshot & fishing BHA f/3500^{m} to surface, 2-3 BPHL, release overshot & L/D fishing BHA & resume POOH rest of the fish w/good condition, RIH w/ 5-7/8" M.M J.mill unsuccessfully due to mill plugged w/debris & pipe coating, unplug same & RIH to 4145^{m}, POOH M.M J.mill to surface, perform pressure test BOP was OK, RIH w/ 7-5/16" polish mill to 3800^{m}, CBU till uniform mud to 103^{pcf}, W/D to 3811.4^{m} & tag LG of 7" SN packer, perform static flow check, commence POOH to surface & L/D polish mill, RIH w/8-1/4" dress mill to 3800^{m}, W/D to 3809.6^{m} (7" L.lap), POOH to surface, perform static flow check at different depths, L/D dress mill, P/U & RIH w/5-7/8" M.M J.mill to 3985^{m}, Circ. mud w/100 GPM for 57.5^{hrs}, commence POOH to surface & P/U same M.M J.mill w/7" scraper & run back to 4010^{m} while scrap inside 7" liner f/3850^{m} to 4010^{m}, POOH & Circ. mud w/100 GPM at 3985^{m} for 21^{hrs}, commence POOH to surface & L/D same, P/U & RIH w/7" RTTS packer, set RTTS PKR at 3860^{m} & pressure test same to 500^{psi} was OK, perform 7" L.lap positive test to 2000^{psi} stepwise & hold pressure for 40 min., unset PKR & POOH to surface & L/D same, RIH w/4-3/4" OEDP & 7" scraper to 3780^{m} & break Circ. every 15 Stds, Circ. mud w/110 GPM for 126.5^{hrs}, resume RIH w/same BHA & scrap f/3780^{m} to 4013^{m} several times, L/U CMT to the string & pressure test same f/500 to 4000^{psi}, was OK, start displace the well w/104^{pcf} completion fluid by 850^{bbl}, commence POOH OEDPs & scraper to surface & L/D same, perform BOP pressure test was OK, P/U & M/U completion string to 33^{m}, R/U Slick line & run 2.75" GS pulling to No-Go depth & check all connections, OK, install 2.75" Check valve in SSD, continue RIH completion string to 262^{m}, R/U Slick line & run 2.8" gauge ring to SSD depth, R/D Slick line equipment & L/U test line to string pressure & test string against check valve to 7000^{psi} Ok, continue

RIH completion string to 3890[m] while pass 7" L.lap without any drag, L/U test line to string pressure & test string against check valve to 7000[psi] stepwise every 50 Jts, was Ok, P/U & M/U 3SV Assy on string, connect control line to TR3SV & pressure test same to 8000[psi], OK, bleed off pressure to 5000[psi], M/U TBG & RIH to 3904[m], L/U test line to string, pressure test string & TR3SV Assy against check valve to 7000[psi] stepwise, OK, bleed off control line pressure to Zero, bleed off pressure above flapper to 1000[psi] & monitor for any pressure build up as inflow test of flapper, Negative, operate TR3SV & check equalizing function, OK, continue RIH completion string f/3904[m] to 3942[m], apply pressure in CILs & rupture both CIVs rupture disk, check string, observe return (indication of successful rupture), M/U 4-1/2" Fluted Swage & TBG HGR, 4-1/2" Running & Pulling tools for Tubing hanger & L/J on HGR, pass CILs & CL thru HGR, cut CILs & CL 1.5[m] above TBG HGR, continue RIH on L/J to 3957[m], check & land TBG HGR in THS, OK, POOH to 3955.5[m], Space out 1.5[m] above THS, L/U test line to string pressure & test string against check valve to 7000[psi] stepwise Ok, R/U Slick Line equipment, install 3.437" Protection Sleeve in TR3SV, retrieve 2.75" WFB-2 check valve f/SSD, run & set 2.75" WRB-2 Check valve in No-Go L/N, POOH to surface & L/D slick line equipment, L/U test line to string, pressure test line f/500 to 7000[psi], OK, set PKR with max 3500[psi] (Observe Shearing indication at 2000[psi]), increasing pressure to 7000[psi] step by step & hold for 15 min, OK, bleed off pressure, perform pull test with 15[klbs] over-pull, OK, land TBG HGR in THS & observe return from string as an indication of travelling Jt shearing, close Tie Down screws, close annular BOP, pressure test TBG HGR & Tie Down screws to 2000[psi], OK, pressure test string to 5000[psi], OK, L/U to annulus & pressure test to 2000[psi], OK, R/U slick line equipment, RIH & retrieve 2.750" check valve f/No-Go, retrieve 3.437" Protection sleeve f/TR3SV, R/D Slick Line equipment, back off L/J, install 4" TWCV in TBG HGR, close SBR & Try pressure test 4" TWCV against SBR, observe leakage f/ram block packer, operate SBR several times & pressure test same, No-Succeed, P/U & M/U 11" Plug tester & 5" DP, RIH in BOP Stack, close top rams, pressure test 4" TWCV against top rams to 2000[psi], OK, L/D 11" Plug tester & 5" DP, change ram block f/SBR to Blind Rams, close Blind rams, pressure test 4" TWCV to 7000[psi], OK, N/U 11"*4 1/16"-10K BTM master valve, while P/U BMV connect & pass 1/4" CL & 3/8" CIL to BMV flange, in order to pressure test TBG HGR metal seal L/U pump truck to BMV, pressure test TBG HGR metal seal to 8000[psi], OK, pressure test 1/4" CL to 8000[psi] for 15min, OK, pressure test 3/8" CILs to 2000[psi], OK, pressure test F-Seal & ring gasket between T.H.S & BMV to 8000[psi] for 15 min, OK, N/U 4-1/16"-10K X-mass tree solid block (MSP - Drilex) & install same on BMV, change positive choke valve direction, tighten all bolts, L/U pump truck & body pressure test X-Mass Tree & BMV against 4" TWCV to 1000-8000[psi] stepwise for 5-15 min, OK, retrieve 4" TWCV f/TBG HGR, close all X-Mass Tree valves except BTM Master valve & Swab valve, L/U separator lines to flare line & oil storage manifold, L/U pump truck to XMT, L/U downstream equipment, R/U well test equipment, press test to pump truck lines, observe, leakages, work on rectify leakages, pressure test pump truck lines to Kill

valve, OK, perform pressure test f/wing valve to ESD, OK, try to pressure test SWT several times, No-Succeed, while observe leakages from data header, needle valve 3"line connections up choke gate valves, pressure test up choke (against kill valve to up choke) to 6500^{psi}, Ok, try to pressure test down choke valve to 6500^{psi}, No-Succeed, while work on rectify leakages of down choke valves, change C/M 2" G.Valve right side behind adjustable choke, continue try to P/T down choke valve, No-Succeed, observe leakage C/M 2" G.Valve right side behind adjustable choke, change 2" G.Valve, perform C/M downstream valves P/T Stepwise to 6500^{psi} for 15min, Ok, perform ESDV function test, Ok, perform separator & separator inlet & outlet valves P/T stepwise to 500^{psi} for 15min, Ok, perform Separator bypass to gas manifold P/T stepwise to 500^{psi} for 15min, Ok, perform Separator bypass to surge tank inlet valve P/T Stepwise to 500^{psi} for 15min, Ok, perform dynamic test of oil storage tank hoses, Ok, perform green burner & hydro test of Burning pit oil line function test by pump truck, observe leakages, work on leakages by welding connection & pressure test same, OK, function test green burner System, OK, R/U Slickline equipment & BOP & lubricator while open Btm & Top master valve, 2SV, Swab valve & L/U to well test C/M, apply 6000^{psi} in control line, open TR3SV & monitor pressure for 15 min; OK, close Swab valve, retrieve X-Mas tree Cap, press test slickline Equipment to 6500^{psi}, Ok, run 3.3" Gauge ring to TR3SV depth & POOH to surface, RIH w/Protection sleeve running tools & set PS in TR3SV, POOH to surface, RIH w/2.2" gauge ring to 4023^{m}, took Wt & try to pass @ this point, No-Succeed, POOH to Surface, R/D Slickline equipment, R/U 1-1/2" CT BOP & Injector Head, perform pull test CT reel w/10^{klbs}, OK, perform CT blind ram BOP press test to 6500^{psi}, observe leakage several points, work on leakages & pressure test same w/6500^{psi}, OK, perform CT body & stripper BOP pressure test to 6500^{psi}, observe drop pressure several times, work on same, Ok, perform CT pipe ram BOP & check valve pressure test to 6500^{psi}, Ok, prepare & pump 5bbl Acid as pickling thru CT reel, displace reel w/26^{bbl} (CT reel Cap) $CaBr_2$ & got Acid in IBC tank, RIH 1-1/2" CT to 4000^{m}, displace well w/NaCl brine 75^{pcf}, POOH CT to 1000^{m} while apply back pressure through well test C/M (200-2800^{psi}), observe up stream pressure decrease f/2600 to 300^{psi} w/Choke size 16/32", return on surface NaCl brine 75^{pcf}, clean up the well on well test C/M size f/ 20/64" to 24/64" & got 60^{bbl} NaCl brine to D.sander tank for 53.5^{hrs} & R/D CT injector head, R/U Slickline service, perform pressure test to 6000^{psi}, observe leakages in several points, Work on leakages & pressure test same w/6000^{psi}, OK, run 1-1/2" memory gauge to 4029^{m}, take flowing survey, POOH to surface & R/D Slickline, R/U CT service, perform Equipment pressure test w/ 6500^{psi}, Ok, F/U CT reel w/26^{bbl} old Diesel, commence RIH 1-1/2" CT to 4117^{m} while pump 20^{bbl} old Diesel, spot 50^{bbl} old Diesel, POOH CT to 4030^{m}, inject 20^{bbl} fresh Diesel (0.3-0.2 bpm, 5700-6000^{psi}) as injectivity test, RIH CT to perforated interval, perform acidizing job & displace CT reel by 26^{bbl} old Diesel, bullhead 43^{bbl} Diesel w/ 3-5 bpm, 5500 to 6000^{psi}, clean up the well on choke size 8/64" to 48/64" for 18^{hrs} while POOH CT to surface, pump 20^{bbl} fresh Diesel thru CT reel, stop clean up the well, R/D CT injector head, R/U Slickline service, perform Slickline

equipment pressure test to 6000[psi], Ok, continue clean up the well on Fix choke size 8/64" to 24/64" for 15[hrs] till constant Up stream Press for take flow rate & run memory gauge to 4032[m], take flowing survey, POOH to surface, observe high amount of petroleum asphalt inside slickline BOP & stafing box, R/D Slickline equipment & N/D lubricator, N/U CT riser & R/U equipment, perform CT BOP pressure test w/6500[psi], OK, open the well, commence RIH w/ 1-1/2" CT to 4030[m] while pump 45[bbl] diesel, spot & bullhead 20 ADR, displace the well w/274[bbl] diesel while POOH CT to surface, bullhead 20 diesel by kill line, R/U Slickline & perform equipment pressure test, Ok, retrieve protection sleeve, R/D Slickline equipment, Perform X-mass tree valves inflow test, perform Top master valve & 2SV & Swab valve & Production valve & Kill valve inflow test, Ok, close TR3SV & perform TR3SV inflow test, Ok, N/U Lubricator & pressure test w/ 5000[psi], OK, set BPV in TBG HGR, N/D Lubricator, install Choke body as per client instruction, close all X-mass tree valves except BMV & SSV & secure wellhead, RELEASE THE RIG @ 12:00, 14-DEC-2023.

2.4 AZNS-040 Well Montage & Well Sketch

General Information:
Well Name: AZNS-040
Pad No.: 50
Well Type: IIF
Well Progile: Vertical
Block: II
Field: South Azadegan
Country / Area: Iran / Khuzestan
Reservoir: Fahlian

Wellhead Coordinate:
Grid Geodetic System: (UTM39N, WGS84)
E: 194,235
N: 3,450,715

Nearby Wells:
AZN-010
AZNS-045 ST1
AZNS-127
AZNS-017 (KZ-Horizontal)

Parties/Refrence:
Client: PPL/PEDEC
Contractor: MSA/OEOC
Rig Cont: OEOC
Rig: OEOC 201

Well Montage

AZNS-040 Re-Entry

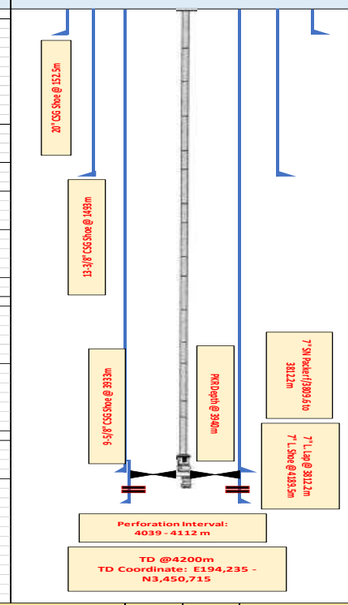
(PAD-50), IIF

Drilling Depth:
TD (MD): 4200m
TD (TVD): 4200m
GLE: 3.95m
RTE: 14.45m

Timing:
Moving Days: 10 Days
Spud Date: 13-07-2023
Total Rig Days: 155
Rig Released Date: 14-12-2023

Field Location/Objective:
***Field Location:** The AZNS oilfield is located in Khuzestan province, approximately 80 Km to the west of Ahwaz city.
***Well Objective:** De-suspend the well, vertical drilling to Fahliyan formation, complete well and produce oil from Fahliyan reservoir..

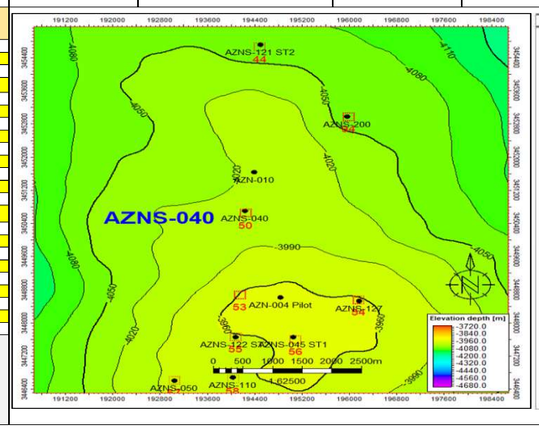
Formation	Formation TOP MD (M)	Formation TOP TVD ss (M)	Formation Thickness (M)
Aghajari	Surface	3.95	1123.1
Gachsaran	1123.1	-1108.6	372.2
Asmari	1495.3	-1480.9	571.4
Pabdeh	1766.89	-1752.3	432.6
Gurpi	2199.53	-2184.9	125.4
Ilam	2536.91	-2522.3	104.7
Lafan	2641.59	-2627	6.9
Sarvak	2648.53	-2633.9	19.7
Kazhdumi	3296.84	-3282.2	140.9
Dariyan	3504.14	-3489.5	188.5
Gadvan	3692.65	-3678	9.5
Fahliyan	3985.7	-3971.1	36.2
	4200	-4186.6	

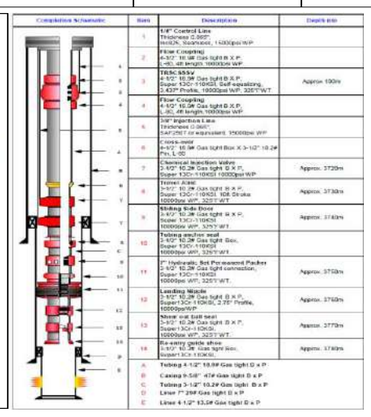


Hole Size / Bit Type / Nozzle Sizes	Drilling Fluid	Logging/ Testing/ Coring	CSG/Liner Design	Cementing	BOP and Well Head	Remarks (Drilling Issues/Lesson Learned)
26" 26" MT Bit MSDGH 135 (open)	Bentonite Mud 65-72 PCF	NIL	0-152.5m: 20", 117 ppf, K-55, Big Omega	**Lead:** Class "G" CMT S.V: 345 bbl, SW: 117.5 pcf *50 bbl return @ surface.	20-3/4", 3K CHH (Weld On 20" CSG)	1. Bit Balled-up with 65 pcf Bentonite Mud in AZNS-045. 2. Cement Losses in AZNS-017 with 115 pcf Sw. 3. Bit balled-up with bentonite mud in AZNS-127.
17-1/2" 17-1/2" TCI Bit MS05SV 425 (open) 2No. 17-1/2" PDC Bit CK5605F (5x15 + 5 x 13) (2x16 + 5x15 + 3x13) 1No. 17-1/2" PDC Bit SKS76D 5111 (9x18) 1No. 17-1/2" PDC Bit MS1953SS 5223 (4x20 +3x16)	Salt Saturated Kcl Polymer WBM 71-86 PCF	CDR/GR/Cal	0-1492m: 13-3/8", 68 ppf, N-80, BTC	**Lead:** Class "G" CMT S.V: 910 bbl, S.W: 90.5 pcf **Tail:** Class "G" CMT S.V: 297 bbl, S.W: 115.5 pcf *347 bbl return @ surface.	20-3/4", 3K BOPs 20-3/4", 3K x 13-5/8", 5K CHS	1. Bit and Stabilizer balling with 67-86 pcf Kcl/Polymer mud in GS formation in AZNS-045. 2. Calcium ion contamination in GS formation. 3. Frequent Bit/Stb balling and tight hole in AZNS-017 due to low Mw and improper mud properties. 4. Wash down 13-3/8" CSG and set CSG off bottom due to bad hole condition in AZNS-017 (mainly low Mw in compare to hole condition). 5. Obs back flow after Cementing 13-3/8" CSG, apply pressure and close well in several steps in AZNS-045. 6. String stuck while POOH & B.Ream in GS formation with 84 pcf Mw in AZNS-045 well. 7. Frequent tight hole and hole stability problem in GS formation in well AZNS-017 (12-1/4" Hole). 8. WL logging tool held up in GS Fm., in well AZNS-017 (12-1/4" Hole). 9. Bit balled-up in base of GS formation in AZNS-127. 10. Frequent tight hole and string washout in GS in AZN-010 with 75-90 pcf Mw.
12-1/4" 1No. MT Bit 217 (3 * 20/32) 4No. PDC Bit M422 (6 * 15/32) M323 (8 * 16/32 + 1 * 13/32) --- (4 * 12/32 + 4 * 14/32) --- (6 * 15/32) 1No. B.Bit 517 (3 * 20/32)	Salt Saturated KCL/Glycol/Polymer/CaCo3 WBM: 73-85PCF	Open Hole Logs: Full Suite/CDR/GR Cased Hole Logs: SBT or CAST/CCL/CBL/VDL/GR from 9-5/8" Casing	0-3933m (3933m): 9-5/8", 53.5 ppf (SD), 110SS, Premium Gas Tight Con. ID: 8.535" (SD) Burst: 10900 psi Colapse: 7950 psi Body Yield: 1,710 Klbs B.SF: 1.17 C.SF: 1.99 A.SF: 2.33 TA.SF: 1.42	**Lead:** Class "G" CMT S.V: 1154 bbl, S.W: 95pcf **Tail:** Class "G" CMT S.V: 116 bbl, S.W: 118pcf	13-5/8", 10K BOPs 13-5/8", 5K x 11", 10K CHH	**Main drilling Challenges: Losses in AS/PD/SV/KZ, Chert formation in base of Pabdeh, Tight hole and string stuck, Washout string, Oil/Gas Flow, CMT Losses** 1. Frequent tight hole while drilling/ripping through GU (L) in AZNS-045 well. 2. Seepage losses 2-6 BPH with 75-87 pcf Mw in AS/PD/GU formation in AZNS-045. Drilling F/ SV to GD with 0.2 BPH losses w/ 84-87.5 pcf Mw. 3. Differential String stuck in KZ of AZNS-045 w/ 87.5 pcf Mw, free same with HCL Acid and cut MW to 81 pcf. 4. Perform top cement job in 9-5/8" Ann due to failure while displacing cement and remained cement inside casing in AZNS-045. 5. Obs2 BPH losses in KZ Fm., in AZNS-017 and frequent string stuck while drilling horizontal hole in KZ with 76-78 pcf Mw. 6. Obs 2-12 BPH losses while drilling through SV in AZNS-017 with 75-77 pcf Mw. 7. Obs 3-5 BPH losses with 80-81 pcf Mw in AZNS-127, losses inc to 12 BPH in KZ & 15 BPH in GD 8. Frequent string washout and twist off string in AZNS-127 while performance drilling with 80-81 pcf Mw. 9. Noticeable CMT losses while cementing 9-5/8" CSG w/ 88 pcf Lead CMT Sw in AZNS-127. 10. Low ROP in Cherty Zone (PD) from 1920-1989m in AZNS-127. 11. Obs 40-50 BPH losses while drilling with 80 pcf Mw in top of SV in AZN-010. 12. Low ROP @191 to 2197, 2900 to 2945 in AZNS-40 13. Stuck String @ 2503m in AZNS-40 14. Low ROP @ 2961.5 & 3069 in AZNS-40 15. Low ROP @3546 to 3656 in AZNS-40 16. Twist off string at 1556.5m, 2115m, 2197m, 3552m and recovered all.
8-1/2" 1No. 8-1/2" MT Bit 517 (3 x 20/32) 1No. 8-1/2" PDC Bit M413 (2 * 13/32 + 4 * 14/32) 1No. S-7/8" MT Bit 217 (open)	Salt Saturated KCL/Polymer/CaCo3 WBM: 100-104 PCF	Open Hole Logs: Full Suite/CDR/GR Cased Hole Logs: SBT or CAST/CCL/CBL/VDL/GR from 7" Liner	3812.2 - 4189.5m (377.3m): 7", 32ppf, 110SS, Premium Gas Tight Con. ID: 6.094" Burst: 12460 psi Colapse: 10780 psi Body Yield: 1,025 Klbs B.SF: 1.24 C.SF: 1.06 A.SF: 1.51 TA.SF: 1.49	**Lead:** Class "G" CMT S.V: 53 bbl, SW:118pcf	13-5/8", 10K BOPs 4-1/16", 10K X-Tree	**Main drilling Challenges: Tight hole and string stuck, Oil/Gas Flow.** 1. String stuck in FH formation with 105-106 pcf MW, free same with Overpull/Jarring/Pipe-Lax in AZNS-045. 2. String stuck while coring in FH., LH and side track well in AZNS-045. 3. 7" L.Lap leakage as Dry Test result in AZNS-127 even with good amount of hard CMT on TOL. Run T.Back Liner and cement same. 4. Seepage losses 2-8 BPH in AZNS-045 with 100-106 pcf Mw, 1-3 BPH w/ 104-105 pcf Mw in AZNS-127 & AZN-010 w/ 100-110 pcf Mw. 5. Low ROP f/3934 to TD in AZNS-40

Time Breakdown

Hole Size	De-Suspend	12-1/4"	8-1/2"	Completion			Sum (Hrs)	Total %	Total (Days)
Drilling		914	117.5				1031.5	27.7%	43.0
Round Trip/BHA		195	10				205	5.5%	8.5
BOP Test		9.5	5				14.5	0.4%	0.6
Hole Condition, Reaming, Circ		49	8.5				57.5	1.5%	2.4
Deviation Survey		4.5	-				4.5	0.1%	0.2
Directional Drilling		-	-				0	0.0%	0.0
Casing/Liner		50	104.5				154.5	4.2%	6.4
Cementing		46	9.5				55.5	1.5%	2.3
WOC		-	-				0	0.0%	0.0
N/U&D BOP		14.5	-				14.5	0.4%	0.6
N/U WH		7	-				7	0.2%	0.3
Drill Out		46.5	63				109.5	2.9%	4.6
Logging		70	86				156	4.2%	6.5
Testing (PIT, LOT, CIT, Dry Test, etc.)		3	161				164	4.4%	6.8
DST		-	-				0	0.0%	0.0
Coring		-	-				0	0.0%	0.0
Plug Back		-	-				0	0.0%	0.0
Completion		-	-	1660.5			1660.5	44.6%	69.2
Well Testing		-	-				0	0.0%	0.0
Rig Repair		-	-				0	0.0%	0.0
Slip & Cut		5	5				10	0.3%	0.4
De-suspend	75.5	-	-				0	0.0%	0.0
Workover		-	-				0	0.0%	0.0
Total Hrs per Hole	75.5	1414.0	570.0	1660.5		3720.0			
Total Days per Hole	3.1	58.9	23.8	69.2		155.0			
Total Days per Hole with 10% Contingency	3.5	64.8	26.1	76.1		170.5			
Cummulative Time (Days)	3.1	62.1	85.8	155.0					
Cummulative Time (Days) with 10% Contingency	3.5	68.3	94.4	170.5					





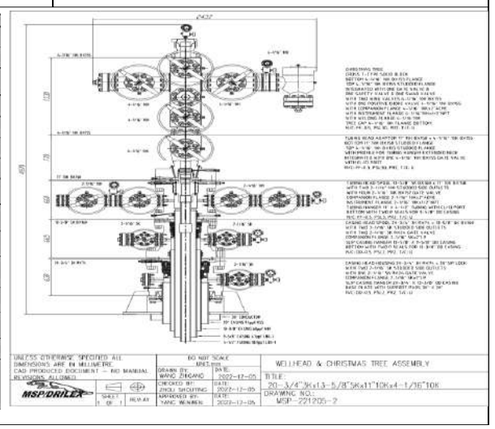


Figure 2-9: AZNS-040 Well Montage

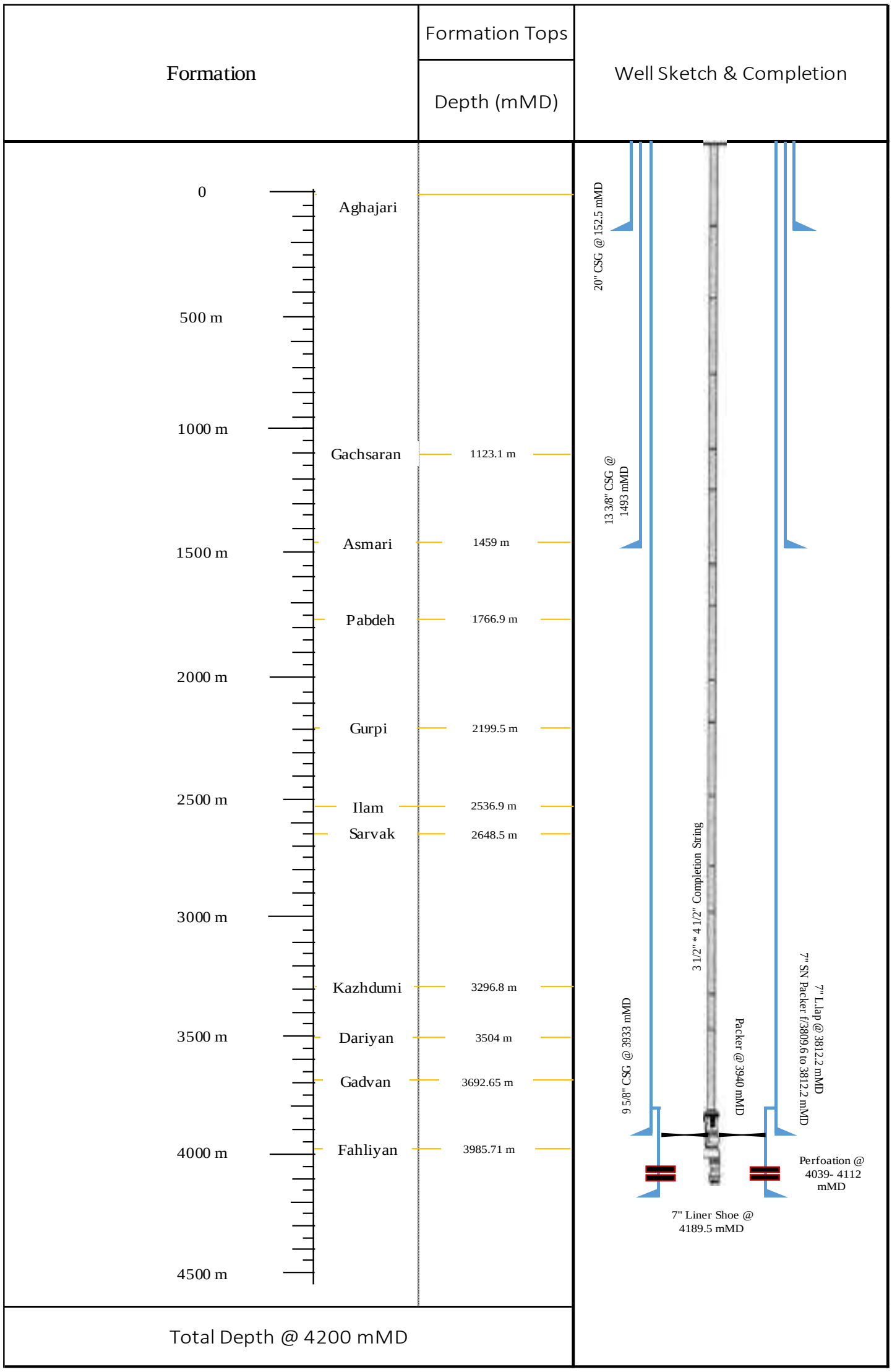


Figure 2-10: AZNS-040 Well Sketch

2.5 ROP Chart

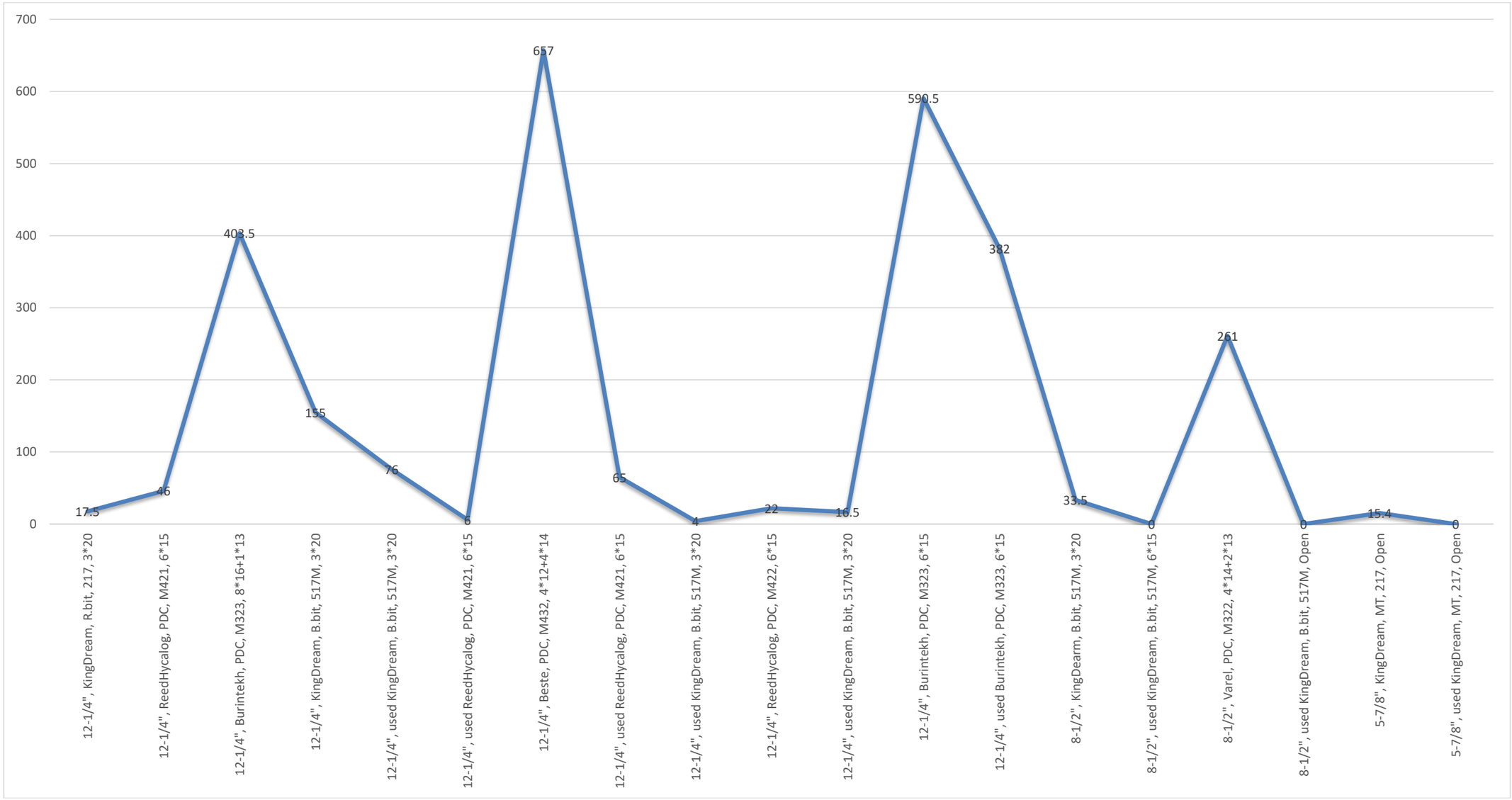


Figure 2-11: AZNS-040 ROP Chart

2.6 Mud Weight vs. Depth

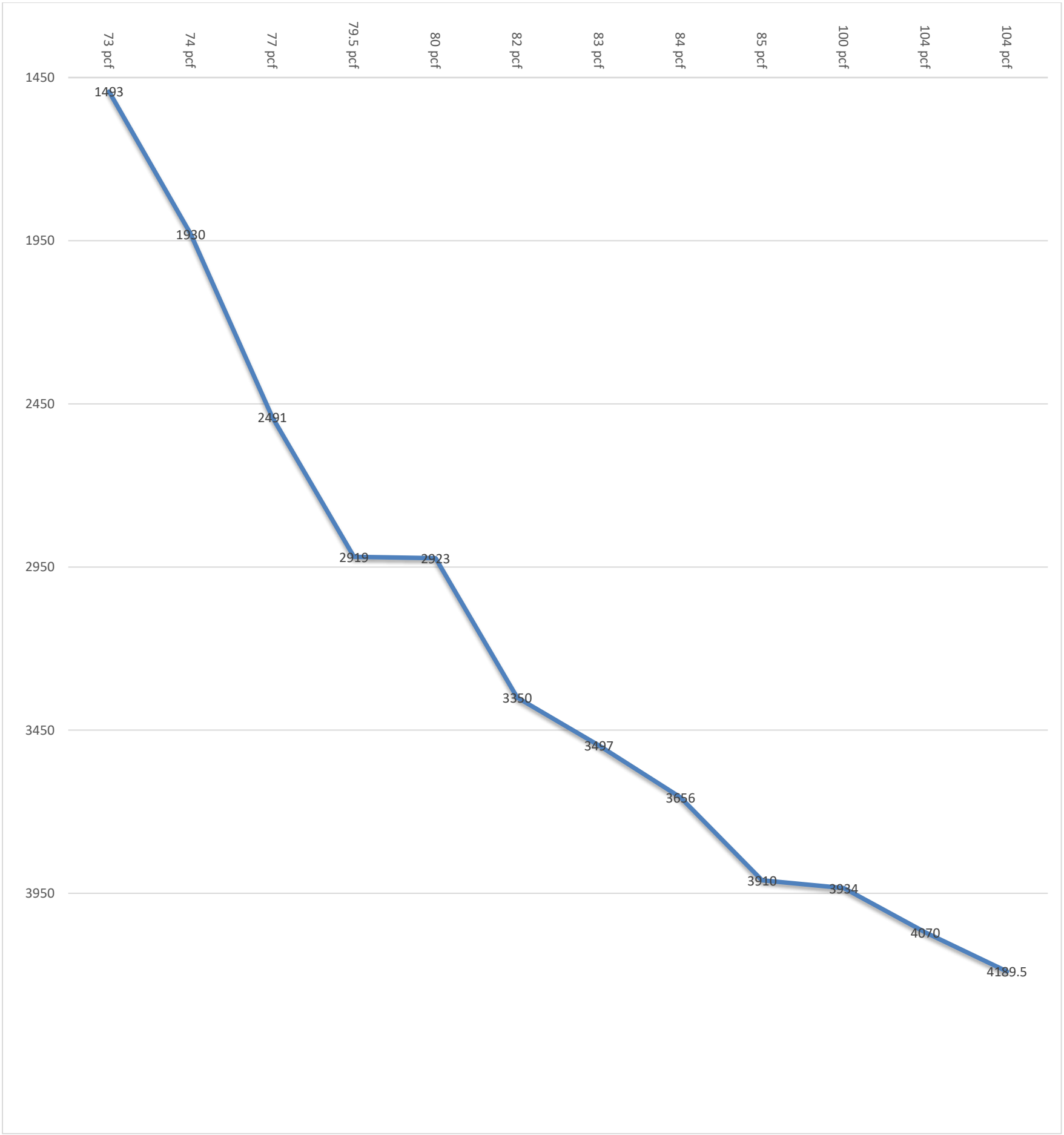


Figure 2-12: AZNS-040 Mw vs. Depth Chart

2.7 Bit Record Table

Bit No	Size	Bit Manufacture	BHA No	Type	IADC Code	Serial No	Jets	CMT	Depth		Formation (Lithology)	Depth		Metre	Hrs	ROP	WOB (klb)	Surf RPM		Krev (For all run)	Pump Press	GPM		Remark
1	12-1/4"	Kingdream	1	R.Bit (HJT217GL)	217	1917438JH	3 20/32	Yes	47-171	1468-1493	S.ST, LST	1493	1510.5		25.5		0-27	40	80	108	0-1500	200	600	Run #1: D/O & Ream: 149m-20.5 Hrs / Form.: 17.5m-5Hrs. TBR:108krev
2	12-1/4"	ReedHycalog	2	PDC (DSX616-A28)	M421	124344	6 15/32	No	-	-	S.ST, LST	1510.5	1556.5	46	2.5	13-20	5-17	120	140	-	800-1400	430	650	Run #1: Formation Drlg
3	12-1/4"	Burintekh	4	PDC (B616YCM382)	M323	400261	8*16/32" + 1*13/32"	No	-	-	Lst, Clst, Shale, Mrl, Cht	1556.5	1960	403.5	41.5	2-20	1-20	80	130	-	550-1880	450	800	Run #1: Formation Drlg
4	12-1/4"	Kingdream	5	B.Bit (HJT517GL)	517M	1912451JH	3 20/32	No	-	-	Lst, Cht	1960	2115	155	37	2-5	10-20	60	80	205	1930-2350	700	800	Run #1: Formation Drlg
4 RR1	12-1/4"	Kingdream	7	B.Bit (HJT517GL)	517M	1912451JH	3 20/32	No	-	-	Lst, Arg lst, Cht, Sh	2115	2191	76	31	1-5	5-20	60	120	221	1650-2150	650	750	Run #2: Formation Drlg
2 RR1	12-1/4"	ReedHycalog	8	PDC (DSX616-A28)	M421	124344	6 15/32	No	-	-	Arg lst, Sh	2191	2197	6	1.5	4	5-15	100	180	-	1850-2370	690	810	Run #2: Formation Drlg
5	12-1/4"	Beste	10	PDC (M1385)	M432	224110	4*12/32" + 4*14/32"	No	-	-	Lst, Arg lst, Sh	2197	2854	657	102	6.44	0-25	80	180	-	2400-2550	500	750	Run #1: Formation Drlg
2 RR2	12-1/4"	ReedHycalog	12	PDC (DSX616-A28)	M421	124344	6 15/32	No	-	-	Lst	2854	2919	65	29	2.24	5-20	80	150	-	2400-2650	680	730	Run #3: Drlg Time: 29 Hrs (Totally 33 Hrs @ 3Run) / Depth Drilled: 65 m (Toally 171m @ 3Run)
4 RR2	12-1/4"	Kingdream	13	B.Bit (HJT517GL)	517M	1912451JH	3 20/32	No	-	-	Lst	2919	2923	4	4.5	0.9	0-12	50	80	24	2450-2650	680	720	Totally in 3 Run: Formation: 235m - 72.5 Hrs, TBR (Totally): 446 Krev
6	12-1/4"	ReedHycalog	14	PDC (DSX133DGJSW)	M422	204049	6 15/32	No	-	-	Lst	2923	2945	22	12.5	1.76	5-15	80	180	-	2450-2650	700	750	Run1: Depth Drilled: 22 m (12.5 Hrs.)
4 RR3	12-1/4"	Kingdream	15	B.Bit (HJT517GL)	517M	1912451JH	3 20/32	No	-	-	Lst	2945	2961.5	16.5	17	0.5-2	5-20	60	85	69 (Totally: 519)	1650-2550	550	700	Run #4: Drlg Time: 16 Hrs (Totally 92.5 Hrs @ 4 Run) / Depth Drilled: 16m (Toally 251m @ 4 Run)
7	12-1/4"	Burintekh	16	PDC (BT616YCM)	M323	40029	6 15/32	No	-	-	Lst	2961.5	3552	590.5	118	1-15	5-20	90	170	-	1900-2700	610	740	Run #1: Formation: 590.5m - 118 Hrs
7 RR1	12-1/4"	Burintekh	16	PDC (BT616YCM)	M323	40029	6 15/32	No	-	-	Lst, Arg Lst, Mrl	3552	3934	382	103	1-10	5-25	90	170	-	2500-2750	600	630	Run #1: Formation: 590.5m - 118 Hrs, Run #2: Formation: 970m -221Hrs
8	8-1/2"	Kingdream	19	B.Bit (HJT517GL)	517M	2003235JH	3 20/32	Yes	3905.5	3934	Lst	3934	3939	33.5	8	5-18	5-18	50	60	28	400-1200	250	400	Run #1: D/O & Ream: 28.5m-6.5 Hrs / Form.: 5m-1.5Hrs. TBR:28krev
8 RR1	8-1/2"	Kingdream	19	(B.Bit) HJT517GL	517M	2003235JH	3 20/32	No	-	-	-	-	-	-	-	-	-	-	-	28	400-1200	250	400	Run #2: Perform Scraper trip (included Junk Sub) TBR: 28 Krev @ 2 Run
9	8-1/2"	Varel	21	PDC(VTD616DGX)	M322	6000554	4*14/32" + 2*13/32"	No	-	-	Lst	3939	4200	261	96	1-6	10-20	140	170	-	2500-2600	400	450	Run #1: Formation
8 RR2	8-1/2"	Kingdream	22	(B.Bit) HJT517GL	517M	2003235JH	Open	No	-	-	-	-	-	-	-	-	-	-	-	28	-	-	-	RIH to 7" L.Lap W/O cement, TBR: 28 Krev @ 3 Runs.
10	5-7/8"	Kingdream	23	MT (HAT217G)	217	2204638JH	Open	Yes	3811	3816.4	-	-	-	15.4	7.5	2	5-10	40	60	25	500-1600	120	250	Drill OUT POB & CMT INSIDE 7" LINER". 15.4 m, 7.5 Hrs CMT DRLG/ &R
10 RR1	5-7/8"	Kingdream	24	MT (HAT217G)	217	2204638JH	Open	No	-	-	-	-	-	-	-	-	-	-	-	25	-	-	-	Run tandem scraper

Table 2-10: AZNS-040 Bit Record

2.8 Bit Parameters Plot

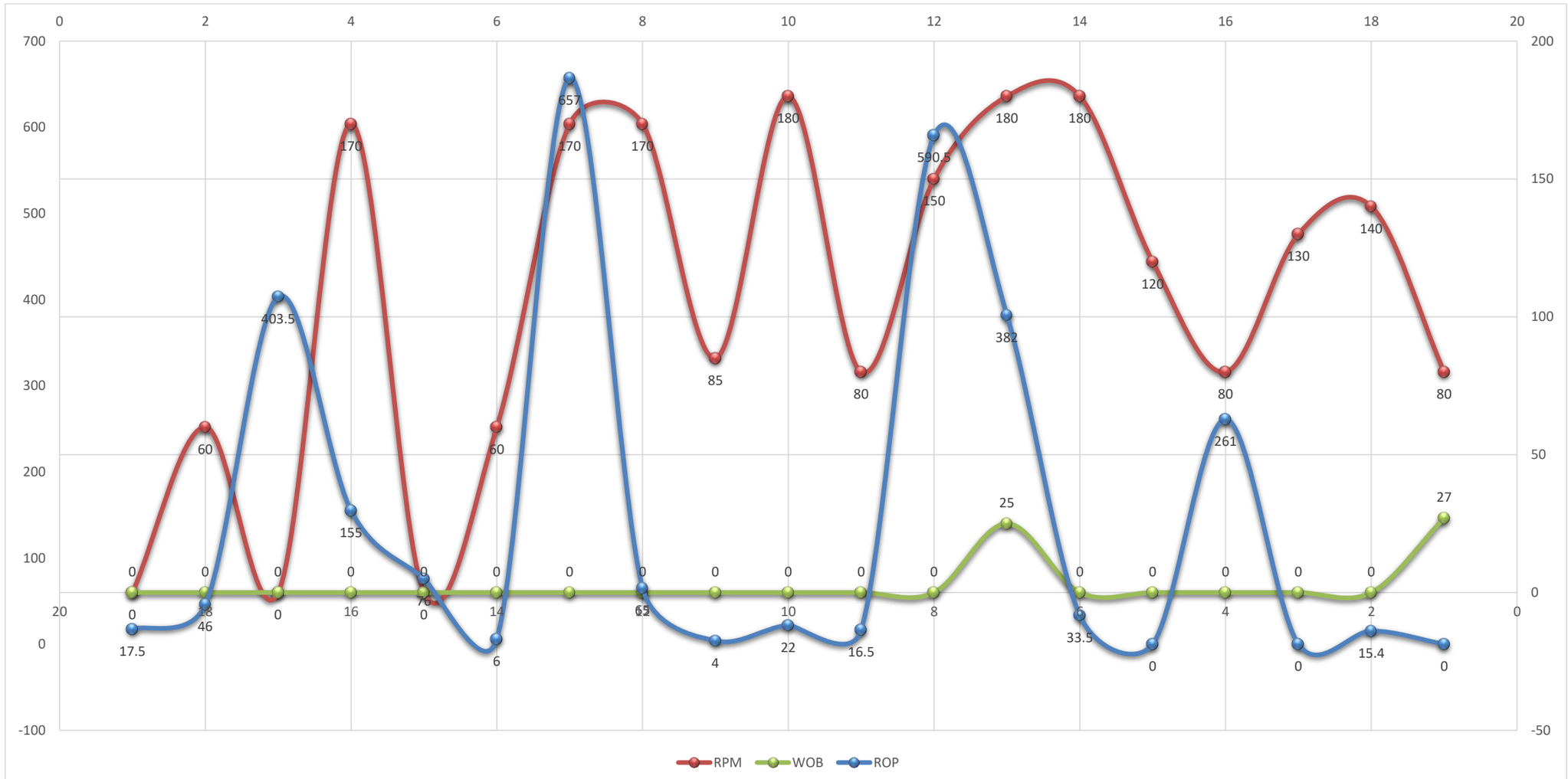


Figure 2-13: Bit Parameters Chart

2.9 Hole Problem History

Hole Size	Depth	Problem Description
12-1/4"	1556.5m	Twist off string / Recovered
	2115m	Twist off string / Recovered
	2191 – 2197m	Low ROP / add KCl solution to active mud system
	2197m	Twist off string / Recovered
	۲۸۵۲	Abnormal drag & stall string
	2900 – 2919m	Low ROP / change bit & work on junk
	2919 – 2923m	Low ROP & bit bouncing / increase Mw
	۲۹۴۵	Low ROP
	2503m	Stuck string while POOH
	۲۹۶۱٫۵m	Low ROP / CBU & work on junk
	3069m	Low ROP / pumping 70bbl low vis pill
	3125m	Observe Over pull while POOH
	3152m	Observe Over pull while POOH
	3552m	Twist off string / Recovered
	3546 – 3656m	Low ROP & high torque / increase Mw
	3854m	Stuck string while POOH
8-1/2"	3934 – 3939m	Low ROP & drlg parameters / change bit & work on junk
	3923 – 4070m	Low ROP & drlg Parameters / increase Mw, B/R & R/D each drill stand

Table 2-11: AZNS-040 Hole Problems

3. Time Analysis Report

3.1 Time vs. Depth Chart

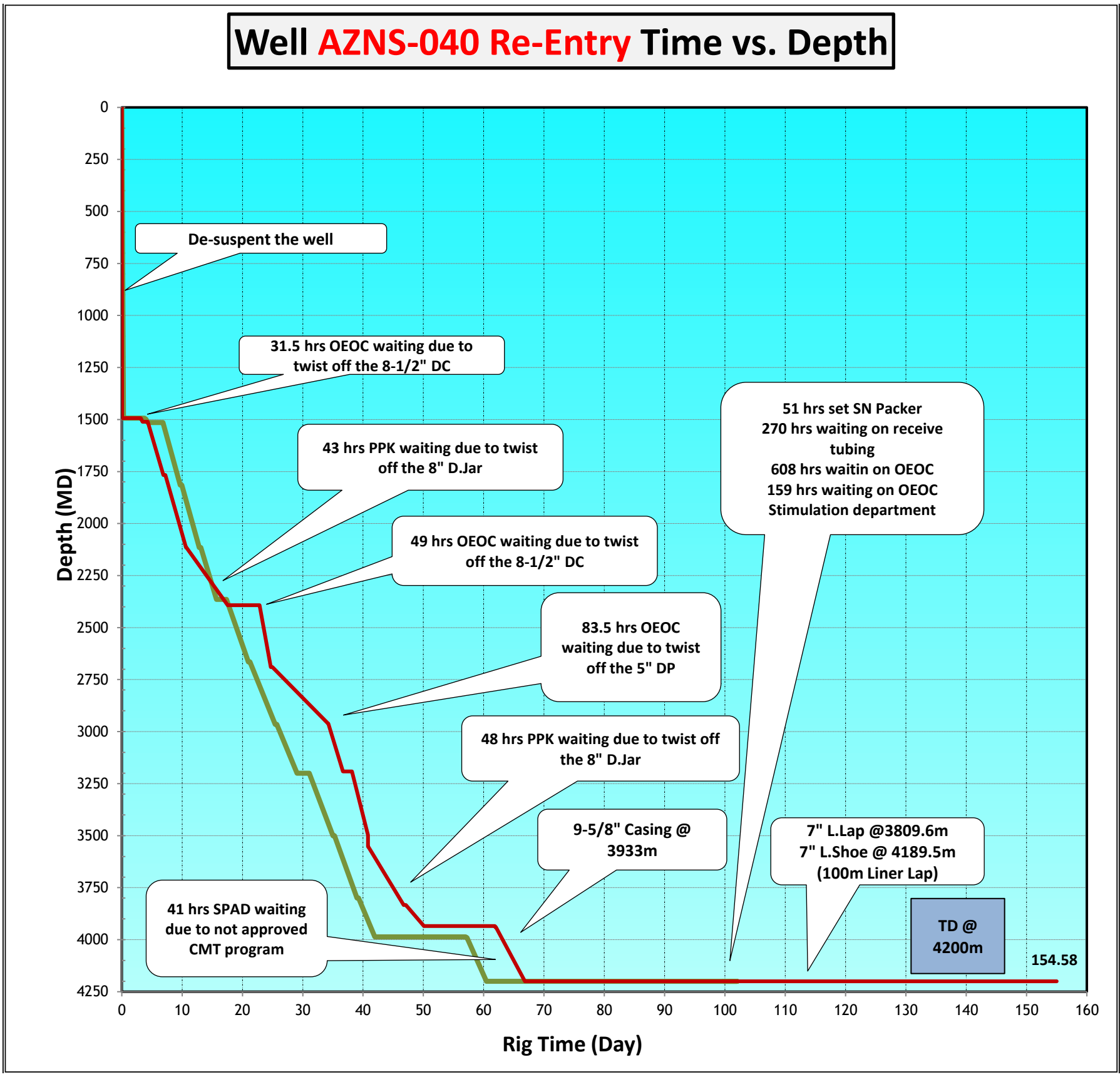


Figure 3-1: AZNS-040 Re-Entry Well Time vs. Depth

3.2 Hole Duration Graph

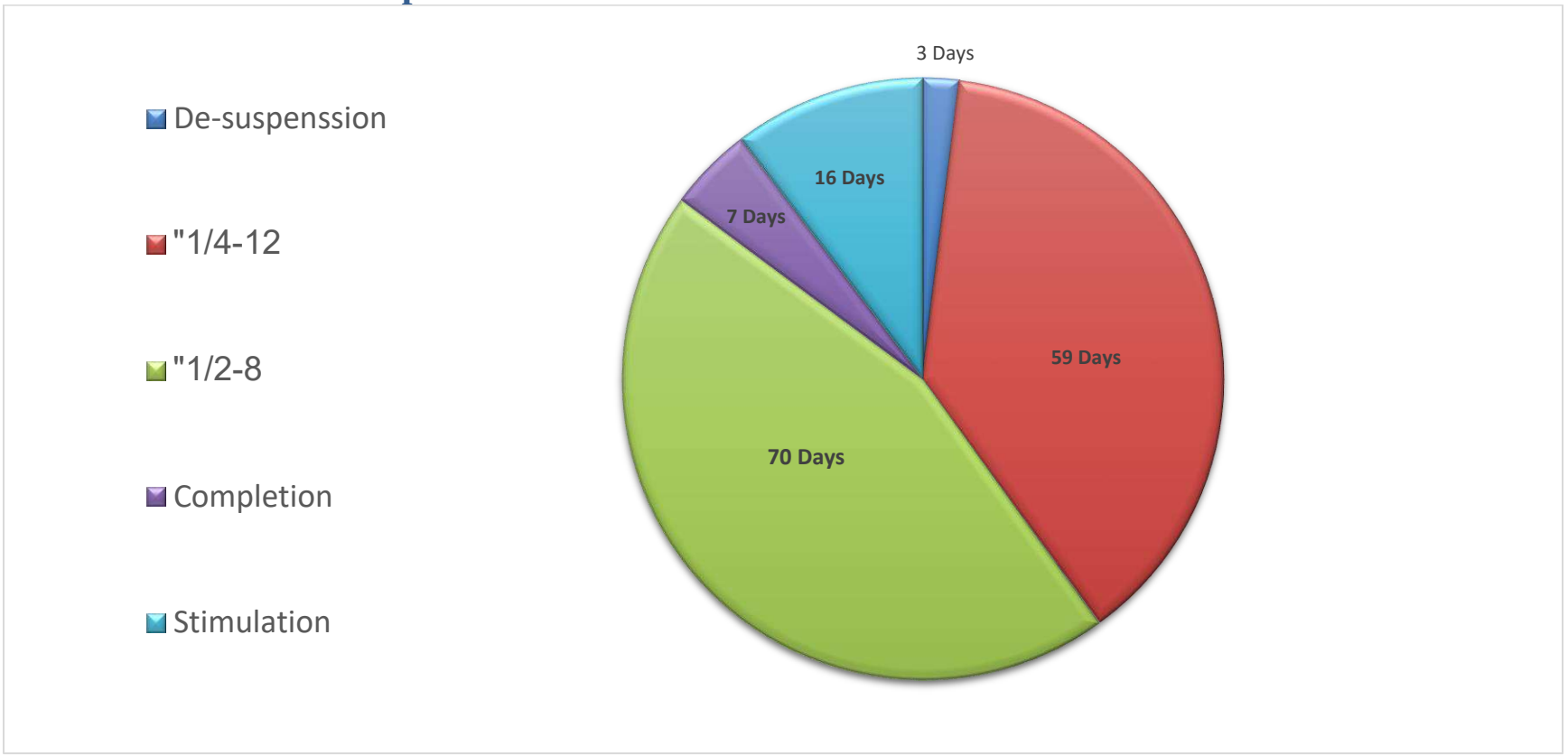


Figure 3-2: AZNS-040 Hole Duration Time Chart

3.3 Total Time Break-Down of Operation

De-Suspend

Held P.JSM. Spud the Rig. Stand Back required 5" DPs, 5" HWDPs and 8-1/2" DC.	De-Suspend		GS	0		0.0	0.00	0.0	0.0	0.0	0.00	0.00	0.00	0.00	0.00	0.00
Record well head pressure, function well head components and fix wellhead if necessary. Bleed off any wellhead/annulus pressure to Zero and monitor same. Remove WH cover. Note: If annulus pressure is not Zero after bleed off, remedial plan will be prepare and follow after Client confirmation.			GS	1493		6.0	0.25	0.3	0.3	6.6	0.28	0.3	1493.0	1.00	0.04	0.04
N/U 13-5/8", 10M BOPs & related lines.			GS	1493		18.0	0.75	1.0	0.8	19.8	0.83	1.1	1493.0	18.00	0.75	0.79
Test Accumulator unit, BOPs, choke manifold & WC equipment as per NIOC policy.			GS	1493		8.0	0.33	1.3	0.4	8.8	0.37	1.5	1493.0	5.00	0.21	1.00
Perform WH pressure integrity test with cup tester.			GS	1493		2.0	0.08	1.4	0.1	2.2	0.09	1.6	1493.0	3.00	0.13	1.13
P/U & M/U 12-1/4" TCI Bit BHA (including junk sub), RIH to tag CMT plug @ 55m. D/Out CMT plug f/ 55-100m. RIH to tag 13-3/8" CSG Float Collar @ 1468m jt by jt. D/Out plugs, F.C, CMT to 3m above CSG shoe @ 1489m.			GS	1493		36.0	1.50	2.9	1.7	39.6	1.65	3.2	1493.0	45.00	1.88	3.00
Displace well with 75 pcf Salt Saturted Kcl/Polymer mud, Circ and homogenize mud properties.			GS	1493		4.0	0.17	3.1	0.2	4.4	0.18	3.4	1493.0	1.50	0.06	3.06
R/U CMT surface line. Perform 13-3/8" CIT. with 1500 psi for 10 min, R/D CMT surface lines.			GS	1493		2.0	0.08	3.2	0.1	2.2	0.09	3.5	1493.0	0.50	0.02	3.08
D/Out CMT, shoe and rat hole to 1493m.			GS	1493		6.0	0.25	3.4	0.3	6.6	0.28	3.8	1493.0	1.50	0.06	3.15
Total Duration of De-Suspend the Well:							*3.4*	*3.4*			*3.8*	*3.8*		*75.5*	*3.1*	*3.15*

12-1/4" Hole Section

12-1/4" Hole Section / 9-5/8" Casing																	
Drilling 12-1/4" Hole to 1498m.	Drilling	5	AS	1498	2.0	2.5	0.10	3.5	0.1	2.8	0.11	3.9	1498	0.5	0.02	3.17	
CBU and balance Mw. perform FIT with Required Pressure 410 psi (EMW: 87pcf, should be equal to max. planned Mw for this hole section plus 2 pcf)	FIT		AS	1498		3.0	0.13	3.6	0.1	3.3	0.14	4.0	1498	1.5	0.06	3.23	
Cont. Drilling 12-1/4" Hole to 1515m.	Drilling	17	AS	1515	2.7	6.3	0.26	3.9	0.3	6.9	0.29	4.3	1510.5	5	0.21	3.44	
POOH to surface, L/D TCI Bit and junk sub.	Trip		AS	1515		7.0	0.29	4.2	0.3	7.7	0.32	4.6	1510.5	7.5	0.31	3.75	
Change Ram, Install shooting nipple, R/U WL Logging Eq. Perform Logging (CBL/VDL logs) from Surface to 1492 m in 13-3/8" CSG, R/D Logging Eq, change Ram.	Log		AS	1515		8.0	0.33	4.5	0.4	8.8	0.37	5.0	1510.5	0	0.00	3.75	
P/U & M/U 8" DCs and 5" DPs/HWDPs stands for drilling 12-1/4" hole section, R/Back same in derrick.	Trip		AS	1515		32.0	1.33	5.5	1.5	35.2	1.47	6.5	1510.5	6	0.25	4.00	
P/U & M/U 12-1/4" PDC Bit (St. STB @ 20m), RIH to Btm @ 1515m.	Trip		AS	1515		8.0	0.33	5.9	0.4	8.8	0.37	6.8	1510.5	7	0.29	4.29	
Cont. Drilling 12-1/4" hole f/ 1515m to 1815m.	Drilling	300	AS PD	1815	4.8	62.5	2.60	8.5	2.9	68.8	2.86	9.7	1767.0	62.5	2.60	6.90	31.5 hrs OEOC waiting due to twist off the 8-1/2" DC and fish left in hole@ 1547m
Take single-shot survey every 100m and multi-shot survey every 300m.	Survey		PD	1815		2.0	0.08	8.6	0.1	2.2	0.09	9.8	1767.0	1	0.04	6.94	
Perform condition trip every drilled 300m (CBU, POH, RIH, CBU)	Hole Cond/Circ		PD	1815		4.0	0.17	8.7	0.2	4.4	0.18	10.0	1767.0	7	0.29	7.23	
Cont. Drilling 12-1/4" hole f/ 1815m to 2115m.	Drilling	300	PD	2115	4.8	62.5	2.60	11.3	2.9	68.8	2.86	12.8	2115.0	82	3.42	10.65	
Take single-shot survey every 100m and multi-shot survey every 300m.	Survey		PD	2115		2.0	0.08	11.4	0.1	2.2	0.09	12.9	2115.0	0	0.00	10.65	
Perform condition trip every drilled 300m (CBU, POH, RIH, CBU)	Hole Cond/Circ		PD	2115		4.0	0.17	11.6	0.2	4.4	0.18	13.1	2115.0	1.5	0.06	10.71	
Cont. Drilling 12-1/4" hole f/ 2115m to 2365m.	Drilling	250	PD GU	2365	4.4	56.8	2.37	13.9	2.6	62.5	2.60	15.7	2392.0	164	6.83	17.54	43 hrs PPK waiting due to twist off mandrel of 8" D.Jar 49 hrs OEOC waiting due to twist off the 8-1/2" DC and fish left in hole@ 2124m
Take single-shot survey every 100m and multi-shot survey every 300m.	Survey		GU	2365		2.0	0.08	14.0	0.1	2.2	0.09	15.8	2392.0	2.5	0.10	17.65	
Perform condition trip every drilled 300m (CBU, POH, RIH, CBU)	Hole Cond/Circ		GU	2365		4.0	0.17	14.2	0.2	4.4	0.18	16.0	2392.0	11.5	0.48	18.13	
POOH to surface, L/D PDC Bit.	Trip		GU	2365		12.0	0.50	14.7	0.6	13.2	0.55	16.5	2392.0	48.5	2.02	20.15	
BOP Test.	BOP Test		GU	2365		6.0	0.25	14.9	0.3	6.6	0.28	16.8	2392.0	6	0.25	20.40	
P/U & M/U New 12-1/4" PDC Bit and same BHA, RIH to Btm.	Trip		GU	2365		12.0	0.50	15.4	0.6	13.2	0.55	17.4	2392.0	59	2.46	22.85	
Cont. Drilling 12-1/4" hole f/ 2365m to 2665m.	Drilling	300	LA SV	2665	3.8	78.9	3.29	18.7	3.6	86.8	3.62	21.0	2690.0	44	1.83	24.69	
Take single-shot survey every 100m and multi-shot survey every 300m.	Survey		SV	2665		2.0	0.08	18.8	0.1	2.2	0.09	21.1	2690.0	1	0.04	24.73	
Perform condition trip every drilled 300m (CBU, POH, RIH, CBU)	Hole Cond/Circ		SV	2665		4.0	0.17	19.0	0.2	4.4	0.18	21.2	2690.0	5.5	0.23	24.96	
Cont. Drilling 12-1/4" hole f/ 2665m to 2965m.	Drilling	300	SV	2965	3.3	90.9	3.79	22.8	4.2	100.0	4.17	25.4	2961.5	221.5	9.23	34.19	83/5 hrs OEOC waiting due to string twist off the string and fish left in hole from 749 m 2 hrs OEOC waiting due to work on wight indicator.
Take single-shot survey every 100m and multi-shot survey every 300m.	Survey		SV	2965		2.0	0.08	22.9	0.1	2.2	0.09	25.5	2961.5		0.00	34.19	
Perform condition trip every drilled 300m (CBU, POH, RIH, CBU)	Hole Cond/Circ		SV	2965		4.0	0.17	23.0	0.2	4.4	0.18	25.7	2961.5	1	0.04	34.23	
Cont. Drilling 12-1/4" hole f/ 2965m to 3200m.	Drilling	235	SV	3200	3.2	73.4	3.06	26.1	3.4	80.8	3.37	29.1	3191.0	58	2.42	36.65	
Take single-shot survey every 100m and multi-shot survey every 300m.	Survey		SV	3200		2.0	0.08	26.2	0.1	2.2	0.09	29.1	3191.0		0.00	36.65	
Perform condition trip every drilled 300m (CBU, POH, RIH, CBU)	Hole Cond/Circ		SV	3200		4.0	0.17	26.3	0.2	4.4	0.18	29.3	3191.0	8	0.33	36.98	
POOH to surface, L/D PDC Bit.	Trip		SV	3200		16.0	0.67	27.0	0.7	17.6	0.73	30.1	3191.0	14	0.58	37.56	1.5 hrs OEOC waiting due to change sensator
BOP Test.	BOP Test		SV	3200		6.0	0.25	27.2	0.3	6.6	0.28	30.3	3191.0		0.00	37.56	
P/U & M/U New 12-1/4" PDC Bit and same BHA, RIH to Btm.	Trip		SV	3200		16.0	0.67	27.9	0.7	17.6	0.73	31.1	3191.0	14	0.58	38.15	2 hrs OEOC waiting due to change sensator
Cont. Drilling 12-1/4" hole f/ 3200m to 3500m.	Drilling	300	KI SV	3500	3.5	85.7	3.57	31.5	3.9	94.3	3.93	35.0	3497.0	64	2.67	40.81	
Take single-shot survey every 100m and multi-shot survey every 300m.	Survey		DR	3500		2.0	0.08	31.6	0.1	2.2	0.09	35.1	3552.0		0.00	40.81	
Perform condition trip every drilled 300m (CBU, POH, RIH, CBU)	Hole Cond/Circ		DR	3500		4.0	0.17	31.7	0.2	4.4	0.18	35.3	3552.0	1	0.04	40.85	
Cont. Drilling 12-1/4" hole f/ 3500m to 3800m.	Drilling	300	DR GD	3800	3.7	81.1	3.38	35.1	3.7	89.2	3.72	39.0	3833.0	141	5.88	46.73	48 hrs PPK waiting due to twist off f/shock sub of 8" D.Jar 4.5 hrs stop operation due to bad weather. 2 hrs OEOC waiting due to poor performance solid control sys, out of service centrifuge #3, lack of shaker screen #200 API
Take single-shot survey every 100m and multi-shot survey every 300m.	Survey		GD	3800		2.0	0.08	35.2	0.1	2.2	0.09	39.1	3833.0		0.00	46.73	
Perform condition trip every drilled 300m (CBU, POH, RIH, CBU)	Hole Cond/Circ		GD	3800		4.0	0.17	35.4	0.2	4.4	0.18	39.3	3833.0	9	0.38	47.10	
Cont. Drilling 12-1/4" hole f/ 3800m to section TD @ 3986m.	Drilling	186	GD	3986	3.2	58.1	2.42	37.8	2.7	63.9	2.66	41.9	3934.0	71.5	2.98	50.08	
Take single-shot survey every 100m and multi-shot survey every 300m.	Survey		GD	3986		2.0	0.08	37.9	0.1	2.2	0.09	42.0	3934.0		0.00	50.08	
CBU, Perform condition trip to last CSG Shoe prior to logging, RIH to Btm, CBU, Spot pill.	Hole Cond/Circ		GD	3986		30.0	1.25	39.1	1.4	33.0	1.38	43.4	3934.0	4.5	0.19	50.27	
POOH to surface, L/D PDC Bit.	Trip		GD	3986		19.0	0.79	39.9	0.9	20.9	0.87	44.3	3934.0	21.5	0.90	51.17	5.5 hrs OEOC waiting due to L/D 5" stds DP's Buckled.
Change Ram, Install shooting nipple, R/U WL Logging Eq. Perform Logging (Full Suite/CDR/GR), R/D Logging Eq, change Ram.	Log		GD	3986		14.0	0.58	40.5	0.6	15.4	0.64	44.9	3934.0	28.5	1.19	52.35	
BOP Test.	BOP Test		GD	3986		6.0	0.25	40.7	0.3	6.6	0.28	45.2	3934.0		0.00	52.35	
Condition hole prior to run casing w/ last BHA. (RIH to Btm, CBU, Spot pill, POOH to surface)	Hole Cond/Circ		GD	3986		40.0	1.67	42.4	1.8	44.0	1.83	47.0	3934.0		0.00	52.35	
R/U CSG running gear, Run 9-5/8" casing to Btm, R/D CSG running gear.	CSG		GD	3986		55.0	2.29	44.7	2.5	60.5	2.52	49.5	3934.0	50	2.08	54.44	
Perform 9-5/8" casing cementing operation.	CMT		GD	3986		8.0	0.33	45.0	0.4	8.8	0.37	49.9	3934.0	46	1.92	56.35	41 hrs SPAD Energy waiting due to not approved CMT program, Lack of pump truck & CMT add.
Set 9-5/8" casing slip and test same w/ 2000 psi. Hang BOPs, Cut & L/D Landing Joint.	N/D BOP		GD	3986		10.0	0.42	45.5	0.5	11.0	0.46	50.4	3934.0	7	0.29	56.65	
Dress & bevel CSG neck, Install 13-5/8", 5K * 11", 10K THS. Inject plastic packing & test P-Seal w/ 4000 psi.	N/U WH		GD	3986		6.0	0.25	45.7	0.3	6.6	0.28	50.6	3934.0	7	0.29	56.94	
N/U 13-5/8", 10K BOP's & related lines.	N/U BOP		GD	3986		8.0	0.33	46.0	0.4	8.8	0.37	51.0	3934.0	7.5	0.31	57.25	
Test Accumulator unit, BOPs, choke manifold & WC equipments as per NIOC policy.	BOP TEST		GD	3986		8.0	0.33	46.4	0.4	8.8	0.37	51.4	3934.0	3.5	0.15	57.40	
L/D 8-1/2" drill Collars, P/U & M/U 6-1/2" DCs and 5" DPs/HWDPs stands for drilling next hole section, R/Back same in derrick.	Trip		GD	3986		7.0	0.29	46.7	0.3	7.7	0.32	51.7	3934.0	5	0.21	57.60	
P/U & M/U 8-1/2" MT Bit BHA, RIH to tag Plugs.	D/Out		GD	3986		17.0	0.71	47.4	0.8	18.7	0.78	52.5	3934.0	21	0.88	58.48	3.5 hrs OEOC waiting due to L/D totally 15 Jts 5" DP Buckled.
D/Out Plugs, F.Collar & CMT to 7m above CSG Shoe, work on Junk sub.	D/Out		GD	3986		5.0	0.21	47.6	0.2	5.5	0.23	52.7	3934.0	10	0.42	58.90	
Circ and balance Mw. R/U CMT surface lines, close well and Perform 9-5/8" CIT. with 1000 psi for 10 min, R/D CMT surface lines.	Testing		GD	3986		3.0	0.13	47.7	0.1	3.3	0.14	52.8	3934.0	1.5	0.06	58.96	
POOH to surface.	D/Out		GD	3986		17.0	0.71	48.4	0.8	18.7	0.78	53.6	3934.0	13.5	0.56	59.52	
RIH 8-1/2" MT Bit with 9-5/8" Scraper to Btm (7m above shoe), CBU, Scrap Casing prior to run CMT quality Log, CBU, Spot pill, POOH.	Log		GD	3986		36.0	1.50	49.9	1.7	39.6	1.65	55.3	3934.0	26	1.08	60.60	3 hrs OEOC waiting due to work on drawworks.
Change Ram, Install shooting nipple, R/U WL Logging Eq. Perform Logging (SBT or CAST/GR/CCL/CBL/VDL logs) from Surface to 3986 m in 9-5/8" CSG, R/D Logging	Log		GD	3986		13.0	0.54	50.5	0.6	14.3	0.60	55.9	3934.0	15.5	0.65	61.25	
P/U & M/U 8-1/2" PDC Bit BHA (including MWD & St. STB @ 20m), RIH to tag CMT (7m above shoe).	Trip		GD	3986		18.0	0.75	51.2	0.8	19.8	0.83	56.7	3934.0	12.5	0.52	61.77	
Displace well volume with 100 pcf Salt Saturated Kcl/Polymer Mud.	D/Out		GD	3986		4.0	0.17	51.4	0.2	4.4	0.18	56.9	3934.0	2	0.08	61.85	
Cont' D/Out CMT, Shoe and Rat Hole.	Hole Cond/Circ		GD	3986		3.0	0.13	51.5	0.1	3.3	0.14	57.0	3934.0		0.00	61.85	

8-1/2" Hole Section

8-1/2" Hole Section / 7" Liner																	
Drilling 8-1/2" hole to 3990m.	Drilling	4	FH	3990	2	2.0	0.08	51.6	0.1	2.2	0.09	57.1	3939.0	1.5	0.06	61.92	
CBU and balance Mw. perform FIT with Required Pressure 728 psi (EMW: 108pcf, should be equal to max. planned Mw for this hole section plus 2 pcf)	FIT		FH	3990		3	0.13	51.7	0.1	3.3	0.14	57.2	3939.0	2	0.08	62.00	
Drilling 8-1/2" hole f/ 3990m to well TD @ 4200m.	Drilling	210	FH	4200	3.0	70.0	2.92	54.6	3.2	77.0	3.21	60.4	4200.0	116	4.83	66.83	1 hrs OEOC waiting due to failure of wash pipe.
Take MWD survey every drilled stand.	Survey		FH	4200		3.5	0.15	54.8	0.2	3.9	0.16	60.6	4200.0		0.00	66.83	
CBU, Perform condition trip to last CSG Shoe prior to Logging, RIH to Btm, CBU, Spot pill.	Hole Cond/Circ		FH	4200		6.0	0.25	55.0	0.3	6.6	0.28	60.9	4200.0	8.5	0.35	67.19	
POOH to surface.	Trip		FH	4200		19.0	0.79	55.8	0.9	20.9	0.87	61.8	4200.0	10	0.42	67.60	
BOP Test.	BOP Test		FH	4200		6.0	0.25	56.1	0.3	6.6	0.28	62.0	4200.0		0.00	67.60	
Change Ram, Install shooting nipple, R/U WL Logging Eq. Perform Logging (Full Suite/CDR/GR Logs), R/D Logging Eq., change Ram.	Log		FH	4200		11.0	0.46	56.5	0.5	12.1	0.50	62.5	4200.0	45	1.88	69.48	8.5 hrs Seland waiting due to reject data run #3 by client.
RIH w/ 8-1/2" PDC Bit BHA (Without MWD), condition hole prior to 7" Liner running (RIH to Btm, CBU, Spot pill, POOH to surface)	Hole Cond/Circ		FH	4200		38.0	1.58	58.1	1.7	41.8	1.74	64.3	4200.0		0.00	69.48	
R/U 7" Liner running Eq., Run 7" liner (+/- 314m LNR) to Btm @ 4200m, R/U CMT lines, Set L.Hanger +/- 3886m (100m overlap), release L.Hanger setting tool.	CSG		FH	4200		30.0	1.25	59.3	1.4	33.0	1.38	65.6	4200.0	95.5	3.98	73.46	1 hrs OEOC rig contractor waiting due to work on drawworks. 1 hrs OEOC LNR Running waiting due to work on jam system. 51 hrs. SN Packer 7 hrs. OEOC 9-5/8" dry test service waiting due to wait on rec. spare part od DCIP valve
Cement 7" Liner as per program.	CMT		FH	4200		5.0	0.21	59.6	0.2	5.5	0.23	65.9	4200.0	9.5	0.40	73.85	
POOH w/ L.Hanger setting tool to 3500m, Flush string & CBU, POH to 3300m.	CSG		FH	4200		4.0	0.17	59.7	0.2	4.4	0.18	66.1	4200.0		0.00	73.85	
Close the well and WOC.	WOC		FH	4200		4.0	0.17	59.9	0.2	4.4	0.18	66.2	4200.0		0.00	73.85	
Open the Well. Cont' POH to surface & L/D L.Hanger setting tool.	CSG		FH	4200		12.0	0.50	60.4	0.6	13.2	0.55	66.8	4200.0	9	0.38	74.23	
BOP Test.	BOP Test		FH	4200		6.0	0.25	60.6	0.3	6.6	0.28	67.1	4200.0	5	0.21	74.44	
P/U & M/U 8-1/2" MT Bit BHA, RIH to L.Lap @ +/- 3886m while D/Out cement (If any cement: Considered 75TD wet for CMT Job), Circ & clean the well, POOH.	D/Out		FH	4200		40.0	1.67	62.3	1.8	44.0	1.83	68.9	4200.0	22	0.92	75.35	
L/D excess 5" DPs/HWDP.	D/Out		FH	4200		5.0	0.21	62.5	0.2	5.5	0.23	69.1	4200.0	4	0.17	75.52	
P/U & M/U 4-3/4" DCs & 3-1/2" DPs/HWDPS, R/Back same in derrick.	D/Out		FH	4200		7.0	0.29	62.8	0.3	7.7	0.32	69.5	4200.0	5	0.21	75.73	
P/U & M/U 5-7/8" MT Bit BHA, RIH to tag POB, D/Out POB, RIH to LDC depth @ +/- 4170m while D/Out any cement, work on junk sub and CBU. Perform 7" Liner CIT. (80% of Burst press) with 1500 psi,for 10 min, (If not obs bumping). POOH to top of Liner Lap @ +/- 3886m, CBU, perform Liner lap positive test w/ 2000 psi, POOH to Surface.	D/Out		FH	4200		40.0	1.67	64.5	1.8	44.0	1.83	71.3	4200.0	32	1.33	77.06	1 hrs. OEOC rig contractor waiting due to poor proviosion service.
RIH with 5-7/8" MT Bit and tandem scraper (7" & 9-5/8") to LDC depth, Scrap inside 7" Liner and 9-5/8" casing before Dry test, CBU, POOH to Surface.	Testing		FH	4200		36.0	1.50	66.0	1.7	39.6	1.65	72.9	4200.0	35	1.46	78.52	4 hrs. OEOC waiting due to lack of D.water.
BOP Test.	BOP Test		FH	4200		6.0	0.25	66.2	0.3	6.6	0.28	73.2	4200.0		0.00	78.52	
RIH with 9-5/8" Dry test BHA, RIH to L.Lap @ +/- 3886m, Perform 7" Liner lap Dry Test (based on fahliyan Reservoir Datum Pressure), POOH.	Testing		FH	4200		68.0	2.83	69.1	3.1	74.8	3.12	76.3	4200.0	124	5.17	83.69	2 hrs. OEOC waiting due to lack of PC monitor in WSA company-man office.
Change Ram, Install shooting nipple, R/U WL Logging Eq. Perform CMT quality Logging of 7" Liner from 7" liner shoe @ 4200 m to 7" Liner Lap @ 3886 m (SBT or CAST/CBL/VDL), R/D Logging Eq, change Ram.	Log		FH	4200		11.0	0.46	69.5	0.5	12.1	0.50	76.8	4200.0	41	1.71	85.40	5 hrs. Sealand waiting due to fail signal in 2nd run.

Completion Section

P/U & M/U 5-7/8" MT Bit and C/Out BHA, RIH to LDC depth. Displace 7" Liner volume with CaBr2 Perforation fluid.	Completion		FH	4200		36.0	1.50	71.0	1.7	39.6	1.65	78.5	4200.0	6	0.25	85.65	6 hrs. OEOC waiting due to transfer 40bbl CaCl2 brine to active system.
P/U & M/U DST/TCP Perforation Assy, RIH to perforation depth, correlate depth with WL and perforate Guns as program. POOH DST/TCP Perforation Assy.	Completion		FH	4200		64.0	2.67	73.7	2.9	70.4	2.93	81.4	4200.0	85	3.54	89.19	28 hrs. wait on daylight
P/U & M/U 5-7/8" Mill BHA, RIH to top of LDC depth while smoothing Perforated intervals.	Completion		FH	4200		36.0	1.50	75.2	1.7	39.6	1.65	83.1	4200.0	86.5	3.60	92.79	20 hrs. wait on receive 3-1/2"4-1/2" tubing.
Change Ram, Install shooting nipple, R/U WL Eq. RIH WL with CCL/GR Log to confirm Perforated intervals, POOH and R/D Logging Eq, change Ram.	Completion		FH	4200		11.0	0.46	75.6	0.5	12.1	0.50	83.6	4200.0	19	0.79	93.58	
P/U and M/U 5-7/8" Mill and 7" Scraper. RIH to Perforation depth while Scrap 7" PKR setting depth Interval, CBU. Displace well volume with CaBr2 Completion fluid.	Completion		FH	4200		24.0	1.00	76.6	1.1	26.4	1.10	84.7	4200.0	892	37.17	130.75	250 hrs. wait on receive 3-1/2"4-1/2" tubing. 608 hrs. OEOC waiting
POOH to surface while L/D all Tubulars.	Completion		FH	4200		56.0	2.33	79.0	2.6	61.6	2.57	87.2	4200.0	22.5	0.94	131.69	
BOP Test.	BOP TEST		FH	4200		6.0	0.25	79.2	0.3	6.6	0.28	87.5	4200.0	6.5	0.27	131.96	
R/U Completion running Eq, Run 4-1/2" * 3-1/2", 10K Completion string as per program, R/D completion running Eq.	Completion		FH	4200		124.0	5.17	84.4	5.7	136.4	5.68	93.2	4200.0	131.5	5.48	137.44	4.5 hrs. Spad Energy waiting. (hidden NPT) 3 hrs. OEOC TBG running service waiting.
Install BPV in TBG Hanger, N/D BOPs.	Completion		FH	4200		14.0	0.58	85.0	0.6	15.4	0.64	93.8	4200.0	15.5	0.65	138.08	3 hrs. OEOC waiting due to SBR ram block packer has leakage.
Retrieve BPV, set TWCV in TBG Hanger. N/U X-mas Tree, perform required test as per program, Retrieve TWCV.	Completion		FH	4200		8.0	0.33	85.3	0.4	8.8	0.37	94.2	4200.0	12	0.50	138.58	
R/U Slick Line, Set protection sleeve in TRSSSV, R/D Slick Line.	Completion		FH	4200		4.0	0.17	85.5	0.2	4.4	0.18	94.4	4200.0	9	0.38	138.96	
R/U CTU, Stimulation and Well Testing Eq. RIH w/ CT to Btm, displace the well with Diesel, POOH CT to surface.	Completion		FH	4200		24.0	1.00	86.5	1.1	26.4	1.10	95.5	4200.0	178	7.42	146.38	155 hrs. OEOC Stimulation Dep. Waitingdue to lack of well test choke manifold, surge tank, CTU & delay in operation.
R/U Slick Line, Retrieve protection sleeve from TRSSSV, R/D Slick Line.	Completion		FH	4200		4.0	0.17	86.6	0.2	4.4	0.18	95.7	4200.0		0.00	146.38	
Flow the well as Pre-Acid clean up till acceptable BSW and reading stable flow rate on 2 different choke size for 72 hrs.	Completion		FH	4200		77.0	3.21	89.8	3.5	84.7	3.53	99.2	4200.0	60.5	2.52	148.90	
R/U Slick Line, Set protection sleeve in TRSSSV, R/D Slick Line.	Completion		FH	4200		Rig Less	0.00	89.8	0.0	0.0	0.00	99.2	4200.0	16.5	0.69	149.58	4 hrs. OEOC Slick Line Service waiting due to delay in Operation. (perform eqp press test) Run memory gauge & take flowing survey
RIH w/ CT and perform Acid Wash as per program, POOH CT to surface.	Completion		FH	4200		Rig Less	0.00	89.8	0.0	0.0	0.00	99.2	4200.0	19	0.79	150.38	
R/U Slick Line, Retrieve protection sleeve from TRSSSV, R/D Slick Line.	Completion		FH	4200		Rig Less	0.00	89.8	0.0	0.0	0.00	99.2	4200.0		0.00	150.38	
Flow the well as Post-Acid clean up till acceptable BSW and reading stable flow rate on 2 different choke size for 96 hrs.	Completion		FH	4200		Rig Less	0.00	89.8	0.0	0.0	0.00	99.2	4200.0	44	1.83	152.21	2.5 hrs. OEOC Well Test Sevice waiting due to failure of separator & C/M.
R/U Slick Line, Set protection sleeve in TRSSSV, R/D Slick Line.	Completion		FH	4200		Rig Less	0.00	89.8	0.0	0.0	0.00	99.2	4200.0	14.5	0.60	152.81	R/U & perfprm press test eqp & run memory gauge
RIH w/ CT to Btm, displace well with Diesel, POOH CT to surface, R/D CTU and well Testing Eq.	Completion		FH	4200		Rig Less	0.00	89.8	0.0	0.0	0.00	99.2	4200.0	12.5	0.52	153.33	
R/U Slick Line, RIH w/ SL and Install Plug and Prong into NO-GO L.NPL, POOH.	Completion		FH	4200		Rig Less	0.00	89.8	0.0	0.0	0.00	99.2	4200.0		0.00	153.33	
RIH w/ SL and Retrieve protection sleeve from TRSSSV, R/D Slick Line.	Completion		FH	4200		Rig Less	0.00	89.8	0.0	0.0	0.00	99.2	4200.0	5	0.21	153.54	
Secure the well by set BPV in TBG Hanger, close all X-mas Tree valves except BMV, Release the Rig.	Completion		FH	4200		Rig Less	0.00	89.8	0.0	0.0	0.00	99.2	4200.0	25	1.04	154.58	
Total Duration of 8-1/2" Hole Section:							***38.4***	***89.8***			***42.2***	***99.2***					
NOTE:																	
Total Duration for Cut & Slip DL in all hole section.	Cut & Slip		FH	4200		12.5	0.52	90.4	0.6	13.8	0.57	99.8	4200.0	10	0.42	155.00	
Total Duration for Rig Service (Consider 0.5hr per day) in all hole section.	Rig Service		FH	4200		48.8	2.03	92.4	2.2	53.7	2.24	102.0	4200.0		0.00	155.00	

3.4 NPT Break-Down

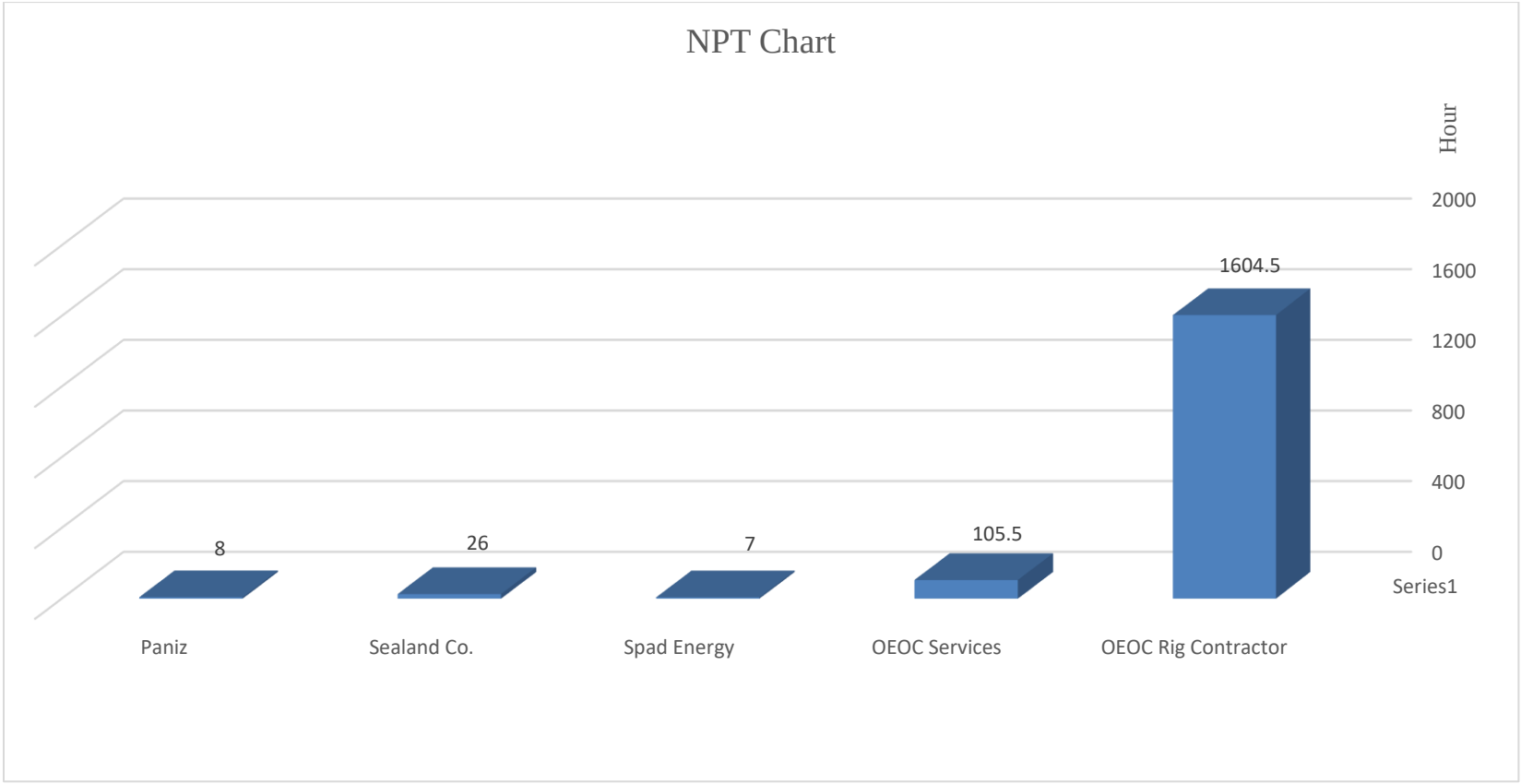


Figure 3-3: AZNS-040 NPT Break-Down Chart

Item	Company	Date	Problem event	Lost Time (Hrs)
1	OEOC	1402.04.26	Twist off 8-1/2" DC	11.5
		1402.04.27		22.5
		1402.05.02		15
2	OEOC	1402.05.03	Fishing & not using drilling jar in BHA	24
		1402.05.04		24
		1402.05.05		24
		1402.05.06		24
		1402.05.07		24
		1402.05.08		11.5
		1402.05.14		22
		1402.05.15		24
3	OEOC	1402.05.16	L/D damaged & buckled pipes	24
4	OEOC	1402.05.17	M/U DPs jt by jt	24
5	OEOC	1402.05.17	Stop drilling due to low ROP	8
6	OEOC	1402.05.20	RIH w/ 12 – ¼” B.bit & 9 – 5/8” Junk sub	21
7	OEOC	1402.05.21	Due to work on junk sub, Low ROP, POOH, Change bit	24
		1402.05.22		6
		1402.05.23		24
		1402.05.24		24

		1402.05.25		24
		1402.05.26		14.5
8	OEOC	1402.06.01	Fishing, Twist off string from 8” shock sub of Jar	5
		1402.06.02		24
		1402.06.03		22
9	OEOC	1402.06.04	Run bak to bottom w/ drilling BHA after fishing	2
10	OEOC	1402.06.10	L/D 5” DPs buckled	2
		1402.06.11		4
11	OEOC	1402.06.14	Approve CMT program & prepare solution for 9 – 5/8” casing cement job	8
		1402.06.15		24
		1402.06.16		10
12	OEOC	1402.06.23	Change wash pipe	2
13	Sealand	1402.06.29	Repeat log	15
14	OEOC	1402.06.29	Work on drawworks	1
15	OEOC Service	1402.06.29	Work on jam system	2
16	Spad Energy	1402.06.30	Delay in operation	7
17	OEOC	1402.07.04	Lack of drill water	5
18	Sealand	1402.07.08	Contamination mud, Logging tools failure	11
19	OEOC	1402.07.09	Polish inside tubings	24
20	OEOC	1402.07.10	Set SN Packer	24
21	OEOC	1402.07.11	Dry test	24

		1402.07.12		24
		1402.07.13		24
		1402.07.14		24
22	OEOC	1402.07.15	Wait on daylight	4
		1402.07.16		9
		1402.07.17		9
		1402.07.18		6
23	Paniz	1402.07.20	Wait on logging group	8
24	OEOC	1402.07.21	Wait on displace the well w/ $CaBr_2$ completion fluid	10
		1402.07.22		24
		1402.07.23		24
		1402.07.24		24
		1402.07.25		24
		1402.07.26		24
		1402.07.27		24
		1402.07.28		24
		1402.07.29		24
		1402.07.30		24
		1402.08.01		24
25	OEOC	1402.08.02	Wait on receive tubings & completion acc.	24
		1402.08.03		24

26	OEOC	1402.08.04	Twist off string & fishing	24
27	OEOC	1402.08.05	Close the well & monitor well pressure	24
27	OEOC	1402.08.06	Shut in the well & monitor annular pressure	24
28	OEOC	1402.08.07	Displace completion in well w/ poly. Mud	24
29	OEOC	1402.08.08	POOH w/ 7” scraper	24
30	OEOC	1402.08.09	RIH w/ 5 – 7/8” junk mill	24
		1402.08.10		24
31	OEOC	1402.08.11	RIH & POOH w/ 7 – 5/16” Polish mill	24
		1402.08.12		24
32	OEOC	1402.08.13	POOH 7 – 5/16” dress mill	24
33	OEOC	1402.08.14	Wait on receive 9 – 5/8” dry test tools	24
34	OEOC	1402.08.15	Wait on receive completion fluid ($CaBr_2$)	24
		1402.08.16		24
		1402.08.17		24
		1402.08.18		24
		1402.08.19		24
		1402.08.20		24
		1402.08.21		24
		1402.08.22		24
		1402.08.23		24
		1402.08.24		24

		1402.08.25		24
		1402.08.26		24
		1402.08.27		24
		1402.08.28		24
35	OEOC	1402.08.29	Wait on replacing completion fluid & delay in operation	6
36	OEOC Services	1402.09.09	Wait on R/U well test equipment & coiled tubing	24
		1402.09.10		24
		1402.09.12		24
		1402.09.13		24
		1402.0914		5
37	OEOC Services	1402.09.20	Work on separator	2.5
Total NPT (hrs.)				1751.5
Total NPT (days)				73

Table 3-1: AZNS-040 NPT Time Break-Down

4. Geological Reports

4.1 Logging and Testing Reports

Wireline Logging Reports

Well AZNS-040 will undergo wireline logging operation in all required intervals. Since this well logging plan is targeted to evaluate the hydrocarbon potential, the full suite Logs should be run in 12-¼" and 8-½" open hole to collect reservoir and required data.
Logging is limited to GR/CDR logs in 12-¼" and 8-½" open holes and CAST log in 9-5/8" casing and 7" Liner that must be carried out by wire line.
The elaborative Logging Data in Well AZNS-040 are as below:

Table 4-1: Logging Table

Hole Size	Casing/Liner Size	Run No.	Log	Interval (m)	Date
12-1/4"	9-5/8"	1	Full Suite: DAS – DLL – MSFL – CALIP – GR	1492 – 3934	02 – Sep – 23
		2	Neutron/Density: LDT – CNT – NGRS	1492 – 3934	03 – Sep – 23
		1	CCL/GR/SBT/VDL	1385 – 3928	12 – Sep – 23
		1	Full Suite: Resistivity – Sonic – DCR – GR	3933 – 4200	19 – Sep - 23
		2	LDT/CNT/NGS	3933 – 3936	19 – Sep – 23
		3	LDT/CNT/NGS	3933 – 4200	20 – Sep – 23
8-1/2"	7"	1	CCL/GR/SBT/VDL	4150 – 3812	30 – Sep – 23
		2	CCL/GR/SBT/VDL	4150 – 3812	01 – Oct – 23
		3	CCL/GR/SBT/VDL	4150 – 3812	01 – Oct - 23

4.2 9-5/8" Casing Cement Evaluation Log Result

Results - Summary

Total Pass (M)	684.80
Total Fail (M)	1244.05
Total Partial Pass (M)	508.20

Results - Filtered Intervals

Show Pass Zones	Yes
Show Fail Zones	No
Show Partial Passes	No
Minimum Interval	5

Isolation Key					
	Pass		Partial		Fail

Top Depth	Bottom Depth	Interval (M)	Min (dB/ft)	Avg (dB/ft)	Max (dB/ft)	Bonded	Derivative	Channel Risk Pass	Formation	Isolated
1491.95	1497.05	5.10	6.39	8.77	15.39	✓	—	✓	✓	
1540.75	1556.65	15.90	6.31	9.28	13.97	✓	—	✓	✓	
1609.75	1614.85	5.10	6.31	9.38	12.47	✓	—	✓	✓	
1618.15	1632.55	14.40	6.35	10.34	15.77	✓	—	✓	✓	
1702.05	1711.05	9.00	6.37	9.84	14.84	✓	—	✓	✓	
2313.65	2320.25	6.60	6.33	8.42	13.94	✓	—	✓	✓	
2363.65	2370.25	6.60	5.95	10.72	17.87	✓	—	✓	✓	
2409.95	2446.95	37.00	6.24	9.92	17.99	✓	—	✓	✓	
2468.35	2483.85	15.50	6.42	8.08	14.33	✓	—	✓	✓	
2487.95	2493.55	5.60	6.35	8.56	14.71	✓	—	✓	✓	
2662.35	2671.15	8.80	6.35	9.21	15.98	✓	—	✓	✓	
2688.35	2694.35	6.00	6.33	9.65	12.82	✓	—	✓	✓	
2695.05	2700.95	5.90	5.71	12.64	17.82	✓	—	✓	✓	
3098.65	3118.05	19.40	6.15	9.15	15.20	✓	—	✓	✓	
3216.55	3223.65	7.10	6.24	9.00	11.91	✓	—	✓	✓	
3259.75	3287.35	27.60	6.35	9.96	16.01	✓	—	✓	✓	
3293.75	3303.85	10.10	6.34	10.02	14.95	✓	—	✓	✓	
3433.65	3439.85	6.20	7.89	10.63	13.54	✓	—	✓	✓	
3468.75	3487.85	19.10	7.56	11.99	19.81	✓	—	✓	✓	
3501.75	3509.15	7.40	7.42	11.39	20.64	✓	—	✓	✓	
3509.45	3518.05	8.60	7.62	10.73	12.99	✓	—	✓	✓	
3519.55	3525.35	5.80	7.77	10.15	12.00	✓	—	✓	✓	
3535.35	3540.75	5.40	7.54	11.26	17.94	✓	—	✓	✓	
3696.45	3702.55	6.10	7.48	10.90	12.81	✓	—	✓	✓	
3708.55	3724.45	15.90	6.22	10.80	17.93	✓	—	✓	✓	
3729.35	3737.05	7.70	7.06	10.75	12.61	✓	—	✓	✓	
3746.75	3758.45	11.70	7.51	11.40	19.74	✓	—	✓	✓	
3777.25	3787.85	10.60	7.56	11.12	20.48	✓	—	✓	✓	
3788.15	3800.75	12.60	7.09	11.49	18.92	✓	—	✓	✓	
3805.45	3811.55	6.10	7.58	11.82	13.96	✓	—	✓	✓	
3820.75	3829.65	8.90	7.63	11.68	19.28	✓	—	✓	✓	

Results - All Intervals

Top Depth	Bottom Depth	Interval (M)	Min (dB/ft)	Avg (dB/ft)	Max (dB/ft)	Bonded	Derivative	Channel Risk Pass	Formation	Isolated
1491.95	1497.05	5.10	6.39	8.77	15.39	✓	–	✓	✓	
1497.05	1498.05	1.00	5.28	6.77	8.76	✗	–	–	✓	
1498.05	1504.85	6.80	4.83	6.45	12.30	✗	–	–	✗	
1504.85	1505.25	0.40	6.07	8.75	11.73	✓	–	✓	✗	
1505.25	1510.75	5.50	4.96	5.81	7.58	✗	–	–	✗	
1510.75	1515.25	4.50	5.10	7.51	11.28	✗	–	–	✓	
1515.25	1515.35	0.10	6.73	8.56	10.66	✓	–	✓	✓	
1515.35	1522.45	7.10	5.07	7.29	14.31	✗	–	–	✗	
1522.45	1523.05	0.60	5.93	8.43	11.14	✗	–	–	✓	
1523.05	1527.65	4.60	5.98	9.50	13.47	✓	–	✓	✓	
1527.65	1528.25	0.60	5.39	8.35	11.07	✗	–	–	✓	
1528.25	1532.05	3.80	4.84	6.41	10.79	✗	–	–	✗	
1532.05	1532.75	0.70	6.36	7.85	9.30	✓	–	✓	✗	
1532.75	1535.75	3.00	5.03	6.82	9.09	✗	–	–	✗	
1535.75	1536.35	0.60	5.43	8.10	10.36	✗	–	–	✓	
1536.35	1539.65	3.30	6.33	10.62	16.26	✓	–	✓	✓	
1539.65	1540.75	1.10	5.08	7.38	10.99	✗	–	–	✓	
1540.75	1556.65	15.90	6.31	9.28	13.97	✓	–	✓	✓	
1556.65	1560.15	3.50	5.61	8.83	13.97	✗	–	–	✓	
1560.15	1560.35	0.20	6.35	8.12	10.53	✓	–	✓	✓	
1560.35	1560.85	0.50	5.95	7.58	9.92	✗	–	–	✓	
1560.85	1561.15	0.30	6.41	7.81	10.06	✓	–	✓	✓	
1561.15	1561.95	0.80	6.04	7.73	10.72	✗	–	–	✓	
1561.95	1564.85	2.90	5.46	6.77	9.95	✗	–	–	✗	
1564.85	1564.95	0.10	6.20	7.22	9.19	✗	–	–	✓	
1564.95	1565.25	0.30	6.40	7.44	9.86	✓	–	✓	✓	
1565.25	1565.45	0.20	6.38	7.80	10.81	✗	–	–	✓	
1565.45	1565.85	0.40	6.32	8.21	11.74	✓	–	✓	✓	
1565.85	1567.15	1.30	5.49	7.86	11.74	✗	–	–	✓	
1567.15	1581.05	13.90	4.86	6.56	13.00	✗	–	–	✗	
1581.05	1582.25	1.20	6.06	10.52	14.48	✓	–	✓	✗	
1582.25	1582.65	0.40	5.95	7.72	10.42	✗	–	–	✗	
1582.65	1583.25	0.60	6.12	8.05	10.44	✓	–	✓	✗	
1583.25	1586.55	3.30	5.23	6.83	9.78	✗	–	–	✗	
1586.55	1586.65	0.10	6.37	7.24	8.28	✓	–	✓	✗	
1586.65	1587.05	0.40	5.90	7.59	9.49	✗	–	–	✗	
1587.05	1588.35	1.30	5.77	8.37	11.26	✗	–	–	✓	
1588.35	1589.35	1.00	5.94	8.48	11.24	✓	–	✓	✓	
1589.35	1590.05	0.70	5.28	7.56	9.29	✗	–	–	✓	
1590.05	1591.45	1.40	5.34	7.81	10.70	✗	–	–	✗	
1591.45	1593.15	1.70	5.84	9.68	14.31	✓	–	✓	✗	
1593.15	1595.95	2.80	5.49	7.59	10.58	✗	–	–	✗	
1595.95	1596.55	0.60	5.85	8.54	10.58	✗	–	–	✓	
1596.55	1597.45	0.90	6.34	9.75	12.11	✓	–	✓	✓	
1597.45	1598.05	0.60	5.74	8.24	11.61	✗	–	–	✓	
1598.05	1598.65	0.60	6.51	8.01	9.86	✓	–	✓	✓	
1598.65	1598.75	0.10	6.39	7.62	9.06	✓	–	✓	✗	

1598.75	1599.25	0.50	5.66	6.95	8.75	✗	–	·	✗	
1599.25	1600.15	0.90	5.63	7.83	10.09	✗	–	·	✓	
1600.15	1601.75	1.60	6.32	8.84	11.20	✓	–	✓	✓	
1601.75	1602.65	0.90	5.47	8.16	11.66	✗	–	·	✓	
1602.65	1603.95	1.30	5.49	7.70	11.47	✗	–	·	✗	
1603.95	1605.15	1.20	6.26	10.26	14.27	✓	–	✓	✗	
1605.15	1609.65	4.50	5.17	7.99	10.97	✗	–	·	✗	
1609.65	1609.75	0.10	6.57	8.48	10.07	✓	–	✓	✗	
1609.75	1614.85	5.10	6.31	9.38	12.47	✓	–	✓	✓	
1614.85	1615.05	0.20	6.21	8.61	11.01	✗	–	·	✓	
1615.05	1615.25	0.20	6.45	9.59	11.82	✓	–	✓	✓	
1615.25	1616.25	1.00	6.39	10.65	13.63	✓	–	✓	✗	
1616.25	1617.05	0.80	6.12	7.65	9.83	✗	–	·	✗	
1617.05	1617.55	0.50	6.44	8.34	10.74	✓	–	✓	✗	
1617.55	1617.65	0.10	6.43	8.33	10.74	✓	–	✓	✓	
1617.65	1618.15	0.50	6.30	7.90	10.58	✗	–	·	✓	
1618.15	1632.55	14.40	6.35	10.34	15.77	✓	–	✓	✓	
1632.55	1633.05	0.50	5.97	10.18	12.72	✗	–	·	✓	
1633.05	1633.45	0.40	6.00	10.06	12.88	✓	–	✓	✓	
1633.45	1634.35	0.90	5.56	9.99	12.75	✗	–	·	✓	
1634.35	1637.95	3.60	6.32	10.38	15.74	✓	–	✓	✓	
1637.95	1640.45	2.50	5.53	8.40	10.83	✗	–	·	✓	
1640.45	1640.75	0.30	6.36	8.85	10.36	✓	–	✓	✓	
1640.75	1642.45	1.70	5.33	8.12	10.68	✗	–	·	✓	
1642.45	1647.25	4.80	6.34	9.82	12.40	✓	–	✓	✓	
1647.25	1647.45	0.20	6.33	8.47	10.09	✗	–	·	✓	
1647.45	1649.95	2.50	6.27	10.04	14.61	✓	–	✓	✓	
1649.95	1651.35	1.40	5.33	8.49	11.44	✗	–	·	✓	
1651.35	1651.85	0.50	4.99	8.24	10.63	✗	–	·	✗	
1651.85	1654.05	2.20	5.31	8.37	10.99	✗	–	·	✓	
1654.05	1654.65	0.60	5.63	9.00	10.99	✗	–	·	✗	
1654.65	1656.05	1.40	5.28	8.64	10.86	✗	–	·	✓	
1656.05	1657.05	1.00	5.31	8.13	10.59	✗	–	·	✗	
1657.05	1657.55	0.50	5.73	8.89	10.66	✗	–	·	✓	
1657.55	1661.05	3.50	6.34	9.47	15.20	✓	–	✓	✓	
1661.05	1661.55	0.50	6.02	8.22	11.28	✗	–	·	✓	
1661.55	1665.25	3.70	6.41	9.23	11.65	✓	–	✓	✓	
1665.25	1666.75	1.50	5.59	9.12	11.74	✗	–	·	✓	
1666.75	1671.15	4.40	5.69	9.47	13.57	✓	–	✓	✓	
1671.15	1671.95	0.80	5.42	7.71	9.12	✗	–	·	✓	
1671.95	1672.35	0.40	6.33	7.89	9.46	✓	–	✓	✓	
1672.35	1672.75	0.40	6.06	8.04	9.42	✗	–	·	✓	
1672.75	1673.25	0.50	6.31	8.05	10.05	✓	–	✓	✓	
1673.25	1673.75	0.50	5.46	7.93	10.62	✗	–	·	✓	
1673.75	1674.65	0.90	5.46	8.24	10.81	✗	–	·	✗	
1674.65	1674.85	0.20	6.16	8.33	10.57	✗	–	·	✓	
1674.85	1677.45	2.60	6.42	8.76	10.43	✓	–	✓	✓	
1677.45	1678.85	1.40	5.29	7.84	9.91	✗	–	·	✓	
1678.85	1679.75	0.90	5.08	7.51	10.19	✗	–	·	✗	
1679.75	1680.25	0.50	5.13	8.81	12.66	✗	–	·	✓	

1680.25	1681.65	1.40	5.96	11.78	17.97	✓	–	✓	✓	
1681.65	1682.05	0.40	5.82	8.31	10.20	✗	–	–	✓	
1682.05	1682.75	0.70	6.23	8.27	10.39	✓	–	✓	✓	
1682.75	1685.15	2.40	5.50	8.26	11.52	✗	–	–	✓	
1685.15	1687.15	2.00	5.47	7.70	10.77	✗	–	–	✗	
1687.15	1691.35	4.20	5.32	8.49	11.51	✗	–	–	✓	
1691.35	1694.35	3.00	6.34	10.31	15.11	✓	–	✓	✓	
1694.35	1695.55	1.20	5.28	8.43	11.38	✗	–	–	✓	
1695.55	1695.65	0.10	6.29	7.74	8.69	✓	–	✓	✓	
1695.65	1696.65	1.00	5.13	7.14	8.86	✗	–	–	✓	
1696.65	1697.75	1.10	5.42	6.99	8.86	✗	–	–	✗	
1697.75	1698.25	0.50	6.20	7.37	9.11	✗	–	–	✓	
1698.25	1698.55	0.30	6.42	7.87	9.48	✓	–	✓	✓	
1698.55	1699.55	1.00	6.16	7.69	9.47	✗	–	–	✓	
1699.55	1700.05	0.50	6.39	7.67	9.03	✓	–	✓	✓	
1700.05	1700.35	0.30	6.38	7.62	8.87	✗	–	–	✓	
1700.35	1701.75	1.40	6.34	8.16	9.96	✓	–	✓	✓	
1701.75	1702.05	0.30	5.87	8.15	9.96	✗	–	–	✓	
1702.05	1711.05	9.00	6.37	9.84	14.84	✓	–	✓	✓	
1711.05	1712.35	1.30	5.39	8.03	11.11	✗	–	–	✓	
1712.35	1715.65	3.30	5.10	7.96	13.99	✗	–	–	✗	
1715.65	1715.75	0.10	6.41	7.71	10.57	✓	–	✓	✗	
1715.75	1715.95	0.20	6.15	7.72	10.58	✗	–	–	✗	
1715.95	1717.95	2.00	5.47	7.87	11.27	✗	–	–	✓	
1717.95	1718.35	0.40	6.33	8.33	10.71	✓	–	✓	✓	
1718.35	1719.45	1.10	5.37	7.68	10.63	✗	–	–	✓	
1719.45	1723.85	4.40	4.67	6.78	9.06	✗	–	–	✗	
1723.85	1724.25	0.40	6.31	7.49	8.74	✓	–	✓	✗	
1724.25	1724.55	0.30	6.23	7.81	9.06	✗	–	–	✗	
1724.55	1725.95	1.40	6.29	11.32	16.65	✓	–	✓	✗	
1725.95	1727.75	1.80	5.26	6.54	8.64	✗	–	–	✗	
1727.75	1728.35	0.60	6.34	7.16	7.89	✓	–	✓	✗	
1728.35	1728.75	0.40	6.10	6.98	7.61	✗	–	–	✗	
1728.75	1728.85	0.10	6.40	6.98	7.44	✓	–	✓	✗	
1728.85	1731.05	2.20	5.23	6.86	9.90	✗	–	–	✗	
1731.05	1732.05	1.00	6.08	7.55	9.58	✗	–	–	✓	
1732.05	1733.75	1.70	6.29	7.82	9.40	✓	–	✓	✓	
1733.75	1734.35	0.60	6.01	7.27	8.61	✗	–	–	✓	
1734.35	1735.85	1.50	6.36	7.62	8.88	✓	–	✓	✓	
1735.85	1736.05	0.20	6.36	7.27	8.53	✗	–	–	✓	
1736.05	1736.45	0.40	6.58	8.46	11.30	✓	–	✓	✓	
1736.45	1737.65	1.20	6.31	10.14	13.97	✓	–	✓	✗	
1737.65	1739.55	1.90	5.30	6.79	8.68	✗	–	–	✗	
1739.55	1740.55	1.00	5.34	6.85	8.43	✗	–	–	✓	
1740.55	1740.75	0.20	6.42	7.42	8.22	✓	–	✓	✓	
1740.75	1743.35	2.60	5.58	7.55	9.08	✗	–	–	✓	
1743.35	1743.45	0.10	6.33	7.33	8.80	✓	–	✓	✓	
1743.45	1744.65	1.20	5.30	7.01	8.73	✗	–	–	✓	
1744.65	1747.15	2.50	5.09	6.48	8.45	✗	–	–	✗	
1747.15	1748.15	1.00	5.84	9.67	13.65	✓	–	✓	✗	

1748.15	1758.55	10.40	4.90	6.80	14.42	✗	–	-	✗	
1758.55	1758.75	0.20	5.78	7.05	8.87	✓	–	✓	✗	
1758.75	1762.45	3.70	4.84	5.85	7.67	✗	–	-	✗	
1762.45	1767.05	4.60	5.18	6.62	8.38	✗	–	-	✓	
1767.05	1782.45	15.40	4.87	6.68	13.71	✗	–	-	✗	
1782.45	1783.25	0.80	4.92	6.41	7.91	✗	–	-	✓	
1783.25	1783.75	0.50	6.36	7.00	7.88	✓	–	✓	✓	
1783.75	1784.35	0.60	5.95	6.94	8.21	✗	–	-	✓	
1784.35	1784.45	0.10	6.23	6.75	7.40	✓	–	✓	✓	
1784.45	1788.25	3.80	5.58	6.81	8.29	✗	–	-	✓	
1788.25	1788.35	0.10	6.39	7.27	8.19	✓	–	✓	✓	
1788.35	1788.85	0.50	6.20	7.33	8.26	✗	–	-	✓	
1788.85	1789.15	0.30	6.26	7.29	8.44	✓	–	✓	✓	
1789.15	1789.85	0.70	5.58	6.80	9.07	✗	–	-	✓	
1789.85	1791.15	1.30	5.25	6.54	9.32	✗	–	-	✗	
1791.15	1792.05	0.90	6.15	10.09	13.69	✓	–	✓	✗	
1792.05	1801.75	9.70	4.62	5.53	8.65	✗	–	-	✗	
1801.75	1802.75	1.00	6.31	10.02	14.11	✓	–	✓	✗	
1802.75	1806.15	3.40	4.67	5.84	7.48	✗	–	-	✗	
1806.15	1807.55	1.40	5.69	6.96	8.96	✗	–	-	✓	
1807.55	1807.75	0.20	6.28	7.05	8.76	✓	–	✓	✓	
1807.75	1809.45	1.70	5.56	6.72	8.45	✗	–	-	✓	
1809.45	1809.95	0.50	6.37	7.50	8.95	✓	–	✓	✓	
1809.95	1810.05	0.10	6.37	7.33	8.84	✗	–	-	✓	
1810.05	1810.15	0.10	6.05	6.86	7.87	✓	–	✓	✓	
1810.15	1810.85	0.70	5.33	6.36	7.49	✗	–	-	✓	
1810.85	1811.25	0.40	5.97	6.84	7.83	✗	–	-	✗	
1811.25	1811.65	0.40	6.34	6.80	7.62	✓	–	✓	✗	
1811.65	1812.25	0.60	5.75	6.82	8.68	✗	–	-	✗	
1812.25	1813.15	0.90	6.33	9.94	13.24	✓	–	✓	✗	
1813.15	1814.15	1.00	5.27	6.52	8.07	✗	–	-	✗	
1814.15	1815.35	1.20	5.33	6.88	10.08	✗	–	-	✓	
1815.35	1815.65	0.30	6.43	7.43	10.08	✓	–	✓	✓	
1815.65	1816.75	1.10	5.06	6.81	9.36	✗	–	-	✓	
1816.75	1818.65	1.90	5.42	6.85	8.77	✗	–	-	✗	
1818.65	1819.45	0.80	5.89	7.83	9.61	✗	–	-	✓	
1819.45	1819.85	0.40	6.42	8.11	9.54	✓	–	✓	✓	
1819.85	1822.55	2.70	4.74	6.84	9.31	✗	–	-	✓	
1822.55	1822.65	0.10	6.90	8.19	9.96	✓	–	✓	✓	
1822.65	1827.15	4.50	4.83	6.39	12.65	✗	–	-	✗	
1827.15	1827.85	0.70	5.42	6.24	7.14	✗	–	-	✓	
1827.85	1827.95	0.10	6.31	6.56	6.78	✓	–	✓	✓	
1827.95	1830.25	2.30	5.61	7.20	8.76	✗	–	-	✓	
1830.25	1830.85	0.60	6.41	7.31	8.16	✓	–	✓	✓	
1830.85	1831.25	0.40	6.37	6.95	7.48	✗	–	-	✓	
1831.25	1831.35	0.10	6.23	6.68	7.03	✓	–	✓	✓	
1831.35	1832.55	1.20	5.82	6.51	7.34	✗	–	-	✓	
1832.55	1832.65	0.10	6.68	7.29	7.68	✓	–	✓	✓	
1832.65	1833.85	1.20	5.86	9.34	12.79	✓	–	✓	✗	
1833.85	1843.55	9.70	4.61	5.59	7.46	✗	–	-	✗	

1843.55	1844.65	1.10	6.18	8.88	12.61	✓	–	✓	✗	
1844.65	1865.45	20.80	4.76	6.46	11.82	✗	–	–	✗	
1865.45	1866.55	1.10	5.70	10.76	15.70	✓	–	✓	✗	
1866.55	1876.25	9.70	4.48	5.46	7.83	✗	–	–	✗	
1876.25	1876.45	0.20	6.87	8.22	10.22	✓	–	✓	✗	
1876.45	1887.55	11.10	4.69	6.28	12.31	✗	–	–	✗	
1887.55	1888.65	1.10	5.92	9.97	14.01	✓	–	✓	✗	
1888.65	1890.85	2.20	4.80	6.44	8.48	✗	–	–	✗	
1890.85	1890.95	0.10	6.26	6.93	7.93	✗	–	–	✓	
1890.95	1892.15	1.20	6.32	7.60	8.82	✓	–	✓	✓	
1892.15	1893.05	0.90	5.07	6.21	7.75	✗	–	–	✓	
1893.05	1898.65	5.60	4.82	5.87	8.28	✗	–	–	✗	
1898.65	1899.55	0.90	5.53	9.03	11.57	✓	–	✓	✗	
1899.55	1908.95	9.40	4.64	6.07	8.30	✗	–	–	✗	
1908.95	1910.15	1.20	6.25	9.65	13.11	✓	–	✓	✗	
1910.15	1913.65	3.50	4.83	6.17	7.81	✗	–	–	✗	
1913.65	1914.55	0.90	4.92	7.24	8.97	✗	–	–	✓	
1914.55	1916.65	2.10	6.32	8.03	9.56	✓	–	✓	✓	
1916.65	1917.45	0.80	5.71	6.56	8.34	✗	–	–	✓	
1917.45	1921.75	4.30	4.70	6.43	12.12	✗	–	–	✗	
1921.75	1921.85	0.10	6.03	7.61	9.40	✓	–	✗	✗	
1921.85	1931.55	9.70	4.29	5.89	10.00	✗	–	–	✗	
1931.55	1932.45	0.90	6.32	10.00	13.06	✓	–	✓	✗	
1932.45	1942.95	10.50	4.85	5.85	8.91	✗	–	–	✗	
1942.95	1943.75	0.80	5.55	9.72	13.85	✓	–	✓	✗	
1943.75	1965.15	21.40	4.33	5.74	12.77	✗	–	–	✗	
1965.15	1966.35	1.20	6.29	9.82	13.04	✓	–	✓	✗	
1966.35	1975.35	9.00	4.95	6.43	9.73	✗	–	–	✗	
1975.35	1976.55	1.20	6.42	11.21	16.08	✓	–	✓	✗	
1976.55	1981.15	4.60	5.37	6.17	7.79	✗	–	–	✗	
1981.15	1981.95	0.80	6.34	7.18	7.80	✓	–	✓	✗	
1981.95	1985.45	3.50	5.46	6.38	7.68	✗	–	–	✗	
1985.45	1986.65	1.20	5.80	9.75	13.40	✓	–	✓	✗	
1986.65	1990.05	3.40	4.97	6.72	8.56	✗	–	–	✗	
1990.05	1990.25	0.20	6.15	6.94	7.44	✓	–	✓	✗	
1990.25	1992.25	2.00	5.16	6.77	7.86	✗	–	–	✗	
1992.25	1994.45	2.20	5.16	6.64	7.76	✗	–	–	✓	
1994.45	1996.15	1.70	4.85	6.20	8.31	✗	–	–	✗	
1996.15	1996.25	0.10	6.89	8.23	10.09	✓	–	✓	✗	
1996.25	2012.55	16.30	4.85	6.67	15.25	✗	–	–	✗	
2012.55	2012.85	0.30	6.07	7.01	7.94	✗	–	–	✓	
2012.85	2013.05	0.20	6.29	7.14	8.36	✓	–	✓	✓	
2013.05	2013.95	0.90	6.09	7.18	8.39	✗	–	–	✓	
2013.95	2014.75	0.80	6.42	7.22	8.23	✓	–	✓	✓	
2014.75	2016.05	1.30	5.59	6.85	8.56	✗	–	–	✓	
2016.05	2018.35	2.30	5.35	6.51	8.74	✗	–	–	✗	
2018.35	2019.55	1.20	6.36	10.64	14.99	✓	–	✓	✗	
2019.55	2027.95	8.40	5.14	6.49	10.11	✗	–	–	✗	
2027.95	2029.35	1.40	5.99	10.19	14.35	✓	–	✓	✗	
2029.35	2039.55	10.20	4.79	6.23	8.27	✗	–	–	✗	

2039.55	2040.05	0.50	5.53	7.06	9.34	✗	–		✓	
2040.05	2041.15	1.10	6.20	10.29	15.09	✓	–	✓	✓	
2041.15	2042.85	1.70	5.98	7.57	10.14	✗	–		✓	
2042.85	2045.85	3.00	6.31	9.00	11.85	✓	–	✓	✓	
2045.85	2048.35	2.50	5.14	7.21	9.49	✗	–		✓	
2048.35	2048.55	0.20	6.27	7.42	9.39	✓	–	✓	✓	
2048.55	2055.35	6.80	4.74	6.99	13.87	✗	–		✓	
2055.35	2055.45	0.10	6.35	7.43	8.67	✓	–	✓	✓	
2055.45	2058.65	3.20	5.46	7.69	11.28	✗	–		✓	
2058.65	2059.95	1.30	6.18	9.32	12.68	✓	–	✓	✓	
2059.95	2062.35	2.40	5.16	8.28	15.29	✗	–		✓	
2062.35	2066.25	3.90	4.90	6.18	7.77	✗	–		✗	
2066.25	2066.65	0.40	5.78	7.37	9.12	✗	–		✓	
2066.65	2067.15	0.50	6.29	7.81	9.12	✓	–	✓	✓	
2067.15	2068.75	1.60	5.27	6.73	8.56	✗	–		✓	
2068.75	2071.35	2.60	5.20	5.97	7.15	✗	–		✗	
2071.35	2072.05	0.70	5.92	6.91	8.38	✗	–		✓	
2072.05	2074.35	2.30	5.99	8.70	14.37	✓	–	✓	✓	
2074.35	2075.35	1.00	5.69	6.56	8.03	✗	–		✓	
2075.35	2080.35	5.00	5.24	6.02	8.07	✗	–		✗	
2080.35	2080.45	0.10	6.12	7.12	8.29	✗	–		✓	
2080.45	2081.65	1.20	6.12	7.33	8.94	✓	–	✓	✓	
2081.65	2081.85	0.20	6.41	7.69	8.96	✗	–		✓	
2081.85	2082.75	0.90	6.28	7.49	8.87	✓	–	✓	✓	
2082.75	2085.45	2.70	5.27	8.03	14.45	✗	–		✓	
2085.45	2085.95	0.50	6.30	7.16	8.04	✓	–	✓	✓	
2085.95	2090.75	4.80	5.29	6.82	9.18	✗	–		✓	
2090.75	2093.65	2.90	5.20	6.39	8.05	✗	–		✗	
2093.65	2094.55	0.90	5.93	7.45	9.53	✗	–		✓	
2094.55	2098.65	4.10	6.43	8.82	15.76	✓	–	✓	✓	
2098.65	2099.45	0.80	6.02	7.36	8.38	✗	–		✓	
2099.45	2100.05	0.60	6.38	6.93	7.64	✓	–	✓	✓	
2100.05	2104.45	4.40	5.80	6.96	9.22	✗	–		✓	
2104.45	2105.95	1.50	6.23	8.52	13.69	✓	–	✓	✓	
2105.95	2110.05	4.10	5.46	6.50	10.13	✗	–		✓	
2110.05	2113.05	3.00	5.24	6.16	7.68	✗	–		✗	
2113.05	2116.55	3.50	5.34	6.55	8.42	✗	–		✓	
2116.55	2117.05	0.50	5.32	6.94	9.97	✗	–		✗	
2117.05	2118.05	1.00	6.37	10.40	14.48	✓	–	✓	✗	
2118.05	2118.25	0.20	6.14	7.26	8.36	✗	–		✗	
2118.25	2118.55	0.30	6.18	7.26	7.84	✗	–		✓	
2118.55	2119.25	0.70	6.08	7.40	8.48	✓	–	✓	✓	
2119.25	2119.85	0.60	5.75	7.06	8.20	✗	–		✓	
2119.85	2121.05	1.20	6.23	7.39	8.17	✓	–	✓	✓	
2121.05	2121.25	0.20	5.83	6.67	7.21	✗	–		✓	
2121.25	2123.85	2.60	5.45	6.41	7.65	✗	–		✗	
2123.85	2124.25	0.40	5.67	6.86	7.94	✗	–		✓	
2124.25	2124.35	0.10	6.41	6.91	7.92	✓	–	✓	✓	
2124.35	2124.85	0.50	5.88	6.66	7.80	✗	–		✓	
2124.85	2132.55	7.70	5.13	6.94	12.08	✓	–	✓	✗	

2132.55	2135.05	2.50	5.78	6.90	9.23	✗	–	–	✓	
2135.05	2135.55	0.50	6.00	6.58	7.70	✗	–	–	✗	
2135.55	2135.75	0.20	6.00	6.92	7.94	✓	–	✓	✗	
2135.75	2139.45	3.70	5.04	7.15	12.39	✗	–	–	✗	
2139.45	2139.65	0.20	6.18	7.63	9.61	✓	–	✓	✗	
2139.65	2141.15	1.50	5.38	6.33	8.07	✗	–	–	✗	
2141.15	2144.95	3.80	4.96	6.40	8.41	✗	–	–	✓	
2144.95	2149.75	4.80	5.08	6.99	15.31	✗	–	–	✗	
2149.75	2150.15	0.40	6.37	7.62	9.55	✓	–	✓	✗	
2150.15	2160.15	10.00	5.10	6.36	8.62	✗	–	–	✗	
2160.15	2161.35	1.20	6.26	9.69	14.36	✓	–	✓	✗	
2161.35	2165.65	4.30	4.98	5.98	9.11	✗	–	–	✗	
2165.65	2166.05	0.40	6.33	6.77	7.39	✓	–	✓	✗	
2166.05	2167.45	1.40	5.91	6.68	7.67	✗	–	–	✗	
2167.45	2167.65	0.20	6.38	6.89	7.68	✓	–	✓	✗	
2167.65	2171.95	4.30	5.27	6.40	7.68	✗	–	–	✗	
2171.95	2173.65	1.70	6.30	9.71	13.82	✓	–	✓	✗	
2173.65	2182.65	9.00	4.78	6.08	8.22	✗	–	–	✗	
2182.65	2182.75	0.10	6.05	7.75	9.01	✓	–	✓	✗	
2182.75	2189.15	6.40	4.87	6.57	13.35	✗	–	–	✗	
2189.15	2190.65	1.50	5.45	6.31	7.45	✗	–	–	✓	
2190.65	2200.05	9.40	4.90	6.08	11.87	✗	–	–	✗	
2200.05	2202.55	2.50	5.34	6.55	8.59	✗	–	–	✓	
2202.55	2205.45	2.90	5.22	5.91	8.92	✗	–	–	✗	
2205.45	2206.45	1.00	5.98	9.85	14.81	✓	–	✓	✗	
2206.45	2216.45	10.00	4.72	6.05	8.44	✗	–	–	✗	
2216.45	2217.55	1.10	6.04	9.03	12.75	✓	–	✓	✗	
2217.55	2245.35	27.80	4.75	6.06	13.41	✗	–	–	✗	
2245.35	2247.45	2.10	5.46	6.61	7.83	✗	–	–	✓	
2247.45	2249.25	1.80	5.28	6.09	7.48	✗	–	–	✗	
2249.25	2250.55	1.30	6.32	9.98	14.02	✓	–	✓	✗	
2250.55	2260.05	9.50	5.06	6.07	9.33	✗	–	–	✗	
2260.05	2261.15	1.10	6.30	10.36	15.22	✓	–	✓	✗	
2261.15	2270.25	9.10	4.74	5.90	9.49	✗	–	–	✗	
2270.25	2271.15	0.90	6.14	10.46	14.88	✓	–	✓	✗	
2271.15	2292.45	21.30	4.75	5.99	14.18	✗	–	–	✗	
2292.45	2293.65	1.20	5.73	9.43	13.95	✓	–	✓	✗	
2293.65	2303.55	9.90	4.73	5.86	9.15	✗	–	–	✗	
2303.55	2304.45	0.90	5.92	10.11	14.15	✓	–	✗	✗	
2304.45	2313.65	9.20	4.95	5.96	7.63	✗	–	–	✗	
2313.65	2320.25	6.60	6.33	8.42	13.94	✓	–	✓	✓	
2320.25	2320.75	0.50	6.06	7.61	8.66	✗	–	–	✓	
2320.75	2321.35	0.60	6.36	7.43	8.31	✓	–	✓	✓	
2321.35	2321.65	0.30	5.58	6.81	7.61	✗	–	–	✓	
2321.65	2330.75	9.10	4.51	6.28	12.84	✗	–	–	✗	
2330.75	2331.75	1.00	4.92	6.16	7.51	✗	–	–	✓	
2331.75	2333.15	1.40	4.91	5.79	6.72	✗	–	–	✗	
2333.15	2333.55	0.40	5.18	6.41	8.72	✗	–	–	✓	
2333.55	2334.45	0.90	6.13	8.75	10.95	✓	–	✓	✓	
2334.45	2335.55	1.10	5.00	6.89	10.74	✗	–	–	✓	

2335.55	2336.95	1.40	4.97	5.92	8.77	✗	–	·	✗	
2336.95	2338.05	1.10	5.85	9.50	14.51	✓	–	✓	✗	
2338.05	2347.75	9.70	4.95	5.88	9.85	✗	–	·	✗	
2347.75	2348.85	1.10	5.87	10.21	14.56	✓	–	✓	✗	
2348.85	2358.95	10.10	4.40	6.11	13.51	✗	–	·	✗	
2358.95	2359.25	0.30	6.56	7.32	8.74	✓	–	✓	✗	
2359.25	2361.15	1.90	6.33	8.08	9.36	✓	–	✓	✓	
2361.15	2361.45	0.30	6.17	7.80	9.15	✗	–	·	✓	
2361.45	2363.45	2.00	6.21	8.29	9.75	✓	–	✓	✓	
2363.45	2363.65	0.20	6.21	7.64	9.10	✗	–	·	✓	
2363.65	2370.25	6.60	5.95	10.72	17.87	✓	–	✓	✓	
2370.25	2373.55	3.30	4.48	6.35	11.17	✗	–	·	✓	
2373.55	2373.85	0.30	6.41	7.23	8.32	✓	–	✓	✓	
2373.85	2374.55	0.70	6.27	7.17	9.08	✗	–	·	✓	
2374.55	2374.85	0.30	6.42	7.55	9.39	✓	–	✓	✓	
2374.85	2376.45	1.60	4.98	7.00	9.39	✗	–	·	✓	
2376.45	2379.75	3.30	4.67	6.08	7.66	✗	–	·	✗	
2379.75	2380.15	0.40	5.02	7.22	9.83	✗	–	·	✓	
2380.15	2384.55	4.40	6.03	9.12	14.61	✓	–	✓	✓	
2384.55	2384.95	0.40	6.32	7.07	8.13	✗	–	·	✓	
2384.95	2385.15	0.20	6.34	6.98	8.13	✓	–	✓	✓	
2385.15	2385.95	0.80	4.97	6.42	7.92	✗	–	·	✓	
2385.95	2400.95	15.00	4.74	6.52	14.82	✗	–	·	✗	
2400.95	2401.15	0.20	6.28	7.22	8.18	✓	–	✓	✗	
2401.15	2401.95	0.80	5.66	6.88	8.90	✗	–	·	✗	
2401.95	2403.45	1.50	6.26	10.03	16.90	✓	–	✓	✗	
2403.45	2405.25	1.80	5.03	6.30	8.48	✗	–	·	✗	
2405.25	2408.95	3.70	6.34	8.11	9.84	✓	–	✓	✓	
2408.95	2409.95	1.00	6.01	7.08	8.44	✗	–	·	✓	
2409.95	2446.95	37.00	6.24	9.92	17.99	✓	–	✓	✓	
2446.95	2447.35	0.40	6.24	7.25	8.75	✗	–	·	✓	
2447.35	2451.15	3.80	6.40	7.91	9.44	✓	–	✓	✓	
2451.15	2451.35	0.20	6.38	7.27	7.88	✗	–	·	✓	
2451.35	2454.05	2.70	6.40	7.54	8.34	✓	–	✓	✓	
2454.05	2454.25	0.20	6.37	7.22	7.95	✗	–	·	✓	
2454.25	2458.35	4.10	6.40	9.05	16.17	✓	–	✓	✓	
2458.35	2458.55	0.20	6.40	7.45	8.32	✗	–	·	✓	
2458.55	2463.45	4.90	6.24	7.91	9.06	✓	–	✓	✓	
2463.45	2463.85	0.40	6.16	6.91	7.67	✗	–	·	✓	
2463.85	2463.95	0.10	6.35	6.95	7.67	✓	–	✓	✓	
2463.95	2464.35	0.40	6.09	6.82	7.67	✗	–	·	✓	
2464.35	2464.55	0.20	6.16	7.03	7.68	✗	–	·	✗	
2464.55	2465.35	0.80	6.70	7.78	8.78	✓	–	✓	✗	
2465.35	2467.15	1.80	6.37	7.81	8.77	✓	–	✓	✓	
2467.15	2468.35	1.20	6.18	9.77	15.75	✗	–	·	✓	
2468.35	2483.85	15.50	6.42	8.08	14.33	✓	–	✓	✓	
2483.85	2484.75	0.90	6.20	6.88	7.58	✗	–	·	✓	
2484.75	2486.55	1.80	6.36	7.53	8.86	✓	–	✓	✓	
2486.55	2487.25	0.70	5.97	6.76	7.70	✗	–	·	✓	
2487.25	2487.75	0.50	6.37	7.52	8.40	✓	–	✓	✓	

2487.75	2487.95	0.20	6.36	7.57	8.52	✗	–	▪	✓	
2487.95	2493.55	5.60	6.35	8.56	14.71	✓	–	✓	✓	
2493.55	2493.85	0.30	6.33	7.31	8.31	✗	–	▪	✓	
2493.85	2493.95	0.10	6.38	7.41	8.45	✓	–	✓	✓	
2493.95	2494.35	0.40	6.18	7.26	8.50	✗	–	▪	✓	
2494.35	2495.35	1.00	6.12	7.13	8.30	✗	–	▪	✗	
2495.35	2495.45	0.10	6.48	7.50	8.41	✓	–	✓	✗	
2495.45	2496.35	0.90	6.27	7.69	8.45	✓	–	✓	✓	
2496.35	2497.15	0.80	5.92	7.47	8.52	✗	–	▪	✓	
2497.15	2497.95	0.80	6.26	7.76	8.76	✓	–	✓	✓	
2497.95	2498.25	0.30	6.16	7.48	8.67	✗	–	▪	✓	
2498.25	2500.55	2.30	6.82	7.81	9.05	✓	–	✓	✓	
2500.55	2501.85	1.30	6.05	10.22	13.40	✓	–	✓	✗	
2501.85	2502.85	1.00	5.62	6.87	8.54	✗	–	▪	✗	
2502.85	2503.15	0.30	5.97	7.02	8.29	✗	–	▪	✓	
2503.15	2504.35	1.20	6.33	7.37	8.54	✓	–	✓	✓	
2504.35	2504.65	0.30	6.16	6.82	7.45	✗	–	▪	✓	
2504.65	2512.45	7.80	5.16	7.22	12.88	✗	–	▪	✗	
2512.45	2512.65	0.20	6.61	7.62	8.47	✓	–	✓	✗	
2512.65	2512.85	0.20	6.35	7.51	8.72	✓	–	✓	✓	
2512.85	2513.35	0.50	6.30	7.60	8.86	✗	–	▪	✓	
2513.35	2514.25	0.90	6.41	7.51	8.93	✓	–	✓	✓	
2514.25	2514.35	0.10	6.16	6.80	7.39	✗	–	▪	✓	
2514.35	2520.15	5.80	5.49	6.59	8.49	✗	–	▪	✗	
2520.15	2520.65	0.50	6.17	7.27	8.42	✓	–	✓	✗	
2520.65	2523.05	2.40	5.76	7.01	8.49	✗	–	▪	✗	
2523.05	2523.25	0.20	6.50	8.07	10.16	✓	–	✓	✗	
2523.25	2529.55	6.30	5.22	6.92	14.61	✗	–	▪	✗	
2529.55	2530.45	0.90	6.43	7.32	8.19	✓	–	✓	✗	
2530.45	2531.35	0.90	6.38	7.14	8.05	✗	–	▪	✗	
2531.35	2531.45	0.10	6.41	7.01	7.61	✓	–	✓	✗	
2531.45	2532.85	1.40	6.01	7.12	8.66	✗	–	▪	✗	
2532.85	2533.25	0.40	6.70	7.54	8.85	✓	–	✓	✗	
2533.25	2534.65	1.40	6.28	9.02	11.98	✓	–	✓	✓	
2534.65	2535.35	0.70	5.67	7.03	8.41	✗	–	▪	✓	
2535.35	2539.45	4.10	6.40	9.43	12.56	✓	–	✓	✓	
2539.45	2540.45	1.00	4.70	6.77	8.52	✗	–	▪	✓	
2540.45	2542.15	1.70	4.69	6.00	7.01	✗	–	▪	✗	
2542.15	2542.55	0.40	5.80	6.58	7.42	✗	–	▪	✓	
2542.55	2544.05	1.50	6.16	8.05	10.07	✓	–	✓	✓	
2544.05	2546.65	2.60	4.96	7.55	15.94	✗	–	▪	✓	
2546.65	2547.45	0.80	6.18	7.53	8.66	✓	–	✓	✓	
2547.45	2548.35	0.90	5.44	6.88	8.63	✗	–	▪	✓	
2548.35	2555.15	6.80	5.14	6.74	8.62	✗	–	▪	✗	
2555.15	2556.35	1.20	6.00	9.73	13.77	✓	–	✓	✗	
2556.35	2560.55	4.20	5.38	6.64	9.07	✗	–	▪	✗	
2560.55	2560.75	0.20	6.03	6.93	8.12	✓	–	✓	✗	
2560.75	2565.85	5.10	4.99	6.35	9.04	✗	–	▪	✗	
2565.85	2567.05	1.20	6.19	10.27	13.14	✓	–	✓	✗	
2567.05	2568.45	1.40	5.25	6.48	7.90	✗	–	▪	✗	

2568.45	2568.55	0.10	5.52	7.03	8.54	✗	–	–	✓	
2568.55	2569.85	1.30	6.36	8.19	9.81	✓	–	✓	✓	
2569.85	2570.35	0.50	5.28	7.14	8.64	✗	–	–	✓	
2570.35	2576.55	6.20	4.99	6.31	8.29	✗	–	–	✗	
2576.55	2577.15	0.60	5.45	7.16	10.00	✗	–	–	✓	
2577.15	2579.85	2.70	6.17	8.91	13.66	✓	–	✓	✓	
2579.85	2580.05	0.20	5.70	7.68	8.87	✗	–	–	✓	
2580.05	2587.75	7.70	5.12	6.73	10.84	✗	–	–	✗	
2587.75	2589.05	1.30	6.31	10.52	13.05	✓	–	✓	✗	
2589.05	2591.45	2.40	4.90	7.04	8.60	✗	–	–	✗	
2591.45	2591.75	0.30	6.06	7.29	8.38	✓	–	✓	✓	
2591.75	2594.25	2.50	5.40	7.70	9.83	✗	–	–	✓	
2594.25	2597.45	3.20	6.43	8.36	10.05	✓	–	✓	✓	
2597.45	2597.75	0.30	4.90	7.87	9.91	✗	–	–	✓	
2597.75	2600.55	2.80	4.71	8.43	14.96	✗	–	–	✗	
2600.55	2601.85	1.30	5.30	6.28	7.42	✗	–	–	✓	
2601.85	2609.95	8.10	4.49	5.58	7.91	✗	–	–	✗	
2609.95	2611.25	1.30	5.73	9.93	13.91	✓	–	✓	✗	
2611.25	2617.95	6.70	5.06	5.89	8.68	✗	–	–	✗	
2617.95	2618.15	0.20	6.10	7.54	9.57	✓	–	✓	✗	
2618.15	2619.55	1.40	6.33	8.58	10.22	✓	–	✓	✓	
2619.55	2622.05	2.50	5.69	8.13	12.79	✗	–	–	✓	
2622.05	2622.45	0.40	5.53	6.62	7.81	✗	–	–	✗	
2622.45	2623.15	0.70	5.53	6.85	8.23	✗	–	–	✓	
2623.15	2624.05	0.90	6.29	7.37	8.65	✓	–	✓	✓	
2624.05	2624.85	0.80	5.92	6.74	7.97	✗	–	–	✓	
2624.85	2632.95	8.10	5.62	6.99	14.23	✗	–	–	✗	
2632.95	2633.05	0.10	6.35	8.60	10.64	✓	–	✓	✗	
2633.05	2641.05	8.00	5.12	6.28	8.84	✗	–	–	✗	
2641.05	2643.85	2.80	5.60	8.32	13.99	✗	–	–	✓	
2643.85	2643.95	0.10	6.30	7.44	8.66	✓	–	✓	✓	
2643.95	2645.05	1.10	5.84	7.17	9.27	✗	–	–	✓	
2645.05	2646.75	1.70	6.52	7.63	10.02	✓	–	✓	✓	
2646.75	2646.85	0.10	6.40	7.31	8.25	✓	–	✓	✗	
2646.85	2649.55	2.70	5.37	6.69	8.67	✗	–	–	✗	
2649.55	2650.05	0.50	5.91	7.38	9.56	✗	–	–	✓	
2650.05	2650.75	0.70	6.42	8.04	9.74	✓	–	✓	✓	
2650.75	2651.25	0.50	5.58	6.94	9.33	✗	–	–	✓	
2651.25	2656.55	5.30	4.57	6.91	13.05	✗	–	–	✗	
2656.55	2656.85	0.30	6.42	7.55	9.08	✓	–	✓	✓	
2656.85	2657.55	0.70	6.31	7.18	9.07	✗	–	–	✓	
2657.55	2658.65	1.10	5.24	7.12	9.41	✗	–	–	✗	
2658.65	2658.85	0.20	5.73	7.87	9.49	✗	–	–	✓	
2658.85	2659.35	0.50	6.38	8.10	9.44	✓	–	✓	✓	
2659.35	2660.05	0.70	6.16	7.42	9.01	✗	–	–	✓	
2660.05	2660.75	0.70	6.41	7.24	7.90	✓	–	✓	✓	
2660.75	2660.85	0.10	6.20	7.08	7.83	✗	–	–	✓	
2660.85	2662.25	1.40	5.67	6.75	9.22	✗	–	–	✗	
2662.25	2662.35	0.10	6.32	7.63	9.58	✗	–	–	✓	
2662.35	2671.15	8.80	6.35	9.21	15.98	✓	–	✓	✓	

2671.15	2672.05	0.90	6.05	7.93	9.68	✗	–	–	✓	
2672.05	2673.55	1.50	6.18	7.75	9.26	✓	–	✓	✓	
2673.55	2674.25	0.70	5.61	6.59	7.72	✗	–	–	✓	
2674.25	2676.25	2.00	5.40	6.29	7.95	✗	–	–	✗	
2676.25	2678.45	2.20	6.39	9.26	13.87	✓	–	✓	✗	
2678.45	2678.85	0.40	6.10	7.21	8.36	✗	–	–	✗	
2678.85	2679.55	0.70	6.18	7.81	9.28	✓	–	✓	✗	
2679.55	2679.65	0.10	5.96	7.79	9.25	✗	–	–	✗	
2679.65	2682.05	2.40	5.46	6.76	9.25	✗	–	–	✓	
2682.05	2682.35	0.30	6.42	7.24	8.12	✓	–	✓	✓	
2682.35	2685.55	3.20	5.56	6.73	9.32	✗	–	–	✓	
2685.55	2686.15	0.60	5.38	7.39	10.02	✗	–	–	✗	
2686.15	2686.65	0.50	5.38	7.65	10.02	✗	–	–	✓	
2686.65	2687.75	1.10	5.92	10.05	12.93	✓	–	✓	✓	
2687.75	2688.35	0.60	5.48	6.94	8.66	✗	–	–	✓	
2688.35	2694.35	6.00	6.33	9.65	12.82	✓	–	✓	✓	
2694.35	2694.65	0.30	6.23	8.10	10.28	✗	–	–	✓	
2694.65	2694.75	0.10	6.14	7.70	9.68	✓	–	✓	✓	
2694.75	2695.05	0.30	5.68	7.86	10.66	✗	–	–	✓	
2695.05	2700.95	5.90	5.71	12.64	17.82	✓	–	✓	✓	
2700.95	2701.85	0.90	4.84	6.16	9.66	✗	–	–	✓	
2701.85	2702.45	0.60	5.86	6.51	7.50	✗	–	–	✗	
2702.45	2702.95	0.50	6.37	6.84	7.47	✓	–	✓	✗	
2702.95	2709.25	6.30	5.16	6.27	7.47	✗	–	–	✗	
2709.25	2710.45	1.20	5.08	10.60	17.41	✗	–	–	✓	
2710.45	2714.55	4.10	6.23	10.13	12.53	✓	–	✓	✓	
2714.55	2715.55	1.00	4.99	6.64	9.26	✗	–	–	✓	
2715.55	2719.15	3.60	4.89	6.38	8.19	✗	–	–	✗	
2719.15	2721.65	2.50	6.20	8.86	14.72	✓	–	✓	✓	
2721.65	2723.35	1.70	5.52	7.00	9.62	✗	–	–	✓	
2723.35	2723.85	0.50	6.40	7.42	8.50	✓	–	✓	✓	
2723.85	2724.45	0.60	5.98	6.82	8.09	✗	–	–	✓	
2724.45	2725.15	0.70	6.30	7.41	8.55	✓	–	✓	✓	
2725.15	2725.45	0.30	6.34	7.59	8.48	✗	–	–	✓	
2725.45	2725.95	0.50	6.31	7.28	8.31	✓	–	✓	✓	
2725.95	2728.15	2.20	5.20	6.47	7.76	✗	–	–	✓	
2728.15	2729.35	1.20	5.22	6.57	7.71	✗	–	–	✗	
2729.35	2730.15	0.80	6.43	6.97	7.93	✓	–	✓	✗	
2730.15	2730.25	0.10	6.43	7.12	7.98	✗	–	–	✗	
2730.25	2730.75	0.50	6.36	7.06	7.98	✓	–	✓	✗	
2730.75	2731.75	1.00	5.71	6.91	8.82	✗	–	–	✗	
2731.75	2731.85	0.10	6.75	8.00	9.94	✓	–	✓	✗	
2731.85	2738.45	6.60	4.75	6.77	14.39	✗	–	–	✗	
2738.45	2738.55	0.10	6.24	6.91	7.75	✗	–	–	✓	
2738.55	2738.75	0.20	6.37	7.21	8.43	✓	–	✓	✓	
2738.75	2739.45	0.70	5.94	7.33	8.75	✗	–	–	✓	
2739.45	2739.65	0.20	6.24	7.00	8.08	✓	–	✓	✓	
2739.65	2740.65	1.00	5.55	6.30	7.69	✗	–	–	✓	
2740.65	2746.35	5.70	4.87	6.61	12.10	✗	–	–	✗	
2746.35	2746.95	0.60	5.76	6.77	8.20	✗	–	–	✓	

2746.95	2750.35	3.40	6.29	8.44	11.23	✓	–	✓	✓	
2750.35	2751.05	0.70	6.23	6.98	8.21	✗	–	·	✓	
2751.05	2752.85	1.80	6.29	7.82	9.10	✓	–	✓	✓	
2752.85	2753.55	0.70	5.87	6.82	9.16	✗	–	·	✓	
2753.55	2754.85	1.30	6.12	10.35	14.38	✓	–	✓	✓	
2754.85	2755.45	0.60	6.03	6.74	7.78	✗	–	·	✓	
2755.45	2764.25	8.80	5.21	6.21	8.24	✗	–	·	✗	
2764.25	2764.35	0.10	6.38	7.44	9.55	✗	–	·	✓	
2764.35	2765.85	1.50	6.37	10.15	14.45	✓	–	✓	✓	
2765.85	2766.75	0.90	5.89	7.18	8.58	✗	–	·	✓	
2766.75	2767.85	1.10	6.39	7.68	9.13	✓	–	✓	✓	
2767.85	2769.85	2.00	5.29	6.66	7.86	✗	–	·	✓	
2769.85	2774.25	4.40	4.86	5.90	7.48	✗	–	·	✗	
2774.25	2774.65	0.40	6.31	7.34	8.76	✓	–	✓	✗	
2774.65	2775.15	0.50	5.99	7.44	8.76	✗	–	·	✗	
2775.15	2775.35	0.20	6.24	8.20	10.51	✓	–	✓	✗	
2775.35	2776.75	1.40	5.71	9.36	15.15	✗	–	·	✗	
2776.75	2777.95	1.20	6.04	6.75	7.90	✗	–	·	✓	
2777.95	2778.75	0.80	6.26	7.27	8.25	✓	–	✓	✓	
2778.75	2780.55	1.80	4.99	6.72	8.24	✗	–	·	✓	
2780.55	2784.55	4.00	4.76	5.91	7.67	✗	–	·	✗	
2784.55	2784.85	0.30	5.84	6.97	8.39	✗	–	·	✓	
2784.85	2785.75	0.90	6.39	8.05	9.59	✓	–	✓	✓	
2785.75	2786.15	0.40	6.34	7.65	9.43	✗	–	·	✓	
2786.15	2786.55	0.40	6.40	7.55	8.84	✓	–	✓	✓	
2786.55	2807.95	21.40	4.76	6.32	14.61	✗	–	·	✗	
2807.95	2809.05	1.10	6.34	8.90	11.75	✓	–	✓	✗	
2809.05	2818.85	9.80	5.01	5.93	8.62	✗	–	·	✗	
2818.85	2820.05	1.20	5.95	10.11	13.38	✓	–	✓	✗	
2820.05	2830.05	10.00	4.21	5.68	8.39	✗	–	·	✗	
2830.05	2831.05	1.00	5.56	10.22	14.43	✓	–	✓	✗	
2831.05	2862.75	31.70	4.41	5.99	15.66	✗	–	·	✗	
2862.75	2863.75	1.00	6.07	8.67	11.36	✓	–	✗	✗	
2863.75	2873.75	10.00	4.24	5.61	9.56	✗	–	·	✗	
2873.75	2873.85	0.10	6.63	8.63	11.07	✓	–	✗	✗	
2873.85	2917.35	43.50	4.39	6.08	15.18	✗	–	·	✗	
2917.35	2918.75	1.40	5.85	9.74	13.10	✓	–	✓	✗	
2918.75	2928.75	10.00	4.92	5.95	8.70	✗	–	·	✗	
2928.75	2930.05	1.30	6.03	9.73	13.06	✓	–	✓	✗	
2930.05	2939.75	9.70	4.82	6.00	8.51	✗	–	·	✗	
2939.75	2940.65	0.90	5.61	9.44	12.80	✓	–	✗	✗	
2940.65	2947.35	6.70	4.59	5.79	8.46	✗	–	·	✗	
2947.35	2948.25	0.90	5.48	7.25	9.87	✗	–	·	✓	
2948.25	2948.75	0.50	6.37	7.77	9.82	✓	–	✓	✓	
2948.75	2949.35	0.60	5.47	6.89	8.49	✗	–	·	✓	
2949.35	2950.15	0.80	5.44	6.74	9.39	✗	–	·	✗	
2950.15	2951.35	1.20	5.84	10.48	15.02	✓	–	✓	✗	
2951.35	2956.35	5.00	4.78	6.22	7.63	✗	–	·	✗	
2956.35	2956.75	0.40	6.41	6.86	7.32	✓	–	✓	✗	
2956.75	2956.85	0.10	6.41	6.88	7.42	✗	–	·	✗	

2956.85	2957.35	0.50	6.38	7.17	8.06	✓	–	✓	✗	
2957.35	2964.25	6.90	4.60	6.97	15.05	✗	–	–	✗	
2964.25	2966.25	2.00	5.52	6.82	8.73	✗	–	–	✓	
2966.25	2970.35	4.10	5.12	6.26	8.29	✗	–	–	✗	
2970.35	2970.55	0.20	6.01	6.90	8.66	✗	–	–	✓	
2970.55	2971.85	1.30	6.53	8.76	10.55	✓	–	✓	✓	
2971.85	2973.55	1.70	4.59	8.99	15.56	✓	–	✓	✓	
2973.55	2983.45	9.90	4.58	6.37	14.35	✗	–	–	✗	
2983.45	2983.55	0.10	5.84	8.21	10.72	✓	–	✓	✗	
2983.55	3002.95	19.40	4.66	6.21	14.59	✗	–	–	✗	
3002.95	3003.05	0.10	6.29	6.93	7.62	✗	–	–	✓	
3003.05	3003.15	0.10	6.39	7.04	7.91	✓	–	✓	✓	
3003.15	3003.55	0.40	6.22	7.25	8.64	✗	–	–	✓	
3003.55	3004.05	0.50	6.38	7.73	9.23	✓	–	✓	✓	
3004.05	3004.45	0.40	6.26	7.68	9.24	✗	–	–	✓	
3004.45	3004.75	0.30	6.32	7.39	8.79	✓	–	✓	✓	
3004.75	3004.85	0.10	6.32	7.36	8.10	✗	–	–	✓	
3004.85	3006.15	1.30	6.35	11.14	16.33	✓	–	✓	✓	
3006.15	3007.05	0.90	5.29	6.72	9.72	✗	–	–	✓	
3007.05	3014.75	7.70	4.65	6.05	9.14	✗	–	–	✗	
3014.75	3015.55	0.80	6.33	7.43	8.90	✓	–	✓	✗	
3015.55	3016.05	0.50	5.59	6.65	7.60	✗	–	–	✗	
3016.05	3016.15	0.10	6.31	6.72	7.12	✗	–	–	✓	
3016.15	3018.05	1.90	6.31	10.16	15.86	✓	–	✓	✓	
3018.05	3018.85	0.80	4.58	6.85	9.24	✗	–	–	✓	
3018.85	3019.65	0.80	4.62	6.59	8.46	✗	–	–	✗	
3019.65	3020.35	0.70	5.48	7.55	9.35	✗	–	–	✓	
3020.35	3020.75	0.40	6.42	8.38	9.78	✓	–	✓	✓	
3020.75	3020.95	0.20	6.42	8.39	9.78	✗	–	–	✓	
3020.95	3022.55	1.60	6.42	8.20	9.75	✓	–	✓	✓	
3022.55	3023.35	0.80	5.62	7.13	8.38	✗	–	–	✓	
3023.35	3023.55	0.20	6.38	7.22	8.33	✗	–	–	✗	
3023.55	3023.65	0.10	6.40	7.16	8.00	✓	–	✓	✗	
3023.65	3026.45	2.80	5.43	7.31	9.44	✗	–	–	✗	
3026.45	3026.65	0.20	5.98	8.56	9.78	✗	–	–	✓	
3026.65	3030.45	3.80	6.02	9.64	17.10	✓	–	✓	✓	
3030.45	3030.65	0.20	5.55	6.73	7.48	✗	–	–	✓	
3030.65	3038.15	7.50	4.75	5.78	7.36	✗	–	–	✗	
3038.15	3038.25	0.10	5.80	7.12	8.64	✓	–	✓	✗	
3038.25	3049.45	11.20	4.63	6.64	12.77	✗	–	–	✗	
3049.45	3050.65	1.20	5.82	10.14	13.58	✓	–	✓	✗	
3050.65	3059.65	9.00	4.50	6.16	9.70	✗	–	–	✗	
3059.65	3061.45	1.80	5.52	7.02	8.21	✗	–	–	✓	
3061.45	3061.75	0.30	6.36	8.08	10.37	✓	–	✓	✓	
3061.75	3062.75	1.00	6.23	10.43	13.79	✓	–	✓	✗	
3062.75	3063.65	0.90	5.86	6.61	7.28	✗	–	–	✗	
3063.65	3064.25	0.60	6.17	6.81	7.26	✓	–	✓	✗	
3064.25	3065.35	1.10	5.86	6.63	7.63	✗	–	–	✗	
3065.35	3065.55	0.20	6.37	7.04	8.15	✓	–	✓	✗	
3065.55	3066.55	1.00	6.18	7.14	8.32	✗	–	–	✗	

3066.55	3066.95	0.40	6.54	7.19	8.12	✓	–	✓	✗	
3066.95	3069.35	2.40	6.38	7.36	8.89	✓	–	✓	✓	
3069.35	3069.55	0.20	6.38	7.55	8.95	✗	–	-	✓	
3069.55	3071.75	2.20	6.50	7.96	9.63	✓	–	✓	✓	
3071.75	3073.05	1.30	6.28	10.11	15.07	✗	–	-	✓	
3073.05	3073.75	0.70	6.29	8.05	10.17	✓	–	✓	✓	
3073.75	3073.95	0.20	6.29	7.41	9.30	✗	–	-	✓	
3073.95	3076.35	2.40	6.32	7.96	9.71	✓	–	✓	✓	
3076.35	3077.15	0.80	5.94	6.99	8.83	✗	–	-	✓	
3077.15	3080.45	3.30	6.24	7.70	9.49	✓	–	✓	✓	
3080.45	3081.05	0.60	5.98	7.11	8.30	✗	–	-	✓	
3081.05	3081.15	0.10	6.40	7.35	8.45	✓	–	✓	✓	
3081.15	3081.55	0.40	6.31	7.31	8.45	✗	–	-	✓	
3081.55	3083.45	1.90	6.41	8.38	12.74	✓	–	✓	✓	
3083.45	3084.35	0.90	6.38	7.07	7.93	✗	–	-	✗	
3084.35	3084.75	0.40	6.40	7.01	7.56	✓	–	✓	✗	
3084.75	3091.35	6.60	5.30	6.37	8.08	✗	–	-	✗	
3091.35	3091.85	0.50	6.35	6.92	8.08	✓	–	✓	✗	
3091.85	3095.35	3.50	4.87	7.29	13.75	✗	–	-	✗	
3095.35	3095.55	0.20	6.40	7.27	7.97	✓	–	✓	✗	
3095.55	3098.05	2.50	5.31	6.24	7.65	✗	–	-	✗	
3098.05	3098.65	0.60	5.63	6.30	7.22	✗	–	-	✓	
3098.65	3118.05	19.40	6.15	9.15	15.20	✓	–	✓	✓	
3118.05	3118.55	0.50	4.93	7.78	11.55	✗	–	-	✓	
3118.55	3119.15	0.60	4.93	6.86	8.79	✗	–	-	✗	
3119.15	3119.35	0.20	6.11	7.38	8.54	✓	–	✓	✗	
3119.35	3120.75	1.40	6.37	8.09	9.40	✓	–	✓	✓	
3120.75	3121.35	0.60	5.98	7.45	8.77	✗	–	-	✓	
3121.35	3121.95	0.60	5.98	7.09	8.06	✗	–	-	✗	
3121.95	3122.35	0.40	6.35	7.18	7.75	✓	–	✓	✗	
3122.35	3125.85	3.50	5.74	7.11	8.98	✗	–	-	✗	
3125.85	3129.75	3.90	4.98	7.34	15.22	✗	–	-	✓	
3129.75	3134.55	4.80	4.96	6.14	8.67	✗	–	-	✗	
3134.55	3135.25	0.70	5.30	6.92	9.86	✗	–	-	✓	
3135.25	3136.35	1.10	6.30	8.47	11.05	✓	–	✓	✓	
3136.35	3136.75	0.40	5.79	7.61	11.34	✗	–	-	✓	
3136.75	3137.85	1.10	6.34	10.28	16.32	✓	–	✓	✓	
3137.85	3138.85	1.00	5.92	6.97	8.51	✗	–	-	✓	
3138.85	3139.55	0.70	6.28	7.52	9.44	✓	–	✓	✓	
3139.55	3140.45	0.90	6.04	7.94	10.85	✗	–	-	✓	
3140.45	3141.25	0.80	6.22	8.43	11.02	✓	–	✓	✓	
3141.25	3142.15	0.90	5.54	7.31	9.65	✗	–	-	✓	
3142.15	3143.35	1.20	6.27	8.01	9.68	✓	–	✓	✓	
3143.35	3145.15	1.80	5.98	7.03	9.59	✗	–	-	✓	
3145.15	3145.85	0.70	6.27	7.46	9.29	✓	–	✓	✓	
3145.85	3147.65	1.80	5.31	6.68	10.00	✗	–	-	✓	
3147.65	3147.75	0.10	6.39	7.69	10.23	✓	–	✓	✓	
3147.75	3149.05	1.30	5.84	10.84	15.67	✓	–	✓	✗	
3149.05	3150.55	1.50	5.76	7.05	8.43	✗	–	-	✗	
3150.55	3150.85	0.30	6.18	7.09	8.51	✗	–	-	✓	

3150.85	3154.15	3.30	6.33	8.08	10.15	✓	–	✓	✓	
3154.15	3154.65	0.50	6.14	7.64	8.93	✗	–	-	✓	
3154.65	3157.25	2.60	6.41	8.10	9.95	✓	–	✓	✓	
3157.25	3157.95	0.70	5.01	6.61	7.63	✗	–	-	✓	
3157.95	3162.65	4.70	4.96	7.59	16.32	✗	–	-	✗	
3162.65	3163.15	0.50	5.69	7.14	9.89	✗	–	-	✓	
3163.15	3165.95	2.80	6.22	8.61	11.43	✓	–	✓	✓	
3165.95	3166.75	0.80	5.75	7.24	10.00	✗	–	-	✓	
3166.75	3170.45	3.70	4.91	7.05	14.45	✗	–	-	✗	
3170.45	3170.55	0.10	6.22	7.89	10.71	✓	–	✓	✗	
3170.55	3174.45	3.90	5.26	6.42	9.25	✗	–	-	✗	
3174.45	3174.75	0.30	5.94	7.31	9.10	✓	–	✓	✗	
3174.75	3180.55	5.80	5.15	6.57	9.95	✗	–	-	✗	
3180.55	3181.65	1.10	5.98	11.21	15.04	✓	–	✓	✗	
3181.65	3191.05	9.40	5.25	6.51	10.22	✗	–	-	✗	
3191.05	3191.15	0.10	6.14	7.78	9.65	✗	–	-	✓	
3191.15	3192.45	1.30	6.14	10.13	13.72	✓	–	✓	✓	
3192.45	3192.95	0.50	5.73	7.40	9.06	✗	–	-	✓	
3192.95	3193.75	0.80	6.16	7.61	9.47	✓	–	✓	✓	
3193.75	3194.85	1.10	5.73	6.89	9.06	✗	–	-	✓	
3194.85	3201.95	7.10	5.44	6.55	8.49	✗	–	-	✗	
3201.95	3203.05	1.10	6.10	10.10	15.48	✓	–	✓	✗	
3203.05	3205.85	2.80	5.38	6.44	7.84	✗	–	-	✗	
3205.85	3206.25	0.40	6.24	7.17	8.12	✓	–	✓	✗	
3206.25	3207.85	1.60	5.37	6.73	8.71	✗	–	-	✗	
3207.85	3207.95	0.10	6.45	7.53	9.05	✓	–	✓	✗	
3207.95	3208.65	0.70	6.38	7.77	9.27	✓	–	✓	✓	
3208.65	3208.75	0.10	6.38	7.44	8.31	✗	–	-	✓	
3208.75	3209.75	1.00	6.31	7.78	10.10	✓	–	✓	✓	
3209.75	3210.55	0.80	6.14	7.60	10.10	✗	–	-	✓	
3210.55	3211.35	0.80	6.41	7.33	9.14	✓	–	✓	✓	
3211.35	3211.65	0.30	6.40	7.13	8.21	✗	–	-	✓	
3211.65	3212.05	0.40	6.42	7.37	8.48	✓	–	✓	✓	
3212.05	3212.55	0.50	6.36	7.28	8.58	✓	–	✓	✗	
3212.55	3213.05	0.50	6.05	6.77	7.95	✗	–	-	✗	
3213.05	3214.35	1.30	6.38	10.41	15.55	✓	–	✓	✗	
3214.35	3216.45	2.10	5.71	6.84	8.63	✗	–	-	✗	
3216.45	3216.55	0.10	6.13	7.16	8.21	✗	–	-	✓	
3216.55	3223.65	7.10	6.24	9.00	11.91	✓	–	✓	✓	
3223.65	3224.35	0.70	5.92	8.40	11.23	✗	–	-	✓	
3224.35	3226.15	1.80	6.09	9.94	15.73	✓	–	✓	✓	
3226.15	3227.15	1.00	5.57	6.98	9.26	✗	–	-	✓	
3227.15	3228.75	1.60	6.31	8.31	10.94	✓	–	✓	✓	
3228.75	3229.25	0.50	5.97	7.59	9.85	✗	–	-	✓	
3229.25	3229.55	0.30	6.08	7.37	9.21	✗	–	-	✗	
3229.55	3230.15	0.60	6.38	7.65	9.57	✓	–	✓	✓	
3230.15	3230.25	0.10	6.38	7.73	9.57	✗	–	-	✓	
3230.25	3231.05	0.80	6.46	8.23	9.80	✓	–	✓	✓	
3231.05	3233.05	2.00	6.35	8.29	10.42	✓	–	✓	✗	
3233.05	3234.05	1.00	5.65	7.62	10.11	✗	–	-	✗	

3234.05	3236.65	2.60	5.94	10.02	16.13	✓	–	✓	✗	
3236.65	3237.15	0.50	5.75	7.37	9.29	✗	–	▪	✗	
3237.15	3238.15	1.00	6.15	7.84	9.32	✓	–	✓	✗	
3238.15	3239.35	1.20	5.80	7.12	9.11	✗	–	▪	✗	
3239.35	3240.25	0.90	6.23	7.60	8.64	✓	–	✓	✗	
3240.25	3240.65	0.40	6.12	7.13	8.17	✗	–	▪	✗	
3240.65	3242.45	1.80	6.29	7.44	8.30	✓	–	✓	✗	
3242.45	3243.35	0.90	5.83	6.83	7.75	✗	–	▪	✗	
3243.35	3243.55	0.20	5.90	7.08	7.95	✗	–	▪	✓	
3243.55	3245.55	2.00	6.20	8.00	9.40	✓	–	✓	✓	
3245.55	3245.85	0.30	6.00	6.90	7.96	✗	–	▪	✓	
3245.85	3257.55	11.70	4.46	6.92	15.42	✗	–	▪	✗	
3257.55	3257.65	0.10	5.80	7.85	9.52	✗	–	▪	✓	
3257.65	3259.25	1.60	5.80	10.20	15.13	✓	–	✓	✓	
3259.25	3259.75	0.50	5.61	7.62	9.06	✗	–	▪	✓	
3259.75	3287.35	27.60	6.35	9.96	16.01	✓	–	✓	✓	
3287.35	3287.85	0.50	6.26	7.85	9.46	✗	–	▪	✓	
3287.85	3288.65	0.80	6.15	7.93	9.77	✓	–	✓	✓	
3288.65	3288.95	0.30	5.21	7.49	9.54	✗	–	▪	✓	
3288.95	3289.55	0.60	5.19	7.44	10.48	✗	–	▪	✗	
3289.55	3290.55	1.00	6.21	9.97	14.22	✓	–	✓	✗	
3290.55	3291.55	1.00	6.01	7.24	8.97	✗	–	▪	✗	
3291.55	3292.05	0.50	6.31	7.84	9.34	✓	–	✓	✗	
3292.05	3293.65	1.60	5.84	7.01	9.37	✗	–	▪	✗	
3293.65	3293.75	0.10	6.09	7.23	8.45	✗	–	▪	✓	
3293.75	3303.85	10.10	6.34	10.02	14.95	✓	–	✓	✓	
3303.85	3304.35	0.50	5.90	7.29	9.28	✗	–	▪	✓	
3304.35	3306.65	2.30	5.32	7.41	10.27	✗	–	▪	✗	
3306.65	3307.55	0.90	5.64	7.79	10.32	✗	–	▪	✓	
3307.55	3307.95	0.40	6.10	8.64	11.44	✓	–	✓	✓	
3307.95	3308.85	0.90	4.76	7.27	11.26	✗	–	▪	✓	
3308.85	3310.15	1.30	4.66	6.58	8.81	✗	–	▪	✗	
3310.15	3310.35	0.20	5.32	7.63	9.70	✗	–	▪	✓	
3310.35	3311.65	1.30	5.84	10.27	14.46	✓	–	✓	✓	
3311.65	3312.45	0.80	4.59	5.73	8.77	✗	–	▪	✓	
3312.45	3321.15	8.70	4.55	5.82	8.97	✗	–	▪	✗	
3321.15	3322.45	1.30	5.98	10.39	14.84	✓	–	✓	✗	
3322.45	3323.25	0.80	5.34	7.15	9.58	✗	–	▪	✗	
3323.25	3323.85	0.60	5.88	7.12	8.41	✗	–	▪	✓	
3323.85	3323.95	0.10	6.24	6.97	7.99	✓	–	✓	✓	
3323.95	3328.95	5.00	5.49	7.54	10.92	✗	–	▪	✓	
3328.95	3393.05	64.10	3.98	6.46	18.68	✗	–	▪	✗	
3393.05	3394.65	1.60	4.27	6.74	10.42	✗	–	▪	✓	
3394.65	3428.55	33.90	4.27	6.77	19.07	✗	–	▪	✗	
3428.55	3429.05	0.50	6.19	8.22	10.97	✗	–	▪	✓	
3429.05	3432.45	3.40	7.78	11.38	18.86	✓	–	✓	✓	
3432.45	3433.15	0.70	6.41	8.96	11.17	✗	–	▪	✓	
3433.15	3433.55	0.40	7.87	9.88	11.17	✓	–	✓	✓	
3433.55	3433.65	0.10	7.87	9.72	10.95	✗	–	▪	✓	
3433.65	3439.85	6.20	7.89	10.63	13.54	✓	–	✓	✓	

3439.85	3439.95	0.10	7.89	10.48	12.65	✗	–	–	✓	
3439.95	3442.15	2.20	7.40	12.83	15.83	✓	–	✓	✓	
3442.15	3442.45	0.30	6.89	10.74	12.56	✗	–	–	✓	
3442.45	3443.75	1.30	7.12	12.30	18.52	✓	–	✓	✓	
3443.75	3444.25	0.50	7.09	8.40	10.23	✗	–	–	✓	
3444.25	3447.05	2.80	7.09	10.35	13.39	✓	–	✓	✓	
3447.05	3448.15	1.10	5.36	7.46	10.41	✗	–	–	✓	
3448.15	3465.75	17.60	5.08	7.68	17.24	✗	–	–	✗	
3465.75	3465.85	0.10	7.70	10.58	14.78	✓	–	✓	✗	
3465.85	3466.35	0.50	7.08	8.20	9.00	✗	–	–	✗	
3466.35	3466.75	0.40	7.79	8.69	9.20	✓	–	✓	✗	
3466.75	3468.05	1.30	6.91	8.15	9.07	✗	–	–	✗	
3468.05	3468.75	0.70	6.98	8.82	10.90	✗	–	–	✓	
3468.75	3487.85	19.10	7.56	11.99	19.81	✓	–	✓	✓	
3487.85	3491.45	3.60	5.92	9.07	12.76	✗	–	–	✓	
3491.45	3491.85	0.40	7.75	10.23	11.92	✓	–	✓	✓	
3491.85	3491.95	0.10	7.89	10.57	12.26	✗	–	–	✓	
3491.95	3493.55	1.60	7.76	10.17	12.48	✓	–	✓	✓	
3493.55	3495.15	1.60	5.91	8.61	11.09	✗	–	–	✓	
3495.15	3496.65	1.50	7.75	9.88	12.39	✓	–	✓	✓	
3496.65	3497.65	1.00	7.25	13.88	21.45	✗	–	–	✓	
3497.65	3500.95	3.30	7.59	11.42	14.88	✓	–	✓	✓	
3500.95	3501.75	0.80	6.26	9.02	11.06	✗	–	–	✓	
3501.75	3509.15	7.40	7.42	11.39	20.64	✓	–	✓	✓	
3509.15	3509.45	0.30	7.58	10.01	12.25	✗	–	–	✓	
3509.45	3518.05	8.60	7.62	10.73	12.99	✓	–	✓	✓	
3518.05	3518.15	0.10	7.84	9.75	11.35	✗	–	–	✓	
3518.15	3518.35	0.20	8.09	9.58	11.40	✓	–	✓	✓	
3518.35	3519.55	1.20	6.95	10.81	19.74	✓	–	✓	✓	
3519.55	3525.35	5.80	7.77	10.15	12.00	✓	–	✓	✓	
3525.35	3526.55	1.20	5.70	8.68	10.83	✗	–	–	✓	
3526.55	3527.05	0.50	7.88	9.62	11.31	✓	–	✓	✓	
3527.05	3527.45	0.40	7.81	9.61	11.64	✗	–	–	✓	
3527.45	3528.45	1.00	7.74	9.60	11.88	✓	–	✓	✓	
3528.45	3532.95	4.50	6.23	9.47	18.74	✗	–	–	✓	
3532.95	3533.65	0.70	7.67	9.32	11.07	✓	–	✓	✓	
3533.65	3535.35	1.70	5.89	8.50	10.97	✗	–	–	✓	
3535.35	3540.75	5.40	7.54	11.26	17.94	✓	–	✓	✓	
3540.75	3541.05	0.30	7.50	8.67	10.73	✗	–	–	✓	
3541.05	3541.35	0.30	7.70	9.30	11.25	✓	–	✓	✓	
3541.35	3541.75	0.40	7.64	9.83	11.53	✗	–	–	✓	
3541.75	3544.35	2.60	7.64	9.55	11.53	✓	–	✓	✓	
3544.35	3544.75	0.40	7.57	9.22	10.87	✗	–	–	✓	
3544.75	3545.55	0.80	6.69	9.48	11.10	✓	–	✓	✓	
3545.55	3546.25	0.70	5.45	7.35	8.26	✗	–	–	✓	
3546.25	3551.55	5.30	4.54	7.28	17.63	✗	–	–	✗	
3551.55	3552.85	1.30	5.71	8.30	11.16	✗	–	–	✓	
3552.85	3555.25	2.40	7.77	9.55	11.32	✓	–	✓	✓	
3555.25	3555.65	0.40	7.49	8.69	9.99	✗	–	–	✓	
3555.65	3557.65	2.00	7.65	9.21	11.27	✓	–	✓	✓	

3557.65	3558.55	0.90	6.51	8.14	10.30	✗	–	–	✓	
3558.55	3558.85	0.30	7.89	9.19	10.32	✓	–	✓	✓	
3558.85	3560.05	1.20	6.81	7.89	9.36	✗	–	–	✓	
3560.05	3560.85	0.80	7.18	9.03	10.61	✓	–	✓	✓	
3560.85	3561.75	0.90	5.92	7.58	9.48	✗	–	–	✓	
3561.75	3562.85	1.10	7.21	11.83	17.03	✓	–	✓	✓	
3562.85	3563.65	0.80	7.15	8.78	10.35	✗	–	–	✓	
3563.65	3564.05	0.40	7.62	8.65	9.98	✓	–	✓	✓	
3564.05	3564.45	0.40	7.35	8.11	9.57	✗	–	–	✓	
3564.45	3564.55	0.10	7.54	8.46	9.69	✗	–	–	✗	
3564.55	3564.65	0.10	7.56	8.55	9.69	✓	–	✓	✗	
3564.65	3573.75	9.10	5.85	8.29	18.71	✗	–	–	✗	
3573.75	3573.95	0.20	7.39	11.70	16.93	✓	–	✓	✗	
3573.95	3578.05	4.10	6.24	8.42	10.97	✗	–	–	✗	
3578.05	3578.55	0.50	7.91	9.06	10.36	✓	–	✓	✗	
3578.55	3579.05	0.50	7.79	9.14	10.04	✓	–	✓	✓	
3579.05	3580.15	1.10	7.21	9.21	10.76	✗	–	–	✓	
3580.15	3581.05	0.90	7.87	9.80	11.69	✓	–	✓	✓	
3581.05	3582.95	1.90	6.82	8.60	10.60	✗	–	–	✓	
3582.95	3583.35	0.40	7.65	9.16	10.98	✓	–	✓	✓	
3583.35	3586.35	3.00	5.16	9.77	18.90	✗	–	–	✓	
3586.35	3586.75	0.40	7.39	8.86	10.15	✓	–	✓	✓	
3586.75	3592.55	5.80	5.88	8.62	11.57	✗	–	–	✓	
3592.55	3592.65	0.10	7.87	9.12	11.26	✓	–	✓	✓	
3592.65	3594.85	2.20	6.73	8.92	11.45	✗	–	–	✓	
3594.85	3596.65	1.80	7.69	11.44	18.15	✓	–	✓	✓	
3596.65	3598.85	2.20	6.92	9.07	11.33	✗	–	–	✓	
3598.85	3598.95	0.10	7.66	8.92	10.31	✓	–	✓	✓	
3598.95	3600.65	1.70	6.66	8.88	11.17	✗	–	–	✓	
3600.65	3602.35	1.70	7.73	9.39	10.78	✓	–	✓	✓	
3602.35	3603.25	0.90	7.37	8.20	9.62	✗	–	–	✓	
3603.25	3606.25	3.00	7.52	9.59	11.11	✓	–	✓	✓	
3606.25	3608.65	2.40	6.59	9.73	18.27	✗	–	–	✓	
3608.65	3612.05	3.40	7.84	10.12	12.20	✓	–	✓	✓	
3612.05	3612.45	0.40	7.35	9.22	10.59	✗	–	–	✓	
3612.45	3613.25	0.80	7.62	9.23	10.59	✓	–	✓	✓	
3613.25	3614.45	1.20	7.39	8.46	10.06	✗	–	–	✓	
3614.45	3614.75	0.30	7.85	8.93	10.31	✓	–	✓	✓	
3614.75	3615.45	0.70	7.31	8.93	10.49	✗	–	–	✓	
3615.45	3615.65	0.20	7.81	9.16	10.71	✓	–	✓	✓	
3615.65	3615.75	0.10	7.84	9.37	10.68	✗	–	–	✓	
3615.75	3616.75	1.00	7.59	9.49	11.07	✓	–	✓	✓	
3616.75	3617.75	1.00	6.91	11.14	17.13	✗	–	–	✓	
3617.75	3620.45	2.70	7.47	9.78	11.89	✓	–	✓	✓	
3620.45	3621.15	0.70	6.73	9.19	11.20	✗	–	–	✓	
3621.15	3621.25	0.10	7.30	9.71	11.20	✓	–	✓	✓	
3621.25	3621.55	0.30	7.10	9.76	11.66	✗	–	–	✓	
3621.55	3622.85	1.30	7.42	10.51	12.41	✓	–	✓	✓	
3622.85	3623.15	0.30	7.42	10.48	12.21	✗	–	–	✓	
3623.15	3624.35	1.20	7.82	10.47	12.73	✓	–	✓	✓	

3624.35	3624.95	0.60	7.50	9.47	11.15	✗	–	▪	✓	
3624.95	3627.25	2.30	7.68	10.51	11.92	✓	–	✓	✓	
3627.25	3628.75	1.50	6.98	10.45	16.67	✗	–	▪	✓	
3628.75	3628.85	0.10	7.77	10.06	13.52	✓	–	✓	✓	
3628.85	3629.55	0.70	7.63	8.74	10.17	✗	–	▪	✓	
3629.55	3629.65	0.10	7.82	8.72	9.97	✓	–	✓	✓	
3629.65	3630.15	0.50	7.49	8.64	10.07	✗	–	▪	✓	
3630.15	3630.35	0.20	7.89	8.86	10.31	✓	–	✓	✓	
3630.35	3630.45	0.10	7.89	9.05	10.33	✗	–	▪	✓	
3630.45	3633.45	3.00	7.88	10.23	12.85	✓	–	✓	✓	
3633.45	3633.75	0.30	7.69	8.67	9.66	✗	–	▪	✓	
3633.75	3633.95	0.20	7.72	8.43	9.25	✓	–	✓	✓	
3633.95	3634.95	1.00	6.88	8.46	10.14	✗	–	▪	✓	
3634.95	3636.85	1.90	7.81	9.53	11.23	✓	–	✓	✓	
3636.85	3637.05	0.20	7.75	8.89	10.26	✗	–	▪	✓	
3637.05	3639.95	2.90	7.81	11.19	18.26	✓	–	✓	✓	
3639.95	3640.95	1.00	5.27	8.16	13.25	✗	–	▪	✓	
3640.95	3642.35	1.40	5.34	7.67	10.07	✗	–	▪	✗	
3642.35	3643.45	1.10	6.45	8.48	10.21	✗	–	▪	✓	
3643.45	3643.75	0.30	6.86	8.99	10.53	✓	–	✓	✓	
3643.75	3644.15	0.40	6.06	7.83	10.29	✗	–	▪	✓	
3644.15	3645.25	1.10	5.59	7.10	8.70	✗	–	▪	✗	
3645.25	3646.05	0.80	5.95	7.85	10.02	✗	–	▪	✓	
3646.05	3646.45	0.40	7.15	8.97	10.21	✓	–	✓	✓	
3646.45	3646.75	0.30	5.65	7.58	9.41	✗	–	▪	✓	
3646.75	3656.95	10.20	4.69	6.88	19.05	✗	–	▪	✗	
3656.95	3658.85	1.90	5.15	7.19	9.74	✗	–	▪	✓	
3658.85	3661.35	2.50	5.08	6.44	8.95	✗	–	▪	✗	
3661.35	3661.55	0.20	6.20	8.71	10.38	✗	–	▪	✓	
3661.55	3662.55	1.00	7.87	12.75	16.34	✓	–	✓	✓	
3662.55	3663.75	1.20	5.97	7.94	10.44	✗	–	▪	✓	
3663.75	3664.35	0.60	5.77	7.10	9.71	✗	–	▪	✗	
3664.35	3665.65	1.30	6.06	7.87	9.78	✗	–	▪	✓	
3665.65	3666.05	0.40	7.76	9.04	10.64	✓	–	✓	✓	
3666.05	3666.25	0.20	7.76	9.47	10.76	✗	–	▪	✓	
3666.25	3668.45	2.20	7.45	9.62	11.25	✓	–	✓	✓	
3668.45	3668.75	0.30	6.64	8.64	10.12	✗	–	▪	✓	
3668.75	3671.35	2.60	5.12	8.17	10.89	✗	–	▪	✗	
3671.35	3671.95	0.60	7.83	9.75	10.89	✓	–	✓	✗	
3671.95	3672.25	0.30	7.67	9.45	10.89	✗	–	▪	✗	
3672.25	3674.65	2.40	5.47	10.29	16.65	✓	–	✓	✓	
3674.65	3676.25	1.60	7.70	9.69	11.26	✓	–	✓	✓	
3676.25	3678.15	1.90	5.96	7.81	10.49	✗	–	▪	✓	
3678.15	3678.75	0.60	6.75	9.34	11.51	✓	–	✓	✓	
3678.75	3679.05	0.30	6.75	8.31	9.94	✗	–	▪	✓	
3679.05	3680.95	1.90	7.61	10.59	12.12	✓	–	✓	✓	
3680.95	3683.45	2.50	4.96	7.66	11.18	✗	–	▪	✓	
3683.45	3684.15	0.70	6.81	8.96	10.57	✓	–	✓	✓	
3684.15	3687.45	3.30	4.90	8.97	15.69	✗	–	▪	✓	
3687.45	3690.85	3.40	7.64	10.43	11.98	✓	–	✓	✓	

3690.85	3691.65	0.80	6.56	9.45	11.48	✗	–		✓	
3691.65	3694.85	3.20	8.04	10.57	12.15	✓	–	✓	✓	
3694.85	3695.95	1.10	5.72	11.83	18.47	✗	–		✓	
3695.95	3696.35	0.40	6.99	10.66	12.11	✓	–	✓	✓	
3696.35	3696.45	0.10	7.79	10.75	12.16	✗	–		✓	
3696.45	3702.55	6.10	7.48	10.90	12.81	✓	–	✓	✓	
3702.55	3702.85	0.30	6.65	9.61	12.01	✗	–		✓	
3702.85	3703.65	0.80	7.23	9.79	11.89	✓	–	✓	✓	
3703.65	3703.85	0.20	7.06	9.59	11.09	✗	–		✓	
3703.85	3707.25	3.40	7.43	11.80	18.85	✓	–	✓	✓	
3707.25	3707.65	0.40	7.25	9.03	10.90	✗	–		✓	
3707.65	3707.75	0.10	7.56	8.84	10.10	✓	–	✓	✓	
3707.75	3708.55	0.80	6.94	9.32	10.85	✗	–		✓	
3708.55	3724.45	15.90	6.22	10.80	17.93	✓	–	✓	✓	
3724.45	3724.65	0.20	5.22	7.62	10.38	✗	–		✓	
3724.65	3726.35	1.70	4.80	7.02	10.45	✗	–		✗	
3726.35	3726.75	0.40	7.46	8.78	10.14	✓	–	✓	✓	
3726.75	3727.25	0.50	7.33	8.79	10.30	✗	–		✓	
3727.25	3727.65	0.40	7.73	8.88	10.30	✓	–	✓	✓	
3727.65	3729.35	1.70	6.54	10.45	16.64	✗	–		✓	
3729.35	3737.05	7.70	7.06	10.75	12.61	✓	–	✓	✓	
3737.05	3737.35	0.30	6.94	10.25	11.89	✗	–		✓	
3737.35	3741.15	3.80	7.14	11.82	20.18	✓	–	✓	✓	
3741.15	3741.75	0.60	5.88	9.02	11.13	✗	–		✓	
3741.75	3746.55	4.80	7.42	10.59	13.07	✓	–	✓	✓	
3746.55	3746.75	0.20	7.46	9.32	10.36	✗	–		✓	
3746.75	3758.45	11.70	7.51	11.40	19.74	✓	–	✓	✓	
3758.45	3758.85	0.40	6.83	9.17	10.95	✗	–		✓	
3758.85	3759.25	0.40	7.63	9.46	10.72	✓	–	✓	✓	
3759.25	3763.55	4.30	5.44	9.43	17.67	✗	–		✓	
3763.55	3765.25	1.70	7.65	9.60	11.22	✓	–	✓	✓	
3765.25	3765.75	0.50	6.44	9.39	11.34	✗	–		✓	
3765.75	3766.45	0.70	7.83	9.64	11.98	✓	–	✓	✓	
3766.45	3766.65	0.20	7.82	9.40	11.55	✗	–		✓	
3766.65	3766.75	0.10	7.65	9.10	10.42	✓	–	✓	✓	
3766.75	3767.45	0.70	5.81	9.23	11.48	✗	–		✓	
3767.45	3768.25	0.80	7.57	9.97	11.39	✓	–	✓	✓	
3768.25	3769.35	1.10	6.78	10.59	12.91	✗	–		✓	
3769.35	3772.85	3.50	6.69	11.65	18.93	✓	–	✓	✓	
3772.85	3773.65	0.80	5.79	7.78	10.29	✗	–		✓	
3773.65	3776.65	3.00	4.57	6.99	11.81	✗	–		✗	
3776.65	3777.25	0.60	5.99	8.15	10.21	✗	–		✓	
3777.25	3787.85	10.60	7.56	11.12	20.48	✓	–	✓	✓	
3787.85	3788.15	0.30	7.39	9.71	10.66	✗	–		✓	
3788.15	3800.75	12.60	7.09	11.49	18.92	✓	–	✓	✓	
3800.75	3801.25	0.50	5.83	8.77	11.65	✗	–		✓	
3801.25	3804.15	2.90	7.44	12.26	20.12	✓	–	✓	✓	
3804.15	3804.65	0.50	7.09	8.79	10.93	✗	–		✓	
3804.65	3804.85	0.20	7.83	9.12	10.93	✓	–	✓	✓	
3804.85	3805.45	0.60	6.74	9.46	11.97	✗	–		✓	

3805.45	3811.55	6.10	7.58	11.82	13.96	✓	–	✓	✓	
3811.55	3811.75	0.20	7.58	9.99	12.61	✗	–	–	✓	
3811.75	3812.95	1.20	7.63	10.02	12.61	✓	–	✓	✓	
3812.95	3813.05	0.10	7.76	9.80	11.72	✗	–	–	✓	
3813.05	3815.15	2.10	7.44	11.18	17.76	✓	–	✓	✓	
3815.15	3816.85	1.70	6.18	8.61	11.20	✗	–	–	✓	
3816.85	3819.75	2.90	7.65	10.90	13.52	✓	–	✓	✓	
3819.75	3820.75	1.00	6.57	9.90	12.65	✗	–	–	✓	
3820.75	3829.65	8.90	7.63	11.68	19.28	✓	–	✓	✓	
3829.65	3830.05	0.40	7.32	10.50	12.29	✗	–	–	✓	
3830.05	3830.95	0.90	6.88	11.05	12.31	✓	–	✓	✓	
3830.95	3833.15	2.20	5.38	8.24	11.79	✗	–	–	✓	
3833.15	3834.55	1.40	4.95	8.01	13.79	✗	–	–	✗	
3834.55	3835.45	0.90	7.12	13.52	16.57	✗	–	–	✓	
3835.45	3836.45	1.00	5.54	7.97	14.04	✗	–	–	✗	
3836.45	3836.95	0.50	5.73	7.47	10.60	✗	–	–	✓	
3836.95	3837.55	0.60	7.18	9.61	11.29	✓	–	✓	✓	
3837.55	3838.25	0.70	7.00	9.27	11.14	✗	–	–	✓	
3838.25	3839.65	1.40	7.25	10.03	11.79	✓	–	✓	✓	
3839.65	3840.55	0.90	5.22	7.04	10.06	✗	–	–	✓	
3840.55	3857.15	16.60	4.22	7.15	16.92	✗	–	–	✗	
3857.15	3858.15	1.00	4.48	7.18	9.42	✗	–	–	✓	
3858.15	3858.35	0.20	7.76	8.70	9.35	✓	–	✓	✓	
3858.35	3859.15	0.80	6.03	8.25	10.02	✗	–	–	✓	
3859.15	3859.65	0.50	7.68	9.28	10.69	✓	–	✓	✓	
3859.65	3860.45	0.80	6.53	8.96	11.62	✗	–	–	✓	
3860.45	3861.85	1.40	7.69	10.32	12.59	✓	–	✓	✓	
3861.85	3863.15	1.30	5.62	7.59	11.49	✗	–	–	✓	
3863.15	3863.65	0.50	7.83	10.35	12.09	✓	–	✓	✓	
3863.65	3864.15	0.50	7.43	9.38	11.46	✗	–	–	✓	
3864.15	3866.85	2.70	7.87	10.57	13.25	✓	–	✓	✓	
3866.85	3869.45	2.60	6.35	10.59	19.38	✓	–	✓	✓	
3869.45	3869.65	0.20	7.37	9.65	11.63	✓	–	✓	✓	
3869.65	3869.85	0.20	7.37	9.36	11.15	✗	–	–	✓	
3869.85	3870.55	0.70	7.80	10.10	12.02	✓	–	✓	✓	
3870.55	3870.65	0.10	7.80	9.94	11.46	✗	–	–	✓	
3870.65	3871.25	0.60	7.61	10.03	11.56	✓	–	✓	✓	
3871.25	3873.75	2.50	6.08	8.93	12.05	✗	–	–	✓	
3873.75	3885.15	11.40	4.64	7.14	17.54	✗	–	–	✗	
3885.15	3885.55	0.40	5.67	7.65	10.16	✗	–	–	✓	
3885.55	3885.75	0.20	7.60	9.18	10.17	✓	–	✓	✓	
3885.75	3886.35	0.60	7.27	9.03	10.17	✗	–	–	✓	
3886.35	3887.45	1.10	7.80	9.09	11.15	✓	–	✓	✓	
3887.45	3887.65	0.20	7.75	9.67	11.15	✗	–	–	✓	
3887.65	3888.05	0.40	7.43	9.34	10.65	✓	–	✓	✓	
3888.05	3888.35	0.30	7.42	8.64	10.60	✗	–	–	✓	
3888.35	3888.45	0.10	7.00	8.32	9.52	✓	–	✓	✓	
3888.45	3890.15	1.70	5.73	10.46	17.70	✗	–	–	✓	
3890.15	3890.25	0.10	7.65	9.60	10.69	✓	–	✓	✓	
3890.25	3890.45	0.20	7.65	9.49	10.69	✗	–	–	✓	

3890.45	3892.15	1.70	7.51	9.44	11.02	✓	–	✓	✓	
3892.15	3892.85	0.70	6.14	9.17	10.68	✗	–	▪	✓	
3892.85	3892.95	0.10	7.85	9.64	10.68	✓	–	✓	✓	
3892.95	3893.85	0.90	6.84	9.10	10.76	✗	–	▪	✓	
3893.85	3893.95	0.10	7.71	9.63	10.67	✓	–	✓	✓	
3893.95	3894.45	0.50	6.62	9.29	10.86	✗	–	▪	✓	
3894.45	3894.95	0.50	7.54	9.24	12.33	✓	–	✓	✓	
3894.95	3897.75	2.80	4.47	7.74	12.14	✗	–	▪	✓	
3897.75	3899.45	1.70	5.29	7.80	10.72	✗	–	▪	✗	
3899.45	3900.25	0.80	5.85	8.36	10.52	✗	–	▪	✓	
3900.25	3900.35	0.10	7.80	9.17	10.72	✓	–	✓	✓	
3900.35	3902.45	2.10	6.52	11.11	19.51	✗	–	▪	✓	
3902.45	3903.15	0.70	7.76	9.75	11.97	✓	–	✓	✓	
3903.15	3903.25	0.10	7.77	10.05	11.97	✗	–	▪	✓	
3903.25	3903.35	0.10	7.65	9.92	11.97	✓	–	✓	✓	
3903.35	3904.45	1.10	5.77	8.51	11.03	✗	–	▪	✓	
3904.45	3904.85	0.40	5.54	8.04	9.95	✗	–	▪	✗	
3904.85	3905.05	0.20	7.52	9.07	10.54	✓	–	✓	✗	
3905.05	3905.25	0.20	6.98	8.55	10.59	✗	–	▪	✗	
3905.25	3906.55	1.30	5.71	8.42	12.09	✗	–	▪	✓	
3906.55	3906.75	0.20	7.73	9.54	12.21	✓	–	✓	✓	
3906.75	3908.15	1.40	6.51	8.64	12.21	✗	–	▪	✓	
3908.15	3909.25	1.10	5.39	6.98	9.16	✗	–	▪	✗	
3909.25	3909.85	0.60	6.16	8.19	10.15	✗	–	▪	✓	
3909.85	3910.15	0.30	7.48	9.48	10.61	✓	–	✓	✓	
3910.15	3912.45	2.30	5.39	10.46	19.02	✗	–	▪	✓	
3912.45	3912.75	0.30	7.80	11.24	17.87	✓	–	✓	✓	
3912.75	3913.35	0.60	6.77	8.54	12.48	✗	–	▪	✓	
3913.35	3915.15	1.80	7.83	10.26	12.15	✓	–	✓	✓	
3915.15	3916.05	0.90	6.41	8.83	11.05	✗	–	▪	✓	
3916.05	3916.55	0.50	7.53	10.05	11.79	✓	–	✓	✓	
3916.55	3916.75	0.20	7.53	9.66	11.01	✗	–	▪	✓	
3916.75	3916.95	0.20	7.56	9.39	11.25	✓	–	✓	✓	
3916.95	3917.45	0.50	6.79	9.42	11.61	✗	–	▪	✓	
3917.45	3917.55	0.10	7.81	9.52	11.61	✓	–	✓	✓	
3917.55	3919.65	2.10	5.36	8.08	12.08	✗	–	▪	✓	
3919.65	3919.95	0.30	7.82	9.40	10.81	✓	–	✓	✓	
3919.95	3920.35	0.40	7.11	8.93	10.60	✗	–	▪	✓	
3920.35	3920.55	0.20	7.24	8.69	9.89	✓	–	✓	✓	
3920.55	3921.45	0.90	5.72	7.57	9.96	✗	–	▪	✓	
3921.45	3922.85	1.40	5.25	9.45	17.38	✗	–	▪	✗	
3922.85	3923.65	0.80	5.44	7.45	10.84	✗	–	▪	✓	
3923.65	3924.15	0.50	7.41	9.53	11.53	✓	–	✓	✓	
3924.15	3925.65	1.50	5.29	7.45	9.49	✗	–	▪	✓	
3925.65	3929.00	3.35	4.94	6.94	9.82	✗	–	▪	✗	

Figure 4-1: AZNS-040 9-5/8" CMT Evaluation Log

4.3 7" Liner Cement Evaluation Log Result

Results - Summary

Total Pass (M)	150.40
Total Fail (M)	13.59
Total Partial Pass (M)	54.42

Results - Filtered Intervals

Show Pass Zones	Yes
Show Fail Zones	No
Show Partial Passes	No
Minimum Interval	3

Isolation Key					
	Pass		Partial		Fail

Top Depth (M)	Bottom Depth (M)	Interval (M)	Min (dB/ft)	Avg (dB/ft)	Max (dB/ft)	Bonded	Derivative	Channel Risk Pass	Formation	Isolated
3938.33	4011.36	73.02	4.49	10.56	17.20	✓	–	✓	✓	
4012.28	4016.48	4.20	4.80	8.49	13.29	✓	–	✓	✓	
4017.41	4022.31	4.90	4.65	8.15	11.08	✓	–	✓	✓	
4028.06	4032.41	4.35	5.18	8.92	11.41	✓	–	✓	✓	
4050.43	4055.46	5.02	5.32	8.19	11.12	✓	–	✓	✓	
4056.56	4059.96	3.40	5.93	9.22	13.65	✓	–	✓	✓	
4110.53	4123.76	13.22	5.36	10.56	15.93	✓	–	✓	✓	
4143.73	4151.08	7.35	5.35	9.22	15.56	✓	–	✓	✓	

Results - All Intervals

Top Depth	Bottom Depth	Interval (M)	Min (dB/ft)	Avg (dB/ft)	Max (dB/ft)	Bonded	Derivative	Channel Risk Pass	Formation	Isolated
3932.98	3935.01	2.02	3.53	5.11	8.76	✗	–	–	✗	
3935.01	3935.91	0.90	4.65	9.30	12.43	✓	–	✓	✗	
3935.91	3937.58	1.67	4.18	5.52	7.99	✗	–	–	✗	
3937.58	3938.33	0.75	5.17	6.25	8.04	✗	–	–	✓	
3938.33	4011.36	73.02	4.49	10.56	17.20	✓	–	✓	✓	
4011.36	4012.28	0.92	5.04	7.28	9.19	✗	–	–	✓	
4012.28	4016.48	4.20	4.80	8.49	13.29	✓	–	✓	✓	
4016.48	4017.41	0.92	4.08	6.84	9.38	✓	–	✗	✓	
4017.41	4022.31	4.90	4.65	8.15	11.08	✓	–	✓	✓	
4022.31	4023.46	1.15	4.57	6.91	10.18	✗	–	–	✓	
4023.46	4026.38	2.92	4.65	8.49	12.49	✓	–	✓	✓	
4026.38	4028.06	1.67	5.05	6.79	9.59	✗	–	–	✓	
4028.06	4032.41	4.35	5.18	8.92	11.41	✓	–	✓	✓	
4032.41	4033.48	1.07	4.02	7.04	9.12	✓	–	✗	✓	
4033.48	4033.63	0.15	4.55	6.32	7.31	✓	–	✓	✓	
4033.63	4034.71	1.07	4.75	6.35	8.92	✗	–	–	✓	
4034.71	4036.66	1.95	5.18	8.10	11.75	✓	–	✓	✓	
4036.66	4037.48	0.82	5.67	6.79	8.42	✗	–	–	✓	
4037.48	4038.06	0.57	5.99	7.43	10.26	✓	–	✓	✓	
4038.06	4038.11	0.05	5.86	7.41	10.33	✗	–	–	✓	
4038.11	4038.51	0.40	5.42	7.60	10.68	✓	–	✓	✓	
4038.51	4041.56	3.05	4.61	6.59	10.34	✗	–	–	✓	
4041.56	4043.16	1.60	5.11	7.83	10.41	✓	–	✓	✓	
4043.16	4045.38	2.22	4.78	6.73	9.33	✗	–	–	✓	
4045.38	4046.31	0.92	4.20	6.02	8.65	✗	–	–	✗	
4046.31	4047.63	1.32	4.29	9.32	13.45	✓	–	✓	✗	
4047.63	4049.73	2.10	4.47	6.21	9.37	✗	–	–	✗	
4049.73	4049.93	0.20	4.97	6.38	7.36	✗	–	–	✓	
4049.93	4050.08	0.15	5.32	6.61	7.72	✓	–	✓	✓	
4050.08	4050.43	0.35	5.76	7.26	9.33	✗	–	–	✓	
4050.43	4055.46	5.02	5.32	8.19	11.12	✓	–	✓	✓	
4055.46	4056.56	1.10	5.36	7.33	9.90	✗	–	–	✓	
4056.56	4059.96	3.40	5.93	9.22	13.65	✓	–	✓	✓	
4059.96	4060.16	0.20	5.79	7.94	9.45	✗	–	–	✓	
4060.16	4061.73	1.57	5.86	7.97	10.55	✓	–	✓	✓	
4061.73	4064.83	3.10	4.08	6.61	10.47	✗	–	–	✓	
4064.83	4066.16	1.32	4.55	7.52	10.12	✓	–	✓	✓	
4066.16	4067.13	0.97	4.81	7.01	9.99	✗	–	–	✓	
4067.13	4069.81	2.67	5.53	9.35	16.03	✓	–	✓	✓	
4069.81	4070.31	0.50	5.57	7.97	10.52	✗	–	–	✓	
4070.31	4071.11	0.80	5.84	7.69	10.35	✓	–	✓	✓	
4071.11	4071.61	0.50	5.68	6.72	7.84	✗	–	–	✓	
4071.61	4073.41	1.80	5.66	7.82	10.12	✓	–	✓	✓	
4073.41	4077.31	3.90	4.85	6.41	9.41	✗	–	–	✓	
4077.31	4078.16	0.85	5.89	7.76	10.29	✓	–	✓	✓	
4078.16	4079.91	1.75	3.99	6.34	9.29	✗	–	–	✓	
4079.91	4081.01	1.10	4.58	8.77	12.19	✓	–	✓	✓	
4081.01	4081.93	0.92	4.42	6.42	8.95	✗	–	–	✓	
4081.93	4082.46	0.52	5.40	8.04	10.46	✓	–	✓	✓	
4082.46	4082.71	0.25	5.36	7.44	9.98	✗	–	–	✓	

4082.71	4083.38	0.67	5.56	7.35	9.70	✓	–	✓	✓	
4083.38	4083.43	0.05	5.63	6.99	8.55	✗	–	–	✓	
4083.43	4083.61	0.17	5.28	6.82	8.45	✓	–	✓	✓	
4083.61	4084.21	0.60	4.96	6.80	8.91	✗	–	–	✓	
4084.21	4085.06	0.85	5.29	8.08	10.22	✓	–	✓	✓	
4085.06	4089.38	4.32	4.78	6.21	9.31	✗	–	–	✓	
4089.38	4090.08	0.70	5.67	7.11	8.38	✓	–	✓	✓	
4090.08	4090.18	0.10	5.98	7.07	8.45	✗	–	–	✓	
4090.18	4091.58	1.40	6.18	10.38	14.11	✓	–	✓	✓	
4091.58	4092.11	0.52	5.05	6.67	9.73	✗	–	–	✓	
4092.11	4092.18	0.07	5.42	6.75	8.75	✓	–	✓	✓	
4092.18	4092.96	0.77	5.50	6.98	9.91	✗	–	–	✓	
4092.96	4093.43	0.47	5.60	7.48	10.58	✓	–	✓	✓	
4093.43	4094.16	0.72	5.60	7.31	11.30	✗	–	–	✓	
4094.16	4095.18	1.02	5.71	7.65	11.43	✓	–	✓	✓	
4095.18	4095.46	0.27	5.80	7.42	10.56	✗	–	–	✓	
4095.46	4096.01	0.55	5.83	7.39	10.67	✓	–	✓	✓	
4096.01	4096.31	0.30	5.82	7.80	10.92	✗	–	–	✓	
4096.31	4098.91	2.60	5.64	7.97	11.67	✓	–	✓	✓	
4098.91	4099.31	0.40	5.64	7.56	11.47	✗	–	–	✓	
4099.31	4099.66	0.35	5.55	7.33	10.47	✓	–	✓	✓	
4099.66	4101.03	1.37	4.22	6.82	9.70	✗	–	–	✓	
4101.03	4102.48	1.45	5.55	9.59	14.74	✓	–	✓	✓	
4102.48	4102.81	0.32	5.54	7.01	8.50	✗	–	–	✓	
4102.81	4103.36	0.55	5.31	7.21	8.76	✓	–	✓	✓	
4103.36	4104.51	1.15	5.31	6.96	8.72	✗	–	–	✓	
4104.51	4105.38	0.87	5.37	7.31	8.86	✓	–	✓	✓	
4105.38	4106.26	0.87	4.99	6.48	9.16	✗	–	–	✓	
4106.26	4108.88	2.62	5.12	7.89	10.68	✓	–	✓	✓	
4108.88	4110.53	1.65	4.85	7.01	10.44	✗	–	–	✓	
4110.53	4123.76	13.22	5.36	10.56	15.93	✓	–	✓	✓	
4123.76	4124.56	0.80	6.22	10.45	12.57	✓	–	✓	✓	
4124.56	4125.96	1.40	3.49	5.49	8.97	✗	–	–	✗	
4125.96	4129.46	3.50	3.45	6.03	9.72	✗	–	–	✓	
4129.46	4130.11	0.65	4.02	6.06	8.72	✗	–	–	✗	
4130.11	4130.31	0.20	4.10	6.62	9.21	✓	–	✗	✗	
4130.31	4132.58	2.27	3.65	6.34	9.24	✗	–	–	✗	
4132.58	4133.23	0.65	3.62	6.34	8.23	✗	–	–	✓	
4133.23	4133.41	0.17	5.54	6.95	8.03	✓	–	✓	✓	
4133.41	4134.63	1.22	4.62	6.75	8.31	✗	–	–	✓	
4134.63	4134.83	0.20	4.24	6.93	9.11	✓	–	✓	✓	
4134.83	4135.21	0.37	3.54	6.93	9.34	✗	–	–	✓	
4135.21	4136.11	0.90	4.05	9.54	13.89	✗	–	–	✗	
4136.11	4136.36	0.25	5.79	8.00	11.52	✓	–	✓	✗	
4136.36	4137.96	1.60	4.01	5.51	7.92	✗	–	–	✗	
4137.96	4141.08	3.12	3.28	6.25	8.74	✗	–	–	✓	
4141.08	4142.41	1.32	4.58	7.10	9.13	✓	–	✓	✓	
4142.41	4143.23	0.82	5.12	6.68	8.41	✗	–	–	✓	
4143.23	4143.68	0.45	5.10	6.99	8.39	✓	–	✓	✓	
4143.68	4143.73	0.05	5.27	7.13	8.46	✗	–	–	✓	
4143.73	4151.08	7.35	5.35	9.22	15.56	✓	–	✓	✓	
4151.08	4151.36	0.27	4.36	6.79	10.52	✗	–	–	✓	
4151.36	4151.39	0.04	4.32	6.26	8.85	✗	–	–	✗	

Figure 4-2: AZNS-040 7" CMT Evaluation Log

5. Bottom Hole Assembly Record

Tool Description	Length (m)	Cum. Length (m)
BHA No. #1		
12 – ¼" R. bit	0.29	0.29
Bit Sub (float in)	1.11	1.40
8 – ½" DC	96.66	98.06
XOS	0.85	98.91
Total Length (m)		**98.91**
Drilling & Mud Properties		
WOB (lb)		0 – 20
RPM		40 – 70
Torque (klb)		0.5 – 1.2
SPP (psi)		100 – 300
Flow (gpm)		300 – 400
Mw (pcf)		67.00
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #1		
12 – ¼" R. bit	0.29	0.29
Bit Sub (float in)	1.11	1.40
8 – ½" DC	105.44	106.84
8" D. Jar C/W flex jt	10.93	117.77

8 – ½” DC	26.69	144.46
XOS	0.85	145.31
5” HWDP	83.32	228.63
Total Length (m)		**228.63**
Drilling & Mud Properties		
WOB (lb)		0 – 25
RPM		40 – 70
Torque (klb)		0.5 – 5
SPP (psi)		1300 – 1500
Flow (gpm)		300 – 600
Mw (pcf)		73.00
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #2		
12 – ¼” R. bit	0.29	0.29
Bit Sub (float in)	1.11	1.40
8 – ½” DC	105.44	106.84
8” D. Jar C/W flex jt	10.93	117.77
8 – ½” DC	26.69	144.46
XOS	0.85	145.31
5” HWDP	83.32	228.63
Total Length (m)		**228.63**

Drilling & Mud Properties		
WOB (lb)		5 – 17
RPM		70 – 140
Torque (klb)		5 – 9
SPP (psi)		800 – 1400
Flow (gpm)		400 – 650
Mw (pcf)		73.00
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #3		
10 – 1/8” Overshot	1.18	1.18
8 – ½” DC	8.90	10.08
XOS	0.85	10.93
5” HWDP	83.32	94.25
Total Length (m)		**94.25**
Drilling & Mud Properties		
WOB (lb)		1 – 10
RPM		90 – 130
Torque (klb)		0.5 – 3
SPP (psi)		550 – 800
Flow (gpm)		450 – 550
Mw (pcf)		73.00

Tool Description	Length (m)	Cum. Length (m)
BHA No. #4		
12 – ¼" PDC bit	0.40	0.40
Bit Sub	1.10	1.50
8 – ½" DC	17.83	19.33
S. stab	2.16	21.49
8 – ½" DC	88.38	109.87
8" D. Jar C/W flex jt	10.93	120.80
8 – ½" DC	26.69	147.49
XOS	0.85	148.34
5" HWDP	83.32	231.66
Total Length (m)		**231.66**
Drilling & Mud Properties		
WOB (lb)		1 – 10
RPM		90 – 130
Torque (klb)		0.5 – 3
SPP (psi)		550 – 800
Flow (gpm)		450 – 550
Mw (pcf)		73.00
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #5		

12 – ¼” PDC bit	0.40	0.40
Bit Sub	1.10	1.50
8 – ½” DC	17.83	19.33
S. stab	2.16	21.49
8 – ½” DC	88.38	109.87
8” D. Jar C/W flex jt	10.93	120.80
8 – ½” DC	26.69	147.49
XOS	0.85	148.34
5” HWDP	83.32	231.66
Total Length (m)		**213.66**
Drilling & Mud Properties		
WOB (lb)		10 – 20
RPM		60 – 80
Torque (klb)		1 – 4
SPP (psi)		1930 – 2350
Flow (gpm)		700 – 800
Mw (pcf)		75.00
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #6		
10 – 1/8” Overshot	2.15	2.15
8 – ½” DC	8.68	10.83

8” D. Jar	2.68	13.51
8 – ½” DC	8.05	21.56
XOS	0.85	22.41
5” HWDP	83.32	105.73
Total Length (m)		**105.73**
Drilling & Mud Properties		
WOB (lb)		10 – 20
RPM		60 – 80
Torque (klb)		1 – 4
SPP (psi)		1930 – 2350
Flow (gpm)		700 – 800
Mw (pcf)		75.00
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #7		
12 – ¼” B. bit	0.29	0.29
Bit Sub	1.10	1.39
8 – ½” DC	17.83	19.22
S. stab	2.16	21.38
8 – ½” DC	88.38	109.76
XOS	0.85	110.61
5” HWDP	83.32	193.93

Total Length (m)		**193.93**
Drilling & Mud Properties		
WOB (lb)		5 – 20
RPM		60 – 120
Torque (klb)		1 – 4
SPP (psi)		1650 – 2150
Flow (gpm)		650 – 750
Mw (pcf)		75.00
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #8		
12 – ¼” B. bit	0.29	0.29
Bit Sub	1.10	1.39
8 – ½” DC	17.83	19.22
S. stab	2.16	21.38
8 – ½” DC	88.38	109.76
XOS	0.85	110.61
5” HWDP	83.32	193.93
Total Length (m)		**193.93**
Drilling & Mud Properties		
WOB (lb)		5 – 20
RPM		60 – 120

Torque (klb)		1 – 4
SPP (psi)		1650 – 2150
Flow (gpm)		650 – 750
Mw (pcf)		75.00
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #9		
10 – 1/8" Overshot	1.18	1.18
8 – ½" DC	8.68	9.86
8" D. Jar	2.68	2.26
8 – ½" DC	8.05	20.59
XOS	0.85	21.44
5" HWDP	83.32	104.76
Total Length (m)		**104.76**
Drilling & Mud Properties		
WOB (lb)		5 – 15
RPM		110 – 180
Torque (klb)		1 – 4
SPP (psi)		1850 – 2370
Flow (gpm)		690 – 810
Mw (pcf)		75.00
Tool Description	**Length (m)**	**Cum. Length (m)**

BHA No. #10		
12 – ¼” PDC bit	0.35	0.35
Bit Sub	1.11	1.46
8 – ½” DC	17.97	19.43
S. stab	2.16	21.59
8 – ½” DC	90.19	111.78
XOS	0.85	112.63
5” HWDP	83.32	195.95
Total Length (m)		**195.95**
Drilling & Mud Properties		
WOB (lb)		10 – 15
RPM		160 – 170
Torque (klb)		8 – 12
SPP (psi)		2450 – 2550
Flow (gpm)		680 – 700
Mw (pcf)		78.00
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #11		
10 – 1/8” Overshot	1.18	1.18
8 – ½” DC	8.68	9.86
8” D. Jar	2.68	12.54

8 – ½” DC	8.05	20.59
XOS	0.85	21.44
Total Length (m)		**21.44**
Drilling & Mud Properties		
WOB (lb)		-
RPM		-
Torque (klb)		-
SPP (psi)		2450 – 2550
Flow (gpm)		680 – 700
Mw (pcf)		78.00
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #12		
12 – ¼” PDC bit	0.35	0.35
Bit sub	1.11	1.46
8 – ½” DC	17.77	19.23
S. stab	2.01	21.24
8 – ½” DC	89.52	110.76
8” mech D. Jar C/W flex jt	10.33	121.09
8 – ½” DC	16.95	138.04
XOS	0.98	139.02
5” HWDP	93.16	232.18

Total Length (m)		232.18
Drilling & Mud Properties		
WOB (lb)		0 – 17
RPM		80 – 150
Torque (klb)		4 – 12
SPP (psi)		2250 – 2650
Flow (gpm)		680 – 730
Mw (pcf)		79.00
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #13		
12 – ¼” TCI bit	0.29	0.29
9 – 5/8” Junk sub	0.92	1.21
8 – ½” D. pin	0.64	1.85
Bit sub	1.11	2.96
8 – ½” DC	17.77	20.73
S. stab	2.01	22.74
8 – ½” DC	89.52	112.26
8” mech D. Jar C/W flex jt	10.33	122.59
8 – ½” DC	16.95	139.54
XOS	0.98	140.52
5” HWDP	83.94	224.46

Total Length (m)		**224.46**
Drilling & Mud Properties		
WOB (lb)		0 – 17
RPM		90 – 150
Torque (klb)		3 – 14
SPP (psi)		2450 – 2650
Flow (gpm)		700 – 730
Mw (pcf)		79.50
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #14		
12 – ¼” PDC bit	0.27	0.27
Bit Sub	1.38	1.65
8 – ½” DC	17.77	19.42
S. stab	2.01	21.43
8 – ½” DC	62.81	84.24
8” mech D. Jar C/W flex jt	10.33	94.57
8 – ½” DC	16.95	111.52
XOS	0.98	112.50
5” HWDP	93.16	205.66
Total Length (m)		**205.66**
Drilling & Mud Properties		

WOB (lb)	0 – 12
RPM	60 – 80
Torque (klb)	1 – 3
SPP (psi)	2550 – 2650
Flow (gpm)	700 – 720
Mw (pcf)	80.00

Tool Description	Length (m)	Cum. Length (m)
BHA No. #15		
12 – ¼” TCI bit	0.29	0.29
9 – 5/8” Junk sub	2.02	2.31
8 – ½” D. pin	0.64	2.95
Bit sub	1.11	4.06
8 – ½” DC	17.77	21.83
S. stab	2.01	23.84
8 – ½” DC	62.81	86.65
8” mech D. Jar C/W flex jt	10.33	96.98
8 – ½” DC	16.95	113.93
XOS	0.98	114.91
5” HWDP	93.16	208.07
Total Length (m)		**208.07**
Drilling & Mud Properties		

WOB (lb)		5 – 15
RPM		80 – 180
Torque (klb)		5 – 9
SPP (psi)		2450 – 2650
Flow (gpm)		720 – 760
Mw (pcf)		80.50
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #16		
12 – ¼” PDC bit	0.36	0.36
Bit sub	1.11	1.47
8 – ½” DC	17.77	19.24
S. stab	2.01	21.25
8 – ½” DC	62.81	84.06
8” mech D. Jar C/W flex jt	10.33	94.39
8 – ½” DC	16.95	111.34
XOS	0.98	112.32
5” HWDP	93.16	205.48
Total Length (m)		**205.48**
Drilling & Mud Properties		
WOB (lb)		10 – 20
RPM		140 – 170

Torque (klb)		5 – 9
SPP (psi)		2600 – 2700
Flow (gpm)		675
Mw (pcf)		83.00
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #17		
10 – 1/8" Overshot	2.15	2.15
8 – ½" DC	8.89	11.04
8" D. Jar	2.68	13.72
8 – ½" DC	8.06	21.78
XOS	0.98	22.76
5" HWDP	93.16	115.92
Total Length (m)		**115.92**
Drilling & Mud Properties		
WOB (lb)		-
RPM		-
Torque (klb)		-
SPP (psi)		2600 – 2700
Flow (gpm)		675
Mw (pcf)		83.00
Tool Description	**Length (m)**	**Cum. Length (m)**

BHA No. #18		
12 – ¼” PDC bit	0.36	0.36
Bit sub	1.11	1.47
8 – ½” DC	17.77	19.24
S. stab	2.01	21.25
8 – ½” DC	62.81	84.06
8” mech D. Jar C/W flex jt	10.17	94.23
8 – ½” DC	16.95	111.18
XOS	0.98	112.16
5” HWDP	93.16	205.32
Total Length (m)		**205.32**
Drilling & Mud Properties		
WOB (lb)		5 – 15
RPM		120 – 170
Torque (klb)		7 – 13
SPP (psi)		2500 – 2700
Flow (gpm)		610 – 640
Mw (pcf)		84.00
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #19		
8 – ½” TCI bit	0.23	0.23

6 – ½" Bit Sub	1.23	1.46
6 – ½" DC	17.74	19.20
8 – ½" S. stab	1.54	20.74
6 – ½" DC	90.30	111.04
6 – ½" mech D. Jar	7.42	118.46
6 – ½" DC	27.46	145.92
5" HWDP	83.73	229.65
Total Length (m)		**229.65**
Drilling & Mud Properties		
WOB (lb)		5 – 18
RPM		50 – 60
Torque (klb)		2 – 8
SPP (psi)		400 – 1200
Flow (gpm)		250 – 400
Mw (pcf)		85.00
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #20		
8 – ½" TCI bit	0.23	0.23
6 – ½" Junk sub	0.92	1.15
5" pup jt	4.61	5.76
6 – ½" D. pin	0.67	6.43

9 – 5/8” scraper	1.39	7.82
6 – ½” bit sub (float in)	1.23	9.05
5” HWDP	55.85	64.90
6 – ½” D. Jar	7.20	72.10
5” HWDP	27.88	99.98
Total Length (m)		**99.98**
Drilling & Mud Properties		
WOB (lb)		5 – 18
RPM		50 – 60
Torque (klb)		2 – 8
SPP (psi)		400 – 1200
Flow (gpm)		250 – 400
Mw (pcf)		85.00
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #21		
8 – ½” PDC bit	0.30	0.30
Bit Sub	1.10	1.40
6 – ½” DC	17.74	19.14
S. stab	1.54	20.68
6 – ½” DC	90.30	110.98
6 – ½” mech D. Jar C/W flex jt	7.37	118.35

6 – ½” DC	27.46	145.81
5” HWDP	83.73	229.54
Total Length (m)		**229.54**
Drilling & Mud Properties		
WOB (lb)		10 – 20
RPM		100 – 170
Torque (klb)		4 – 7
SPP (psi)		2500 – 2600
Flow (gpm)		450
Mw (pcf)		101.00
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #22		
8 – ½” TCI bit	0.23	0.23
6 – ½” Bit Sub (float in)	1.10	1.33
6 – ½” DC	135.50	136.83
5” HWDP	83.73	220.56
Total Length (m)		**220.56**
Drilling & Mud Properties		
WOB (lb)		-
RPM		-
Torque (klb)		-

SPP (psi)		-
Flow (gpm)		-
Mw (pcf)		104.00
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #23		
5 – 7/8” MT bit	0.18	0.18
4 – ¾” bit sub (float in)	0.95	1.13
4 – ¾” DC	110.64	111.77
3 – ½” HWDP	83.72	195.49
3 – ½” DP	292.43	487.92
Total Length (m)		**487.92**
Drilling & Mud Properties		
WOB (lb)		-
RPM		-
Torque (klb)		-
SPP (psi)		-
Flow (gpm)		-
Mw (pcf)		104.00
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #24		
5 – 7/8” MT bit	0.18	0.18

7” scraper	1.01	1.19
4 – ¾” bit sub (float in)	0.95	2.14
4 – ¾” DC	110.64	112.78
3 – ½” HWDP	83.72	196.50
3 – ½” DP	146.19	342.69
XOS	0.87	343.56
5” DP	9.55	353.11
XOS	0.67	353.78
9 – 5/8” scraper	1.39	355.17
6 – ½” bit sub	1.10	356.27
Total Length (m)		**356.27**
Drilling & Mud Properties		
WOB (lb)		-
RPM		-
Torque (klb)		-
SPP (psi)		-
Flow (gpm)		-
Mw (pcf)		104.00
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #25		
5” B. bit	1.26	1.26

5” anchor pipe	1.53	2.79
XOS	0.40	3.19
7 – 4/5” packer	2.09	5.28
XOS	0.40	5.68
XOS	0.40	6.08
5” S. jt, 5” Hydraulic Jar	1.73	7.81
5” middle BT	1.25	9.06
5” hydrospring tester valve, 5” DCIp	3.20	12.26
XOS	0.25	12.51
5” DP, 5” R. sub	9.96	22.47
6 – ½” DC	135.50	157.97
Total Length (m)		**157.97**
Drilling & Mud Properties		
WOB (lb)		-
RPM		-
Torque (klb)		-
SPP (psi)		-
Flow (gpm)		-
Mw (pcf)		104.00
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #26		

7 – 5/16” polish mill	0.55	0.55
6 – ½” D. pin	0.66	1.21
6 – ½” bit sub (float in)	1.10	2.31
6 – ½” DC	53.51	55.82
5” HWDp	55.85	111.67
Total Length (m)		**111.67**
Drilling & Mud Properties		
WOB (lb)		-
RPM		-
Torque (klb)		-
SPP (psi)		-
Flow (gpm)		-
Mw (pcf)		104.00
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #27		
SN packer Assy.	5.21	5.21
6 – ½” DC	108.04	113.25
5” HWDP	83.73	196.98
Total Length (m)		**196.98**
Drilling & Mud Properties		
WOB (lb)		-

RPM	-
Torque (klb)	-
SPP (psi)	-
Flow (gpm)	-
Mw (pcf)	104.00

Tool Description	Length (m)	Cum. Length (m)
BHA No. #28		
5” B. bit	1.26	1.26
5” anchor pipe	1.53	2.79
XOS	0.40	3.19
7 – 4/5” packer	2.09	5.28
XOS	0.40	5.68
XOS	0.40	6.08
5” S. jt, 5” Hydraulic jar	1.73	7.81
5” middle BT	1.25	9.06
5” hydrospring tester valve, 5” DCIP	3.20	12.26
XOS	0.25	12.51
5” DP, 5” R. sub	9.96	22.47
6 – ½” DC	108.04	130.51
Total Length (m)		**130.51**
Drilling & Mud Properties		

WOB (lb)	-
RPM	-
Torque (klb)	-
SPP (psi)	-
Flow (gpm)	-
Mw (pcf)	104.00

Tool Description	Length (m)	Cum. Length (m)
BHA No. #29		
5 – 7/8" M.M. Junk mill	0.93	0.93
4 – ¾" bit sub (float in)	0.95	1.88
4 – ¾" DC	110.64	112.52
3 – ½" HWDP	83.72	196.24
3 – ½" DP	292.43	488.67
Total Length (m)		**488.67**

Drilling & Mud Properties	
WOB (lb)	-
RPM	-
Torque (klb)	-
SPP (psi)	-
Flow (gpm)	-
Mw (pcf)	104.00

Tool Description	Length (m)	Cum. Length (m)
BHA No. #30		
5 – 7/8" M.M. Junk mill	0.93	0.93
4 – ¾" bit sub (float in)	0.95	1.88
4 – ¾" DC	110.64	112.52
3 – ½" HWDP	83.72	196.24
3 – ½" DP	292.43	488.67
Total Length (m)		**488.67**
Drilling & Mud Properties		
WOB (lb)		-
RPM		-
Torque (klb)		-
SPP (psi)		-
Flow (gpm)		-
Mw (pcf)		104.00
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #31		
4 – ¾" DC	82.95	82.95
3 – ½" HWDP	55.74	138.69
D. pin	0.32	139.01
7" scraper	1.00	140.01

4 – ¾" it sub (float in)	0.95	140.96
3 – ½" HWDP	27.98	168.94
3 – ½" DP	292.43	461.37
XOS	0.87	462.24
Total Length (m)		**462.24**
Drilling & Mud Properties		
WOB (lb)		-
RPM		-
Torque (klb)		-
SPP (psi)		-
Flow (gpm)		-
Mw (pcf)		102.5
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #32		
7 – 5/8" Overshot	1.18	1.18
6 – ½" DC	17.80	18.98
6 – ½" fishing Jar	2.75	21.73
6 – ½" DC	9.22	30.95
5" HWDP	28.14	59.09
Total Length (m)		**59.09**
Drilling & Mud Properties		

WOB (lb)	-
RPM	-
Torque (klb)	-
SPP (psi)	-
Flow (gpm)	-
Mw (pcf)	101.5

Tool Description	Length (m)	Cum. Length (m)
BHA No. #33		
5 – 7/8" M.M Junk mill	0.93	0.93
4 – ¾" bit sub (float in)	1.23	2.16
4 – ¾" DC	81.55	83.71
3 – ½" HWDP	83.62	167.33
XOS	0.87	168.20
Total Length (m)		**168.20**

Drilling & Mud Properties	
WOB (lb)	-
RPM	-
Torque (klb)	-
SPP (psi)	-
Flow (gpm)	-
Mw (pcf)	102.00

Tool Description	Length (m)	Cum. Length (m)
BHA No. #34		
7 – 5/16” dress mill	0.55	0.55
D. pin	0.67	1.22
Bit sub	1.17	2.39
6 – ½” DC	27.02	29.41
5” HWDP	55.81	85.22
Total Length (m)		**85.22**
Drilling & Mud Properties		
WOB (lb)		-
RPM		-
Torque (klb)		-
SPP (psi)		-
Flow (gpm)		-
Mw (pcf)		103.00
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #35		
7 – 5/16” dress mill	0.55	0.55
D. pin	0.67	1.22
Bit sub	1.17	2.39
6 – ½” DC	27.02	29.41

5” HWDP	55.81	85.22
Total Length (m)		**85.22**
Drilling & Mud Properties		
WOB (lb)		-
RPM		-
Torque (klb)		-
SPP (psi)		-
Flow (gpm)		-
Mw (pcf)		103.00
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #36		
5 – 7/8” M.M Junk mill	0.93	0.93
4 – ¾” bit sub (float in)	1.23	2.16
4 – ¾” DC	54.48	56.64
3 – ½” HWDP	83.72	140.36
3 – ½” DP	292.43	432.79
Total Length (m)		**432.79**
Drilling & Mud Properties		
WOB (lb)		-
RPM		-
Torque (klb)		-

SPP (psi)		-
Flow (gpm)		-
Mw (pcf)		103.00
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #37		
5 – 7/8” M.M Junk mill	0.93	0.93
7” scraper	1.00	1.93
4 – ¾” bit sub (float in)	1.23	3.16
4 – ¾” DC	82.95	86.11
3 – ½” HWDP	83.72	169.83
3 – ½” DP	292.43	462.26
Total Length (m)		**462.26**
Drilling & Mud Properties		
WOB (lb)		-
RPM		-
Torque (klb)		-
SPP (psi)		-
Flow (gpm)		-
Mw (pcf)		103.00
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #38		

Mule shoe	1.99	1.99
7” packer	1.49	3.48
S. jt	1.17	4.65
XOS	0.26	4.91
4 – ¾” DC	27.54	32.45
XOS	0.26	32.71
C. valve	2.24	34.95
XOS	0.30	35.25
3 – ½” HWDP	83.72	118.97
3 – ½” DP	292.43	411.40
Total Length (m)		**411.40**
Drilling & Mud Properties		
WOB (lb)		-
RPM		-
Torque (klb)		-
SPP (psi)		-
Flow (gpm)		-
Mw (pcf)		103.00
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #39		
4 – ¾” DC	9.02	9.02

D. pin	0.32	9.34
7” scraper	1.00	10.34
Bit sub (float in)	1.23	11.57
4 – ¾” DC	101.62	113.19
3 – ½” HWDP	83.72	196.91
3 – ½” DP	292.43	489.34
Total Length (m)		**489.34**

Drilling & Mud Properties	
WOB (lb)	-
RPM	-
Torque (klb)	-
SPP (psi)	-
Flow (gpm)	-
Mw (pcf)	103.00

6. Casing Information

6.1 9-5/8" Casing

MSA

Field :	Azadegan	Well NO:	AZNS-040	Rig NO:	208 OEOC	Drlg CO:	OEOC

9-5/8" Casing

Grade	ID "	Drift "	Weight lb/ft	Burst psi	Collaspse psi	Tensile Strenght Klbs	Coupling OD in	TQ ft x lb	Nominal Dia. OD
TP110SS	8.535	8.5	53.5	10,900	7,950	1,710	10.625	17170-19080-20990	9 5/8

Total Depth :	3934.0 m	Formation Name :	Gadvan	Hook Weight :	60 k lb
Shoe Depth :	3933.5	Formation Top :	3696	STRG CAP (bbl):	914
Float Collar :	3911.00			Mud Weight (pcf) :	85

Number on jt	Grade	Order No.	Jt Length (m)	Cum Length (m)	Dist to RT (m)	M/D weight (Klbs)	STRG CAP. (bbl)	Remarks	
								Centralizers	IN / OUT
					3,933.00	60.00	0.00		
Float Shoe	Float Shoe 9-5/8", 53.5#,C-110, TPCQ (PPA)		0.44	0.44	3,932.56	60.07	0.08		
B1	9-5/8" CSG, 53.5#,TP110SS, TPCQ	1	10.84	11.18	3,921.82	61.66	2.60	BOW	
A1	9-5/8" CSG, 53.5#,TP110SS, TPCQ	2	10.52	21.70	3,911.30	63.21	5.04	BOW	
Float Collar	Float Collar 9-5/8", 53.5#,C-110, C110 (PPA)		0.30	22.00	3,911.00	63.27	5.11		
A2	9-5/8" CSG, 53.5#,TP110SS, TPCQ	3	11.14	33.14	3,899.86	64.91	7.70		
A3	9-5/8" CSG, 53.5#,TP110SS, TPCQ	4	11.15	44.29	3,888.71	66.55	10.28	FU	
A4	9-5/8" CSG, 53.5#,TP110SS, TPCQ	5	11.06	55.34	3,877.66	68.17	12.85		
A5	9-5/8" CSG, 53.5#,TP110SS, TPCQ	6	11.07	66.41	3,866.59	69.80	15.42		
A6	9-5/8" CSG, 53.5#,TP110SS, TPCQ	7	11.07	77.47	3,855.53	71.42	17.99	BOW	
A7	9-5/8" CSG, 53.5#,TP110SS, TPCQ	8	10.77	88.24	3,844.76	73.00	20.49		
A8	9-5/8" CSG, 53.5#,TP110SS, TPCQ	9	10.62	98.86	3,834.15	74.56	22.95		
A9	9-5/8" CSG, 53.5#,TP110SS, TPCQ	10	10.54	109.39	3,823.61	76.11	25.40		
A10	9-5/8" CSG, 53.5#,TP110SS, TPCQ	11	10.88	120.27	3,812.73	77.71	27.92		
A11	9-5/8" CSG, 53.5#,TP110SS, TPCQ	12	10.81	131.07	3,801.93	79.30	30.43	BOW	
A12	9-5/8" CSG, 53.5#,TP110SS, TPCQ	13	10.64	141.72	3,791.28	80.86	32.90	FU	
A13	9-5/8" CSG, 53.5#,TP110SS, TPCQ	14	10.70	152.42	3,780.58	82.43	35.39		
1	9-5/8" CSG, 53.5#,TP110SS, TPCQ	15	10.66	163.08	3,769.92	84.00	37.86		
2	9-5/8" CSG, 53.5#,TP110SS, TPCQ	16	10.42	173.50	3,759.50	85.53	40.28		
3	9-5/8" CSG, 53.5#,TP110SS, TPCQ	17	10.00	183.50	3,749.50	87.00	42.60	BOW	
4	9-5/8" CSG, 53.5#,TP110SS, TPCQ	18	10.55	194.05	3,738.95	88.55	45.06		
5	9-5/8" CSG, 53.5#,TP110SS, TPCQ	19	10.67	204.72	3,728.28	90.12	47.53		
6	9-5/8" CSG, 53.5#,TP110SS, TPCQ	20	11.05	215.77	3,717.23	91.74	50.10		
7	9-5/8" CSG, 53.5#,TP110SS, TPCQ	21	11.04	226.81	3,706.19	93.37	52.66		
8	9-5/8" CSG, 53.5#,TP110SS, TPCQ	22	11.14	237.95	3,695.05	95.00	55.25	BOW	
9	9-5/8" CSG, 53.5#,TP110SS, TPCQ	23	11.14	249.09	3,683.91	96.64	57.83	Break Circ	
10	9-5/8" CSG, 53.5#,TP110SS, TPCQ	24	11.16	260.25	3,672.75	98.28	60.42		
11	9-5/8" CSG, 53.5#,TP110SS, TPCQ	25	11.14	271.39	3,661.61	99.91	63.01		
12	9-5/8" CSG, 53.5#,TP110SS, TPCQ	26	11.07	282.46	3,650.54	101.54	65.58		
13	9-5/8" CSG, 53.5#,TP110SS, TPCQ	27	11.06	293.51	3,639.49	103.16	68.15	BOW	
14	9-5/8" CSG, 53.5#,TP110SS, TPCQ	28	11.06	304.57	3,628.43	104.79	70.72		
15	9-5/8" CSG, 53.5#,TP110SS, TPCQ	29	10.64	315.21	3,617.79	106.35	73.19		
16	9-5/8" CSG, 53.5#,TP110SS, TPCQ	30	10.85	326.06	3,606.94	107.95	75.71		
17	9-5/8" CSG, 53.5#,TP110SS, TPCQ	31	11.16	337.22	3,595.78	109.58	78.30		
18	9-5/8" CSG, 53.5#,TP110SS, TPCQ	32	11.15	348.37	3,584.64	111.22	80.88	BOW	
19	9-5/8" CSG, 53.5#,TP110SS, TPCQ	33	11.14	359.50	3,573.50	112.86	83.47	FU	
20	9-5/8" CSG, 53.5#,TP110SS, TPCQ	34	11.14	370.64	3,562.36	114.49	86.06		
21	9-5/8" CSG, 53.5#,TP110SS, TPCQ	35	11.15	381.80	3,551.20	116.13	88.65		
22	9-5/8" CSG, 53.5#,TP110SS, TPCQ	36	11.06	392.86	3,540.15	117.76	91.21		
23	9-5/8" CSG, 53.5#,TP110SS, TPCQ	37	10.48	403.34	3,529.66	119.30	93.65	BOW	
24	9-5/8" CSG, 53.5#,TP110SS, TPCQ	38	10.56	413.90	3,519.10	120.85	96.10		
25	9-5/8" CSG, 53.5#,TP110SS, TPCQ	39	10.83	424.73	3,508.27	122.44	98.61		
26	9-5/8" CSG, 53.5#,TP110SS, TPCQ	40	10.86	435.59	3,497.41	124.04	101.14		
27	9-5/8" CSG, 53.5#,TP110SS, TPCQ	41	10.67	446.27	3,486.73	125.61	103.62		
28	9-5/8" CSG, 53.5#,TP110SS, TPCQ	42	10.76	457.02	3,475.98	127.19	106.11	BOW	
29	9-5/8" CSG, 53.5#,TP110SS, TPCQ	43	10.75	467.77	3,465.23	128.77	108.61	Break Circ	
30	9-5/8" CSG, 53.5#,TP110SS, TPCQ	44	11.06	478.84	3,454.16	130.39	111.18		
31	9-5/8" CSG, 53.5#,TP110SS, TPCQ	45	11.07	489.90	3,443.10	132.02	113.75		
32	9-5/8" CSG, 53.5#,TP110SS, TPCQ	46	11.07	500.97	3,432.03	133.64	116.32		
33	9-5/8" CSG, 53.5#,TP110SS, TPCQ	47	11.04	512.01	3,420.99	135.26	118.88	BOW	
34	9-5/8" CSG, 53.5#,TP110SS, TPCQ	48	11.06	523.07	3,409.93	136.89	121.45		
35	9-5/8" CSG, 53.5#,TP110SS, TPCQ	49	11.04	534.10	3,398.90	138.51	124.01		
36	9-5/8" CSG, 53.5#,TP110SS, TPCQ	50	11.04	545.14	3,387.86	140.13	126.57		
37	9-5/8" CSG, 53.5#,TP110SS, TPCQ	51	10.67	555.80	3,377.20	141.70	129.05		
38	9-5/8" CSG, 53.5#,TP110SS, TPCQ	52	11.16	566.96	3,366.04	143.34	131.64	BOW	
39	9-5/8" CSG, 53.5#,TP110SS, TPCQ	53	11.15	578.11	3,354.89	144.97	134.23	FU	
40	9-5/8" CSG, 53.5#,TP110SS, TPCQ	54	10.48	588.59	3,344.41	146.51	136.66		
41	9-5/8" CSG, 53.5#,TP110SS, TPCQ	55	11.05	599.64	3,333.36	148.14	139.23		
42	9-5/8" CSG, 53.5#,TP110SS, TPCQ	56	11.06	610.70	3,322.30	149.76	141.79		
43	9-5/8" CSG, 53.5#,TP110SS, TPCQ	57	10.68	621.38	3,311.62	151.33	144.27	BOW	
44	9-5/8" CSG, 53.5#,TP110SS, TPCQ	58	11.05	632.43	3,300.57	152.95	146.84		
45	9-5/8" CSG, 53.5#,TP110SS, TPCQ	59	10.42	642.85	3,290.15	154.49	149.26		
46	9-5/8" CSG, 53.5#,TP110SS, TPCQ	60	10.58	653.43	3,279.57	156.04	151.71		
47	9-5/8" CSG, 53.5#,TP110SS, TPCQ	61	10.67	664.10	3,268.90	157.61	154.19		
48	9-5/8" CSG, 53.5#,TP110SS, TPCQ	62	10.53	674.63	3,258.37	159.16	156.64	BOW	
49	9-5/8" CSG, 53.5#,TP110SS, TPCQ	63	10.46	685.09	3,247.91	160.69	159.07	Break Circ	
50	9-5/8" CSG, 53.5#,TP110SS, TPCQ	64	10.87	695.97	3,237.03	162.29	161.59		
51	9-5/8" CSG, 53.5#,TP110SS, TPCQ	65	11.05	707.02	3,225.98	163.92	164.16		
52	9-5/8" CSG, 53.5#,TP110SS, TPCQ	66	11.05	718.07	3,214.93	165.54	166.72		
53	9-5/8" CSG, 53.5#,TP110SS, TPCQ	67	11.05	729.12	3,203.88	167.16	169.29	BOW	
54	9-5/8" CSG, 53.5#,TP110SS, TPCQ	68	11.05	740.18	3,192.83	168.79	171.86		
55	9-5/8" CSG, 53.5#,TP110SS, TPCQ	69	10.52	750.70	3,182.30	170.33	174.30		
56	9-5/8" CSG, 53.5#,TP110SS, TPCQ	70	11.11	761.81	3,171.19	171.96	176.88		
57	9-5/8" CSG, 53.5#,TP110SS, TPCQ	71	11.16	772.97	3,160.03	173.60	179.47		
58	9-5/8" CSG, 53.5#,TP110SS, TPCQ	72	11.05	784.02	3,148.98	175.23	182.04	BOW	
59	9-5/8" CSG, 53.5#,TP110SS, TPCQ	73	11.05	795.07	3,137.93	176.85	184.60	FU	
60	9-5/8" CSG, 53.5#,TP110SS, TPCQ	74	10.72	805.79	3,127.21	178.43	187.09		
61	9-5/8" CSG, 53.5#,TP110SS, TPCQ	75	11.06	816.85	3,116.15	180.05	189.66		
62	9-5/8" CSG, 53.5#,TP110SS, TPCQ	76	11.06	827.91	3,105.09	181.68	192.23		
63	9-5/8" CSG, 53.5#,TP110SS, TPCQ	77	10.70	838.62	3,094.38	183.25	194.71	BOW	
64	9-5/8" CSG, 53.5#,TP110SS, TPCQ	78	11.07	849.68	3,083.32	184.87	197.28		

65	9-5/8" CSG, 53.5#,TP110SS, TPCQ	79	10.94	860.63	3,072.38	186.48	199.82		
66	9-5/8" CSG, 53.5#,TP110SS, TPCQ	80	11.15	871.77	3,061.23	188.12	202.41		
67	9-5/8" CSG, 53.5#,TP110SS, TPCQ	81	10.97	882.74	3,050.26	189.73	204.96		
68	9-5/8" CSG, 53.5#,TP110SS, TPCQ	82	11.06	893.80	3,039.20	191.35	207.53	BOW	
69	9-5/8" CSG, 53.5#,TP110SS, TPCQ	83	11.06	904.87	3,028.14	192.98	210.09	Break Circ	
70	9-5/8" CSG, 53.5#,TP110SS, TPCQ	84	10.70	915.57	3,017.44	194.55	212.58		
71	9-5/8" CSG, 53.5#,TP110SS, TPCQ	85	11.04	926.61	3,006.39	196.17	215.14		
72	9-5/8" CSG, 53.5#,TP110SS, TPCQ	86	11.06	937.67	2,995.33	197.80	217.71		
73	9-5/8" CSG, 53.5#,TP110SS, TPCQ	87	11.07	948.73	2,984.27	199.42	220.28	BOW	
75	9-5/8" CSG, 53.5#,TP110SS, TPCQ	88	11.06	959.79	2,973.21	201.05	222.85		
76	9-5/8" CSG, 53.5#,TP110SS, TPCQ	89	10.70	970.49	2,962.51	202.62	225.33		
77	9-5/8" CSG, 53.5#,TP110SS, TPCQ	90	10.91	981.41	2,951.59	204.22	227.87		
78	9-5/8" CSG, 53.5#,TP110SS, TPCQ	91	11.05	992.46	2,940.54	205.85	230.43		
79	9-5/8" CSG, 53.5#,TP110SS, TPCQ	92	10.66	1,003.12	2,929.89	207.41	232.91	BOW	
80	9-5/8" CSG, 53.5#,TP110SS, TPCQ	93	11.06	1,014.18	2,918.82	209.04	235.47	FIU	
81	9-5/8" CSG, 53.5#,TP110SS, TPCQ	94	11.16	1,025.34	2,907.67	210.68	238.06		
82	9-5/8" CSG, 53.5#,TP110SS, TPCQ	95	11.06	1,036.39	2,896.61	212.30	240.63		
83	9-5/8" CSG, 53.5#,TP110SS, TPCQ	96	11.06	1,047.45	2,885.55	213.93	243.20		
84	9-5/8" CSG, 53.5#,TP110SS, TPCQ	97	11.05	1,058.49	2,874.51	215.55	245.76	BOW	
85	9-5/8" CSG, 53.5#,TP110SS, TPCQ	98	11.06	1,069.55	2,863.45	217.17	248.33		
86	9-5/8" CSG, 53.5#,TP110SS, TPCQ	99	10.49	1,080.04	2,852.96	218.71	250.77		
87	9-5/8" CSG, 53.5#,TP110SS, TPCQ	100	10.46	1,090.50	2,842.50	220.25	253.20		
88	9-5/8" CSG, 53.5#,TP110SS, TPCQ	101	11.06	1,101.57	2,831.44	221.88	255.76		
89	9-5/8" CSG, 53.5#,TP110SS, TPCQ	102	11.03	1,112.60	2,820.41	223.50	258.32	BOW	
90	9-5/8" CSG, 53.5#,TP110SS, TPCQ	103	11.06	1,123.66	2,809.35	225.12	260.89	Break Circ	
91	9-5/8" CSG, 53.5#,TP110SS, TPCQ	104	10.58	1,134.24	2,798.77	226.68	263.35		
92	9-5/8" CSG, 53.5#,TP110SS, TPCQ	105	10.59	1,144.82	2,788.18	228.23	265.81		
93	9-5/8" CSG, 53.5#,TP110SS, TPCQ	106	10.94	1,155.77	2,777.23	229.84	268.35		
94	9-5/8" CSG, 53.5#,TP110SS, TPCQ	107	10.81	1,166.57	2,766.43	231.43	270.86	BOW	
95	9-5/8" CSG, 53.5#,TP110SS, TPCQ	108	11.16	1,177.73	2,755.27	233.07	273.45		
96	9-5/8" CSG, 53.5#,TP110SS, TPCQ	109	11.05	1,188.78	2,744.22	234.69	276.01		
97	9-5/8" CSG, 53.5#,TP110SS, TPCQ	110	11.05	1,199.83	2,733.17	236.31	278.58		
98	9-5/8" CSG, 53.5#,TP110SS, TPCQ	111	10.68	1,210.51	2,722.49	237.88	281.06		
99	9-5/8" CSG, 53.5#,TP110SS, TPCQ	112	11.15	1,221.65	2,711.35	239.52	283.65	BOW	
100	9-5/8" CSG, 53.5#,TP110SS, TPCQ	113	11.06	1,232.71	2,700.29	241.14	286.21	FIU	
101	9-5/8" CSG, 53.5#,TP110SS, TPCQ	114	11.06	1,243.78	2,689.22	242.77	288.78		
102	9-5/8" CSG, 53.5#,TP110SS, TPCQ	115	11.06	1,254.84	2,678.16	244.39	291.35		
103	9-5/8" CSG, 53.5#,TP110SS, TPCQ	116	11.05	1,265.88	2,667.12	246.02	293.92		
104	9-5/8" CSG, 53.5#,TP110SS, TPCQ	117	11.06	1,276.94	2,656.06	247.64	296.48	BOW	
105	9-5/8" CSG, 53.5#,TP110SS, TPCQ	118	11.06	1,288.00	2,645.00	249.27	299.05		
106	9-5/8" CSG, 53.5#,TP110SS, TPCQ	119	11.06	1,299.06	2,633.94	250.89	301.62		
107	9-5/8" CSG, 53.5#,TP110SS, TPCQ	120	11.05	1,310.11	2,622.89	252.51	304.18		
108	9-5/8" CSG, 53.5#,TP110SS, TPCQ	121	11.06	1,321.17	2,611.83	254.14	306.75		
109	9-5/8" CSG, 53.5#,TP110SS, TPCQ	122	10.67	1,331.84	2,601.16	255.70	309.23	BOW	
110	9-5/8" CSG, 53.5#,TP110SS, TPCQ	123	11.16	1,342.99	2,590.01	257.34	311.82	Break Circ	
111	9-5/8" CSG, 53.5#,TP110SS, TPCQ	124	10.78	1,353.77	2,579.23	258.93	314.32		
112	9-5/8" CSG, 53.5#,TP110SS, TPCQ	125	11.16	1,364.93	2,568.07	260.57	316.91		
113	9-5/8" CSG, 53.5#,TP110SS, TPCQ	126	10.80	1,375.73	2,557.27	262.15	319.42		
114	9-5/8" CSG, 53.5#,TP110SS, TPCQ	127	10.64	1,386.37	2,546.64	263.72	321.89	BOW	
115	9-5/8" CSG, 53.5#,TP110SS, TPCQ	128	11.05	1,397.42	2,535.58	265.34	324.46		
116	9-5/8" CSG, 53.5#,TP110SS, TPCQ	129	11.06	1,408.48	2,524.52	266.96	327.02		
117	9-5/8" CSG, 53.5#,TP110SS, TPCQ	130	10.93	1,419.42	2,513.59	268.57	329.56		
118	9-5/8" CSG, 53.5#,TP110SS, TPCQ	131	11.05	1,430.46	2,502.54	270.19	332.13		
119	9-5/8" CSG, 53.5#,TP110SS, TPCQ	-	11.24	1,430.46	OUT	270.19	332.13		OUT
120	9-5/8" CSG, 53.5#,TP110SS, TPCQ	132	10.59	1,441.05	2,491.95	271.75	334.59	BOW	
121	9-5/8" CSG, 53.5#,TP110SS, TPCQ	133	10.78	1,451.83	2,481.17	273.33	337.09	FIU	
122	9-5/8" CSG, 53.5#,TP110SS, TPCQ	134	11.05	1,462.88	2,470.12	274.96	339.66		
123	9-5/8" CSG, 53.5#,TP110SS, TPCQ	135	10.89	1,473.78	2,459.23	276.56	342.18		
124	9-5/8" CSG, 53.5#,TP110SS, TPCQ	136	11.16	1,484.94	2,448.07	278.20	344.78		
125	9-5/8" CSG, 53.5#,TP110SS, TPCQ	137	11.05	1,495.99	2,437.02	279.82	347.34	BOW	
126	9-5/8" CSG, 53.5#,TP110SS, TPCQ	138	10.91	1,506.89	2,426.11	281.42	349.87		
127	9-5/8" CSG, 53.5#,TP110SS, TPCQ	-	11.26	1,506.89	OUT	281.42	349.87		OUT
128	9-5/8" CSG, 53.5#,TP110SS, TPCQ	139	10.96	1,517.86	2,415.14	283.03	352.42		
129	9-5/8" CSG, 53.5#,TP110SS, TPCQ	140	10.48	1,528.34	2,404.66	284.57	354.85		
130	9-5/8" CSG, 53.5#,TP110SS, TPCQ	141	10.76	1,539.09	2,393.91	286.15	357.35		
131	9-5/8" CSG, 53.5#,TP110SS, TPCQ	142	11.15	1,550.24	2,382.76	287.79	359.94	BOW	
132	9-5/8" CSG, 53.5#,TP110SS, TPCQ	143	11.14	1,561.38	2,371.62	289.43	362.52	Break Circ	
133	9-5/8" CSG, 53.5#,TP110SS, TPCQ	144	10.97	1,572.35	2,360.65	291.04	365.07		
134	9-5/8" CSG, 53.5#,TP110SS, TPCQ	145	10.32	1,582.67	2,350.33	292.56	367.47		
135	9-5/8" CSG, 53.5#,TP110SS, TPCQ	146	10.76	1,593.43	2,339.57	294.14	369.97		
136	9-5/8" CSG, 53.5#,TP110SS, TPCQ	-	11.74	1,593.43	OUT	294.14	369.97		OUT
137	9-5/8" CSG, 53.5#,TP110SS, TPCQ	147	11.14	1,604.57	2,328.43	295.77	372.55	BOW	
138	9-5/8" CSG, 53.5#,TP110SS, TPCQ	148	11.14	1,615.71	2,317.29	297.41	375.14		
139	9-5/8" CSG, 53.5#,TP110SS, TPCQ	149	11.04	1,626.76	2,306.25	299.03	377.70		
140	9-5/8" CSG, 53.5#,TP110SS, TPCQ	150	11.06	1,637.82	2,295.19	300.66	380.27		
141	9-5/8" CSG, 53.5#,TP110SS, TPCQ	151	11.06	1,648.88	2,284.13	302.28	382.84		
142	9-5/8" CSG, 53.5#,TP110SS, TPCQ	152	10.96	1,659.84	2,273.16	303.89	385.38	BOW	
143	9-5/8" CSG, 53.5#,TP110SS, TPCQ	153	10.41	1,670.25	2,262.75	305.42	387.90	FIU	
144	9-5/8" CSG, 53.5#,TP110SS, TPCQ	154	10.46	1,680.71	2,252.29	306.96	390.23		
145	9-5/8" CSG, 53.5#,TP110SS, TPCQ	155	11.15	1,691.86	2,241.15	308.60	392.82		
146	9-5/8" CSG, 53.5#,TP110SS, TPCQ	156	11.14	1,703.00	2,230.00	310.23	395.41		
147	9-5/8" CSG, 53.5#,TP110SS, TPCQ	157	11.07	1,714.07	2,218.93	311.86	397.98	BOW	
148	9-5/8" CSG, 53.5#,TP110SS, TPCQ	158	11.05	1,725.12	2,207.88	313.48	400.54		
149	9-5/8" CSG, 53.5#,TP110SS, TPCQ	159	11.16	1,736.27	2,196.73	315.12	403.13		
150	9-5/8" CSG, 53.5#,TP110SS, TPCQ	160	11.06	1,747.33	2,185.67	316.74	405.70		
151	9-5/8" CSG, 53.5#,TP110SS, TPCQ	161	11.16	1,758.49	2,174.51	318.38	408.29		
152	9-5/8" CSG, 53.5#,TP110SS, TPCQ	162	11.05	1,769.53	2,163.47	320.01	410.85	BOW	
153	9-5/8" CSG, 53.5#,TP110SS, TPCQ	163	11.06	1,780.59	2,152.41	321.63	413.42	Break Circ	
154	9-5/8" CSG, 53.5#,TP110SS, TPCQ	164	11.06	1,791.65	2,141.35	323.26	415.99		
155	9-5/8" CSG, 53.5#,TP110SS, TPCQ	165	11.14	1,802.79	2,130.21	324.89	418.58		
156	9-5/8" CSG, 53.5#,TP110SS, TPCQ	166	11.16	1,813.95	2,119.05	326.53	421.17		
157	9-5/8" CSG, 53.5#,TP110SS, TPCQ	167	10.97	1,824.92	2,108.08	328.14	423.71	BOW	
158	9-5/8" CSG, 53.5#,TP110SS, TPCQ	168	10.68	1,835.60	2,097.40	329.71	426.19		
159	9-5/8" CSG, 53.5#,TP110SS, TPCQ	169	11.06	1,846.65	2,086.35	331.33	428.76		
160	9-5/8" CSG, 53.5#,TP110SS, TPCQ	170	10.88	1,857.53	2,075.47	332.93	431.28		
161	9-5/8" CSG, 53.5#,TP110SS, TPCQ	171	11.16	1,868.68	2,064.32	334.57	433.88		
162	9-5/8" CSG, 53.5#,TP110SS, TPCQ	172	10.73	1,879.41	2,053.59	336.15	436.37	BOW	
163	9-5/8" CSG, 53.5#,TP110SS, TPCQ	173	11.07	1,890.48	2,042.52	337.77	438.94	FIU	

164	9-5/8" CSG, 53.5#,TP110SS, TPCQ	174	11.15	1,901.62	2,031.38	339.41	441.52		
165	9-5/8" CSG, 53.5#,TP110SS, TPCQ	175	10.40	1,912.02	2,020.98	340.94	443.94		
166	9-5/8" CSG, 53.5#,TP110SS, TPCQ	-	11.26	1,912.02	OUT	340.94	443.94		OUT
167	9-5/8" CSG, 53.5#,TP110SS, TPCQ	176	11.04	1,923.06	2,009.94	342.56	446.50		
168	9-5/8" CSG, 53.5#,TP110SS, TPCQ	177	11.15	1,934.21	1,998.79	344.20	449.09	BOW	
169	9-5/8" CSG, 53.5#,TP110SS, TPCQ	178	10.43	1,944.64	1,988.36	345.73	451.51		
170	9-5/8" CSG, 53.5#,TP110SS, TPCQ	179	9.93	1,954.57	1,978.43	347.19	453.82		
171	9-5/8" CSG, 53.5#,TP110SS, TPCQ	180	10.46	1,965.03	1,967.97	348.73	456.25		
172	9-5/8" CSG, 53.5#,TP110SS, TPCQ	181	11.07	1,976.10	1,956.90	350.35	458.81		
173	9-5/8" CSG, 53.5#,TP110SS, TPCQ	182	11.06	1,987.16	1,945.84	351.98	461.38	BOW	
174	9-5/8" CSG, 53.5#,TP110SS, TPCQ	183	10.75	1,997.91	1,935.09	353.56	463.88	Break Circ	
175	9-5/8" CSG, 53.5#,TP110SS, TPCQ	184	11.05	2,008.96	1,924.04	355.18	466.44		
176	9-5/8" CSG, 53.5#,TP110SS, TPCQ	185	11.15	2,020.10	1,912.90	356.82	469.03		
177	9-5/8" CSG, 53.5#,TP110SS, TPCQ	186	11.15	2,031.25	1,901.75	358.46	471.62		
178	9-5/8" CSG, 53.5#,TP110SS, TPCQ	187	11.15	2,042.40	1,890.60	360.09	474.21	BOW	
179	9-5/8" CSG, 53.5#,TP110SS, TPCQ	188	10.88	2,053.28	1,879.72	361.69	476.74		
180	9-5/8" CSG, 53.5#,TP110SS, TPCQ	189	10.89	2,064.17	1,868.83	363.29	479.26		
181	9-5/8" CSG, 53.5#,TP110SS, TPCQ	190	11.14	2,075.31	1,857.69	364.93	481.85		
182	9-5/8" CSG, 53.5#,TP110SS, TPCQ	191	10.69	2,086.00	1,847.00	366.50	484.33		
183	9-5/8" CSG, 53.5#,TP110SS, TPCQ	192	11.05	2,097.04	1,835.96	368.12	486.90	BOW	
184	9-5/8" CSG, 53.5#,TP110SS, TPCQ	193	10.19	2,107.23	1,825.77	369.62	489.26	F/U	
185	9-5/8" CSG, 53.5#,TP110SS, TPCQ	194	10.39	2,117.61	1,815.39	371.15	491.67		
186	9-5/8" CSG, 53.5#,TP110SS, TPCQ	195	10.31	2,127.92	1,805.08	372.66	494.07		
187	9-5/8" CSG, 53.5#,TP110SS, TPCQ	196	10.77	2,138.69	1,794.32	374.24	496.56		
188	9-5/8" CSG, 53.5#,TP110SS, TPCQ	197	11.05	2,149.74	1,783.26	375.87	499.13	BOW	
189	9-5/8" CSG, 53.5#,TP110SS, TPCQ	198	11.05	2,160.78	1,772.22	377.49	501.70		
190	9-5/8" CSG, 53.5#,TP110SS, TPCQ	199	11.06	2,171.85	1,761.16	379.11	504.26		
191	9-5/8" CSG, 53.5#,TP110SS, TPCQ	200	10.63	2,182.48	1,750.52	380.68	506.73		
192	9-5/8" CSG, 53.5#,TP110SS, TPCQ	201	11.02	2,193.50	1,739.50	382.30	509.29		
193	9-5/8" CSG, 53.5#,TP110SS, TPCQ	202	11.07	2,204.57	1,728.44	383.92	511.86	BOW	
194	9-5/8" CSG, 53.5#,TP110SS, TPCQ	203	11.01	2,215.58	1,717.43	385.54	514.42	Break Circ	
195	9-5/8" CSG, 53.5#,TP110SS, TPCQ	204	11.15	2,226.73	1,706.28	387.18	517.01		
196	9-5/8" CSG, 53.5#,TP110SS, TPCQ	205	11.16	2,237.88	1,695.12	388.81	519.60		
197	9-5/8" CSG, 53.5#,TP110SS, TPCQ	206	11.05	2,248.93	1,684.07	390.44	522.16		
198	9-5/8" CSG, 53.5#,TP110SS, TPCQ	207	10.46	2,259.39	1,673.61	391.98	524.59	BOW	
199	9-5/8" CSG, 53.5#,TP110SS, TPCQ	208	10.90	2,270.29	1,662.71	393.58	527.12		
200	9-5/8" CSG, 53.5#,TP110SS, TPCQ	209	11.16	2,281.45	1,651.55	395.22	529.71		
201	9-5/8" CSG, 53.5#,TP110SS, TPCQ	210	11.16	2,292.61	1,640.39	396.85	532.30		
202	9-5/8" CSG, 53.5#,TP110SS, TPCQ	211	11.06	2,303.68	1,629.32	398.48	534.87		
203	9-5/8" CSG, 53.5#,TP110SS, TPCQ	212	11.04	2,314.72	1,618.28	400.10	537.44	BOW	
204	9-5/8" CSG, 53.5#,TP110SS, TPCQ	213	11.16	2,325.88	1,607.13	401.74	540.03	F/U	
205	9-5/8" CSG, 53.5#,TP110SS, TPCQ	214	11.16	2,337.03	1,595.97	403.38	542.62		
206	9-5/8" CSG, 53.5#,TP110SS, TPCQ	215	11.15	2,348.18	1,584.82	405.02	545.21		
207	9-5/8" CSG, 53.5#,TP110SS, TPCQ	216	11.04	2,359.22	1,573.78	406.64	547.77	BOW	
208	9-5/8" CSG, 53.5#,TP110SS, TPCQ	217	11.07	2,370.29	1,562.71	408.26	550.34		
209	9-5/8" CSG, 53.5#,TP110SS, TPCQ	218	10.85	2,381.13	1,551.87	409.86	552.86		
210	9-5/8" CSG, 53.5#,TP110SS, TPCQ	219	10.96	2,392.10	1,540.90	411.47	555.40		
211	9-5/8" CSG, 53.5#,TP110SS, TPCQ	220	10.97	2,403.06	1,529.94	413.08	557.95		
212	9-5/8" CSG, 53.5#,TP110SS, TPCQ	221	10.96	2,414.02	1,518.98	414.69	560.49	BOW	
213	9-5/8" CSG, 53.5#,TP110SS, TPCQ	222	11.04	2,425.06	1,507.94	416.31	563.06	Break Circ	
214	9-5/8" CSG, 53.5#,TP110SS, TPCQ	223	11.06	2,436.13	1,496.88	417.94	565.63		
215	9-5/8" CSG, 53.5#,TP110SS, TPCQ	224	11.06	2,447.19	1,485.81	419.56	568.19	F/U Tools, Change Elevator	
216	9-5/8" CSG, 53.5#,TP110SS, TPCQ	225	10.97	2,458.16	1,474.84	421.17	570.74	Rigid	
217	9-5/8" CSG, 53.5#,TP110SS, TPCQ	226	9.77	2,467.93	1,465.08	422.61	573.01		
218	9-5/8" CSG, 53.5#,TP110SS, TPCQ	227	9.54	2,477.46	1,455.54	424.01	575.22		
219	9-5/8" CSG, 53.5#,TP110SS, TPCQ	228	11.16	2,488.62	1,444.39	425.65	577.81		
220	9-5/8" CSG, 53.5#,TP110SS, TPCQ	229	11.05	2,499.67	1,433.34	427.27	580.38		
221	9-5/8" CSG, 53.5#,TP110SS, TPCQ	230	10.92	2,510.58	1,422.42	428.88	582.91		
222	9-5/8" CSG, 53.5#,TP110SS, TPCQ	231	11.06	2,521.64	1,411.36	430.50	585.48		
223	9-5/8" CSG, 53.5#,TP110SS, TPCQ	232	10.96	2,532.60	1,400.40	432.11	588.03		
224	9-5/8" CSG, 53.5#,TP110SS, TPCQ	233	11.06	2,543.67	1,389.33	433.74	590.59		
225	9-5/8" CSG, 53.5#,TP110SS, TPCQ	234	10.31	2,553.98	1,379.02	435.25	592.99		
226	9-5/8" CSG, 53.5#,TP110SS, TPCQ	235	10.30	2,564.28	1,368.72	436.77	595.38	Rigid	
227	9-5/8" CSG, 53.5#,TP110SS, TPCQ	236	11.05	2,575.33	1,357.67	438.39	597.95	F/U	
228	9-5/8" CSG, 53.5#,TP110SS, TPCQ	237	10.83	2,586.16	1,346.84	439.98	600.46		
229	9-5/8" CSG, 53.5#,TP110SS, TPCQ	238	10.80	2,596.95	1,336.05	441.57	602.97		
230	9-5/8" CSG, 53.5#,TP110SS, TPCQ	239	11.06	2,608.01	1,324.99	443.19	605.53		
231	9-5/8" CSG, 53.5#,TP110SS, TPCQ	240	11.06	2,619.07	1,313.93	444.82	608.10		
232	9-5/8" CSG, 53.5#,TP110SS, TPCQ	241	11.06	2,630.13	1,302.87	446.44	610.67		
233	9-5/8" CSG, 53.5#,TP110SS, TPCQ	242	11.16	2,641.29	1,291.71	448.08	613.26		
234	9-5/8" CSG, 53.5#,TP110SS, TPCQ	243	11.16	2,652.44	1,280.56	449.72	615.85		
235	9-5/8" CSG, 53.5#,TP110SS, TPCQ	244	11.16	2,663.60	1,269.40	451.36	618.44		
236	9-5/8" CSG, 53.5#,TP110SS, TPCQ	245	11.05	2,674.64	1,258.36	452.98	621.00	Rigid	
237	9-5/8" CSG, 53.5#,TP110SS, TPCQ	246	11.04	2,685.68	1,247.32	454.60	623.57	Break Circ	
238	9-5/8" CSG, 53.5#,TP110SS, TPCQ	247	11.15	2,696.83	1,236.17	456.24	626.16		
239	9-5/8" CSG, 53.5#,TP110SS, TPCQ	248	11.00	2,707.83	1,225.17	457.85	628.71		
240	9-5/8" CSG, 53.5#,TP110SS, TPCQ	249	11.16	2,718.99	1,214.01	459.49	631.30		
241	9-5/8" CSG, 53.5#,TP110SS, TPCQ	250	10.82	2,729.81	1,203.19	461.08	633.81		
242	9-5/8" CSG, 53.5#,TP110SS, TPCQ	251	11.05	2,740.86	1,192.14	462.71	636.38		
243	9-5/8" CSG, 53.5#,TP110SS, TPCQ	252	11.06	2,751.92	1,181.08	464.33	638.95		
244	9-5/8" CSG, 53.5#,TP110SS, TPCQ	253	11.06	2,762.98	1,170.02	465.96	641.52		
245	9-5/8" CSG, 53.5#,TP110SS, TPCQ	254	11.05	2,774.04	1,158.96	467.58	644.08		
246	9-5/8" CSG, 53.5#,TP110SS, TPCQ	255	10.70	2,784.73	1,148.27	469.15	646.57	Rigid	
247	9-5/8" CSG, 53.5#,TP110SS, TPCQ	256	10.33	2,795.06	1,137.94	470.67	648.96	F/U	
248	9-5/8" CSG, 53.5#,TP110SS, TPCQ	257	11.06	2,806.12	1,126.88	472.29	651.53		
249	9-5/8" CSG, 53.5#,TP110SS, TPCQ	258	11.06	2,817.18	1,115.82	473.92	654.10		
250	9-5/8" CSG, 53.5#,TP110SS, TPCQ	259	10.55	2,827.73	1,105.27	475.47	656.55		
251	9-5/8" CSG, 53.5#,TP110SS, TPCQ	260	10.85	2,838.58	1,094.42	477.06	659.07		
252	9-5/8" CSG, 53.5#,TP110SS, TPCQ	261	11.04	2,849.62	1,083.38	478.69	661.63		
253	9-5/8" CSG, 53.5#,TP110SS, TPCQ	262	11.16	2,860.78	1,072.22	480.33	664.22		
254	9-5/8" CSG, 53.5#,TP110SS, TPCQ	263	11.06	2,871.84	1,061.17	481.95	666.79		
255	9-5/8" CSG, 53.5#,TP110SS, TPCQ	264	10.94	2,882.78	1,050.22	483.56	669.33		
256	9-5/8" CSG, 53.5#,TP110SS, TPCQ	265	10.49	2,893.26	1,039.74	485.10	671.76	Rigid	
257	9-5/8" CSG, 53.5#,TP110SS, TPCQ	266	10.43	2,903.69	1,029.31	486.63	674.19	Break Circ	
258	9-5/8" CSG, 53.5#,TP110SS, TPCQ	267	10.97	2,914.66	1,018.34	488.24	676.73		
259	9-5/8" CSG, 53.5#,TP110SS, TPCQ	268	11.00	2,925.66	1,007.35	489.86	679.29		
260	9-5/8" CSG, 53.5#,TP110SS, TPCQ	269	10.99	2,936.65	996.35	491.47	681.84		
261	9-5/8" CSG, 53.5#,TP110SS, TPCQ	270	10.74	2,947.38	985.62	493.05	684.33		

262	9-5/8" CSG, 53.5#,TP110SS, TPCQ	271	10.88	2,958.26	974.74	494.65	686.86		
263	9-5/8" CSG, 53.5#,TP110SS, TPCQ	272	10.90	2,969.16	963.84	496.25	689.39		
264	9-5/8" CSG, 53.5#,TP110SS, TPCQ	273	11.07	2,980.23	952.77	497.87	691.96		
265	9-5/8" CSG, 53.5#,TP110SS, TPCQ	274	11.16	2,991.39	941.61	499.51	694.55		
266	9-5/8" CSG, 53.5#,TP110SS, TPCQ	275	11.06	3,002.45	930.55	501.14	697.12	Rigid	
267	9-5/8" CSG, 53.5#,TP110SS, TPCQ	276	11.05	3,013.50	919.50	502.76	699.68	FIU	
268	9-5/8" CSG, 53.5#,TP110SS, TPCQ	277	11.06	3,024.56	908.44	504.39	702.25		
269	9-5/8" CSG, 53.5#,TP110SS, TPCQ	278	10.93	3,035.49	897.51	505.99	704.79		
270	9-5/8" CSG, 53.5#,TP110SS, TPCQ	279	11.06	3,046.55	886.45	507.62	707.36		
271	9-5/8" CSG, 53.5#,TP110SS, TPCQ	280	10.68	3,057.24	875.76	509.19	709.84		
272	9-5/8" CSG, 53.5#,TP110SS, TPCQ	281	10.42	3,067.66	865.34	510.72	712.26		
273	9-5/8" CSG, 53.5#,TP110SS, TPCQ	282	11.04	3,078.70	854.30	512.34	714.82		
274	9-5/8" CSG, 53.5#,TP110SS, TPCQ	283	11.05	3,089.75	843.25	513.96	717.39		
275	9-5/8" CSG, 53.5#,TP110SS, TPCQ	284	10.39	3,100.14	832.86	515.49	719.80		
276	9-5/8" CSG, 53.5#,TP110SS, TPCQ	285	10.06	3,110.20	822.80	516.97	722.13	Rigid	
277	9-5/8" CSG, 53.5#,TP110SS, TPCQ	286	10.05	3,120.25	812.75	518.45	724.47	Break Circ	
278	9-5/8" CSG, 53.5#,TP110SS, TPCQ	287	10.35	3,130.60	802.40	519.97	726.87		
279	9-5/8" CSG, 53.5#,TP110SS, TPCQ	288	11.15	3,141.75	791.25	521.61	729.46		
280	9-5/8" CSG, 53.5#,TP110SS, TPCQ	289	11.14	3,152.89	780.11	523.24	732.04		
281	9-5/8" CSG, 53.5#,TP110SS, TPCQ	290	11.14	3,164.03	768.97	524.88	734.63		
282	9-5/8" CSG, 53.5#,TP110SS, TPCQ	291	11.04	3,175.07	757.93	526.50	737.20		
283	9-5/8" CSG, 53.5#,TP110SS, TPCQ	292	11.15	3,186.22	746.78	528.14	739.78		
284	9-5/8" CSG, 53.5#,TP110SS, TPCQ	293	10.38	3,196.61	736.39	529.67	742.20		
285	9-5/8" CSG, 53.5#,TP110SS, TPCQ	294	10.72	3,207.33	725.67	531.24	744.69		
286	9-5/8" CSG, 53.5#,TP110SS, TPCQ	295	10.68	3,218.02	714.98	532.81	747.17	Rigid	
287	9-5/8" CSG, 53.5#,TP110SS, TPCQ	296	10.46	3,228.47	704.53	534.35	749.59	FIU	
288	9-5/8" CSG, 53.5#,TP110SS, TPCQ	297	10.79	3,239.26	693.74	535.93	752.10		
289	9-5/8" CSG, 53.5#,TP110SS, TPCQ	298	11.04	3,250.29	682.71	537.56	754.66		
290	9-5/8" CSG, 53.5#,TP110SS, TPCQ	299	11.05	3,261.34	671.66	539.18	757.23		
291	9-5/8" CSG, 53.5#,TP110SS, TPCQ	300	10.88	3,272.22	660.78	540.78	759.75		
292	9-5/8" CSG, 53.5#,TP110SS, TPCQ	301	11.05	3,283.27	649.73	542.40	762.32		
293	9-5/8" CSG, 53.5#,TP110SS, TPCQ	302	10.62	3,293.89	639.11	543.96	764.78		
294	9-5/8" CSG, 53.5#,TP110SS, TPCQ	303	11.04	3,304.93	628.07	545.58	767.35		
295	9-5/8" CSG, 53.5#,TP110SS, TPCQ	304	11.04	3,315.98	617.02	547.21	769.91		
296	9-5/8" CSG, 53.5#,TP110SS, TPCQ	305	11.03	3,327.00	606.00	548.83	772.47	Rigid	
297	9-5/8" CSG, 53.5#,TP110SS, TPCQ	306	11.06	3,338.06	594.94	550.45	775.04	Break Circ	
298	9-5/8" CSG, 53.5#,TP110SS, TPCQ	307	11.07	3,349.13	583.87	552.08	777.61		
299	9-5/8" CSG, 53.5#,TP110SS, TPCQ	308	11.04	3,360.17	572.83	553.70	780.17		
300	9-5/8" CSG, 53.5#,TP110SS, TPCQ	309	10.74	3,370.91	562.09	555.28	782.67		
301	9-5/8" CSG, 53.5#,TP110SS, TPCQ	310	11.05	3,381.96	551.04	556.90	785.23		
302	9-5/8" CSG, 53.5#,TP110SS, TPCQ	311	11.15	3,393.11	539.89	558.54	787.82		
303	9-5/8" CSG, 53.5#,TP110SS, TPCQ	312	11.07	3,404.17	528.83	560.16	790.39		
304	9-5/8" CSG, 53.5#,TP110SS, TPCQ	313	11.06	3,415.23	517.77	561.79	792.96		
305	9-5/8" CSG, 53.5#,TP110SS, TPCQ	314	11.06	3,426.29	506.71	563.41	795.52		
306	9-5/8" CSG, 53.5#,TP110SS, TPCQ	315	11.06	3,437.35	495.65	565.03	798.09	Rigid	
307	9-5/8" CSG, 53.5#,TP110SS, TPCQ	316	11.06	3,448.41	484.59	566.66	800.66	FIU	
308	9-5/8" CSG, 53.5#,TP110SS, TPCQ	317	11.06	3,459.47	473.53	568.28	803.23		
309	9-5/8" CSG, 53.5#,TP110SS, TPCQ	318	10.81	3,470.27	462.73	569.87	805.74		
310	9-5/8" CSG, 53.5#,TP110SS, TPCQ	319	10.68	3,480.96	452.04	571.44	808.22		
311	9-5/8" CSG, 53.5#,TP110SS, TPCQ	320	10.97	3,491.93	441.07	573.05	810.76		
312	9-5/8" CSG, 53.5#,TP110SS, TPCQ	321	10.82	3,502.75	430.25	574.64	813.28		
313	9-5/8" CSG, 53.5#,TP110SS, TPCQ	322	10.85	3,513.60	419.40	576.24	815.80		
314	9-5/8" CSG, 53.5#,TP110SS, TPCQ	323	10.85	3,524.45	408.55	577.83	818.32		
315	9-5/8" CSG, 53.5#,TP110SS, TPCQ	324	10.58	3,535.03	397.97	579.39	820.77		
316	9-5/8" CSG, 53.5#,TP110SS, TPCQ	325	10.87	3,545.90	387.10	580.98	823.30	Rigid	
317	9-5/8" CSG, 53.5#,TP110SS, TPCQ	326	10.95	3,556.85	376.15	582.59	825.84	Break Circ	
318	9-5/8" CSG, 53.5#,TP110SS, TPCQ	327	11.06	3,567.90	365.10	584.21	828.40		
319	9-5/8" CSG, 53.5#,TP110SS, TPCQ	328	11.06	3,578.96	354.04	585.84	830.97		
320	9-5/8" CSG, 53.5#,TP110SS, TPCQ	329	11.06	3,590.02	342.98	587.46	833.54		
321	9-5/8" CSG, 53.5#,TP110SS, TPCQ	330	10.66	3,600.68	332.32	589.03	836.01		
322	9-5/8" CSG, 53.5#,TP110SS, TPCQ	331	11.13	3,611.81	321.19	590.67	838.60		
323	9-5/8" CSG, 53.5#,TP110SS, TPCQ	332	10.47	3,622.28	310.72	592.20	841.03		
324	9-5/8" CSG, 53.5#,TP110SS, TPCQ	333	11.04	3,633.33	299.67	593.83	843.59		
325	9-5/8" CSG, 53.5#,TP110SS, TPCQ	334	11.06	3,644.39	288.62	595.45	846.16		
326	9-5/8" CSG, 53.5#,TP110SS, TPCQ	335	10.73	3,655.12	277.88	597.03	848.65	Rigid	
327	9-5/8" CSG, 53.5#,TP110SS, TPCQ	336	10.66	3,665.78	267.23	598.59	851.13	FIU	
328	9-5/8" CSG, 53.5#,TP110SS, TPCQ	337	10.49	3,676.27	256.73	600.14	853.56		
329	9-5/8" CSG, 53.5#,TP110SS, TPCQ	338	10.61	3,686.88	246.12	601.70	856.03		
330	9-5/8" CSG, 53.5#,TP110SS, TPCQ	-	11.14	3,686.88	OUT	601.70	856.03		OUT
331	9-5/8" CSG, 53.5#,TP110SS, TPCQ	-	11.15	3,686.88	OUT	601.70	856.03		OUT
332	9-5/8" CSG, 53.5#,TP110SS, TPCQ	-	11.15	3,686.88	OUT	601.70	856.03		OUT
333	9-5/8" CSG, 53.5#,TP110SS, TPCQ	339	11.07	3,697.94	235.06	603.32	858.60		
334	9-5/8" CSG, 53.5#,TP110SS, TPCQ	340	11.06	3,709.01	223.99	604.95	861.17		
335	9-5/8" CSG, 53.5#,TP110SS, TPCQ	341	11.00	3,720.01	212.99	606.56	863.72		
336	9-5/8" CSG, 53.5#,TP110SS, TPCQ	342	11.06	3,731.07	201.93	608.19	866.29		
337	9-5/8" CSG, 53.5#,TP110SS, TPCQ	343	10.62	3,741.68	191.32	609.75	868.75		
338	9-5/8" CSG, 53.5#,TP110SS, TPCQ	344	10.48	3,752.16	180.84	611.29	871.19		
339	9-5/8" CSG, 53.5#,TP110SS, TPCQ	345	10.64	3,762.80	170.20	612.85	873.65	Rigid	
340	9-5/8" CSG, 53.5#,TP110SS, TPCQ	346	10.62	3,773.42	159.58	614.41	876.12	Break Circ	
341	9-5/8" CSG, 53.5#,TP110SS, TPCQ	347	10.94	3,784.36	148.64	616.02	878.66		
342	9-5/8" CSG, 53.5#,TP110SS, TPCQ	-	11.14	3,784.36	OUT	616.02	878.66		OUT
343	9-5/8" CSG, 53.5#,TP110SS, TPCQ	348	11.12	3,795.48	137.52	617.65	881.24		
344	9-5/8" CSG, 53.5#,TP110SS, TPCQ	349	11.13	3,806.62	126.39	619.29	883.83		
345	9-5/8" CSG, 53.5#,TP110SS, TPCQ	-	11.14	3,806.62	OUT	619.29	883.83		OUT
346	9-5/8" CSG, 53.5#,TP110SS, TPCQ	350	11.03	3,817.64	115.36	620.91	886.39		
347	9-5/8" CSG, 53.5#,TP110SS, TPCQ	351	11.03	3,828.67	104.33	622.53	888.95		
348	9-5/8" CSG, 53.5#,TP110SS, TPCQ	352	10.88	3,839.55	93.45	624.13	891.48		
349	9-5/8" CSG, 53.5#,TP110SS, TPCQ	353	10.88	3,850.43	82.57	625.72	894.00		
350	9-5/8" CSG, 53.5#,TP110SS, TPCQ	354	11.02	3,861.45	71.55	627.34	896.56		
351	9-5/8" CSG, 53.5#,TP110SS, TPCQ	355	11.13	3,872.59	60.42	628.98	899.15	Rigid	
352	9-5/8" CSG, 53.5#,TP110SS, TPCQ	356	11.02	3,883.61	49.40	630.60	901.70	FIU	
353	9-5/8" CSG, 53.5#,TP110SS, TPCQ	357	11.01	3,894.62	38.38	632.21	904.26		
354	9-5/8" CSG, 53.5#,TP110SS, TPCQ	358	11.02	3,905.64	27.36	633.83	906.82	Rigid	
355	9-5/8" CSG, 53.5#,TP110SS, TPCQ	359	11.05	3,916.68	16.32	635.46	909.38	Rigid	
356	9-5/8" CSG, 53.5#,TP110SS, TPCQ	360	11.03	3,927.71	5.29	637.08	911.95		
74	9-5/8" CSG, 53.5#,TP110SS, TPCQ	361	8.73	3,936.44	-3.44	638.36	913.97	Landing Joint (#74)	

Figure 6-1: AZNS-040 9-5/8" CSG Tally

South Azadegan Oilfield Project

Casing Report

General Data

FIELD	AZNS	WELL NO.	040/F-21	LOCATION	194,235.000		N
RIG	208	WSS	S.Jahani / A.Eyvazi		3,450,715.000		E
DATE	1402.06.18	HOLE DEPTH	3934 m MD		MUD WT.	85	PCF
HOLE SIZE	12-1/4" inches		3934 m TVD		MUD TYPE	S.KCl.Poly.Lst Mud	

	DEPTH metres	O.D. in	WEIGHT lb/ft	GRADE	CONNECTION TYPE	COMMENTS
PREVIOUS CASING	1492.0	13-3/8"	68#	N-80	BTC	

CASING STRING COMPOSITION

Qty.	O.D. in	I.D.	ITEM DESCRIPTION / MANUFACTURER	Wt. lb/ft	GRADE	CONNECTION	Make Up Torque	LENGTH m	DEPTH m MD
1	9-5/8"	8.535	FLOAT SHOE (PPA)	53.5	C-110	TPCQ	17170-19080-20990	0.480	3932.66
1	9-5/8"	8.535	9-5/8" CSG	53.5	TP110SS	TPCQ	17170-19080-20990	10.980	3921.82
1	9-5/8"	8.535	9-5/8" CSG	53.5	TP110SS	TPCQ	17170-19080-20990	10.663	3911.30
1	9-5/8"	8.535	FLOAT COLLAR (PPA)	53.5	C-110	TPCQ	17170-19080-20990	0.440	3911.00
359	9-5/8"	8.535	9-5/8" CSG	53.5	TP110SS	TPCQ	17170-19080-20990	3964.701	-3.44

FINAL SHOE DEPTH	3933.00 m		CASING WEIGHT IN AIR	700	klbs
TOTAL LENGTH OF CSG RUN	3929.56 m		CASING WEIGHT IN MUD	580	klbs
LENGTH OF CSG ABOVE RT	-3.44 m		WEIGHT SET ON HANGER	200	klbs

PERFORMANCE	DATE/TIME
STARTED RUNNING CSG	1402.06.12 -18:00
FINISHED RUNNING CSG	1402.06.14 - 17:00
TOTAL TIME (hours)	47
DOWNTIME (hours)	-

REASON(S) FOR DOWNTIME:

CASING RUNNING COMPANY: OEOC

CENTRALISER / SCRATCHER PLACEMENT

MANUFACTURER	PPA CO
TYPE	BOW / RIGIDTYPE
PLACEMENT	Install as per CSG Tally.

COMMENTS:

HPJSM, P/U 9-5/8" F.Shoe (53.5#, PPA, C110,PDC Drillable) & M/U on 1 Jt 9-5/8" CSG (53.5#, TP110S, TPCQ), 1 Jt 9-5/8" CSG (53.5#, TP110S, TPCQ), 9-5/8" F.Collar (53.5#, PPA, C110, PDC Drillable) & RIH w/ 2 Jts 9-5/8"CSG to 40 m. M/U Circ. Swage on CSG string, Establish Circ. Mud & Check F.Shoe & F.Collar, Ok. Cont. Run 9-5/8"CSG to 3933m (Totally: 361 Jts), F/U String every 10 Jts & Break Circ. Every 20 Jts; 1-5 BPH Losses.
While string stuck @ 3925, 3928 & 3930.8m. Work on stuck; Got free.
Install centralizer as per CSG Tally.

Figure 6-2: AZNS-040 9-5/8" CSG Report

6.2 7" Liner

7″ LNR Tally

Field:	SOUTH AZADEGAN	Well NO:	SAZN-040	Rig NO:	OEOC 208	Drlg CO:	MSA
Client:	PETRO PARS	Job:	7" Liner			Date:	30-Jan-24

Grade	ID "	Drift "	Weight lb/ft	Burst psi	Collaspse psi	Yield klbs	Coupling OD in	TQ ft x lb	Nominal Dia. OD
T110S	6.094	5.969	32	12,457	10,780	1,024	7.717	12470-13860-15250	7

Total Depth :	4200.0 m	Formation Name:	Fahliyan	Hook Weight :	58 k lb
Shoe Depth :	4189.50	Formation Top:	3989 *mt*	STRG CAP (bbl):	260
LDC :	4154.54	Rig Floor to Spool:		Mud Weight (pcf) :	105
L.LNR	3812. 21	0. LAP:	120.79		

Number on jt	Grade	Order No.	Jt Length (m)	Cum Length (m)	Dist to RT (m)	M/D weight (Klbs)	STRG CAP. (bbl)	Remarks	
								Centralizers	IN / OUT
					4,189.400	58.000	0.000		
Float Shoe	7" REAMER F.SHOE (32#,110SS, TPCQ) PPA	-	0.820	0.820	4188.58	58.068	0.097		
1	7" LNR, 32#, TP110S, TPCQ THD	1	11.245	12.065	4,177.335	59.006	1.428	Spirolizer	
2	7" LNR, 32#, TP110S, TPCQ THD	2	11.242	23.307	4166.093	59.944	2.759	Spirolizer	
FLOAT COLLAR	7" F.Collar (32#, 110SS, TPCQ) PPA	-	0.252	23.559	4,165.841	59.976	2.789		
3	7" LNR, 32#, TP110S, TPCQ THD	3	11.050	34.609	4,154.791	60.898	4.097	Spirolizer	
LDC	7" LDC (32#, 110SS, TPCQ) PPA	-	0.252	34.861	4,154.539	60.930	4.126		
4	7" LNR, 32#, TP110S, TPCQ THD	4	11.242	46.103	4,143.297	61.868	5.457		
5	7" LNR, 32#, TP110S, TPCQ THD	5	11.242	57.345	4,132.055	62.806	6.788	Spirolizer	
6	7" LNR, 32#, TP110S, TPCQ THD	6	11.243	68.588	4,120.812	63.744	8.118		
7	7" LNR, 32#, TP110S, TPCQ THD	7	11.241	79.829	4,109.571	64.682	9.449		
8	7" LNR, 32#, TP110S, TPCQ THD	8	11.230	91.059	4,098.341	65.619	10.778	Spirolizer	
9	7" LNR, 32#, TP110S, TPCQ THD	9	11.239	102.298	4,087.102	66.557	12.109		
10	7" LNR, 32#, TP110S, TPCQ THD	10	11.237	113.535	4,075.865	67.495	13.439		
11	7" LNR, 32#, TP110S, TPCQ THD	11	11.248	124.783	4,064.617	68.434	14.770	Spirolizer	
12	7" LNR, 32#, TP110S, TPCQ THD	12	11.250	136.033	4,053.367	69.372	16.102		
13	7" LNR, 32#, TP110S, TPCQ THD	13	11.247	147.280	4,042.120	70.311	17.433		
14	7" LNR, 32#, TP110S, TPCQ THD	14	11.248	158.528	4,030.872	71.249	18.764	Spirolizer	
15	7" LNR, 32#, TP110S, TPCQ THD	15	11.241	169.769	4,019.631	72.187	20.095		
16	7" LNR, 32#, TP110S, TPCQ THD	16	11.250	181.019	4,008.381	73.126	21.426		
17	7" LNR, 32#, TP110S, TPCQ THD	17	11.242	192.261	3,997.139	74.064	22.757	Spirolizer	
18	7" LNR, 32#, TP110S, TPCQ THD	18	11.247	203.508	3,985.892	75.003	24.088		
19	7" LNR, 32#, TP110S, TPCQ THD	19	11.113	214.621	3,974.779	75.930	25.404		
20	7" LNR, 32#, TP110S, TPCQ THD	20	11.244	225.865	3,963.535	76.869	26.735	Spirolizer	
21	7" LNR, 32#, TP110S, TPCQ THD	21	11.245	237.110	3,952.290	77.807	28.066		
22	7" LNR, 32#, TP110S, TPCQ THD	22	11.111	248.221	3,941.179	78.734	29.381		
23	7" LNR, 32#, TP110S, TPCQ THD	23	11.249	259.470	3,929.930	79.673	30.712	Spirolizer	
24	7" LNR, 32#, TP110S, TPCQ THD	24	11.089	270.559	3,918.841	80.598	32.025		
25	7" LNR, 32#, TP110S, TPCQ THD	25	11.245	281.804	3,907.596	81.537	33.356		
26	7" LNR, 32#, TP110S, TPCQ THD	26	11.243	293.047	3,896.353	82.475	34.687		
27	7" LNR, 32#, TP110S, TPCQ THD	27	11.237	304.284	3,885.116	83.413	36.017		
28	7" LNR, 32#, TP110S, TPCQ THD	28	11.242	315.526	3,873.874	84.351	37.348		
29	7" LNR, 32#, TP110S, TPCQ THD	29	11.087	326.613	3,862.787	85.276	38.660		
30	7" LNR, 32#, TP110S, TPCQ THD	30	11.243	337.856	3,851.544	86.214	39.991		
31	7" LNR, 32#, TP110S, TPCQ THD	31	11.249	349.105	3,840.295	87.153	41.322		
32	7" LNR, 32#, TP110S, TPCQ THD	32	10.863	359.968	3,829.432	88.060	42.608		
33	7" LNR, 32#, TP110S, TPCQ THD	33	11.242	371.210	3,818.190	88.998	43.939		
Rotational Hyd HGR	7" Rotational Hyd HGR (PPA)	-	1.820	373.030	3,816.370	89.148	44.154		
POB	7" Pack off bushing (PPA)	-	0.400	373.430	3,815.970	89.181	44.201		
LG & Top packer	7" LG & Top packer (PPA)	-	0.760	374.190	3,815.210	89.244	44.291		
Tie back (PBR)	7" Tie back sleeve (PPA)	-	3.000	377.190	3,812.210	89.491	44.646		
bols neck	setting tools neck	-	2.120	379.310	3,810.090	89.611	44.767		
5" DP STD	133STD + 1 JTS	-	3,818.800	4,198.110	-8.710	307.927	259.909		
5" Pup joint DP	5" Pup joint DP	-	3.100	4,201.210	-11.810	308.104	260.086		

Figure 6-3: AZNS-040 7" CSG Tally

South Azadegan Oilfield Project

Casing Report

General Data

FIELD	AZNS	WELL NO.	040/F-21	LOCATION	194,235.000		N
RIG	208	WSS	A.Zamani		3,450,715.000		E
DATE	1402.06.31	HOLE DEPTH	4200 m MD	MUD WT.	105	PCF	
HOLE SIZE	8-1/2" inches		4200 m TVD	MUD TYPE	S.Poly.Lst Mud		

	DEPTH metres	O.D. in	WEIGHT lb/ft	GRADE	CONNECTION TYPE	COMMENTS
PREVIOUS CASING	3393.0	9-5/8"	53.5	TP110SS / C-110	TPCQ	

LNR STRING COMPOSITION

Qty.	O.D. in	I.D.	ITEM DESCRIPTION / MANUFACTURER	Wt. lb/ft	GRADE	CONNECTION	Make Up Torque	LENGTH m	DEPTH m MD
1	7"	6.094	7" REAMER F.SHOE (PPA)	32	110SS	TPCQ	12470-13860-15250	0.820	4188.75-4189.57
2	7"	6.094	LNR	32	TP110S	TPCQ	12470-13860-15250	22.487	4166.26-4188.75
1	7"	6.094	F.COLLAR (PPA)	32	110SS	TPCQ	12470-13860-15250	0.252	4166.01-4166.26
1	7"	6.094	LNR	32	TP110S	TPCQ	12470-13860-15250	11.237	4154.77-4166.01
1	7"	6.094	LDC (PPA)	32	110SS	TPCQ	12470-13860-15250	0.252	4154.42-4154.77
30	7"	6.094	LNR	32	TP110S	TPCQ	12470-13860-15250	336.349	3818.71-4154.42
1	7"	6.094	Rotating Hyd HGR Assy (PPA)	32	110SS	TPCQ	12470-13860-15250	5.98	3812.2-3818.71

FINAL SHOE DEPTH	4189.57 m		LNR WEIGHT IN AIR	40.1	klbs
TOTAL LENGTH OF LNR RUN	377.38 m		LNR WEIGHT IN MUD	31.5	klbs
7" L.LAP	3812.20 m				klbs

PERFORMANCE	DATE/TIME
STARTED RUNNING LNR	21-Sep-2023, 0:00
FINISHED RUNNING LNR	22-Sep-2023,09:90
TOTAL TIME (hours)	33.5
DOWNTIME (hours)	-

REASON(S) FOR DOWNTIME:

CASING RUNNING COMPANY: OEOC

CENTRALISER / SCRATCHER PLACEMENT

MANUFACTURER	PPA CO
TYPE	7" x 8-1/2" Spirolizer
PLACEMENT	Install as per CSG Tally.

COMMENTS:

P/U & M/U 7" Reamer type F.Shoe (PPA, 32#, 110SS, TPCQ) & RIH w/ 2 Jts 7" LNR (32#, TP110S, TPCQ), 7" F.Collar (PPA, 32#, 110SS, TPCQ) & 1 Jt 7" LNR (32#, TP110S, TPCQ) to 34.8m while Check shoe & collar for float holding; OK. F/U string, P/U & M/U 7" LDC (PPA, 32#, 110SS, TPCQ) & Cont. RIH w/ 30 Jts (Totally 33 Jts) 7" LNR (32#, 110SS, TPCQ) to 371m, while F/U every 7 Jts & Break Circ. every 14 Jts w/ 1-3 BPHL, Install 7" x 8-1/2" Spirolizer as per program. P/U & M/U PPA Rotational Hyd set / Mech release 7" LNR HGR Assy (Tandem cone) include Tie back sleeve (Len: 3m, OD: 8-1/4",ID: 7-3/8") RIH w/ 7" LNR on 5" DP Stds to 3910m (Above 9-5/8" CSG Shoe) while Drift all Stds, F/U & Break Circ every 10 Stds, 1-3 BPHL. Establish Circ as 1.2 x String volume w/ 135 GPM / 500 Psi, 1-3 BPHL. Record P/U & S/O String Wt on M/D: 310/300 Kips. Cont. RIH w/ 7" LNR on 5" DP Stds f/ 3910 to 4033m while Drift all Stds, NL-NG. Took Wt @ 4033m. P/U Stirng, Observed string Stuck. Work on stuck upward to 25 Kips Overpull, String got free, W/D f/ 4033 to 4187m, Took Wt @ 4187m, Try to wash this point, No-Succ, Rotate stirng w/ 10-14 RPM/100-110 GPM/400-500 Psi & W&R to 4190m, Observe No-Progress @ this point. P/U String to 4189.5m. Record P/U & S/O String Wt on M/D: 335/310 Kips, Drop ball, Chase the ball to seat in LDC & Increase Press to 1800 Psi, Hold Press & Set 7" LNR HGR @ 4189.5m by apply 70 Kips Wt, P/U Stirng to 270 Wt on M/D, Increase Press to 2800 Psi, Observed shear ball seat & got return, Release setting tools by rotate 26 round forward, P/U String 0.8m & confirm string be free (Lost 32 Kips of LNR Wt on M/D).

Figure 6-4: AZNS-040 7" CSG Report

7. Cement Program & Reports

7.1 9-5/8" Casing

Cement Report

General Data

FIELD	AZNS	WELL NO.	040/F21	LOCATION	194,235.00	N	
RIG	208	DRILL REP	S.Jahani/A.Eyvazi		3,450,715.00	E	
DATE	1402.06.16	HOLE DEPTH	3,934 m MD	MUD WT.	85	PCF	
HOLE SIZE	12-1/4" inches		3,934 m TVD	MUD TYPE	S.KCl.Poly. Lst Mud		
CEMENTING CONTRACTOR	SPAD Energy CO						

TYPE OF CEMENT JOB (casing, liner, squeeze, plug etc)	CASING	CASING / LINER INFORMATION				
		SHOE DEPTH:	3933.00 m	HANGER:		m
IF PLUGS		DEPTH m	O.D. in	I.D. in	WEIGHT lb/ft	GRADE
CEMENT PLACEMENT TOP	m	3933	9-5/8"	8.535"	53.5#	TP110SS
BOTTOM	m					
CEMENT PLACEMENT TOP	m					
BOTTOM	m					

PREVIOUS CASING	DEPTH m	O.D. in	WEIGHT lb/ft	GRADE	CONNECTION TYPE	COMMENTS
	1522	13-3/8"	68#	N-80	BTC	

Cement Job	VOL bbl	WT. pcf	Cement Sx.	RECIPE	Additive
Chemical Wash	40	62.4		FRESH WATER, CHEMICAL WASH, ANTI-FOAM AGENT	1632 GAL F.WATER ,200 GAL CHEMICAL WASH ,8 GAL ANTI-FOAM AGENT
Spacer	60	90.0		FRESH WATER, CHEMICAL SPACER, NACL, ANTI-FOAM AGENT, SURFACTANT, BARITE	3150 GAL FRESH WATER, 900 LB CHEMICAL SPACER, 11250 LB NACL, 9 GAL ANTI-FOAM AGENT, 180 GAL SURFACTANT, 16650 LB BARITE
First Stage					
Lead Cement Slurry	1154	95	2768	CLASS "G" DELIJAN	11217 LB MICRO SILICA, 4530 LB RETARDER R-02, 1329 LB FLC F-09, 1661 LB DISPERSANT, 90 GAL ANTI-FOAM AGENT
Tail Cement Slurry	116	118	370.00	CLASS "G" DELIJAN + SILICA FLOUR	13570 LB SILICA FLOUR, 541 LB RETARDER R-02,108 LB RETARDER R-12, 216 LB FLC F-05, 541 LB DISPERSANT, 270 LB MICRO SILICA, 108 GAL ANTI-GAS MIGRATION, 16 GAL ANTI-FOAM AGENT
Total Pumped	1270				

Operational Comments / Time Breakdown

OPERATION	TIME / DATE	REASON(S) FOR DOWNTIME:					
L/U CMT LINE & TEST IT	10:00-10:30/1402.06.16	Due to not approved CMT Program; 41 Hrs downtime.					
PUMP CHEMICAL WASH	10:30-10:40/1402.06.16						
PUMP SPACER	10:40-11:00/1402.06.16						
PUMP CMT LEAD	11:00-12:30/1402.06.16	DISPLACEMENT					
PUMP CMT TAIL	12:30-13:00/1402.06.16	RATE bbl/min	VOL bbl	PRESS psi	PLUG BUMP (Yes/No, N/A) Top	Int.	Bttm	FINAL PRESS psi
DISPLACMENT	13:00-15:00/1402.06.16	6-10	905	200-2000				

DOWNTIME	41 hrs	BACKFLOW (Yes/No):	Yes

POST JOB INFORMATION	CEMENT TAGGED @	3905.5 m	CEMENT EXPECTED @	3900 m
	CEMENT DRILLED @	4-6 m/hr	TIME SINCE PLACEMENT	49 hrs

COMMENTS:

Held PJSM, L/U CMT line to CMT head, Press test CMT line to 500/3500 psi; OK. Pump 40 bbl 62.4 pcf Chemical wash, 60 bbl 90 pcf Spacer ahead. Mix & Pump 2768 sxs CMT G Delijan (SV: 1154 bbl, SW: 95pcf) as lead slurry, Mix & Pump 370 sxs CMT G Delijan blended w/ 36.63 lb/sxs Silica Flour (SV: 116 bbl, SW: 118 pcf) as tail slurry, Drop top plug. Displace w/ 908 bbl 85 pcf mud, No observe bumping. Check back flow, not OK. Got 10 bbl Contaminated CMT & 30 bbl Net CMT slurry on return. 156 bbl Loss during pumping CMT & 149 bbl Loss during displace.

Figure 7-1: AZNS-040 9-5/8" CSG Cement Report

Operation Information

Operation Type: 9-5/8" Casing Cementing Job
Open Hole Size: 12-1/4”
Casing Depth: 3933 m
Total Volume
- Pump 40 bbls 62.4 pcf Chemical wash
- Pump 60 bbls 90 pcf spacer
- Mix & pump 1154 bbls 95 pcf Lead Cement
- Mix & pump 116 bbls 118 pcf Tail cement

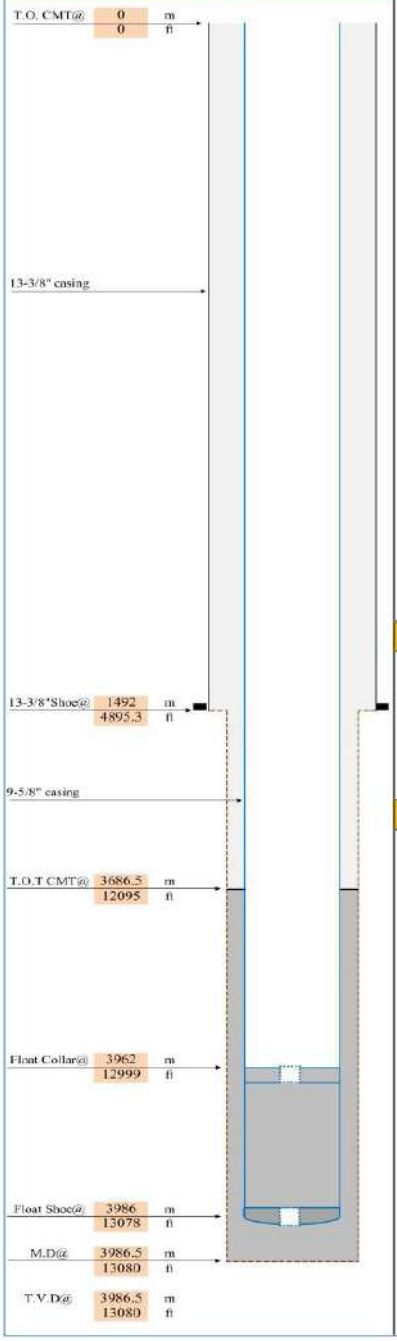
Total Displacement Volume: Displace cement with 908 bbls 85 pcf mud
Slurry Data: Lead Vol. 1154^{bbl} & 95^{pcf}, Tail Vol. 116^{bbl}, 118^{pcf}

Cementing Program

CEMENTING JOB DESIGN AND CALCULATIONS — CM-FO-001

Client	MSA	Rig Name	208 OEOC	Job Type	9-5/8"	REV: 04
Field	Azadegan	Well Name	AZNS-40	Design Date	6-Sep-23	

Well Schematic



Cement Operation Volume

Previous Casing				
OD "	ID "	Weight (ppf)	Grade	Length (m)
13 3/8	12.415	68	N-80	1492

New Casing				
OD "	ID "	Weight (ppf)	Grade	Length (m)
9 5/8	8.535	53.5	110SS	3986

Open Hole	
ID"	Length (m)
12 1/4	2494.5

Excess	
Open Hole	Cased Hole
100%	20%

Volume		
Lead Slurry	1154.18	bbl
Tail Slurry	116.79	bbl
Displacement	919.91	bbl

Temperature

Temperature Gradient	1.42	°F/100ft
BHST	266.0	°F
BHCT	209.0	°F

Pumping Schedule

	DESCRIPTION	Volume (bbl)	Rate (bpm)	Pump	Job Time (min)
1	Chemical Wash	40	5	CMT	8
2	Spacer	60	6	RIG	10
3	Drop Bottom Plug			CMT	5
4	Mix & Pump Lead Cement Slurry	1,154.18	6	CMT	192
5	Mix & Pump Tail Cement Slurry	116.79	6	CMT	19
6	Drop Top Plug			CMT	5
7	Displacement	829.91	6	RIG	138
8	Displacement	90	4	RIG	23

DESCRIPTION	(MIN)
Total Job Time	401
Lead Cementing Time	378
Tail Cementing Time	185

CEMENTING JOB DESIGN AND CALCULATIONS

CM-FO-001 | REV: 04

Client	MSA	Rig Name	208 OEOC	Job Type	9-5/8"
Field	Azadegan	Well Name	AZNS-40	Design Date	6-Sep-23

Mud Removal

Chemical Wash

Material	PDKC Code	Conc.		Qty for Job	
Fresh Water		40.8	gal/bbl	1,632	gal
Chemical Wsh	SE-CW10	5.0	gal/bbl	200	gal
Antifoam Agent	SE-A20	0.2	gal/bbl	8	gal

DV	0	bbl
CW Volume	40	bbl
CW Density	62.4	pcf

Spacer

Material	PDKC Code	Conc.		Qty for Job	
Fresh Water		35	gal/bbl	3150	gal
Chemical Spacer	SE-P03	10	lb/bbl	900	lb
NaCl		125	lb/bbl	11250	lb
Antifoam Agent	SE-A20	0.1	gal/bbl	9	gal
Surfactant	SE-S40	2	gal/bbl	180	gal
Barite		185	lb/bbl	16650	lb

DV	30	bbl
Spacer Volume	90	bbl
Spacer Density	90	pcf
M.W. Density	87	pcf

* If required, control PH between 9.2 - 9.5
* Note: The spacer weight and materials will change with refer to the compatibility test (on Rig Site)

Slurry Design

Lead Cement Slurry

Material	PDKC CODE	Density lb/ft3	S.G	Conc.	UOM	Qty for Job*	
Cement	Class G Delijan	195.94	3.14	110.00	lb/sk	2,770.41	sk
Micro Silica	DS94	70.512	1.13	5.70	lb/sk	17,231.34	lb
Retarder	SE-R02	81.12	1.30	1.25	lb/sk	3,778.80	lb
Fluid Loss Control	SE-F09	94.85	1.52	0.44	lb/sk	1,330.14	lb
Dispersant	SE-D01	81.12	1.30	0.55	lb/sk	1,662.67	lb
Antifoam Agent	SE-A20	62.40	1.00	0.03	gal/sk	90.69	gal

Slurry Parameters		
Slurry Volume	6,480.26	Cu ft
	1,154.18	bbl
Slurry Density	95	pcf
Slurry Yield	2.3391	cu ft/sk
Mix Water	12.4648	gal/sk
Mix Fluid	13.2998	gal/sk
Dead Volume	80	bbl

Mix Water Type	Fresh Water

Mix Water	897.18	bbl
Mix Fluid	957.28	bbl

Tail Cement Slurry

Material	PDKC CODE	Density lb/ft3	S.G	Conc.	UOM	Qty for Job*	
Cement	Class G Delijan	195.94	3.14	110.00	lb/sk	370.48	sk
Silica Flour	FS95	164.11	2.63	36.63	lb/sk	13,570.76	lb
Retarder	SE-R02	81.12	1.30	1.00	lb/sk	541.35	lb
Retarder	SE-R12	64.27	1.03	0.20	lb/sk	108.27	lb
Fluid Loss Control	SE-F05	94.85	1.52	0.40	lb/sk	216.54	lb
Dispersant	SE-D01	81.12	1.30	1.00	lb/sk	541.35	lb
Micro Silica	DS94	70.512	1.13	0.50	lb/sk	270.68	lb
Anti Gas Migration	SE-G10	62.40	1.00	0.20	gal/sk	108.27	gal
Antifoam Agent	SE-A20	62.40	1.00	0.03	gal/sk	16.24	gal

Slurry Parameters		
Slurry Volume	655.75	cu ft
	116.79	bbl
Slurry Density	118	pcf
Slurry Yield	1.7700	cu ft/sk
Mix Water	6.8520	gal/sk
Mix Fluid	7.3741	gal/sk
Dead Volume	30	bbl

Mix Water Type	Fresh Water

Mix Water	88.32	bbl
Mix Fluid	95.05	bbl

Figure 7-2: AZNS-040 9-5/8" Cement Design & Calculation

CEMENTING JOB DESIGN AND CALCULATIONS					CM-FO-001
Client	MSA	Rig Name	208 OEOC	Job Type	9-5/8"
Field	Azadegan	Well Name	AZNS-40	Design Date	6-Sep-23

REV: 04

JOB PROCEDURE

1	Checking equipment and ensuring of proper condition
2	Cleaning mud pits and solution tank and preparing cement additives
3	Taking work permission before end of running the Casing
4	Holding pre job safety meeting & tool box under supervision of HSE officer
5	Filling mud pits with fresh water and start mixing of required additives for making Spacer & Solution based on the program
5_1	Lead solution mixing procedure: Mix Water : **897.18** bbl Add Micro Silica-DS94 to fresh water and allow mixing for 60 min Add Retarder-R02 and allow mixing for 15 min Add F.L.C-F09 and allow mixing for 15 min Add Dispersant-D01 and allow mixing for 15 min Add Anti Foam-A20 and allow mixing for 5 min
5_2	Tail solution mixing procedure: Mix Water : **88.32** bbl Add Micro Silica-DS94 to fresh water and allow mixing for 60 min Add Retarder-R02 and allow mixing for 15 min Add Retarder-R12 and allow mixing for 15 min Add F.L.C-F05 and allow mixing for 15 min Add Dispersant-D01 and allow mixing for 15 min Add Anti Gas-G10 and allow mixing for 15 min Add Anti Foam-A20 and allow mixing for 5 min
6	Transfer Lead solution to the solution tank
7	Setting cement pumps in suitable position
8	Connecting siloes containing dry blended material to cement pumps
9	Connecting solution tank/mud pits to cement pump trucks
10	Starting the circulation of drilling fluid by 1.5 times of well capacity for well cleaning and consider well conditions
11	Start mixing **40** bbl chemical wash in pump truck
12	Line up pipes from cement pumps to cement head 9-5/8"
13	Starting the air compressor and reach to the working pressure of silos (25 psi)
14	Pressure testing at 3000 psi for 10 min.
15	Pumping **40** bbl chemical wash by pump truck
16	Pumping **60** bbl spacer by rig pump
17	Release bottom plug
18	Mixing and Pumping slurry down to the well with cement pump trucks Lead: **1154.18** bbl , Tail: **116.79** bbl
19	Release top plug
20	Displacement slurry with Drilling fluids **829.91** BBL Rate **6** BPM & **90** BBL Rate **4** BPM
21	Landing & Bumping the plug, bleed off pressure and check for float shoe and collar
22	Equipment, pipe lines and tools disassembled
23	End of operation

Figure 7-3: AZNS-040 9-5/8" Cement Job Procedure

Cementing Lab Report

LA-FO-00
Rev.04

Report No:						Field:	Azadegan	Job Type :	9-5/8"
Date:	6-Sep-23		1402/06/15			Client:	MSA	OH Size:	12-1/4" inch
Well Depth:	(TVD):	3986.5	m	13079.7065	ft	Well:	AZNS-40	Last Csg/Liner:	13-3/8 inch
	(MD):	3986.5	m	13079.7065	ft	Rig:	208 OEOC	Mud Weight:	87 pcf

Cementing Additives	PDKC Code	Conc.	Units
Cement	Class "G Delijan	110.00	lb/sk
Micro Silica	DS94	5.70	lb/sk
Retarder	SE-R02	1.25	lb/sk
F.L.C	SE-F09	0.44	lb/sk
Dispersant	SE-D01	0.55	lb/sk
Anti-foam Agent	SE-A20	0.03	gal/sk

Cement and Mixing Water

Cement Type: Class G Delijan

Slurry Density:	12.701	ppg	95	pcf
Slurry Yield:	0.417	bbl/sk	2.3391	cuft/sk
Mixing Water Required:	0.2968	bbl/sk	12.4648	gal/sk
Total Mixing Fluid Required:	0.3167	bbl/sk	13.2998	gal/sk

Temperature	BHCT:	98.3 °C	BHST:	130.0 °C	
Temperature	BHCT:	209 °F	BHST:	266 °F	Temperature Gradient: 1.42 °F / 100 ft

API Free Fluid:	0	%	@ 2 hrs	0 @250 CC		
Fluid Loss:	44	cc/30 min @	209 °F	and	1000 Psi	
API Fluid Loss:	88	cc/30 min @	209 °F	and	1000 Psi	

Rheology Data

°F	°c	300	200	100	60	30	6	3	10 sec Gel	10 min Gel	PV CP	YP lbs/100ft²
80	26.7	33	18.5	14			4.5	3.5	5	18	28.50	4.50
209	98.3	16	12	8			4	3	6	9	12.00	4.00

Pumping Time:	30(BC):	8:27	Hrs:Min	50(BC):	8:37	Hrs:Min	70(BC):	8:45 Hrs:Min

Compressive Strength:	--	Psi	in	24 Hrs @	266 °F	3000 psi
Crash test C.S:	In progress	Psi	in	24 Hrs @	266 °F	3000 psi

Water Annalysis: Chlorides: 1000 mg/liter

PH: Prehydrated DB: Dry Blended % bwoc: % by weight of cement % bwow: % by weight of water

Note: Calc. based on 110 lbs sack of dry cement.

Remarks:

Pilot test data

Function	Code	Concentration		Lab QTY (gr)
Cement	G Delijan	110.00	lb/sk	451.98
Micro Silica	DS94	5.70	lb/sk	23.42
Retarder	SE-R02	1.25	lb/sk	5.14
F.L.C	SE-F09	0.44	lb/sk	1.81
Dispersant	SE-D01	0.55	lb/sk	2.260
Anti-foam Agent	SE-A20	0.03	gal/sk	1.03
Water		12.465	gal/sk	427.420

Lab Volume (ml)	600
Slurry Density (pcf)	95
Yield (cuft/sk)	2.3391
Mix Water (gal/sk)	12.4648
Mix Fluid (gal/sk)	13.2998

Cementing Lab Report

LA-FO-00

Rev.04

Report No:						Field:	Azadegan	Job Type :	9-5/8"
Date:	6-Sep-23		1402/06/15			Client:	MSA	OH Size:	12-1/4" inch
Well Depth:	(TVD):	3986.5	m	13079.7065	ft	Well:	AZNS-40	Last Csg/Liner:	13-3/8 inch
	(MD):	3986.5	m	13079.7065	ft	Rig:	208 OEOC	Mud Weight:	87 pcf

Cementing Additives	PDKC Code	Conc.	Units
Cement	Class "G Delijan	110.00	lb/sk
Silica Flour	FS95	36.63	lb/sk
Retarder	SE-R02	1.00	lb/sk
Retarder	SE-R12	0.20	lb/sk
F.L.C	SE-F05	0.40	lb/sk
Dispersant	SE-D01	1.00	lb/sk
Micro Silica	DS94	0.50	lb/sk
Anti Gas Migration	SE-G10	0.20	gal/sk
Anti-foam Agent	SE-A20	0.03	gal/sk

Cement and Mixing Water

Cement Type: Class G Delijan

Slurry Density:	15.775 ppg	118 pcf
Slurry Yield:	0.315 bbl/sk	1.7700 cuft/sk
Mixing Water Required:	0.1631 bbl/sk	6.8520 gal/sk
Total Mixing Fluid Required:	0.1756 bbl/sk	7.3741 gal/sk

Temperature	BHCT:	98.3 °C	BHST:	130.0 °C	
Temperature	BHCT:	209 °F	BHST:	266 °F	Temperature Gradient: 1.42 °F / 100 ft

API Free Fluid:	0	%	@ 2 hrs	0 @250 CC	
Fluid Loss:	17	cc/30 min	@	209 °F	and 1000 Psi
API Fluid Loss:	34	cc/30 min	@	209 °F	and 1000 Psi

Rheology Data

°F	°C	300	200	100	60	30	6	3	10 sec Gel	10 min Gel	PV CP	YP lbs/100ft²
80	26.7	147	104	59			8	6.5	11	34	132.00	15.00
209	98.3	31	26.5	13.5			2	2	2	6	26.25	4.75

Pumping Time:	30(BC):	4:54	Hrs:Min	50(BC):	5:07	Hrs:Min	70(BC): 5:19 Hrs:Min

Compressive Strength:	4880	Psi	in	24 Hrs @	266 °F	3000 psi
Crash test C.S:	In progress	Psi	in	24 Hrs @	266 °F	3000 psi

Water Annalysis: Chlorides: 1000 mg/liter

PH: Prehydrated DB: Dry Blended % bwoc: % by weight of cement % bwow: % by weight of water

Note: Calc. based on 110 lbs sack of dry cement.

Remarks:

Pilot test data

Function	Code	Concentration		Lab QTY (gr)
Cement	G Delijan	110.00	lb/sk	597.29
Silica Flour	FS95	36.63	lb/sk	198.90
Retarder	SE-R02	1	lb/sk	5.43
Retarder	SE-R12	0.2	lb/sk	1.09
F.L.C	SE-F05	0.4	lb/sk	2.17
Dispersant	SE-D01	1	lb/sk	5.43
Micro Silica	DS94	0.50	lb/sk	2.71
Anti-foam Agent	SE-A20	0.03	gal/sk	1.36
Anti Gas Migration	SE-G10	0.20	gal/sk	9.24
Water		6.85	gal/sk	310.49

Lab Volume (ml)	600
Slurry Density (pcf)	118
Yield (cuft/sk)	1.7700
Mix Water (gal/sk)	6.8520
Mix Fluid (gal/sk)	7.3741

Figure 7-4: AZNS-040 9-5/8" Cement Lab Report

Execution of Operation

Item	Equipment used for the job	Qty
1	Cementing pump truck & related connections (PD-01, PD-03, PD-05, SE-R-01)	4 ea.
2	Cementing head 9-5/8" single	1 ea.
3	Swage 9-5/8"	1 ea.
4	Cross over 9-5/8"	1 ea.
5	Air compressor (SE-C-01, PD-C-04, SE-RC-03)	3 ea.
6	Solution tank (PD-ST-02, PD-ST-03)	1 ea.
7	Silo (PD-S-07, SE-S-07, PD-S-03, SE-S-01)	5 ea.

Table 7-1: AZNS-040 Used Equipments

Item	Operation crew	Qty
1	Cementing supervisor	1
2	Cementing engineer	1
3	Cementing operator	4
4	Cementing helper	5
5	Cementing helper bulk	2

Table 7-2: AZNS-040 Operation Crew

Operaton Ticket

FIELD SERVICES TICKET

SPAD ENERGY — ماهان صنعت آریا MSA EPC CONTRACTOR

Client:	MSA	Contract No:		Invoice No:	
Service Company:	SPAD ENERGY				
Field: AZADGAN	Well No: AZNS-40	Rig No:	QEOC 208	Date:	7- Sep- 2023
Division:		Period:	25-Jul-2023 till 7-Sep-2023		
Service Requested:	PERSONEL-EQUIPMENT				

ITEM	Description	Amount		STB		OPT		Total
		Qty	Unit	Qty	Unit	Qty	Unit	
1	Cementing Supervisor	1	per	2	Day	1	Day	3
2	Cement Engineer	1	per	0	Day	3	Day	3
3	Cementing Operator	2	per	2	Day	1	Day	3
4	Cementing Operator	2	per	1	Day	1	Day	2
5	Helper	1	per	1	Day	1	Day	2
6	Helper	6	per	2	Day	1	Day	3
7	PumpTruck PD - 01	1	UNIT	3	Day	0	Day	3
8	PumpTruck PD - 05	1	UNIT	1	Day	1	Day	2
9	PumpTruck PD - 03	1	UNIT	2	Day	0	Day	2
10	PumpTruck SE-R 01	1	UNIT	1	Day	1	Day	2
11	Silo PD-S-07	1	UNIT	39	Day	1	Day	40
12	Silo SE-S-07	1	UNIT	39	Day	1	Day	40
13	Silo PD-S-03	1	UNIT	42	Day	1	Day	43
14	Silo PD-S-01	1	UNIT	25	Day	1	Day	26
15	Air Compressor SE-C-01	1	UNIT	6	Day	1	Day	7
16	Air Compressor PD-C-04	1	UNIT	9	Day	1	Day	10
17	Air Compressor SE-RC-02	1	UNIT	2	Day	1	Day	3
18	Solution Tank PD-ST 02	1	UNIT	5	Day	2	Day	7
19	Solution Tank PD-ST 03	1	UNIT	4	Day	2	Day	6
21	Cement head single 9 5\8 "	1	UNIT	3	Day	1	Day	4
22	X-over 9 5\8 "	2	UNIT	15	Day	1	Day	16
20	Swage 9 5\8 "	1	UNIT	16	Day	0	Day	16
23	MOB/ DMOB PumpTruck PD - 01	1	UNIT	1				
24	MOB/ DMOB PumpTruck PD - 05	1	UNIT	1				
25	MOB/ DMOB PumpTruck PD - 03	1	UNIT	1				
26	MOB/ DMOB PumpTruck SE-R01	1	UNIT	1				
27	MOB/ DMOB Silo SE-S-07	1	UNIT	1				
28	MOB/ DMOB Silo PD-S-07	1	UNIT	1				
29	MOB/ DMOB Silo PD-S-03	1	UNIT	1				
30	MOB/ DMOB Silo PD-S-01	1	UNIT	1				
31	MOB/ DMOB Air Compressor SE-C-01	1	UNIT	1				
32	MOB/ DMOB Air Compressor PD-C-04	1	UNIT	1				
33	MOB/ DMOB Air Compressor SE-RC-02	1	UNIT	1				
34	MOB/ DMOB Solution Tank PD-ST 02	1	UNIT	1				
35	MOB/ DMOB Solution Tank PD-ST 03	1	UNIT	1				
36	Cement head single 9 5\8 "	1	UNIT	1				
37	X-over 9 5\8 "	2	UNIT	1				
38	MOB/ DMOB Swage 9 5\8 "	1	UNIT	1				

SPAD Energy supervisor: از طرف

Client Representative: Drilling Supervisor

Date: South Azadegan Project

Rig No:

CONTRACTOR Date:

	DATE			TIME
Operation	5	Sep	2023	14:00
Operation Start	6	Sep	2023	
End of operation	7	Sep	2023	
Left Location	7	Sep	2023	
On Base				

Job Log & Summary

JOB SUMMARY

Client:	MSA	Rig No:	98	JOB PERIOD DATE	
Service Company:	SPAD ENERGY	Well No:	AZNS-40	From:	2023,Sep,5
Field:	AZADEGAN	Hole Section:	8.5"	To:	2023,Sep,7
Service Requested:	9 5/8" SQ. CEMENTING OPR.				

JOB LOG

Date	Time	Volume (bbl)	Density (pcf)	Rate (bpm)	Pressure (psi)	Description
6-Sep-23	5:30					Start mixing solution
6-Sep-23	11:00					Start line up
7-Sep-23	5:00					End of Mixing Solution
7-Sep-23	7:30					Safty meeting
7-Sep-23	8:47				3500	Test the High Pressure Line Up W 3500 psi
7-Sep-23	9:00					Start pump 40 bbl ch. wash w/both pump truck
7-Sep-23	9:01	40	62.4	10		End of pumping 40 bbl ch wash w/both pump truck
7-Sep-23	9:25	60	90	6		Start to Pumping spacer by Rig Pump
7-Sep-23	9:50		95			Start to Mix and pump lead Slurry 95 pcf
7-Sep-23	11:45	1154	95	6-12	400	End of pumping after 1154 bbl lead Slurry 95 pcf
7-Sep-23	11:47		118			Start mix and pump Tail slurry 118 pcf
7-Sep-23	12:20	116.8	118			End of pumping after 116.8 bbl Tail slurry 118 pcf
7-Sep-23	12:23					Release plug and Start Displacement W/Rig pump
7-Sep-23	15:00					Observe 95 pcf CMT slurry returne & End of Job

Material Chemical wash (Preflush)

Consume			Mixing Information		
Materials	Unit	QTY for Job		Chemical wash (Preflush) Volume	40 bbl
Fresh water	Gal	1632		Chemical wash (Preflush) Density	62.4 PCF
Chemical wash SE-CW10	Gal	200.00			
Antifoam(SE-A20)	Gal	8.00	Total Dead Volume		
			0		

Material Spacer

Consume			Mixing Information		
Materials	Unit	QTY for Job		Spacer Volume	90
Fresh water	Gal	3150		Spacer Density	90 PCF
Chemical Spacer SE-P03	Lb	900			
Nacl	Lb	11250			
Antifoam(SE-A20)	Gal	9			
Surfactant	Gal	180	Total Dead Volume		
Barite	Lb	18650	90 bbl		

Material Lead Slurry

Consume			Mixing Information		
Materials	Unit	QTY for Job	Total Mixing Water in Mud Tanks	Cement Slurry Volume	1154 bbl
Class "G" Cement Delijan	Sack	2729.61	591.19 bbl	Cement Slurry Density	95 PCF
Micro Silica	Lb	17,211.14	Total Mixing Fluid in Mud Tanks	Slurry Yield	2.3391cu ft/sk
Retarder SE-R-02	Lb	1,776.80	957.22 bbl	Mixing Water Required	12.4648 gal/sk
Fluid loss Control SE-F09	Lb	1,330.14		Mixing Fluid Required	13.2998 gal/sk
Dispersant SE-D01	Lb	1,064.47			
Antifoam(SE-A20)	Gal	76.60	Total Dead Volume		
			35 bbl		

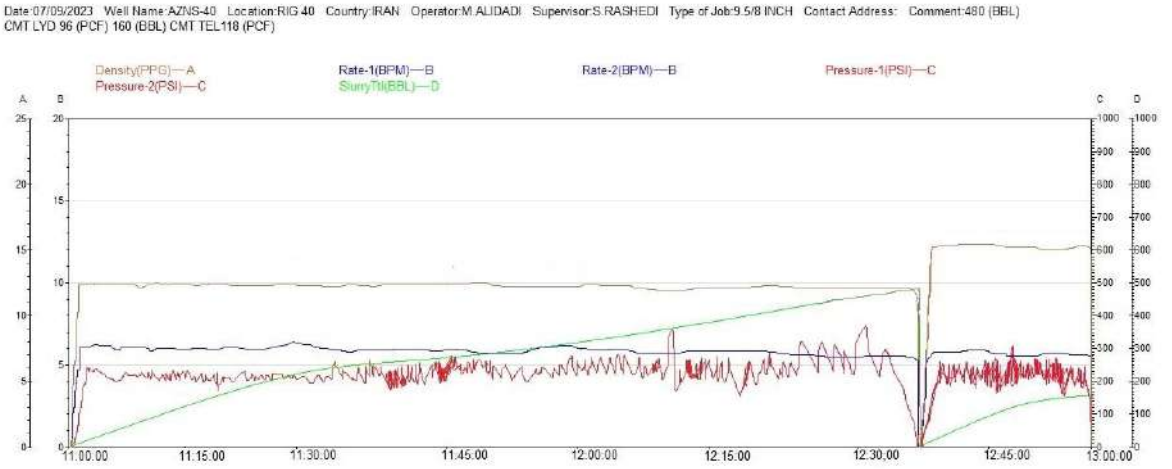
Material Tail Slurry

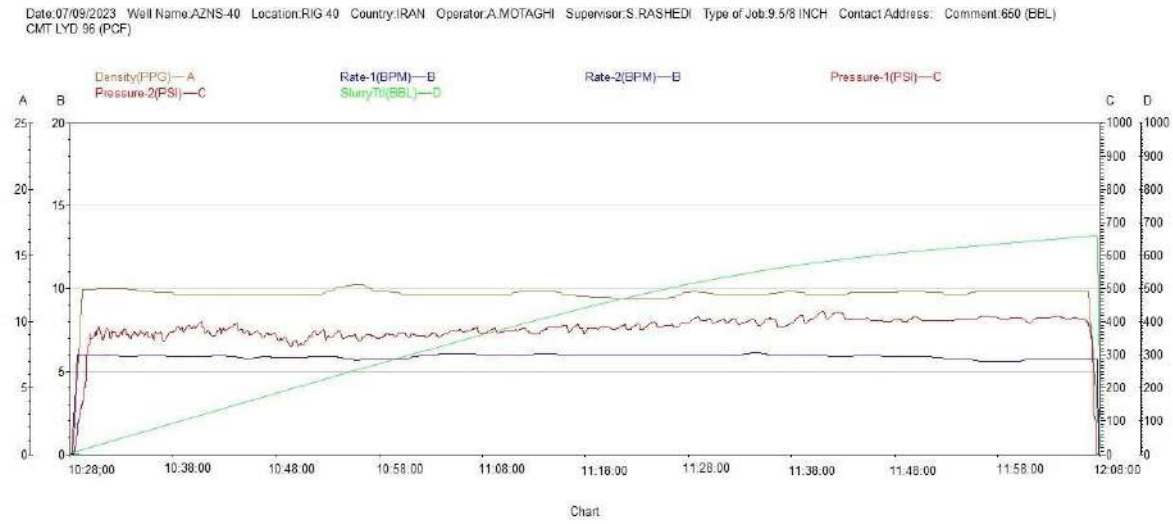
Consume			Mixing Information		
Materials	Unit	QTY for Job	Total Mixing Water in Mud Tanks	Cement Slurry Volume	116.8 bbl
Class "G" Cement Delijan	Sack	370.48	98.22 bbl	Cement Slurry Density	118PCF
Silica Flour	Lb	13,570.76	Total Mixing Fluid in Mud Tanks	Slurry Yield	17700 cu ft/sk
Retarder SE-R-02	Lb	541.35	95.85 bbl	Mixing Water Required	5.8529 gal/sk
Retarder SE-R-012	Lb	108.27		Mixing Fluid Required	7.3741 gal/sk
Fluid loss Control SE-F05	Lb	216.54			
Dispersant SE-D01	Lb	541.35			
Micro Silica	Lb	270.68			
Anti Gas Migration SE-G10	Gal	108.27	Total Dead Volume		
Antifoam(SE-A20)	Gal	10.24	10 bbl		

Client Representative: ماهان صنعت آریا Drilling Supervisor South Azadegan Project Rig No: SPDS CONTRACTOR Date:

SPAD ENERGY supervisor:

Job Graph





Operation Instruction

Pre Job Preparation

1. Bulk plant R/U and preparations (prepare the checklist)
2. Cementing Pump truck R/U and preparations (prepare the checklist)
3. Preparing mixing fluid (Solution) (with good cooperation with the rig crew)
4. Cementing High-Pressure Line R/U to the rig floor (secure them with safety cable)

5. Setting up HSE measurements for the operation

Job Description

1. R/U 9-5/8" CMT head, top plug already loaded. L/U CMT head to pump trucks, Prepare total 900 bbl CMT Solution & Transfer to CMT solution Tanks.
2. L/U CMT Line to CMT Head, Pressure test lines w/ 500-3500 psi for 5-15 Min; Ok.
3. Pumped 40 bbls chemical wash (62.4 pcf) by pump trucks & 60 bbls spacer (90 pcf) by rig pump.
4. Mixed & Pumped 2770 sks CMT "G" Delijan (S.V: 1154 bbls, S.W: 95 pcf) as lead slurry W/ 6 bpm.
5. Mixed & Pumped 370 sks CMT "G" Delijan blended W/ 36.63 lb/sks Silica Flour (S.V: 116 bbls, S.W: 118 pcf) as tail slurry W/ 6 bpm.
6. Drop 9-5/8" Top Plug.
7. Displace CMT slurry W/908 bbls 85 pcf mud, Not observed Bumping. Check back-flow, not OK. Got 10 bbls contaminated CMT & 30 bbls Net CMT on the surface. 156 bbls Loss during pumping CMT & 149 bbls Loss during Displace.

Pumping Schedule			
Name	Flow Rate (bbl/min)	Volume (bbl)	Stage Time (min)
Chemical Wash	5	40	8
Spacer	6	60	10
Lead Cement Slurry	6	1154	193
Tail Cement Slurry	6	116	20
Displacement	6 – 8	908	120

Table 7-3: AZNS-040 Pumping Schedule

Slurry type: Chemical Wash				
Weight: 62.4^{pcf}		**Volume: 40^{bbl}**		
Row	Materials Name	Consumble	Unit	Description
1	Fresh Water	1632	gal	Pumped by pump truck
2	Chemical Wash (SE-CW10)	200	gal	
3	Anti Foam (SE-A20)	8	gal	
Slurry type: Spacer				
Weight: 90^{pcf}		**Volume: 90^{bbl}**		**D.V: 30^{bbl}**
Row	**Materials Name**	**Consumble**	**Unit**	**Description**
1	Fresh Water	3150	gal	
2	Chemical Spacer (SE-P03)	900	lb	
3	NaCl	11250	lb	
4	Anti Foam (SE-A20)	9	gal	
5	Surfactant (SE-S40)	180	gal	
6	Barite	16650	lb	
Slurry type: Lead				
Weight: 95^{pcf}	**Slurry Volume: 1154^{bbl}**	**Mixed Water Vol.: 897.18^{bbl}**	**Mixed Fluid Vol.: 957.28^{bbl}**	**Slurry yield: 2.3391 $^{cu.ft/sk}$**
Row	**Materials Name**	**Consumble**	**Unit**	**Description**
1	Class "G" cement Delijan	2770.41	sks	

2	Micro Silica (DS94)	17231.34	lb	
3	Retarder (SE-R02)	3778.80	lb	
4	F.L.C (SE-F09)	1330.14	lb	
5	Dispersant (SE-D01)	1662.67	lb	
6	Ani Foam (SE-A20)	90.69	gal	
Slurry type: Tail				
Weight: 118^{pcf}	**Slurry Volume: 116^{bbl}**	**Mixed Water Vol.: 69.37^{bbl}**	**Mixed Fluid Vol.: 75.92^{bbl}**	**Slurry yield: 1.3552 $^{cu.ft/sk}$**
Row	**Materials Name**	**Consumble**	**Unit**	**Description**
1	Class "G" cement Delijan	370.48	sks	
2	Silica Flour (FS95)	13570.76	lb	
3	Retarder (SE-R02)	541.35	lb	
4	Retarder (SE-R12)	108.27	Lb	
5	F.L.C (SE-F05)	216.54	lb	
6	Dispersant (SE-D01)	541.35	lb	
7	Micro Silica (DS94)	270.68	lb	
8	Anti Gas Migration (SE-G10)	108.27	gal	
9	Ani Foam (SE-A20)	16.24	gal	

Table 7-4: AZNS-040 Material Information

7.2 7" Liner

AZNS-040 7" Cement Report — Page 1 of 1

Project	P/T	Zone	Dep.	Disc.	Type	Sequence	Rev.
SAC	P	IIF 40	DR	DE	REP	0019	00

Cement Report

General Data

FIELD	AZNS	WELL NO.	040/F21	LOCATION	194,235.00	N	
RIG	208	DRILL REP	A.Zamani/ H.Karami		3,450,715.00	E	
DATE	1402.06.31	HOLE DEPTH	4,190 m MD	MUD WT.	104	PCF	
HOLE SIZE	8-1/2" inches		4,190 m TVD	MUD TYPE	S.KCl.Poly. Lst Mud		
CEMENTING CONTRACTOR	SPAD Energy Co.						

TYPE OF CEMENT JOB (casing, liner, squeeze, plug etc)	CASING

CASING / LINER INFORMATION

SHOE DEPTH:	3933.00 m	HANGER:	m

IF PLUGS			DEPTH m	O.D. in	I.D. in	WEIGHT lb/ft	GRADE
CEMENT PLACEMENT	TOP	m	3812-4189.5	7"	6.094"	32#	TP110SS
	BOTTOM	m					
CEMENT PLACEMENT	TOP	m					
	BOTTOM	m					

	DEPTH m	O.D. in	WEIGHT lb/ft	GRADE	CONNECTION TYPE	COMMENTS
PREVIOUS CASING	3933	9-5/8"	53#	TP110SS	TPCQ	

Cement Job	VOL bbl	WT. pcf	Cement Sx.	RECIPE	Additive
Chemical Wash	10	62.4		FRESH WATER, CHEMICAL WASH, ANTI-FOAM AGENT	408GAL F.WATER, 10GAL CHEMICAL WASH, 2 GAL ANTI-FOAM AGENT
Spacer	40	110.0		FRESH WATER, CHEMICAL SPACER, NACL, ANTI-FOAM AGENT, SURFACTANT, BARITE	1400 GAL FRESH WATER, 400 LB CHEMICAL SPACER, 5000 LB NACL, 4 GAL ANTI-FOAM AGENT, 80 GAL SURFACTANT, 30800 LB BARITE
First Stage					
Lead Cement Slurry	55.96	118	166	CLASS "G" DELIJAN, SILICA FLOUR,MICRO SILICA,RETARDER R-02,DISPERSANT, FLC F-19,ANTI-GAS MIGRATION , ANTI-FOAM AGENT	174.19 SK CMT CLASS "G" DELIJAN, 6380.59 SILICA FLOUR, 348.38 LB MICRO SILICA, 174.19 LB RETARDER R-02, 34.82 LB DISPERSANT, 174.19 LB FLC F-19, 52.26 GAL ANTI-GAS MIGRATION , 5.23GAL
Tail Cement Slurry					
Total Pumped	55.96				

Operational Comments / Time Breakdown

OPERATION	TIME / DATE	REASON(S) FOR DOWNTIME:
L/U CMT LINE & TEST IT	09:30-15:00/1402.06.31	
PUMP CHEMICAL WASH	15:00-17:30/1402.06.31	
PUMP SPACER	15:00-17:30/1402.06.31	
PUMP CMT LEAD	15:00-17:30/1402.06.31`	
DISPLACMENT	17:30-19:00/1402.06.31	

DISPLACEMENT

RATE bbl/min	VOL bbl	PRESS psi	PLUG BUMP (Yes/No, N/A) Top	Int.	Bttm	FINAL PRESS psi
4-5	256	800-1300				1300

DOWNTIME	hrs	BACKFLOW (Yes/No):	Yes

POST JOB INFORMATION					
CEMENT TAGGED @	m	CEMENT EXPECTED @	m		
CEMENT DRILLED @	m/hr	TIME SINCE PLACEMENT	hrs		

COMMENTS:

M/U CMT Head on string (Pump down plug already loaded), L/U CMT line & stand pipe manifold to CMT Head. Establish Circ thru CMT Head by Rig mump up to 5 BPM/ 1200 Psi w/ 1-3 BPHL, while preparing CMT solution in CMT unit.Pump 166 sxs CMT "G" Delijan blend w/ 36.63 lb/sxs silica flour (SV: 53 bbl, SW: 118 Pcf) w/ 3-4 BPM/ 800-1200 Psi, 1 BPHL. Release Pump down plug, Displace w/ 20 bbl 104 Pcf mud, Observed leakage from CMT line on rig floor, Switch to rig pump, Cont. Displace w/ totally 215 bbl 104 pcf mud & observe shearing by 1800 Psi Press.Cont. Displace w/ totally 256 bbl 104 Pcf mud w/ 0-1 BPHL, Observe bumping by 2000 Psi (Final pumping Press: 1300 Psi), Hold Press for 5 min; Ok. Bleed of Press to Zero, Check back flow; No-return. (Total formation loss during Job: 2 bbl).

Figure 7-5: AZNS-040 7" CSG Cement Report

Operation Information

Operation Type: 7" Casing Cementing Job
Open Hole Size: 8-1/2”
Liner Depth: 4189.5 m
Total Volume
- Pump 10^{bbl} 62.4^{pcf} Chemical wash
- Pump 40^{bbl} 110^{pcf} spacer
- Mix & pump 54.96^{bbl} 118^{pcf} Cement Slurry

Total Displacement Volume: Displace cement with 256 bbls 104 pcf mud
Slurry Data: 54 bbl & 118 pcf

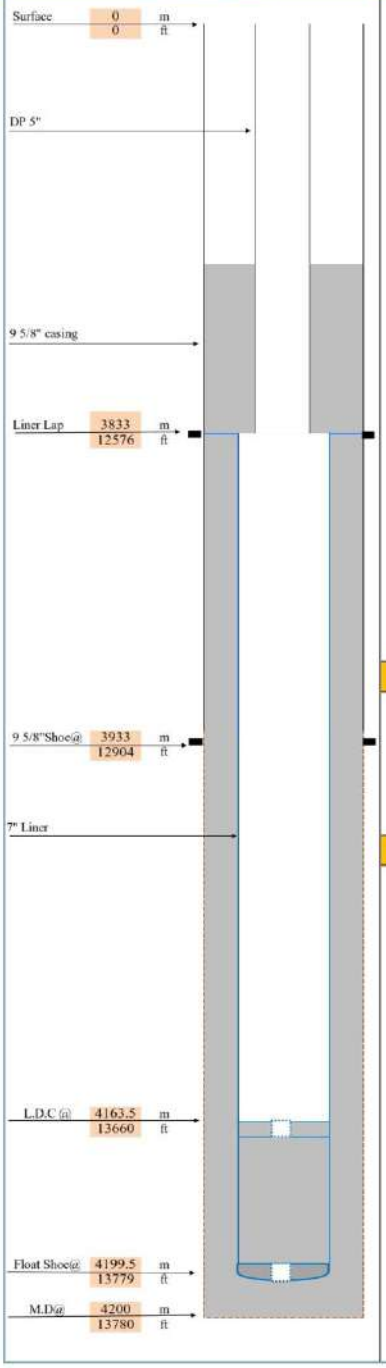
Cementing Program

CEMENTING JOB DESIGN AND CALCULATIONS

CM-FO-001 | REV: 04

Client	MSA	Rig Name	208 OEOC	Job Type	7" Liner
Field	Azadegan	Well Name	AZNS-40	Design Date	19-Sep-23

Well Schematic



Cement Operation Volume

Previous Casing				
OD "	ID "	Weight (ppf)	Grade	Length (m)
9 5/8	8.535	53.5	110SS	3933

New Liner				
OD "	ID "	Weight (ppf)	Grade	Length (m)
7	6.094	32	110SS	367

Drill Pipe				
OD "	ID "	Weight (ppf)	Grade	Length (m)
5	4.276	19.5	G-105	3833

Open Hole	
ID"	Length (m)
8 1/2	267

Excess	
Open Hole	Cased Hole
100%	20%

Volume		
Slurry	55.96	bbl
Displacement	259.70	bbl

Temperature

Temperature Gradient	1.42	°F/100ft
BHST	275.7	°F
BHCT	216.6	°F

Pumping Schedule

	DESCRIPTION	Volume (bbl)	Rate (bpm)	Pump	Job Time (min)
1	Chemical Wash	10	5	CMT	2
2	Spacer	10	6	RIG	2
3	Mix & Pump Cement Slurry	53.96	5	CMT	11
4	Drop Dart Plug			RIG	5.0
5	Mix & Pump Cement Slurry	2.00	5	CMT	0.4
6	Displacement	169.70	5	CMT	34
7	Displacement	90.00	3	CMT	30
8	Pull Out Drill Pipe			RIG	40

DESCRIPTION	(MIN)
Total Job Time	124
Cementing Time	120

CEMENTING JOB DESIGN AND CALCULATIONS

CM-FO-001

REV: 04

Client	MSA	Rig Name	208 OEOC	Job Type	7" Liner
Field	Azadegan	Well Name	AZNS-40	Design Date	19-Sep-23

Mud Removal

Chemical Wash

Material	PDKC Code	Conc.		Qty for Job	
Fresh Water		40.8	gal/bbl	408	gal
Chemical Wash	SE-CW10	1.0	gal/bbl	10	gal
Anti-foam Agent	SE-A20	0.2	gal/bbl	2	gal

DV	0	bbl
CW Volume	10	bbl
CW Density	62.4	pcf

Spacer

Material	PDKC Code	Conc.		Qty for Job	
Fresh Water		35	gal/bbl	1,400	gal
Chemical Spacer	SE-P03	10	lb/bbl	400	lb
NaCl		125	lb/bbl	5,000	lb
Anti-foam Agent	SE-A20	0.1	gal/bbl	4	gal
Surfactant	SE-S40	2	gal/bbl	80	gal
Barite		280	lb/bbl	30,800	lb

DV	30	bbl
Spacer Volume	40	bbl
Spacer Density	110	pcf
M.W. Density	106	pcf

* If required, control PH between 9.2 - 9.5
* Note: The spacer weight and materials will change with refer to the compatibility test (on Rig Site)

Slurry Design

Cement Slurry

Material	PDKC CODE	Density lb/ft3	S.G	Conc.	UOM	Qty for Job*	
Cement	Class G Delijan	195.94	3.14	110.00	lb/sk	174.19	sk
Silica Flour	FS95	164.11	2.63	36.63	lb/sk	6,380.59	lb
Micro Silica	DS94	70.512	1.13	2.00	lb/sk	348.38	lb
Retarder	SE-R02	81.12	1.30	1.00	lb/sk	174.19	lb
Dispersant	SE-D41	69.88	1.12	0.20	lb/sk	34.84	lb
F.L.C	SE-F19	94.85	1.52	1.00	lb/sk	174.19	lb
Anti Gas Migration	SE-G10	63.65	1.02	0.30	gal/sk	52.26	gal
Anti-foam Agent	SE-A20	62.40	1.00	0.03	gal/sk	5.23	gal

Slurry Parameters		
Slurry Volume	314.20	Cu ft
	55.96	bbl
Slurry Density	118	pcf
Slurry Yield	1.8038	cu ft/sk
Mix Water	6.8793	gal/sk
Mix Fluid	7.6267	gal/sk
Dead Volume	0	bbl

Mix Water Type	Fresh Water

Mix Water	28.53	bbl
Mix Fluid	31.63	bbl

Figure 7-6: AZNS-040 7" Cement Design & Calculation

CEMENTING JOB DESIGN AND CALCULATIONS						CM-FO-001
Client	MSA	Rig Name	208 OEOC	Job Type	7" Liner	REV: 04
Field	Azadegan	Well Name	AZNS-40	Design Date	19-Sep-23	

JOB PROCEDURE	
1	Checking equipment and ensuring of proper condition
2	Cleaning mud pits and preparing cement additives
3	Taking work permission before the end of running the Casing
4	Holding pre job safety meeting & tool box under supervision of HSE officer
5	Filling mud pits & pump truck displacement tanks with fresh water and start mixing of required additives for making Spacer & Solution based on the program
5_1	Slurry solution mixing procedure: Mix Water : **28.53** bbl Add Micro Silica-DS94 to fresh water and allow mixing for 60 min Add Retarder-R02 and allow mixing for 15 min Add Dispersant-D41 and allow mixing for 15 min Add F.L.C-F19 and allow mixing for 15 min Add Anti Gas-G10 and allow mixing for 15 min Add Anti Foam-A20 and allow mixing for 5 min
6	Setting cement pumps in suitable position
7	Connecting silos containing dry cement to pump trucks
8	Starting the circulation of drilling fluid by 1.5 times of well capacity for well cleaning and consider well conditions
9	Starting the air compressor and reach to the working pressure of silos (25 psi)
10	Pressure testing at 3000 psi for 10 min.
11	Pumping **10** bbl chemical wash by pump truck
12	Pumping **10** bbl spacer by rig pump
13	Mixing and Pumping slurry down to the well with cement pump truck **53.96** bbl Rate **5** BPM
14	Release Dart plug
15	Mixing and Pumping slurry down to the well with cement pump truck **2.00** bbl Rate **5** BPM
16	Displacement slurry with Drilling fluids by pump truck **169.70** BBL Rate **5** BPM & **90** BBL Rate **3** BPM
17	Landing Plug, bleed off pressure and check for float shoe and collar (Shearing Displacement V. **223.38** bbl) (Bumping Displacement V. **36.33** bbl)
18	Never over-displace more than half the shoe track volume if bumping is not observed (**2.5** bbl)
19	Pull out 10 stands drill pipe, flush string and hold 50 psi pressure on top of the cement to prevent gas migration until cement sets
20	Equipment, pipe lines and tools disassembled
21	End of operation

Figure 7-7: AZNS-040 7" Cement Job Procedure

Cementing Lab Report

LA-FO-00
Rev.04

Report No:						Field:	Azadegan	Job Type :	7	Liner	inch
Date:	19-Sep-23		1402/06/28			Client:	MSA	OH Size:	8 1/2		inch
Well Depth:	(TVD):	4200.0	m	13780.2	ft	Well:	AZNS-40	Last Csg/Liner:	9 5/8		inch
	(MD):	4200.0	m	13780.2	ft	Rig:	208 OEOC	Mud Weight:	106		pcf

Cementing Additives	PDKC Code	Conc.	Units
Cement	Class "G Delijan	110.00	lb/sk
Silica Flour	FS95	36.63	lb/sk
Micro Silica	DS94	2.00	lb/sk
Retarder	SE-R02	1.00	lb/sk
Dispersant	SE-D41	0.20	lb/sk
F.L.C	SE-F19	1.00	lb/sk
Anti Gas Migration	SE-G10	0.30	gal/sk
Anti-foam Agent	SE-A20	0.03	gal/sk

Cement and Mixing Water

Cement Type: Class G Delijan

Slurry Density:	15.775	ppg	118	pcf
Slurry Yield:	0.321	bbl/sk	1.8038	cuft/sk
Mixing Water Required:	0.1638	bbl/sk	6.8793	gal/sk
Total Mixing Fluid Required:	0.1816	bbl/sk	7.6267	gal/sk

Temperature	BHCT:	102.6 °C	BHST:	135.4 °C		
Temperature	BHCT:	216.6 °F	BHST:	275.7 °F	Temperature Gradient:	1.42 °F / 100 ft

API Free Fluid:	0	%	@ 2 hrs	0 @250 CC	
Fluid Loss:	25	cc/30 min @	216.6 °F	and	1000 Psi
API Fluid Loss:	50	cc/30 min @	216.6 °F	and	1000 Psi

Rheology Data

°F	°c	300	200	100	60	30	6	3	10 sec Gel	10 min Gel
80	26.7									
216.6	102.6	70	45	22			13	8	3	2

Pumping Time:	30(BC):	3:40	Hrs:Min	50(BC):	3:50	Hrs:Min	70(BC):	3:55	Hrs:Min

Compressive Strength:	3415	Psi	in	24 Hrs @	275.7 °F		3500 psi
Crash test C.S:	--	Psi	in	24 Hrs @	275.7 °F		3500 psi

Water Annalysis: Chlorides: 400 mg/liter

PH: Prehydrated DB: Dry Blended % bwoc: % by weight of cement % bwow: % by weight of water

Note: Calc. based on 110 lbs sack of dry cement.

Remarks:

Pilot test data

Function	Code	Concentration		Lab QTY (gr)
Cement	G Delijan	110.00	lb/sk	586.11
Silica Flour	FS95	36.63	lb/sk	195.17
Micro Silica	DS94	2.00	lb/sk	10.66
Retarder	SE-R02	1	lb/sk	5.33
Dispersant	SE-D41	0.2	lb/sk	1.07
F.L.C	SE-F19	1	lb/sk	5.33
Anti Gas Migration	SE-G10	0.3	lb/sk	13.61
Anti-foam Agent	SE-A20	0.03	gal/sk	1.33
Water		6.88	gal/sk	305.90

Lab Volume (ml)	600
Slurry Density (pcf)	118
Yield (cuft/sk)	1.8038
Mix Water (gal/sk)	6.8793
Mix Fluid (gal/sk)	7.6267

Figure 7-8: AZNS-040 7" Cement Lab Report

Execution of Operation

Item	Equipment used for the job	Qty
1	Cementing pump truck & related connections (PD-03, PD-05)	2 ea.
2	Air compressor (SE-C-01, PD-C-04)	2 ea.
3	Silo (PD-S-07)	1 ea.

Table 7-5: AZNS-040 Used Equipments

Item	Operation crew	Qty
1	Cementing supervisor	1
2	Cementing engineer	1
3	Cementing operator	2
4	Cementing helper	4
5	Cementing helper bulk	1

Table 7-6: AZNS-040 Operation Crew

Operaton Ticket

FIELD SERVICES TICKET

ماهان صنعت آریا MSA EPC CONTRACTOR

Client:	MSA	Contract No:	Invoice No:
Service Company:	SPAD ENERGY		
Field : AZADGAN	Well No: AZNS-40	Rig No: OEOC 208	Date: 22- Sep- 2023
Division:		Period:	13-Sep-2023 till 22-Sep-2023
Service Requested:	PERSONEL-EQUIPMENT Liner 7" Operation		

ITEM	Description	Amount		STB		OPT		Total
		Qty	Unit	Qty	Unit	Qty	Unit	
1	Cementing Supervisor	1	per	2	Day	1	Day	3
2	Cement Engineer	1	per	2	Day	1	Day	3
3	Cementing Operator	2	per	2	Day	1	Day	3
4	Helper	5	per	2	Day	1	Day	3
5	PumpTruck PD - 05	1	UNIT	3	Day	0	Day	3
6	PumpTruck PD - 03	1	UNIT	2	Day	1	Day	3
7	Silo PD-S-07	1	UNIT	1	Day	1	Day	2
8	Air Compressor SE-C-01	1	UNIT	1	Day	1	Day	2
9	Air Compressor PD-C-04	1	UNIT	3	Day	0	Day	3
11	MOB/ DMOB PumpTruck PD - 03	1	UNIT	1				
12	MOB/ DMOB Silo SE-S-07	1	UNIT	1				
13	MOB/ DMOB Air Compressor SE-C-01	1	UNIT	1				
14	MOB/ DMOB Air Compressor PD-C-04	1	UNIT	1				

مقدار ۱۰۰۰ لیتر گازوئیل تحویل و در سیلو نگهداری گردیده است

SPAD Energy supervisor:

Client Representative:

ماهان صنعت آریا
Drilling Supervisor
South Azadegan Project
Rig No:
PDS CONTRACTOR Date:

	DATE			TIME
On location	20	Sep	2023	
Operation Start	22	Sep	2023	
End of operation	22	Sep	2023	
Left Location	23	Sep	2023	12:00
On Base				

۱- بدلیل تأخیر در ساخت سلوشن و عملیات سیمانکاری ۵ ساعت ویتینگ لحاظ شده است.
۲- عملکرد پرسنل بسیار ضعیف بوده است.
۳- وزن دوغاب بدست آمده حین پایلوت تست و همچنین حین عملیات سیمانکاری به مراتب پایین تر از وزن اعلام شده در برنامه سیمان بوده است و اصلا قابل قبول نمی باشد.
۴- عدم رعایت موارد ایمنی و همچنین باعث آلودگی در لوکیشن توسط پرسنل کاملا مشهود بوده است.
۵- فقط ۲۰ بشکه از ۲۵۶ بشکه از کل دیسپلیس سیمان توسط پمپ تراک انجام شد که بدلیل نشتی شدید از لاین سیمان کاری ناچارا ادامه دیسپلیس با پمپ های دکل انجام شده است.
۶- حین پمپاژ دوغاب بدلیل مشکل فنی پمپ تراک چندین بار پمپاژ قطع شده است.

- تایید منوط به بررسی حرفه ای توان می باشد.

Job Log & Summary

JOB SUMMARY

Client:	MSA	Rig No:	208	JOB PERIOD DATE	
Service Company:	SPAD ENERGY	Well No:	AZNS-40	From:	20 Sep 2023
Field:	S. AZADEGAN	Hole Section:	8 1/2"	To:	22 Sep 2023
Service Requested:	7" Liner CEMENTING OPR. 22 Sep. 2023				

JOB LOG

Date	Time	Volume (bbl)	Density (pcf)	Rate (bpm)	Pressure (psi)	Description
21-Sep-23	22:15					Start Mix spacer
22-Sep-23	11:00					Fill up water in CMT unit for solution
22-Sep-23	13:00					Start mix solution
22-Sep-23	14:30					End of mix the solution
22-Sep-23	14:45					Safty and operation meeting
22-Sep-23	15:05				3000	Test the line and both unit W500/3000 psi
22-Sep-23	15:27	10	110	6		Pump 10 bbl spacer W/Rig pump
22-Sep-23	15:35	10	62.4	5		End of Spacer and Start pump pre flush by CMT unit #5
22-Sep-23	15:50		118			Start mix CMT slurry 118 pcf by CMT unit #3
22-Sep-23	16:38	54	118	5	Max 1000	End of pumping after 54 bbl CMT slurry 118pcf
22-Sep-23	16:39					Release wiper plug
22-Sep-23						Displace W/Rig pump

Material Chemical wash (Preflush)

Consume			Mixing Information		
Materials	Unit	QTY for Job		Chemical wash (Preflush) Volume	10 bbl
Fresh water	Gal	408		Chemical wash (Preflush) Density	62.4 PCF
Chemical wash SE-CW10	Gal / 5gal can	10 gal / 2 can			
Antifoam(SE-A20)	Gal / 5gal can	2gal / 1 can	Total Dead Volume		
			0		

Material Spacer

Consume			Mixing Information		
Materials	Unit	QTY for Job		Spacer Volume	40 bbl
Fresh water	Gal	1400		Spacer Density	110 pcf
Chemical Spacer SE-P03	Lb / 55Lb sks	400 / (8sks)			
Nacl (Provided by MSA)	Lb	5000			
Antifoam(SE-A20)	Gal / 5gal can	4 gal / 1 can			
Surfactant	Gal/55gal Drum	80gal /2 Drum	Total Dead Volume		
Barite (Provided by MSA)	Lb	30600	30 bbl		

Material CMT Slurry

Consume			Mixing Information		
Materials	Unit	QTY for Job	Total Mixing Water in CMT unit	Cement Slurry Volume	54 bbl
Class "G" Cement Delijan	110 Lb /Sack	174.19	28.53 bbl	Cement Slurry Density	118 pcf
Silica Flour	Lb	6,380.59	Total Mixing Fluid in CMT unit	Slurry Yield	1.8038 Cu ft/sks
Micro Silica DS94	Lb	348.38	31.63 bbl	Mixing Water Required	6.8793 gal / sks
Retarder SE-R-02	Lb / 55Lb sks	174.19 / 4 sks		Mixing Fluid Required	7.6267 gal / sks
Dispersant SE-D41	Lb / 55Lb sks	34.84 / 1 sks			
Fluid loss Control SE-F19	Lb / 55Lb sks	174.19 / 4 sks			
Antifoam(SE-A20)	Gal / 5gal can	5.23 gal	Total Dead Volume		
			0		

Client Representative:

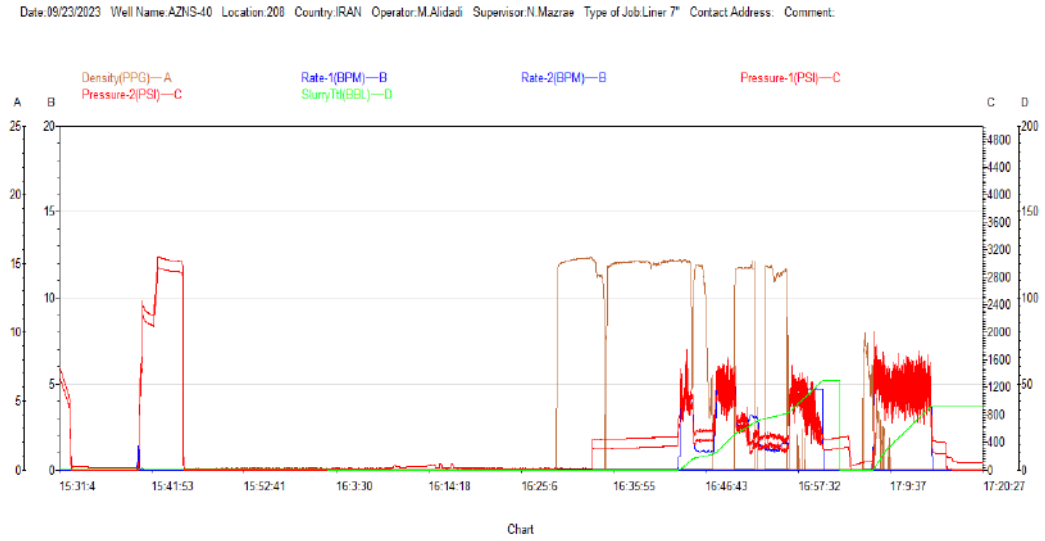
ماهان صنعت آریا
Drilling Supervisor
South Azadegan Project
Rig 208
LPDS CONTRACTOR Date:

SPAD ENERGY supervisor:

فرشاد تقی نژاد

۱- بدلیل تأخیر در ساخت سلوشن و عملیات سیمانکاری ۵ ساعت ویتینگ لحاظ شده است.
۲- عملکرد پرسنل بسیار ضعیف بوده است.
۳- وزن دوغاب بدست آمده حین پایلوت تست و همچنین حین عملیات سیمانکاری به مراتب پایین تر از وزن اعلام شده در برنامه سیمان بوده است و اصلا قابل قبول نمی باشد.
۴- عدم رعایت موارد ایمنی و همچنین باعث آلودگی در لوکیشن توسط پرسنل کاملا مشهود بوده است.
۵- فقط ۲۰ بشکه از کل ۲۵۶ بشکه از کل دیسپلیس سیمان توسط پمپ تراک انجام شد که بدلیل نشتی شدید از لاین سیمان کاری ناچارا ادامه دیسپلیس با پمپ های دکل انجام شده است.
۶- حین پمپاژ دوغاب بدلیل مشکل فنی پمپ تراک چندین بار پمپاژ قطع شده است.

Job Graph



Operation Instruction

Pre Job Preparation

1. Bulk plant R/U and preparations (prepare the checklist)
2. Cementing pump truck R/U and preparations (prepare the checklist)
3. Preparing mixing fluid (Solution) (with good cooperation with the rig crew)
4. Cementing high-pressure line R/U to he rig floor (secure them with safety cable)
5. Setting up HSE measurments for the operation.

Job Description

1. M/U CMT Head on String (Pump Down Plug already loaded), L/U CMT line & stand pipe manifold to CMT Head.
2. Pressure test CMT lines W/3000 psi for 10 Min; Ok.
3. Pumped 10 bbls Spacer (110 pcf) by Rig Pump & 10 bbls Chemical Wash (62.4 pcf) by CMT Pump Truck.
4. Mixed & Pumped 168 sks CMT "G" Delijan blended W/ 36.63 lb/sks Silica Flour (S.V: 54 bbls, S.W: 118 pcf) as slurry W/ 3-4 bpm / 800-1200 psi, 1 BPHL.
5. Release Pump down Plug.
6. Displace W/20 bbls 104 pcf mud, Observe Leakage from CMT line on rig floor, switch torig Pump, Continue Displace W/ totally 215 bbls 104 pcf Mud &

observe Shearing by1800 psi Pressure. Continue Displace W/ totally 256 bbls 104 pcf Mud W/0-1 BPHL, Observe Bumping by 2000 psi (Final bumping Pressure: 1300 Psi), Hold Pressure for 5 min; OK.

7. Bleed of Pressure to Zero, Check back-flow; No-Return. (Total formation loss during job: 2bbl)

Pumping Schedule			
Name	Flow Rate (bbl/min)	Volume (bbl)	Stage Time (min)
Chemical Wash	5	10	2
Spacer	6	10	2
Cement Slurry	3 – 4	54	15
Displacement	3 – 5	256	64

Table 7-7: AZNS-040 Pumping Schedule

Slurry type: Chemical Wash				
Weight: 62.4^{pcf}		**Volume: 10^{bbl}**		
Row	Materials Name	Consumble	Unit	Description
1	Fresh Water	408	gal	Pumped by pump truck
2	Chemical Wash (SE-CW10)	10	gal	
3	Anti Foam (SE-A20)	2	gal	
Slurry type: Spacer				
Weight: 90^{pcf}		**Volume: 40^{bbl}**		**D.V: 30^{bbl}**
Row	**Materials Name**	**Consumble**	**Unit**	**Description**
1	Fresh Water	1400	gal	Pumped by

2	Chemical Spacer (SE-P03)	400	lb	rig pump
3	NaCl	5000	lb	
4	Anti Foam (SE-A20)	4	gal	
5	Surfactant (SE-S40)	80	gal	
6	Barite	30800	lb	
Cement Slurry				
Weight: 118^{pcf}	**Slurry Volume: 54^{bbl}**	**Mixed Water Vol.: 28.53^{bbl}**	**Mixed Fluid Vol.: 31.63^{bbl}**	**Slurry yield: 1.8038$^{cu.ft/sk}$**
Row	**Materials Name**	**Consumble**	**Unit**	**Description**
1	Class "G" cement Delijan	168.08	sks	
2	Silica Flour (FS95)	6380.59	lb	
3	Micro Silica (DS94)	348.38	lb	
4	Retarder (SE-R02)	174.19	lb	
5	F.L.C (SE-F19)	174.19	lb	
6	Dispersant (SE-D41)	38.84	lb	
7	Anti Gas Migration (SE-G10)	52.26	gal	
8	Ani Foam (SE-A20)	5.23	gal	

Table 7-8: AZNS-040 Material Information

Operation Problem

Although success in pressure test of line before the job, during displacing and after pumping 20^{bbl} 104^{pcf} mud, the swivel joint failed and caused leakage from CMT line on the rig floor. Since it was in a bad hand place on the floor and the time shortage during the operation, changing the leaked swivel was time-consuming. Thus the displacement was continued by the rig pump.

8. Completion Report

8.1 Tubing Information

Tubing Size(in)	Weight (ppf)	Grade	Ultimate Strengths	Connection	Collapse Pressure (psi)	Burst Pressure (psi)	Interval (m)
4-1/2	18.9	L-80	233 KLB	HSM3	13830	13380	0-3703
3-1/2	10.2	L-80	440KLB	HSM3	12120	11560	3703-3955

Table 8-1: AZNS-040 Tubing Information

8.2 Completion Tools Specification Checklist

Item	Equipment Description	
1	**CONTROL LINE "WEIQIXIN":**	
	THINKNESS:	0.065"
	WORK PRESSURE:	15000 PSI
	O.D.:	0.25"
	TYPE:	SEAMLESS
	MATERIAL:	INC 825
	LENGTH:	120m/Roll
2	**UPPER FLOW COUPLING " WEIQIXIN" :**	
	OD:	**5.5"**
	ID:	3.640"
	THREADS:	4-1/2" 18.9 PPF, HSM3
	WORKING PRESSURE:	10000 PSI
	LENGTH:	6 FT
	MATERIAL:	13CR-110KSI
	LOWER FLOW COUPLING " WEIQIXIN" :	
	OD:	5.550"
	ID:	3.640"
	THREADS:	4-1/2" 18.9 PPF, HSM3
	WORKING PRESSURE:	10000 PSI
	LENGTH:	4 FT
	MATERIAL:	13CR-110KSI
3	**TUBING RETRIEVABL SAFETY VALVE " WEIQIXIN" :**	
	SETTING DEPTH:	66 m

Item	Equipment Description	
	WORKING PRESSURE:	10000 PSI
	TEMPERATURE:	325 DEG F
	MAX O.D.:	7.5"
	MIN. I.D.:	3.437"
	PROFILE:	AF
	SELF-EQUALIZING FEATURE	HYDRAULIC CONTROL
	MATERIAL:	13CR-110KSI
	THREADS:	4-1/2" 18.9 PPF HSM3
	PROTECTION SLEEVE	
	PROFILE	AF
	MIN ID	2.9"
4	**TRAVELING JOINT " WEIQIXIN" :**	
	STROKE LENGTH	10 FT
	SHEAR PIN SET:	15000LB
	ELASTOMER:	AFLAS
	WORKING PRESSURE:	10000 psi
	TEMPERATURE:	300 DEG F
	TOOL OD:	5.315"
	TOOL ID:	2.922"
	THREADS:	3-1/2" 10.20 PPF HSM3 B*P
	MATERIAL:	13CR-110KSI
	TRANSFER TORQUE FEATURE	YES
5	**SLIDING SIDE DOOR " WEIQIXIN" :**	
	PROFILE:	X
	MAX.OD:	4.551"
	MIN ID:	2.750"
	PRESSURE RATING:	10000 PSI
	TEMPERATURE:	300 DEG F
	THREADS:	3-1/2" 10.20 PPF HSM3 B*P
	MATERIAL:	13CR-110KSI
	OPEN METHOD	DOWNWARD OPEN
6	**COMPLETION PACKER " WEIQIXIN" :**	
	TYPE:	PERMANENT
	CASING :	7" 32 PPF
	OD:	5.689"
	ID:	2.922"
	WORKING PRESSURE:	10000 PSI
	TEMPERATURE:	300 DEG F
	THREADS:	3-1/2" 10.20 PPF HSM3 B*P
	MATERIAL:	13CR-110KSI
	ELEMENT MATERIAL:	AFLAS
	4.0" ANCHOR SEAL	

Item	Equipment Description	
	MAX OD (PACKING):	4"
	ID:	2.922"
	WORKING PRESSURE:	10000PSI
	TEMPERATURE:	300 DEG F
	THREADS:	3-1/2" 10.20 PPF HSM3 B
	TYPE:	RATCH LATCH
7	**LANDING NIPPLE " WEIQIXIN" :**	
	PROFILE TYPE:	XN
	WORKING PRESSURE:	10000 PSI
	MAX O.D.:	4.5"
	SEAL BORE:	2.75”
	MIN ID:	2.635"
	THREADS:	3-1/2" 10.20 PPF HSM3 B*P
	TEMPERATURE:	300 DEG F
	MATERIAL:	13CR-110KSI
8	**RE-ENTRY GUIDE SHOE " WEIQIXIN" :**	
	OD:	4.5"
	MIN ID:	2.922"
	TOP CONNECTION	3-1/2" 10.20 PPF HSM3 BOX
	MATERIAL:	13CR-110KSI
9	**3-1/2" CHEMICAL INJECTION" WEIQIXIN" :**	
	TYPE	TUBING RETRIVABLE
	MAX OD	5.00"
	MIN ID	2.922"
	MATERIAL:	13CR-110KSI
	TEMPERATURE	300 DEG F
	CONNECTION	3-1/2" 10.20 PPF HSM3 B*P
	CIV CONNECTION	NPT 3/8" BOX FOR CIV
	1.00"CIV VALVE	
	TYPE	DUAL CHECK VALVE WITH RUPTURE DISC
	WORKING PRESSURE	15000PSI
	MARERIAL	13CR-110KSI
	TEMPERATURE:	300 DEG F
	TOP CONNECTION	3/8" AUTOCLAVE
	BOTTOM CONNECTION	3/8" NPT
	RUPTURE DISC PRESSURE	6000PSI

Item	Name	Description
1	**Tubing Hanger**	11" BOWL x 4-1/2" 18.9 PPF BOX TPCQ-EX CONNECTION BOTTOM & 4-1/2" 4-2G ACME BOX TOP WITH 4-1/2" EXTENDED NECK, WITH ONE CONTINUOUS 1/4" NPT C.LINE EXIT FOR SSSV, ONE 1/4" LINE EXIT FOR DHPG, INSTALLATION COMPLETE WITH RELATED FITTINGS AND PLUGS, WITH TWO CHANNEL FOR 3/8" CHEMICAL, C/W ALIGNMENT PREPATION FOR ESP. BPV: 4" TYPE H THREAD PREPARATION, THREE FS SEAL TO PACK OFF THE UPPER NECK AND TWO S SEAL TO PACK OFF THE BODY OF HANGER, MATERIAL: 410SS. M.C: FF-0.5 T.C: U PSL:3G PR:2 COMPANY: MSP
2	**Fluted Adaptor**	4 1/2" 18.9 PPF, L-80 HSM3 PIN * TPCQ EX PIN, OD: 5.5'' ID: 3.64'' SLOTS: 6 * 1 INCH WIDTH

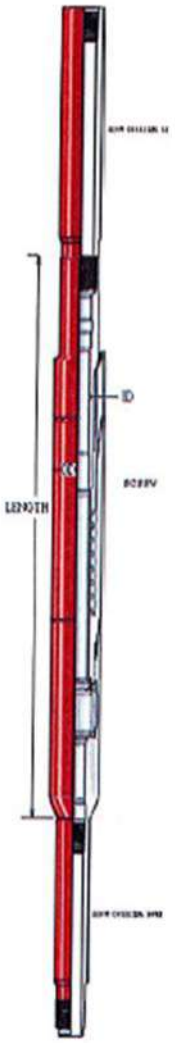
Table 8-2: AZNS-040 Tools Specification

8.3 Assembling Test Report

WEIQIXIN PETROLEUM MACHINERY(TIANJIN)CO.LTD

惟其信 WEIQIXIN

SCSSV Test Report



Type:	210W 343-AF04	
Serial Number:	2306 23038-01	
Connection Size:	4-1/2" 18.90 PPF HSM3 BOX × PIN	
Working Pressure	10,000Psi	
OD:	7.500"	
ID:	3.437"	
Length:	1.780 m	
Material:	S13CR-110 KSI	
Hydraulic Control Pressure	Full Open	Full Closed
	1800 Psi	900 Psi
8,000 Psi Hydraulic Control Pressure No Drop in 15 Minutes	ok	
8,000 Psi Pressure No Drop in 15 Minutes	ok	
8,000 Psi flapper Pressure No Drop in 15 Minutes	ok	
Top Connection	Length:	1.620 m
Flow Coupling UP:	Serial Number:	2306 Y230414-01
	Length:	1.725 m
Flow Coupling DOWN:	Serial Number:	2306 Y230415-02
	Length:	1.115 m
Bottom Connection:	Length:	1.621m
Well Name and Location:	AZNS-40	
MSA Project Manager: Name/Title: Date:	E-purushresh	
WQX: Name/Title: Date:	2023.12.11	

Figure 8-1: AZNS-040 TRSSSV Test Report

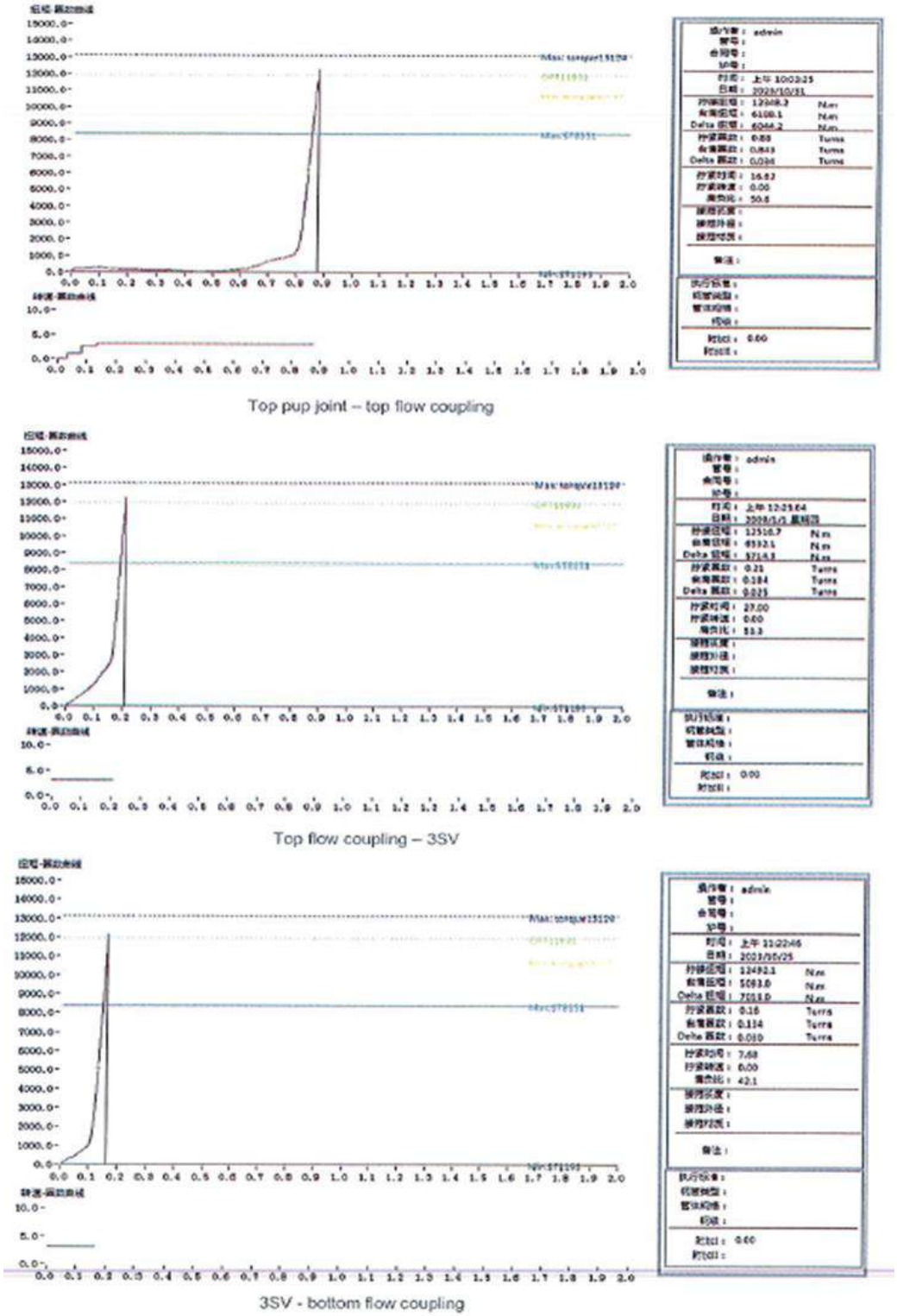
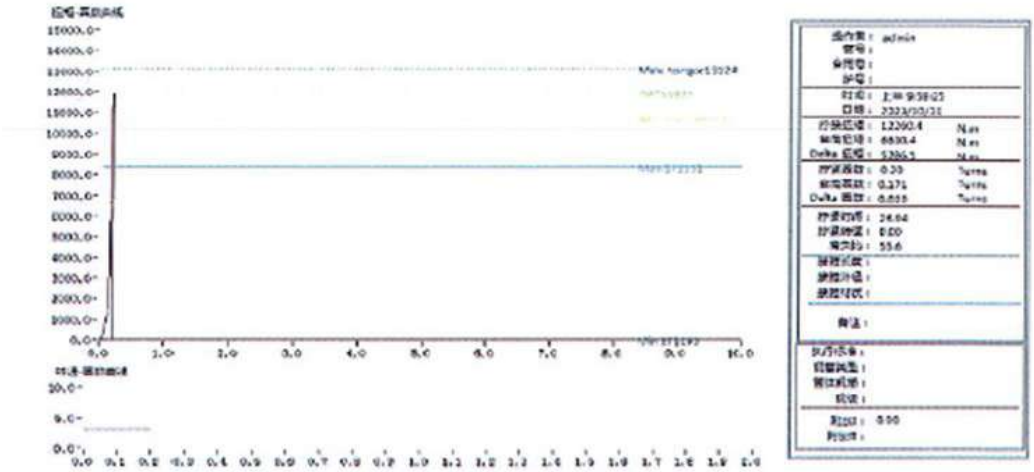
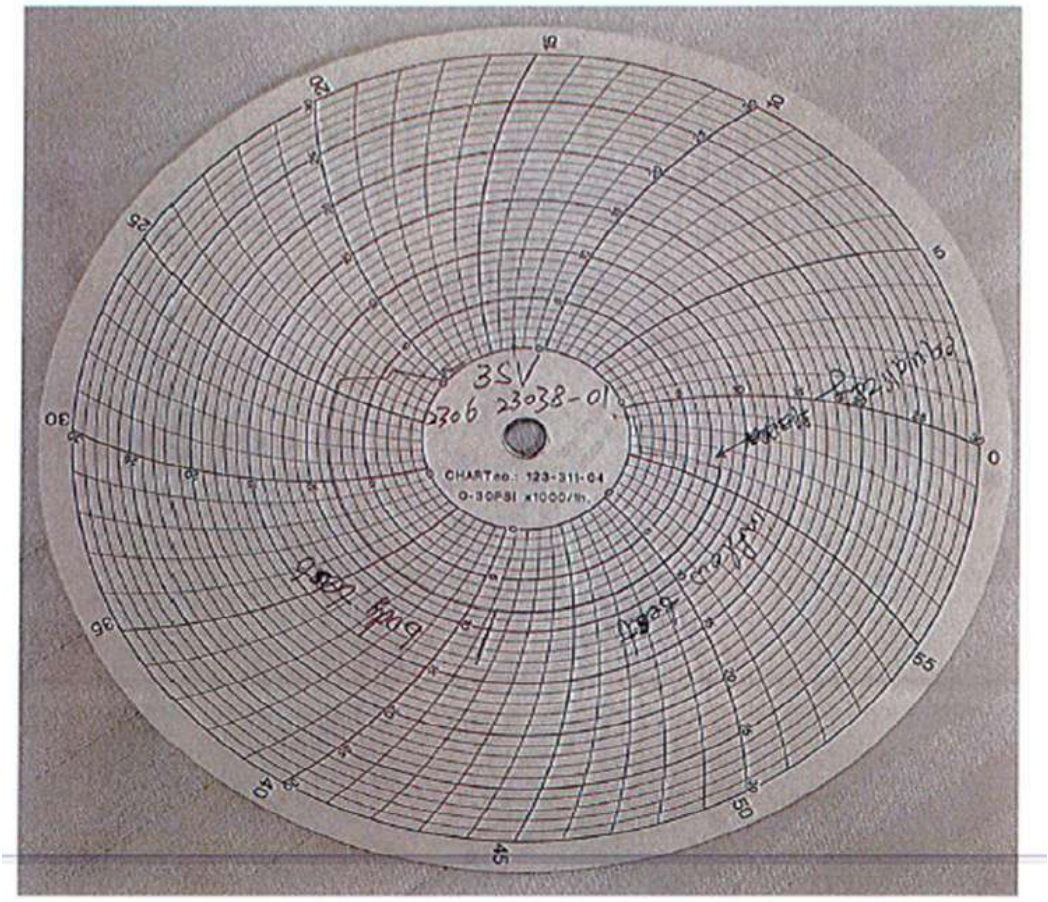


Figure 8-2: AZNS-040 TRSSSV Connections Graph



Bottom flow coupling - bottom pup joint

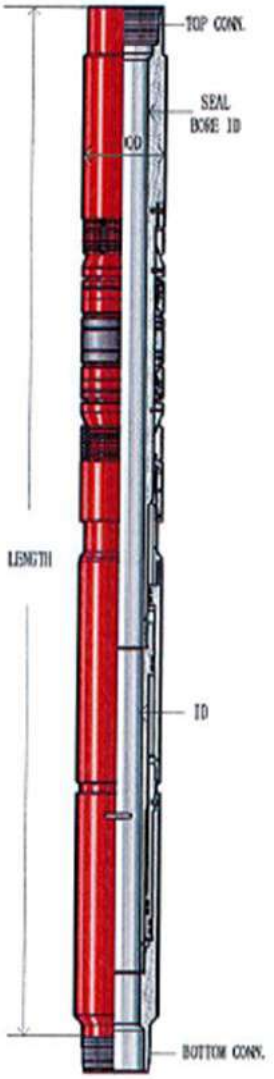


Pressure test curve

Figure 8-3: AZNS-040 TRSSSV Pressure Test Graph

WEIQIXIN PETROLEUM MACHINERY(TIANJIN)CO.LTD

WSDP Hydraulic Set Permanent Packer Test Report



Type:	336W144-002	
Serial Number:	2306 23055-01	
Connection Size:	LEFT HAND X 3-1/2" 10.20 PPF HSM3 PIN	
Working Pressure	10,000 Psi	
OD:	5.689"	
ID:	2.922"	
Length:	4.570m	
Material:	S13CR-110KSI	
Temperature:	300 DEG F	
Element :	Aflas	
Setting Pressure:	3,500 Psi	
Seal Bore Size:	Upper Seal Bore ID:4.00" & Lower Seal Bore ID:2.922"	
Top Connection:	Length:	1.824m
Bottom Connection:	Length:	1.822 m
Well Name and Location:	AZNS-40	

MSA Project Manager:
Name/Title:
Date:

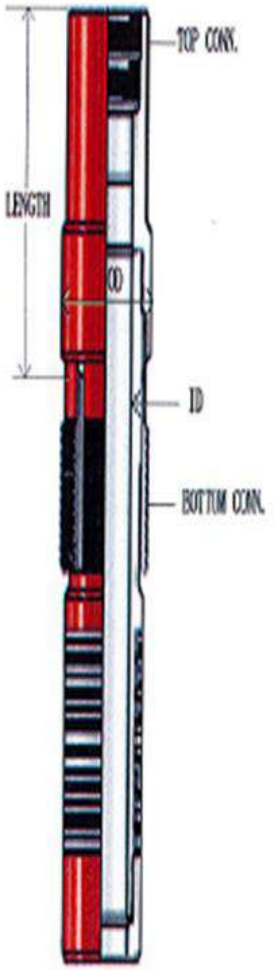
WQX:
Name/Title:
Date: 2023.12.11

Figure 8-4: AZNS-040 Permanent Packer Test Report

WEIQIXIN PETROLEUM MACHINERY(TIANJIN) CO.LTD

惟其信 WEIQIXIN

Model WK Anchor Tubing Seal Report

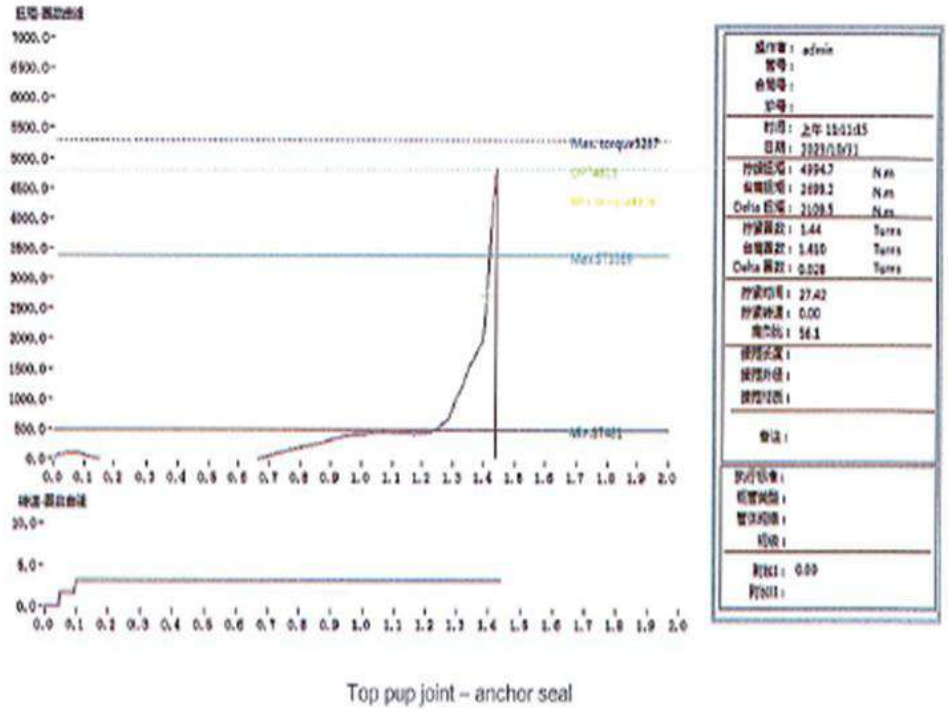


Type:	360W150-000
Serial Number:	2306 23063-02
Connection Size:	3-1/2" 10.20 PPF HSM3 BOX X MULE SHOE GUIDE
OD:	5.689"
ID:	2.922"
Seal OD:	4.000"
Working pressure:	10,000Psi
Length:	230mm
Material:	S13CR-110 KSI
Well Name and Location	AZNS-40

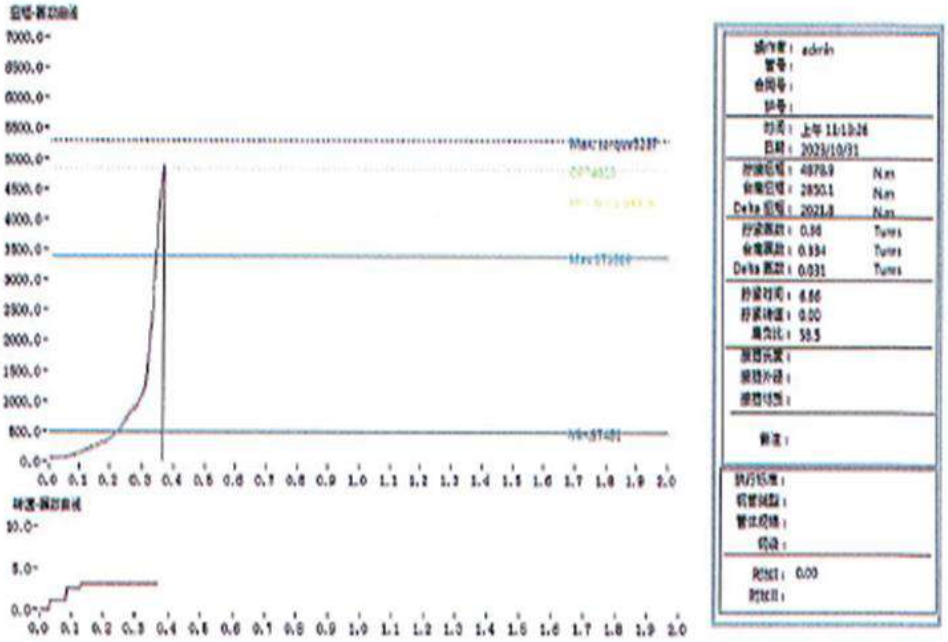
MSA Project Manager:
Name/Title:
Date: E-Purparesh

WQX:
Name/Title:
Date: 2023.12.11

Figure 8-5: AZNS-040 Anchor Seal Test Report



Top pup joint – anchor seal

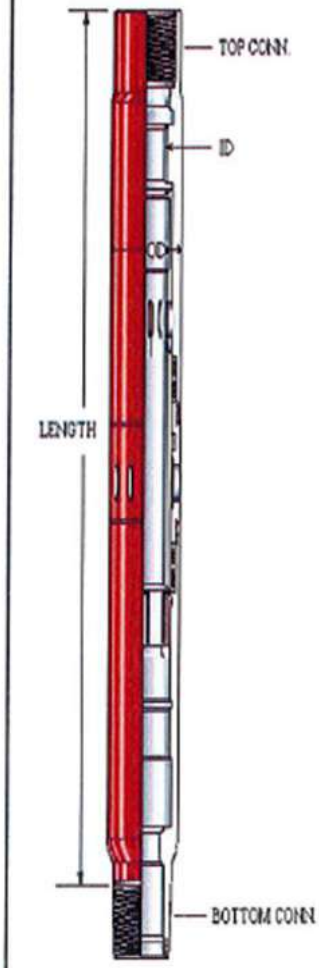


Packer – bottom pup joint

Figure 8-6: AZNS-040 Permanent Packer Connections Graph

WEIQIXIN PETROLEUM MACHINERY(TIANJIN) CO.LTD

Sliding Side Door Test Report



Type:	750W275-X39	
Serial Number:	2306 23039-01	
Connection Size:	3-1/2" 10.20 PPF HSM3 BOX× PIN	
Working Pressure	10,000 Psi	
OD:	4.551"	
ID:	2.750"	
Seal Bore ID :	2.750"	
Length:	1.180m	
Temperature	300 DEG F	
Material:	S13CR-110KSI	
Profile:	X	
Shifting Direction:	DOWNWARD OPEN	
Top Connection:	Length:	1.625 m
Bottom Connection:	Length:	1.617 m
8,000 Psi Pressure No Drop in 15 Minutes	Ok	
8,000 Psi Check Valve Pressure No Drop in 15 Minutes	Ok	
Well Name and Location:	AZNS-40	

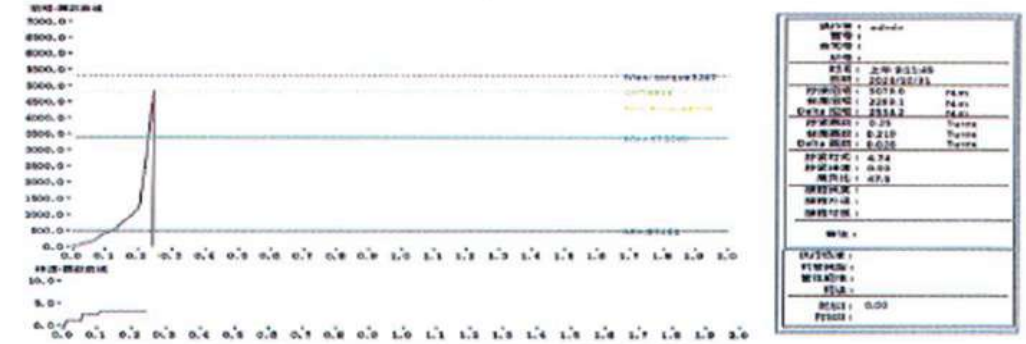
MSA Project Manager:
Name/Title:
Date:

WQX:
Name/Title:
Date: 2023.12.11

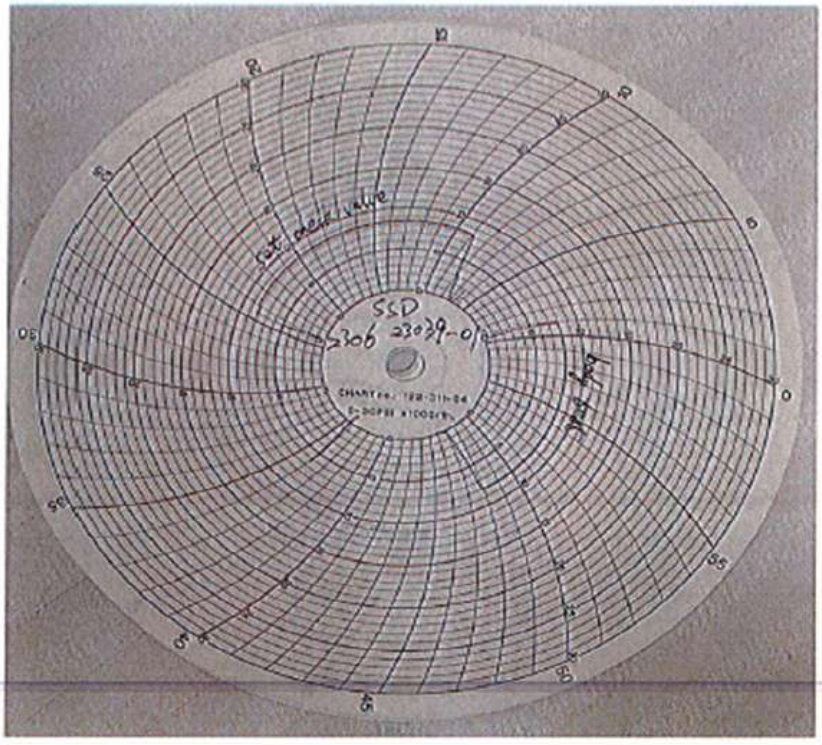
Figure 8-7: AZNS-040 SSD Test Report



Top pup joint – SSD



SSD – bottom pup joint



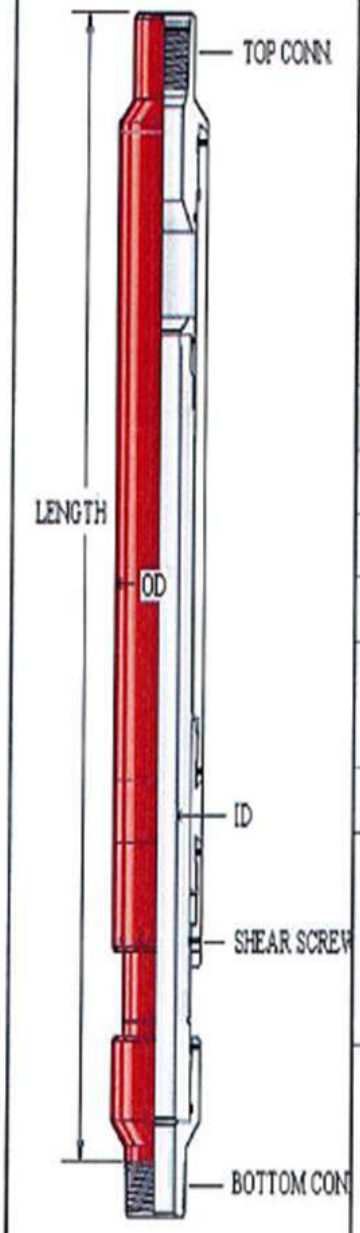
Pressure test curve

Figure 8-8: AZNS-040 SSD Connection & Pressure Test Graph

WEIQIXIN PETROLEUM MACHINERY(TIANJIN) CO.LTD

惟其信 WEIQIXIN

Expansion Joint Test Report

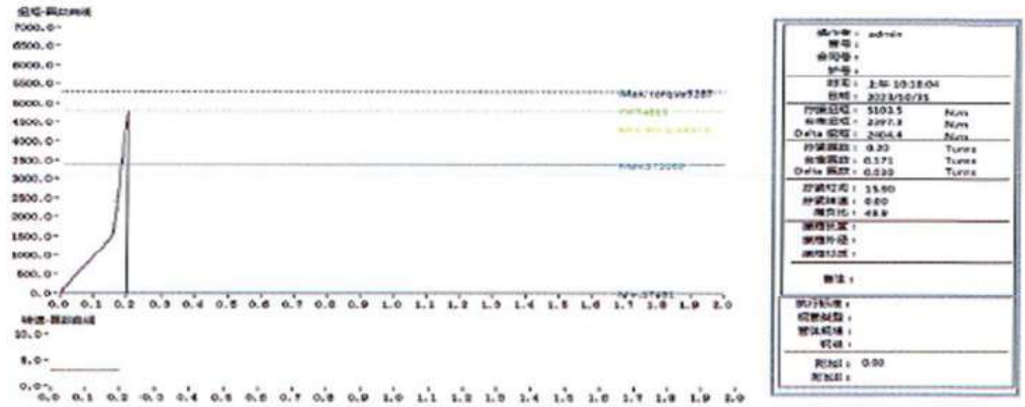


Type:	370W350-030	
Serial Number:	2306 23040-02	
Connection Size:	3-1/2" 10.20 PPF HSM3 BOX× PIN	
Working Pressure	10,000 Psi	
OD:	5.315"	
ID:	2.922"	
Length:	7.320 m	
Material	S13CR-110KSI	
Share Screw:	3	
Top Connection:	Length:	1.820 m
Bottom Connection:	Length:	1.622 m
8,000 Psi Pressure No Drop in 15 Minutes	ok	
Well Name and Location:	AZNS-40	

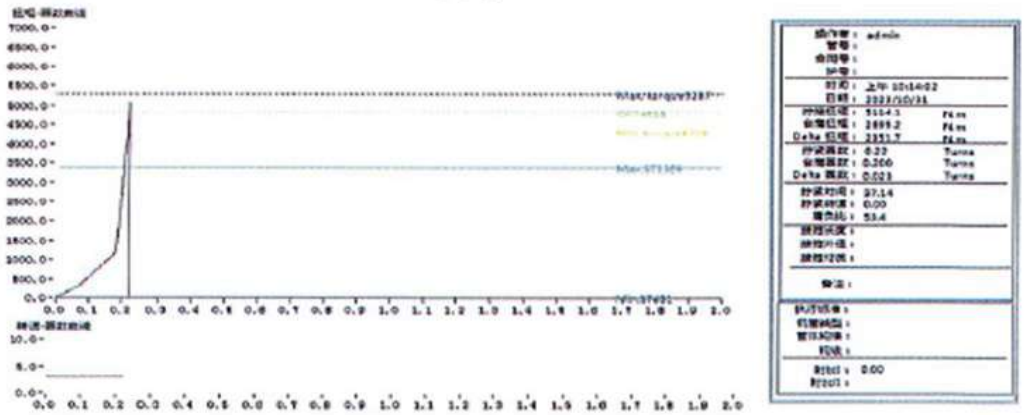
MSA Project Manager:
Name/Title:
Date:

WQX:
Name/Title:
Date: 2013.12.11

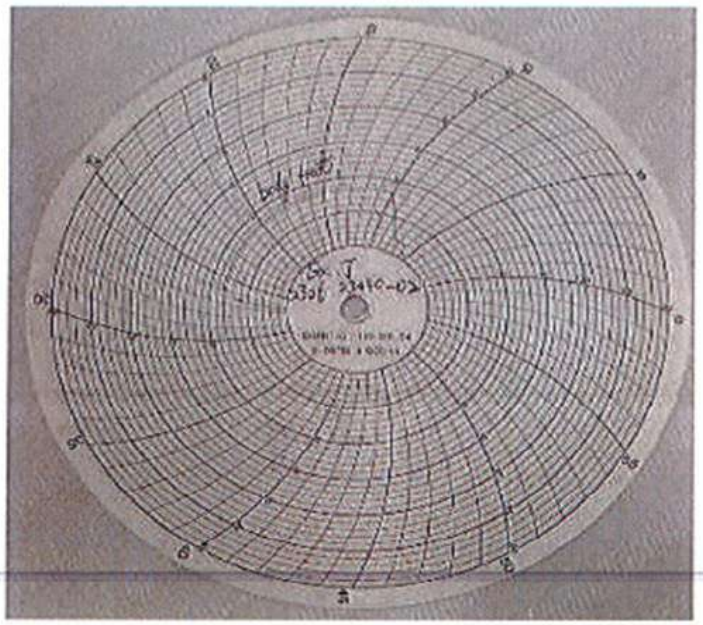
Figure 8-9: AZNS-040 Expansion Joint Test Report



Top pup joint – EX.J



EX.J – bottom pup joint



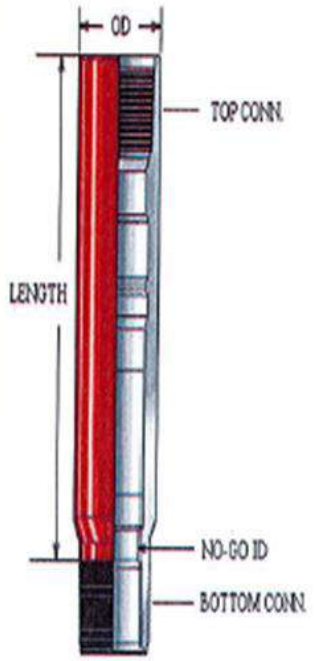
Pressure test curve

Figure 8-10: AZNS-040 Expansion Joint Connection & Pressure Test Graph

WEIQIXIN PETROLEUM MACHINERY(TIANJIN) CO.LTD

维其信 WEIQIXIN

Landing Nipple Test Report

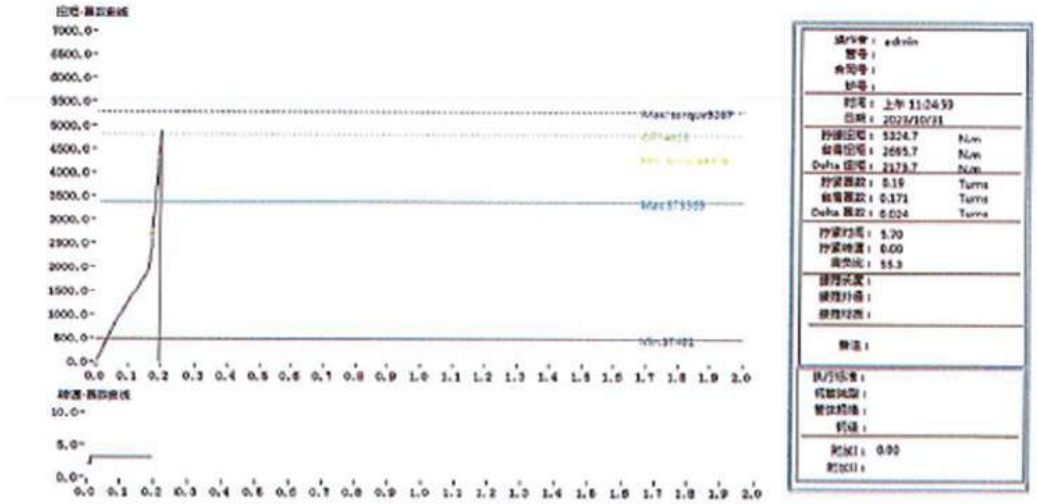


Type:	706W275-N24	
Serial Number:	2306 Y230416-01	
Connection Size:	3-1/2"10.22 PPF HSM3 BOX × PIN	
OD:	4.50"	
ID:	2.75"	
NO-GO ID:	2.635"	
Length:	0.370m	
Material:	S13CR-110 KSI	
Top Connection:	Length:	1.620m
8,000 Psi blanking plug Pressure No Drop in 15 Minutes	Ok	
Well Name and Location	AZNS-40	

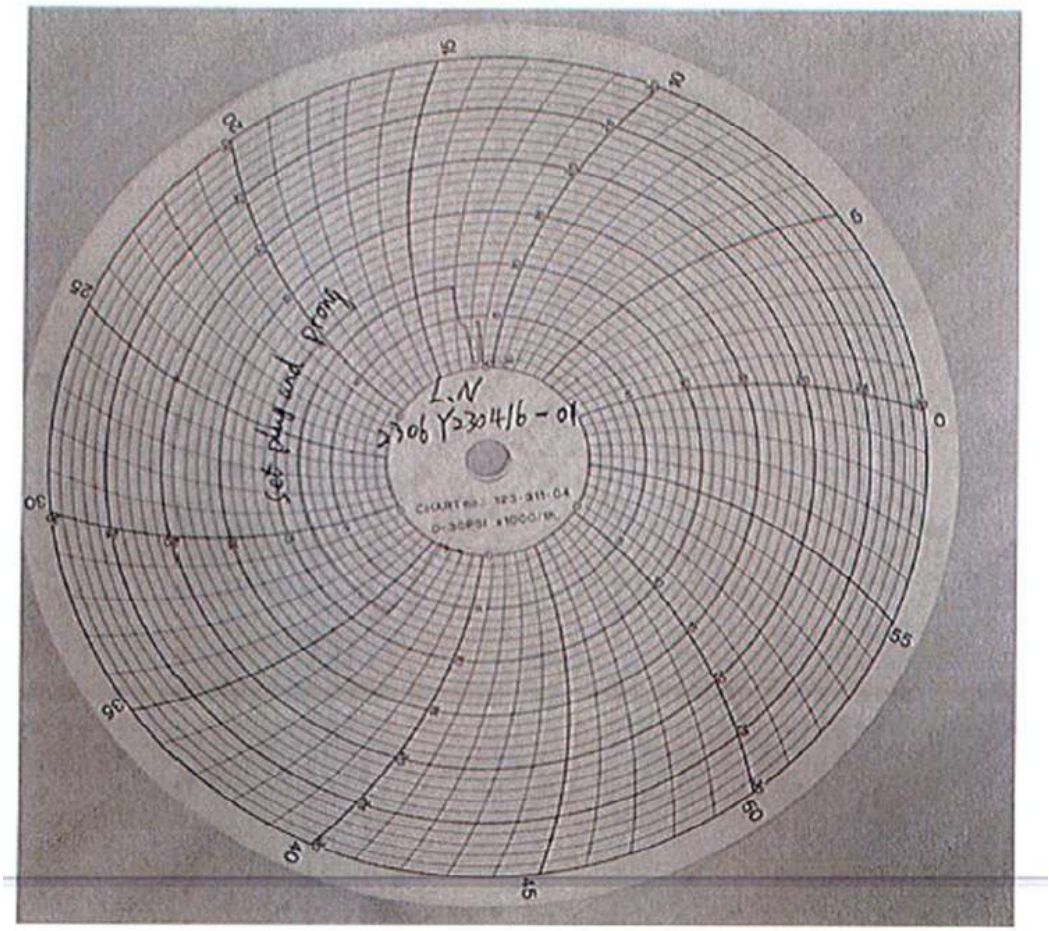
MSA Project Manager:
Name/Title:
Date: E-Parvaresh

WQX:
Name/Title:
Date: 2023.12.11

Figure 8-11: AZNS-040 Landing Nipple Test Report



Top pup point – L.N



Pressure test curve

Figure 8-12: AZNS-040 Landing Nipple Connection & Pressure Test Graph

WEIQIXIN PETROLEUM MACHINERY(TIANJIN) CO.LTD

惟其信 WEIQIXIN

Chemical Injection Valve Test Report



Type:	802W100-601
Serial Number:	2306 23068-02
Connection Size:	3/8" JAM NUT BOX x 3/8" NPT PIN
OD:	1"
ID:	0.299"
Rupture DISK:	6000Psi
Spring open pressure:	500Psi
Length:	310mm
Material:	S13CR-110 KSI
8,000 Psi Pressure No Drop in 15 Minutes	Ok
Well Name and Location	AZNS-40

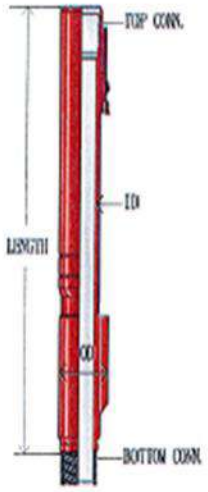
MSA Project Manager:
Name/Title:
Date: E. Pantaresh

WQX:
Name/Title:
Date: 2023.12.11

Figure 8-13: AZNS-040 First CIV Test Report

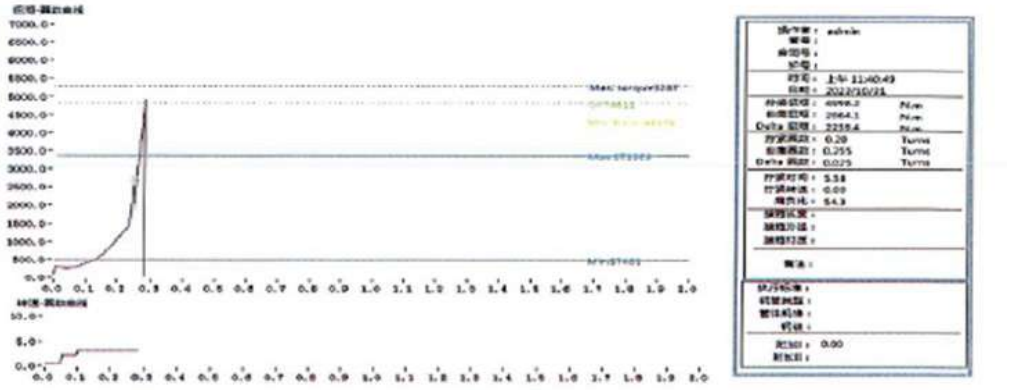
WEIQIXIN PETROLEUM MACHINERY(TIANJIN) CO.LTD

Chemical Injection Mandrel Test Report

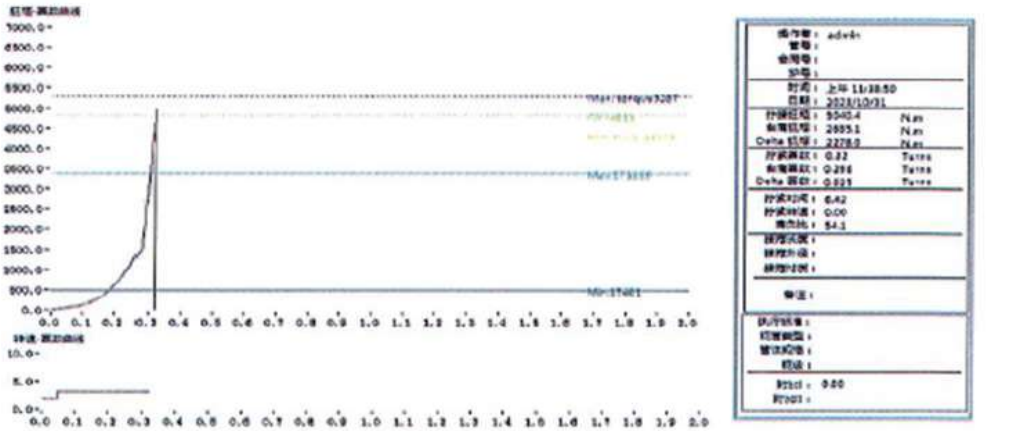


Type:	800W350-028	
Serial Number:	2306 23067-01	
Connection Size:	3-1/2" 10.2PPF HSM3 BOX × PIN	
OD:	5.000"	
ID:	2.922"	
Working pressure:	10,000Psi	
Temperature:	300 DEG F	
Length:	0.725m	
Top Connection:	Length:	1.618 m
Bottom Connection:	Length:	1.622 m
Material:	S13CR-110KSI	
8,000 Psi Pressure No Drop in 15 Minutes	Ok	
Well Name and Location	AZNS-40	
MSA Project Manager: Name/Title: Date:	E-pourarjesm	
WQX: Name/Title: Date:	吴晓伟 2023.12.11	

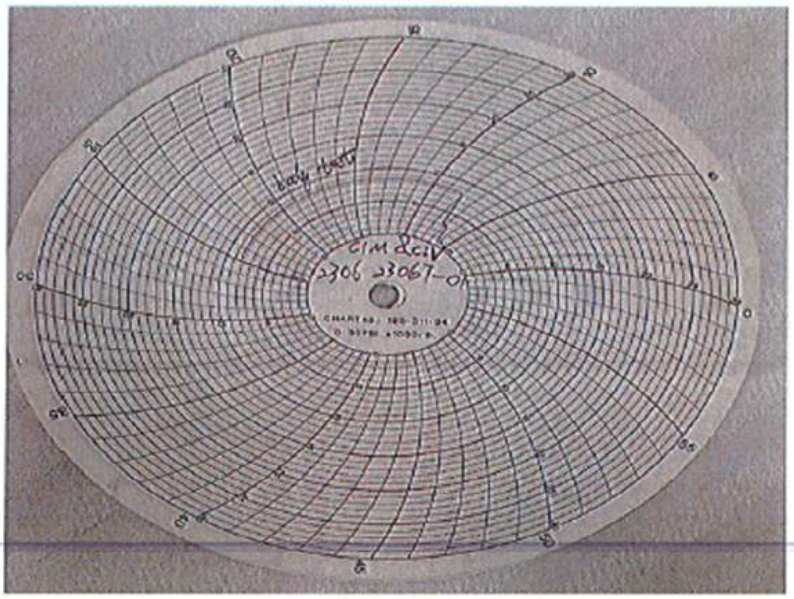
Figure 8-14: AZNS-040 First CIM Test Report



Top pup joint - CIM



CIM – bottom pup joint



Pressure test curve

Figure 8-15: AZNS-040 First CIM Connection & Pressure Test Graph

WEIQIXIN PETROLEUM MACHINERY(TIANJIN) CO.LTD

惟其信 WEIQIXIN

Chemical Injection Valve Test Report



Type:	802W100-601
Serial Number:	2306 23068-01
Connection Size:	3/8" JAM NUT BOX x 3/8" NPT PIN
OD:	1"
ID:	0.299"
Rupture DISK:	6000Psi
Spring open pressure:	500Psi
Length:	310mm
Material:	S13CR-110 KSI
8,000 Psi Pressure No Drop in 15 Minutes	Ok
Well Name and Location	AZNS-40

MSA Project Manager:
Name/Title:
Date: E.Parvaresh

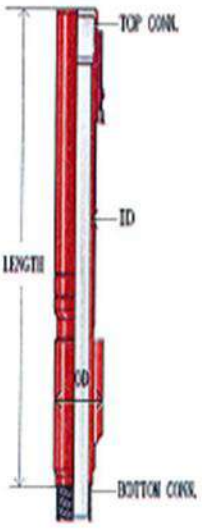
WQX:
Name/Title:
Date: 2023.12.11

Figure 8-16: AZNS-040 Second CIV Test Report

WEIQIXIN PETROLEUM MACHINERY(TIANJIN) CO.LTD

惟其信 WEIQIXIN

Chemical Injection Mandrel Test Report



Type:	800W350-028	
Serial Number:	2306 23067-02	
Connection Size:	3-1/2" 10.2PPF HSM3 BOX × PIN	
OD:	5.000"	
ID:	2.922"	
Working pressure:	15,000Psi	
Temperature:	300 DEG F	
Length:	0.725m	
Top Connection:	Length:	1.625 m
Bottom Connection:	Length:	1.622 m
Material:	S13CR-110KSI	
8,000 Psi Pressure No Drop in 15 Minutes	Ok	
Well Name and Location	AZNS-40	
MSA Project Manager: Name/Title: Date:	E. Parvaresh	
WQX: Name/Title: Date:	2023.12.11	

Figure 8-17: AZNS-040 Second CIM Test Report

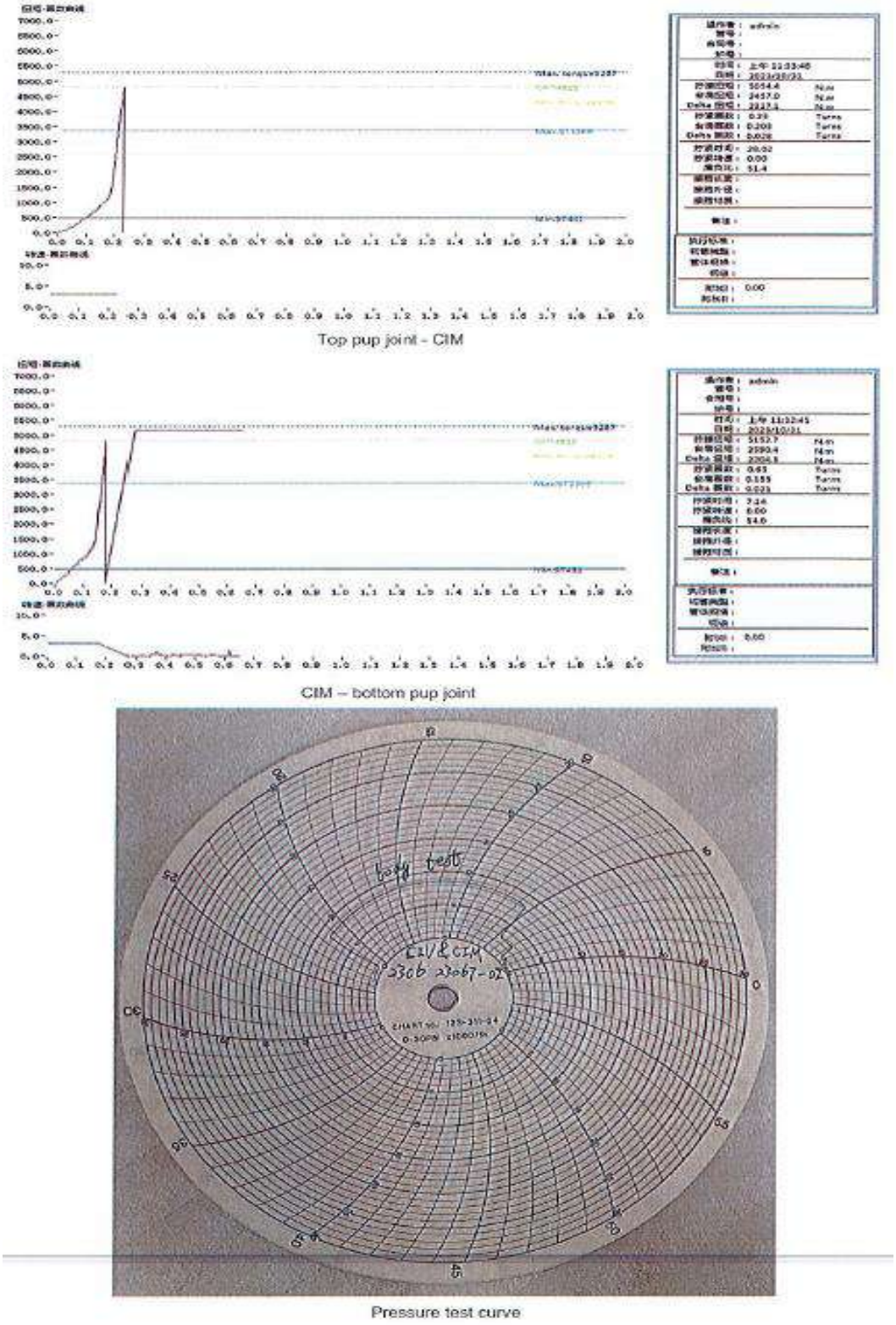


Figure 8-18: AZNS-040 Second CIM Connection & Pressure Test Graph

8.4 Completion Operation Checklist

Item	Operation Name	Remark	Check Mark✓
1	Scraper Trip	Packer depth should be scraped	√
2	Pre-job preparation	Pre-connection All Down Hole Equipment With Upper & Lower Pup Joints On Confirmed Thread-makeup at Workshop, And Pressure Test Prior To Being Transported To Well Site.	√
		Retrieve Wear Bushing, M/U Jet Sub And Flush Bop Stack & Well Head With Fresh Water.	√
		All Crossovers, Pup Joints For Space Out Measured Put In Right Place.	√
		Lift Control Line And Protectors On Rig Floor.	√
		Hold A Pre-Job Safety Meeting For Running Upper Completion String.	√
3	Assembling and RIH	M/U 3-1/2" Re-entry Guide Shoe And 3-1/2" BTM No Go Landing Nipple	√
		Check The Conditions Landing Nipple To Make Sure There Is No Damage, Observe There Is No Any Trash Trapped Inside.	√
		Assemble Landing Nipple.	√
		RIH 1 Joint Tubing As Per Tally	√
		P/U And M/U Hydraulic-Set PERMANENT Production Packer.	√
		Much More Care Should Be Taken When Packer Is Passing Well Head And Liner Hanger While RIH,	√
		Assemble Ratch Latch If Packer	-
		Slack Off Packer Assembly, Double Check The Quantity Of Shear Pins,	√

		Remove The Tape On The Element, Check If Any Damages On It.	
		RIH 1 Joint Tubing As Per Tally	√
		M/U Pup Joint And SSD (close Position) To Tubing String With check valve Inside.	√
		Pressure Test Tubing String 7000psi/15min Every 50 JTs.	√
		RIH Completion String and M/U traveling joint as per running tally.	√
		M/U 2 NO. CIV as per running tally& Press Test to 2000 Psi.	-
		Cont. Running Tubing as per Tally while Install Cross protector for the 2 NO. 3/8"control line.	-
		M/U TRSSSV	√
		Install control line to TRSSSV and pressure test control line 8000psi x 15mins. And then keep 5000psi pressure while running to ensure TRSSSV open. Install TRSSSV protector for the 1/4" control line. Continue to run 4-1/2" tubing as per running tally.	√
4	M/U and Space out Tubing Hanger	Pick up & M/U tubing hanger with double pin crossover. M/U landing joint. Check string weight upward and downward.	√
		Roughly cut control line about 2 meters above tubing hanger, wrap 2-3 turns and then feed through Tubing Hanger.	√
		Check string weight upward and downward and Space out Tubing Hanger 1.5 m (Half stroke) above tubing head spool	√
Item	**Operation Name**	**Remark**	**Check Mark✓**
5	Set production	R/Up Slick Line. Remove Check valve inside SSD	√
		RIH and Set check vlve in Landing Nipple. POOH to surface.	√

	packer	Pressurize inside tubing to 500psi to ensure traveling joint is full open. Increase Pressure gradually to SET PACKER. Increase pressure to 7000 Psi to Test String.	√
7	Land Tubing hanger	Move down completion string and apply required weight to shear traveling joint load pins and Land Tubing hanger in T.H.S. Slack off completely & Record the weight.	√
		Drain the BOP stack through the gate valves of tubing head before landing hanger in THS. Leave the valves open.	√
		Tie Down Screws on THS should be fully open before Operation	√
8	N/D BOP & N/U X-mas tree	Tighten tie down screws.	√
		Test inside string w/ 5000 psi, 15 min.	√
		R/Up Slick Line. RIH with slick line and Ret. Check valve from Landing Nipple. POOH to surface.	√
		Test Annulus w/ 2000 Psi 15min.	√
		Close BOP ram and test above tubing hanger (annulus between BOP and landing joint) to 1000psi.	√
		B/O & L/D landing joint.	√
		Set 4” 2Way in TBG Hanger & Test it, Back off and L/D Running Tool.	√
		N/D BOP.	√
		Feed control line, through head bonnet.	√
		Check hanger for any damage and pressure test control line to 5000psi	√
		Pressure test chemical injection line & rapture the disc after installing	-

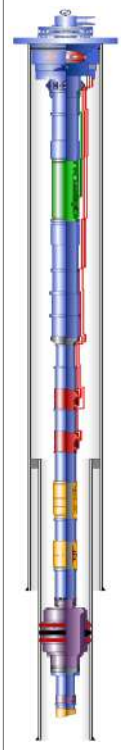
		Bonnet.	
		N/U x-mass tree as program.(Bottom Master)	√
		Perform required tests for X-mass tree. (Bottom Master)	√
		Connect from X-mass tree production valve to the pump unit & well test choke manifold with T-type connection & test all lines to 7500 Psi.	√
		Remove 4" two way, install 4" one way	√
		Close TRSSSV	√
		Release the rig. Clean the well site.	√

Table 8-3: AZNS-040 Operation Checklist

8.5 Completion Sketch

AZNS-40 Completion Sketch

Contractor	MSA / OEOC	Liner & Tubing Data					Client	PEDEC / PPL
Field	AZADEGAN	Description	7"Liner	3 1/2" Tubing	4 1/2" Tubing	Tubing	7'' SN PKR:	3809.6
Well Number:	AZNS-40	OD[in]	7.00	3.50	4.50		7'' Liner Lap:	3812.2
Rig :	208-OEOC	Nominal weight [lb/ft]	32.00	10.20	18.90		7'' Liner Shoe:	4189.5
RTE-GLE	14.45	Grade	TP110SS	L-80	L-80			
RTE-THF	11.33	Thread	TPCQ	HSM3	HSM3		Total Depth:	4200
Rig Spudded Date :	13/07/2023	Inside diameter [in]	6.094	2.915	3.640		X-mass Tree:	MSP-1 Type
End of Operation	28/11/2023	API Drift Diameter	5.969	2.797	3.515		Size & Rating:	4-1/16" 10K

Schematic		Description	Max OD	Min ID	Length(m)	BTM(m)	TOP(m)
	1	Elevation (Floor - TBG Head Flange)				11.576	0.000
	2	TBG HANGER MSP, 10K, 11" * 4 1/2'' ACME Box up * 4 1/2'' TPCQ-EX Box Down , M/C: FF-0.5 T/C: UPSL: 3G PR: 2, 4" Type ''H'' Two Way	11	3.68	0.598	12.174	11.576
	3	4-1/2" FLUTED SWAGE ADAPTER D/P, 18.9# TPCQ-EX PIN ON TOP X HSM3 PIN ON DOWN* SUPER 13CR-110KSI, WEIQIXIN Co	6.1	3.64	1.010	13.184	12.174
	4	5 JOINTs TUBING 4 1/2",HSM3 ,18.9#, L-80, Type-1	5.118	3.64	47.619	60.803	13.184
	5	PUP JOINT 4 1/2",HSM3 ,18.9#, L-80, Type-1	5.118	3.64	1.621	62.424	60.803
	6	4-1/2" FLOW COUPLING,18.9#,SUPER 13CR-110KSI,HSM3 , 6 FT, WEIQIXIN Co	5.5	3.64	1.725	64.149	62.424
	7	TRSSV-10 Tubing Retrievable Saftey Valve, 4 1/2", 18.9# W/ 3.437" Seal bore,HSM3 , SUPER 13CR-110KSI, 10K WP With AF Profile, Flapper Type, WEIQIXIN Co	7.5	3.437	1.780	65.929	64.149
	8	4-1/2" FLOW COUPLING, 18.9#,SUPER 13CR-110KSI,HSM3 , 4' FT, WEIQIXIN Co	5.5	3.64	1.115	67.044	65.929
	9	PUP JOINT 4 1/2",HSM3 ,18.9#, L-80, Type-1	5.118	3.64	1.620	68.664	67.044
	10	382 JOINTs TUBING 4-1/2"HSM3 18.9#, L-80	5.118	3.64	3634.503	3703.167	68.664
	11	X-over 4-1/2"HSM3, 18.9# BOX * 3-1/2"HSM3 , 10.2# PIN,	5.118	2.922	1.120	3704.287	3703.167
	12	18 JOINTs TUBING 3-1/2"HSM3 , 10.2#, L-80.Type-1	4.25	2.922	170.582	3874.869	3704.287
	13	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.618	3876.487	3874.869
	14	3 1/2" CIV TUBING RETRIEVABLE CHEMICAL INJECTION MANDRELHSM3 , 10.2#,SUPER 13CR-110KSI W/CHECK VALVE, DUAL TYPE & RUPTURE DISC, WEIQIXIN Co	5	2.922	0.725	3877.212	3876.487
	15	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.622	3878.834	3877.212
	16	1 JOINT TUBING 3-1/2"HSM3 , 10.2#, L-80.Type-1	4.25	2.922	9.505	3888.339	3878.834
	17	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.898	3890.237	3888.339
	18	3 1/2" CIV TUBING RETRIEVABLE CHEMICAL INJECTION MANDRELHSM3 , 10.2#,SUPER 13CR-110KSI W/CHECK VALVE, DUAL TYPE & RUPTURE DISC, WEIQIXIN Co	5	2.922	0.725	3890.962	3890.237
	19	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.898	3892.860	3890.962
	20	1 JOINT TUBING 3-1/2"HSM3 , 10.2#, L-80.Type-1	4.25	2.922	9.499	3902.359	3892.860
	21	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.820	3904.179	3902.359
	22	3-1/2" EXP. JOINT,HSM3 , SUPER 13CR-110KSI, 10'STK (Fully Open) Splind Rotational Lock System Any Position, Shear Pin Set 20,000 LBS, WEIQIXIN Co	5.135	2.922	5.760	3909.939	3904.179
	23	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.822	3911.761	3909.939
	24	1 JOINT TUBING 3-1/2"HSM3 , 10.2#, L-80.Type-1	4.25	2.922	9.491	3921.252	3911.761
	25	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.625	3922.877	3921.252
	26	3-1/2" NON-ELASTOMERIC SLIDING SIDE DOOR, SUPER 13CR-110KSI, HSM3 With Type X Selective Profile (PROFILE: 2.750" X, DOWNWARD OPEN), WEIQIXIN Co	4.551	2.75	1.180	3924.057	3922.877
	27	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.617	3925.674	3924.057
	28	1 JOINT TUBING 3-1/2"HSM3 , 10.2#, L-80.Type-1	4.25	2.922	9.495	3935.169	3925.674
	29	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.824	3936.993	3935.169
	30	3-1/2" ANCHOR SEAL ASSEMBLY,HSM3, 10.2#,SUPER 13CR-110KSI, 4" PACKING OD, WEIQIXIN Co	4/00	2.922	0.230	3937.223	3936.993
	31	7'' HYDRAULIC SET PERMANENT PRODUCTION PACKER, MODEL WSDP TO BE RUN IN 7.00" 29-32 PPF CASING, SUPER 13CR-110KSI, UPPER SEAL BORE: 4", WEIQIXIN Co	5//689	2.922	4.570	3941.793	3937.223
	33	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.250	2.922	1.822	3943.615	3941.793
	34	1 JOINT TUBING 3-1/2"HSM3 , 10.2#, L-80.Type-1	4.25	2.922	9.508	3953.123	3943.615
	35	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.620	3954.743	3953.123
	36	13-1/2" BOTTOM NO-GO LANDING,HSM3 , 10.2#, SUPER 13CR-110KSI W/ Seal Bore 2.75" TYPE XN, WEIQIXIN Co	4.5	2.562	0.370	3955.113	3954.743
	37	3-1/2" WIRLINE RE-ENTRY GUIDE, 10.2 #,HSM3 Con,SUPER 13CR-110KSI, WEIQIXIN Co	4.25	2.922	0.300	3955.413	3955.113
		END OF COMPLETION STRING					3955.413

Tubing Retrievable Saftey Valve	65.929	64.149
Traveling Joint	3909.939	3904.179
CIV 1	3877.212	3876.487
CIV 2	3890.962	3890.237
SSD	3924.057	3922.877
7'' HYDRAULIC SET PERMANENT PRODUCTION PACKER	3941.793	3937.223
NO-GO Landing Nipple	3955.113	3954.743
Entry Guide	3955.413	3955.113

Figure 8-19: AZNS-040 Completion Sketch

8.6 Completion Tally

MSA

Field :	Azadegan	Well NO:	AZNS40	Rig NO:	OEOC208	Drig CO:	OEOC
PEDEC, PPL		Completion String					

Grade	ID "	Drift "	Weight lb/ft	Burst psi	Collaspse psi	Yield klbs	Coupling "	TQ ft x lb	Nominal Dia. OD
L-80	2.922	2.797	10.2	11,560.0	12,120.0	233	3.962	3200-3550-3900	3-1/2"
L-80	3.640	3.515	18.9	13,380.0	13,820.0	440	5.137	7920-8800-9680	4-1/2"

BTM NO GO Depth:	3954.74	Formation Name :	Fahliyan	Hook Weight :	58 k lb
BTM of String	3955.41	Formation Top :	3989	VOLUME (TOTAL)	163.26
Packer Depth :	3937.22			Mud Weight (pcf) :	104

Number on jt	Grade	Run In No.	Jt Length (m)	Cum Length (m)	Dist to RT (m)	M/D weight (Klbs)	VOL(BBL)	Remarks	
								joint no	IN / OUT
					3955.413	58.00			
3-1/2" WIRLINE RE-ENTRY GUIDE, 10.2 #,HSM3 Con,SUPER 13CR-110KSI, WEIQIXIN Co	3-1/2" Mule Shoe		0.300	0.30	3955.113	58.01	0.01		
13-1/2" BOTTOM NO-GO LANDING,HSM3 , 10.2#, SUPER 13CR-110KSI W/ Seal Bore 2.75" TYPE XN, WEIQIXIN Co	3-1/2" BTM NO GO		0.370	0.67	3954.743	58.02	0.02		
3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	Pup Joint 3-1/2", 10.2#,HSM3		1.620	2.29	3953.123	58.06	0.06		
410	TUBING 3-1/2", 10.2#, L-80,HSM3	1	9.508	11.80	3943.615	58.31	0.32	1	
3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	Pup Joint 3-1/2", 10.2#,HSM3		1.822	13.62	3941.793	58.36	0.37		
7" HYDRAULIC SET PERMANENT PRODUCTION PACKER, MODEL WSDP TO BE RUN IN 7.00" 29-32 PPF CASING, SUPER 13CR-110KSI, UPPER SEAL BORE: 4", WEIQIXIN Co	PKR		4.570	18.19	3937.223	58.48	0.50		
3-1/2" ANCHOR SEAL ASSEMBLY,HSM3, 10.2#,SUPER 13CR-110KSI, 4" PACKING OD, WEIQIXIN Co	Ratch Lach		0.230	18.42	3936.993	58.49	0.50		
3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	Pup Joint 3-1/2", 10.2#,HSM3		1.824	20.24	3935.169	58.53	0.55		
409	TUBING 3-1/2", 10.2#, L-80,HSM3	2	9.495	29.74	3925.674	58.78	0.81	2	
3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	Pup Joint 3-1/2", 10.2#,HSM3		1.617	31.36	3924.057	58.83	0.85		
3-1/2" NON-ELASTOMERIC SLIDING SIDE DOOR, SUPER 13CR-110KSI, HSM3 With Type X Selective Profile (PROFILE: 2.750" X, DOWNWARD OPEN), WEIQIXIN Co	SSD		1.180	32.54	3922.877	58.86	0.89		
3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	Pup Joint 3-1/2", 10.2#,HSM3		1.625	34.16	3921.252	58.90	0.93		
408	TUBING 3-1/2", 10.2#, L-80,HSM3	3	9.491	43.65	3911.761	59.15	1.19	3	
3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	Pup Joint 3-1/2", 10.2#,HSM3		1.822	45.47	3909.939	59.20	1.24		
3-1/2" EXP. JOINT,HSM3 , SUPER 13CR-110KSI, 10'STK (Fully Open) Splind Rotational Lock System Any Position, Shear Pin Set 20,000 LBS, WEIQIXIN Co	Traveling Jt		5.760	51.23	3904.179	59.35	1.39		
3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	Pup Joint 3-1/2", 10.2#,HSM3		1.820	53.05	3902.359	59.40	1.44		
407	TUBING 3-1/2", 10.2#, L-80,HSM3	4	9.499	62.55	3892.860	59.65	1.70	4	
3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	Pup Joint 3-1/2", 10.2#,HSM3		1.898	64.45	3890.962	59.70	1.75		
3 1/2" CIV TUBING RETRIEVABLE CHEMICAL INJECTION MANDRELHSM3 , 10.2#,SUPER 13CR-110KSI W/CHECK VALVE, DUAL TYPE & RUPTURE DISC, WEIQIXIN Co	CIV		0.725	65.18	3890.237	59.72	1.77		
3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	Pup Joint 3-1/2", 10.2#,HSM3		1.898	67.07	3888.339	59.77	1.83		
406	TUBING 3-1/2", 10.2#, L-80,HSM3	5	9.505	76.58	3878.834	60.02	2.08	5	
3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	Pup Joint 3-1/2", 10.2#,HSM3		1.622	78.20	3877.212	60.06	2.13		
3 1/2" CIV TUBING RETRIEVABLE CHEMICAL INJECTION MANDRELHSM3 , 10.2#,SUPER 13CR-110KSI W/CHECK VALVE, DUAL TYPE & RUPTURE DISC, WEIQIXIN Co	CIV		0.725	78.93	3876.487	60.08	2.15		
3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	Pup Joint 3-1/2", 10.2#,HSM3		1.618	80.54	3874.869	60.12	2.19		Protector

405	TUBING 3-1/2", 10.2#, L-80,HSM3	6	9.202	89.75	3865.667	60.36	2.44	6	Protector
404	TUBING 3-1/2", 10.2#, L-80,HSM3	7	9.496	99.24	3856.171	60.61	2.70	7	Protector
403	TUBING 3-1/2", 10.2#, L-80,HSM3	8	9.493	108.74	3846.678	60.87	2.96	8	Protector
402	TUBING 3-1/2", 10.2#, L-80,HSM3	9	9.479	118.21	3837.199	61.11	3.22	9	Protector
401	TUBING 3-1/2", 10.2#, L-80,HSM3	10	9.491	127.71	3827.708	61.36	3.48	10	Protector
400	TUBING 3-1/2", 10.2#, L-80,HSM3	11	9.495	137.20	3818.213	61.62	3.73	11	Protector
399	TUBING 3-1/2", 10.2#, L-80,HSM3	12	9.483	146.68	3808.730	61.86	3.99	12	Protector
398	TUBING 3-1/2", 10.2#, L-80,HSM3	13	9.498	156.18	3799.232	62.12	4.25	13	Protector
397	TUBING 3-1/2", 10.2#, L-80,HSM3	14	9.493	165.67	3789.739	62.37	4.51	14	Protector
396	TUBING 3-1/2", 10.2#, L-80,HSM3	15	9.490	175.16	3780.249	62.62	4.77	15	Protector
395	TUBING 3-1/2", 10.2#, L-80,HSM3	16	9.503	184.67	3770.746	62.87	5.03	16	Protector
394	TUBING 3-1/2", 10.2#, L-80,HSM3	17	9.492	194.16	3761.254	63.12	5.28	17	Protector
393	TUBING 3-1/2", 10.2#, L-80,HSM3	18	9.504	203.66	3751.750	63.37	5.54	18	Protector
392	TUBING 3-1/2", 10.2#, L-80,HSM3	19	9.488	213.15	3742.262	63.62	5.80	19	Protector
391	TUBING 3-1/2", 10.2#, L-80,HSM3	20	9.496	222.65	3732.766	63.87	6.06	20	Protector
390	TUBING 3-1/2", 10.2#, L-80,HSM3	21	9.487	232.13	3723.279	64.12	6.32	21	Protector
389	TUBING 3-1/2", 10.2#, L-80,HSM3	22	9.487	241.62	3713.792	64.37	6.58	22	Protector
388	TUBING 3-1/2", 10.2#, L-80,HSM3	23	9.505	251.13	3704.287	64.62	6.83	23	Protector
X-over 4-1/2"HSM3, 18.9# BOX * 3-1/2"HSM3 , 10.2# PIN,	XOS 4-1/2", HSM3 B X 3-1/2" HSM3, P	0	1.120	252.25	3703.167	64.67	6.86		

387	TUBING 4-1/2", 18.9#, L-80,HSM3	1	9.526	261.77	3693.641	65.14	7.27	1	Protector / Press Test
386	TUBING 4-1/2", 18.9#, L-80,HSM3	2	9.530	271.30	3684.111	65.60	7.67	2	Protector
385	TUBING 4-1/2", 18.9#, L-80,HSM3	3	9.528	280.83	3674.583	66.07	8.07	3	Protector
384	TUBING 4-1/2", 18.9#, L-80,HSM3	4	9.526	290.36	3665.057	66.53	8.47	4	Protector
383	TUBING 4-1/2", 18.9#, L-80,HSM3	5	9.526	299.88	3655.531	67.00	8.88	5	Protector
382	TUBING 4-1/2", 18.9#, L-80,HSM3	6	9.529	309.41	3646.002	67.46	9.28	6	Protector
381	TUBING 4-1/2", 18.9#, L-80,HSM3	7	9.527	318.94	3636.475	67.93	9.68	7	Protector
380	TUBING 4-1/2", 18.9#, L-80,HSM3	8	9.528	328.47	3626.947	68.39	10.08	8	Protector
379	TUBING 4-1/2", 18.9#, L-80,HSM3	9	9.522	337.99	3617.425	68.86	10.49	9	Protector
378	TUBING 4-1/2", 18.9#, L-80,HSM3	10	9.526	347.51	3607.899	69.32	10.89	10	Protector
377	TUBING 4-1/2", 18.9#, L-80,HSM3	11	9.523	357.04	3598.376	69.79	11.29	11	Protector
376	TUBING 4-1/2", 18.9#, L-80,HSM3	12	9.528	366.57	3588.848	70.25	11.69	12	Protector
375	TUBING 4-1/2", 18.9#, L-80,HSM3	13	9.524	376.09	3579.324	70.72	12.09	13	Protector
374	TUBING 4-1/2", 18.9#, L-80,HSM3	14	9.525	385.61	3569.799	71.18	12.50	14	Protector
373	TUBING 4-1/2", 18.9#, L-80,HSM3	15	9.526	395.14	3560.273	71.65	12.90	15	Protector
372	TUBING 4-1/2", 18.9#, L-80,HSM3	16	9.528	404.67	3550.745	72.11	13.30	16	Protector
371	TUBING 4-1/2", 18.9#, L-80,HSM3	17	9.522	414.19	3541.223	72.58	13.70	17	Protector
370	TUBING 4-1/2", 18.9#, L-80,HSM3	18	9.524	423.71	3531.699	73.04	14.11	18	Protector
369	TUBING 4-1/2", 18.9#, L-80,HSM3	19	9.518	433.23	3522.181	73.51	14.51	19	Protector
368	TUBING 4-1/2", 18.9#, L-80,HSM3	20	9.525	442.76	3512.656	73.97	14.91	20	Protector
367	TUBING 4-1/2", 18.9#, L-80,HSM3	21	9.524	452.28	3503.132	74.44	15.31	21	Protector
366	TUBING 4-1/2", 18.9#, L-80,HSM3	22	9.525	461.81	3493.607	74.90	15.71	22	Protector
365	TUBING 4-1/2", 18.9#, L-80,HSM3	23	9.526	471.33	3484.081	75.37	16.12	23	Protector
364	TUBING 4-1/2", 18.9#, L-80,HSM3	24	9.530	480.86	3474.551	75.83	16.52	24	Protector
363	TUBING 4-1/2", 18.9#, L-80,HSM3	25	9.520	490.38	3465.031	76.30	16.92	25	Protector
362	TUBING 4-1/2", 18.9#, L-80,HSM3	26	9.526	499.91	3455.505	76.76	17.32	26	Protector
361	TUBING 4-1/2", 18.9#, L-80,HSM3	27	9.522	509.43	3445.983	77.23	17.73	27	Protector
360	TUBING 4-1/2", 18.9#, L-80,HSM3	28	9.526	518.96	3436.457	77.69	18.13	28	Protector
359	TUBING 4-1/2", 18.9#, L-80,HSM3	29	9.525	528.48	3426.932	78.16	18.53	29	Protector
358	TUBING 4-1/2", 18.9#, L-80,HSM3	30	9.523	538.00	3417.409	78.62	18.93	30	Protector
357	TUBING 4-1/2", 18.9#, L-80,HSM3	31	9.529	547.53	3407.880	79.09	19.33	31	Protector
356	TUBING 4-1/2", 18.9#, L-80,HSM3	32	9.526	557.06	3398.354	79.55	19.74	32	Protector
355	TUBING 4-1/2", 18.9#, L-80,HSM3	33	9.528	566.59	3388.826	80.02	20.14	33	Protector
354	TUBING 4-1/2", 18.9#, L-80,HSM3	34	9.526	576.11	3379.300	80.48	20.54	34	Protector
353	TUBING 4-1/2", 18.9#, L-80,HSM3	35	9.526	585.64	3369.774	80.95	20.94	35	Protector

352	TUBING 4-1/2", 18.9#, L-80,HSM3	36	9.529	595.17	3360.245	81.41	21.35	36	Protector
351	TUBING 4-1/2", 18.9#, L-80,HSM3	37	9.528	604.70	3350.717	81.88	21.75	37	Protector
350	TUBING 4-1/2", 18.9#, L-80,HSM3	38	9.528	614.22	3341.189	82.34	22.15	38	Protector
349	TUBING 4-1/2", 18.9#, L-80,HSM3	39	9.528	623.75	3331.661	82.81	22.55	39	Protector
348	TUBING 4-1/2", 18.9#, L-80,HSM3	40	9.527	633.28	3322.134	83.27	22.96	40	Protector
347	TUBING 4-1/2", 18.9#, L-80,HSM3	41	9.529	642.81	3312.605	83.74	23.36	41	Protector
346	TUBING 4-1/2", 18.9#, L-80,HSM3	42	9.535	652.34	3303.070	84.21	23.76	42	Protector
345	TUBING 4-1/2", 18.9#, L-80,HSM3	43	9.533	661.88	3293.537	84.67	24.16	43	Protector
344	TUBING 4-1/2", 18.9#, L-80,HSM3	44	9.555	671.43	3283.982	85.14	24.57	44	Protector
343	TUBING 4-1/2", 18.9#, L-80,HSM3	45	9.535	680.97	3274.447	85.60	24.97	45	Protector
342	TUBING 4-1/2", 18.9#, L-80,HSM3	46	9.544	690.51	3264.903	86.07	25.37	46	Protector
341	TUBING 4-1/2", 18.9#, L-80,HSM3	47	9.528	700.04	3255.375	86.53	25.77	47	Protector
340	TUBING 4-1/2", 18.9#, L-80,HSM3	48	9.544	709.58	3245.831	87.00	26.18	48	Protector
339	TUBING 4-1/2", 18.9#, L-80,HSM3	49	9.544	719.13	3236.287	87.47	26.58	49	Protector
338	TUBING 4-1/2", 18.9#, L-80,HSM3	50	9.554	728.68	3226.733	87.93	26.98	50	Protector
337	TUBING 4-1/2", 18.9#, L-80,HSM3	51	9.553	738.23	3217.180	88.40	27.39	51	Protector / Press Test
336	TUBING 4-1/2", 18.9#, L-80,HSM3	52	9.558	747.79	3207.622	88.87	27.79	52	Protector
335	TUBING 4-1/2", 18.9#, L-80,HSM3	53	9.549	757.34	3198.073	89.33	28.19	53	Protector
334	TUBING 4-1/2", 18.9#, L-80,HSM3	54	9.566	766.91	3188.507	89.80	28.60	54	Protector
333	TUBING 4-1/2", 18.9#, L-80,HSM3	55	9.404	776.31	3179.103	90.26	29.00	55	Protector
332	TUBING 4-1/2", 18.9#, L-80,HSM3	56	9.554	785.86	3169.549	90.72	29.40	56	Protector
331	TUBING 4-1/2", 18.9#, L-80,HSM3	57	9.426	795.29	3160.123	91.18	29.80	57	Protector
330	TUBING 4-1/2", 18.9#, L-80,HSM3	58	9.418	804.71	3150.705	91.64	30.19	58	Protector
329	TUBING 4-1/2", 18.9#, L-80,HSM3	59	9.468	814.18	3141.237	92.11	30.59	59	Protector
328	TUBING 4-1/2", 18.9#, L-80,HSM3	60	9.528	823.70	3131.709	92.57	31.00	60	Protector
327	TUBING 4-1/2", 18.9#, L-80,HSM3	61	9.526	833.23	3122.183	93.04	31.40	61	Protector
326	TUBING 4-1/2", 18.9#, L-80,HSM3	62	9.524	842.75	3112.659	93.50	31.80	62	Protector
325	TUBING 4-1/2", 18.9#, L-80,HSM3	63	9.448	852.20	3103.211	93.96	32.20	63	Protector
324	TUBING 4-1/2", 18.9#, L-80,HSM3	64	9.527	861.73	3093.684	94.43	32.60	64	Protector
323	TUBING 4-1/2", 18.9#, L-80,HSM3	65	9.522	871.25	3084.162	94.89	33.00	65	Protector
322	TUBING 4-1/2", 18.9#, L-80,HSM3	66	9.525	880.78	3074.637	95.36	33.41	66	Protector
321	TUBING 4-1/2", 18.9#, L-80,HSM3	67	9.525	890.30	3065.112	95.82	33.81	67	Protector
320	TUBING 4-1/2", 18.9#, L-80,HSM3	68	9.527	899.83	3055.585	96.29	34.21	68	Protector
319	TUBING 4-1/2", 18.9#, L-80,HSM3	69	9.522	909.35	3046.063	96.75	34.61	69	Protector
318	TUBING 4-1/2", 18.9#, L-80,HSM3	70	9.525	918.88	3036.538	97.22	35.02	70	Protector
317	TUBING 4-1/2", 18.9#, L-80,HSM3	71	9.398	928.27	3027.140	97.68	35.41	71	Protector
316	TUBING 4-1/2", 18.9#, L-80,HSM3	72	9.521	937.79	3017.619	98.14	35.82	72	Protector
315	TUBING 4-1/2", 18.9#, L-80,HSM3	73	9.525	947.32	3008.094	98.61	36.22	73	Protector
314	TUBING 4-1/2", 18.9#, L-80,HSM3	74	9.529	956.85	2998.565	99.07	36.62	74	Protector
313	TUBING 4-1/2", 18.9#, L-80,HSM3	75	9.527	966.38	2989.038	99.54	37.02	75	Protector

312	TUBING 4-1/2", 18.9#, L-80,HSM3	76	9.273	975.65	2979.765	99.99	37.41	76	Protector
311	TUBING 4-1/2", 18.9#, L-80,HSM3	77	9.527	985.18	2970.238	100.45	37.82	77	Protector
310	TUBING 4-1/2", 18.9#, L-80,HSM3	78	9.529	994.70	2960.709	100.92	38.22	78	Protector
309	TUBING 4-1/2", 18.9#, L-80,HSM3	79	9.528	1004.23	2951.181	101.39	38.62	79	Protector
308	TUBING 4-1/2", 18.9#, L-80,HSM3	80	9.527	1013.76	2941.654	101.85	39.02	80	Protector
307	TUBING 4-1/2", 18.9#, L-80,HSM3	81	9.529	1023.29	2932.125	102.32	39.43	81	Protector
306	TUBING 4-1/2", 18.9#, L-80,HSM3	82	9.527	1032.82	2922.598	102.78	39.83	82	Protector
305	TUBING 4-1/2", 18.9#, L-80,HSM3	83	9.529	1042.34	2913.069	103.25	40.23	83	Protector
304	TUBING 4-1/2", 18.9#, L-80,HSM3	84	9.528	1051.87	2903.541	103.71	40.63	84	Protector
303	TUBING 4-1/2", 18.9#, L-80,HSM3	85	9.418	1061.29	2894.123	104.17	41.03	85	Protector
302	TUBING 4-1/2", 18.9#, L-80,HSM3	86	9.524	1070.81	2884.599	104.64	41.43	86	Protector
301	TUBING 4-1/2", 18.9#, L-80,HSM3	87	9.431	1080.25	2875.168	105.10	41.83	87	Protector
300	TUBING 4-1/2", 18.9#, L-80,HSM3	88	9.325	1089.57	2865.843	105.55	42.22	88	Protector
299	TUBING 4-1/2", 18.9#, L-80,HSM3	89	9.391	1098.96	2856.452	106.01	42.62	89	Protector
298	TUBING 4-1/2", 18.9#, L-80,HSM3	90	9.410	1108.37	2847.042	106.47	43.02	90	Protector
297	TUBING 4-1/2", 18.9#, L-80,HSM3	91	9.400	1117.77	2837.642	106.93	43.42	91	Protector
296	TUBING 4-1/2", 18.9#, L-80,HSM3	92	9.406	1127.18	2828.236	107.39	43.81	92	Protector
295	TUBING 4-1/2", 18.9#, L-80,HSM3	93	9.525	1136.70	2818.711	107.85	44.21	93	Protector
294	TUBING 4-1/2", 18.9#, L-80,HSM3	94	9.524	1146.23	2809.187	108.32	44.62	94	Protector
293	TUBING 4-1/2", 18.9#, L-80,HSM3	95	9.522	1155.75	2799.665	108.78	45.02	95	Protector
292	TUBING 4-1/2", 18.9#, L-80,HSM3	96	9.522	1165.27	2790.143	109.25	45.42	96	Protector
291	TUBING 4-1/2", 18.9#, L-80,HSM3	97	9.532	1174.80	2780.611	109.71	45.82	97	Protector
290	TUBING 4-1/2", 18.9#, L-80,HSM3	98	9.528	1184.33	2771.083	110.18	46.23	98	Protector
289	TUBING 4-1/2", 18.9#, L-80,HSM3	99	9.530	1193.86	2761.553	110.64	46.63	99	Protector
288	TUBING 4-1/2", 18.9#, L-80,HSM3	100	9.530	1203.39	2752.023	111.11	47.03	100	Protector
287	TUBING 4-1/2", 18.9#, L-80,HSM3	101	9.534	1212.92	2742.489	111.57	47.43	101	Protector / Press Test
286	TUBING 4-1/2", 18.9#, L-80,HSM3	102	9.531	1222.46	2732.958	112.04	47.84	102	Protector
285	TUBING 4-1/2", 18.9#, L-80,HSM3	103	9.530	1231.99	2723.428	112.50	48.24	103	Protector
284	TUBING 4-1/2", 18.9#, L-80,HSM3	104	9.530	1241.52	2713.898	112.97	48.64	104	Protector
283	TUBING 4-1/2", 18.9#, L-80,HSM3	105	9.528	1251.04	2704.370	113.44	49.04	105	Protector
282	TUBING 4-1/2", 18.9#, L-80,HSM3	106	9.531	1260.57	2694.839	113.90	49.45	106	Protector
281	TUBING 4-1/2", 18.9#, L-80,HSM3	107	9.525	1270.10	2685.314	114.37	49.85	107	Protector
280	TUBING 4-1/2", 18.9#, L-80,HSM3	108	9.527	1279.63	2675.787	114.83	50.25	108	Protector
279	TUBING 4-1/2", 18.9#, L-80,HSM3	109	9.525	1289.15	2666.262	115.30	50.65	109	Protector
278	TUBING 4-1/2", 18.9#, L-80,HSM3	110	9.527	1298.68	2656.735	115.76	51.06	110	Protector
277	TUBING 4-1/2", 18.9#, L-80,HSM3	111	9.528	1308.21	2647.207	116.23	51.46	111	Protector
276	TUBING 4-1/2", 18.9#, L-80,HSM3	112	9.527	1317.73	2637.680	116.69	51.86	112	Protector
275	TUBING 4-1/2", 18.9#, L-80,HSM3	113	9.523	1327.26	2628.157	117.16	52.26	113	Protector
274	TUBING 4-1/2", 18.9#, L-80,HSM3	114	9.523	1336.78	2618.634	117.62	52.66	114	Protector
273	TUBING 4-1/2", 18.9#, L-80,HSM3	115	9.527	1346.31	2609.107	118.09	53.07	115	Protector

272	TUBING 4-1/2", 18.9#, L-80,HSM3	116	9.522	1355.83	2599.585	118.55	53.47	116	Protector
271	TUBING 4-1/2", 18.9#, L-80,HSM3	117	9.522	1365.35	2590.063	119.02	53.87	117	Protector
270	TUBING 4-1/2", 18.9#, L-80,HSM3	118	9.528	1374.88	2580.535	119.48	54.27	118	Protector
269	TUBING 4-1/2", 18.9#, L-80,HSM3	119	9.552	1384.43	2570.983	119.95	54.68	119	Protector
268	TUBING 4-1/2", 18.9#, L-80,HSM3	120	9.553	1393.98	2561.430	120.41	55.08	120	Protector
267	TUBING 4-1/2", 18.9#, L-80,HSM3	121	9.532	1403.52	2551.898	120.88	55.48	121	Protector
266	TUBING 4-1/2", 18.9#, L-80,HSM3	122	9.533	1413.05	2542.365	121.34	55.89	122	Protector
265	TUBING 4-1/2", 18.9#, L-80,HSM3	123	9.540	1422.59	2532.825	121.81	56.29	123	Protector
264	TUBING 4-1/2", 18.9#, L-80,HSM3	124	9.532	1432.12	2523.293	122.28	56.69	124	Protector
263	TUBING 4-1/2", 18.9#, L-80,HSM3	125	9.527	1441.65	2513.766	122.74	57.09	125	Protector
262	TUBING 4-1/2", 18.9#, L-80,HSM3	126	9.527	1451.17	2504.239	123.21	57.50	126	Protector
261	TUBING 4-1/2", 18.9#, L-80,HSM3	127	9.531	1460.71	2494.708	123.67	57.90	127	Protector
260	TUBING 4-1/2", 18.9#, L-80,HSM3	128	9.533	1470.24	2485.175	124.14	58.30	128	Protector
259	TUBING 4-1/2", 18.9#, L-80,HSM3	129	9.528	1479.77	2475.647	124.60	58.70	129	Protector
258	TUBING 4-1/2", 18.9#, L-80,HSM3	130	9.531	1489.30	2466.116	125.07	59.11	130	Protector
257	TUBING 4-1/2", 18.9#, L-80,HSM3	131	9.522	1498.82	2456.594	125.53	59.51	131	Protector
256	TUBING 4-1/2", 18.9#, L-80,HSM3	132	9.522	1508.34	2447.072	126.00	59.91	132	Protector
255	TUBING 4-1/2", 18.9#, L-80,HSM3	133	9.527	1517.87	2437.545	126.46	60.31	133	Protector
254	TUBING 4-1/2", 18.9#, L-80,HSM3	134	9.527	1527.40	2428.018	126.93	60.71	134	Protector
253	TUBING 4-1/2", 18.9#, L-80,HSM3	135	9.531	1536.93	2418.487	127.39	61.12	135	Protector
252	TUBING 4-1/2", 18.9#, L-80,HSM3	136	9.527	1546.45	2408.960	127.86	61.52	136	Protector
251	TUBING 4-1/2", 18.9#, L-80,HSM3	137	9.530	1555.98	2399.430	128.32	61.92	137	Protector
250	TUBING 4-1/2", 18.9#, L-80,HSM3	138	9.435	1565.42	2389.995	128.78	62.32	138	Protector
249	TUBING 4-1/2", 18.9#, L-80,HSM3	139	9.531	1574.95	2380.464	129.25	62.72	139	Protector
248	TUBING 4-1/2", 18.9#, L-80,HSM3	140	9.531	1584.48	2370.933	129.71	63.12	140	Protector
247	TUBING 4-1/2", 18.9#, L-80,HSM3	141	9.530	1594.01	2361.403	130.18	63.53	141	Protector
246	TUBING 4-1/2", 18.9#, L-80,HSM3	142	9.381	1603.39	2352.022	130.64	63.92	142	Protector
245	TUBING 4-1/2", 18.9#, L-80,HSM3	143	9.529	1612.92	2342.493	131.10	64.33	143	Protector
244	TUBING 4-1/2", 18.9#, L-80,HSM3	144	9.528	1622.45	2332.965	131.57	64.73	144	Protector
243	TUBING 4-1/2", 18.9#, L-80,HSM3	145	9.529	1631.98	2323.436	132.03	65.13	145	Protector
242	TUBING 4-1/2", 18.9#, L-80,HSM3	146	9.531	1641.51	2313.905	132.50	65.53	146	Protector
241	TUBING 4-1/2", 18.9#, L-80,HSM3	147	9.530	1651.04	2304.375	132.96	65.94	147	Protector
240	TUBING 4-1/2", 18.9#, L-80,HSM3	148	9.530	1660.57	2294.845	133.43	66.34	148	Protector
239	TUBING 4-1/2", 18.9#, L-80,HSM3	149	9.528	1670.10	2285.317	133.89	66.74	149	Protector
238	TUBING 4-1/2", 18.9#, L-80,HSM3	150	9.530	1679.63	2275.787	134.36	67.14	150	Protector / Press Test
237	TUBING 4-1/2", 18.9#, L-80,HSM3	151	9.527	1689.15	2266.260	134.82	67.54	151	Protector
236	TUBING 4-1/2", 18.9#, L-80,HSM3	152	9.530	1698.68	2256.730	135.29	67.95	152	Protector
235	TUBING 4-1/2", 18.9#, L-80,HSM3	153	9.529	1708.21	2247.201	135.76	68.35	153	Protector
234	TUBING 4-1/2", 18.9#, L-80,HSM3	154	9.533	1717.75	2237.668	136.22	68.75	154	Protector
233	TUBING 4-1/2", 18.9#, L-80,HSM3	155	9.528	1727.27	2228.140	136.69	69.15	155	Protector

232	TUBING 4-1/2", 18.9#, L-80,HSM3	156	9.528	1736.80	2218.612	137.15	69.56	156	Protector
231	TUBING 4-1/2", 18.9#, L-80,HSM3	157	9.529	1746.33	2209.083	137.62	69.96	157	Protector
230	TUBING 4-1/2", 18.9#, L-80,HSM3	158	9.523	1755.85	2199.560	138.08	70.36	158	Protector
229	TUBING 4-1/2", 18.9#, L-80,HSM3	159	9.525	1765.38	2190.035	138.55	70.76	159	Protector
228	TUBING 4-1/2", 18.9#, L-80,HSM3	160	9.528	1774.91	2180.507	139.01	71.17	160	Protector
227	TUBING 4-1/2", 18.9#, L-80,HSM3	161	9.525	1784.43	2170.982	139.48	71.57	161	Protector
226	TUBING 4-1/2", 18.9#, L-80,HSM3	162	9.530	1793.96	2161.452	139.94	71.97	162	Protector
225	TUBING 4-1/2", 18.9#, L-80,HSM3	163	9.525	1803.49	2151.927	140.41	72.37	163	Protector
224	TUBING 4-1/2", 18.9#, L-80,HSM3	164	9.523	1813.01	2142.404	140.87	72.78	164	Protector
223	TUBING 4-1/2", 18.9#, L-80,HSM3	165	9.432	1822.44	2132.972	141.33	73.17	165	Protector
222	TUBING 4-1/2", 18.9#, L-80,HSM3	166	9.555	1832.00	2123.417	141.80	73.58	166	Protector
221	TUBING 4-1/2", 18.9#, L-80,HSM3	167	9.399	1841.40	2114.018	142.26	73.97	167	Protector
220	TUBING 4-1/2", 18.9#, L-80,HSM3	168	9.546	1850.94	2104.472	142.72	74.38	168	Protector
219	TUBING 4-1/2", 18.9#, L-80,HSM3	169	9.408	1860.35	2095.064	143.18	74.77	169	Protector
218	TUBING 4-1/2", 18.9#, L-80,HSM3	170	9.558	1869.91	2085.506	143.65	75.18	170	Protector
217	TUBING 4-1/2", 18.9#, L-80,HSM3	171	9.555	1879.46	2075.951	144.12	75.58	171	Protector
216	TUBING 4-1/2", 18.9#, L-80,HSM3	172	9.526	1888.99	2066.425	144.58	75.98	172	Protector
215	TUBING 4-1/2", 18.9#, L-80,HSM3	173	9.532	1898.52	2056.893	145.05	76.39	173	Protector
214	TUBING 4-1/2", 18.9#, L-80,HSM3	174	9.521	1908.04	2047.372	145.51	76.79	174	Protector
213	TUBING 4-1/2", 18.9#, L-80,HSM3	175	9.527	1917.57	2037.845	145.98	77.19	175	Protector
212	TUBING 4-1/2", 18.9#, L-80,HSM3	176	9.522	1927.09	2028.323	146.44	77.59	176	Protector
211	TUBING 4-1/2", 18.9#, L-80,HSM3	177	9.525	1936.62	2018.798	146.91	78.00	177	Protector
210	TUBING 4-1/2", 18.9#, L-80,HSM3	178	9.531	1946.15	2009.267	147.37	78.40	178	Protector
209	TUBING 4-1/2", 18.9#, L-80,HSM3	179	9.527	1955.67	1999.740	147.84	78.80	179	Protector
208	TUBING 4-1/2", 18.9#, L-80,HSM3	180	9.528	1965.20	1990.212	148.30	79.20	180	Protector
207	TUBING 4-1/2", 18.9#, L-80,HSM3	181	9.527	1974.73	1980.685	148.77	79.60	181	Protector
206	TUBING 4-1/2", 18.9#, L-80,HSM3	182	9.532	1984.26	1971.153	149.23	80.01	182	Protector
205	TUBING 4-1/2", 18.9#, L-80,HSM3	183	9.528	1993.79	1961.625	149.70	80.41	183	Protector
204	TUBING 4-1/2", 18.9#, L-80,HSM3	184	9.525	2003.31	1952.100	150.16	80.81	184	Protector
203	TUBING 4-1/2", 18.9#, L-80,HSM3	185	9.525	2012.84	1942.575	150.63	81.21	185	Protector
202	TUBING 4-1/2", 18.9#, L-80,HSM3	186	9.529	2022.37	1933.046	151.09	81.62	186	Protector
201	TUBING 4-1/2", 18.9#, L-80,HSM3	187	9.531	2031.90	1923.515	151.56	82.02	187	Protector
200	TUBING 4-1/2", 18.9#, L-80,HSM3	188	9.519	2041.42	1913.996	152.02	82.42	188	Protector
199	TUBING 4-1/2", 18.9#, L-80,HSM3	189	9.529	2050.95	1904.467	152.49	82.82	189	Protector
198	TUBING 4-1/2", 18.9#, L-80,HSM3	190	9.525	2060.47	1894.942	152.95	83.23	190	Protector
197	TUBING 4-1/2", 18.9#, L-80,HSM3	191	9.525	2070.00	1885.417	153.42	83.63	191	Protector
196	TUBING 4-1/2", 18.9#, L-80,HSM3	192	9.522	2079.52	1875.895	153.88	84.03	192	Protector
195	TUBING 4-1/2", 18.9#, L-80,HSM3	193	9.523	2089.04	1866.372	154.35	84.43	193	Protector
194	TUBING 4-1/2", 18.9#, L-80,HSM3	194	9.522	2098.56	1856.850	154.81	84.83	194	Protector
193	TUBING 4-1/2", 18.9#, L-80,HSM3	195	9.522	2108.09	1847.328	155.28	85.24	195	Protector

192	TUBING 4-1/2", 18.9#, L-80,HSM3	196	9.525	2117.61	1837.803	155.74	85.64	196	Protector
191	TUBING 4-1/2", 18.9#, L-80,HSM3	197	9.527	2127.14	1828.276	156.21	86.04	197	Protector
190	TUBING 4-1/2", 18.9#, L-80,HSM3	198	9.525	2136.66	1818.751	156.67	86.44	198	Protector
189	TUBING 4-1/2", 18.9#, L-80,HSM3	199	9.537	2146.20	1809.214	157.14	86.85	199	Protector
188	TUBING 4-1/2", 18.9#, L-80,HSM3	200	9.519	2155.72	1799.695	157.60	87.25	200	Protector / Press Test
187	TUBING 4-1/2", 18.9#, L-80,HSM3	201	9.526	2165.24	1790.169	158.07	87.65	201	Protector
186	TUBING 4-1/2", 18.9#, L-80,HSM3	202	9.526	2174.77	1780.643	158.53	88.05	202	Protector
185	TUBING 4-1/2", 18.9#, L-80,HSM3	203	9.537	2184.31	1771.106	159.00	88.46	203	Protector
184	TUBING 4-1/2", 18.9#, L-80,HSM3	204	9.532	2193.84	1761.574	159.46	88.86	204	Protector
183	TUBING 4-1/2", 18.9#, L-80,HSM3	205	9.531	2203.37	1752.043	159.93	89.26	205	Protector
182	TUBING 4-1/2", 18.9#, L-80,HSM3	206	9.531	2212.90	1742.512	160.40	89.66	206	Protector
181	TUBING 4-1/2", 18.9#, L-80,HSM3	207	9.531	2222.43	1732.981	160.86	90.07	207	Protector
180	TUBING 4-1/2", 18.9#, L-80,HSM3	208	9.533	2231.97	1723.448	161.33	90.47	208	Protector
179	TUBING 4-1/2", 18.9#, L-80,HSM3	209	9.526	2241.49	1713.922	161.79	90.87	209	Protector
178	TUBING 4-1/2", 18.9#, L-80,HSM3	210	9.532	2251.02	1704.390	162.26	91.27	210	Protector
177	TUBING 4-1/2", 18.9#, L-80,HSM3	211	9.527	2260.55	1694.863	162.72	91.67	211	Protector
176	TUBING 4-1/2", 18.9#, L-80,HSM3	212	9.527	2270.08	1685.336	163.19	92.08	212	Protector
175	TUBING 4-1/2", 18.9#, L-80,HSM3	213	9.528	2279.61	1675.808	163.65	92.48	213	Protector
174	TUBING 4-1/2", 18.9#, L-80,HSM3	214	9.526	2289.13	1666.282	164.12	92.88	214	Protector
173	TUBING 4-1/2", 18.9#, L-80,HSM3	215	9.405	2298.54	1656.877	164.58	93.28	215	Protector
172	TUBING 4-1/2", 18.9#, L-80,HSM3	216	9.293	2307.83	1647.584	165.03	93.67	216	Protector
171	TUBING 4-1/2", 18.9#, L-80,HSM3	217	9.570	2317.40	1638.014	165.50	94.08	217	Protector
170	TUBING 4-1/2", 18.9#, L-80,HSM3	218	9.415	2326.81	1628.599	165.96	94.47	218	Protector
169	TUBING 4-1/2", 18.9#, L-80,HSM3	219	9.405	2336.22	1619.194	166.42	94.87	219	Protector
168	TUBING 4-1/2", 18.9#, L-80,HSM3	220	9.409	2345.63	1609.785	166.88	95.27	220	Protector
167	TUBING 4-1/2", 18.9#, L-80,HSM3	221	9.324	2354.95	1600.461	167.33	95.66	221	Protector
166	TUBING 4-1/2", 18.9#, L-80,HSM3	222	9.530	2364.48	1590.931	167.80	96.06	222	Protector
165	TUBING 4-1/2", 18.9#, L-80,HSM3	223	9.558	2374.04	1581.373	168.26	96.47	223	Protector
164	TUBING 4-1/2", 18.9#, L-80,HSM3	224	9.531	2383.57	1571.842	168.73	96.87	224	Protector
163	TUBING 4-1/2", 18.9#, L-80,HSM3	225	9.557	2393.13	1562.285	169.19	97.27	225	Protector
162	TUBING 4-1/2", 18.9#, L-80,HSM3	226	9.545	2402.67	1552.740	169.66	97.68	226	Protector
161	TUBING 4-1/2", 18.9#, L-80,HSM3	227	9.530	2412.20	1543.210	170.13	98.08	227	Protector
160	TUBING 4-1/2", 18.9#, L-80,HSM3	228	9.532	2421.74	1533.678	170.59	98.48	228	Protector
159	TUBING 4-1/2", 18.9#, L-80,HSM3	229	9.525	2431.26	1524.153	171.06	98.88	229	Protector
158	TUBING 4-1/2", 18.9#, L-80,HSM3	230	9.525	2440.79	1514.628	171.52	99.29	230	Protector
157	TUBING 4-1/2", 18.9#, L-80,HSM3	231	9.525	2450.31	1505.103	171.99	99.69	231	Protector
156	TUBING 4-1/2", 18.9#, L-80,HSM3	232	9.522	2459.83	1495.581	172.45	100.09	232	Protector
155	TUBING 4-1/2", 18.9#, L-80,HSM3	233	9.530	2469.36	1486.051	172.92	100.49	233	Protector
154	TUBING 4-1/2", 18.9#, L-80,HSM3	234	9.530	2478.89	1476.521	173.38	100.90	234	Protector
153	TUBING 4-1/2", 18.9#, L-80,HSM3	235	9.524	2488.42	1466.997	173.85	101.30	235	Protector

152	TUBING 4-1/2", 18.9#, L-80,HSM3	236	9.529	2497.95	1457.468	174.31	101.70	236	Protector
151	TUBING 4-1/2", 18.9#, L-80,HSM3	237	9.529	2507.47	1447.939	174.78	102.10	237	Protector
150	TUBING 4-1/2", 18.9#, L-80,HSM3	238	9.408	2516.88	1438.531	175.24	102.50	238	Protector
149	TUBING 4-1/2", 18.9#, L-80,HSM3	239	9.352	2526.23	1429.179	175.69	102.89	239	Protector
148	TUBING 4-1/2", 18.9#, L-80,HSM3	240	9.530	2535.76	1419.649	176.16	103.30	240	Protector
147	TUBING 4-1/2", 18.9#, L-80,HSM3	241	9.528	2545.29	1410.121	176.62	103.70	241	Protector
146	TUBING 4-1/2", 18.9#, L-80,HSM3	242	9.532	2554.82	1400.589	177.09	104.10	242	Protector
145	TUBING 4-1/2", 18.9#, L-80,HSM3	243	9.531	2564.36	1391.058	177.55	104.50	243	Protector
144	TUBING 4-1/2", 18.9#, L-80,HSM3	244	9.530	2573.89	1381.528	178.02	104.91	244	Protector
143	TUBING 4-1/2", 18.9#, L-80,HSM3	245	9.528	2583.41	1372.000	178.48	105.31	245	Protector
142	TUBING 4-1/2", 18.9#, L-80,HSM3	246	9.529	2592.94	1362.471	178.95	105.71	246	Protector
141	TUBING 4-1/2", 18.9#, L-80,HSM3	247	9.527	2602.47	1352.944	179.42	106.11	247	Protector
140	TUBING 4-1/2", 18.9#, L-80,HSM3	248	9.530	2612.00	1343.414	179.88	106.52	248	Protector
139	TUBING 4-1/2", 18.9#, L-80,HSM3	249	9.530	2621.53	1333.884	180.35	106.92	249	Protector
138	TUBING 4-1/2", 18.9#, L-80,HSM3	250	9.527	2631.06	1324.357	180.81	107.32	250	Protector / Press Test
137	TUBING 4-1/2", 18.9#, L-80,HSM3	251	9.527	2640.58	1314.830	181.28	107.72	251	Protector
136	TUBING 4-1/2", 18.9#, L-80,HSM3	252	9.532	2650.12	1305.298	181.74	108.13	252	Protector
135	TUBING 4-1/2", 18.9#, L-80,HSM3	253	9.526	2659.64	1295.772	182.21	108.53	253	Protector
134	TUBING 4-1/2", 18.9#, L-80,HSM3	254	9.531	2669.17	1286.241	182.67	108.93	254	Protector
133	TUBING 4-1/2", 18.9#, L-80,HSM3	255	9.525	2678.70	1276.716	183.14	109.33	255	Protector
132	TUBING 4-1/2", 18.9#, L-80,HSM3	256	9.524	2688.22	1267.192	183.60	109.74	256	Protector
131	TUBING 4-1/2", 18.9#, L-80,HSM3	257	9.524	2697.75	1257.668	184.07	110.14	257	Protector
130	TUBING 4-1/2", 18.9#, L-80,HSM3	258	9.524	2707.27	1248.144	184.53	110.54	258	Protector
129	TUBING 4-1/2", 18.9#, L-80,HSM3	259	9.523	2716.79	1238.621	185.00	110.94	259	Protector
128	TUBING 4-1/2", 18.9#, L-80,HSM3	260	9.526	2726.32	1229.095	185.46	111.34	260	Protector
127	TUBING 4-1/2", 18.9#, L-80,HSM3	261	9.524	2735.84	1219.571	185.93	111.75	261	Protector
126	TUBING 4-1/2", 18.9#, L-80,HSM3	262	9.530	2745.37	1210.041	186.39	112.15	262	Protector
125	TUBING 4-1/2", 18.9#, L-80,HSM3	263	9.524	2754.90	1200.517	186.86	112.55	263	Protector
124	TUBING 4-1/2", 18.9#, L-80,HSM3	264	9.531	2764.43	1190.986	187.32	112.95	264	Protector
123	TUBING 4-1/2", 18.9#, L-80,HSM3	265	9.530	2773.96	1181.456	187.79	113.36	265	Protector
122	TUBING 4-1/2", 18.9#, L-80,HSM3	266	9.530	2783.49	1171.926	188.25	113.76	266	Protector
121	TUBING 4-1/2", 18.9#, L-80,HSM3	267	9.525	2793.01	1162.401	188.72	114.16	267	Protector
120	TUBING 4-1/2", 18.9#, L-80,HSM3	268	9.521	2802.53	1152.880	189.18	114.56	268	Protector
119	TUBING 4-1/2", 18.9#, L-80,HSM3	269	9.528	2812.06	1143.352	189.65	114.97	269	Protector
118	TUBING 4-1/2", 18.9#, L-80,HSM3	270	9.518	2821.58	1133.834	190.11	115.37	270	Protector
117	TUBING 4-1/2", 18.9#, L-80,HSM3	271	9.527	2831.11	1124.307	190.58	115.77	271	Protector
116	TUBING 4-1/2", 18.9#, L-80,HSM3	272	9.517	2840.62	1114.790	191.04	116.17	272	Protector
115	TUBING 4-1/2", 18.9#, L-80,HSM3	273	9.527	2850.15	1105.263	191.51	116.57	273	Protector
114	TUBING 4-1/2", 18.9#, L-80,HSM3	274	9.527	2859.68	1095.736	191.97	116.98	274	Protector
113	TUBING 4-1/2", 18.9#, L-80,HSM3	275	9.530	2869.21	1086.206	192.44	117.38	275	Protector

112	TUBING 4-1/2", 18.9#, L-80,HSM3	276	9.531	2878.74	1076.675	192.90	117.78	276	Protector
111	TUBING 4-1/2", 18.9#, L-80,HSM3	277	9.530	2888.27	1067.145	193.37	118.18	277	Protector
110	TUBING 4-1/2", 18.9#, L-80,HSM3	278	9.530	2897.80	1057.615	193.83	118.59	278	Protector
109	TUBING 4-1/2", 18.9#, L-80,HSM3	279	9.532	2907.33	1048.083	194.30	118.99	279	Protector
108	TUBING 4-1/2", 18.9#, L-80,HSM3	280	9.546	2916.88	1038.537	194.77	119.39	280	Protector
107	TUBING 4-1/2", 18.9#, L-80,HSM3	281	9.530	2926.41	1029.007	195.23	119.79	281	Protector
106	TUBING 4-1/2", 18.9#, L-80,HSM3	282	9.531	2935.94	1019.476	195.70	120.20	282	Protector
105	TUBING 4-1/2", 18.9#, L-80,HSM3	283	9.527	2945.46	1009.949	196.16	120.60	283	Protector
104	TUBING 4-1/2", 18.9#, L-80,HSM3	284	9.532	2955.00	1000.417	196.63	121.00	284	Protector
103	TUBING 4-1/2", 18.9#, L-80,HSM3	285	9.524	2964.52	990.893	197.09	121.40	285	Protector
102	TUBING 4-1/2", 18.9#, L-80,HSM3	286	9.534	2974.05	981.359	197.56	121.81	286	Protector
101	TUBING 4-1/2", 18.9#, L-80,HSM3	287	9.532	2983.59	971.827	198.02	122.21	287	Protector
100	TUBING 4-1/2", 18.9#, L-80,HSM3	288	9.529	2993.12	962.298	198.49	122.61	288	Protector
99	TUBING 4-1/2", 18.9#, L-80,HSM3	289	9.531	3002.65	952.767	198.95	123.01	289	Protector
98	TUBING 4-1/2", 18.9#, L-80,HSM3	290	9.528	3012.17	943.239	199.42	123.42	290	Protector
97	TUBING 4-1/2", 18.9#, L-80,HSM3	291	9.528	3021.70	933.711	199.88	123.82	291	Protector
96	TUBING 4-1/2", 18.9#, L-80,HSM3	292	9.530	3031.23	924.181	200.35	124.22	292	Protector
95	TUBING 4-1/2", 18.9#, L-80,HSM3	293	9.527	3040.76	914.654	200.81	124.62	293	Protector
94	TUBING 4-1/2", 18.9#, L-80,HSM3	294	9.532	3050.29	905.122	201.28	125.03	294	Protector
93	TUBING 4-1/2", 18.9#, L-80,HSM3	295	9.532	3059.82	895.590	201.74	125.43	295	Protector
92	TUBING 4-1/2", 18.9#, L-80,HSM3	296	9.528	3069.35	886.062	202.21	125.83	296	Protector
91	TUBING 4-1/2", 18.9#, L-80,HSM3	297	9.530	3078.88	876.532	202.67	126.23	297	Protector
90	TUBING 4-1/2", 18.9#, L-80,HSM3	298	9.528	3088.41	867.004	203.14	126.64	298	Protector
89	TUBING 4-1/2", 18.9#, L-80,HSM3	299	9.530	3097.94	857.474	203.61	127.04	299	Protector
88	TUBING 4-1/2", 18.9#, L-80,HSM3	300	9.531	3107.47	847.943	204.07	127.44	300	Protector / Press Test
87	TUBING 4-1/2", 18.9#, L-80,HSM3	301	9.530	3117.00	838.413	204.54	127.84	301	Protector
86	TUBING 4-1/2", 18.9#, L-80,HSM3	302	9.560	3126.56	828.853	205.00	128.25	302	Protector
85	TUBING 4-1/2", 18.9#, L-80,HSM3	303	9.557	3136.12	819.296	205.47	128.65	303	Protector
84	TUBING 4-1/2", 18.9#, L-80,HSM3	304	9.561	3145.68	809.735	205.94	129.05	304	Protector
83	TUBING 4-1/2", 18.9#, L-80,HSM3	305	9.558	3155.24	800.177	206.40	129.46	305	Protector
82	TUBING 4-1/2", 18.9#, L-80,HSM3	306	9.533	3164.77	790.644	206.87	129.86	306	Protector
81	TUBING 4-1/2", 18.9#, L-80,HSM3	307	9.558	3174.33	781.086	207.33	130.26	307	Protector
80	TUBING 4-1/2", 18.9#, L-80,HSM3	308	9.557	3183.88	771.529	207.80	130.67	308	Protector
79	TUBING 4-1/2", 18.9#, L-80,HSM3	309	9.390	3193.27	762.139	208.26	131.06	309	Protector
78	TUBING 4-1/2", 18.9#, L-80,HSM3	310	9.376	3202.65	752.763	208.72	131.46	310	Protector
77	TUBING 4-1/2", 18.9#, L-80,HSM3	311	9.421	3212.07	743.342	209.18	131.86	311	Protector
76	TUBING 4-1/2", 18.9#, L-80,HSM3	312	9.397	3221.47	733.945	209.64	132.25	312	Protector
75	TUBING 4-1/2", 18.9#, L-80,HSM3	313	9.239	3230.71	724.706	210.09	132.64	313	Protector
74	TUBING 4-1/2", 18.9#, L-80,HSM3	314	9.396	3240.10	715.310	210.55	133.04	314	Protector
73	TUBING 4-1/2", 18.9#, L-80,HSM3	315	9.527	3249.63	705.783	211.01	133.44	315	Protector

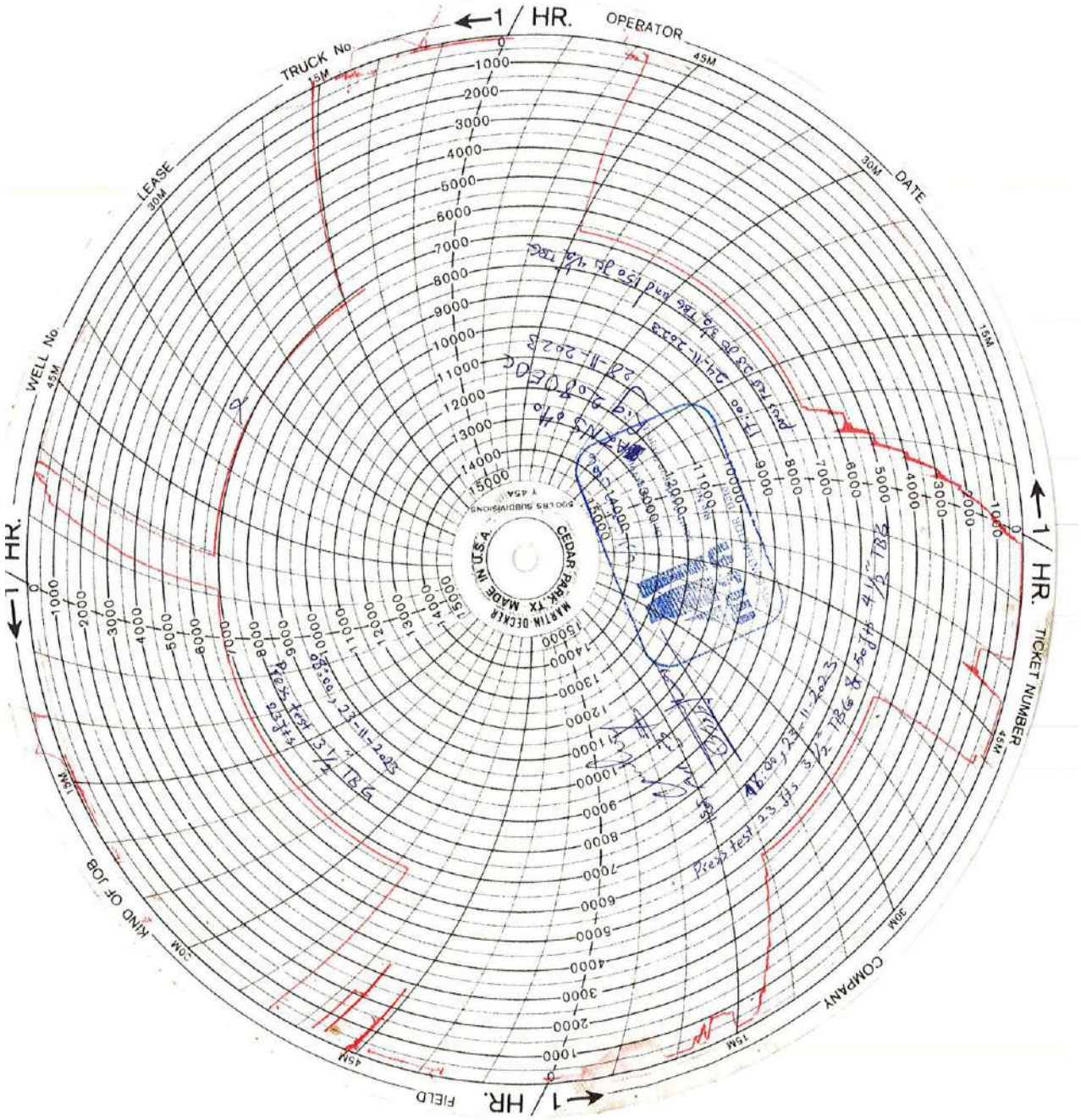
72	TUBING 4-1/2", 18.9#, L-80,HSM3	316	9.533	3259.16	696.250	211.48	133.85	316	Protector
71	TUBING 4-1/2", 18.9#, L-80,HSM3	317	9.530	3268.69	686.720	211.94	134.25	317	Protector
70	TUBING 4-1/2", 18.9#, L-80,HSM3	318	9.532	3278.23	677.188	212.41	134.65	318	Protector
69	TUBING 4-1/2", 18.9#, L-80,HSM3	319	9.533	3287.76	667.655	212.87	135.05	319	Protector
68	TUBING 4-1/2", 18.9#, L-80,HSM3	320	9.531	3297.29	658.124	213.34	135.46	320	Protector
67	TUBING 4-1/2", 18.9#, L-80,HSM3	321	9.525	3306.81	648.599	213.80	135.86	321	Protector
66	TUBING 4-1/2", 18.9#, L-80,HSM3	322	9.531	3316.35	639.068	214.27	136.26	322	Protector
65	TUBING 4-1/2", 18.9#, L-80,HSM3	323	9.534	3325.88	629.534	214.73	136.66	323	Protector
64	TUBING 4-1/2", 18.9#, L-80,HSM3	324	9.534	3335.41	620.000	215.20	137.07	324	Protector
63	TUBING 4-1/2", 18.9#, L-80,HSM3	325	9.534	3344.95	610.466	215.66	137.47	325	Protector
62	TUBING 4-1/2", 18.9#, L-80,HSM3	326	9.190	3354.14	601.276	216.11	137.86	326	Protector
61	TUBING 4-1/2", 18.9#, L-80,HSM3	327	9.531	3363.67	591.745	216.58	138.26	327	Protector
60	TUBING 4-1/2", 18.9#, L-80,HSM3	328	9.531	3373.20	582.214	217.04	138.66	328	Protector
59	TUBING 4-1/2", 18.9#, L-80,HSM3	329	9.531	3382.73	572.683	217.51	139.06	329	Protector
58	TUBING 4-1/2", 18.9#, L-80,HSM3	330	9.531	3392.26	563.152	217.97	139.47	330	Protector
57	TUBING 4-1/2", 18.9#, L-80,HSM3	331	9.533	3401.79	553.619	218.44	139.87	331	Protector
56	TUBING 4-1/2", 18.9#, L-80,HSM3	332	9.534	3411.33	544.085	218.91	140.27	332	Protector
55	TUBING 4-1/2", 18.9#, L-80,HSM3	333	9.534	3420.86	534.551	219.37	140.67	333	Protector
54	TUBING 4-1/2", 18.9#, L-80,HSM3	334	9.532	3430.39	525.019	219.84	141.08	334	Protector
53	TUBING 4-1/2", 18.9#, L-80,HSM3	335	9.531	3439.93	515.488	220.30	141.48	335	Protector
52	TUBING 4-1/2", 18.9#, L-80,HSM3	336	9.534	3449.46	505.954	220.77	141.88	336	Protector
51	TUBING 4-1/2", 18.9#, L-80,HSM3	337	9.418	3458.88	496.536	221.23	142.28	337	Protector
50	TUBING 4-1/2", 18.9#, L-80,HSM3	338	9.409	3468.29	487.127	221.69	142.68	338	Protector
49	TUBING 4-1/2", 18.9#, L-80,HSM3	339	9.428	3477.71	477.699	222.15	143.08	339	Protector
48	TUBING 4-1/2", 18.9#, L-80,HSM3	340	9.426	3487.14	468.273	222.61	143.47	340	Protector
47	TUBING 4-1/2", 18.9#, L-80,HSM3	341	9.420	3496.56	458.853	223.07	143.87	341	Protector
46	TUBING 4-1/2", 18.9#, L-80,HSM3	342	9.401	3505.96	449.452	223.53	144.27	342	Protector
45	TUBING 4-1/2", 18.9#, L-80,HSM3	343	9.412	3515.37	440.040	223.99	144.67	343	Protector
44	TUBING 4-1/2", 18.9#, L-80,HSM3	344	9.530	3524.90	430.510	224.45	145.07	344	Protector
43	TUBING 4-1/2", 18.9#, L-80,HSM3	345	9.528	3534.43	420.982	224.92	145.47	345	Protector
42	TUBING 4-1/2", 18.9#, L-80,HSM3	346	9.529	3543.96	411.453	225.38	145.87	346	Protector
41	TUBING 4-1/2", 18.9#, L-80,HSM3	347	9.532	3553.49	401.921	225.85	146.28	347	Protector
40	TUBING 4-1/2", 18.9#, L-80,HSM3	348	9.528	3563.02	392.393	226.31	146.68	348	Protector
39	TUBING 4-1/2", 18.9#, L-80,HSM3	349	9.528	3572.55	382.865	226.78	147.08	349	Protector
38	TUBING 4-1/2", 18.9#, L-80,HSM3	350	9.522	3582.07	373.343	227.24	147.48	350	Protector / Press Test
37	TUBING 4-1/2", 18.9#, L-80,HSM3	351	9.528	3591.60	363.815	227.71	147.89	351	Protector
36	TUBING 4-1/2", 18.9#, L-80,HSM3	352	9.526	3601.12	354.289	228.17	148.29	352	Protector
35	TUBING 4-1/2", 18.9#, L-80,HSM3	353	9.188	3610.31	345.101	228.62	148.68	353	Protector
34	TUBING 4-1/2", 18.9#, L-80,HSM3	354	9.528	3619.84	335.573	229.09	149.08	354	Protector
33	TUBING 4-1/2", 18.9#, L-80,HSM3	355	9.528	3629.37	326.045	229.55	149.48	355	Protector

32	TUBING 4-1/2", 18.9#, L-80,HSM3	356	9.531	3638.90	316.514	230.02	149.88	356	Protector
31	TUBING 4-1/2", 18.9#, L-80,HSM3	357	9.520	3648.42	306.994	230.48	150.28	357	Protector
30	TUBING 4-1/2", 18.9#, L-80,HSM3	358	9.528	3657.95	297.466	230.95	150.69	358	Protector
29	TUBING 4-1/2", 18.9#, L-80,HSM3	359	9.523	3667.47	287.943	231.41	151.09	359	Protector
28	TUBING 4-1/2", 18.9#, L-80,HSM3	360	9.531	3677.00	278.412	231.88	151.49	360	Protector
27	TUBING 4-1/2", 18.9#, L-80,HSM3	361	9.525	3686.53	268.887	232.34	151.89	361	Protector
26	TUBING 4-1/2", 18.9#, L-80,HSM3	362	9.525	3696.05	259.362	232.81	152.30	362	Protector
25	TUBING 4-1/2", 18.9#, L-80,HSM3	363	9.538	3705.59	249.824	233.27	152.70	363	Protector
24	TUBING 4-1/2", 18.9#, L-80,HSM3	364	9.533	3715.12	240.291	233.74	153.10	364	Protector
23	TUBING 4-1/2", 18.9#, L-80,HSM3	365	9.538	3724.66	230.753	234.20	153.50	365	Protector
22	TUBING 4-1/2", 18.9#, L-80,HSM3	366	9.536	3734.20	221.217	234.67	153.91	366	Protector
21	TUBING 4-1/2", 18.9#, L-80,HSM3	367	9.543	3743.74	211.674	235.13	154.31	367	Protector
20	TUBING 4-1/2", 18.9#, L-80,HSM3	368	9.538	3753.28	202.136	235.60	154.71	368	Protector
19	TUBING 4-1/2", 18.9#, L-80,HSM3	369	9.536	3762.81	192.600	236.07	155.12	369	Protector
18	TUBING 4-1/2", 18.9#, L-80,HSM3	370	9.569	3772.38	183.031	236.53	155.52	370	Protector
17	TUBING 4-1/2", 18.9#, L-80,HSM3	371	9.568	3781.95	173.463	237.00	155.92	371	Protector
16	TUBING 4-1/2", 18.9#, L-80,HSM3	372	9.557	3791.51	163.906	237.47	156.33	372	Protector
15	TUBING 4-1/2", 18.9#, L-80,HSM3	373	9.430	3800.94	154.476	237.93	156.73	373	Protector
14	TUBING 4-1/2", 18.9#, L-80,HSM3	374	9.569	3810.51	144.907	238.39	157.13	374	Protector
13	TUBING 4-1/2", 18.9#, L-80,HSM3	375	9.543	3820.05	135.364	238.86	157.53	375	Protector
12	TUBING 4-1/2", 18.9#, L-80,HSM3	376	9.530	3829.58	125.834	239.33	157.94	376	Protector
11	TUBING 4-1/2", 18.9#, L-80,HSM3	377	9.530	3839.11	116.304	239.79	158.34	377	Protector
10	TUBING 4-1/2", 18.9#, L-80,HSM3	378	9.529	3848.64	106.775	240.26	158.74	378	Protector
9	TUBING 4-1/2", 18.9#, L-80,HSM3	379	9.528	3858.17	97.247	240.72	159.14	379	Protector
8	TUBING 4-1/2", 18.9#, L-80,HSM3	380	9.529	3867.70	87.718	241.19	159.54	380	Protector
7	TUBING 4-1/2", 18.9#, L-80,HSM3	381	9.528	3877.22	78.190	241.65	159.95	381	Protector
6	TUBING 4-1/2", 18.9#, L-80,HSM3	382	9.526	3886.75	68.664	242.12	160.35	382	Protector

PUP JOINT 4 1/2",HSM3 ,18.9#, L-80, Type-1	PUP JOINT 4 1/2",HSM3		1.620	3888.37	67.044	242.20	160.42		
4-1/2" FLOW COUPLING, 18.9#,SUPER 13CR-110KSI,HSM3 , 4' FT, WEIQIXIN Co	Short Flow Coupling		1.115	3889.48	65.929	242.25	160.46		
TRSSV-10 Tubing Retrievable Saftey Valve, 4 1/2", 18.9# W/ 3.437" Seal bore,HSM3 , SUPER 13CR-110KSI, 10K WP With AF Profile, Flapper Type, WEIQIXIN Co	TRSSV		1.780	3891.26	64.149	242.34	160.54		
4-1/2" FLOW COUPLING,18.9#,SUPER 13CR-110KSI,HSM3 , 6 FT, WEIQIXIN Co	Long Flow Coupling		1.725	3892.99	62.424	242.42	160.61		
PUP JOINT 4 1/2",HSM3 ,18.9#, L-80, Type-1	PUP JOINT 4 1/2",HSM3		1.621	3894.61	60.803	242.50	160.68		
5	TUBING 4-1/2", 18.9#, L-80,HSM3	1	9.525	3904.14	51.278	242.97	161.08	1	Protector / Press Test
4	TUBING 4-1/2", 18.9#, L-80,HSM3	2	9.523	3913.66	41.755	243.43	161.49	2	Protector
3	TUBING 4-1/2", 18.9#, L-80,HSM3	3	9.527	3923.19	32.228	243.90	161.89	3	Protector
2	TUBING 4-1/2", 18.9#, L-80,HSM3	4	9.519	3932.70	22.709	244.36	162.29	4	Protector
1	TUBING 4-1/2", 18.9#, L-80,HSM3	5	9.525	3942.23	13.184	244.83	162.69	5	Protector
4-1/2" FLUTED SWAGE ADAPTER D/P, 18.9# TPCQ-EX PIN ON TOP X HSM3 PIN ON DOWN* SUPER 13CR-110KSI, WEIQIXIN Co	Fluted Swage		1.010	3943.24	12.174	244.87	162.73		
TBG HANGER MSP, 10K, 11" * 4 1/2" ACME Box up * 4 1/2" TPCQ-EX Box Down , M/C: FF-0.5 T/C: UPSL: 3G PR: 2, 4" Type "H" Two Way	TBG HGR		0.598	3943.84	11.576	244.90	162.76		
Tubing Hanger Running Tool	Tubing Hanger Running Tool		0.250	3944.09	11.326	244.92	162.77		
L/Joint	L/Joint		11.603	3955.69	-0.277	245.48	163.26		Final Press Test

Figure 8-20: AZNS-040 Completion Tally

8.7 Completion String Pressure Test



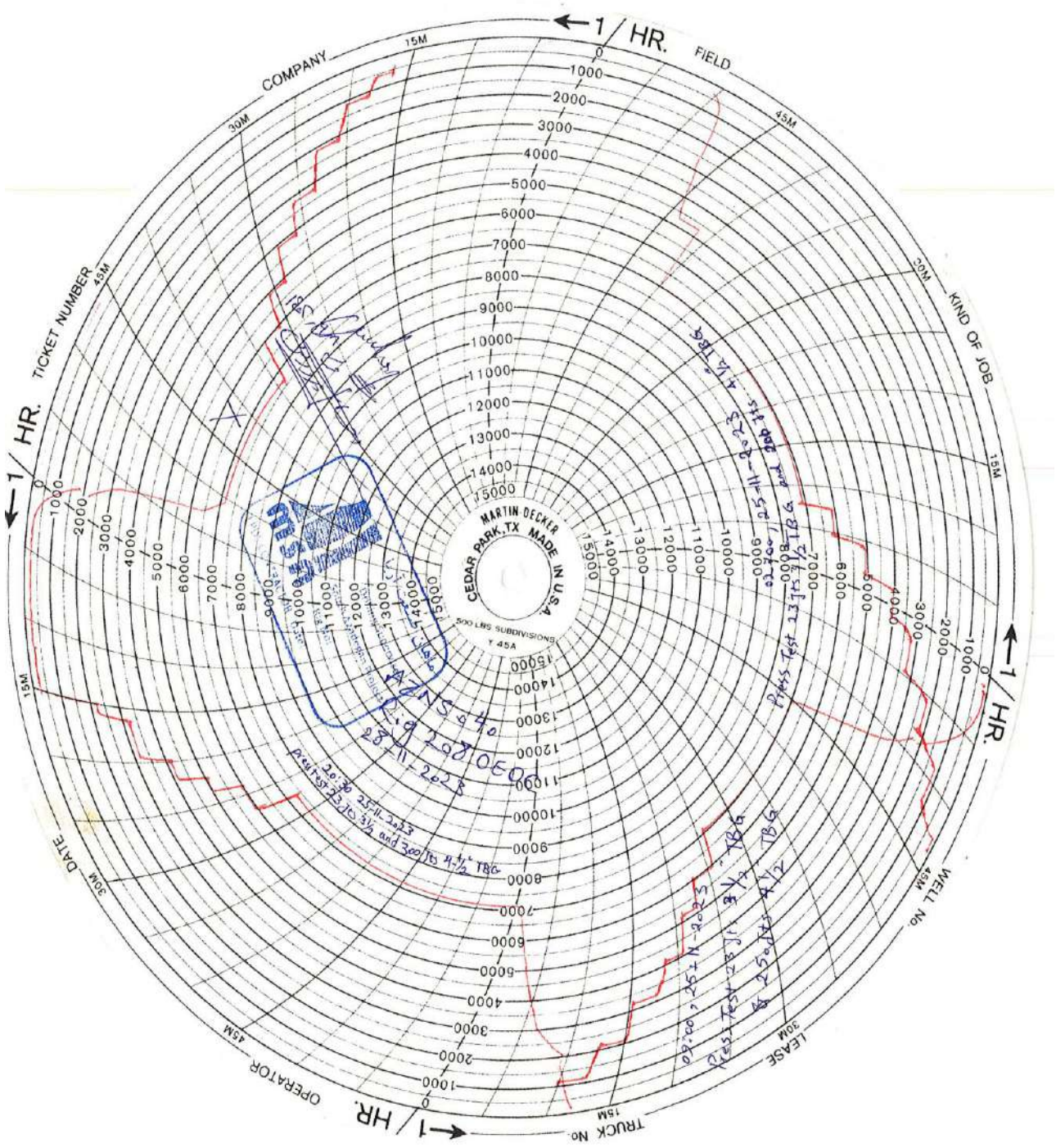


Figure 8-21: AZNS-040 Tubing Pressure Test Graph

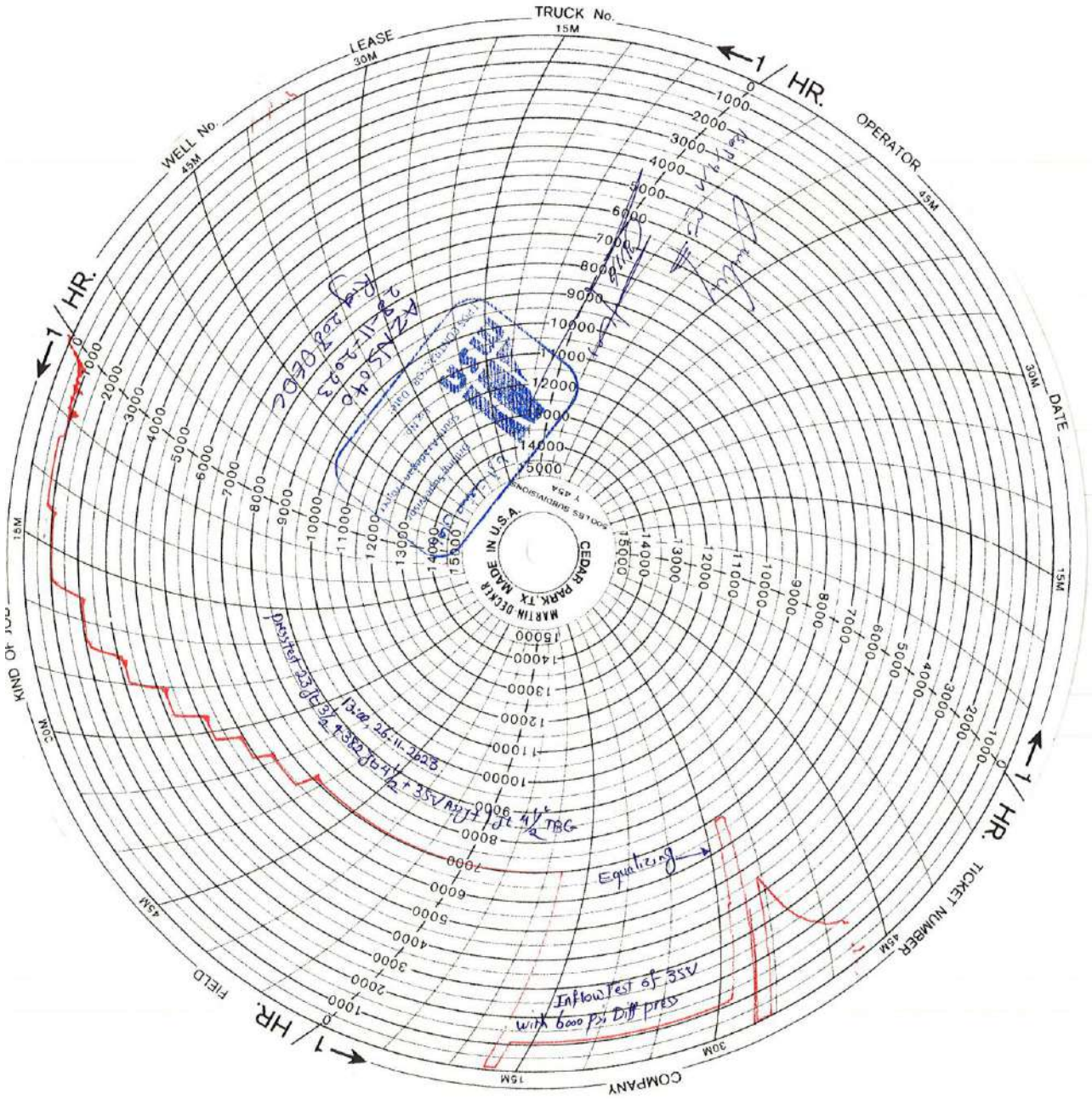


Figure 8-22: AZNS-040 Tubing Pressure & 3SV InflowTest Graph

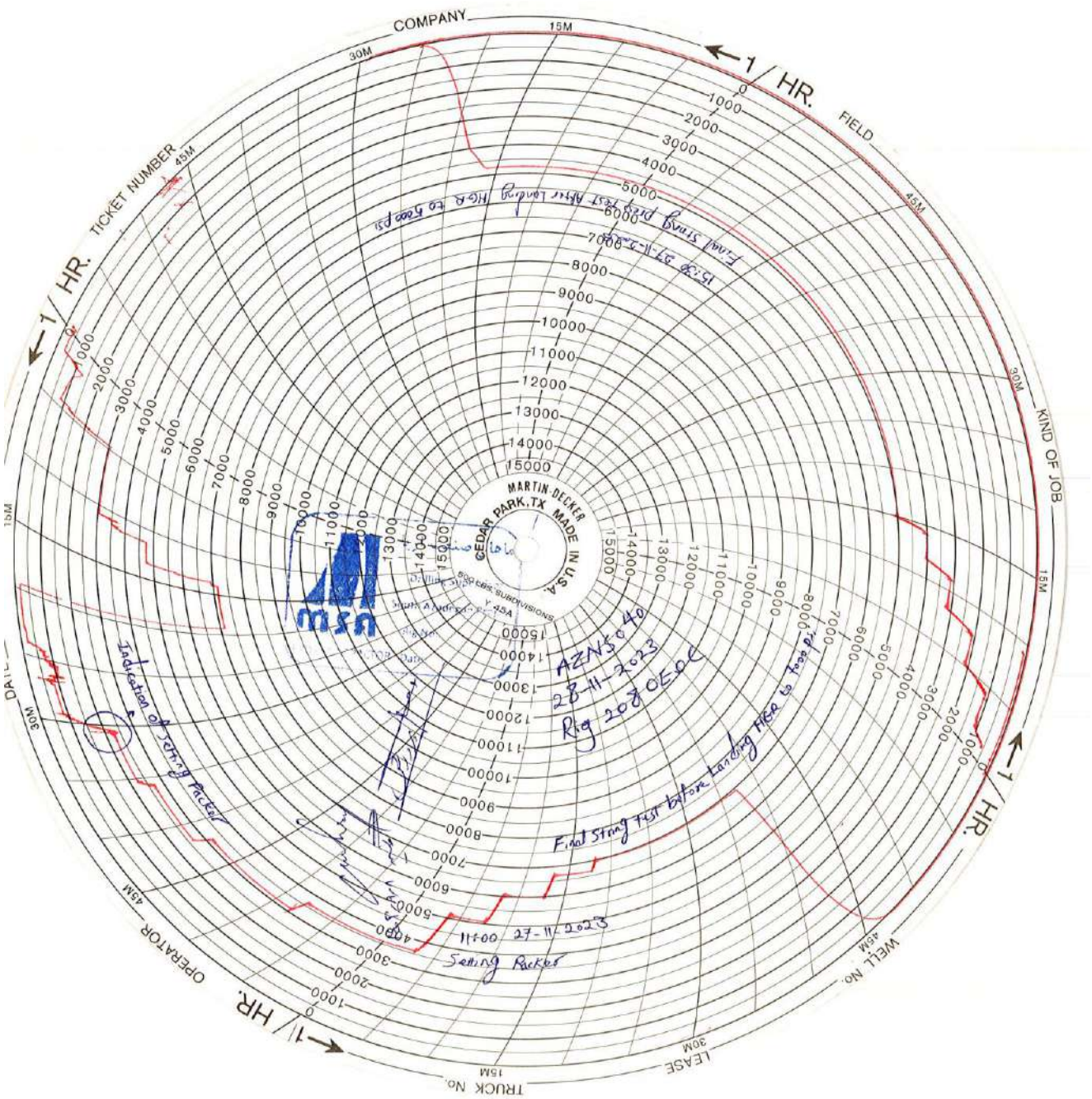


Figure 8-23: AZNS-040 String Pressure Test Graph

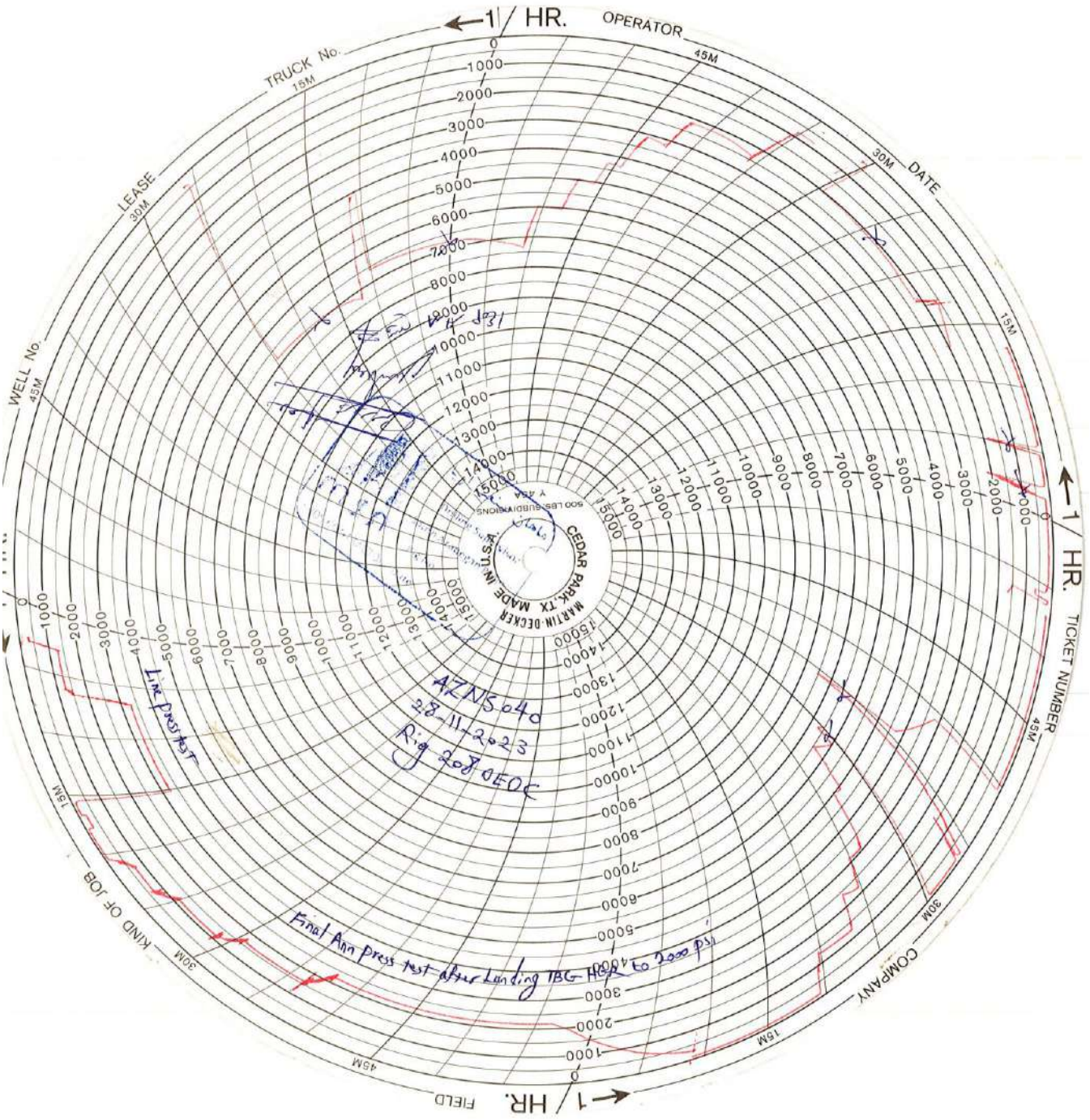


Figure 8-24: AZNS-040 Annulus Pressure Test Grap

9. Wellhead Report

9.1 Final Well Status

Well Head

Item	Specification (size, pressure rating, M/C)	Test Pressure	Test Duration
C.H.H	20-3/4'' 3K, DD-0.5 PSL1, PR1	Already Installed	
C.H.S	20-3/4'' 3K * 13-5/8'' 5K , DD-0.5, PSL1, PR1	Already Installed	
MSP T.H.S.	13-5/8'' 5K * 11''-10K, FF-0.5, PSL3, PR2	4500 psi	15 MIN
MSP X-MASS TREE(BTM)	4-1/16''-10K, FF-0.5, PSL3G, PR2	8000 psi	15 MIN

Wellbore Completion Fluids

Well bore	Fluid Type	Weight (pcf)
Cased Hole	CABR2	103-104

Well Head Pressure Recor

Annulus	Pressure (psi)	Date / Time
Wellhead	0	1402.09.08
Tubing- 9 5/8"	0	1402.09.08
9 5/8" – 13 3/8"	0	1402.09.08
13 3/8" – 20"	0	1402.09.08

Barriers

Barrier Type	Depth	Pressure Test	Date Tested
Close TRSSSV	66 m	500 psi (inflow test after finish flow the well)	1402.09.22

Temporary Installed Equipment:

After Completing the well, Plug & Prong Should have been installed in No-Go landing nipple which was not done by OEOC company due to start flowing operation.

Table 9-1: AZNS-040 Status

9.2 Wellhead & X-mass tree Specification

Well Type	Specification	Check and Remark
MSP Co. Well Head 10000 Psi for Fahliyan Wells	**TUBING HEAD ADAPTOR** BOTTOM STUDDED FLANGE 11", 10000 PSI, BX158 TOP STUDDED FLANGE 4-1/16" 10000 PSI BX155 FULL BORE PREPARED WITH TWO S-SEAL AND ONE METAL TO METAL SEAL TO PACK TOP NECK OF TUBING HANGER C/W RELATED TEST AND INJECTION PORTS PREPARTION. C/W TWO CONTINUOUS FEED THROUGH TYPE 1/4" NPT CONTROL LINE EXITS PREPARTION (ONE FOR 1/4" TRSSSV CL AND ONE FOR 1/4" DOWNHOLE PRESSURE GAUGE) AND TWO 3/8" NPT CONTINUOUS FEED THROUGH TYPE LINE PORTS FOR CHEMCIAL INJECTION LINE EXIT INCLUDING RELATED SWAGE LOCK FITTINGS C/WITH ONE 4-1/16", 10K WP FULL BORE MANUAL GATE VALVE FORGED BODY AND BONNET, ALLOY STAINLESS STEEL 410SS GATE & SEAT AISI 410 COATED WITH TUNGSTEN CARBIDE USING HVOF METHOD STEM: INCONEL 718 WITH BACK SEAT CAPABILITY, NON RISING STEM METAL TO METAL SEAL BETWEEN SEAT AND GATE C/W HAND WHEEL FORGED BODY ALLOY STEEL AISI 410 MAXIMUM HEIGHT: 800MM M/C:FF-0.5 T/C:U PSL:3G PR:2 CROSS T-TYPE SOLID BLOCK CROSS T-TYPE SOLID BLOCK, BOTTOM FLANGED 4-1/16" API 10000 PSI WP TOP FLANGED STUDDED 4-1/16" API 10000 PSI WP FOR TREE CAP WITH TWO STUDDED SIDE OUTLETS 4-1/16" 10000PSI WP ELEVATED AT 900 DEG AND TWO 4-1/16", 10000 PSI MANUAL GATE VALVE AT THE TOP AND BOTTOM AND ONE 4-1/16", 10000 PSI WP SAFETY VALVE EQUIPPED WITH HYDRAULIC WIRE CUT	

	ACTUATOR AS UPPER MASTER VALVE MATERIAL OF BODY,BONNET: AISI 410 SEAT, GATE: AISI 410SS HARD FACED WITH TUNGSTEN CARBIDE UASING HVOP METHOD MATERIAL OF STEM: INCONEL 718 M/C:FF-0.5 T/C:U PSL:3G PR:2G. MANUAL GATE VALVE API 4-1/16" 10000 PSI WP. FLANGED ENDS AND FULL BORE METAL TO METAL SEAL BETWEEN BODY/SEAT/GATE WITH BACK SEAT CAPABILITY, NON RISING STEM C/W HANDWHEELS MATERIAL OF BODY,BONNET,SEAT,GATE: AISI 410SS MATERIAL OF STEM: INCONEL 718 GATE&SEAT HARD FACED WITH TUNGSTEN CARBIDE BY HOVF METHOD M/C:FF-0.5 T/C:U PSL:3G PR:2 X-MAS TREE CAP 4-1/16" API 10000 PSI WP, FLANGED BOTTOM (4-1/16", BORE) 8-3/8"-4ACME-2G THREAD TOP, BLANKING PLUG TAPPED WITH 1/2" NPT THREAD, FORGED BODY, MATERIAL: STAINLESS STEEL 410 M/C:FF-0.5 T/C:U PSL:3G PR:2	

Table 9-2: AZNS-040 Wellhead Specification

9.3 Wellhead sketch

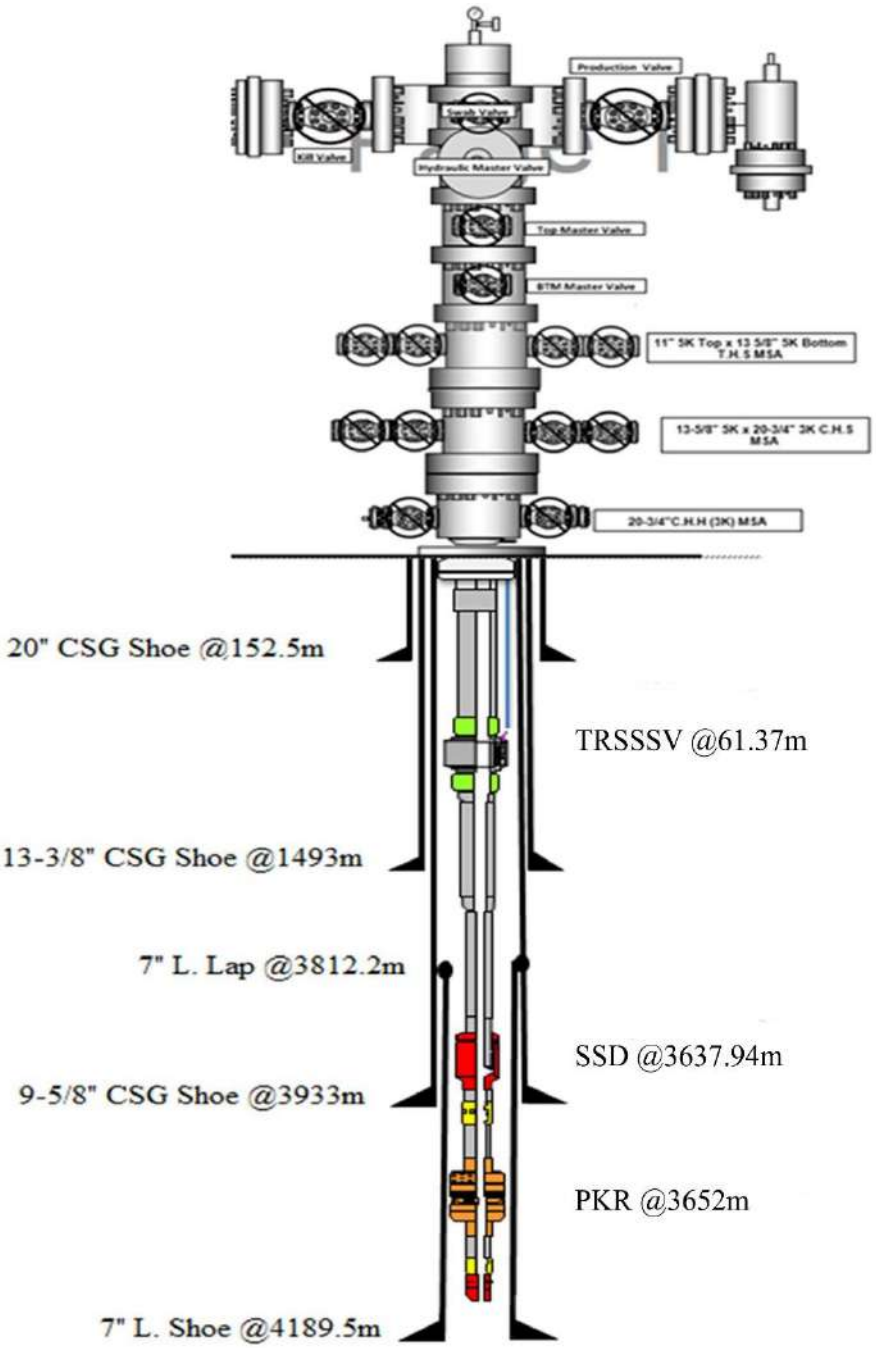


Figure 9-1: AZNS-040 Wellhead Sketch

X-mass Tree Specification (Section D)

Item	Equipment	M/C	T/C	PSL	Qty
1	BTM Master Valve 11” * 4 1/16” 10K	FF-0.5	U	3G	1
2	TOP Master Valve 4 1/16” 10K	FF-0.5	U	3G	1
3	Surface Safety Valve 4 1/16” 10K	FF-0.5	U	3G	1
4	Deeping Valve 4 1/16” 10K	FF-0.5	U	3G	1
5	Wing gate valve-1 (Kill Valve) 4 1/16" 10K	FF-0.5	U	3G	1
6	Wing gate valve-2 (production valve) 10K	FF-0.5	U	3G	1
7	41/16" Choke valve 4 1/16" 10K	FF-0.5	U	3G	1
8	Christ Mas Tree	FF-0.5	U	3G	1

Table 9-3: AZNS-040 Section "D" Specification

9.4 X-mass Tree Pressure Test Report (Section D – befor install)

Item	Test	Check Mark and Result
1	Pressure test for the kill Valve W/ 1000 & 8000 Psi / 15 min	OK
2	Pressure test for the SSV W/ 1000 & 8000 Psi / 15 min	OK
3	Pressure test for the top master valve W/ 1000 & 8000 Psi / 15 min	OK
4	Pressure test for Swab valve W/ 1000 & 8000 Psi / 15 min	OK
5	Pressure test for the Production valve W/ 1000 & 8000 Psi / 15 min	OK

Table 9-4: AZNS-040 Section "D" Pressure Test Before Installation

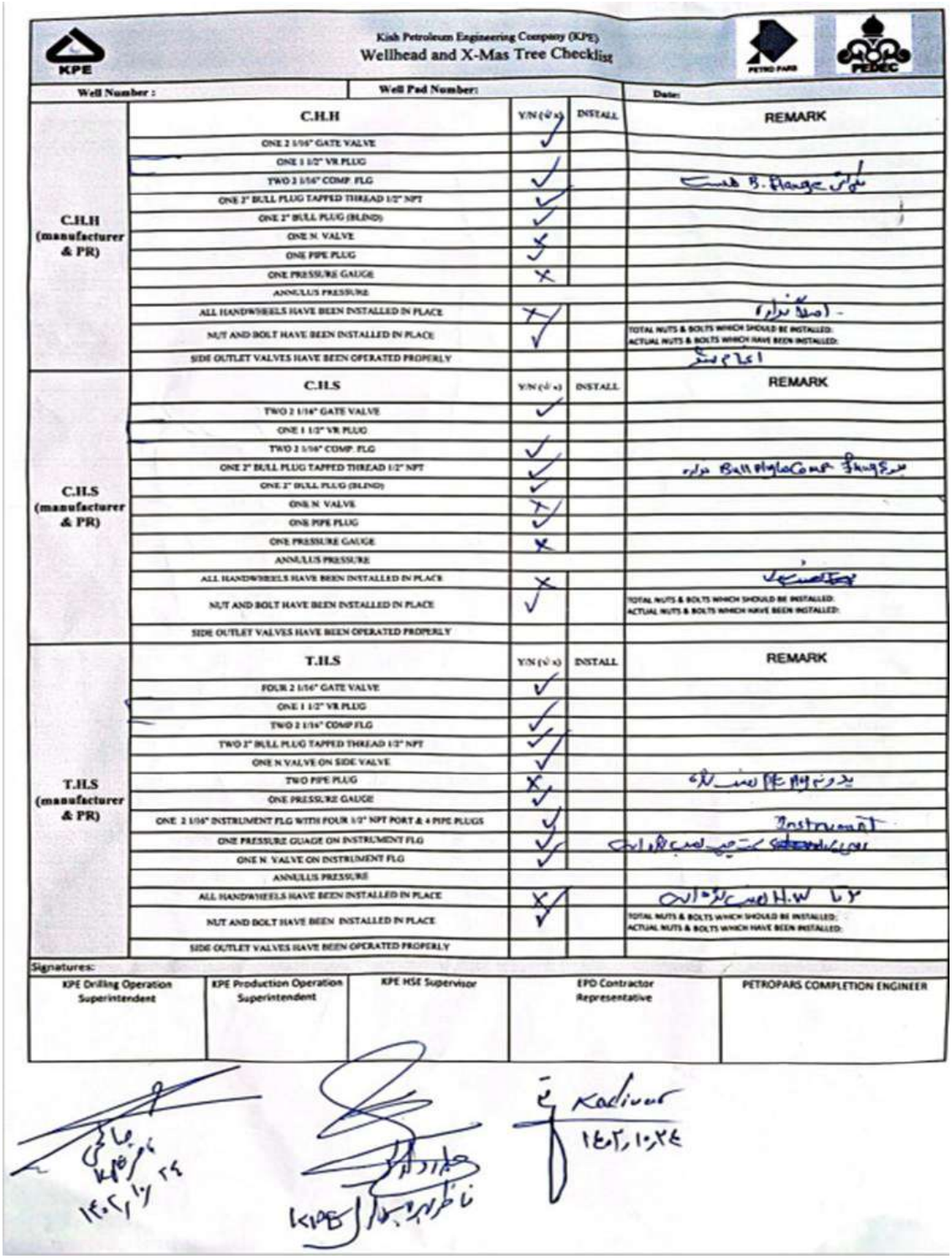
9.5 X-mass Tree Pressure Test Report (Section D – after install)

Item	Test	Check mark and Result
1	Pressure test X-MAS TREE solid block and bottom section w/ 1000 & 8000 psi for 15min	OK
2	Pressure test for the SSV Exit block W/ 8000 Psi / 15 min	OK
3	Pressure test neck of tubing hanger metal seal W/1000 &8000 psi for 15min	OK

4	Pressure test for Swab valve W/ 8000 Psi / 15 min	OK
5	Pressure test for Production valve W/8000 Psi / 15 min	OK
6	Pressure test cavity of FS.seal and Ring Gasket W 8000 Psi / 15 min	OK
7	Pressure test CIV line to 2000psi	OK

Table 9-5: AZNS-040 Section "D" Pressure Test After Installation

9.6 Wellhead Related Test Report



Kish Petroleum Engineering Company (KPE)

Wellhead and X-Mas Tree Checklist

Well Number : | Well Pad Number: | Date:

		Y/N (✓/×)	INSTALL	REMARK
C.H.H (manufacturer & PR)	C.H.H			
	ONE 2 1/16" GATE VALVE	✓		
	ONE 1 1/2" VR PLUG			
	TWO 2 1/16" COMP. FLG	✓		
	ONE 2" BULL PLUG TAPPED THREAD 1/2" NPT	✓		
	ONE 2" BULL PLUG (BLIND)	✓		
	ONE N. VALVE	✓		
	ONE PIPE PLUG	✓		
	ONE PRESSURE GAUGE	×		
	ANNULUS PRESSURE			
	ALL HANDWHEELS HAVE BEEN INSTALLED IN PLACE	✓		
	NUT AND BOLT HAVE BEEN INSTALLED IN PLACE	✓		TOTAL NUTS & BOLTS WHICH SHOULD BE INSTALLED: ACTUAL NUTS & BOLTS WHICH HAVE BEEN INSTALLED:
	SIDE OUTLET VALVES HAVE BEEN OPERATED PROPERLY			

		Y/N (✓/×)	INSTALL	REMARK
C.H.S (manufacturer & PR)	C.H.S			
	TWO 2 1/16" GATE VALVE	✓		
	ONE 1 1/2" VR PLUG			
	TWO 2 1/16" COMP. FLG	✓		
	ONE 2" BULL PLUG TAPPED THREAD 1/2" NPT	✓		
	ONE 2" BULL PLUG (BLIND)	✓		
	ONE N. VALVE	✓		
	ONE PIPE PLUG	✓		
	ONE PRESSURE GAUGE	×		
	ANNULUS PRESSURE			
	ALL HANDWHEELS HAVE BEEN INSTALLED IN PLACE	×		
	NUT AND BOLT HAVE BEEN INSTALLED IN PLACE	✓		TOTAL NUTS & BOLTS WHICH SHOULD BE INSTALLED: ACTUAL NUTS & BOLTS WHICH HAVE BEEN INSTALLED:
	SIDE OUTLET VALVES HAVE BEEN OPERATED PROPERLY			

		Y/N (✓/×)	INSTALL	REMARK
T.H.S (manufacturer & PR)	T.H.S			
	FOUR 2 1/16" GATE VALVE	✓		
	ONE 1 1/2" VR PLUG	✓		
	TWO 2 1/16" COMP FLG	✓		
	TWO 2" BULL PLUG TAPPED THREAD 1/2" NPT	✓		
	ONE N VALVE ON SIDE VALVE	✓		
	TWO PIPE PLUG	✓		
	ONE PRESSURE GAUGE	✓		
	ONE 2 1/16" INSTRUMENT FLG WITH FOUR 1/2" NPT PORT & 4 PIPE PLUGS	✓		
	ONE PRESSURE GUAGE ON INSTRUMENT FLG	✓		
	ONE N VALVE ON INSTRUMENT FLG	✓		
	ANNULUS PRESSURE			
	ALL HANDWHEELS HAVE BEEN INSTALLED IN PLACE	✓		
	NUT AND BOLT HAVE BEEN INSTALLED IN PLACE	✓		TOTAL NUTS & BOLTS WHICH SHOULD BE INSTALLED: ACTUAL NUTS & BOLTS WHICH HAVE BEEN INSTALLED:
	SIDE OUTLET VALVES HAVE BEEN OPERATED PROPERLY			

Signatures:

KPE Drilling Operation Superintendent	KPE Production Operation Superintendent	KPE HSE Supervisor	EPD Contractor Representative	PETROPARS COMPLETION ENGINEER

Kish Petroleum Engineering Company (KPE)

Wellhead and X-Mas Tree Checklist

KPE

Well Number :	Well Pad Number:			Date:
	X-MASS TREE	Y/N (√ s)	INSTALL	REMARK
X-MASS TREE (manufacturer & PR)	ADJUSTABLE/FIXED CHOKE VALVE	√		
	CHECK VALVE JUST FOR WATER DISPOSAL WELL			
	ONE COMP FLG (ON KILL LINE)	√		
	ONE 2" BULL PLUG TAPPED THREAD 1/2" NPT	√		
	ONE N VALVE ON TREE CAP	√		
	ONE PRESS GAUGE ON TREE CAP	√		
	ONE PIPE PLUG	√		
	4 1/16 INSTRUMENT FLG (BEFORE CHOKE VALVE, WITH FOUR 1/2" NPT PORT & 4 PIPE PLUGS)	√		
	ONE PRESSURE GUAGE ON INSTRUMENT FLG	√		
	ONE NEEDLE VALVE ON INSTRUMENT FLG	√		
	4 1/16" WELD NECK FLG	√		
	1/4" FITTING (FOR C.L) SWAGELOK	√		
	ONE 1/2" FITTING FOR TREECAP	√		
	WELLHEAD PRESSURE			
	ALL HANDWHEELS HAVE BEEN INSTALLED IN PLACE	√		
	ALL NUT AND BOLT HAVE BEEN INSTALLED IN PLACE	√		TOTAL NUTS & BOLTS WHICH SHOULD BE INSTALLED: ACTUAL NUTS & BOLTS WHICH HAVE BEEN INSTALLED:
	SSV CAP	√		
	TOP MASTER VALVE HAVE BEEN OPERATED PROPERLY			TURNS TO BE FULLY OPERATED:
	DIPPING VALVE HAVE BEEN OPERATED PROPERLY			TURNS TO BE FULLY OPERATED:
	PRODUCTION VALVE HAVE BEEN OPERATED PROPERLY			TURNS TO BE FULLY OPERATED:
	KILL VALVE HAVE BEEN OPERATED PROPERLY			TURNS TO BE FULLY OPERATED:

Remarks :

1.

2.

Signatures:

KPE Drilling Operation Superintendent	KPE Production Operation Superintendent	KPE HSE Supervisor	EPD Contractor Representative	PETROPARS COMPLETION ENGINEER

Figure 9-2: AZNS-040 Wellhead & Xmass tree Checklist

تاریخ: ۱۴۰۲/۰۹/۰۷
پیوست: دارد

بسمه تعالی

موضوع: صورتجلسه تست سطحی تاج چاه ۰۴۰ ازادگان جنوبی، قبل از نصب در دکل ۲۰۸ عملیات اکتشاف نفت

بدینوسیله در مورخه ۱۴۰۲/۰۹/۰۴ تست های سطحی تاج چاه X-MASS TREE SOLID BLOCK T TYPE 4-1/16" 10K, MSP DRILEX مرتبط به چاه شماره ۰۴۰ آزادگان جنوبی با حضور نماینده محترم کارفرما (پتروپارس) در محل دستگاه ۲۰۸ عملیات اکتشاف نفت، قبل از نصب بر روی تاج چاه به شرح ذیل انجام پذیرفت:

- BODY PRESSURE TEST OF X-MASS TREE SOLID BLOCK W/ 1000PIS, 5MIN & 8000 PSI, 15MIN "OK" SN: 008929K N180003
- INFLOW PRESSURE TEST OF 4-1/16" PRODUCTION (WING) VALVE W/ 1000PIS, 5MIN & 8000 PSI, 15MIN "OK"
- INFLOW PRESSURE TEST OF 4-1/16" DIPPING (SWAB) VALVE W/ 1000PIS, 5MIN & 8000 PSI, 15MIN "OK".
- INFLOW & OUT FLOW PRESSURE TEST OF 4-1/16" KILL VALVE W/ 1000PIS, 5MIN & 8000 PSI, 15MIN "OK" SN: 002929B N180086
- INFLOW PRESSURE TEST OF 4-1/16" SURFACE SAFETY VALVE W/ 1000PIS, 5MIN & 8000 PSI, 15MIN "OK"
- INFLOW & OUT FLOW PRESSURE TEST OF 4-1/16" TOP MASTER VALVE W/ 1000PIS, 5MIN & 8000 PSI, 15MIN "OK"
- PRESSURE TEST OF 2SV ACTUATOR W/ 2500 PSI, 15 MIN "OK"

– گراف مربوط به تست فشار پیوست صورتجلسه گردید.
– لازم به ذکر است طبق قرارداد، روی تاج چاه کاهنده ثابت نصب گردید.

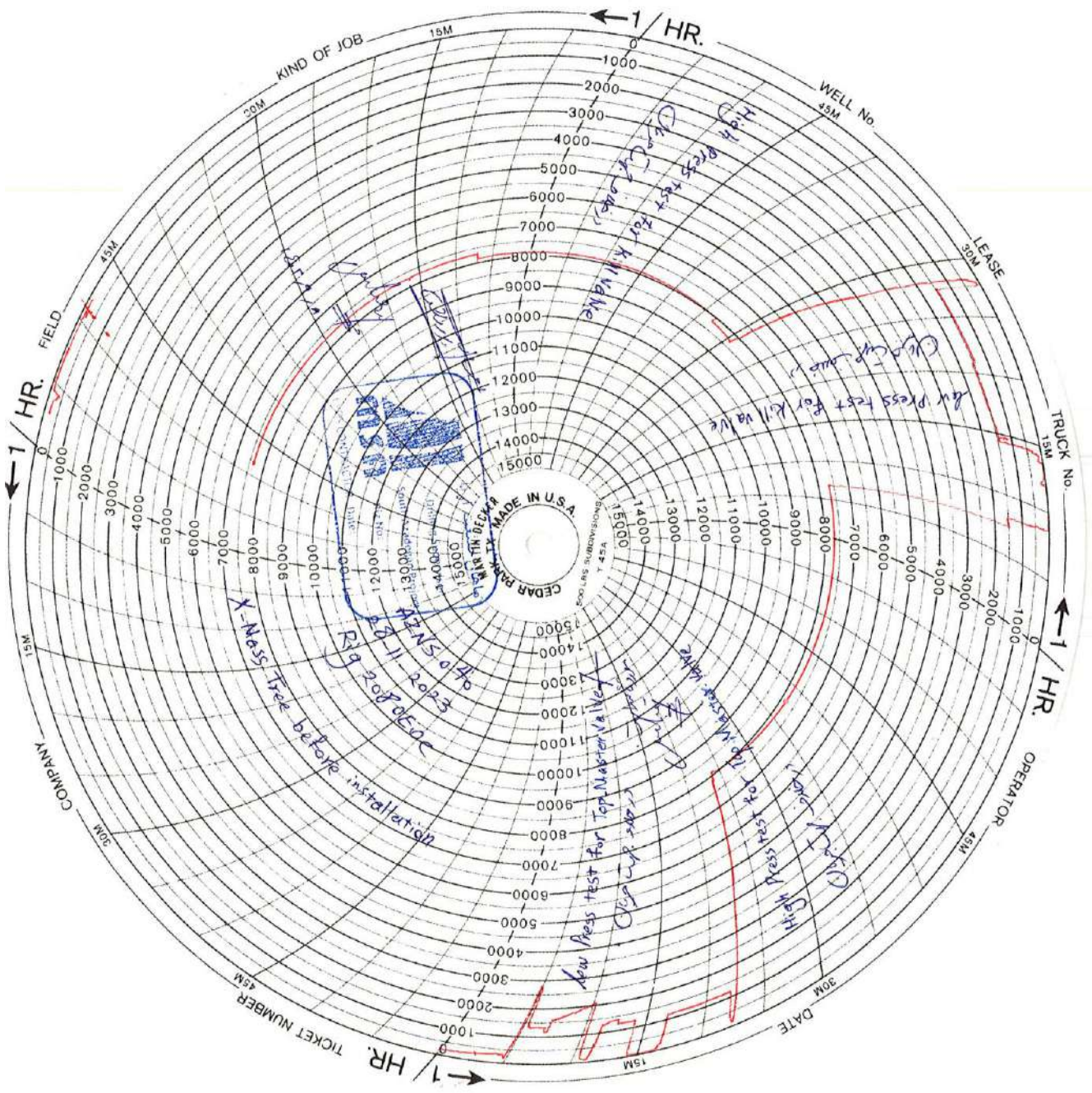
ماهان صنعت آریا
Drilling Supervisor
South Azadegan Project
Rig No:
LPDS CONTRACTOR Date:

ناظر عملیات حفاری
شرکت ماهان صنعت آریا

نماینده محترم تکمیل چاه کارفرما
شرکت نفت کیش

۱۴۰۲/۰۹/۰۸

Figure 9-3: AZNS-040 Wellhead MOM Before Installation


KIND OF JOB
WELL No.
LEASE
TRUCK No.
OPERATOR
DATE
TICKET NUMBER
COMPANY
FIELD
1/HR.
MADE IN U.S.A.
CEDAR PARK, TX
MARTIN DECKER
X-Mas Tree before installation
AZNS-40
20-11-2023
Rig 20 of DEOC

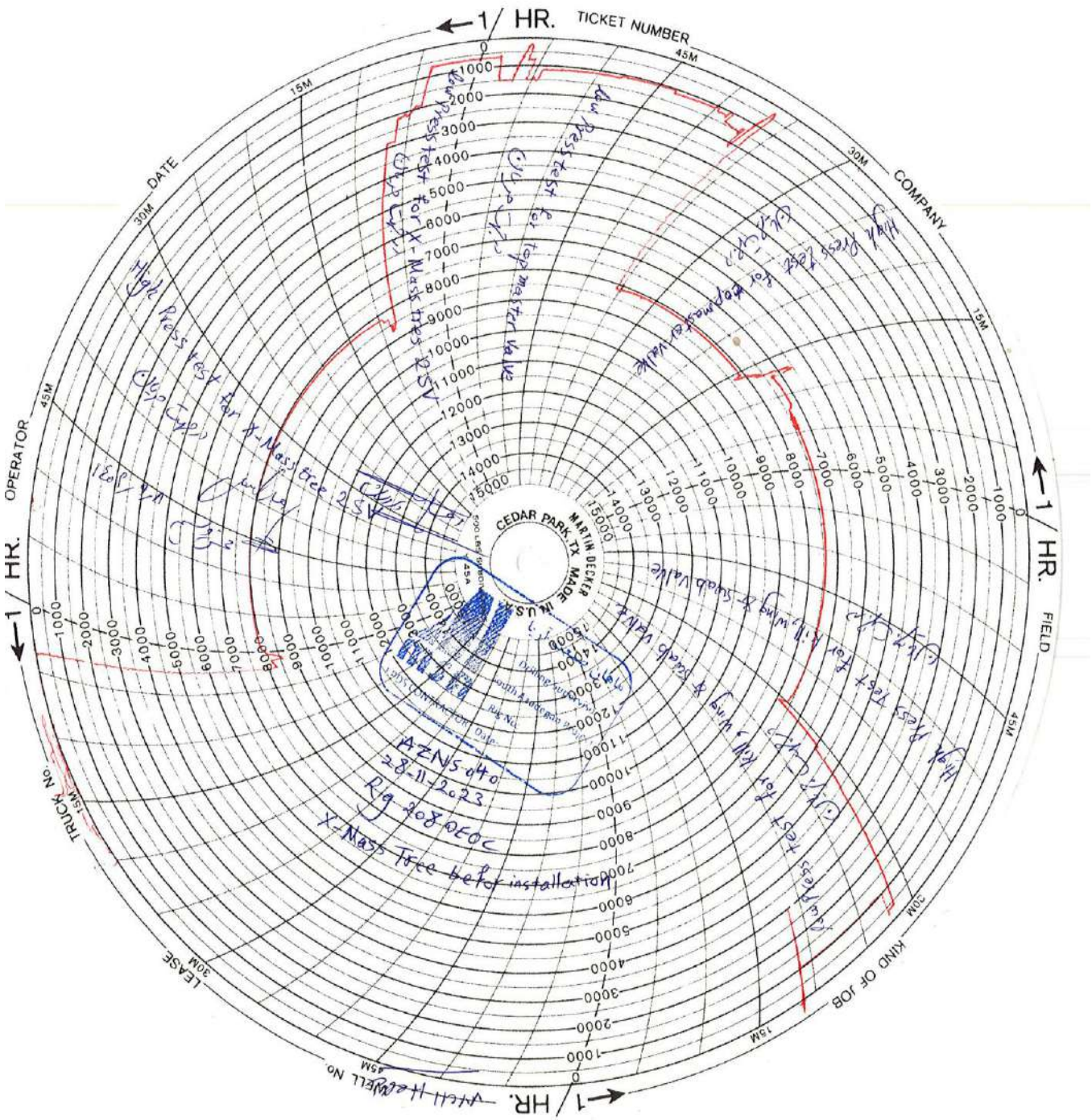


Figure 9-4: AZNS-040 X-Mass Tree Pressure Test Graph Before Installation

تاریخ: ۱۴۰۲/۰۹/۰۷
پیوست: دارد

بسمه تعالی

موضوع: صورتجلسه تست های پس از نصب تاج و ولهد چاه ۰۴۰ ازادگان جنوبی بعد از نصب در دکل ۲۰۸ عملیات اکتشاف نفت

بدینوسیله در مورخه ۱۴۰۲/۰۹/۰۷ تست های تاج چاه X-MASS TREE SOLID BLOCK T TYPE 4-1/16" 10K, MSP و ولهد چاه شماره ۰۴۰ ازادگان جنوبی با حضور نماینده محترم تکمیل چاه کارفرما و نماینده عملیات حفاری ماهان صنعت آریا در محل دستگاه حفاری ۲۰۸ عملیات اکتشاف نفت بعد از نصب بر روی چاه به شرح ذیل انجام پذیرفت:

- BODY PRESSURE TEST OF X-MASS TREE SOLID BLOCK W/ 10K 11" x 4-1/16" 10K TUBING HEAD ADAPTOR (BTM MASTER VALVE) 10K MSP/DRILEX W/ 8000 PSI FOR 15MIN "OK" SN: 008929K N180003
- PRESSURE TEST TBG HGR METAL SEAL TO 800PSI FOR 5MIN, 8000PSI FOR 15MIN "OK"
- PRESS TEST F-SEAL & RING GASKET BETWEEN T.H.S & TUBING HEAD ADAPTOR (BTM MASTER VALVE) TO 8000 PSI FOR 15 MIN "OK"
- PRESSURE TEST NECK OF TBG HGR F-SEAL TO 8000PSI FOR 15MIN "OK"
- PRESS TEST 1/4" CONTROL LINE TO 8000 PSI FOR 15MIN "OK"
- PRESS TEST 3/8"CONTROL INJECTION LINES TO 2000 PSI "OK"

- فشار تمامی دالیزها بررسی گردید که فشار آنها صفر بود.
- بعد از اتمام نصب تاج چاه درون آویزه لوله مغزی TWO WAY CHECK VALVE جهت انگیزش چاه خارج گردید.
- خروجی کنترل لاین شیر ایمنی درون چاهی درون چاهی در جهت شیر اصلی پایینی می باشد و همچنین CIV شماره یک در جهت ویست پیت و CIV شماره دو در خلاف جهت ویست پیت می باشد و همچنین بر روی دو CIV و کنترل لاین پلاگ کپ نصب گردید.
- لازم به ذکر است روی تاج چاه کاهنده ثابت نصب گردید و سایزهای مختلف CHOKE BEAN کاهنده ثابت تاج چاه طبق قرداد می بایست تحویل کارفرما گردد.
- گراف مربوط به تست فشار پیوست صورتجله گردید.

ماهان صنعت آریا
Drilling Supervisor
South Azadegan Project
Rig No:
EPDS CONTRACTOR Date:

ناظر عملیات حفاری
شرکت ماهان صنعت آریا

نماینده محترم تکمیل چاه کارفرما
شرکت نفت کیش

۱۴۰۲/۰۹/۰۸

Figure 9-5: AZNS-040 Wellhead MOM After Installation

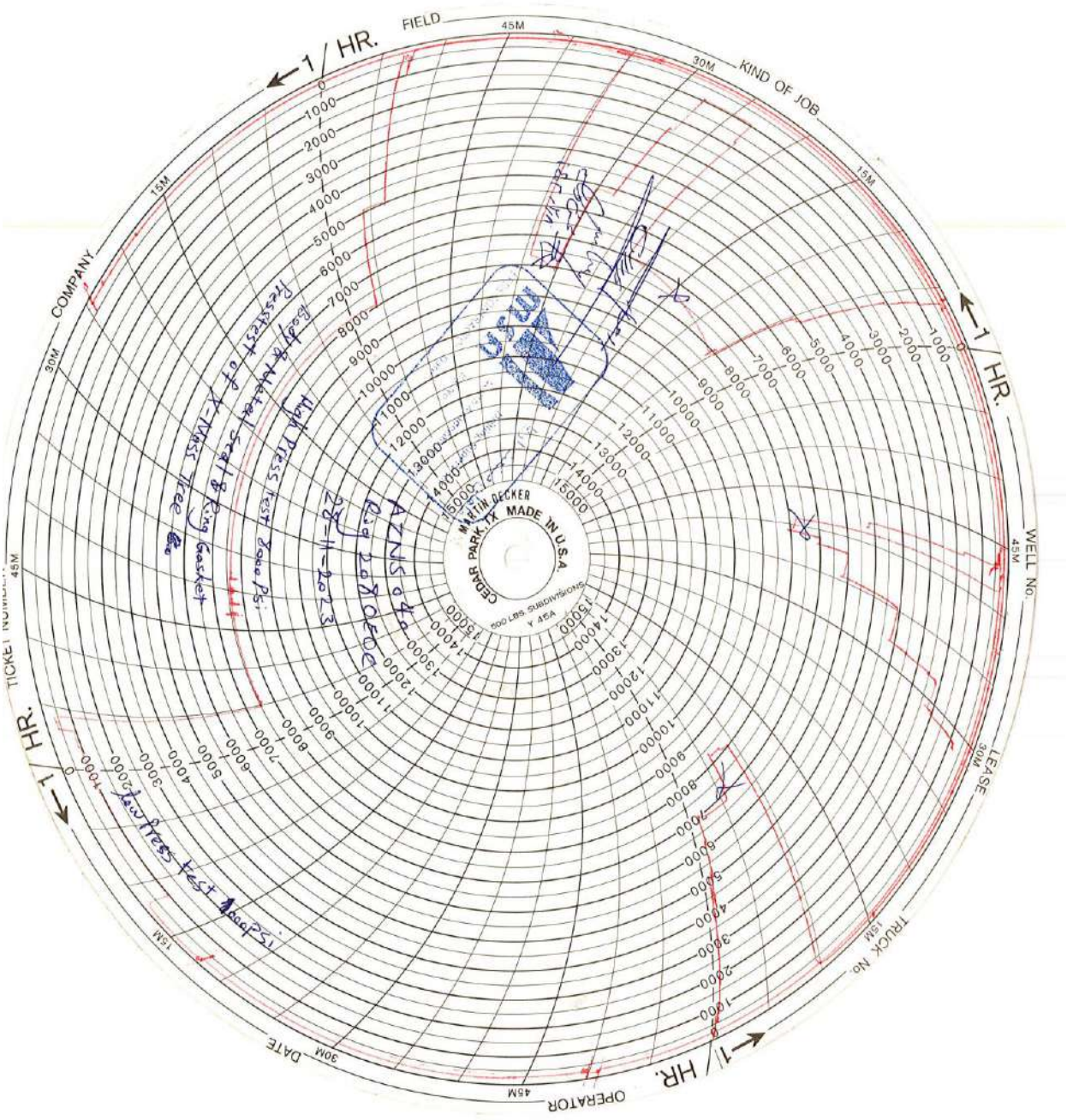


Figure 9-6: AZNS-040 Body & Metal Seal of X-Mass Tree Pressure Test Graph

10. Drilling Fluid Information

Hole Size (in)	Casing Size (in)	Casing Depth (m)	Drilling Fluid Sys.	Mud Properties
26	20	152.5	Already has been drilled.	
17-1/2	13-3/8	1492	Already has been drilled.	
12-1/4	9-5/8	3933.8	KCl Pol. Mud	Mw: 73 – 85^{pcf}
8-1/2	7	3182.2 – 4189	KCl Pol. Mud	Mw: 101 – 104^{pcf}
3628^{m} 4-1/2" completion string + 253^{m} 3-1/2" completion string				

Table 10-1: AZNS-040 Drilling Fluid Information

10.1 12-1/4" Hole Section

Operation Remark: The OEOC 208 rig was spud at 13 July 2023 at AZN#40 and Made up 12 ¼'' D/O BHA then RIH to 160^{m} and performed integrity test against cement plug after that drilled out cement plug from 160^{m} to 171^{m} then RIH to 1468^{m} and drilled out float equipment's with water from 1468^{m} to 1492^{m} then displaced the hole with 73^{pcf} KCl POL mud.

Started drilling formation as 12 ¼'' hole section from 1492^{m} to 1505^{m} then POOH to surface, Made up 12 ¼'' drilling BHA and RIH to 1505^{m} then commenced drilling formation as 12 ¼'' hole section from 1505^{m} to 1556^{m}, at this point observed string twist off indications then POOH to surface.

Made up 10 1/8'' over shot and RIH to 1547^{m} then catch the fish and POOH to surface after that Made up drilling BHA and RIH to 1556^{m} then commenced drilling formation

from 1556[m] to 1930[m] with 4-8BPH loss then performed condition trip with good hole condition, Commenced drilling formation from 1930[m] to 1960[m] with 2-8BPH loss then POOH to surface as change bit.

Made up new BHA and RIH to 1960[m] then commenced drilling formation as 12 ¼'' hole section from 1960[m] to 2115[m] with 3-8BPH loss at this point sting was twisted off after that POOH to surface.

Made 10 1/8'' over shot and fishing BHA and RIH to 1997[m] then catch the fish and POOH to surface, Made up 12 ¼'' drilling BHA and RIH to 2115[m] then commenced drilling formation from 2115[m] to 2191[m] with 2-6BPH loss after that POOH to surface.

Made up new BHA and RIH to 2191m then commenced drilling formation from 2191[m] to 2197[m] with 2-4BPH loss, at this point observed string twisted off then POOH to surface after that made up 10 1/8'' over shot and RIH to 2124[m] then catch the fish and POOH to surface.

Made up 12 ¼'' drilling BHA and RIH to 2197[m] then commenced drilling formation as 12 ¼'' hole section from 2197[m] to 2512[m] with 4-6BPH loss meanwhile increased Mw to 77[pcf] then performed condition trip with good hole condition after that commenced drilling formation from 2512[m] to 2854[m] with 6 -10BPH loss, performed condition trip with good hole condition, During condition trip string was twisted off after that POOH to surface.

Made 10 1/8'' over shot and fishing BHA then RIH to 2100[m] and catch the fish after that POOH to surface, Made up new 12 ¼'' drilling BHA and RIH to 2854[m] then commenced drilling formation from 2854[m] to 2919[m] with 4-8BPH loss then POOH to surface due to low ROP.

Made up new 12 ¼'' drilling BHA and RIH to 2919[m] then commenced drilling formation from 2919[m] to 2923[m] meanwhile worked on junk sub after that POOH to surface, Made up 12 ¼'' drilling BHA and RIH to 2923[m] then commenced drilling formation from 2923[m] to 2945[m] with 4-8BPH loss after that POOH to surface due to low ROP.

Made up 12 ¼'' TCI bit and junk sub then RIH to 2945[m] after that commenced drilling formation from 2945[m] to 2961[m] with 3-6BPH loss meanwhile worked on junk sub then POOH to surface, Made up 12 ¼'' drilling BHA and RIH to 2961[m] then commenced drilling formation from 2961[m] to 3275[m] with 6-12BPH loss meanwhile increased Mw to 82[pcf] after that performed condition trip with good hole condition, commenced drilling

formation as 12 ¼'' hole section from 3275^{m} to 3497^{m} with 4-8BPH loss meanwhile increased Mw to 83^{pcf}, at 3497^{m} observed string was twisted off the POOH to surface.

Made 10 1/8'' over shot and fishing BHA and RIH to 3565^{m} then catch the fish and POOH to surface, Made up drilling BHA and RIH to 3565^{m} then commenced drilling formation as 12 ¼'' hole section from 3565^{m} to 3934^{m} (casing point) with 5-10BPH loss meanwhile increased Mw to 85^{pcf} after that spot 200^{bbl} high viscosity pill then POOH to surface.

Performed logging at several runs then RIH with 9 5/8'' casing to 3933.8^{m} and performed cement job as per program.

Drilling Fluid Remark: The 73^{pcf} fresh KCl/Polymer mud was used for starting this section which was built by using drill water.

It was tried to control mud properties by using all available solid control equipment such as shale shakers, centrifuge was applied as per requirement in order to control Mw. Hole cleaning was improved by pumping tandem pill high viscosity pills.

It was tried to reach optimum shale inhibition by using 3-5 wt. % KCl as main shale inhibitor by decreasing the tendency of to absorb, swell and slough-off.

The Mw increased up to by adding Limestone Powder up to 85^{pcf} at end of this section. SCE such as Shakers and centrifuges were running to discard LGS while drilling this section.
Treated frequently with essential chemical such as Caustic Soda as Alkalinity controller to adjust pH, Starch LV & HT and PAC LV for control FL, Soda Ash for control hardness & Xanthan gum as Viscosifier.

Interval	1492 – 3934^{m}	12-1/4" Hole Interval	9-5/8" Set @ 3933.8^{m}

Table 10-2: AZNS-040 12-1/4" Hole Data

FLUID PROPERTIES RECAP					
KCL Pol Mud		Drilling start	Range		Drilling End
MW (min)	pcf	73	73	85	85
Funnel Viscosity	sec/qt	43	39	46	43
PV	cps	15	15	21	20
YP	lb/100sq.ft	10	8	16	15
API Fluid loss	cc	2.5	2.5	4.8	2.9
pH		9.6	9	9.6	9.5
Solid	%	10	10	19	19
OIL	%	0	0	2	1
KCL	%	2	1	5	5
Cl (NaCl)	mg/L	153,000	153000	176000	176,000
Total Hardness	mg/L	1,240	720	1280	800
Ca Hardness	mg/L	840	600	880	680

Figure 10-1: Drilling Fluid Type

Products	Units Used	Unit wt lbs	Total wt		Effective Concentration	
Barite*	6	3,300	19,800	lbs	-	ppb
Bit Lube	18	55	990	gal	0.07	gpb
Caustic Soda	284	55	15,620	lbs	1.13	ppb
KCL	73	2,200	160,600	lbs	11.62	ppb
Limestone	395	2,200	869,000	lbs	62.88	ppb
Limestone Chips F , M	472	55	25,960	lbs	1.88	ppb
Oyster Shell M	73	55	4,015	lbs	0.29	ppb
Pac LV	377	55	20,735	lbs	1.50	ppb
Salt (NaCl)	459	2,200	1,009,800	lbs	73.06	ppb
Starch LV	1,864	55	102,520	lbs	7.42	ppb
Starch HT	600	55	33,000	lbs	2.39	ppb
Soda ASH	190	55	10,450	lbs	0.76	ppb
Sodium Bicarbonate	2	55	110	lbs	0.01	ppb
Xanthan Gum	95	55	5,225	lbs	0.38	ppb
*6BB Barite was used for cement spacer.						

Initial Volume	0.0 bbl
Total Volume Built	13,820.8 bbl
Volume received	0.0 bbl
Volume Returned	0.0 bbl
Total Volume Consumed	11,489 bbl
Volume Transfer to Next Section	2,332.0 bbl
Meter Drilled	2,442 m

Figure 10-2: Material Consumption Analysis

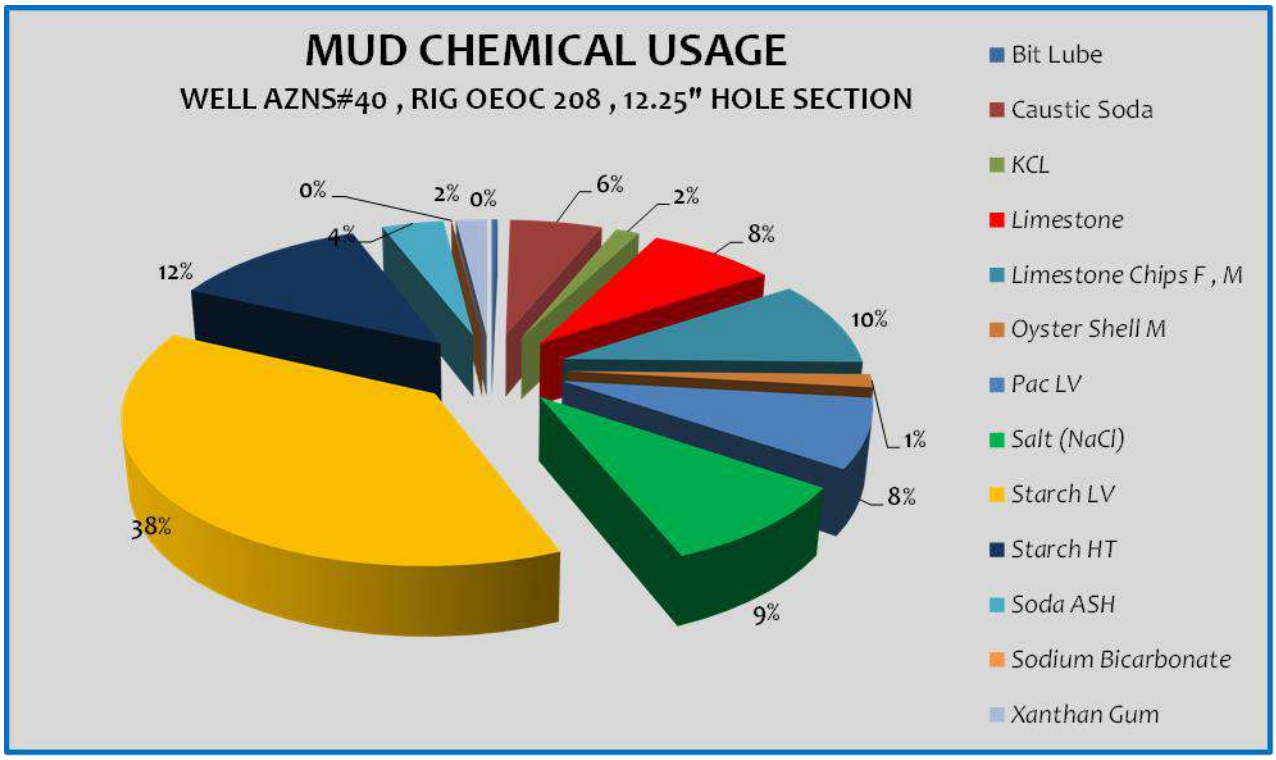


Figure 10-3: Mud Chemical Usage Graph

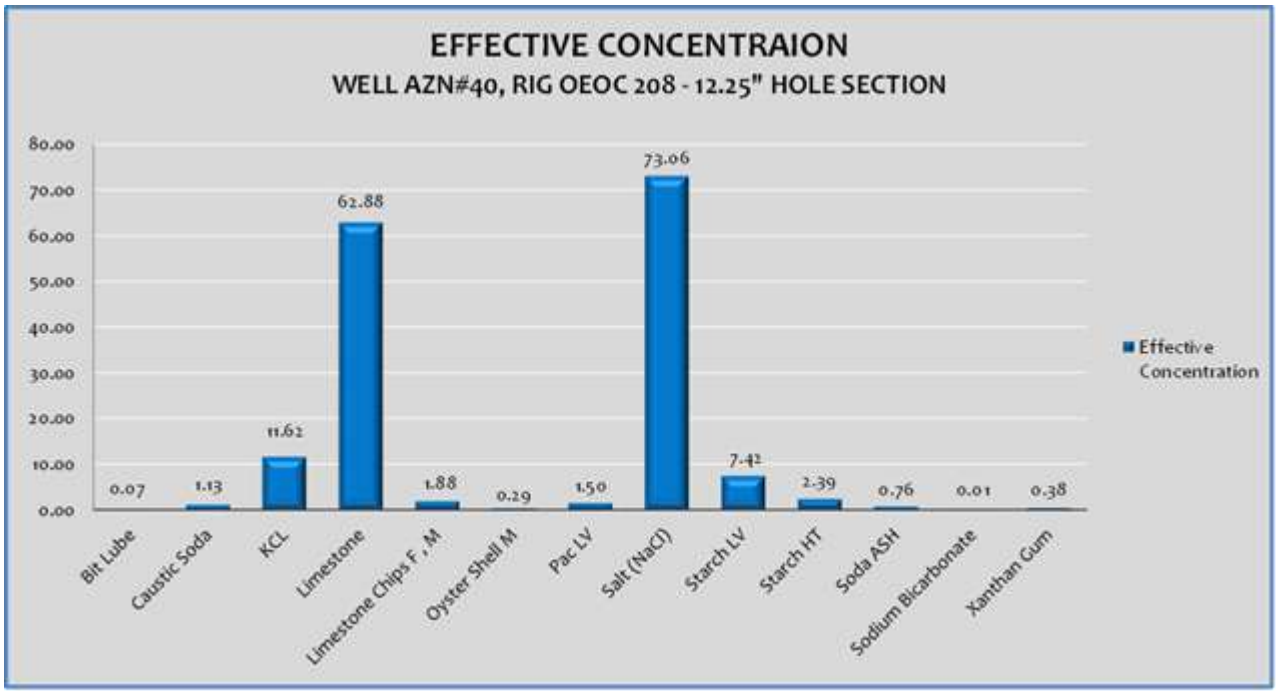


Figure 10-4: Effective Concentration Chart

WELL AZN#40, RIG OEOC 208 , 12.25" HOLE SECTION VOLUME ANALYSIS		
Initial Volume	bbls	0.0
Received Volume	bbls	0.0
Volume Return	bbls	0.0
Volume Built	bbls	13,820.8
Surface Losses/Dumped	bbls	4,963.0
Formation Losses	bbls	6,526.0
Total Losses	bbls	11,489.0
Volume Transfer Out	bbls	2,332

Figure 10-5: Volume Analysis Table

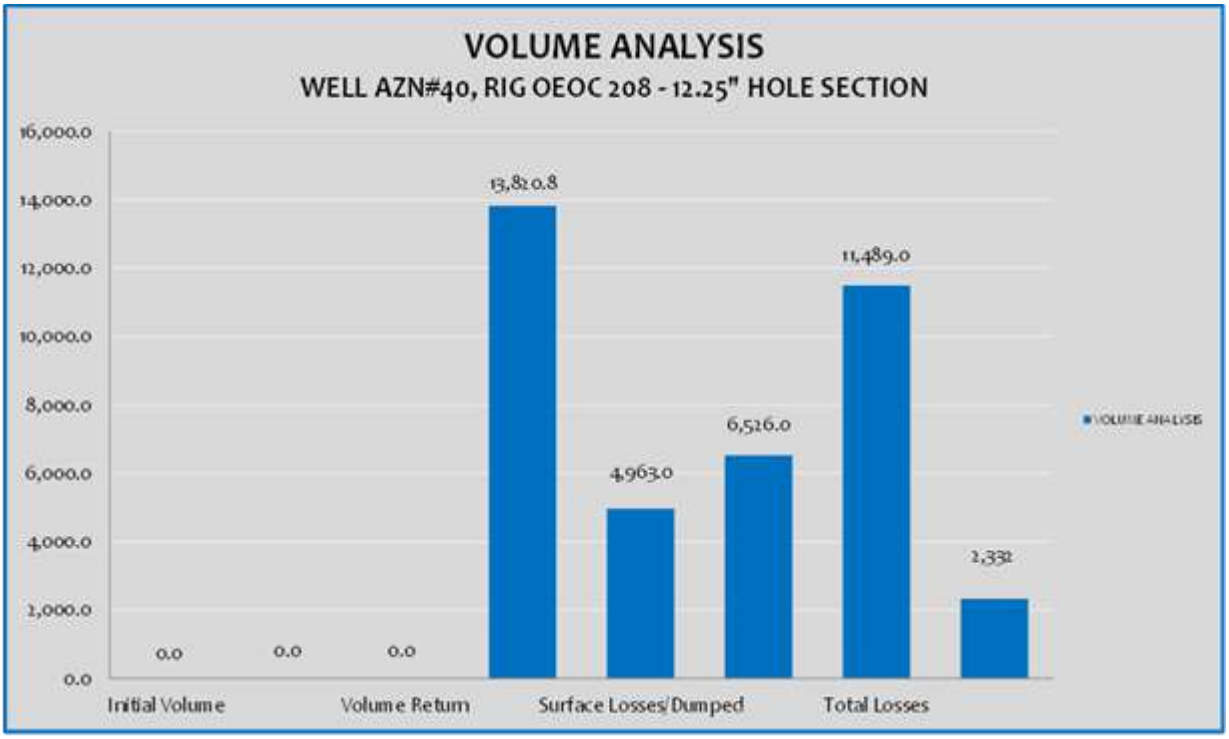


Figure 10-6: Volume Analysis Chart

Date/DMR.N		Volume of Built	Surface Losses	Formation Losses
13-Jul-23	01	1,085.1	0	0
14-Jul-23	02	14.0	0	0
15-Jul-23	03	300.6	140	0
16-Jul-23	04	189.4	38	0
17-Jul-23	05	629.0	84	40
18-Jul-23	06	172.3	60	70
19-Jul-23	07	590.6	159	130
20-Jul-23	08	368.7	189	122
21-Jul-23	09	327.8	234	85
22-Jul-23	10	182.3	104	75
23-Jul-23	11	183.5	123	120
24-Jul-23	12	182.3	92	60
25-Jul-23	13	0.2	88	20
26-Jul-23	14	183.2	74	40
27-Jul-23	15	358.7	107	70
28-Jul-23	16	0.1	106	38
29-Jul-23	17	0.0	39	24
30-Jul-23	18	148.5	87	53
31-Jul-23	19	253.7	94	110
1-Aug-23	20	357.0	150	130
2-Aug-23	21	198.8	103	189
3-Aug-23	22	560.7	199	210
4-Aug-23	23	402.8	116	240
5-Aug-23	24	212.0	65	200
6-Aug-23	25	225.9	20	110
7-Aug-23	26	0.0	20	96
8-Aug-23	27	173.1	19	96
9-Aug-23	28	350.3	166	96
10-Aug-23	29	4.7	78	130
11-Aug-23	30	330.5	54	125

12-Aug-23	31	187.3	34	125
13-Aug-23	32	152.8	59	120
14-Aug-23	33	217.5	10	150
15-Aug-23	34	10.8	82	70
16-Aug-23	35	202.8	83	87
17-Aug-23	36	229.1	37	68
18-Aug-23	59	439.7	60	140
19-Aug-23	60	228.9	83	163
20-Aug-23	36	379.8	77	145
21-Aug-23	59	153.6	78	150
22-Aug-23	60	316.4	86	170
23-Aug-23	36	165.2	78	190
24-Aug-23	59	284.5	59	180
25-Aug-23	60	263.0	23	140
26-Aug-23	36	476.3	136	190
27-Aug-23	59	352.0	143	200
28-Aug-23	60	363.1	156	220
29-Aug-23	36	247.3	123	190
30-Aug-23	59	144.5	51	150
31-Aug-23	60	131.9	69	150
1-Sep-23	36	123.0	74	160
2-Sep-23	59	129.9	15	130
3-Sep-23	60	0.1	200	110
4-Sep-23	36	0.3	20	72
5-Sep-23	59	0.1	23	87
6-Sep-23	60	179.8	40	140
7-Sep-23	36	25.1	235	150
8-Sep-23	59	123.9	0	0
9-Sep-23	60	306.3	21	0
Total (bbl)		Volume of Built	Surface Losses	Formation Losses
		13,820.8	4,963	6,526

Figure 10-7: Daily Volume Breakdown Table

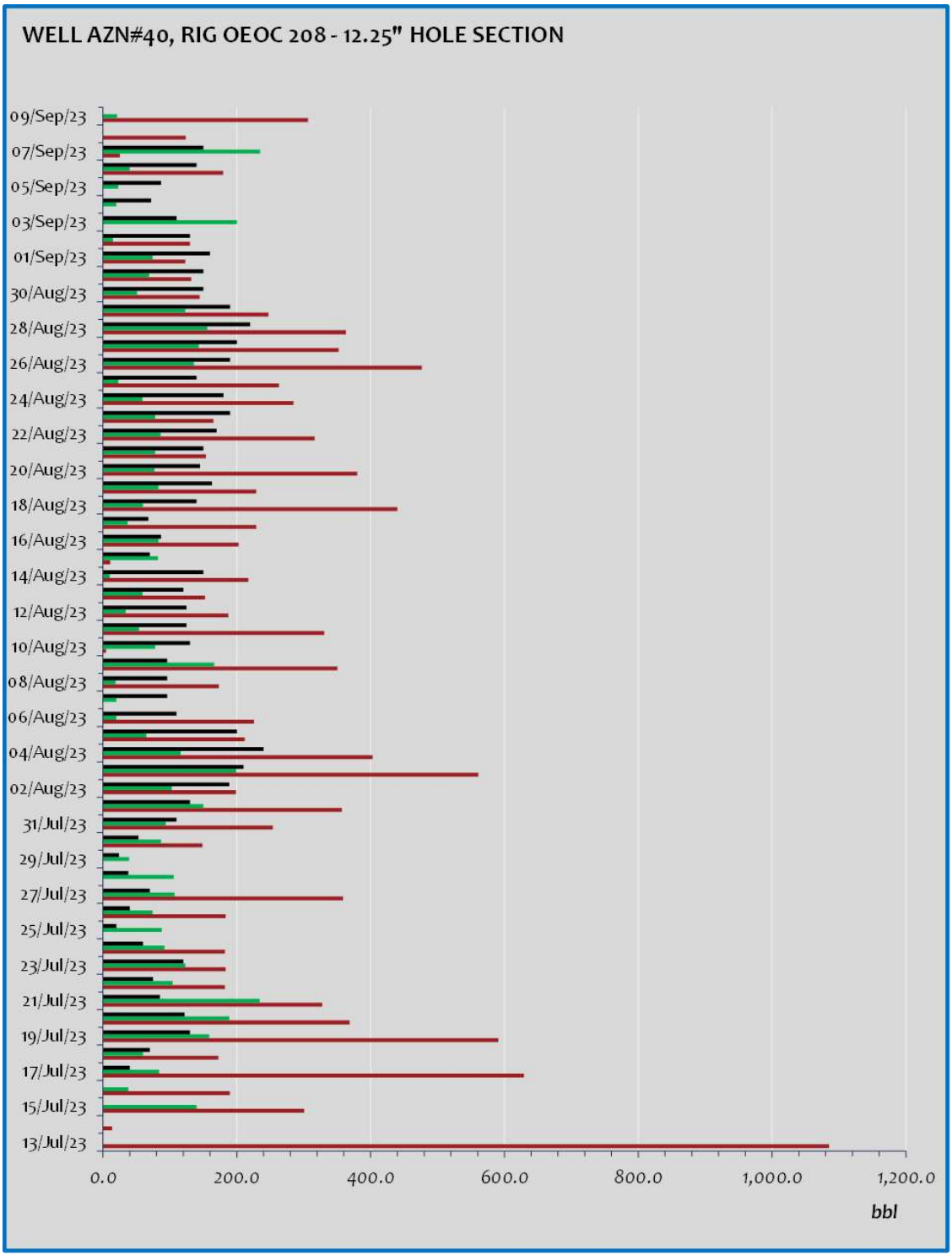


Figure 10-8: Daily Volume Breakdown Chart

Date/DMR.N		Volume of Built	Dump/Contamination	Shakers	Evaporation	Centrifuge	Trip	Mud cleaner	Pit Cleaning	Formation	Other
13-Jul-23	01	1,085.1	0	0	0	0	0	0	0	0	0
14-Jul-23	02	14.0	0	0	0	0	0	0	0	0	0
15-Jul-23	03	300.6	140	0	0	0	0	0	0	0	0
16-Jul-23	04	189.4	0	20	0	0	8	0	10	0	0
17-Jul-23	05	629.0	0	20	10	25	9	0	20	40	0
18-Jul-23	06	172.3	0	10	10	0	15	0	10	70	15
19-Jul-23	07	590.6	0	48	15	55	0	0	0	130	41
20-Jul-23	08	368.7	0	48	15	40	0	48	0	122	38
21-Jul-23	09	327.8	40	32	15	40	20	48	0	85	39
22-Jul-23	10	182.3	0	15	14	33	0	22	0	75	20
23-Jul-23	11	183.5	0	28	14	27	0	24	0	120	30
24-Jul-23	12	182.3	0	10	8	24	10	15	0	60	25
25-Jul-23	13	0.2	0	0	8	24	50	6	0	20	0
26-Jul-23	14	183.2	0	16	12	22	0	14	0	40	10
27-Jul-23	15	358.7	0	24	15	28	0	15	0	70	25
28-Jul-23	16	0.1	40	4	15	22	5	10	0	38	10
29-Jul-23	17	0.0	0	0	15	22	2	0	0	24	0
30-Jul-23	18	148.5	0	12	15	28	0	22	0	53	10
31-Jul-23	19	253.7	0	30	13	27	0	19	0	110	5
1-Aug-23	20	357.0	20	33	15	35	0	22	0	130	25
2-Aug-23	21	198.8	0	36	15	32	0	0	0	189	20
3-Aug-23	22	560.7	74	36	15	44	0	0	0	210	30
4-Aug-23	23	402.8	0	36	16	44	0	0	0	240	20
5-Aug-23	24	212.0	0	10	15	30	0	0	0	200	10
6-Aug-23	25	225.9	0	0	5	15	0	0	0	110	0
7-Aug-23	26	0.0	0	0	5	15	0	0	0	96	0
8-Aug-23	27	173.1	0	0	5	14	0	0	0	96	0
9-Aug-23	28	350.3	0	36	10	120	0	0	0	96	0
10-Aug-23	29	4.7	0	17	10	36	5	0	0	130	10
11-Aug-23	30	330.5	0	16	12	20	3	0	0	125	3

12-Aug-23	31	187.3	0	6	10	0	3	0	10	125	5
13-Aug-23	32	152.8	0	23	9	12	0	0	5	120	10
14-Aug-23	33	217.5	0	5	5	0	0	0	0	150	0
15-Aug-23	34	10.8	0	20	10	12	0	0	20	70	20
16-Aug-23	35	202.8	0	18	10	0	0	0	45	87	10
17-Aug-23	36	229.1	0	10	12	0	0	0	0	68	15
18-Aug-23	37	439.7	0	20	14	16	0	0	0	140	10
19-Aug-23	38	228.9	0	28	16	24	0	0	0	163	15
20-Aug-23	39	379.8	0	28	19	0	0	0	0	145	30
21-Aug-23	40	153.6	0	30	18	0	0	0	20	150	10
22-Aug-23	41	316.4	0	37	19	0	0	0	20	170	10
23-Aug-23	42	165.2	0	30	18	0	5	0	25	190	0
24-Aug-23	43	284.5	0	0	19	0	25	0	15	180	0
25-Aug-23	44	263.0	0	0	13	0	0	0	0	140	10
26-Aug-23	45	476.3	0	35	18	43	0	0	20	190	20
27-Aug-23	46	352.0	0	37	17	44	0	0	45	200	0
28-Aug-23	47	363.1	0	45	17	44	0	0	30	220	20
29-Aug-23	48	247.3	0	47	16	40	0	0	10	190	10
30-Aug-23	49	144.5	0	35	16	0	0	0	0	150	0
31-Aug-23	50	131.9	0	30	15	24	0	0	0	150	0
1-Sep-23	51	123.0	0	4	15	0	0	0	45	160	10
2-Sep-23	52	129.9	0	0	5	0	0	0	0	130	10
3-Sep-23	53	0.1	0	0	5	0	0	0	145	110	50
4-Sep-23	54	0.3	0	0	0	0	0	0	10	72	10
5-Sep-23	55	0.1	0	8	5	0	0	0	0	87	10
6-Sep-23	56	179.8	0	5	5	0	0	0	20	140	10
7-Sep-23	57	25.1	70	0	0	0	0	0	80	150	85
8-Sep-23	58	123.9	0	0	0	0	0	0	0	0	0
9-Sep-23	59	306.3	15	6	0	0	0	0	0	0	0
Total (bbl)		Volume of Built	Dump/Contaminate	Shakers	Evaporation	Centrifuge	Trip	Mud cleaner	Pit Cleaning	Formation	Other
		13,820.8	399	1,044	643	1,081	160	265	605	6,526	766.0

Figure 10-9: Loss Distribution Table

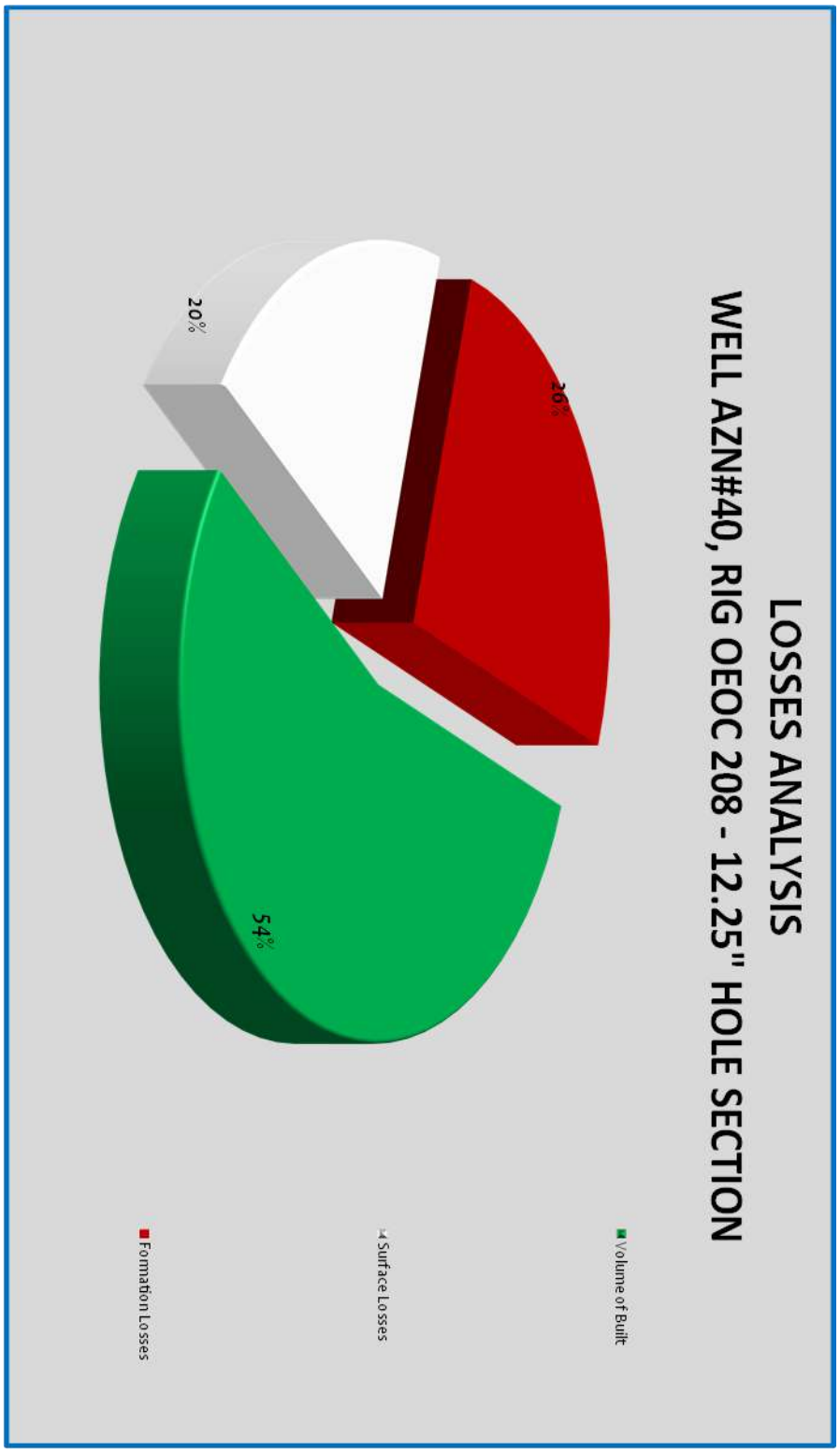


Figure 10-10: Loss Analysis Graph

Date	DEPTH	Density	VIS	PV	YP	Water Loss	pH	MBT	Solid %	Oil %	Chloride	KCL (%Wt)	Calcium	Total Hardness	Mud Type
13-Jul-23	1492	73	50	30	11	2.5	10.5	2.5	10	0	160000	3	600	1000	Kcl Pol Mud
14-Jul-23	1492	73	50	30	11	2.5	9.6	2.5	10	0	160000	3	600	1000	Kcl Pol Mud
15-Jul-23	1492	73	50	30	11	2.5	9.6	2.5	10	0	160000	2	600	1000	Kcl Pol Mud
16-Jul-23	1510	73	43	15	10	2.5	9.6	2.5	10	0	153000	2	840	1240	Kcl Pol Mud
17-Jul-23	1556	73	43	15	10	2.5	9.4	2.5	10	0	153000	1.4	880	1280	Kcl Pol Mud
18-Jul-23	1571	73	41	15	8	2.5	9.3	2.5	10	0	153000	1.4	880	1280	Kcl Pol Mud
19-Jul-23	1756	73	39	17	9	3	9.6	2.5	10	0	160000	2	880	1000	Kcl Pol Mud
20-Jul-23	1908	73	39	17	10	3	9.5	2.5	11	0	160000	2	880	1120	Kcl Pol Mud
21-Jul-23	1960	74	41	18	11	3	9.5	2.5	11	0	160000	2	880	1080	Kcl Pol Mud
22-Jul-23	2010	75	39	16	10	3	9.6	2.5	12	0	160000	2	880	1080	Kcl Pol Mud
23-Jul-23	2084	75	40	17	11	2.8	9.4	2.5	14	0	162000	2	840	1080	Kcl Pol Mud
24-Jul-23	2115	75	42	18	12	2.8	9.5	2.5	14	0	162000	2	760	960	Kcl Pol Mud
25-Jul-23	2115	75	42	18	12	2.8	9.5	2.5	14	0	162000	2	760	960	Kcl Pol Mud
26-Jul-23	2161	75	41	16	13	2.8	9.5	2.5	14	0	160000	2	800	1000	Kcl Pol Mud
27-Jul-23	2191	75	42	17	15	2.9	9.5	2.5	14	0	160000	2	800	1000	Kcl Pol Mud
28-Jul-23	2197	75	42	17	15	2.9	9.5	2.5	14	0	160000	2.5	800	1000	Kcl Pol Mud
29-Jul-23	2197	75	42	17	15	2.9	9.5	2.5	14	0	160000	2.5	800	1000	Kcl Pol Mud
30-Jul-23	2253	76	41	16	15	2.9	9.6	2.5	15	0	160000	3	800	1000	Kcl Pol Mud
31-Jul-23	2394	76	40	16	15	2.7	9.4	2.5	15	0	160000	3	760	960	Kcl Pol Mud
1-Aug-23	2,512	77.0	41	17	15	2.6	9.4	2.5	15.5	0	170,000	3	640	760	Kcl Pol Mud
2-Aug-23	2,689	77.0	41	16	15	2.6	9.3	2.5	15.5	0	174,000	3	640	760	Kcl Pol Mud
3-Aug-23	2,768	78.0	42	17	16	2.8	9.2	2.5	15.5	1	174,000	3	680	780	Kcl Pol Mud
4-Aug-23	2,850	78.0	42	17	16	2.7	9.4	2.5	16.5	1	174,000	3	680	760	Kcl Pol Mud
5-Aug-23	2,854	78.0	43	17	16	2.7	9.4	2.5	16.5	1	174,000	3	680	760	Kcl Pol Mud
6-Aug-23	2,854	78.0	43	17	16	2.7	9.4	2.5	16.5	1	174,000	3	680	760	Kcl Pol Mud
7-Aug-23	2,854	78.0	43	17	16	2.7	9.3	2.5	16.5	1	174,000	3	680	760	Kcl Pol Mud
8-Aug-23	2,854	78.0	43	17	16	2.7	9.2	2.5	16.5	1	174,000	3	680	760	Kcl Pol Mud
9-Aug-23	2,900	79.0	42	17	13	3.6	9.6	2.5	16.5	2	172,000	2.5	680	760	Kcl Pol Mud
10-Aug-23	2,919	79.0	43	18	15	2.7	9.6	2.5	16.5	2	172,000	2.5	680	760	Kcl Pol Mud
11-Aug-23	2,922	79.5	42	16	14	2.7	9.5	2.5	16.5	2	172,000	2	680	760	Kcl Pol Mud
12-Aug-23	2,923	80.0	46	17	15	3.5	9.4	2.5	16.5	1.5	172,000	2	680	760	Kcl Pol Mud
13-Aug-23	2,944	80.5	43	18	15	3.7	9.5	2.5	17	1	174,000	1	680	760	Kcl Pol Mud
14-Aug-23	2,945	80.5	43	17	14	4	9.4	2.5	17	1	172,000	0.5	680	760	Kcl Pol Mud
15-Aug-23	2,952	80.5	43	17	15	4	9.6	2.5	17	1	176,000	0.5	760	840	Kcl Pol Mud
16-Aug-23	2,961	80.5	43	17	15	4	9.5	2.5	17	1	172,000	0.5	760	840	Kcl Pol Mud
17-Aug-23	2,988	80.5	41	16	12	4	9.6	2.5	17	1	172,000	0.5	680	800	Kcl Pol Mud
18-Aug-23	3,069	80.5	41	15	13	4.3	9.5	2.5	17	1	176,000	0.5	640	720	Kcl Pol Mud
19-Aug-23	3,191	80.5	41	15	12	4.3	9.5	2.5	17	1	176,000	0.5	600	720	Kcl Pol Mud
20-Aug-23	3,275	80.5	41	15	13	4.8	9.5	2.5	17	1	176,000	0.5	600	720	Kcl Pol Mud
21-Aug-23	3,349	82.0	41	16	13	3	9.4	2.5	18	1	176,000	5	600	720	Kcl Pol Mud
22-Aug-23	3,497	83.0	42	17	15	3.2	9.4	2.5	18	1	176,000	5	600	720	Kcl Pol Mud
23-Aug-23	3,552	83.0	43	19	16	3.1	9.4	2.5	18	1	174,000	5	600	720	Kcl Pol Mud
24-Aug-23	3,552	83.0	43	19	16	3.1	9.3	2.5	18	1	174,000	5	600	720	Kcl Pol Mud
25-Aug-23	3,552	83.0	43	19	16	3.1	9.3	2.5	18	1	174,000	5	600	720	Kcl Pol Mud
26-Aug-23	3,656	84.0	42	21	15	3.1	9.5	2.5	18.5	0	172,000	5	600	720	Kcl Pol Mud
27-Aug-23	3,776	84.0	42	21	15	3.1	9.5	2.5	18.5	0	172,000	5	620	720	Kcl Pol Mud
28-Aug-23	3,833	84.0	42	19	14	3.1	9.5	2.5	18.5	0	174,000	5	600	720	Kcl Pol Mud
29-Aug-23	3,876	84.5	42	19	14	3	9.4	2.5	18.5	0	174,000	5	680	800	Kcl Pol Mud
30-Aug-23	3,910	85.0	43	20	15	3	9.5	2.5	19	1	174,000	5	680	800	Kcl Pol Mud
31-Aug-23	3,932	85.0	43	20	15	2.9	9.5	2.5	19	1	176,000	5	680	800	Kcl Pol Mud
1-Sep-23	3,934	85.0	43	20	15	2.9	9.5	2.5	19	1	176,000	5	680	800	Kcl Pol Mud
2-Sep-23	3,934	85.0	43	20	15	2.9	9.4	2.5	19	1	176,000	5	680	800	Kcl Pol Mud
3-Sep-23	3,934	85.0	43	20	15	2.9	9.5	2.5	19	1	176,000	5	680	800	Kcl Pol Mud
4-Sep-23	3,934	85.0	43	20	15	2.9	9.5	2.5	19	1	176,000	5	680	800	Kcl Pol Mud
5-Sep-23	3,934	85.0	43	20	15	2.9	9.5	2.5	19	1	176,000	5	680	800	Kcl Pol Mud
6-Sep-23	3,934	85.0	43	20	15	2.9	9.5	2.5	19	1	176,000	5	680	800	Kcl Pol Mud
7-Sep-23	3,934	85.0	43	20	15	2.9	9.5	2.5	19	1	176,000	5	680	800	Kcl Pol Mud
8-Sep-23	3,934	85.0	43	20	15	2.9	9.2	2.5	19	1	176,000	5	680	800	Kcl Pol Mud
9-Sep-23	3,934	85.0	45	20	16	2.9	9.9	2.5	19	1	176,000	5	880	960	Kcl Pol Mud

Drilling Start
Drilling End

Figure 10-11: Fluid Properties Recap

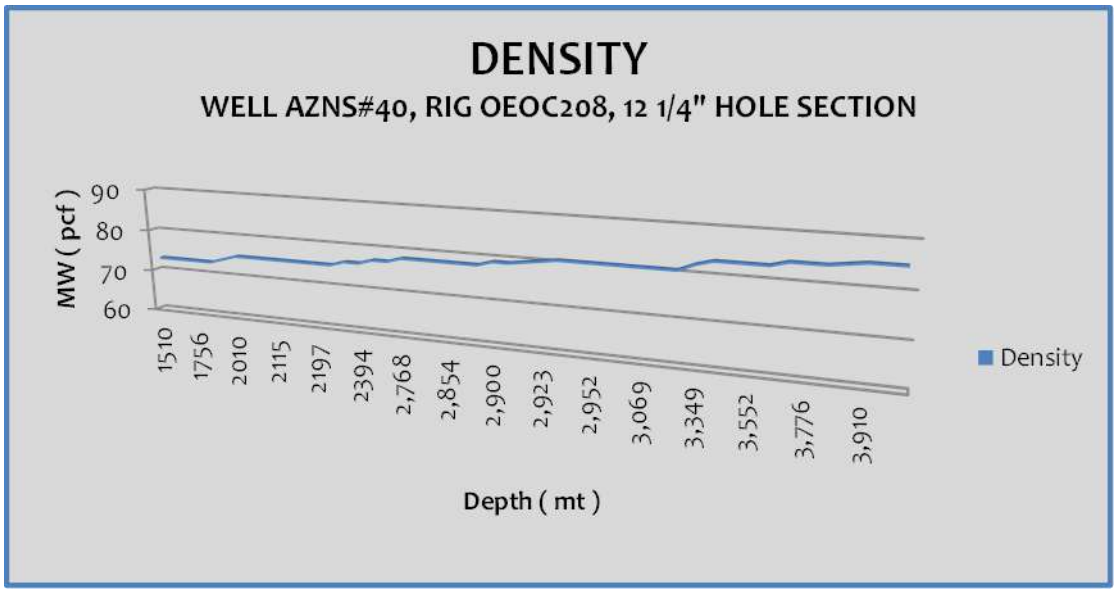


Figure 10-12: Fluid Density Graph

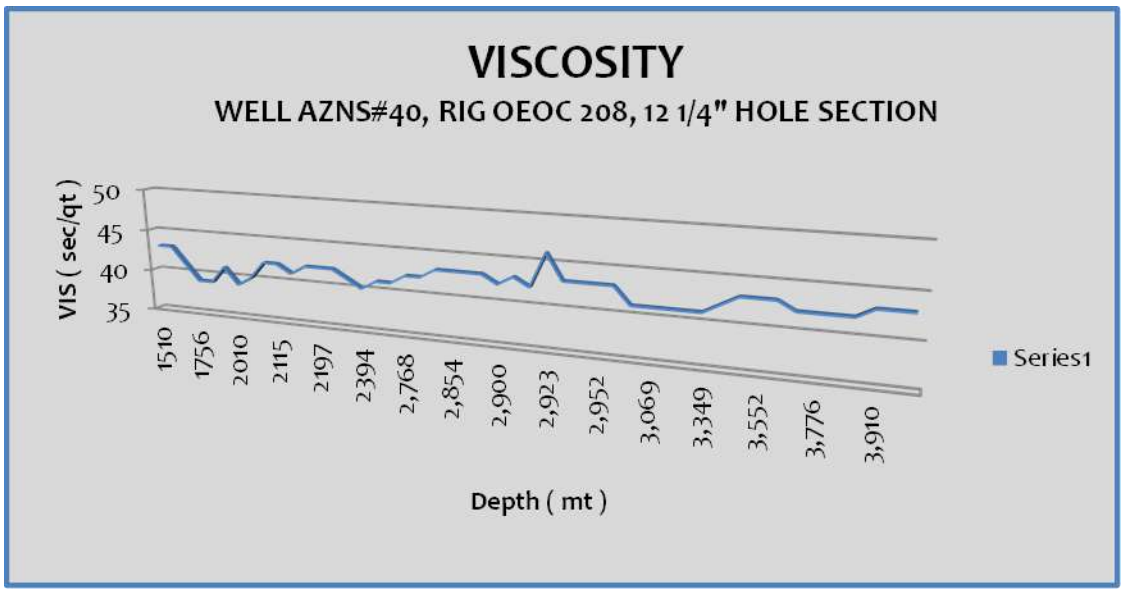


Figure 10-13: Fluid Viscosity Graph

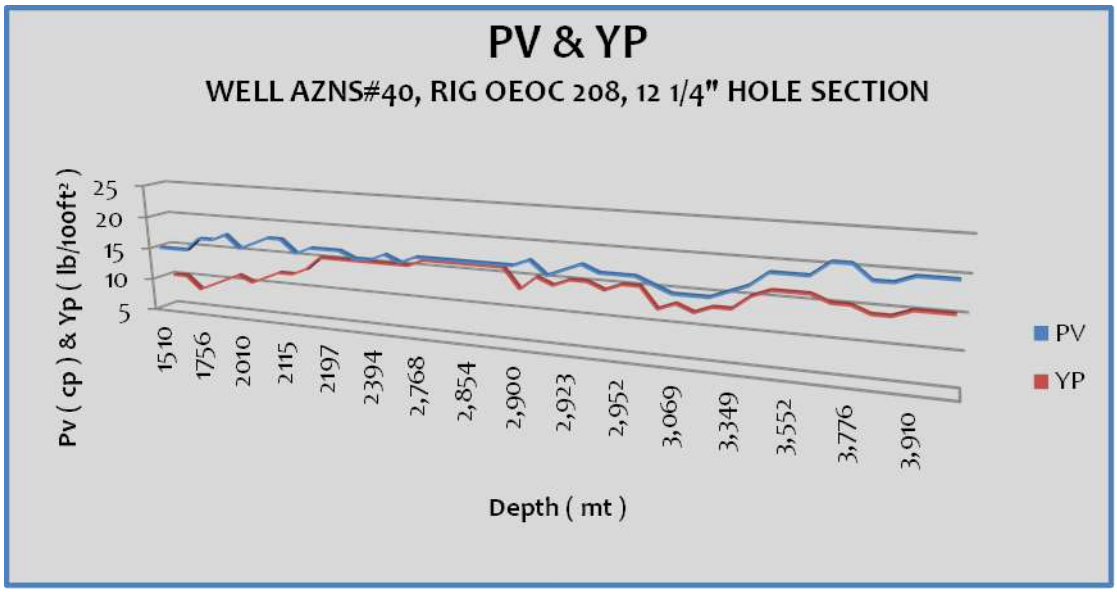


Figure 10-14: Fluid PV & YP Graph

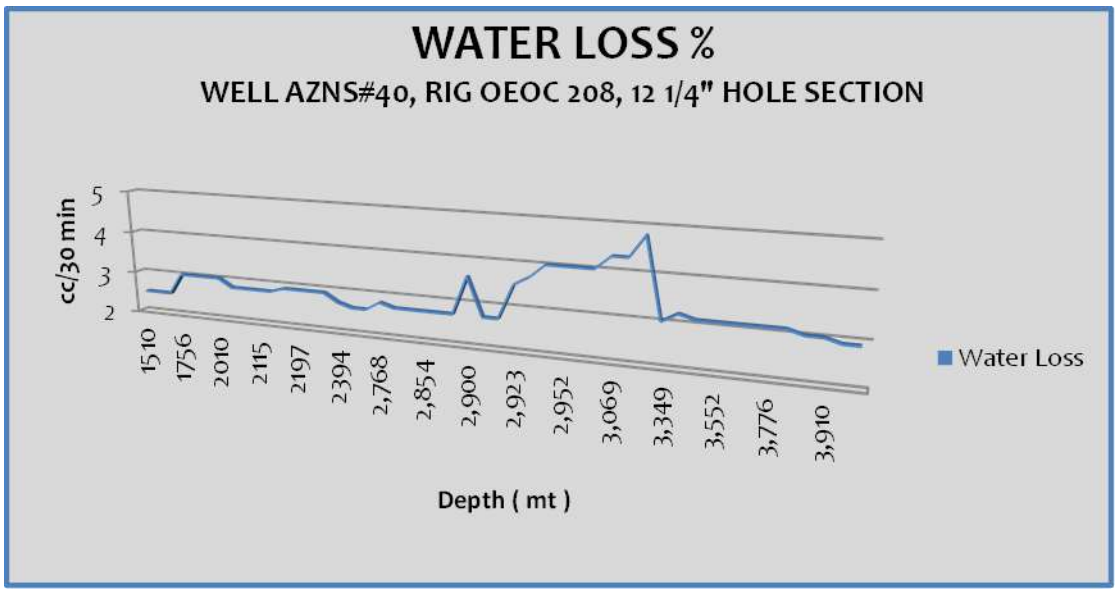


Figure 10-15: Water Loss Graph

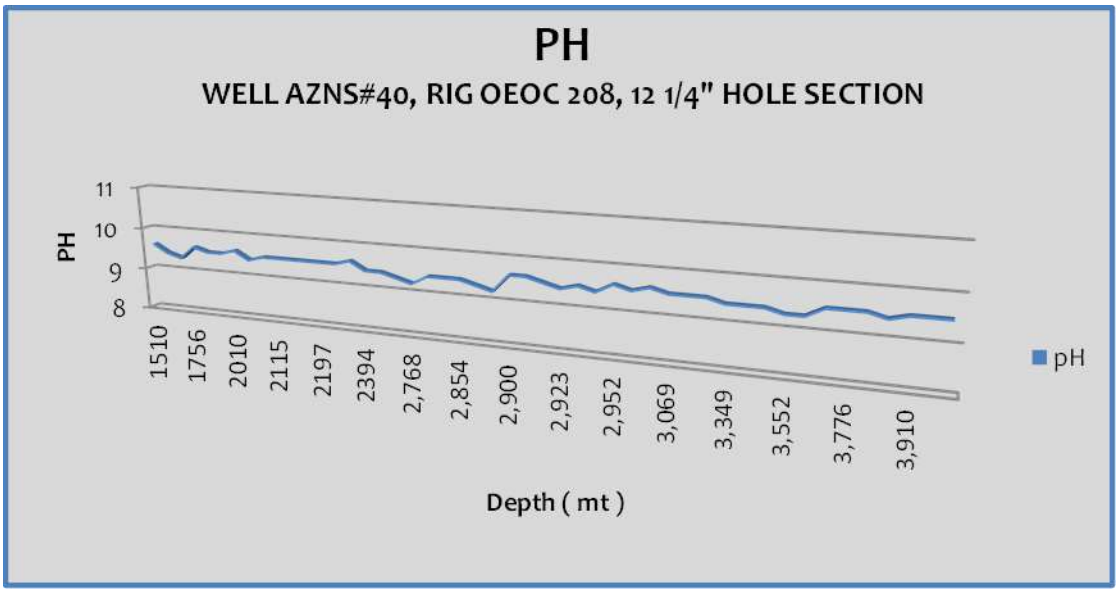


Figure 10-16: PH Graph

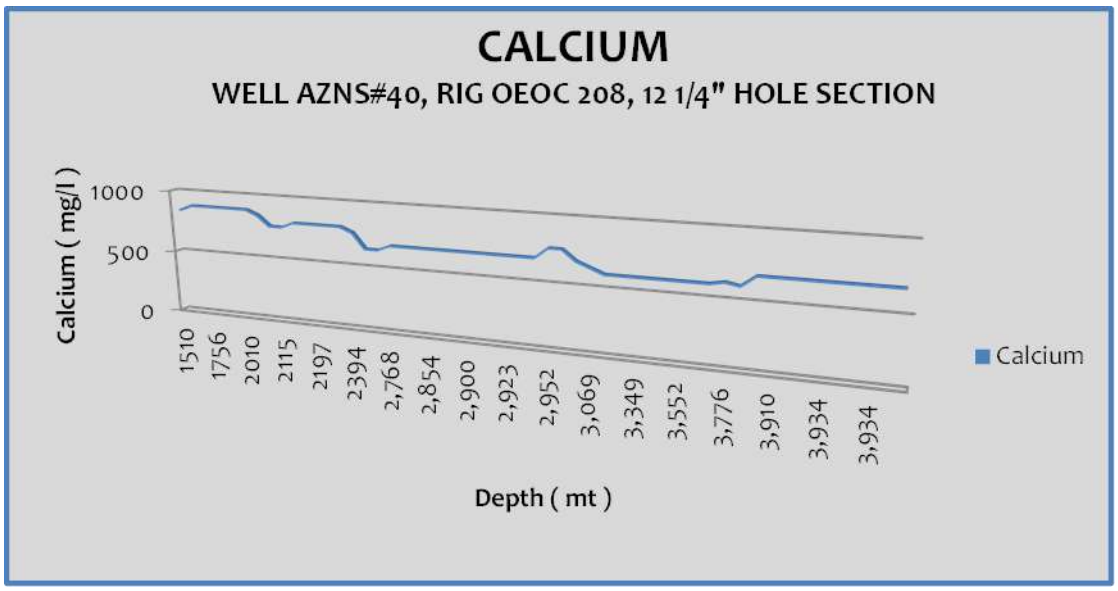


Figure 10-17: Calcium Graph

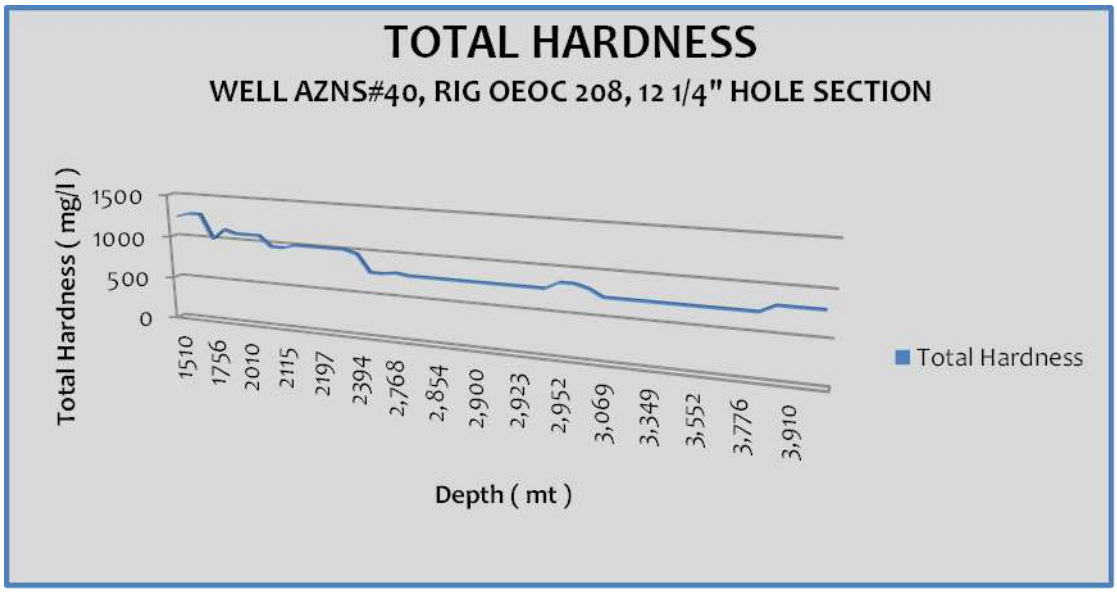


Figure 10-18: Total Hardness Graph

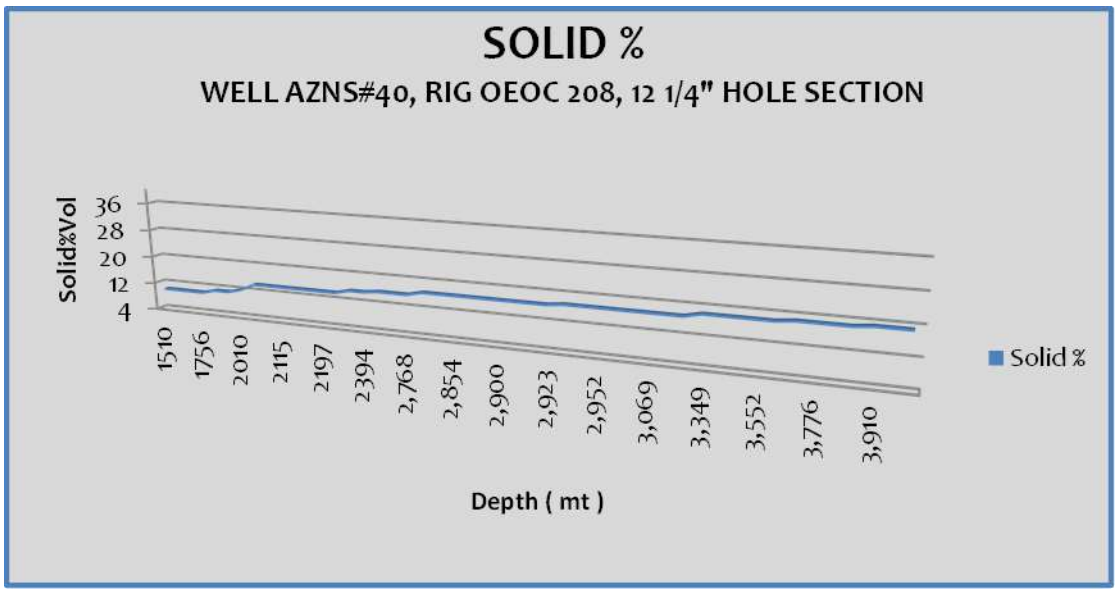


Figure 10-19: Solid Graph

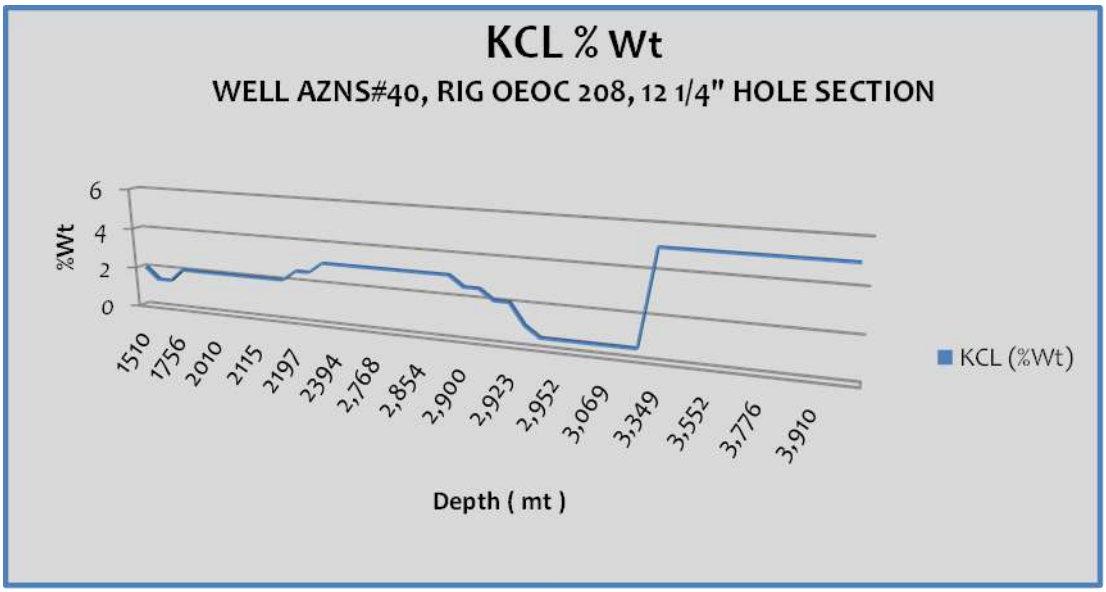


Figure 10-20: KCl% Wt Graph

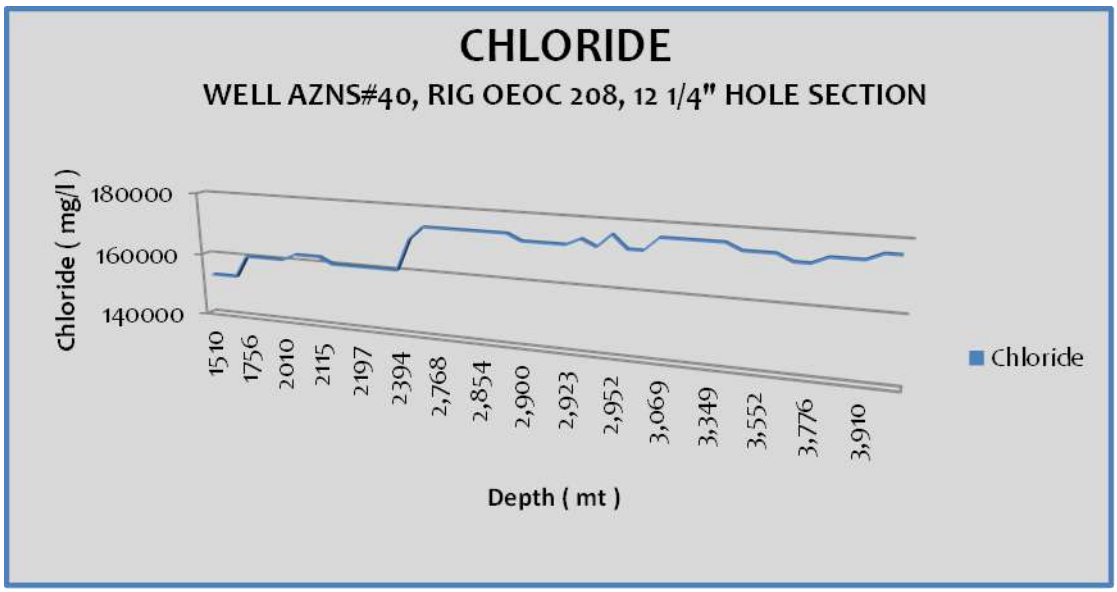


Figure 10-21: Chloride Graph

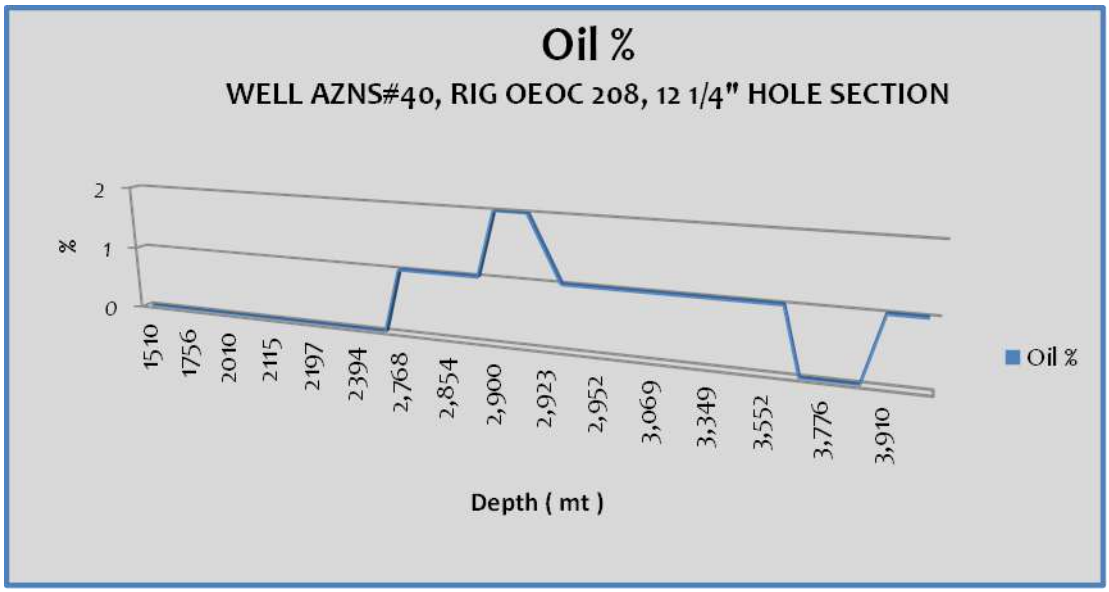


Figure 10-22: Oil% Graph

Rig OEOC 208 has 3 shale shakers were dressed with:

- **Shaker #1:** 200-170 Mesh Screen
- **Shaker #2:** 200-170 Mesh Screen
- **Shaker #3:** 200-170 Mesh Screen

WELL AZN#40, RIG OEOC 208 - 12.25" HOLE SECTION	
Solid Control Equipment Working Hr.s	
Shaker # 01	821
Shaker # 02	800
Shaker # 03	394
Mud Cleaner	318
Centrifuge#1	434
Centrifuge#2	83
Centrifuge#3	90

Figure 10-23: Solid Control Equipment Working Hrs.

10.2 8-1/2" Hole Section

Operation Remark: Made up 8 ½'' drill out BHA and RIH to 3912^{m} then drilled out cement from 3912^{m} to 3933^{m} (9-5/8'' casing shoe) after that started drilling formation as 8 ½'' hole section from 3933^{m} to 3939^{m} with NL-NG then POOH to surface.

Made up 8 ½'' bit and scrapper BHA and RIH to 3939^{m} then performed scrapping from 3939^{m} to 3932^{m} after that POOH to surface, Performed logging (CCL-GR-SBT-VDL).

Made 8 ½'' drilling BHA and RIH to 3939^{m} then displaced with 100^{pcf} KCl/POL mud after that performed FIT test then commenced drilling formation as 8 ½'' hole section from 3939^{m} to 4200^{m} (7'' Liner point) with 1-3 BPH loss meanwhile increased Mw to 104^{pcf}, performed condition trip with good hole condition and spot high viscosity pill and POOH to surface.

Performed logging as per program then RIH with 7'' liner to 4189^{m} after that performed cement job successfully, made 8 ½'' drill out BHA and RIH to 3802^{m} then washed down to 3812.2^{m}, confirmed liner lap at 3812.2^{m} after that POOH to surface.

Made 5 7/8'' clean out BHA and RIH to 3185^{m} then drilled out POB after that RIH to 4180^{m} then POOH to surface, made up tandem scrapper BHA and RIH to 4154^{m} and performed scrapping then POOH to surface.

Performed dry test as per program then performed logging (CCL-GR-SBT-VDL).

Drilling Fluid Remark: The 101^{pcf} KCl/Polymer mud was used for starting this section which was mixed with 12 ¼'' hole 85^{pcf} mud.

It was tried to control mud properties by using all available solid control equipment such as shale shakers was applied as per requirement in order to control Mw.

The Mw increased up to by adding Limestone Powder up to 104^{pcf} at end of this section.

SCE such as Shakers and centrifuges were running to discard LGS while drilling this section.

Treated frequently with essential chemical such as lime as Alkalinity controller to adjust pH, Starch LV & HT for control FL, Soda Ash for control hardness & Xanthan gum as Viscosifier.

Interval II	**3934 - 4200m**	**8-1/2" Hole Interval**	**7" Liner set @ 3182.2 to 4189m**

Table 10-3: AZNS-040 8-1/2" Hole Data

FLUID PROPERTIES RECAP					
KCL Pol Mud		Drilling start	Range		Drilling End
MW (min)	pcf	101	101	104	104
Funnel Viscosity	sec/qt	50	50	60	60
PV	cps	33	33	58	58
YP	lb/100sq.ft	14	11	16	16
API Fluid loss	cc	4.5	2.2	5.5	2.2
pH		9.6	10	11.0	10.3
Solid	%	31	31	40	40
KCL	%	3	3		3
Cl (NaCl)	mg/L	184,000	182000	186000	186,000
Total Hardness	mg/L	860	860	1400	1,400
Ca Hardness	mg/L	720	720	1200	1,200

Figure 10-24: Drilling Fluid Type

Products	Units Used	Unit wt lbs	Total wt	Effective Concentration
Barite*	8	3,300	26,400 lbs	- ppb
CaCl2**	15	2,200	33,000 lbs	275.00 ppb
Caustic Soda	24	55	1,320 lbs	0.44 ppb
Lime	36	55	1,980 lbs	0.66 ppb
Limestone	145	2,200	319,000 lbs	106.89 ppb
Salt (NaCl)	20	2,200	44,000 lbs	14.74 ppb
Starch LV	95	55	5,225 lbs	1.75 ppb
Starch HT	21	55	1,155 lbs	0.39 ppb
Soda ASH	5	55	275 lbs	0.09 ppb
Sodium Bicarbonate	12	55	660 lbs	0.22 ppb
Xanthan Gum	3	55	165 lbs	0.06 ppb

***8BB Barite was used for cement spacer.**
****15BB CaCl2 was used for packer zone.**

Initial Volume	2,329.0 bbl
Total Volume Built	655.3 bbl
Volume received	0.0 bbl
Volume Returned	0.0 bbl
Total Volume Consumed	1,242 bbl
Volume Transfer to Next Section	1,742.0 bbl
Meter Drilled	266 m

Figure 10-25: Material Consumption Analysis

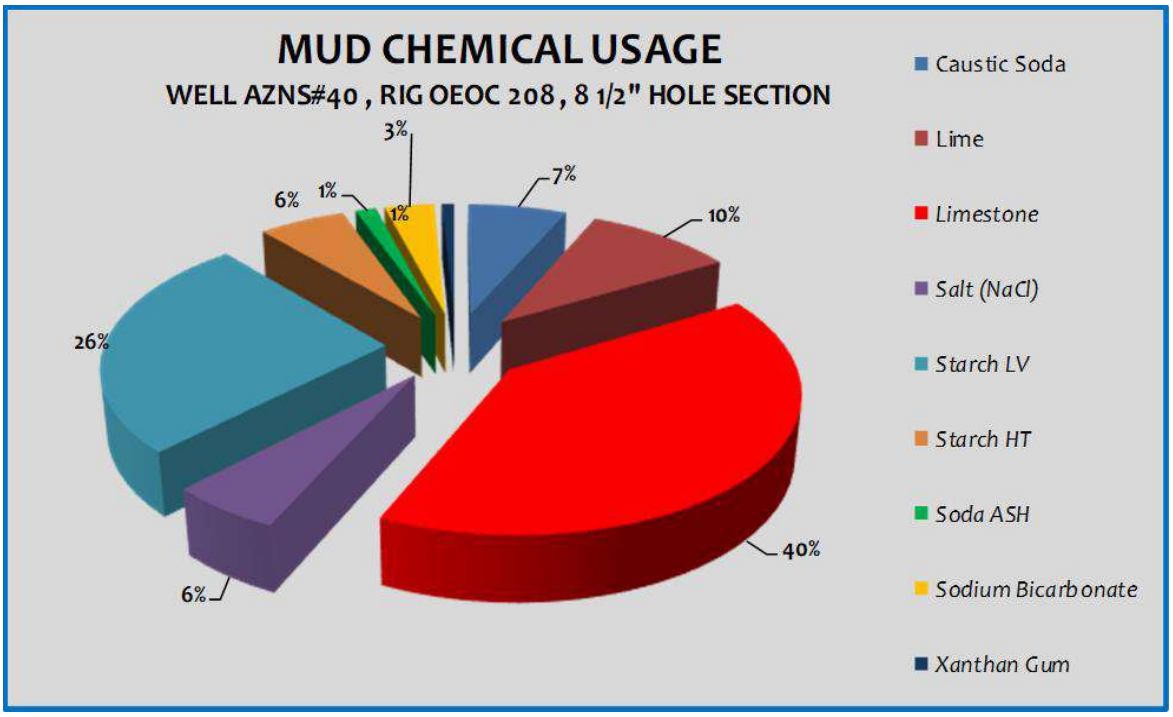


Figure 10-26: Mud Chemical Usage Graph

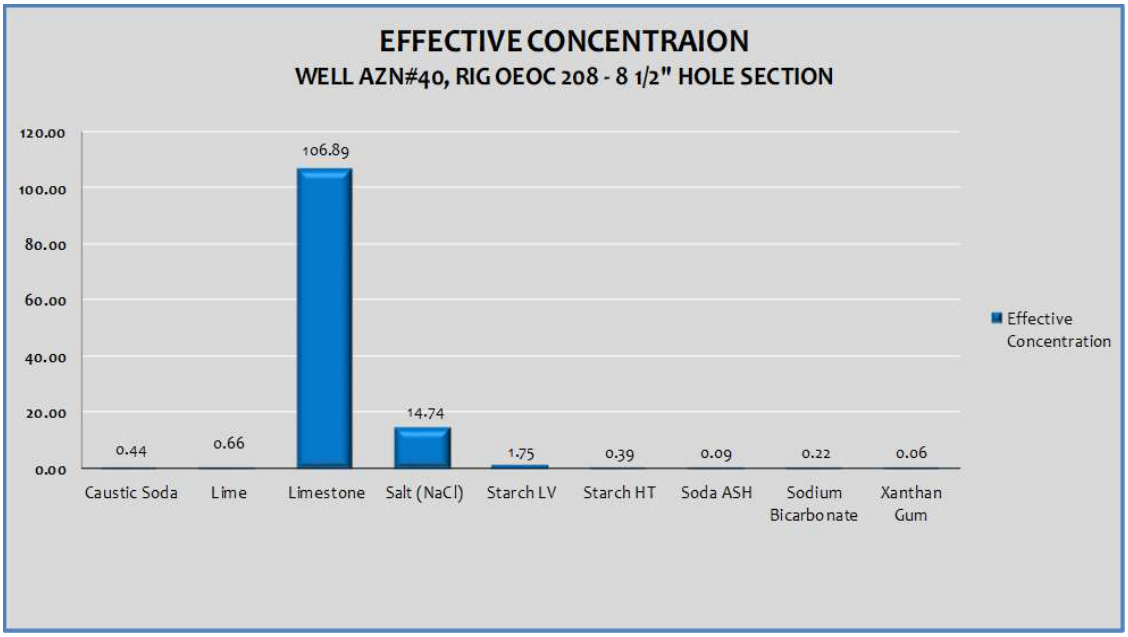


Figure 10-27: Effective Concentration

WELL AZN#40, RIG OEOC 208 , 8 1/2" HOLE SECTION VOLUME ANALYSIS		
Initial Volume	bbls	2,329.0
Received Volume	bbls	0.0
Volume Return	bbls	0.0
Volume Built	bbls	655.3
Surface Losses/Dumped	bbls	861.0
Formation Losses	bbls	381.0
Total Losses	bbls	1,242.0
Volume Transfer Out	bbls	1,742

Figure 10-28: Volume Analysis Table

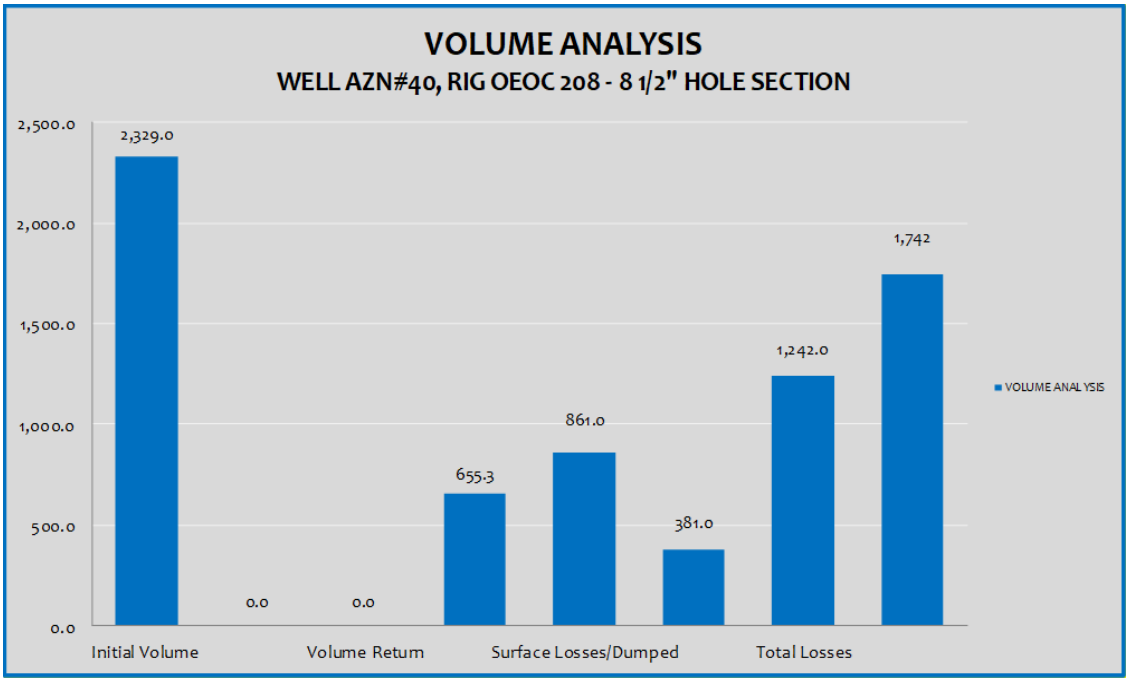


Figure 10-29: Volume Analysis Graph

Date/DMR.N		Volume of Built	Surface Losses	Formation Losses
10-Sep-23	60	147.1	23	0
11-Sep-23	61	23.5	25	0
12-Sep-23	62	0.0	0	0
13-Sep-23	63	68.0	114	0
14-Sep-23	64	76.8	50	24
15-Sep-23	65	3.3	70	48
16-Sep-23	66	12.3	145	70
17-Sep-23	67	16.3	77	71
18-Sep-23	68	12.0	12	50
19-Sep-23	69	0.0	50	47
20-Sep-23	70	0.1	0	45
21-Sep-23	71	18.0	0	24
22-Sep-23	72	0.1	15	2
23-Sep-23	73	0.4	30	0
24-Sep-23	74	0.0	0	0
25-Sep-23	75	12.2	70	0
26-Sep-23	76	155.1	0	0
27-Sep-23	77	0.0	97	0
28-Sep-23	78	0.0	33	0
29-Sep-23	79	0.0	50	0
30-Sep-23	80	35.5	0	0
1-Oct-23	81	58.3	0	0
2-Oct-23	82	16.3	0	0
Total (bbl)		Volume of Built	Surface Losses	Formation Losses
		655.3	861	381

Figure 10-30: Daily Volume Breakdown Table

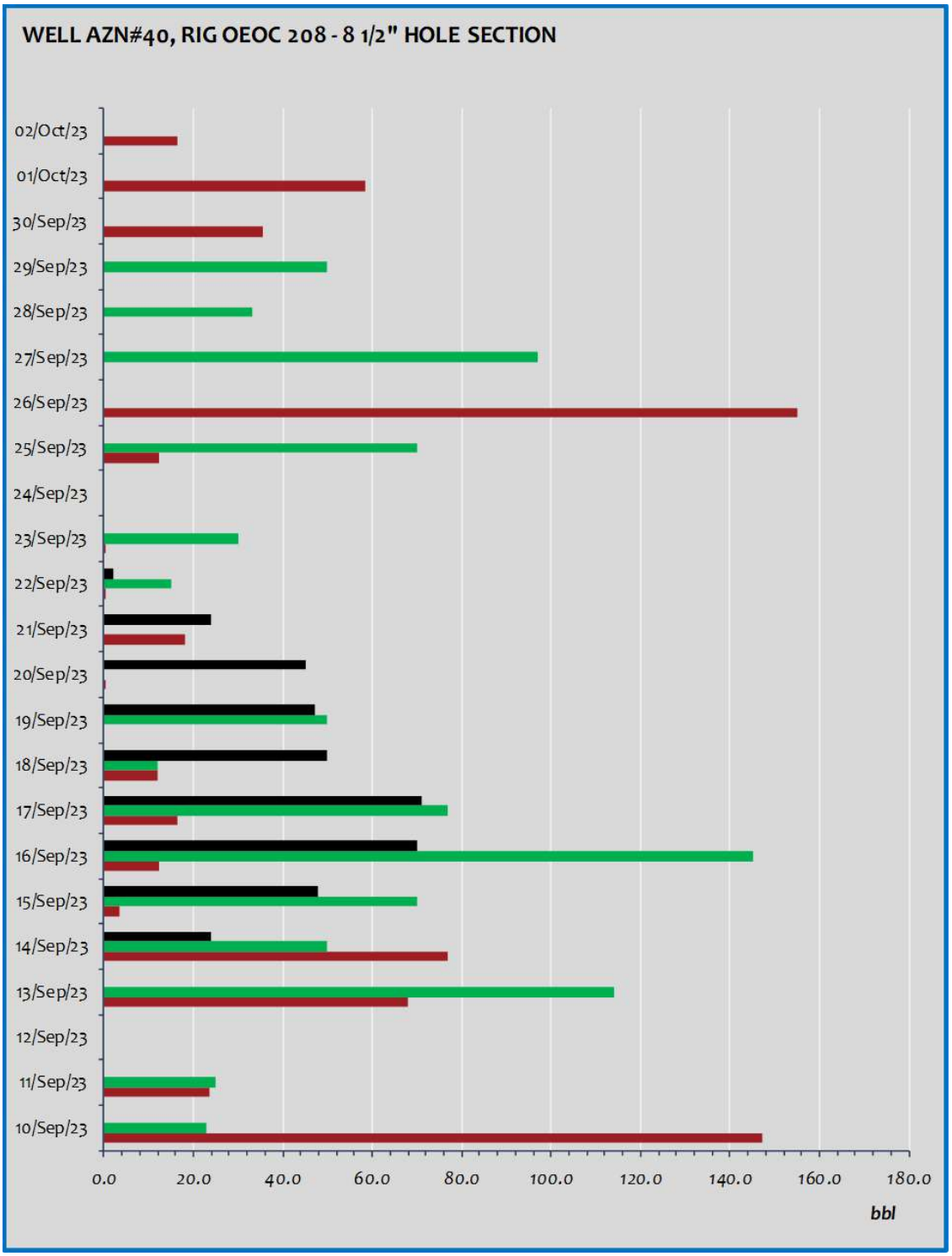


Figure 10-31: Daily Volume Breakdown Chart

Date/DMR.N		Volume of Built	Dump/Contamination	Shakers	Evaporation	Centrifuge	Trip	Mud cleaner	Pit Cleaning	Formation	Other
10-Sep-23	60	147.1	0	8	0	0	15	0	0	0	0
11-Sep-23	61	23.5	0	0	5	0	20	0	0	0	0
12-Sep-23	62	0.0	0	0	0	0	0	0	0	0	0
13-Sep-23	63	68.0	84	10	10	0	0	0	10	0	0
14-Sep-23	64	76.8	0	30	10	0	0	0	10	24	0
15-Sep-23	65	3.3	0	38	12	0	0	0	10	48	10
16-Sep-23	66	12.3	85	40	10	0	0	0	10	70	0
17-Sep-23	67	16.3	0	42	10	0	0	0	15	71	10
18-Sep-23	68	12.0	0	8	4	0	0	0	0	50	0
19-Sep-23	69	0.0	0	0	0	0	0	0	50	47	0
20-Sep-23	70	0.1	0	0	0	0	0	0	0	45	0
21-Sep-23	71	18.0	0	0	0	0	0	0	0	24	0
22-Sep-23	72	0.1	0	0	0	0	15	0	0	2	0
23-Sep-23	73	0.4	30	0	0	0	0	0	0	0	0
24-Sep-23	74	0.0	0	0	0	0	0	0	0	0	0
25-Sep-23	75	12.2	0	0	0	0	20	0	50	0	0
26-Sep-23	76	155.1	0	0	0	0	0	0	0	0	0
27-Sep-23	77	0.0	0	0	0	0	45	0	52	0	0
28-Sep-23	78	0.0	0	0	0	0	0	0	33	0	0
29-Sep-23	79	0.0	0	0	0	0	20	0	30	0	0
30-Sep-23	80	35.5	0	0	0	0	0	0	0	0	0
1-Oct-23	81	58.3	0	0	0	0	0	0	0	0	0
2-Oct-23	82	16.3	0	0	0	0	0	0	0	0	0
Total (bbl)		Volume of Built	Dump/Contaminate	Shakers	Evaporation	Centrifuge	Trip	Mud cleaner	Pit Cleaning	Formation	Other
		655.3	199	176	61	0	135	0	270	381	20.0

Figure 10-32: Loss Distribution Table

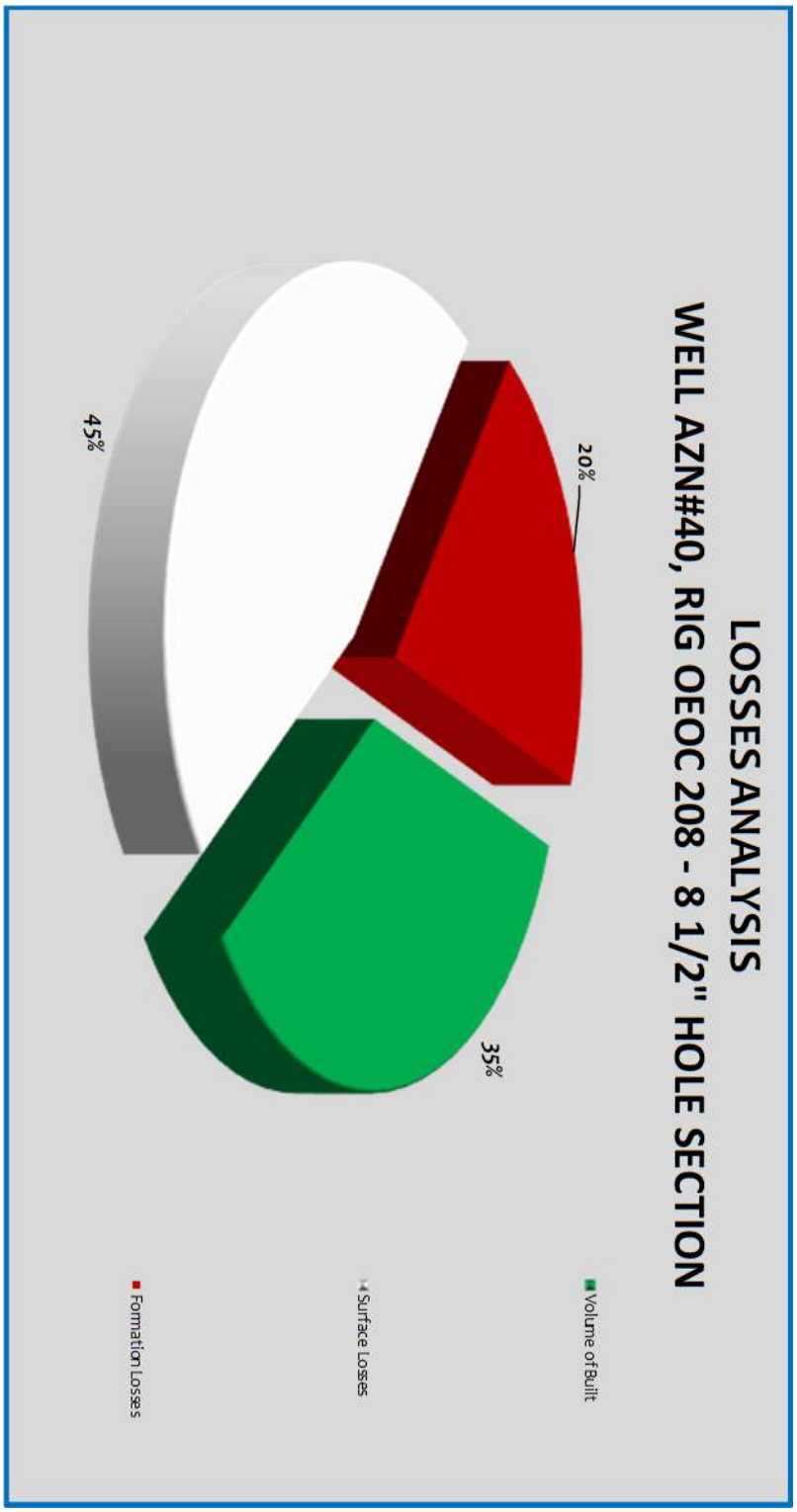


Figure 10-33: Loss Analysis Graph

Date	DEPTH	Density	VIS	PV	YP	Water Loss	pH	MBT	Solid %	Oil %	Chloride	KCL (%Wt)	Calcium	Total Hardness	Mud Type
10-Sep-23	3939	85	43	23	16	3.6	9.6	2.5	19	1	176000	5	720	800	Kd Pol Mud
11-Sep-23	3939	85	43	23	16	3.6	9.6	2.5	19	1	176000	5	720	800	Kd Pol Mud
12-Sep-23	3939	85	43	23	16	3.6	9.6	2.5	19	1	176000	5	720	800	Kd Pol Mud
13-Sep-23	3956	101	50	33	14	4.5	9.6	2.5	31	0	184000	3	720	860	Kd Pol Mud
14-Sep-23	4013	103	55	43	13	5.5	9.6	2.5	40	0	184000	3	720	860	Kd Pol Mud
15-Sep-23	4070	104	55	49	12	4.5	10.3	2.5	40	0	182000	3	1000	1200	Kd Pol Mud
16-Sep-23	4129	104	55	50	13	4.5	10.5	2.5	40	0	183000	3	1000	1200	Kd Pol Mud
17-Sep-23	4186	104	58	57	15	2.2	10.4	2.5	40	0	186000	3	1200	1400	Kd Pol Mud
18-Sep-23	4,200	104.0	60	58	16	2.2	10.3	2.5	40	0	186,000	3	1,200	1,400	Kd Pol Mud
19-Sep-23	4200	104	60	58	16	2.2	10.3	2.5	40	0	186000	3	1200	1400	Kd Pol Mud
20-Sep-23	4200	104	60	58	16	2.2	10.5	2.5	40	0	186000	3	1200	1400	Kd Pol Mud
21-Sep-23	4200	104	60	58	16	2.2	10.5	2.5	40	0	186000	3	1200	1400	Kd Pol Mud
22-Sep-23	4200	104	60	58	16	2.2	10.5	2.5	40	0	186000	3	1200	1400	Kd Pol Mud
23-Sep-23	4200	104	62	58	13	4	11	2.5	40	0	186000	3	1520	1680	Kd Pol Mud
24-Sep-23	4200	104	62	58	13	4	11	2.5	40	0	186000	3	1520	1680	Kd Pol Mud
25-Sep-23	4200	104	59	50	12	18	11	2.5	40	0	186000	3	1520	1680	Kd Pol Mud
26-Sep-23	4200	104	57	50	11	18	11	2.5	40	0	186000	3	1520	1680	Kd Pol Mud
27-Sep-23	4200	104	57	50	11	18	11	2.5	40	0	186000	3	1520	1680	Kd Pol Mud
28-Sep-23	4200	104	57	50	11	18	10.5	2.5	40	0	186000	3	1520	1680	Kd Pol Mud
29-Sep-23	4200	104	60	50	11	18	9	2.5	40	0	186000	3	1520	1680	Kd Pol Mud
30-Sep-23	4200	104	60	50	11	18	9.5	2.5	40	0	186000	3	1520	1680	Kd Pol Mud
1-Oct-23	4200	104	60	50	11	18	9.5	2.5	40	0	186000	3	1520	1680	Kd Pol Mud
2-Oct-23	4200	104	60	50	11	18	9.5	2.5	40	0	186000	3	1520	1680	Kd Pol Mud

Drilling Start

Drilling End

Figure 10-34: Fluid Properties Recap

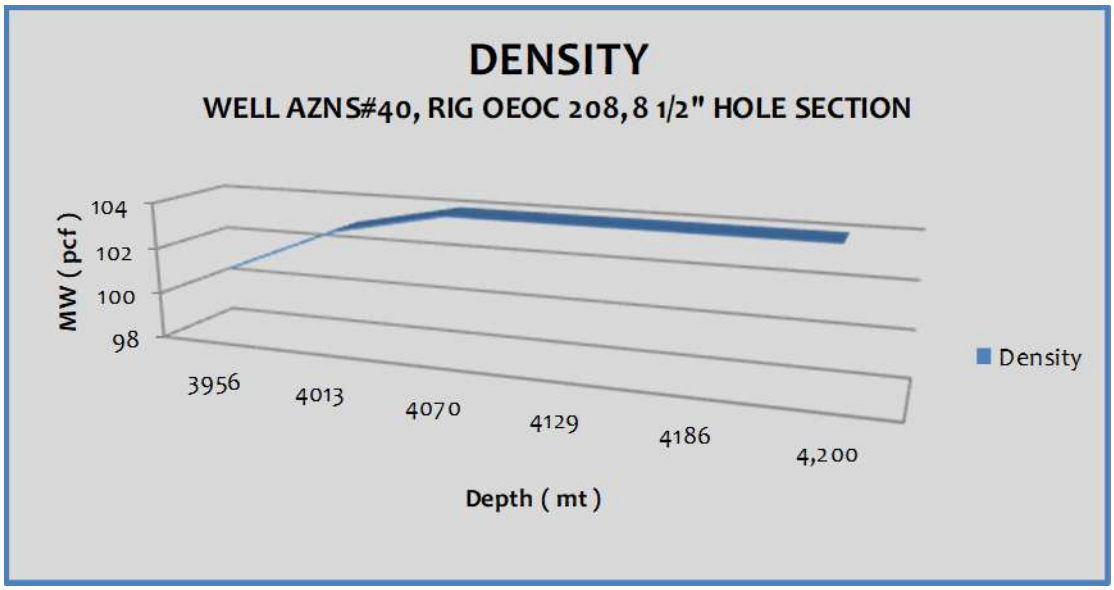


Figure 10-35: Fluid Density Graph

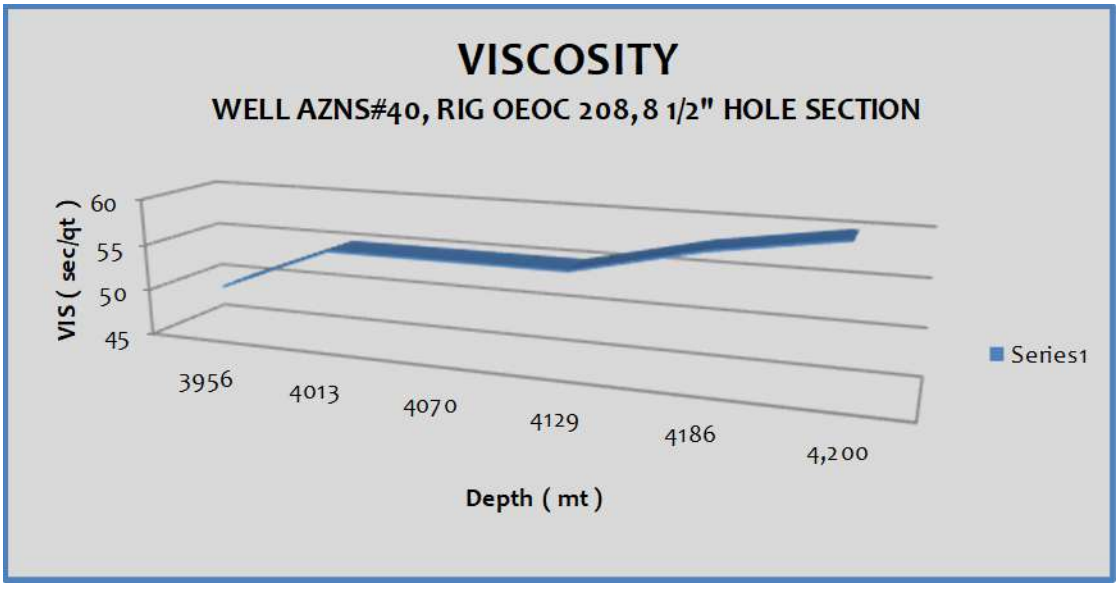


Figure 10-36: Fluid Viscosity Graph

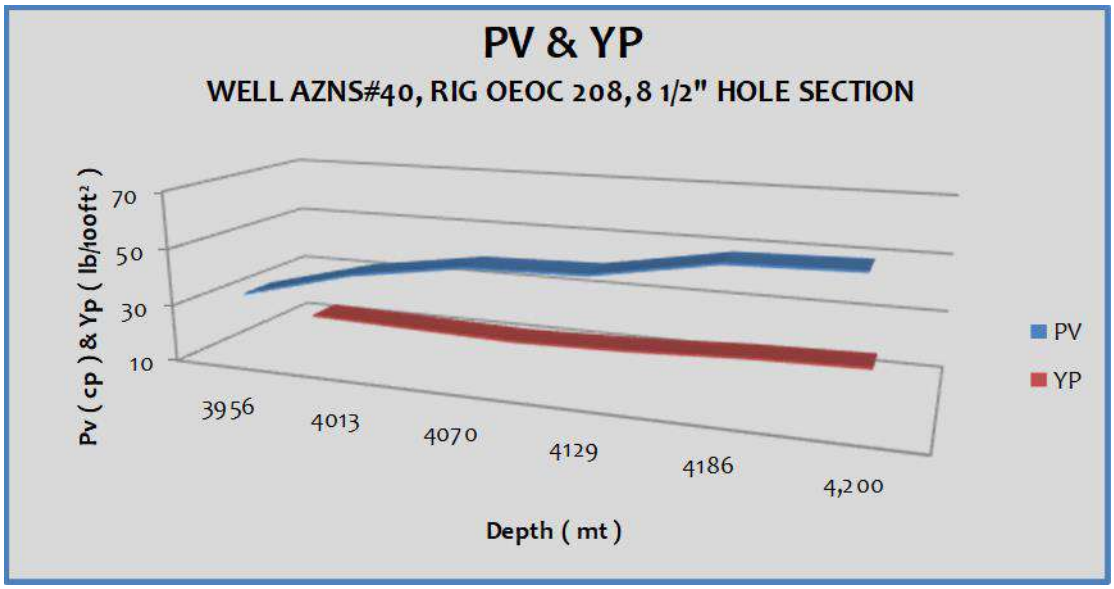


Figure 10-37: Fluid PV & YP Graph

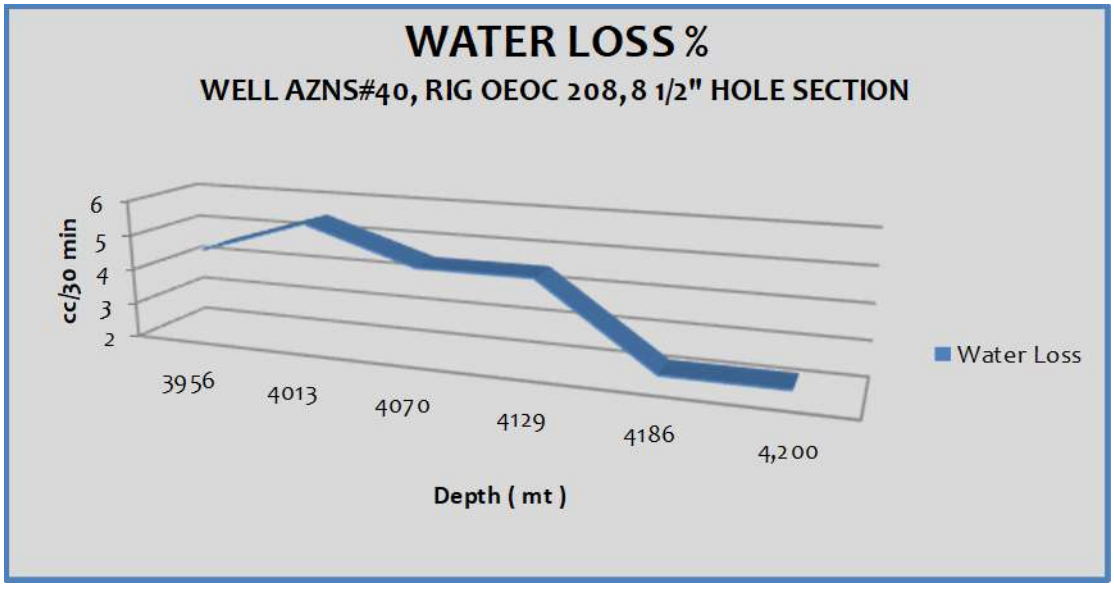


Figure 10-38: Water Loss % Graph

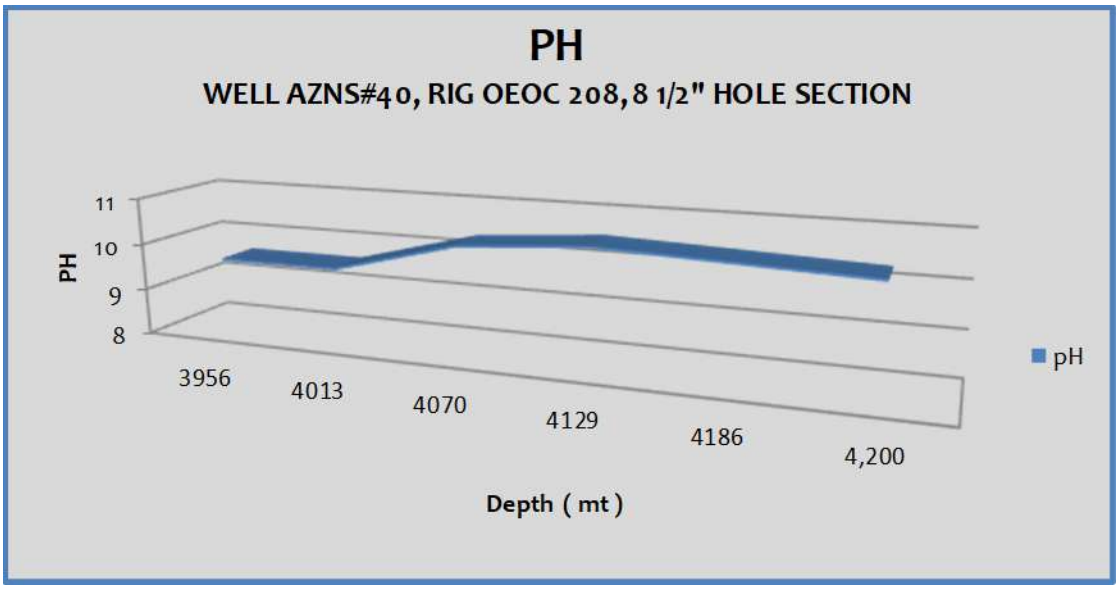


Figure 10-39: PH Graph

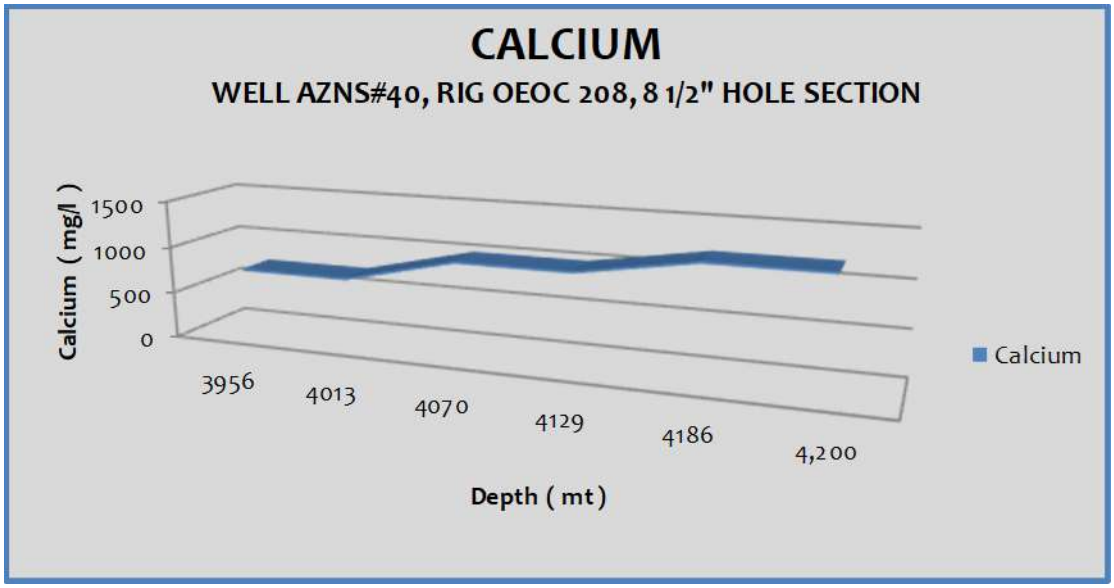


Figure 10-40: Calcium Graph

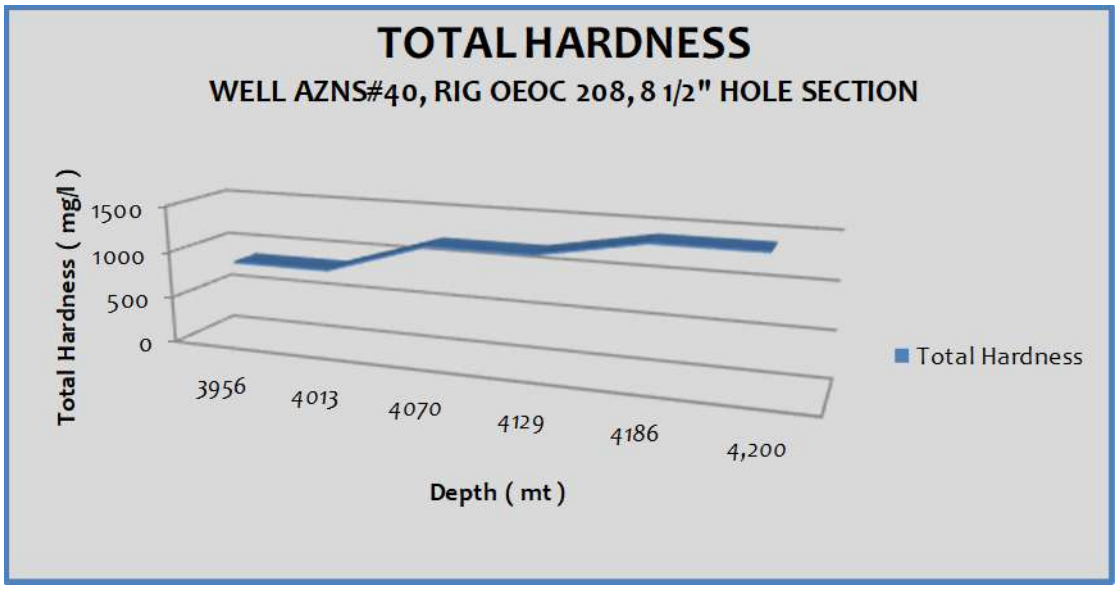


Figure 10-41: Total Hardness Graph

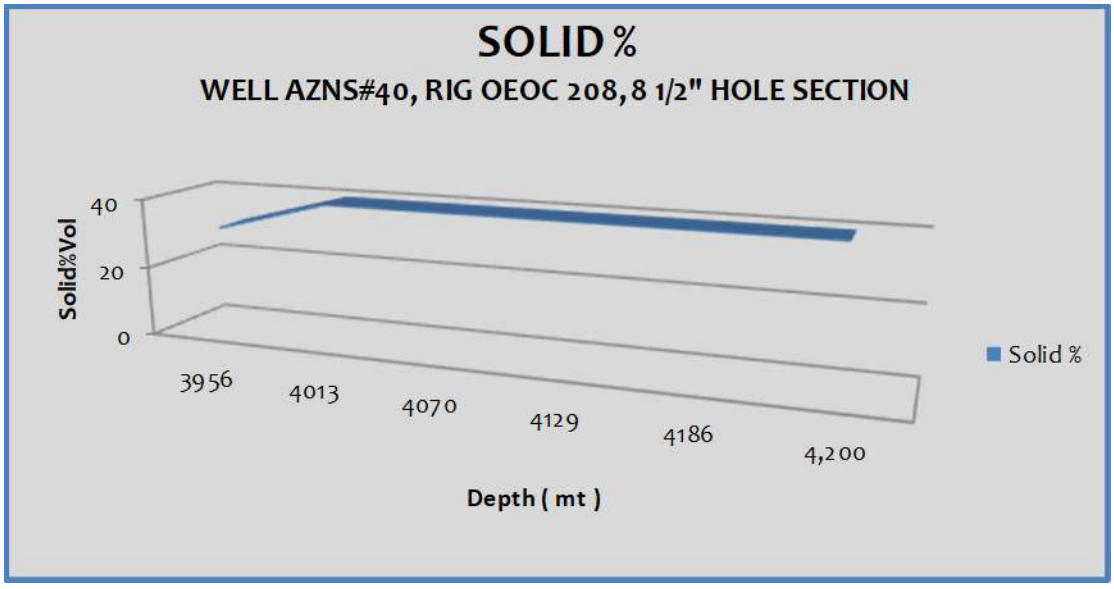


Figure 10-42: Solid % Graph

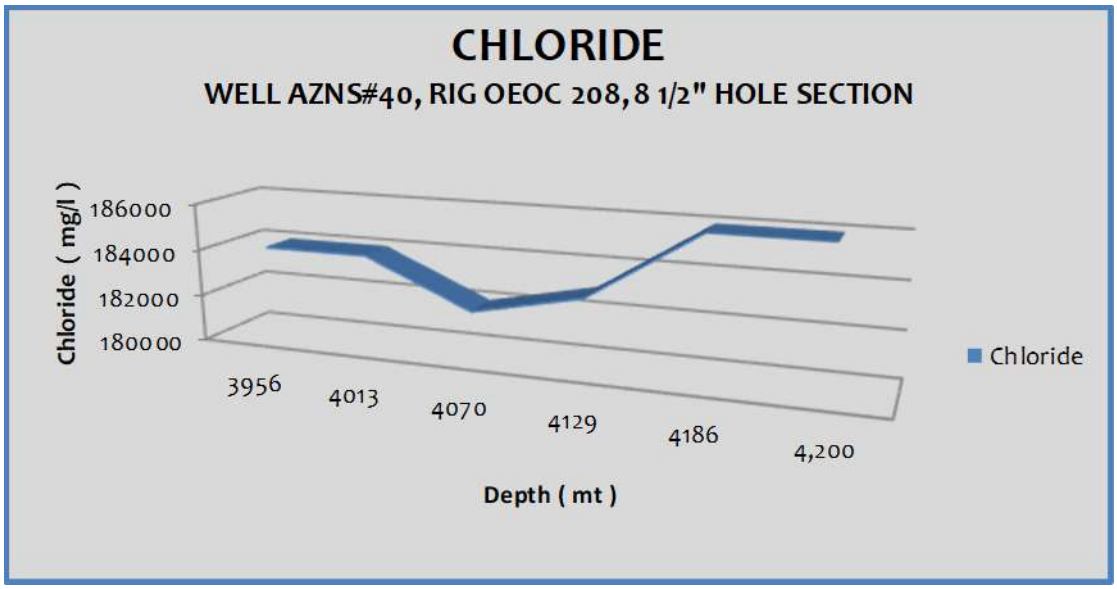


Figure 10-43: Chloride Graph

Rig OEOC 208 has 3 shale shakers were dressed with:

- **Shaker #1:** 200-170-120 Mesh Screen
- **Shaker #2:** 200-170-120 Mesh Screen
- **Shaker #3:** 200-170-140-120 Mesh Screen

WELL AZN#40, RIG OEOC 208 - 8 1/2" HOLE SECTION	
Solid Control Equipment Working Hr.s	
Shaker # 01	189.5
Shaker # 02	224.5
Shaker # 03	131.5
Mud Cleaner	0
Centrifuge#1	0
Centrifuge#2	0
Centrifuge#3	0

Figure 10-44: Solid Control Equipment Working Hrs.

10.3 Completion Section

Operation Remark: Performed perforation in several stages then RIH with 5 7/8'' mill to 4154^{m} after that pumped high viscosity pill and POOH to surface, performed logging after that RIH with 5 7/8'' mill to 4154^{m} and decreased Mw to 102^{pcf} then POOH to surface.

Made up 7'' scrapper and RIH to 4150^{m} then performed slow mud circulation due to shortage of tubing meanwhile prepared 930^{bbl} 101.5^{pcf} inhibited $CaBr_2$ brine mixed with $CaCl_2$ brine (780^{bbl} $CaBr_2$ brine + 150^{bbl} $CaCl_2$ brine + 1%V Corrosion inhibitor + 0.5%V H_2S scavenger + 0.5%V O_2 scavenger) and displaced the hole with 101.5^{pcf} inhibited brine after that POOH to 1700^{m}, at this point observed string drop in the well.

Made 5 7/8'' fishing tools and RIH to 3800^{m} then performed flow check/observed gain after that shut the well and kill the well, displaced the well with 102^{pcf} mud then POOH to surface.

Made up 5 7/8'' mill and RIH to 4154^{m} then increased Mw to 103^{pcf} and POOH to surface, made up 7 5/16'' polish mill and RIH to 3811.4^{m} and dress tie back sleeve then POOH to surface, made up 5 7/8'' mill and RIH to 3985^{m} then performed circulation with NL-NG after that POOH to surface.

Made up 7'' RTTS packer and RIH to 3860^{m} then performed positive test/OK after that POOH to surface, made 4 ¾'' OEDP and 7'' scrapper and RIH to 4013^{m} then performed circulation with NL-NG after that displaced the hole with 104^{pcf} Inhibited $CaBr_2$ brine then POOH to surface.

Made tubing running tools and RIH with 4 ½'' and 3 ½'' tubing to 3935^{m}.

Inhibited $CaBr_2$ Brine: The completion fluid for this section will be $CaCl_2$ brine mixed with $CaBr_2$ brine treated with a scavenger, inhibitor.

Completion						
Products	Units Used	Unit wt lbs	Total wt		Effective Concentration	
1737 bbl Completion fluid (Inhibited CaBr2 brine) has been made						
CaBr2	1257	BBL	52794	gals	30.39	gpb
CaCl2	26	1 MT	57200	lbs	232.52	ppb
Corrision Inhibitor	13	55 Gal	715	gals	0.11	gpb
O2 Scavenger	6	55 Gal	330	gals	0.05	gpb
H2S Scavenger	6	55 Gal	330	gals	0.05	gpb
MW :			102.0			*pcf*
Initial Volume :			0.0			*bbl*
Total Volume Built (Completion Fluid) :			1,737.0			*bbl*

Mud Material consumption						
Products	Units Used	Unit wt lbs	Total wt		Effective Concentration	
2015 bbl 102 to 103 PCF mud has been made						
Limestone	207	1.5 MT	683100	lits	89.64	gpb
Caustic Soda	9	55 lb	495	lbs	0.06	ppb
Salt (NaCl)	43	1 MT	94600	lbs	12.41	ppb
Soda Ash	8	55 lb	440	lbs	0.06	ppb
Sodium Bicarbonate		55 lb	0	lbs	0.00	ppb
Starch LV	30	55 lb	1650	lbs	0.22	ppb
Xanthan gum	19	55 lb	1045	lbs	3536.00	ppb
Lime	30	55 lb	1650	lbs	0.22	ppb
MW :			102 to 103			*pcf*
Initial Volume :			1,720.0			*bbl*
Received volume :			0.0			*bbl*
Return volume :			0.0			*bbl*
Total Volume Built (Mud) :			2,015.0			*bbl*

Figure 10-45: Volume & Material Consumption

11. Well Stimulation & Clean Up

11.1 Operatipon Summary

Objective of operation was to increase injectivity by injecting 49.15 gal/ft of 15 % HCl via Coiled Tubing and conditioning the formation for stimulation operation, and reduce formation damage and wash out the remained debris and residuals of completion and drilling operations. Based on mentioned objective, operation was designed by stimulating well by using 15% HCl, Flowing pressure survey was designed to investigate well condition.

In this operation, the well was cleaned up by flowing with 16/64" to 60/64" choke size while rate measurement in same choke size. To perform acid wash, RIH CT to reach the bottom of producing interval. After displacing the wellbore by diesel, Pre-flush, main acid (15% HCL), and over-flush were injected by CT, and finally, the stimulation fluid was squeezed. After acidizing, the well was cleaned up and a flowing pressure/temperature survey was carried out.

Surface testing data were analyzed to investigate the effect of stimulation operation. According to pre-stimulation clean up results show that wellhead pressure and oil rate were 920 PSI and 995 BPD at 24/64" choke size. On the other hand, data gathered from post stimulation clean up shows that well head pressure and oil rate have increased to 2758 PSI and 2532 BPD at 24/64" choke size. The above data illustrates that stimulation operation has led to a 1838-PSI increase in wellhead pressure and a 1537-BPD increase in oil rate. On top of that, the calculated pre- and post-stimulation PI of the well were about 0.18 and 0.88 STB/ (D.PSI) at the datum depth. Generally, the stimulation operation was carried out successfully.

Pre/Post Acid	Pre – Acid	Post – Acid
Choke Size (1/64" Inch)	24	24
Wellhead Pressure (PSIG)	920	2758
Flow Rate (STB/D)	995	2532
PI (STB/D/PSI)	0.18	0.88

Table 11-1: AZNS-040 Pre and Post Acid Potential of the Well

11.2 General Information

General information	
SERVICE:	Well testing, acidizing, Coiled Tubing, S/L, oil transfer
SERVICE COMPANY:	OEOC
OPERATION DATE:	05/December/2023-14/December/2023
STATE BEFORE THE OPERATION:	New Drill
WELL GEOMETRY:	Deviated
WELL TYPE:	Oil Producer
FLUID TYPE:	OIL & GAS
COMPLETION METHOD:	7” Perforated
CASING SIZE:	20”, 13 3/8”, 9 5/8”, 7”
TUBING SIZE:	4 ½” 18.9 ppf L-80 (0–3714.73 m MD); 3 ½” 10.2 ppf L-80 (3714.73 – 3963.97 m MD)
PRODUCING INTERVAL:	4039.00 – 4112.00 m MD

Table 11-2: AZNS-040 General Information

11.3 Well Basic Information

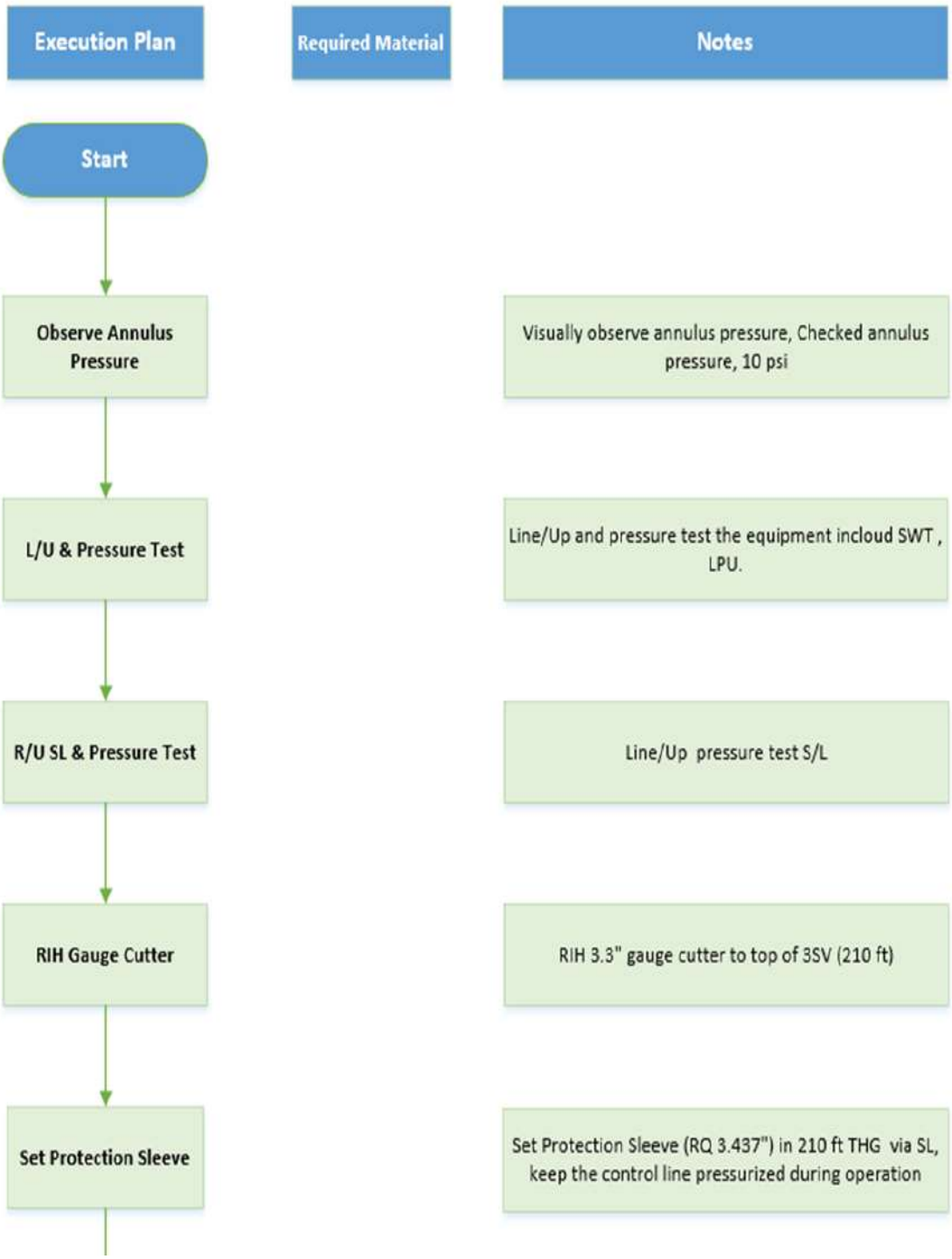
WELL BASIC INFORMATION	
Well Name	AZNS-40
Well Type	Highly Dev. Sarvak Prod. Well
Well Pad	Pad No.50
Well Objective	To remove possible damage and enhance well’ PI

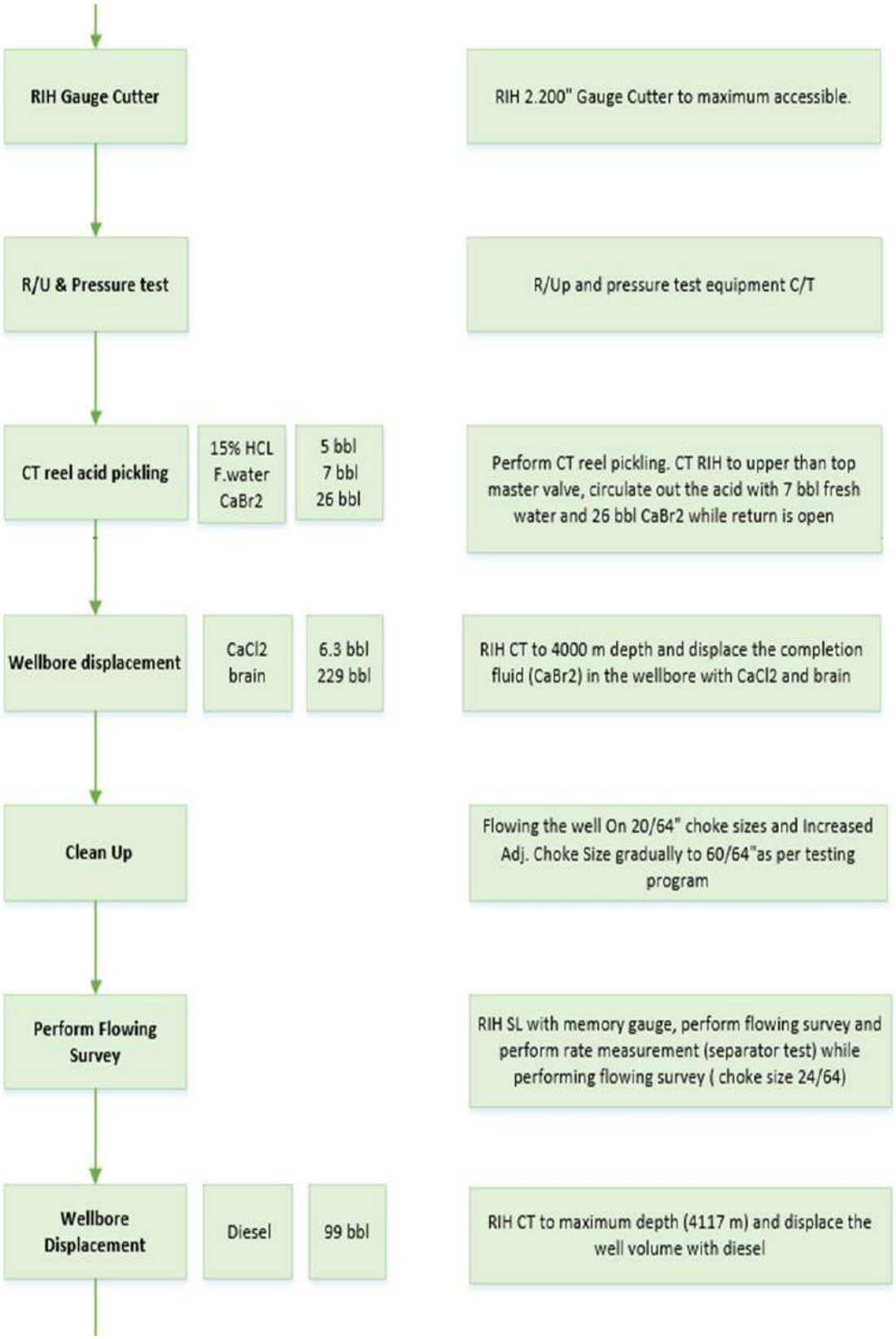
Reference Depth (Rotary Table Elevation from mean sea level)	13.45 MAMSL
Designed TD	4200 m MD
Target Formation	Fahliyan (Fah-1b, Fah-1c, Fah-2b, and Fah-3a)
Oil Density (API)	38 as per last post acid flow
Minimum ID	2.562" @ 3963.02 m MD, 3 ½" Bottom No Go Landing Nipple
Total Depth	4200 MD (-4186.55 m TVDss)
Producing Formation	Fahliyan (Fah-1b, Fah-1c, Fah-2b, and Fah-3a)
Formation Lithology	Limestone 90%, Dolomite 8.5%, Clay 1.5%
Clay Type	Illite, Kaolinite, Montmorillonite
Formation/Layer Permeability (md)	Fah (1b) = 50; Fah (1c) = 20 ; Fah (2b) = 1; Fah (3a) = 5
Formation/Layer Porosity (%)	Fah (1b) = 13; Fah (1c) = 9 ; Fah (2b) = 2; Fah (3a) = 4
H2S Content (%)	0.35 mole %
Asphaltene content (%)	Less than 0.5 wt %
Surface Temperature	About 150 ˚F
Reservoir Temperature	270 ˚F @ Datum Depth (Oil Datum Depth = 4,000 mss)
Reservoir Pressure	9276 psi @ Datum Depth:4000 mss (Based on Dynamic Model)
GOR (SCF/STB)	1030 as per last post acid flow
Formation Fracture Gradient (PSI/ft)	0.962 psi/ft

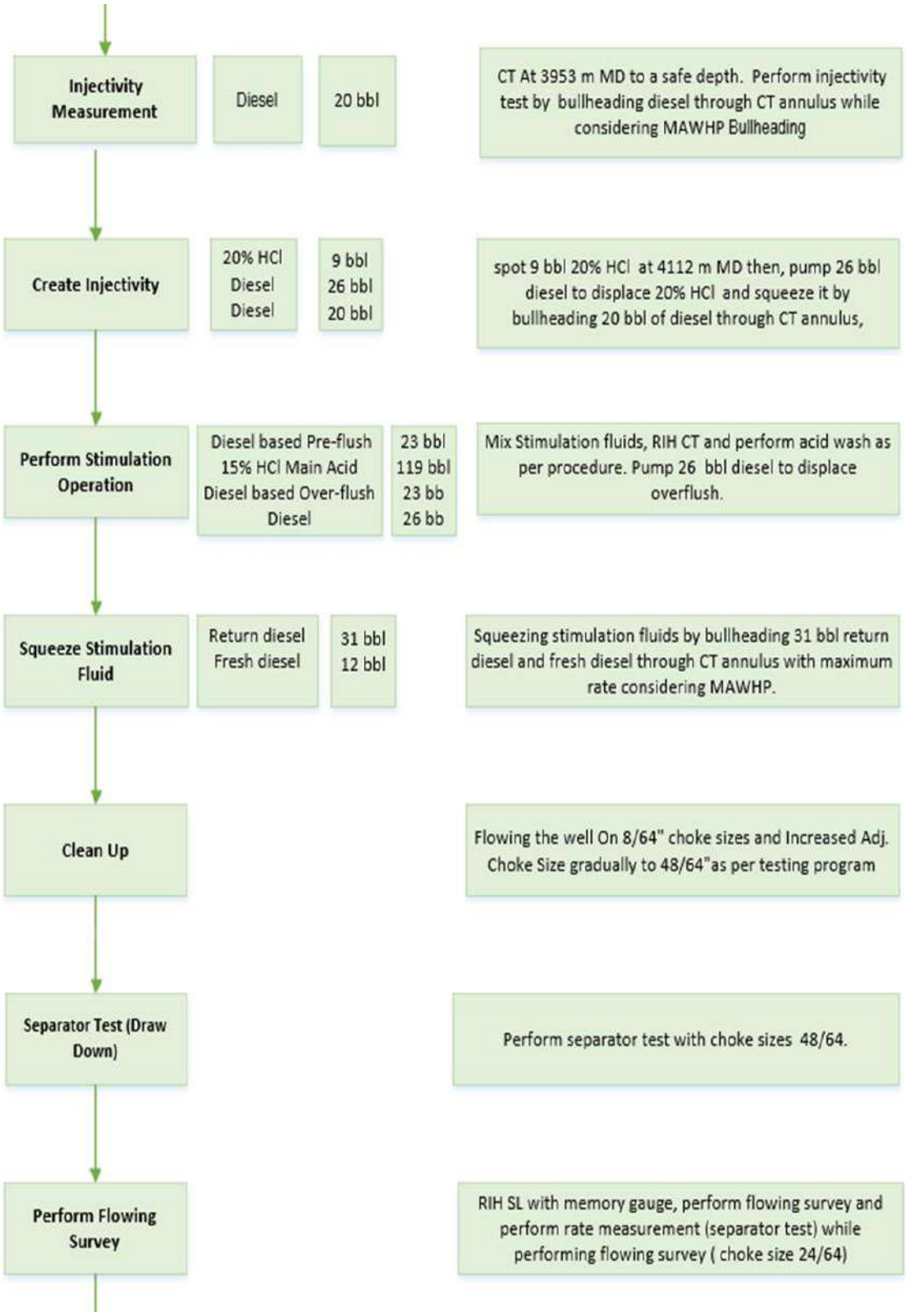
Table 11-3: AZNS-040 Well Basic Information

11.4 Total Operation Result

Executed Operation Flow Chart




RIH Gauge Cutter
RIH 2.200" Gauge Cutter to maximum accessible.
R/U & Pressure test
R/Up and pressure test equipment C/T
CT reel acid pickling
15% HCL
F.water
CaBr2
5 bbl
7 bbl
26 bbl
Perform CT reel pickling. CT RIH to upper than top master valve, circulate out the acid with 7 bbl fresh water and 26 bbl CaBr2 while return is open
Wellbore displacement
CaCl2
brain
6.3 bbl
229 bbl
RIH CT to 4000 m depth and displace the completion fluid (CaBr2) in the wellbore with CaCl2 and brain
Clean Up
Flowing the well On 20/64" choke sizes and Increased Adj. Choke Size gradually to 60/64"as per testing program
Perform Flowing Survey
RIH SL with memory gauge, perform flowing survey and perform rate measurement (separator test) while performing flowing survey (choke size 24/64)
Wellbore Displacement
Diesel
99 bbl
RIH CT to maximum depth (4117 m) and displace the well volume with diesel


Injectivity Measurement
Diesel
20 bbl
CT At 3953 m MD to a safe depth. Perform injectivity test by bullheading diesel through CT annulus while considering MAWHP Bullheading
Create Injectivity
20% HCl
Diesel
Diesel
9 bbl
26 bbl
20 bbl
spot 9 bbl 20% HCl at 4112 m MD then, pump 26 bbl diesel to displace 20% HCl and squeeze it by bullheading 20 bbl of diesel through CT annulus,
Perform Stimulation Operation
Diesel based Pre-flush
15% HCl Main Acid
Diesel based Over-flush
Diesel
23 bbl
119 bbl
23 bb
26 bb
Mix Stimulation fluids, RIH CT and perform acid wash as per procedure. Pump 26 bbl diesel to displace overflush.
Squeeze Stimulation Fluid
Return diesel
Fresh diesel
31 bbl
12 bbl
Squeezing stimulation fluids by bullheading 31 bbl return diesel and fresh diesel through CT annulus with maximum rate considering MAWHP.
Clean Up
Flowing the well On 8/64" choke sizes and Increased Adj. Choke Size gradually to 48/64"as per testing program
Separator Test (Draw Down)
Perform separator test with choke sizes 48/64.
Perform Flowing Survey
RIH SL with memory gauge, perform flowing survey and perform rate measurement (separator test) while performing flowing survey (choke size 24/64)

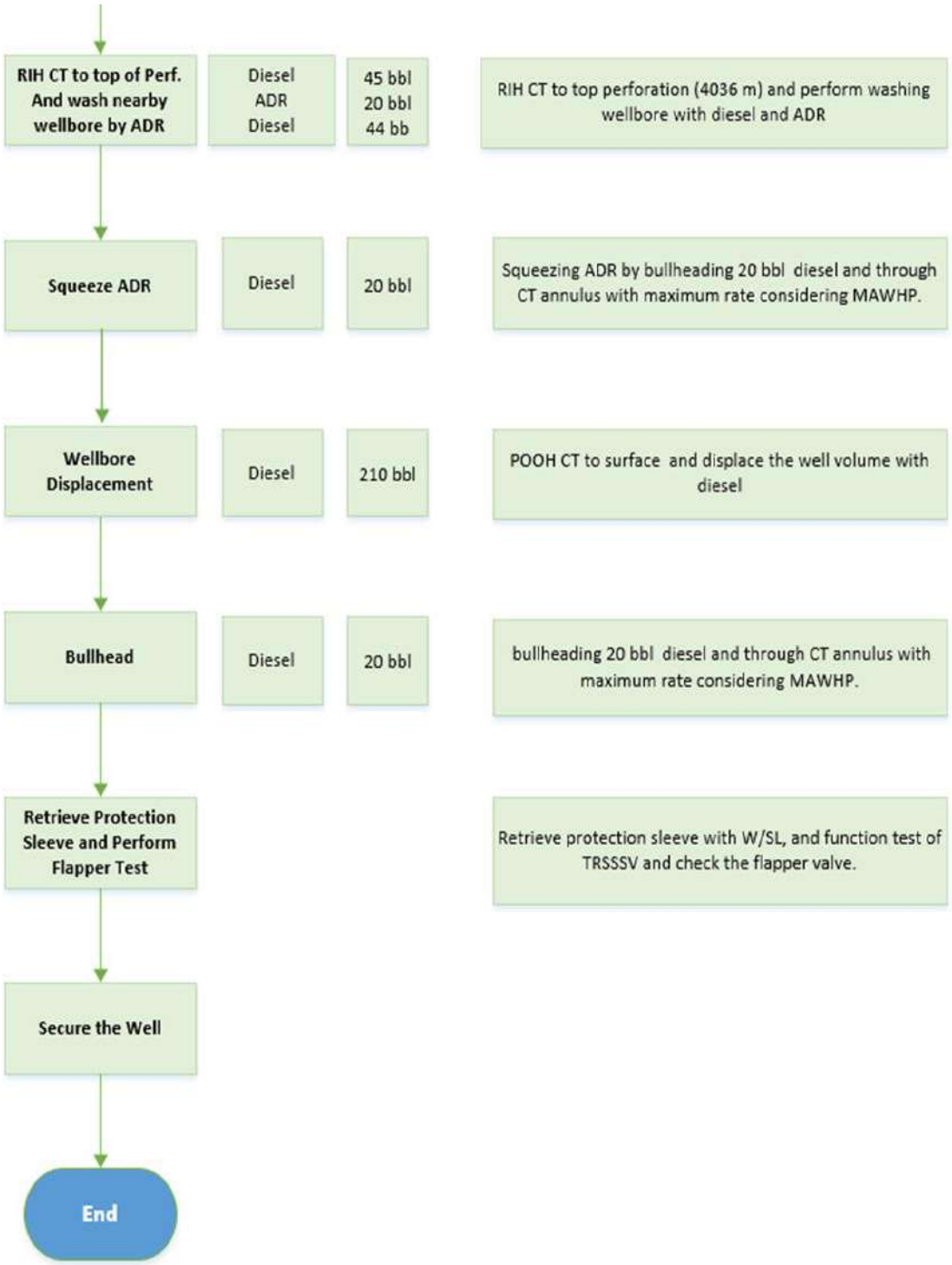


Figure 11-1: Executed Operation Flow Chart

Daily Operation Summary

Date	Operation Summary
05 – Dec – 2023	Rig up and PT of SWT equipment, Rig up and PT of S/L, Open TMV (SIWHP: 0, Annulus pressure: 9 psi), S/L with gauge cutter to the 3SV depth (219 ft.), RIH protection sleeve (RQ profile, ID=2.9"), RIH by gauge ring due to TD check, Rig up and PT of CT.
06 – Dec – 2023	Perform acid pickling, RIH CT while pumping CaBr2 and 6.3 bbl CaCl2 and brine.
07 – Dec – 2023	Continue pumping brine, Divert flow to separator and surge tank.
08 – Dec – 2023	Closed the well of C/M and start rig down CT injector and BOP, Perform P/T S/L in order to FGS.
09 – Dec – 2023	Open the well on adj C/S 8/64" and increase gradually to 24/64" and divert to separator, Perform rate measurement on 24/64 Fix choke during 10 min through surge tank, RIH S/L for flowing survey.
10 – Dec – 2023	CT RIH to maximum depth by pumping 99 bbl return diesel, Perform injectivity by 20 bbl fresh diesel, pumping 9 bbl HCL 20%, Squeezing the HCL 20% in the well with bull heading operation (20 bbl diesel), Pumping 23 bbl preflush, Pumping 119 bbl HCL 15%, Pumping 23 bbl overflash, Displacement CT reel with 26bbl Return diesel, Performed bullheading operation with 31bbl return, Open the well on adj C/S 8/64" and increase gradually to 28/64" and divert to separator, CT POOH to surface with pumping 77 bbl diesel, Increase C/S gradually to 44/64 adj.
11 – Dec – 2023	Perform rate measurement on 48/64 fix choke during 10 min through surge tank. Close the well from C/M (because separator gas-fisher not working).
12 – Dec – 2023	Perform rate measurement on 24/64 Fix choke during 10 min through surge tank. S/L RIH with two memory gauges to perform flowing survey, Perform rate measurement on 24/64 Fix choke during 10 min through surge tank.
13 – Dec – 2023	CT RIH While pumping 45 bbl diesel, Pumping 20 bbl ADR, Pumping 44 bbl diesel, Bullheading the ADR with pumping 20 bbl diesel, displacing well with pumping 210 bbl diesel, Bullheading with 20 bbl diesel, RIH S/L with GS pulling 3.5" to retrieve protection sleeve.
14 – Dec – 2023	Perform Inflow test of 3SV, Perform pressure test of wellhead, Closed all valve and release all equipment's.

Table 11-4: AZNS-040 Daily Operation Summary

Sequence of Operation

DATE (dd/mm/yyyy)	TIME (hh:mm)	Sequence
05-December-2023	00:00	Rig up SWT equipment
05-December-2023	04:15	Pressure test gas line burner with 350 psi for 15 min and function test green burner is (ok).
05-December-2023	05:15	Start R/U S/L Lubricator and B.O.P Equipment.
05-December-2023	05:30	Open the 3SV.
05-December-2023	05:45	Open TMV & production valve (closed C/M) (WHP: 0 psi & Ann p: 9 psi).
05-December-2023	06:10	Function test S/L BOP (Ok).
05-December-2023	06:20	Disconnected X-mass three cap by S/L group.
05-December-2023	06:45	Rig Up Dual B.O.P.
05-December-2023	07:45	Rig Up S/L Lubricator.
05-December-2023	07:50	Perform Pull Test Robe Socket 800 lb (ok).
05-December-2023	08:10	Perform Pressure Test Lubricator by Pump Truck with Water (observed Leakage).
05-December-2023	08:55	Perform P/T Lubricator with 6500 psi Is (Ok).
05-December-2023	08:57	Open TMV & swab valve (SHIWP :0 psi)
05-December-2023	09:00	RIH S/L With gauge cutter 3.3. BHA: Robe Socket 1 ,7/8 +10ft Stem 1, 7/8 + 4 ft mechanical jar 1 7/8+ gauge cutter 3.300).
05-December-2023	09:08	S/L at TRSSV depth (219 ft) and POOH to the surface.
05-December-2023	09:12	S/L at Surface.
05-December-2023	10:18	RIH SL to set P.S. BHA: Rope socket 1-7/8” + 10ft stem 1-7/8” + 4ft M.J 1-7/8” + running tool + protection sleeve (Profile: AF, ID: 2.9”, OD: 3.437”)

05-December-2023	10:26	SL at 210 ft (3SV depth) set protection sleeve and POOH to the surface.
05-December-2023	11:15	SL at surface closed the well from TMV & swab valve.
05-December-2023	11:37	RIH with gauge cutter 2.2" to check maximum accessible depth. Rope socket 1-7/8” + 10ft stem 1-7/8” +4ftM.J 1-7/8” +gauge cutter 2.2
05-December-2023	13:38	S/L at surface.
05-December-2023	13:40	Rig down S/L lubricator & BOP.
05-December-2023	14:30	Rig down S/L completely.
05-December-2023	14:50	Rig up CT riser & BOP.
05-December-2023	16:13	Function test for BOP (ok).
05-December-2023	19:35	Perform Pull Test of CT BOP ok.
05-December-2023	20:15	Perform pressure test of CT blind (Fail to Riser).
05-December-2023	22:20	Perform pressure test CT blind rams (Riser is Ok) Fail blind rams.
06-December-2023	00:40	Perform pressure test CT blind rams is OK.
06-December-2023	01:00	Rig up to Injector head.
06-December-2023	01:30	Finished rig up injector head.
06-December-2023	01:50	Perform pressure test of CT body & stripper is ok.
06-December-2023	03:40	Perform pressure test for CT check valve Flapper is (OK).
06-December-2023	04:50	Start mixed acid pickling (HCL 15%).
06-December-2023	06:20	Finished mixed.
06-December-2023	06:20	Start pumped acid pickling (R: 1 bpm, P:1300 psi, V:5 bbl.).
06-December-2023	06:25	Finished pumped.
06-December-2023	06:25	Displaced acid pickling with 7 bbl water.

06-December-2023	06:35	Start displaced acid pickling by CaBr2 (P:2300 psi, R: 1 bpm).
06-December-2023	07:00	Finished displaced acid pickling by CaBr2 (26 bbl).
06-December-2023	07:30	Opened the c/m on 16/64 adj and increase gradually up to 24/64.
06-December-2023	07:50	Opened the well from TMV.
06-December-2023	07:50	Start CT RIH by CaBr2.
06-December-2023	08:30	Take sample from C/M, weight :102 pcf
06-December-2023	12:12	CT @ 4000 m.
06-December-2023	12:32	finish pumping CaBr2 (26 bbl for CT pipe fill up + 14 bbl for RIH, Total pumping CaBr2: 40 bbl).
06-December-2023	12:43	Start pumping CaCl2 as spacer.
06-December-2023	12:45	Finished pumped CaCl2 (6.3 bbl.) and start pumped brain (75 pcf) P:2500 psi, Rate :0.6, SWHP :238 psi.
06-December-2023	14:30	CT at 3700 m and continue pumped brine (R:0.5, P:3700 psi).
06-December-2023	14:38	Decrease C/S of 24/64 adj to 22/64 adj.
06-December-2023	16:30	Decrease C/S of 22/64 adj to 20/64 adj.
06-December-2023	17:10	Decrease C/S of 20/64 adj to 18/64 adj.
06-December-2023	18:43	Stop pumping brine to CT (Total Volume: 206 bbl).
06-December-2023	18:50	Increase c/s of 18/64 adj to 20/64 adj.
06-December-2023	21:30	CT at 1000 m and continue pumped brine (R:0.5, P:3700 psi).
06-December-2023	21:30	R/U N2 pump and N2 tank.
07-December-2023	00:20	Divert flow of rig trip tanks to storage tank.
07-December-2023	00:25	Start pumping brine by pump truck (R: 0.3 bpm), (P: (1800 psi).
07-December-2023	00:30	Increase C/S of 20/64 adj to C/S 24/64 adj.
07-December-2023	00:33	Decrease C/S of 24/64 adj to 20/64 adj.

07-December-2023	01:02	CT at surface and stopped pumping brine (volume total:6+18=23 bbl).
07-December-2023	01:05	Close swab valve.
07-December-2023	01:12	Increase C/S of 20/64 adj to C/S 24/64 adj.
07-December-2023	04:12	Divert flow to separator and lighting the burner boom to burn the separator gas.
07-December-2023	04:45	Release annulus pressure of 1100 psi to 120 psi (V=1Lit /weight= 98).
07-December-2023	05:00	Take a sample from CM, W=58 PCF, BS&W=18%.
07-December-2023	06:00	Take a sample from CM, W=53 PCF, BS&W=15%.
07-December-2023	07:00	Take a sample from CM, W=53 PCF, BS&W=8%.
07-December-2023	07:53	Increase C/S of 24/64 adj to C/S 28/64 adj.
07-December-2023	08:00	Take a sample from CM, W=53 PCF, BS&W=8%
07-December-2023	09:00	Take a sample from CM, W=53 PCF, BS&W=3%
07-December-2023	09:14	Increase C/S of 24/64 adj to C/S 32/64 adj
07-December-2023	10:25	Divert directional of flow to surge tank.
07-December-2023	10:27	Increase C/S of 32/64 adj to C/S 36/64 adj.
07-December-2023	10:30	Take a sample from CM, W=55 PCF, BS&W=10%
07-December-2023	10:40	Decrease C/S gradually of 36/64 adj to 28/64 adj
07-December-2023	10:45	Increase C/S gradually of 28/64 adj to C/S 36/64 adj
07-December-2023	10:55	Increase C/S gradually of 36/64 adj to C/S 44/64 adj
07-December-2023	11:00	Take a sample from CM, W=55 PCF, BS&W=10%
07-December-2023	11:15	Increase C/S of 44/64 adj to C/S 48/64 adj
07-December-2023	12:10	Release annulus pressure of 720 psi to 200 psi (V=110 Lit / weight= 98) (sediment).
07-December-2023	13:30	Increase C/S of 48/64 adj to C/S 52/64 adj

07-December-2023	14:00	Take a sample from CM, W=53 PCF, BS&W=7%
07-December-2023	14:31	Increase C/S of 52/64 adj to C/S 56/64 adj
07-December-2023	15:48	Increase C/S of 56/64 adj to C/S 60/64 adj
07-December-2023	16:34	Release annulus pressure of 710 psi to 210 psi (V=130 Lit) (W=102PCF).
07-December-2023	17:42	Decrease C/S gradually of 60/64 adj to 56/64 adj
07-December-2023	18:35	Decrease C/S gradually of 56/64 adj to 52/64 adj
07-December-2023	22:00	Release annulus pressure of 670 psi to 220 psi (V=110 Lit) (W=102 PCF).
07-December-2023	23:58	Increase C/S of 52/64 adj to C/S 56/64 adj
08-December-2023	00:00	Take a sample from C/M: W=53 PCF, BS&W=3 %(sediment).
08-December-2023	01:00	Take a sample from C/M: W=53 PCF, BS&W=3 %(sediment).
08-December-2023	02:30	Take a sample from C/M: W=53 PCF, BS&W=3 %(sediment).
08-December-2023	03:00	Take a sample from C/M: W=52 PCF, BS&W=2 %(sediment).
08-December-2023	04:00	Take a sample from C/M: W=52 PCF, BS&W=1 %(sediment).
08-December-2023	04:10	Decrease C/S gradually of 56/64 adj to 52/64 adj
08-December-2023	04:38	Decrease C/S gradually of 52/64 adj to 48/64 adj
08-December-2023	04:55	Decrease C/S gradually of 48/64 adj to 44/64 adj
08-December-2023	05:55	Decrease C/S gradually of 44/64 adj to 40/64 adj
08-December-2023	06:07	Decrease C/S gradually of 40/64 adj to 20/64 adj
08-December-2023	06:30	Closed the well of C/M.
08-December-2023	06:35	Closed the well of TMV.
08-December-2023	06:39	Open the swab valve.
08-December-2023	06:40	Bleed off CTU pressure to zero.

08-December-2023	06:30	Closed the well of C/M and start rig down CT injector and BOP.
08-December-2023	08:00	Finished rig down CT.
08-December-2023	08:00	Start R/U S/L dual BOP & lubricator equipment.
08-December-2023	08:30	Rigged up dual BOP.
08-December-2023	08:45	Function test BOP is (Ok).
08-December-2023	09:45	Power On 2 set memory's gauge.
08-December-2023	10:00	Rigged up S/L lubricator.
08-December-2023	10:07	Perform pull test robe socket 800 lb is (ok).
08-December-2023	10:20	Start pressure test lubricator by pump truck with water.
08-December-2023	11:00	Perform pressure test lubricator by pump truck with 6500 psi (leakage flange and BOP).
08-December-2023	11:45	Perform pressure test excrement flange) by pump truck with 6500 psi is (pass flow from TMV (Closed).
08-December-2023	12:00	TMV (C/O) observe for leakage.
08-December-2023	12:35	Perform inflow test TMV is ok.
08-December-2023	14:10	Perform P/T S/L lubricator & BOP is ok.
08-December-2023	14:37	Open TMV, WHP: 5078 psi.
08-December-2023	14:44	Open the well from choke manifold with adjustable choke size 8/64" and increase choke size gradually to 24/64" adjustable to separator.
08-December-2023	16:00	Take a sample from C/M, W= 52 PCF, BS&W=1% (water).
08-December-2023	22:00	Take a sample from C/M, W= 52 PCF, BS&W=0.5% (water).
09-December-2023	07:13	Perform rate measurement on 24/64 fix choke during 10 min through surge tank at 07:13 - 07:23.
09-December-2023	08:17	Perform rate measurement on 24/64 fix choke during 10 min through surge tank at 08:17 - 08:27.
09-December-2023	09:14	RIH S/L for flowing survey @09:14 (BHA S/L Robe Socket 1 ,7/8 +10ft Stem 1, 7/8 + 3 ft hydraulic jar 1 7/8+ 2 ft shock absorber 1, 7/8 + 2 Set memory gauges).

09-December-2023	11:29	S/L at 13235 ft. @11:29 and P.O.O.H to the surface (2* Step 10 min, 5* Step 7 min, 18 Step 5 min).
09-December-2023	12:51	Perform rate measurement on 24/64 Fix choke during 10 min through surge tank (12:51 - 13:01).
09-December-2023	16:18	S/L at surface.
09-December-2023	16:21	Closed TMV.
09-December-2023	17:00	Power off memory gauges.
09-December-2023	17:15	Start rig down S/L BOP & lubricator.
09-December-2023	18:00	Finished rig down S/L completely and start R/U CT BOP & injector head.
09-December-2023	18:00	R/U CT BOP.
09-December-2023	18:02	Function test for CT BOP.
09-December-2023	18:36	Perform pressure test for BOP flange & BOP body & blind rams (6500 psi), (NOT OK), leakage of TMV.
09-December-2023	19:05	Perform pressure test for BOP flange & BOP body & blind rams (6500 psi) (3 step), (OK).
09-December-2023	21:13	Perform pull test for CT (10000 psi), (OK).
09-December-2023	21:20	Start R/U injector.
09-December-2023	21:55	Perform pressure test for stripper (6500 psi), (OK).
09-December-2023	22:38	Perform pressure test for pipe rams CT (6500 psi), (OK).
09-December-2023	23:17	Perform pressure test for check valve CT (1000 psi), (OK).
09-December-2023	23:45	Pumping return diesel to CT for displacement brine.
10-December-2023	00:20	Finish pumping (10 bbl.).
10-December-2023	00:30	Open the TMV of SSV.
10-December-2023	00:41	Open the well from C/M with C/S 20/64 adj to separator.
10-December-2023	00:45	Start CT RIH by pumping return diesel.
10-December-2023	00:50	Closed the well of C/M (because torn off burner).

10-December-2023	1:10	Open the well of C/M with C/S 20/64 adj.
10-December-2023	02:00	Take sample from CM: W= 52 PCF, BS&W=0.5 %(water).
10-December-2023	05:26	The CT arrived to maximum depth 4117 total pumped return diesel 99 bbl.
10-December-2023	05:35	Start displacing the well with 50 bbl return diesel.
10-December-2023	05:45	Hold PJSM.
10-December-2023	05:45	Start mixing preflush & overflush.
10-December-2023	06:01	Take sample from CM: W= 52 PCF, BS&W=0.5 %(water).
10-December-2023	06:25	Decrease C/S gradually of 20/64 adj to 16/64 adj.
10-December-2023	06:40	Finish mixed preflush & overflush.
10-December-2023	06:48	Closed the well from C/M (for operation enjectivity).
10-December-2023	06:50	POOH C/T to depth 3953m and finish pumping 50 bbl return diesel.
10-December-2023	06:58	CT at 3953 m and start injectivity by fresh diesel (P: Max: 6400 psi R: 1/2 - 0.3 bpm).
10-December-2023	07:15	Mixing HCL (15%) (120bbl).
10-December-2023	08:31	Finish injectivity with pumping 20bbl fresh diesel (Rate:0.2 bpm, P=5700-6700 psi).
10-December-2023	11:30	Perform mixing 9bbl the HCL 20%.
10-December-2023	11:50	Start pumping HCL 20% (V:9 bbl - R :0.2~0.5 - P: 5500~4000 psi).
10-December-2023	12:00	Opened the well from C/M on 8/64 adj and flowing to storage tanks.
10-December-2023	12:05	Increase C/S gradually from 8/64 adj to 24/64 adj.
10-December-2023	12:10	Decrease C/S from 24/64 adj to 20/64 adj.
10-December-2023	12:13	Displacement CT reel with diesel (V: 26bbl - P: 3800 psi - R: 0.7).
10-December-2023	12:13	CT RIH (from 3949 m to 4112 m).
10-December-2023	12:15	Decrease C/S from 20/64 adj to 16/64 adj.

10-December-2023	12:17	Increase C/S gradually from 16/64 adj to 28/64 adj.
10-December-2023	12:40	CT POOH (from 4112 m to 3990 m).
10-December-2023	12:41	Decrease C/S from 28/64 adj to 20/64 adj.
10-December-2023	12:49	Decrease C/S from 20/64 adj to 16/64 adj.
10-December-2023	12:50	Pumping diesel from CT (V: 3 bbl).
10-December-2023	12:55	Closed the well from C/M.
10-December-2023	12:55	Squeezing the HCL 20% in the well with bull heading operation, which has been done by diesel, from kill line and inside the CT reel from 12:55 at 13:15. while the well was closed (MAX rate: 2 - MAX WHP: 6000 psi - V: 20 bbl).
10-December-2023	13:20	Pumping preflush (V: 23 bbl).
10-December-2023	14:10	Finish 23 bbl preflush.
10-December-2023	14:20	Pumping HCL 15% (V=119BBL).
10-December-2023	16:05	Start POOH CT.
10-December-2023	16:15	C/T arrived at depth 4035m.
10-December-2023	16:50	Finish 119 bbl HCL.
10-December-2023	16:51	pumping overflash (total= 23 bbl).
10-December-2023	17:20	Finish pumping 23 bbl overflush.
10-December-2023	17:23	Displacement CT reel with 26bbl Return diesel at 17:23 to 18:06.
10-December-2023	17:24	CT POOH (from 4035 to 3950) at 17:23 to 18:02.
10-December-2023	18:06	Performed bullheading operation with 31bbl return diesel and 12 bbl fresh diesel at 18:06 to 18:29 (Max Rate: 5.1 - Max P: 6350 psi).
10-December-2023	18:40	Open the well from C/M with adj C/S 8/64" and increase C/S gradually 28/64" to storage tank.
10-December-2023	18:36	Start pumping diesel with minimum rate for CT POOH.
10-December-2023	18:45	Start CT POOH (from 3950 m to surface).

10-December-2023	19:10	Increase adj C/S from 24/64 to 28/64 adj.
10-December-2023	19:15	Divert flow to separator by C/S 28/64" adj.
10-December-2023	20:59	Increase adj C/S from 28/64 to 32/64 adj.
10-December-2023	21:30	Finish pumping 77 bbl diesel while CT POOH to surface.
10-December-2023	21:32	increase adj C/S from 32/64 to 36/64 adj.
10-December-2023	21:32	Increase C/S gradually of 28/64 adj to 32/64 adj.
10-December-2023	21:34	Increase C/S gradually of 32/64 adj to 36/64 adj.
10-December-2023	21:45	Started pumping caustic.
10-December-2023	21:45	Increase adj C/S from 36/64 to 40/64 adj.
10-December-2023	22:03	Increase C/S gradually of 40/64 adj to 44/64 adj.
10-December-2023	22:20	Increase adj C/S from 44/64 to 48/64 adj.
10-December-2023	23:50	Release annulus pressure from 700 psi to 220 psi (Vol:160 lit) (W=101 pcf).
10-December-2023	23:59	Divert direction of flow from 48/64" adj to 48/64" fix.
11-December-2023	00:00	Continue Flowing the well through the C/M to the separator and the oil storage tanks.
11-December-2023	00:00	Take a sample from CM, W= 53 PCF, BS&W= 3 %(sediment).
11-December-2023	00:11	Release annulus pressure from 700 psi to 220 psi (Vol:190 lit) (W=102 pcf).
11-December-2023	01:00	Take a sample from CM, W= 52 PCF, BS&W= 6 %(sediment).
11-December-2023	01:00	Pumping Caustic at 01:00 to 03:00 (V: 20 bbl).
11-December-2023	01:48	Release annulus pressure from 1000 psi to 220 psi (Vol:170 lit) (W=101 pcf).
11-December-2023	06:17	Release annulus pressure from 1000psi to 220psi (Vol: 170 lit- W= 101 pcf).
11-December-2023	10:00	Take a sample from CM, W= 52 PCF, BS&W= 0.5 %(sediment)
11-December-2023	10:05	Perform rate measurement on 48/64 fix choke during 10 min through surge tank.

11-December-2023	11:00	Take a sample from CM, W= 52 PCF, BS&W= 0
11-December-2023	11:39	Perform rate measurement on 48/64 Fix choke during 10 min through surge tank.
11-December-2023	12:33	Close the well from TMV for the CT R/D and the S/L R/U.
11-December-2023	12:45	The CT R/D.
11-December-2023	14:45	Performed the S/L dual BOP function test "OK".
11-December-2023	14:50	The SL dual BOP up.
11-December-2023	15:56	Power on 3 set of memory gauge, (lower memory gauge S/N: 90506 PPS, Middle memory gauge S/N: 90504 PPS, upper memory gauge S/N:03019).
11-December-2023	16:15	The S/L lubricator up.
11-December-2023	16:25	Performed the S/L pull test up to 800lb "ok".
11-December-2023	16:35	Start the SL BOP and lubricator pressure test.
11-December-2023	17:20	Open the well from TMV / SIWHP = 4131 psi
11-December-2023	17:25	Open the well from choke manifold with adjustable C/S 8/64 to separator.
11-December-2023	17:34	Close the well from choke manifold (because separator gas-fisher CONTROL not working).
11-December-2023	19:00	Finished repairing the separator gas fisher control.
11-December-2023	19:04	Perform equalizing pressure the TMV via pump truck pressure for opened the TMV (4000 psi).
11-December-2023	19:06	Opened the well from TMV.
11-December-2023	19:12	Opened the well from C/M and flowing to the separator.
11-December-2023	19:17	Closed the well from C/M and TMV for repairing adj chock valve.
11-December-2023	19:50	Perform equalizing pressure the TMV via pump truck pressure for opening the TMV (4000 psi).
11-December-2023	19:55	Opened the well from TMV.
11-December-2023	19:56	Opened the well from C/M on 12/64 adj and flowing to the separator.
11-December-2023	20:00	Increase adj C/S from 12/64 to 20/64 adj.

11-December-2023	20:18	Increase adj C/S from 20/64 to 24/64 adj.
11-December-2023	22:07	Divert direction of flow from 24/64" adj to 24/64" fix.
12-December-2023	00:00	Take a sample from CM, W= 52 PCF, BS&W= nill.
12-December-2023	04:00	Take a sample from CM, W= 53 PCF, BS&W= nill.
12-December-2023	07:06	Perform rate measurement on 24/64 Fix choke during 10 min through surge tank.
12-December-2023	07:51	Perform rate measurement on 24/64 Fix choke during 10 min through surge tank.
12-December-2023	08:20	Opened swab valve.
12-December-2023	08:30	RIH SL, (BHA RUN1: 12ft stem, hydraulic jar 3ft, shock absorber 2ft, memory gauge 3set 5ft).
12-December-2023	10:43	The SL arrived at 13230ft (first step), the flowing survey operation will perform at 25 steps (2 step 10min, 5step 7min, 18 step 5min).
12-December-2023	12:14	Perform rate measurement on 24/64 Fix choke during 10 min through surge tank.
12-December-2023	16:38	The S/L arrived on surface.
12-December-2023	16:40	Close the well from TMV.
12-December-2023	16:50	Close the well from choke manifold and close swab valve & SSV.
12-December-2023	17:28	Power off 3 set the memory gauges.
12-December-2023	17:45	Start rig down the S/L.
12-December-2023	18:00	The memory gauges and the gauge specialist with computers released.
12-December-2023	18:30	Disconnect the riser (due to the protection sleeve not passing through the riser).
12-December-2023	18:30	S/L crew repaired the packing of the hydraulic stuffing box.
12-December-2023	22:50	Connected the CT riser on the wellhead.
13-December-2023	00:00	Rig up the CT BOP.
13-December-2023	00:40	Perform function test of the CT BOP rams (OK)
13-December-2023	00:45	Opened the kill line and swab valve and production valve.

13-December-2023	00:50	Perform pressure test the CT BOP body and blind rams' low pressure (with 400 psi) from (ok).
13-December-2023	01:43	Observe leakage from U/C 3" connection.
13-December-2023	02:05	Perform pressure test for the CT BOP body and blind rams' high pressure during 15 min (with 6500 psi).
13-December-2023	02:20	Rig up the CT injector.
13-December-2023	03:35	Perform pressure test for the Rubber stripper, 500 psi during 5 min and 6500 psi during 15 min (2 step).
13-December-2023	04:20	Perform pressure test for pipe rams, 300 psi during 5 min and 6500 psi during 15 min.
13-December-2023	05:00	Perform pressure test for the CT check valve, during 5 min.
13-December-2023	05:20	Opened the 2SV via hand pump pressure (2000 psi).
13-December-2023	05:35	Performed equalizing pressure the TMV via pump truck pressure for opening the TMV (5000 psi)
13-December-2023	05:40	Opened the well from TMV.
13-December-2023	05:45	Opened the well from C/M to the separator with C/S 12/64.
13-December-2023	05:50	CT RIH while the pump truck was pumping the diesel fluid (with BHA: Cross over + check valve double flapper + perforation sub + wash nuzzle).
13-December-2023	05:50	Increase C/S gradually of 12/64 adj to 28/64 adj.
13-December-2023	10:00	The CT arrived to 4036m (top of perforation) and stopped at this depth.
13-December-2023	09:55	Finish pumping 45 bbl diesel and switch pumping to ADR.
13-December-2023	10:10	The CT arrived at 4039m and stopped at this depth.
13-December-2023	10:20	Decrease adj C/S from 28/64 to 12/64 adj.
13-December-2023	10:25	Finish 20 bbl ADR and switch pumping to diesel.
13-December-2023	10:45	Increase adj C/S from 12/64 to 14/64 adj.
13-December-2023	11:00	Take a sample from CM, W= 52 PCF, BS&W= %0.5 (water).
13-December-2023	11:18	Opened the kill line valve and closed the well from C/M.

13-December-2023	11:20	Finish pumping 44 bbl diesel.
13-December-2023	11:25	Start bullheading the ADR with pumping 20 bbl diesel @ (11:25 Till 11:35) (pumping P= 4800psi, WHP= 4600psi, R= 2Bpm).
13-December-2023	11:35	Closed kill valve.
13-December-2023	11:38	Open the well from C/M with C/S 16/64adj to the separator, surge and the oil storage tanks.
13-December-2023	11:40	Start pumping diesel for displacing the well.
13-December-2023	12:41	The CT start POOH from 4036m and continue pumping diesel.
13-December-2023	14:10	Increase adj C/S from 16/64 to 20/64 adj.
13-December-2023	15:10	C/T arrived at depth 2000m while the LPU is pumping diesel for displacing the well.
13-December-2023	15:34	The CT POOH.
13-December-2023	17:07	C/T arrived at the surface and stop displacing the well with pumping 210bbl diesel.
13-December-2023	17:10	Closed the well from C/M and opened the TMV after equalizing the pressure with pump truck for bullheading 20bbl diesel.
13-December-2023	17:15	Bullheading 20bbl diesel @ 17:15 till 17:23 (pumping pressure: 5000psi, WHP= 4900psi, Rate= 3.1bpm).
13-December-2023	17:25	Closed the well from TMV and released the U/C pressure from C/M to the oil storage tank.
13-December-2023	17:45	The CT start to R/D.
13-December-2023	19:00	Disconnect the riser (Due to the protection sleeve not passing through the riser).
13-December-2023	19:30	Perform function test of the SL BOP.
13-December-2023	19:45	Rig up the SL lubricator and dual BOP.
13-December-2023	٢١:٢٠	Perform pressure test of S/L lubricator and BOP with 500 psi during 5 Minutes and with 6500 psi during 15 min.
13-December-2023	21:50	Opened the 2SV via hand pump pressure (2000 psi).
13-December-2023	22:03	Opened the well from TMV.

13-December-2023	22:10	Slick line started RIH.
13-December-2023	22:25	Slick line reached 214 ft and catch protection sleeve.
13-December-2023	22:30	Slick line at surface and retrieved protection sleeve successfully.
13-December-2023	22:35	Closed the well from TMV.
13-December-2023	22:38	Released the WHP from C/M to the storage tank at 22:38
13-December-2023	22:39	Closed the swab valve.
13-December-2023	22:40	Rig down S/L.
14-December-2023	02:05	Closed TMV and released pressure to 367 psi, pressure didn't build up and decreased to 238 during 15 min.
14-December-2023	02:25	Closed swab valve and apply 6500 psi in X-Mass tree and closed production valve, and kill valve. Disconnect kill valve line to check Inflow test visually for leakage from kill valve, didn't observe any water drops and duo to leakage of C/M valves can't in flow test production line and decided to check visually leakage by opening needle valve on upstream data header, drops of water didn't stop, so Inflow test of production valve not reliable. But inflow test of swab valve and body test of 3SV were Ok.
14-December-2023	05:00	Due to leakage of C/M decided to disconnect pressure transmitter from annulus sec. C on instrument flange of production valve and closed well test SSV and performed pressure test for confirming no pressure drop for inflow test of 3SV with pump truck, Ok.
14-December-2023	05:18	Closed 3SV by released the control line from 5100 to zero. Released WHP 4680 psig to 660 psi and observed for any pressure build up.
14-December-2023	05:28	Pressure Increased to 1590 for 10 minutes, failed.
14-December-2023	05:50	Opened 3SV and WHP increased 4575 psig and then closed 3SV.
14-December-2023	06:05	Increased WHP to 575 with pump truck then closed well test SSV.
14-December-2023	06:30	Monitor pressure buildup of 3SV inflow test, pressure decreased from 272 psig to 268 psig for 25 min.
14-December-2023	06:40	Function test 3SV and pressure test with 8500 psi of control lines by hand pump for 10 minutes, Ok.
14-December-2023	07:10	Performed inflow test production valve with 6500 psi for 20 minutes.
14-December-2023	07:20	Apply 4000 psi on top of TMV to equalize pressure and open it.

14-December-2023	07:25	Open TMV.
14-December-2023	08:45	Installed lubricator on well head.
14-December-2023	09:05	Performed pressure test of lubricator up to 5000 psi for 10 minutes, Ok.
14-December-2023	09:25	Installed one way.
14-December-2023	10:00	Nipple down lubricator and one way installed successfully and closed all well head valves.
14-December-2023	10:15	Flash SWT lines with pumping 120 bbl water.
End Of Job		

Table 11-5: AZNS-040 Detailed Sequence of Operation

Operation Results

11.4.4.1 SUMMARY OF WELL TESTING RESULTS

Well Testing Rate Measurement (Before Acidizing)																			
Date (DD-MM-YYYY)	Time	Choke size (Inch)	UCP (psi)	DCP (psi)	SEP.P (psi)	S.T (F)	D.P (in.H2O)	Orifice Size (inch)	API (oil)	gas SP.GR	H2S (ppm)	CO2 (%)	BS&W (%)	gas outlet (ID)	beta ratio	Ann P(psi)	Weight (PCF)	Q- Oil (BBL/Day)	Q-Gas (SCF/Day)	GOR (SCF/BBL)
٠٩-December-2023	7:13	٢٤/٦٤	953	202	196	78	43	1.25	40	0.768	9000	3	0.5	5.761	0.22	249	52	972	813006	836
٠٩-December-2023	8:17	٢٤/٦٤	951	196	190	83	46	1.25	39.5	0.778	8500	3	0.5	5.761	0.22	263	52	995	819489	824
٠٩-December-2023	12:51	٢٤/٦٤	920	190	183	89	45	1.25	39	0.788	8000	3	0.5	5.761	0.22	296	52	995	787118	791

Table 11-6: AZNS-040 Well Testing Measurment Before Acidizing

Well Testing Rate Measurement (After Acidizing)																			
Date (DD-MM-YYYY)	Time	Choke size (Inch)	UCP (psi)	DCP (psi)	SEP.P (psi)	S.T (F)	D.P (in.H2O)	Orifice Size (inch)	API (oil)	gas SP.GR	H2S (ppm)	CO2 (%)	BS&W (%)	gas outlet (ID)	beta ratio	Ann P(psi)	Weight (PCF)	Q- Oil (BBL/Day)	Q-Gas (SCF/Day)	GOR (SCF/BBL)
١١-December-2023	10:05	٤٨/64	1367	270	235	132	51	3	37	0.8	8000	3	0	5.761	0.52	466	52	5087	5380449	1058
١١-December-2023	11:39	٤٨/64	1355	268	235	135	51	3	37	0.814	8000	3	0	5.761	0.52	508	52	5041	5320508	1055
١٢-December-2023	7:06	٢٤/64	2782	237	222	110	52	2.125	39	0.786	8000	3	0	5.761	0.37	1	52	2532	2651003	1047
١٢-December-2023	7:51	٢٤/64	2778	237	223	110	52	2.125	38.9	0.784	8000	3	0	5.761	0.37	1	52	2532	2660012	1051
١٢-December-2023	12:14	٢٤/64	2758	231	216	115	53	2.125	38	0.8	8000	3	0	5.761	0.37	0	52	2532	2607290	1030

Table 11-7: AZNS-040 Well Testing Measurment After Acidizin

Surface testing data were analyzed to investigate the effect of stimulation operation. According to pre-stimulation clean up results well head pressure and oil rate were observed 920^{psi} & 995 BPD at 24/64" choke size. On the other hand, data gathered from post stimulation clean up shows that well head pressure and oil rate have increased to 2758^{psi} & 2532 BPD at 24/64 choke size. The above data illustrates that stimulation operation has led to an 1838^{psi} increase in wellhead pressure and a 1537-BPD increase in oil rate. Generally, the stimulation operation was carried out successfully.

11.4.4.2 Well Testing Mail Results & Plots

Date (DD-MM-YYYY)	Time HH:MM	U.C.P (PSI)	U.C.T(°F)	D.C.P (PSI)	D.C.T (°F)	SEP.P (PSI)	SEP.T (°F)	Annulus pressure	DP (in.H2O)	Orifice size (inch)	Q-Oil (STB/Day)	Q-Gas (SCF/Day)
05-December-2023	8:15	91	47.6	4	61.3	0	0	9	0	0	0	0
05-December-2023	8:35	5502	47.4	1	63.7	0	0	9	0	0	0	0
05-December-2023	8:55	75	47.2	0	66.4	0	0	10	0	0	0	0
05-December-2023	9:15	24	54.3	0	68.4	0	0	10	0	0	0	0

05-December-2023	9:35	16	58.8	0	70.0	0	0	10	0	0	0	0
05-December-2023	9:55	8	62.1	0	71.7	0	0	9	0	0	0	0
05-December-2023	10:15	5	61.3	0	73.2	0	0	8	0	0	0	0
05-December-2023	10:35	8	60.8	0	74.6	0	0	8	0	0	0	0
05-December-2023	10:55	7	69.1	0	76.0	0	0	9	0	0	0	0
05-December-2023	11:15	5	72.4	3	77.7	0	0	9	0	0	0	0
05-December-2023	11:35	5	72.3	3	78.5	0	0	6	0	0	0	0
05-December-	11:55	9	74.9	3	79.4	0	0	6	0	0	0	0

2023												
05-December-2023	12:15	18	74.1	3	79.8	0	0	8	0	0	0	0
05-December-2023	12:35	50	76.1	3	80.3	0	0	9	0	0	0	0
05-December-2023	12:55	4141	76.1	3	81.9	0	0	7	0	0	0	0
05-December-2023	13:15	5135	77.0	2	83.2	0	0	6	0	0	0	0
05-December-2023	13:35	6527	75.6	3	84.2	0	0	7	0	0	0	0
05-December-2023	13:55	6100	76.4	4	85.4	0	0	7	0	0	0	0
05-December-2023	14:15	5876	78.4	3	86.0	0	0	7	0	0	0	0

05-December-2023	14:35	5486	79.2	3	86.6	0	0	7	0	0	0	0
05-December-2023	14:55	5224	79.4	3	86.9	0	0	7	0	0	0	0
05-December-2023	15:15	5083	79.3	3	87.1	0	0	8	0	0	0	0
05-December-2023	15:35	4878	8024.0	1	86.5	0	0	8	0	0	0	0
05-December-2023	15:55	5	78.7	1	84.2	0	0	7	0	0	0	0
05-December-2023	16:15	4	78.2	1	82.7	0	0	8	0	0	0	0
05-December-2023	16:40	4	77.9	1	81.3	0	0	7	0	0	0	0
05-December-	17:00	3	78.7	0	80.3	0	0	6	0	0	0	0

2023												
05-December-2023	17:20	3	78.7	1	79.3	0	0	8	0	0	0	0
05-December-2023	17:40	3	79.0	1	78.4	0	0	8	0	0	0	0
05-December-2023	18:00	4	79.6	1	77.6	0	0	8	0	0	0	0
05-December-2023	18:20	4	79.0	1	76.8	0	0	8	0	0	0	0
05-December-2023	18:40	3	70.5	1	76.1	0	0	7	0	0	0	0
05-December-2023	19:00	3	69.4	0	75.3	0	0	7	0	0	0	0
05-December-2023	19:20	3	68.3	2	74.6	0	0	8	0	0	0	0

05-December-2023	19:40	2	67.3	1	73.9	0	0	9	0	0	0	0
05-December-2023	20:00	5216	66.7	1	73.2	0	0	8	0	0	0	0
05-December-2023	20:20	5	63.3	3	71.4	0	0	9	0	0	0	0
05-December-2023	20:40	5	61.1	3	70.5	0	0	7	0	0	0	0
05-December-2023	21:00	5	6062.0	3	69.8	0	0	9	0	0	0	0
05-December-2023	21:20	4	59.1	2	69.0	0	0	7	0	0	0	0
05-December-2023	21:40	5	58.3	3	68.3	0	0	7	0	0	0	0
05-December-	22:00	5	57.5	4	67.7	0	0	8	0	0	0	0

2023												
05-December-2023	22:20	3	56.6	2	67.2	0	0	9	0	0	0	0
05-December-2023	22:40	5040	56.2	3	66.6	0	0	9	0	0	0	0
05-December-2023	23:00	16	55.3	3	66.1	0	0	9	0	0	0	0
05-December-2023	23:20	17	54.6	3	65.6	0	0	9	0	0	0	0
05-December-2023	23:40	17	54.1	4	65.2	0	0	9	0	0	0	0
06-December-2023	0:00	16	53.6	3	64.8	0	0	10	0	0	0	0
06-December-2023	0:20	17	52.9	3	64.4	0	0	10	0	0	0	0

06-December-2023	0:40	6423	52.4	4	64.0	0	0	9	0	0	0	0
06-December-2023	1:00	68	51.6	3	63.6	0	0	10	0	0	0	0
06-December-2023	1:20	16	51.4	3	63.2	0	0	8	0	0	0	0
06-December-2023	1:40	17	51.0	4	62.9	0	0	10	0	0	0	0
06-December-2023	2:00	5494	50.6	3	62.6	0	0	8	0	0	0	0
06-December-2023	2:20	6693	50.1	2	62.3	0	0	8	0	0	0	0
06-December-2023	2:40	6563	49.6	2	62.0	0	0	10	0	0	0	0
06-December-	3:00	636	50.9	3	63.4	0	0	10	0	0	0	0

2023												
06-December-2023	3:20	6462	50.2	2	62.8	0	0	9	0	0	0	0
06-December-2023	3:40	1550	50.0	3	65.4	0	0	9	0	0	0	0
06-December-2023	4:00	3	49.5	2	64.5	0	0	8	0	0	0	0
06-December-2023	4:20	8	49.1	3	63.8	0	0	9	0	0	0	0
06-December-2023	4:40	8	48.7	3	63.2	0	0	10	0	0	0	0
06-December-2023	5:05	8	48.3	2	62.6	0	0	8	0	0	0	0
06-December-2023	5:25	7	47.6	3	62.1	0	0	8	0	0	0	0

06-December-2023	5:45	8	47.5	2	61.7	0	0	9	0	0	0	0
06-December-2023	6:05	20	47.3	2	61.4	0	0	9	0	0	0	0
06-December-2023	6:25	5	49.7	4	62.7	0	0	10	0	0	0	0
06-December-2023	6:45	7	56.7	4	66.9	0	0	10	0	0	0	0
06-December-2023	7:05	7	60.4	4	69.4	0	0	9	0	0	0	0
06-December-2023	7:25	1327	62.4	3	69.8	0	0	9	0	0	0	0
06-December-2023	7:45	7	60.8	3	68.9	0	0	10	0	0	0	0
06-December-	8:05	28	65.5	3	68.5	0	0	9	0	0	0	0

2023												
06-December-2023	8:25	26	70.8	3	69.2	0	0	8	0	0	0	0
06-December-2023	8:45	8	72.7	3	70.0	0	0	6	0	0	0	0
06-December-2023	9:05	8	73.1	3	70.6	0	0	6	0	0	0	0
06-December-2023	9:25	5	75.5	3	71.9	0	0	6	0	0	0	0
06-December-2023	9:45	5	76.1	3	72.5	0	0	7	0	0	0	0
06-December-2023	10:05	6	76.4	3	73.8	0	0	7	0	0	0	0
06-December-2023	10:25	10	76.4	3	74.7	0	0	6	0	0	0	0

06-December-2023	10:45	5	76.6	3	75.7	0	0	6	0	0	0	0
06-December-2023	11:05	4	75.6	3	75.8	0	0	6	0	0	0	0
06-December-2023	11:25	5	77.2	3	76.8	0	0	6	0	0	0	0
06-December-2023	11:45	6	78.7	4	78.2	0	0	5	0	0	0	0
06-December-2023	12:05	4	79.0	3	78.9	0	0	5	0	0	0	0
06-December-2023	12:25	238	79.4	3	79.8	0	0	6	0	0	0	0
06-December-2023	12:45	239	79.6	3	80.3	0	0	6	0	0	0	0
06-December-	13:05	245	79.8	4	81.4	0	0	5	0	0	0	0

2023												
06-December-2023	13:25	343	78.6	3	82.2	0	0	5	0	0	0	0
06-December-2023	13:45	476	77.9	3	82.0	0	0	6	0	0	0	0
06-December-2023	14:05	559	78.3	3	81.7	0	0	6	0	0	0	0
06-December-2023	14:25	787	79.5	3	82.9	0	0	5	0	0	0	0
06-December-2023	14:45	850	79.5	3	83.7	0	0.0	6	0	0	0	0
06-December-2023	15:05	1072	79.6	3	83.7	0	0.0	6	0	0	0	0
06-December-2023	15:25	1239	78.7	2	83.2	0	0.0	5	0	0	0	0

06-December-2023	15:45	1546	79.4	3	82.2	0	0.0	5	0	0	0	0
06-December-2023	16:05	1569	78.2	3	81.3	0	70.8	6	0	0	0	0
06-December-2023	16:25	1414	77.0	3	80.4	0	70.5	5	0	0	0	0
06-December-2023	16:50	2065	76.2	3	79.3	0	705.0	5	0	0	0	0
06-December-2023	17:10	2302	75.4	3	78.3	0	69.4	6	0	0	0	0
06-December-2023	17:30	3012	75.5	3	77.4	0	68.7	5	0	0	0	0
06-December-2023	17:50	2851	75.9	3	76.6	0	68.4	5	0	0	0	0
06-December-	18:10	2727	75.3	2	76.0	0	67.9	5	0	0	0	0

2023												
06-December-2023	18:30	2011	74.7	3	75.6	0	67.6	5	0	0	0	0
06-December-2023	18:50	812	75.5	3	75.1	0	66.8	5	0	0	0	0
06-December-2023	19:10	26	77.1	3	74.5	0	66.2	5	0	0	0	0
06-December-2023	19:30	29	78.4	2	73.9	0	65.6	5	0	0	0	0
06-December-2023	19:50	32	79.5	2	73.4	0	65.3	5	0	0	0	0
06-December-2023	20:10	22	79.7	2	72.8	0	64.6	8	0	0	0	0
06-December-2023	20:30	37	80.4	2	72.2	0	63.7	9	0	0	0	0

06-December-2023	20:50	49	81.6	3	71.7	0	63.2	8	0	0	0	0
06-December-2023	21:10	50	83.0	2	71.3	0	62.7	8	0	0	0	0
06-December-2023	21:30	58	84.4	3	70.9	0	62.2	8	0	0	0	0
06-December-2023	21:50	66	85.1	3	70.5	0	61.5	9	0	0	0	0
06-December-2023	22:10	89	86.1	3	70.0	0	60.7	21	0	0	0	0
06-December-2023	22:30	103	87.4	3	69.7	0	60.3	30	0	0	0	0
06-December-2023	22:50	148	88.9	2	69.4	0	59.9	32	0	0	0	0
06-December-	23:10	2107	89.6	3	69.1	0	59.1	405	0	0	0	0

2023												
06-December-2023	23:30	219	90.3	2	68.8	0	58.6	40	0	0	0	0
06-December-2023	23:50	426	90.5	2	68.6	0	58.2	42	0	0	0	0
07-December-2023	0:10	1224	82.2	3	68.4	0	57.6	47	0	0	0	0
07-December-2023	0:30	916	87.6	6	86.6	0	57.1	26	0	0	0	0
07-December-2023	0:50	1120	82.2	5	83.7	0	56.6	31	0	0	0	0
07-December-2023	1:10	1336	74.6	5	79.4	0	56.2	34	0	0	0	0
07-December-2023	1:30	931	88.0	6	86.1	0	55.8	37	0	0	0	0

07-December-2023	1:50	898	89.4	3	87.6	0	55.4	59	0	0	0	0
07-December-2023	2:10	1054	87.1	4	88.5	0	54.5	88	0	0	0	0
07-December-2023	2:30	912	89.8	8	89.6	0	53.7	100	0	0	0	0
07-December-2023	2:50	1028	90.2	9	88.9	0	53.3	125	0	0	0	0
07-December-2023	3:10	1100	92.1	19	89.0	0	52.8	145	0	0	0	0
07-December-2023	3:30	1198	92.2	11	88.9	0	52.5	161	0	0	0	0
07-December-2023	3:50	1237	91.3	16	85.7	0	52.3	178	0	0	0	0
07-December-	4:10	1313	90.2	22	81.7	0	52.0	194	0	0	0	0

2023												
07-December-2023	4:30	1196	90.9	51	79.7	41	51.6	202	0	0	0	0
07-December-2023	4:50	1243	91.1	21	80.6	10	57.3	219	0	0	0	0
07-December-2023	5:10	1631	87.8	25	79.8	14	57.5	148	0	0	0	0
07-December-2023	5:30	1440	89.3	29	79.8	17	61.7	136	0	0	0	0
07-December-2023	5:50	1465	88.1	67	77.0	55	59.9	150	0	0	0	0
07-December-2023	6:10	1162	90.2	34	79.4	22	60.4	147	0	0	0	0
07-December-2023	6:30	1222	91.5	46	76.1	35	59.8	174	0	0	0	0

07-December-2023	6:50	1216	91.4	53	71.6	42	59.5	196	0	0	0	0
07-December-2023	7:10	1221	92.1	29	76.3	18	58.7	219	0	0	0	0
07-December-2023	7:30	1291	91.9	41	73.2	29	58.5	248	0	0	0	0
07-December-2023	7:50	1337	91.7	31	71.8	20	59.4	275	0	0	0	0
07-December-2023	8:10	1124	92.2	60	76.5	49	63.0	278	0	0	0	0
07-December-2023	8:30	1021	92.6	57	77.0	45	64.0	305	0	0	0	0
07-December-2023	8:50	1007	94.4	50	77.9	40	67.1	334	0	0	0	0
07-December-	9:10	1027	95.0	47	77.4	35	68.5	367	0	0	0	0

2023												
07-December-2023	9:30	921	97.1	69	80.5	56	71.8	603	0	0	0	0
07-December-2023	9:50	872	97.2	63	80.0	50	72.1	421	0	0	0	0
07-December-2023	10:10	853	1004.0	61	80.3	49	75.1	456	0	0	0	0
07-December-2023	10:30	813	1006.0	86	82.5	71	78.1	493	0	0	0	0
07-December-2023	10:50	757	102.0	73	83.0	59	80.3	533	0	0	0	0
07-December-2023	11:10	603	104.6	101	86.8	86	82.9	577	0	0	0	0
07-December-2023	11:30	503	105.9	88	88.0	70	83.6	628	0	0	0	0

07-December-2023	11:50	468	106.1	74	87.7	59	82.9	682	0	0	0	0
07-December-2023	12:10	465	106.5	72	87.6	56	83.4	711	0	0	0	0
07-December-2023	12:30	492	105.3	64	86.5	51	82.7	209	0	0	0	0
07-December-2023	12:50	520	105.2	69	86.1	52	81.8	249	0	0	0	0
07-December-2023	13:10	536	106.3	66	86.2	52	82.0	284	0	0	0	0
07-December-2023	13:30	511	106.6	119	88.6	96	82.7	318	0	0	0	0
07-December-2023	13:50	470	107.7	86	88.8	70	84.5	360	0	0	0	0
07-December-	14:10	451	108.2	78	88.9	62	85.2	396	0	0	0	0

2023												
07-December-2023	14:30	450	108.4	80	88.9	64	85.3	434	0	0	0	0
07-December-2023	14:50	372	109.6	89	91.2	71	86.7	474	0	0	0	0
07-December-2023	15:10	363	110.1	87	91.5	70	87.1	519	0	0	0	0
07-December-2023	15:30	345	108.8	90	91.9	75	87.1	561	0	0	0	0
07-December-2023	15:50	329	108.9	93	92.3	76	87.9	600	0	0	0	0
07-December-2023	16:10	309	108.3	94	93.0	77	88.1	645	0	0	0	0
07-December-2023	16:30	305	107.5	99	93.4	81	87.9	699	0	0	0	0

07-December-2023	16:50	305	107.6	97	93.8	79	87.9	215	0	0	0	0
07-December-2023	17:10	307	107.5	100	93.8	81	88.1	262	0	0	0	0
07-December-2023	17:30	300	107.5	98	94.1	81	87.6	305	0	0	0	0
07-December-2023	17:50	313	107.9	89	93.6	74	87.1	348	0	0	0	0
07-December-2023	18:10	323	108.0	97	93.6	79	85.9	387	0	0	0	0
07-December-2023	18:30	323	108.3	96	93.8	79	85.6	423	0	0	0	0
07-December-2023	18:50	354	108.2	81	92.9	65	84.5	451	0	0	0	0
07-December-	19:10	376	108.6	89	92.9	72	84.5	478	0	0	0	0

2023												
07-December-2023	19:30	383	109.1	92	93.3	73	86.2	505	0	0	0	0
07-December-2023	19:50	384	108.6	95	93.4	75	86.0	532	0	0	0	0
07-December-2023	20:10	383	108.7	94	93.4	74	84.9	558	0	0	0	0
07-December-2023	20:30	381	111.4	94	93.6	74	84.0	581	0	0	0	0
07-December-2023	20:50	375	110.5	95	93.7	76	83.7	605	0	0	0	0
07-December-2023	21:10	368	111.8	94	93.9	76	85.0	629	0	0	0	0
07-December-2023	21:30	364	112.3	97	94.2	77	86.4	652	0	0	0	0

07-December-2023	21:50	363	112.2	94	94.2	74	86.4	673	0	0	0	0
07-December-2023	22:10	362	111.9	94	94.3	75	84.9	211	0	0	0	0
07-December-2023	22:30	360	112.1	90	94.5	71	84.1	230	0	0	0	0
07-December-2023	22:50	360	111.8	92	94.7	73	83.9	248	0	0	0	0
07-December-2023	23:10	359	112.1	91	94.6	72	84.1	264	0	0	0	0
07-December-2023	23:30	355	112.8	91	94.9	75	84.0	281	0	0	0	0
07-December-2023	23:50	351	112.5	90	95.0	73	83.6	297	0	0	0	0
08-December-	0:10	332	112.9	101	96.0	83	85.2	314	0	0	0	0

2023												
08-December-2023	0:30	315	113.6	96	96.2	79	86.4	331	0	0	0	0
08-December-2023	0:50	307	112.9	94	96.1	76	84.5	349	0	0	0	0
08-December-2023	1:10	304	113.3	92	96.4	77	84.5	364	0	0	0	0
08-December-2023	1:30	302	113.8	93	96.5	76	84.4	379	0	0	0	0
08-December-2023	1:50	300	113.4	93	96.7	76	84.4	392	0	0	0	0
08-December-2023	2:10	300	113.7	903	96.5	74	84.9	404	0	0	0	0
08-December-2023	2:30	300	114.0	91	96.6	75	85.5	417	0	0	0	0

08-December-2023	2:50	299	113.7	93	96.7	76	85.3	428	0	0	0	0
08-December-2023	3:10	302	113.9	93	96.9	76	84.4	441	0	0	0	0
08-December-2023	3:30	303	113.4	97	96.8	78	83.8	456	0	0	0	0
08-December-2023	3:50	306	113.7	93	96.9	76	83.6	472	0	0	0	0
08-December-2023	4:10	310	113.9	98	97.0	79	83.5	487	0	0	0	0
08-December-2023	4:30	335	114.0	89	96.1	72	83.7	500	0	0	0	0
08-December-2023	4:50	376	113.9	77	95.1	62	82.1	510	0	0	0	0
08-December-	5:10	438	112.8	73	93.6	60	81.5	517	0	0	0	0

2023												
08-December-2023	5:30	481	112.8	80	92.8	66	80.7	525	0	0	0	0
08-December-2023	5:50	503	111.6	88	92.8	69	80.5	534	0	0	0	0
08-December-2023	6:10	574	110.0	38	88.9	30	78.6	542	0	0	0	0
08-December-2023	6:30	860	98.3	0	78.4	0	70.3	544	0	0	0	0
08-December-2023	6:50	1	73.2	0	74.3	0	65.2	558	0	0	0	0
08-December-2023	7:10	1	65.2	0	71.5	0	61.4	586	0	0	0	0
08-December-2023	7:30	1	60.5	0	69.6	0	58.9	629	0	0	0	0

08-December-2023	7:50	1	58.1	0	68.3	0	58.3	542	0	0	0	0
08-December-2023	8:10	2	60.4	0	68.4	0	60.9	457	0	0	0	0
08-December-2023	8:30	2	64.2	0	69.7	0	63.4	369	0	0	0	0
08-December-2023	8:50	2	68.3	0	71.2	0	66.6	277	0	0	0	0
08-December-2023	9:10	2	71.4	0	73.4	0	69.0	200	0	0	0	0
08-December-2023	9:30	2	75.1	0	75.5	0	71.2	127	0	0	0	0
08-December-2023	9:50	2	78.3	0	77.6	0	73.6	62	0	0	0	0
08-December-	10:10	37	81.9	0	79.7	0	76.7	11	0	0	0	0

2023												
08-December-2023	10:30	6347	82.3	0	81.5	0	78.9	-9	0	0	0	0
08-December-2023	10:50	6354	84.0	0	83.2	0	81.1	-11	0	0	0	0
08-December-2023	11:12	2	79.0	0	79.7	0	81.9	-14	0	0	0	0
08-December-2023	11:32	3	83.4	0	81.5	0	82.1	-15	0	0	0	0
08-December-2023	11:52	2	77.9	0	78.4	0	80.1	-16	0	0	0	0
08-December-2023	12:12	21	77.7	0	75.7	0	77.6	-16	0	0	0	0
08-December-2023	12:32	5	78.2	0	78.0	0	77.0	-17	0	0	0	0

08-December-2023	12:52	6243	85.0	0	79.9	0	76.7	-18	0	0	0	0
08-December-2023	13:12	3	83.8	1	81.0	0	76.6	-18	0	0	0	0
08-December-2023	13:32	4	87.1	1	82.8	0	76.6	-17	0	0	0	0
08-December-2023	13:52	4942	89.9	2	84.7	0	76.8	-17	0	0	0	0
08-December-2023	14:12	4	87.3	1	83.4	0	76.9	-18	0	0	0	0
08-December-2023	14:32	2913	86.6	-1	84.2	0	76.9	-18	0	0	0	0
08-December-2023	14:52	2101	1007.0	91	77.9	77	79.3	-17	0	0	0	0
08-December-	15:12	1585	102.3	72	78.9	60	76.5	-18	0	0	0	0

2023												
08-December-2023	15:32	1478	102.3	91	78.6	63	76.9	-18	0	0	0	0
08-December-2023	15:52	1280	102.9	83	81.9	70	78.4	-17	0	0	0	0
08-December-2023	16:12	1216	102.9	62	80.5	45	78.0	-17	0	0	0	0
08-December-2023	16:32	1170	102.4	57	80.4	48	76.8	-17	0	0	0	0
08-December-2023	16:52	1148	101.7	59	80.6	47	76.5	-18	0	0	0	0
08-December-2023	17:12	1127	102.5	59	80.9	47	76.1	-2	0	0	0	0
08-December-2023	17:32	1113	101.0	57	81.1	47	75.6	1	0	0	0	0

08-December-2023	17:52	1101	101.3	57	81.4	47	75.4	5	0	0	0	0
08-December-2023	18:12	1097	100.9	58	81.6	47	75.6	8	0	0	0	0
08-December-2023	18:32	1086	100.3	57	81.6	48	75.2	12	0	0	0	0
08-December-2023	18:52	1076	100.5	58	81.7	48	74.9	17	0	0	0	0
08-December-2023	19:12	1064	103.8	57	81.8	45	74.7	22	0	0	0	0
08-December-2023	19:32	1055	105.1	56	82.0	45	74.7	27	0	0	0	0
08-December-2023	19:52	1050	104.8	56	82.2	46	74.6	34	0	0	0	0
08-December-	20:12	1043	105.1	56	82.3	46	74.5	42	0	0	0	0

2023												
08-December-2023	20:32	1034	105.3	56	82.3	45	74.6	49	0	0	0	0
08-December-2023	20:52	1029	105.4	56	82.5	46	74.4	58	0	0	0	0
08-December-2023	21:12	1021	105.5	56	82.6	46	74.9	66	0	0	0	0
08-December-2023	21:32	1015	105.7	55	82.6	45	74.3	75	0	0	0	0
08-December-2023	21:52	1009	105.9	55	82.7	45	74.1	85	0	0	0	0
08-December-2023	22:12	1003	106.2	54	82.7	45	74.4	93	0	0	0	0
08-December-2023	22:32	998	106.0	55	82.8	44	74.5	100	0	0	0	0

08-December-2023	22:52	994	105.8	55	82.8	45	74.2	108	0	0	0	0
08-December-2023	23:12	987	105.6	50	82.6	42	73.5	116	0	0	0	0
08-December-2023	23:32	983	105.8	51	82.7	41	73.6	124	0	0	0	0
08-December-2023	23:52	979	105.0	51	82.6	42	73.2	131	0	0	0	0
09-December-2023	0:12	974	105.2	52	82.6	42	72.7	137	0	0	0	0
09-December-2023	0:32	971	105.1	53	82.8	43	72.5	144	0	0	0	0
09-December-2023	0:52	966	104.8	52	82.7	43	71.9	148	0	0	0	0
09-December-	1:12	963	105.1	52	82.7	43	72.0	155	0	0	0	0

2023												
09-December-2023	1:37	957	104.0	52	82.6	43	71.5	161	0	0	0	0
09-December-2023	1:57	954	104.4	51	82.7	43	71.2	167	0	0	0	0
09-December-2023	2:17	951	104.5	52	82.6	42	71.6	171	0	0	0	0
09-December-2023	2:37	948	104.7	51	82.7	42	71.8	176	0	0	0	0
09-December-2023	2:57	946	104.4	75	84.0	68	71.7	181	0	0	0	0
09-December-2023	3:17	947	104.1	205	89.4	200	74.7	187	0	0	0	0
09-December-2023	3:37	942	103.9	203	89.5	196	74.6	192	40.54	0	0	822466.67

09-December-2023	3:57	941	102.9	203	89.5	197	74.5	196	41.7	0	0	813740.28
09-December-2023	4:17	935	103.7	198	89.5	192	75.7	197	41.83	0	0	820480
09-December-2023	4:37	933	102.6	199	89.5	191	74.2	200	42.01	0	0	837199.53
09-December-2023	4:57	930	104.4	199	89.7	192	76.3	205	42.15	0	0	815912
09-December-2023	5:17	930	104.5	200	89.7	192	76.2	208	42.38	0	0	835696
09-December-2023	5:37	944	104.1	198	89.5	192	74.8	217	42.33	0	0	831250
09-December-2023	5:57	950	104.1	201	89.8	195	76.2	226	43.45	0	0	813986
09-December-	6:17	952	103.7	202	89.7	195	76.1	231	42.8	0	0	813723

2023												
09-December-2023	6:37	952	102.8	202	89.7	196	75.8	237	42.92	0	0	823236
09-December-2023	6:57	954	103.9	201	89.7	195	77.6	244	43.1	0	0	836346
09-December-2023	7:13	953	103.1	202	89.8	196	78	245	43	1.25	972	813006
09-December-2023	7:17	953	103.1	203	89.8	195	78.0	249	43.05	0	0	815912
09-December-2023	7:37	953	103.5	205	89.6	197	77.0	254	43.45	1.25	0	822152
09-December-2023	7:57	953	104.7	201	89.9	195	79.8	260	43.8	1.25	0	819905
09-December-2023	8:17	951	106	196	90.3	190	83	263	46	1.25	995	816128

09-December-2023	8:37	948	107.5	195	90.3	188	84.2	264	45.61	1.25	0	818701
09-December-2023	8:57	945	108.0	191	90.3	186	83.8	270	45.63	1.25	0	813723
09-December-2023	9:17	941	108.6	191	90.5	183	86.4	272	45.84	1.25	0	809247
09-December-2023	9:37	936	108.6	185	90.4	179	87.7	273	46.47	1.25	0	804100
09-December-2023	9:57	935	109.0	186	90.5	181	88.7	276	46.95	1.25	0	811982
09-December-2023	10:17	932	109.8	188	90.6	182	89.7	280	49.23	1.25	0	832312
09-December-2023	10:37	929	110.0	185	90.7	179	90.0	280	46.48	1.25	0	804049
09-December-	10:57	928	110.5	184	90.9	179	91.4	283	48.09	1.25	0	816758

2023												
09-December-2023	11:17	928	111.0	186	91.1	179	92.5	285	46.75	1.25	0	803026
09-December-2023	11:37	926	111.1	188	91.3	181	92.1	287	47.05	1.25	0	811143
09-December-2023	11:57	925	110.9	185	91.3	178	89.5	289	46.71	1.25	0	804139
09-December-2023	12:17	923	110.4	185	91.2	178	88.4	291	46.69	1.25	0	804013
09-December-2023	12:37	923	109.7	189	91.1	182	88.8	293	45.14	1.25	0	797683
09-December-2023	12:51	920	110	190	91	183	89	304	45	1.25	995	787118
09-December-2023	12:57	920	110.1	189	91.3	183	88.7	297	45.4	1.25	0	802882

09-December-2023	13:17	918	110.8	192	91.4	185	88.9	299	46.27	1.25	0	813375
09-December-2023	13:37	916	110.7	191	91.4	185	88.7	301	45.18	1.25	0	803870
09-December-2023	13:57	916	111.3	190	91.6	183	89.2	304	45.86	1.25	0	806608
09-December-2023	14:17	916	111.1	189	91.6	183	88.9	305	44.36	1.25	0	794279
09-December-2023	14:37	917	111.5	190	91.6	183	89.3	310	44.42	1.25	0	794469
09-December-2023	14:57	918	111.1	193	91.6	186	89.2	312	44.86	1.25	0	803579
09-December-2023	15:17	917	110.8	190	91.5	185	88.8	314	44.42	1.25	0	797052
09-December-	15:37	915	108.4	191	91.2	185	88.3	318	44.59	1.25	0	799640

2023												
09-December-2023	15:57	914	108.5	192	91.0	186	87.5	320	0	0	0	0
09-December-2023	16:17	916	108.2	193	91.0	186	86.9	322	0	0	0	0
09-December-2023	16:37	0	94.2	0	85.6	0	806.0	335	0	0	0	0
09-December-2023	16:57	0	87.0	0	83.0	0	77.5	354	0	0	0	0
09-December-2023	17:17	0	80.8	0	80.6	0	75.7	388	0	0	0	0
09-December-2023	17:37	1	76.2	0	78.5	0	73.9	384	0	0	0	0
09-December-2023	17:57	1	73.6	0	76.9	0	72.6	323	0	0	0	0

09-December-2023	18:17	1	71.9	0	75.6	0	71.6	258	0	0	0	0
09-December-2023	18:37	0	70.3	0	74.6	0	70.3	199	0	0	0	0
09-December-2023	18:57	347	70.8	0	74.0	0	69.5	145	0	0	0	0
09-December-2023	19:17	2081	73.0	0	73.2	0	68.5	101	0	0	0	0
09-December-2023	19:37	6719	72.3	0	72.5	0	67.5	68	0	0	0	0
09-December-2023	19:57	13	70.4	0	72.0	0	66.9	46	0	0	0	0
09-December-2023	20:17	19	69.4	0	71.4	0	66.2	32	0	0	0	0
09-December-	20:37	17	67.7	0	70.9	0	65.4	24	0	0	0	0

2023												
09-December-2023	20:57	17	66.5	0	70.5	0	64.8	18	0	0	0	0
09-December-2023	21:17	18	65.8	0	70.2	0	64.6	15	0	0	0	0
09-December-2023	21:37	18	64.7	0	69.9	0	64.0	11	0	0	0	0
09-December-2023	21:57	969	64.1	0	69.5	0	63.4	10	0	0	0	0
09-December-2023	22:17	6241	65.1	0	69.1	0	62.8	8	0	0	0	0
09-December-2023	22:37	6363	64.9	0	68.9	0	62.4	6	0	0	0	0
09-December-2023	22:57	6333	64.5	0	68.8	0	62.2	4	0	0	0	0

09-December-2023	23:17	845	62.9	0	71.4	0	62.0	4	0	0	0	0
09-December-2023	23:37	2485	62.4	0	71.3	0	61.5	7	0	0	0	0
09-December-2023	23:57	571	65.3	0	73.2	0	60.4	3	0	0	0	0
10-December-2023	0:17	952	69.3	0	75.8	0	59.9	1	0	0	0	0
10-December-2023	0:37	4994	68.2	0	73.9	0	59.9	1	0	0	0	0
10-December-2023	0:57	4049	80.7	0	67.3	0	60.6	0	0	0	0	0
10-December-2023	1:17	2403	87.6	63	72.1	58	63.6	-1	0	0	0	0
10-December-	1:37	2310	86.8	23	65.7	17	60.6	-1	0	0	0	0

2023												
10-December-2023	1:57	3100	80.8	39	64.8	35	59.5	-1	0	0	0	0
10-December-2023	2:17	3253	79.8	55	66.7	50	59.1	0	0	0	0	0
10-December-2023	2:37	3229	79.4	50	66.2	46	58.8	0	0	0	0	0
10-December-2023	2:57	3310	78.9	44	66.6	39	58.9	1	0	0	0	0
10-December-2023	3:17	3432	77.0	53	65.6	49	58.5	0	0	0	0	0
10-December-2023	3:37	3542	76.0	56	66.6	52	58.7	2	0	0	0	0
10-December-2023	3:57	3588	77.7	602	66.2	55	58.5	1	0	0	0	0

10-December-2023	4:17	3073	77.8	68	65.9	63	60.6	2	0	0	0	0
10-December-2023	4:37	3076	82.7	51	70.3	46	60.3	31	0	0	0	0
10-December-2023	4:57	3013	82.5	55	66.6	50	60.4	17	0	0	0	0
10-December-2023	5:17	3904	84.3	76	69.4	70	60.1	6	0	0	0	0
10-December-2023	5:37	3492	82.6	60	68.0	54	60.6	18	0	0	0	0
10-December-2023	5:57	3722	80.9	74	68.9	67	61.2	6	0	0	0	0
10-December-2023	6:17	3572	80.6	74	68.0	67	61.3	6	0	0	0	0
10-December-	6:37	4374	78.2	49	67.1	42	59.7	13	0	0	0	0

2023												
10-December-2023	6:57	4099	75.3	1	70.0	0	58.5	5	0	0	0	0
10-December-2023	7:17	5972	69.5	1	68.7	0	58.4	15	0	0	0	0
10-December-2023	7:37	6030	65.9	1	68.1	0	58.4	7	0	0	0	0
10-December-2023	7:57	6142	64.7	2	67.8	0	58.5	18	0	0	0	0
10-December-2023	8:17	5973	63.8	4	67.9	0	59.1	5	0	0	0	0
10-December-2023	8:37	5362	63.7	5	68.3	0	59.9	6	0	0	0	0
10-December-2023	8:57	4596	64.3	5	68.9	0	61.2	202	0	0	0	0

10-December-2023	9:17	4361	66.9	5	70.5	0	63.8	4	0	0	0	0
10-December-2023	9:37	4282	70.5	3	72.5	0	67.7	9	0	0	0	0
10-December-2023	9:57	4259	73.0	3	74.3	0	70.4	8	0	0	0	0
10-December-2023	10:17	4255	75.0	2	76.0	0	72.2	17	0	0	0	0
10-December-2023	10:37	4279	77.4	3	77.8	0	74.8	9	0	0	0	0
10-December-2023	10:57	4250	79.5	3	79.2	0	76.4	12	0	0	0	0
10-December-2023	11:17	4315	80.8	3	80.3	0	77.7	10	0	0	0	0
10-December-	11:37	4345	82.6	0	81.5	0	79.5	24	0	0	0	0

2023												
10-December-2023	11:57	5315	84.6	0	82.5	0	78.4	29	0	0	0	0
10-December-2023	12:17	2300	83.9	903	65.6	78	72.5	22	0	0	0	0
10-December-2023	12:37	2498	88.3	12	69.0	8	69.0	14	0	0	0	0
10-December-2023	12:57	4264	89.2	0	69.2	0	70.5	12	0	0	0	0
10-December-2023	13:17	4524	90.5	0	74.8	0	72.4	6	0	0	0	0
10-December-2023	13:37	4929	92.0	0	78.6	0	74.0	6	0	0	0	0
10-December-2023	13:57	5067	93.4	0	81.4	0	75.1	12	0	0	0	0

10-December-2023	14:17	4425	93.6	0	83.7	0	75.9	10	0	0	0	0
10-December-2023	14:37	5188	94.3	0	85.2	0	76.4	6	0	0	0	0
10-December-2023	14:57	5579	94.8	0	86.2	0	76.7	-4	0	0	0	0
10-December-2023	15:17	5759	94.1	0	86.7	0	76.9	-3	0	0	0	0
10-December-2023	15:37	5125	90.4	0	86.6	0	76.9	-4	0	0	0	0
10-December-2023	15:57	4925	87.7	0	84.9	0	76.7	-6	0	0	0	0
10-December-2023	16:17	4830	86.1	0	83.5	0	76.3	-6	0	0	0	0
10-December-	16:37	4793	84.5	0	82.3	0	76.0	-7	0	0	0	0

2023												
10-December-2023	16:57	4695	82.0	0	81.0	0	75.0	-5	0	0	0	0
10-December-2023	17:17	4689	802.0	0	79.5	0	73.7	-5	0	0	0	0
10-December-2023	17:37	4684	78.5	0	78.2	0	72.6	-6	0	0	0	0
10-December-2023	17:57	4707	77.0	0	77.1	0	71.4	-4	0	0	0	0
10-December-2023	18:17	5376	75.8	0	75.9	0	709.0	-5	0	0	0	0
10-December-2023	18:37	4796	74.1	0	74.9	0	69.2	-5	0	0	0	0
10-December-2023	18:57	3885	86.7	7	98.6	0	68.4	9	0	0	0	0

10-December-2023	19:17	3500	90.1	228	80.2	219	70.7	12	0	0	0	0
10-December-2023	19:37	3413	98.2	92	99.9	80	78.8	10	0	0	0	0
10-December-2023	19:57	3572	100.9	38	104.4	303	78.9	2	0	0	0	0
10-December-2023	20:17	3753	102.0	190	101.2	182	84.9	0	0	0	0	0
10-December-2023	20:37	3829	103.1	150	100.3	141	88.9	2	0	0	0	0
10-December-2023	20:57	3867	104.9	182	100.0	173	90.9	2	0	0	0	0
10-December-2023	21:22	3781	103.8	136	90.2	117	86.8	6	0	0	0	0
10-December-	21:42	3372	110.9	212	91.3	191	85.6	14	0	0	0	0

2023												
10-December-2023	22:02	2963	114.1	248	91.2	221	86.5	109	0	0	0	0
10-December-2023	22:22	2728	115.2	290	92.0	258	87.6	206	0	0	0	0
10-December-2023	22:53	2433	121.1	282	94.7	251	909.0	581	0	0	0	0
10-December-2023	23:13	2377	122.6	227	94.8	192	89.4	341	0	0	0	0
10-December-2023	23:33	2330	123.8	229	94.5	189	90.2	551	0	0	0	0
10-December-2023	23:53	2287	125.1	226	99.5	196	89.9	742	0	0	0	0
11-December-2023	0:13	1701	130.5	323	105.3	276	98.4	940	0	0	0	0

11-December-2023	0:33	1614	132.8	302	107.2	263	101.0	308	0	0	0	0
11-December-2023	0:53	1572	135.1	296	108.5	257	99.6	526	0	0	0	0
11-December-2023	1:13	1547	137.1	297	109.4	256	100.3	720	52.67	3	0	5397972
11-December-2023	1:33	1525	1407.0	294	110.7	254	103.2	886	52.19	3	0	5389022.54
11-December-2023	1:53	1509	141.9	291	112.4	252	105.1	586	52.42	3	0	5388817.86
11-December-2023	2:13	1496	143.5	292	113.3	250	105.5	283	51.21	3	0	5380458.17
11-December-2023	2:33	1484	144.2	285	105.0	239	100.3	371	48.66	3	0	5390743.18
11-December-	2:53	1476	145.7	284	109.4	240	98.4	463	48.72	3	0	5384273.82

2023												
11-December-2023	3:13	1466	146.9	288	115.7	250	109.1	542	50.95	3	0	5398203.45
11-December-2023	3:33	1458	147.8	286	116.4	248	115.5	629	49.84	3	0	5310633.81
11-December-2023	3:53	1450	148.9	284	117.1	247	117.1	679	51.59	3	0	5355704.16
11-December-2023	4:13	1443	149.3	284	117.7	246	118.7	737	51.03	3	0	5408629.55
11-December-2023	4:33	1436	150.7	282	118.4	245	117.9	789	51.41	3	0	5394269.75
11-December-2023	4:53	1431	151.0	280	118.8	244	119.4	837	50.84	3	0	5387890.57
11-December-2023	5:13	1426	152.0	282	119.3	245	119.3	880	51.86	3	0	5391341.64

11-December-2023	5:33	1420	153.2	281	119.9	244	120.1	923	51.56	3	0	5405362.96
11-December-2023	5:53	1415	153.7	279	120.4	242	121.2	969	51.44	3	0	5408755.64
11-December-2023	6:13	1410	154.2	280	120.7	242	122.2	989	51.45	3	0	5345646.65
11-December-2023	6:33	1405	154.3	276	121.1	242	120.8	234	50.68	3	0	5355373.2
11-December-2023	6:53	1401	154.9	275	121.4	240	122.3	260	51.38	3	0	5345237.43
11-December-2023	7:13	1398	156.0	274	121.6	239	123.0	287	51.21	3	0	5385891.67
11-December-2023	7:33	1394	156.9	274	122.1	238	124.2	312	51.31	3	0	5362087.34
11-December-	7:53	1389	156.3	273	122.4	238	123.2	336	51.11	3	0	5403008.15

2023												
11-December-2023	8:13	1387	157.6	274	122.7	238	124.8	358	51.27	3	0	5410487.1
11-December-2023	8:33	1382	158.2	272	123.0	238	125.6	379	51.01	3	0	5376158.21
11-December-2023	8:53	1379	159.2	271	123.3	237	128.4	400	50.88	3	0	5388772.94
11-December-2023	9:18	1375	160.3	270	123.5	236	129.5	422	50.93	3	0	5364655.01
11-December-2023	9:38	1373	160.4	271	123.9	236	131.0	442	50.91	3	0	5348155.56
11-December-2023	9:58	1369	161.5	269	124.2	236	132.1	462	50.57	3	0	5380458.17
11-December-2023	10:05	1367	162	270	124	235	132	470	51	3	5087	5380449

11-December-2023	10:18	1366	162.1	269	124.2	235	132.5	470	51.16	3	0	5364655.01
11-December-2023	10:38	1363	162.8	270	124.5	236	133.6	491	51.34	3	0	5418837.72
11-December-2023	10:58	1361	162.9	268	124.6	234	134.0	502	50.79	3	0	5368455.73
11-December-2023	11:18	1359	163.3	270	124.9	235	134.9	502	51.35	3	0	5401714.72
11-December-2023	11:39	1355	164.4	268	125.1	235	135	506	51	3	5041	5320508
11-December-2023	11:58	1353	164.6	268	125.2	234	134.1	516	50.86	3	0	5367692.88
11-December-2023	12:18	1350	164.6	267	125.4	233	134.7	526	50.77	3	0	5349153.33
11-December-	12:38	2	153.6	0	121.6	0	125.3	535	0	0	0	0

2023												
11-December-2023	12:58	3	120.0	0	109.7	0	110.4	344	0	0	0	0
11-December-2023	13:18	2	116.4	0	106.5	0	101.5	143	0	0	0	0
11-December-2023	13:38	1	113.6	0	105.2	0	96.4	27	0	0	0	0
11-December-2023	13:58	2	111.0	0	103.6	0	92.8	4	0	0	0	0
11-December-2023	14:18	0	108.8	0	102.4	0	901.0	-4	0	0	0	0
11-December-2023	14:38	0	105.4	0	100.8	0	88.2	-6	0	0	0	0
11-December-2023	14:58	0	103.6	0	97.9	0	87.0	-9	0	0	0	0

11-December-2023	15:18	0	101.5	0	95.5	0	85.8	-8	0	0	0	0
11-December-2023	15:38	0	97.3	0	92.6	0	84.7	-10	0	0	0	0
11-December-2023	15:58	0	93.9	0	89.4	0	83.3	-10	0	0	0	0
11-December-2023	16:18	0	90.0	0	86.8	0	81.6	-10	0	0	0	0
11-December-2023	16:38	2973	89.9	0	85.0	0	80.2	-11	0	0	0	0
11-December-2023	16:58	6398	90.9	0	83.2	0	78.8	-11	0	0	0	0
11-December-2023	17:18	10	83.3	0	81.7	0	77.0	-12	0	0	0	0
11-December-	17:38	2909	142.6	1	108.8	0	105.0	-11	0	0	0	0

2023												
11-December-2023	17:58	2569	115.3	0	101.1	0	91.2	-10	0	0	0	0
11-December-2023	18:18	2414	101.0	0	93.9	0	83.3	-11	0	0	0	0
11-December-2023	18:38	2327	95.7	0	88.9	0	78.2	-11	0	0	0	0
11-December-2023	18:58	2276	92.8	0	85.1	0	75.6	-11	0	0	0	0
11-December-2023	19:18	3380	134.9	298	110.3	296	102.8	-10	0	0	0	0
11-December-2023	19:38	46	104.5	0	94.7	0	81.5	13	0	0	0	0
11-December-2023	19:58	4165	94.0	0	89.1	0	77.5	1	0	0	0	0

11-December-2023	20:18	3643	132.0	166	101.3	149	94.2	-4	0	0	0	0
11-December-2023	20:38	3344	134.2	162	104.0	147	100.4	-6	0	0	0	0
11-December-2023	20:58	3268	134.1	156	103.6	141	101.3	-7	0	0	0	0
11-December-2023	21:18	3260	133.7	143	103.3	129	100.6	-8	0	0	0	0
11-December-2023	21:38	3262	133.8	139	103.3	124	1004.0	-4	0	0	0	0
11-December-2023	21:58	3223	135.0	142	104.0	129	100.9	-5	0	0	0	0
11-December-2023	22:18	3032	135.9	180	106.0	164	102.2	-4	0	0	0	0
11-December-	22:38	2967	136.2	177	106.2	160	101.1	-2	0	0	0	0

2023												
11-December-2023	22:58	2932	138.0	175	106.7	159	103.4	-2	0	0	0	0
11-December-2023	23:18	2908	137.4	175	106.8	158	102.5	-2	0	0	0	0
11-December-2023	23:38	28905	137.6	175	107.0	157	103.3	-1	0	0	0	0
11-December-2023	23:58	2876	137.5	172	107.2	156	101.8	1	0	0	0	0
12-December-2023	0:18	2865	137.8	174	107.3	156	101.8	3	0	0	0	0
12-December-2023	0:47	2852	138.2	173	107.6	156	104.4	6	0	0	0	0
12-December-2023	1:13	2841	138.8	176	108.0	156	107.1	5	0	0	0	0

12-December-2023	1:33	2835	138.8	254	110.2	243	109.4	13	0	0	0	0
12-December-2023	1:53	2829	139.7	239	110.2	225	108.0	3	51.79	2.125	0	2638217.52
12-December-2023	2:13	2824	139.7	238	110.4	225	107.8	2	51.64	2.125	0	2635122.12
12-December-2023	2:33	2820	140.1	239	110.5	224	108.0	2	51.74	2.125	0	2634991.53
12-December-2023	2:53	2815	1407.0	239	110.5	225	107.2	1	51.79	2.125	0	2640140.89
12-December-2023	3:13	2812	140.3	239	110.7	225	107.9	1	51.77	2.125	0	2637923.16
12-December-2023	3:33	2808	140.6	238	110.9	223	110.8	1	51.9	2.125	0	2626532.1
12-December-	3:53	2804	140.6	238	111.0	223	110.0	0	51.97	2.125	0	2628442.02

2023												
12-December-2023	4:13	2802	141.2	238	111.1	224	109.6	1	51.97	2.125	0	2633190.12
12-December-2023	4:33	2799	141.5	238	111.2	223	111.7	2	52.01	2.125	0	2627525.4
12-December-2023	4:53	2795	141.3	237	111.2	223	1109.0	12	51.95	2.125	0	2627561.07
12-December-2023	5:13	2793	141.3	237	111.2	222	108.7	8	51.63	2.125	0	2618541.64
12-December-2023	5:33	2791	141.8	238	111.4	223	111.1	6	51.91	2.125	0	2626091.21
12-December-2023	5:53	2788	141.3	238	111.5	223	110.7	3	51.95	2.125	0	2626257.17
12-December-2023	6:13	2787	141.3	237	111.6	223	110.6	2	51.92	2.125	0	2627566.65

12-December-2023	6:33	2785	141.5	237	111.7	223	112.0	0	51.91	2.125	0	2624071.22
12-December-2023	6:53	2784	141.8	237	111.7	222	111.8	2	51.88	2.125	0	2617674.39
12-December-2023	7:06	2782	142	237	111	222	110	2	52	2.125	2532	26510033
12-December-2023	7:13	2781	142.3	237	111.8	221	109.5	2	51.7	2.125	0	2614415.26
12-December-2023	7:33	2780	140.7	237	111.7	222	107.8	1	51.79	2.125	0	2622549.93
12-December-2023	7:51	2778	142.0	237	112.0	223	110	0	52	2.125	2532	2660012
12-December-2023	8:13	2777	143.1	236	112.1	223	111.0	2	51.62	2.125	0	2618928.76
12-December-	8:33	2771	143.9	236	111.7	222	112.1	2	51.41	2.125	0	2603233.88

2023												
12-December-2023	8:53	2772	143.3	236	112.3	222	110.7	0	51.29	2.125	0	2605164.5
12-December-2023	9:13	2770	141.4	236	112.1	222	111.9	2	51.38	2.125	0	2604888.58
12-December-2023	9:33	2770	141.4	236	112.2	222	109.1	0	51.39	2.125	0	2609614.19
12-December-2023	9:53	2767	143.2	236	112.4	222	110.9	1	51.38	2.125	0	2605333.54
12-December-2023	10:13	2766	144.9	237	112.7	223	113.2	0	51.62	2.125	0	2611892.8
12-December-2023	10:33	2763	144.5	235	112.7	222	113.0	-1	51.62	2.125	0	2608537.17
12-December-2023	10:53	2766	145.7	237	113.0	223	115.2	0	51.71	2.125	0	2644900.78

12-December-2023	11:13	2762	144.6	233	112.7	219	114.4	-1	53.34	2.125	0	2663469.21
12-December-2023	11:33	2761	145.9	231	112.9	217	114.8	0	53.53	2.125	0	2657074.03
12-December-2023	11:53	2760	143.2	231	112.6	216	114.8	0	53.58	2.125	0	2654062.44
12-December-2023	12:14	2758	143.2	231	112.6	216	115	-1	53	2.125	2532	2607290
12-December-2023	12:33	2757	146.3	231	113.0	216	114.6	0	53.64	2.125	0	2622556.83
12-December-2023	12:53	2756	146.5	231	113.2	216	115.2	0	53.54	2.125	0	2616818.36
12-December-2023	13:13	2753	147.8	231	113.5	216	116.5	0	53.63	2.125	0	2618157.16
12-December-	13:33	2753	148.0	231	113.5	216	116.3	0	53.44	2.125	0	2613987.94

2023												
12-December-2023	13:53	2752	147.8	231	113.5	216	116.0	2	53.72	2.125	0	2619339.81
12-December-2023	14:13	2749	147.2	232	113.6	217	116.0	1	0	0	0	0
12-December-2023	14:33	2747	146.2	230	113.4	216	112.7	2	0	0	0	0
12-December-2023	14:53	2745	147.1	231	113.6	216	114.1	2	0	0	0	0
12-December-2023	15:13	2745	146.0	237	113.7	222	114.7	1	0	0	0	0
12-December-2023	15:33	2745	144.6	237	113.6	224	113.3	2	0	0	0	0
12-December-2023	15:53	2744	144.1	239	113.5	225	112.9	2	0	0	0	0

12-December-2023	16:13	2743	144.5	248	114.0	235	112.8	3	0	0	0	0
12-December-2023	16:33	2743	145.0	252	114.1	239	114.2	2	0	0	0	0
12-December-2023	16:53	5	115.4	0	102.1	0	96.2	1	0	0	0	0
12-December-2023	17:13	7	97.3	0	94.5	0	88.9	-3	0	0	0	0
12-December-2023	17:33	11	86.9	0	89.6	0	84.1	-6	0	0	0	0
12-December-2023	17:53	14	80.8	0	85.4	0	80.6	-8	0	0	0	0
12-December-2023	18:13	14	76.1	0	82.3	0	77.2	-9	0	0	0	0
12-December-	18:33	14	73.1	0	80.0	0	74.7	-10	0	0	0	0

2023												
12-December-2023	18:53	14	71.0	0	78.1	0	73.1	-8	0	0	0	0
12-December-2023	19:13	13	69.4	0	76.7	0	71.8	-10	0	0	0	0
12-December-2023	19:33	13	68.0	0	75.4	0	70.6	-8	0	0	0	0
12-December-2023	19:53	13	66.7	0	74.4	0	69.5	-10	0	0	0	0
12-December-2023	20:13	13	65.4	0	73.3	0	68.1	-9	0	0	0	0
12-December-2023	20:33	13	64.6	0	72.4	0	67.3	-9	0	0	0	0
12-December-2023	20:53	12	64.0	0	71.8	0	66.5	-10	0	0	0	0

12-December-2023	21:13	14	66.0	0	71.3	0	65.9	-11	0	0	0	0
12-December-2023	21:33	13	65.4	0	70.7	0	65.0	-10	0	0	0	0
12-December-2023	21:53	14	65.0	0	70.3	0	64.5	-10	0	0	0	0
12-December-2023	22:13	13	64.8	0	70.1	0	64.3	-9	0	0	0	0
12-December-2023	22:33	13	64.6	0	69.9	0	64.0	-7	0	0	0	0
12-December-2023	22:53	13	64.4	0	69.7	0	63.9	-2	0	0	0	0
12-December-2023	23:13	13	64.2	0	69.5	0	63.7	-7	0	0	0	0
12-December-	23:33	13	64.0	0	69.3	0	63.3	-10	0	0	0	0

2023												
12-December-2023	23:53	13	63.8	0	69.1	0	62.7	-9	0	0	0	0
13-December-2023	0:13	13	63.6	0	68.9	0	62.4	-8	0	0	0	0
13-December-2023	0:33	13	63.5	0	68.8	0	62.2	-10	0	0	0	0
13-December-2023	0:53	285	63.3	0	68.6	0	61.8	-10	0	0	0	0
13-December-2023	1:13	17	63.2	0	68.5	0	61.6	-10	0	0	0	0
13-December-2023	1:33	1476	63.2	0	68.5	0	61.8	-10	0	0	0	0
13-December-2023	1:53	46	63.2	0	68.5	0	61.5	-9	0	0	0	0

13-December-2023	2:13	6051	63.2	0	68.5	0	61.4	-10	0	0	0	0
13-December-2023	2:33	16	63.2	0	68.5	0	61.3	-8	0	0	0	0
13-December-2023	2:53	17	63.2	0	68.5	0	61.5	-6	0	0	0	0
13-December-2023	3:13	17	63.3	0	68.6	0	61.5	-9	0	0	0	0
13-December-2023	3:33	17	63.3	0	68.6	0	61.5	-8	0	0	0	0
13-December-2023	3:53	6289	63.3	0	68.6	0	61.4	-10	0	0	0	0
13-December-2023	4:13	6322	63.3	0	68.6	0	61.4	-8	0	0	0	0
13-December-	4:33	5864	64.1	0	69.4	0	61.4	-6	0	0	0	0

2023												
13-December-2023	4:53	1570	66.2	0	71.5	0	61.6	-6	0	0	0	0
13-December-2023	5:13	2356	65.6	0	70.9	0	62.0	-6	0	0	0	0
13-December-2023	5:33	53	64.8	0	70.1	0	62.0	-5	0	0	0	0
13-December-2023	5:53	4013	86.6	192	91.9	164	69.3	-5	0	0	0	0
13-December-2023	6:13	3420	91.4	171	96.7	147	86.8	-6	0	0	0	0
13-December-2023	6:33	3272	91.2	161	96.5	136	86.2	-5	0	0	0	0
13-December-2023	6:53	3116	91.0	152	95.4	136	86.5	-1	0	0	0	0

13-December-2023	7:13	2995	88.2	152	94.2	136	85.7	-8	0	0	0	0
13-December-2023	7:39	2985	81.5	152	93.0	136	84.9	-7	0	0	0	0
13-December-2023	8:07	3206	81.4	148	91.8	132	84.1	-9	0	0	0	0
13-December-2023	8:27	3235	81.6	148	90.6	132	83.3	-7	0	0	0	0
13-December-2023	8:47	3036	81.8	134	89.4	118	82.5	-9	0	0	0	0
13-December-2023	9:07	2896	82.0	134	88.2	118	81.7	-7	0	0	0	0
13-December-2023	9:27	2874	82.2	134	87.0	118	80.9	-6	0	0	0	0
13-December-	9:47	2837	82.4	135	85.8	119	80.1	-6	0	0	0	0

2023												
13-December-2023	10:07	2837	82.6	141	84.6	125	79.3	-6	0	0	0	0
13-December-2023	10:27	3046	82.8	145	83.4	129	78.5	-6	0	0	0	0
13-December-2023	10:47	4267	83.0	146	82.2	130	77.7	-5	0	0	0	0
13-December-2023	11:07	4211	83.2	145	81.0	129	76.9	-4	0	0	0	0
13-December-2023	11:27	4231	83.4	145	79.8	129	76.1	-5	0	0	0	0
13-December-2023	11:47	4267	83.6	145	78.6	129	75.3	-3	0	0	0	0
13-December-2023	12:07	4251	83.8	146	77.4	130	74.5	-1	0	0	0	0

13-December-2023	12:27	4251	84.0	106	76.2	904	73.7	-2	0	0	0	0
13-December-2023	12:47	4211	84.2	61	75.0	45	72.9	-4	0	0	0	0
13-December-2023	13:07	4460	84.4	60	73.8	44	72.1	-4	0	0	0	0
13-December-2023	13:27	4251	84.6	61	72.6	45	71.3	-3	0	0	0	0
13-December-2023	13:47	4455	81.4	49	74.3	45	72.8	-3	0	0	0	0
13-December-2023	14:07	4411	80.6	84	73.7	79	72.2	-3	0	0	0	0
13-December-2023	14:27	3792	95.4	145	83.9	137	76.6	-4	0	0	0	0
13-December-	14:47	3988	92.7	14	79.9	10	75.6	-6	0	0	0	0

2023												
13-December-2023	15:07	4143	84.1	81	76.7	76	72.9	-6	0	0	0	0
13-December-2023	15:27	4234	79.1	99	74.9	97	71.9	-6	0	0	0	0
13-December-2023	15:47	4207	76.8	108	73.8	103	71.2	-8	0	0	0	0
13-December-2023	16:07	4124	75.7	108	72.8	104	70.4	-7	0	0	0	0
13-December-2023	16:27	4101	75.4	108	72.1	104	69.6	-8	0	0	0	0
13-December-2023	16:47	4101	75.9	106	71.7	102	69.1	-7	0	0	0	0
13-December-2023	17:07	4108	74.6	106	71.3	102	68.5	-5	0	0	0	0

13-December-2023	17:27	3424	71.8	0	70.7	45	67.7	-6	0	0	0	0
13-December-2023	17:47	4	68.4	3	72.6	6	66.4	-6	0	0	0	0
13-December-2023	18:13	4	67.9	2	72.8	0	65.4	-7	0	0	0	0
13-December-2023	18:39	4	66.3	2	71.8	0	64.1	-7	0	0	0	0
13-December-2023	18:59	3	65.3	1	71.3	0	63.3	-6	0	0	0	0
13-December-2023	19:19	4	64.5	2	70.9	0	62.7	-5	0	0	0	0
13-December-2023	19:39	4	63.9	2	70.6	0	62.4	-7	0	0	0	0
13-December-	19:59	3	63.5	2	70.3	0	62.2	-6	0	0	0	0

2023												
13-December-2023	20:19	3	62.7	1	70.0	0	61.6	-6	0	0	0	0
13-December-2023	20:39	3	62.4	2	69.8	0	61.5	-6	0	0	0	0
13-December-2023	20:59	4	62.1	1	69.6	0	61.4	-5	0	0	0	0
13-December-2023	21:19	660	61.9	1	69.4	0	61.2	-4	0	0	0	0
13-December-2023	21:39	6714	61.9	2	69.1	0	61.0	-2	0	0	0	0
13-December-2023	21:59	4385	61.3	2	68.9	0	60.8	-2	0	0	0	0
13-December-2023	22:19	4118	61.3	2	68.9	0	60.8	-2	0	0	0	0

13-December-2023	22:39	5	60.6	2	68.8	0	60.7	-1	0	0	0	0
13-December-2023	22:59	8	60.6	2	68.7	0	60.4	-2	0	0	0	0
13-December-2023	23:19	9	60.4	2	68.5	0	60.0	-2	0	0	0	0
13-December-2023	23:39	11	606.0	2	68.2	0	59.5	-2	0	0	0	0
13-December-2023	23:59	11	59.5	2	67.9	0	58.9	-1	0	0	0	0
14-December-2023	0:19	188	62.0	2	69.4	0	58.4	-2	0	0	0	0
14-December-2023	0:39	162	60.8	2	68.8	0	57.9	-2	0	0	0	0
14-December-	0:59	152	59.6	2	68.2	0	57.5	-1	0	0	0	0

2023												
14-December-2023	1:19	4156	58.9	3	67.7	0	57.1	-2	0	0	0	0
14-December-2023	1:39	4141	68.8	3	77.2	0	56.8	-1	0	0	0	0
14-December-2023	1:59	312	62.4	3	70.4	0	56.5	1	0	0	0	0
14-December-2023	2:19	116	60.9	3	69.6	0	56.1	2	0	0	0	0
14-December-2023	2:39	99	59.7	3	68.8	0	55.8	1	0	0	0	0
14-December-2023	2:59	1	58.7	2	68.2	0	55.4	1	0	0	0	0
14-December-2023	3:19	1	58.0	2	67.5	0	55.0	0	0	0	0	0

14-December-2023	3:39	45	57.1	3	67.0	0	54.6	0	0	0	0	0
14-December-2023	3:59	9	57.5	3	66.3	0	54.2	0	0	0	0	0
14-December-2023	4:19	2	57.0	2	65.9	0	53.8	18	0	0	0	0
14-December-2023	4:39	4	56.5	2	65.5	0	53.6	570	0	0	0	0
14-December-2023	4:59	101	55.9	3	65.3	0	53.2	4667	0	0	0	0
14-December-2023	5:19	623	55.2	2	65.1	0	53.1	714	0	0	0	0
14-December-2023	5:39	2726	55.3	3	64.9	0	52.8	2891	0	0	0	0
14-December-	5:59	518	67.7	3	74.7	0	52.6	382	0	0	0	0

2023												
14-December-2023	6:19	4	60.7	3	70.5	0	52.3	272	0	0	0	0
14-December-2023	6:39	5	58.9	3	69.1	0	52.0	4577	0	0	0	0
14-December-2023	6:59	5	57.9	3	66.7	0	51.8	461	0	0	0	0
14-December-2023	7:19	5	56.9	3	66.2	0	51.8	4590	0	0	0	0
14-December-2023	7:39	5	60.4	3	68.5	0	52.1	0	0	0	0	0
14-December-2023	7:59	10	61.6	6	69.0	0	52.5	0	0	0	0	0

Table 11-8: AZNS-040 Main Results

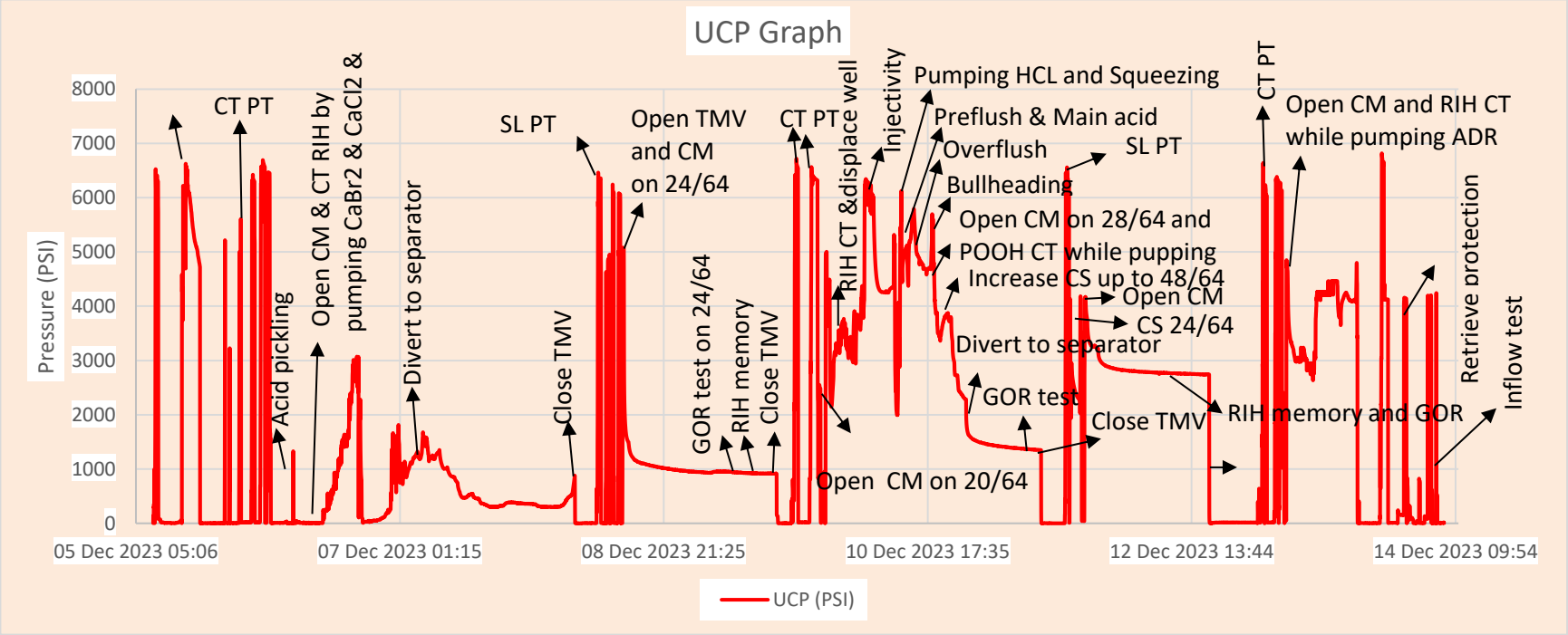


Figure 11-2: AZNS-40 Upstream Pressure Graph

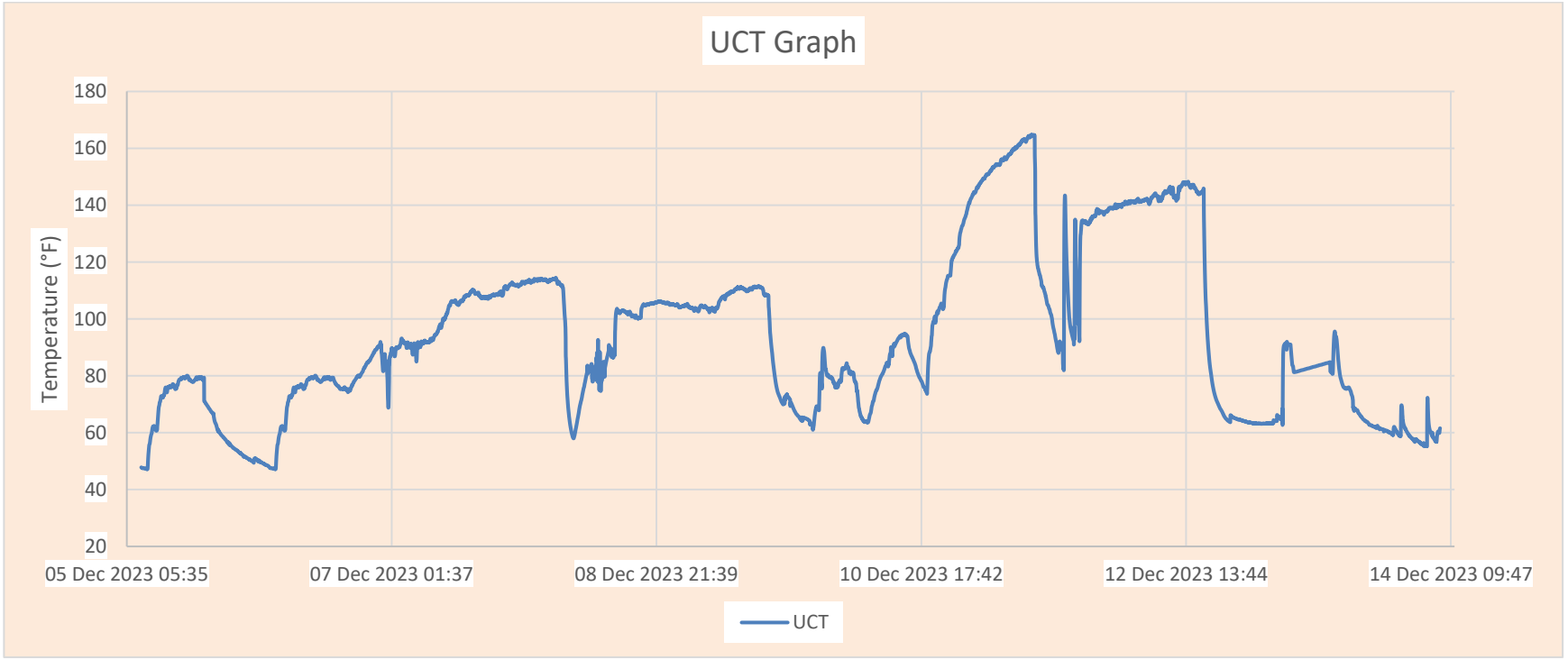


Figure 11-3: AZNS-40 Upstream Temperature Graph

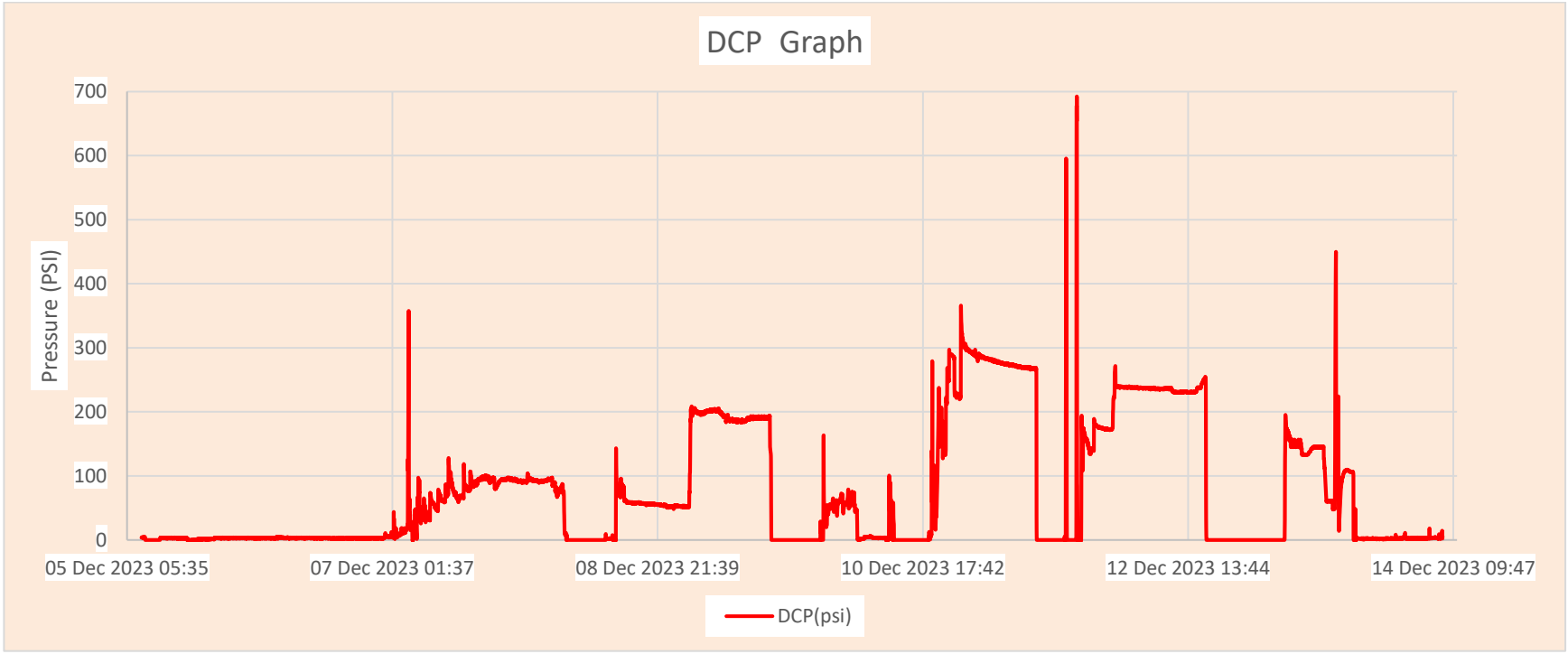


Figure 11-4: AZNS-40 Downsteam Pressure Graph

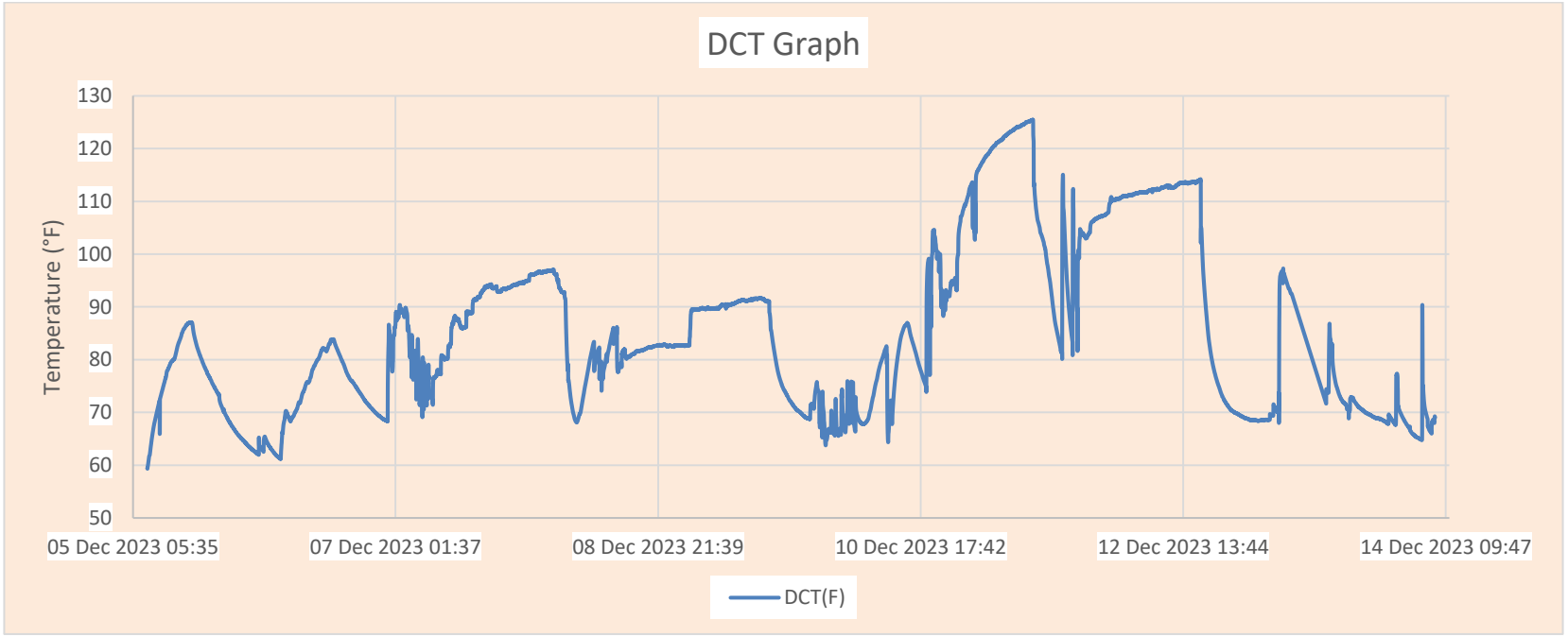


Figure 11-5: AZNS-40 Downsteam Temperature Graph

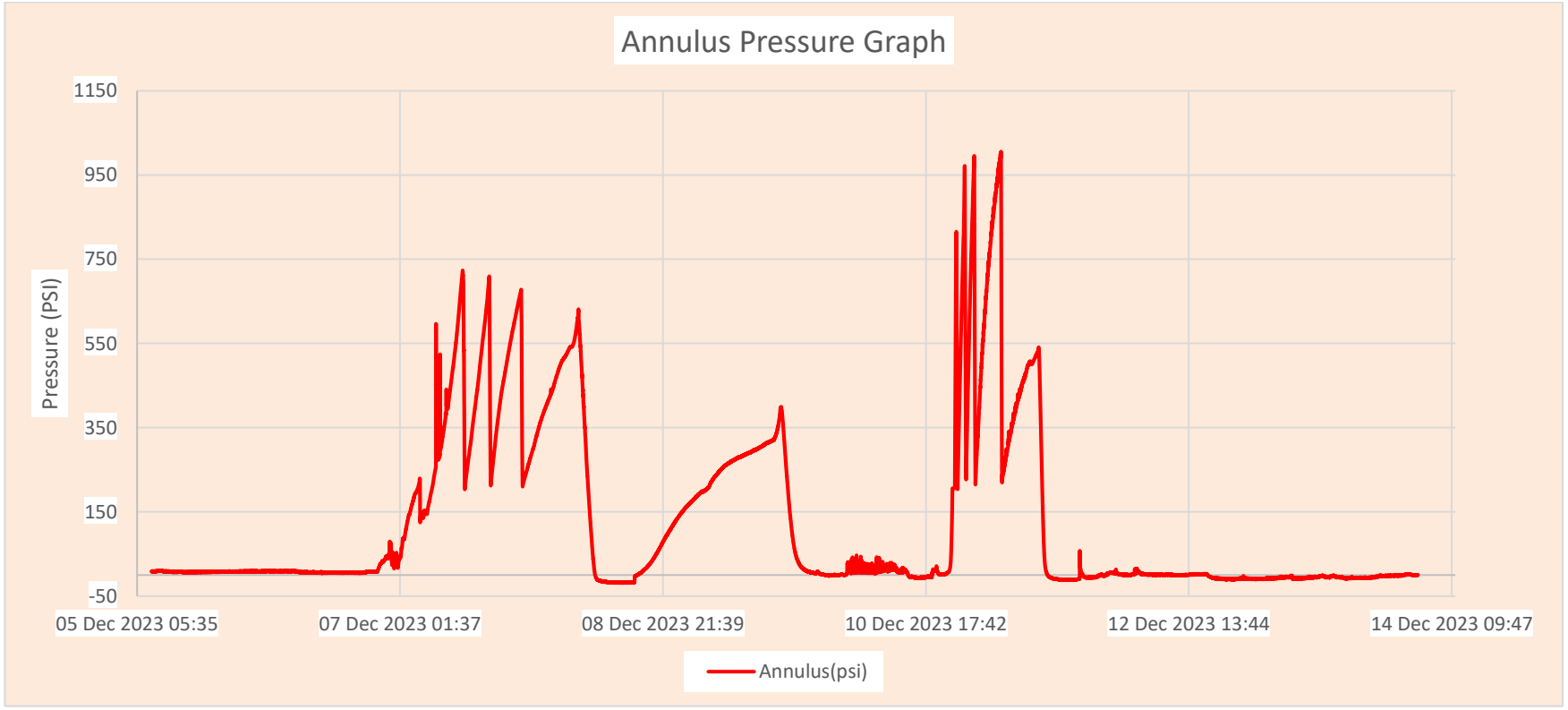


Figure 11-6: AZNS-40 Annulus Pressure Graph

11.4.4.3 Summary of Pumping Data

ITEM	Pumped material	Planned	Total used	Unit
1	Diesel	960	538	BBL
2	HCL 32%	60	59.2	BBL
3	Fresh Water	350	537	BBL
4	Mutual Solvent	4	4	Drum
5	Iron Reducing Agent	3	3	Drum
6	Iron Control Agent	1	1	Drum
7	Corrosion Inhibitor Aid	3	3	Drum
8	Anti-sludge Agent	3	3	Drum
9	Non-Emulsifier Agent	3	3	Drum
10	Corrosion Inhibitor	4	4	Drum
11	H2S Scavenger	3	3	Drum
12	Asphaltene Solvent	50	20	BBL
13	Caustic Soda	8	6	Sacks
14	Liquid nitrogen	7000	…….	USG
15	Anti-foam	55	…….	USG

Table 11-9: AZNS-040 Total Consumable Planned & Used

11.4.4.4 Summary of Pumping Operation Result

Summary Of Pumping Operation Results					
Operation	**Start Date and time DD/MM/YYYY, HH:MM**	**End Date and time DD/MM/YYYY, HH:MM**	**Volume**		**Unit**
			USED	**PLANNED**	
Fill Up Surface Lines with Water	05-December-2023, 04:15	06-December-2023, 03:40	90	90	BBL
acid pickling	06-December-2023, 06:20	06-December-2023, 06:25	5	5	BBL
Displace acid pickling by water	06-December-2023, 06:25	06-December-2023, 06:35	7	7	BBL
Displace acid pickling by CaBr2	06-December-2023, 06:35	06-December-2023, 07:33	26	26	BBL
CT RIH Simultaneously Pumping CaBr2	06-December-2023, 07:50	06-December-2023, 12:32	40	40	BBL
pumping CaCl2 as spacer	06-December-2023, 12:43	06-December-2023, 12:45	6.3	6.3	BBL
Displace wellbore by pumping brine	06-December-2023, 12:45	06-December-2023, 18:43	206	206	BBL
CT POOH Simultaneously Pumping brine	07-December-2023, 00:25	07-December-2023, 01:00	23	23	BBL
Pumping return diesel to CT displacement	09-December-2023, 23:45	10-December-2023, 00:20	10	10	BBL
CT RIH Simultaneously pumping return diesel	10-December-2023, 00:45	10-December-2023, 05:26	99	99	BBL
Displace wellbore by pumping return diesel	10-December-2023, 05:39	10-December-2023, 06:55	50	50	BBL
Bullheading with diesel	10-December-2023, 06:59	10-December-2023, 08:32	20	20	BBL

pumping HCL 20%	10-December-2023, 11:46	10-December-2023, 12:12	9	9	BBL
CT RIH Simultaneously pumping diesel	10-December-2023, 12:18	10-December-2023, 12:56	26	26	BBL
Squeezing the HCL 20% with bullheading operation	10-December-2023, 13:02	10-December-2023, 13:17	20	20	BBL
Pumping pre flush	10-December-2023, 13:26	10-December-2023, 14:16	23	23	BBL
Pumping main acid (HCL 15 %)	10-December-2023, 14:27	10-December-2023, 16:56	119	119	BBL
Pumping over flush	10-December-2023, 16:57	10-December-2023, 17:2٦	23	23	BBL
Displace CT with diesel	10-December-2023, 17:2٩	10-December-2023, 18:14	26	26	BBL
Bullheading with diesel	10-December-2023, 18:14	10-December-2023, 18:29	31	31	BBL
CT POOH Simultaneously pumping diesel	10-December-2023, 18:٤١	10-December-2023, 21:3٢	77	77	BBL
Pumping caustic	11-December-2023, 01:00	11-December-2023, 03:00	100	100	BBL
CT RIH Simultaneously pumping diesel	13-December-2023, 05:50	11-December-2023, 10:00	45	…	BBL
CT at top perforation and Pumping ADR	13-December-2023, 10:00	11-December-2023, 10:28	20	…	BBL
CT at 4039m and pumping diesel	13-December-2023, 10:28	11-December-2023, 11:24	44	….	BBL
Bullheading with diesel	13-December-2023, 11:29	11-December-2023, 11:39	20	…	BBL
CT POOH Simultaneously Pumping diesel	13-December-2023, 11:45	11-December-2023, 17:13	210	….	BBL
Bullheading with diesel	13-December-2023, 17:20	11-December-2023, 17:29	20	….	BBL

Table 11-10: AZNS-040 Pumping Operation Result

11.4.4.5 Pumping Operation Results & Plots

Pre-Acid Bullhead Operation Report				
Date (DD-MM-YYYY)	Time (HH:MM)	Pumping pressure (PSI)	Pump Rate (BPM)	Volume total (BBL)
10-December-2023	7:01	3836	0	0.03
10-December-2023	7:03	3855	0	0.03
10-December-2023	7:04	4461	0.09	0.06
10-December-2023	7:06	5358	0.67	1.62
10-December-2023	7:08	6526	0.59	3.56
10-December-2023	7:09	6650	0	4.1
10-December-2023	7:11	6604	0.43	4.22
10-December-2023	7:13	6674	0	4.66
10-December-2023	7:14	6736	0	4.79
10-December-2023	7:16	6775	0	5.01
10-December-2023	7:18	6538	0	5.01
10-December-2023	7:19	6775	0.43	5.43
10-December-2023	7:21	6690	0	5.61
10-December-2023	7:23	6488	0	5.61
10-December-2023	7:24	6306	0	5.61
10-December-2023	7:26	6570	0.43	6.01
10-December-2023	7:28	6868	0	6.58
10-December-2023	7:29	6616	0	6.58

10-December-2023	7:31	6449	0	6.58
10-December-2023	7:33	6698	0	6.91
10-December-2023	7:34	6500	0	6.91
10-December-2023	7:36	6398	0	6.94
10-December-2023	7:38	6741	0.23	7.32
10-December-2023	7:39	6570	0	7.38
10-December-2023	7:41	6395	0	7.38
10-December-2023	7:43	6418	0	7.41
10-December-2023	7:44	6752	0.34	7.83
10-December-2023	7:46	6635	0	7.96
10-December-2023	7:48	6453	0	7.96
10-December-2023	7:49	6430	0.4	8.02
10-December-2023	7:51	6736	0	8.29
10-December-2023	7:53	6519	0	8.29
10-December-2023	7:54	6321	0	8.29
10-December-2023	7:56	6558	0.4	8.71
10-December-2023	7:58	6690	0	9.07
10-December-2023	7:59	6445	0	9.07
10-December-2023	8:01	6290	0	9.07
10-December-2023	8:03	6570	0.17	9.4
10-December-2023	8:04	6592	0	9.54
10-December-2023	8:06	6367	0	9.54

10-December-2023	8:08	6166	0	9.54
10-December-2023	8:09	6418	0.25	9.91
10-December-2023	8:11	6720	0.1	10.29
10-December-2023	8:13	6550	0	10.35
10-December-2023	8:14	6333	0	10.35
10-December-2023	8:16	6142	0	10.35
10-December-2023	8:18	6422	0.03	10.88
10-December-2023	8:19	6348	0	11.03
10-December-2023	8:21	6158	0	11.03
10-December-2023	8:23	6360	0.48	11.43
10-December-2023	8:24	6500	0	11.92
10-December-2023	8:26	6278	0	11.92
10-December-2023	8:28	6483	0	12.4
10-December-2023	8:29	6278	0	12.4
10-December-2023	8:31	6328	0.53	12.66

- The counter could not record the pumping rate for low volume of pumping, in such cases the pumped volume of fluid is considered from the differential the pumping fluid in its storage tank.

Table 11-11: AZNS-040 Pre-Acid Bullhead Operation Report

Pumping Pre-Flush Operation Report				
Date (DD-MM-YYYY)	Time (HH:MM)	Pumping pressure (PSI)	Pump Rate (BPM)	Volume total (BBL)
10-December-2023	13:26	5716	0.47	0.30
10-December-2023	13:27	5750	0.47	0.78
10-December-2023	13:28	5828	0.46	1.26
10-December-2023	13:29	5909	0.47	1.73
10-December-2023	13:30	5925	0.47	2.18
10-December-2023	13:31	5979	0.47	2.65
10-December-2023	13:32	5929	0.47	3.12
10-December-2023	13:33	5999	0.46	3.58
10-December-2023	13:34	6030	0.47	4.05
10-December-2023	13:35	5983	0.46	4.52
10-December-2023	13:36	5968	0.46	4.99
10-December-2023	13:37	6023	0.47	5.46
10-December-2023	13:38	6053	0.47	5.93
10-December-2023	13:39	6023	0.47	6.39
10-December-2023	13:40	6076	0.47	6.85
10-December-2023	13:41	6060	0.46	7.31
10-December-2023	13:42	5972	0.45	7.77
10-December-2023	13:43	6011	0.46	8.23
10-December-2023	13:44	5983	0.47	8.70

10-December-2023	13:45	6018	0.47	9.16
10-December-2023	13:46	6002	0.46	9.63
10-December-2023	13:47	5936	0.47	10.09
10-December-2023	13:48	6023	0.47	10.56
10-December-2023	13:49	5999	0.45	11.01
10-December-2023	13:50	6057	0.46	11.49
10-December-2023	13:51	6065	0.47	11.95
10-December-2023	13:52	6065	0.46	12.41
10-December-2023	13:53	6038	0.46	12.87
10-December-2023	13:54	6060	0.47	13.33
10-December-2023	13:55	6081	0.46	13.79
10-December-2023	13:56	6030	0.46	14.25
10-December-2023	13:57	6057	0.46	14.71
10-December-2023	13:58	6081	0.46	15.17
10-December-2023	13:59	6123	0.46	15.63
10-December-2023	14:00	6123	0.46	16.09
10-December-2023	14:01	6072	0.46	16.55
10-December-2023	14:02	6118	0.45	17.01
10-December-2023	14:03	6081	0.46	17.47
10-December-2023	14:04	6108	0.46	17.93
10-December-2023	14:05	6065	0.45	18.38
10-December-2023	14:06	6135	0.46	18.84

10-December-2023	14:07	6060	0.46	19.29
10-December-2023	14:08	6127	0.46	19.75
10-December-2023	14:09	6135	0.46	20.20
10-December-2023	14:10	6127	0.46	20.66
10-December-2023	14:11	6142	0.46	21.12
10-December-2023	14:12	6115	0.46	21.57
10-December-2023	14:13	6178	0.46	22.03
10-December-2023	14:14	6135	0.45	22.49
10-December-2023	14:15	6161	0.46	22.94

Table 11-12: AZNS-040 Pumping Pre-flush Operation Report

Pumping Main Acid Operation Report				
Date (DD-MM-YYYY)	Time (HH:MM)	Pumping pressure (PSI)	Pump Rate (BPM)	Volume total (BBL)
10-December-2023	14:28	5758	0.51	0.49
10-December-2023	14:30	6084	0.51	1.78
10-December-2023	14:33	6290	0.51	3.05
10-December-2023	14:35	6325	0.51	4.31
10-December-2023	14:38	6445	0.50	5.58
10-December-2023	14:40	6275	0.44	6.77
10-December-2023	14:43	6325	0.46	7.91
10-December-2023	14:45	6309	0.46	9.06
10-December-2023	14:48	6255	0.46	10.22
10-December-2023	14:50	6243	0.49	11.39
10-December-2023	14:53	6352	0.54	12.68
10-December-2023	14:55	6200	0.53	14.02
10-December-2023	14:58	6239	0.57	15.38
10-December-2023	15:00	6045	0.56	16.81
10-December-2023	15:03	6318	0.67	18.47
10-December-2023	15:05	6270	0.68	20.17
10-December-2023	15:08	6336	0.68	21.84
10-December-2023	15:10	6193	0.67	23.53
10-December-2023	15:13	5979	0.68	25.21
10-December-2023	15:15	6181	0.82	26.78
10-December-2023	15:18	6251	0.83	28.81
10-December-2023	15:20	6297	0.80	30.83
10-December-2023	15:23	6360	0.82	32.85
10-December-2023	15:25	6216	0.81	34.89
10-December-2023	15:28	6267	0.82	36.94
10-December-2023	15:30	6200	0.84	39.03
10-December-2023	15:33	6142	0.84	41.12
10-December-2023	15:35	5941	0.84	43.23
10-December-2023	15:38	5972	0.85	45.33
10-December-2023	15:40	6026	0.84	47.47
10-December-2023	15:43	5983	0.86	49.61
10-December-2023	15:45	5808	0.84	51.77
10-December-2023	15:48	5847	0.84	53.93
10-December-2023	15:50	5890	0.89	56.11
10-December-2023	15:53	5735	0.87	58.31
10-December-2023	15:55	6239	1.00	60.67
10-December-2023	15:58	6091	1.00	63.16

10-December-2023	16:00	6158	0.99	65.66
10-December-2023	16:03	6150	0.99	68.16
10-December-2023	16:05	6139	1.00	70.68
10-December-2023	16:08	6154	1.02	73.20
10-December-2023	16:10	6041	1.01	75.73
10-December-2023	16:13	5963	1.03	78.26
10-December-2023	16:15	6103	1.01	80.82
10-December-2023	16:18	3988	0.00	82.74
10-December-2023	16:20	5614	0.90	84.31
10-December-2023	16:23	5835	0.98	86.67
10-December-2023	16:25	5750	0.98	89.10
10-December-2023	16:28	5823	0.97	91.54
10-December-2023	16:30	6026	0.95	93.94
10-December-2023	16:33	6072	0.93	96.30
10-December-2023	16:35	5936	0.94	98.65
10-December-2023	16:38	5999	0.98	101.00
10-December-2023	16:40	5893	0.99	103.46
10-December-2023	16:43	6127	0.98	105.91
10-December-2023	16:45	6275	0.96	108.33
10-December-2023	16:48	6318	0.95	110.71
10-December-2023	16:50	6193	0.96	113.11
10-December-2023	16:53	5975	1.00	115.56
10-December-2023	16:55	3992	0.43	117.98

Table 11-13: AZNS-040 Pumping Main Acid Operation Report

Pumping Over Flush Operation Report				
Date (DD-MM-YYYY)	Time (HH:MM)	Pumping pressure (PSI)	Pump Rate (BPM)	Volume total (BBL)
10-December-2023	16:57	5560	0.72	0.45
10-December-2023	16:58	5765	0.68	1.03
10-December-2023	16:59	5883	0.74	1.64
10-December-2023	17:00	5929	0.74	2.26
10-December-2023	17:01	6014	0.79	2.90
10-December-2023	17:02	6130	0.81	3.58
10-December-2023	17:02	6294	0.82	4.24
10-December-2023	17:03	6313	0.80	4.91
10-December-2023	17:04	6255	0.79	5.57
10-December-2023	17:05	6371	0.80	6.25
10-December-2023	17:06	6367	0.79	6.90
10-December-2023	17:07	6355	0.79	7.56
10-December-2023	17:07	6313	0.81	8.22
10-December-2023	17:08	6336	0.79	8.88
10-December-2023	17:09	6309	0.80	9.54
10-December-2023	17:10	6425	0.80	10.21
10-December-2023	17:11	6422	0.79	10.87
10-December-2023	17:12	6395	0.80	11.55
10-December-2023	17:12	6395	0.80	12.21
10-December-2023	17:13	6270	0.81	12.87
10-December-2023	17:14	6348	0.80	13.54
10-December-2023	17:15	6275	0.80	14.21
10-December-2023	17:16	6224	0.80	14.88
10-December-2023	17:17	6258	0.81	15.55
10-December-2023	17:17	6282	0.80	16.22
10-December-2023	17:18	6263	0.80	16.88
10-December-2023	17:19	6379	0.80	17.55
10-December-2023	17:20	6278	0.80	18.22
10-December-2023	17:21	6425	0.80	18.89
10-December-2023	17:22	6430	0.79	19.54
10-December-2023	17:22	6367	0.78	20.21
10-December-2023	17:23	6464	0.79	20.86
10-December-2023	17:24	6414	0.79	21.52
10-December-2023	17:25	6495	0.77	22.18
10-December-2023	17:26	6464	0.77	22.82

Table 11-14: AZNS-040 Pumping Over Flush Operation Report

Post Acid Bullhead Operation Report				
Date (DD-MM-YYYY)	Time (HH:MM)	Pumping pressure (PSI)	Pump Rate (BPM)	Volume total (BBL)
10-December-2023	18:14	5117	0.97	0.97
10-December-2023	18:15	5145	1.44	2.04
10-December-2023	18:16	5191	1.45	3.27
10-December-2023	18:16	5257	1.45	4.46
10-December-2023	18:17	5400	1.92	5.88
10-December-2023	18:18	5315	2.05	7.59
10-December-2023	18:19	5412	2.05	9.28
10-December-2023	18:20	5528	2.08	11.01
10-December-2023	18:21	5630	2.37	12.92
10-December-2023	18:21	5653	2.52	14.97
10-December-2023	18:22	5661	2.60	17.03
10-December-2023	18:23	5859	3.10	19.58
10-December-2023	18:24	6018	3.22	22.15
10-December-2023	18:25	6088	3.50	24.96
10-December-2023	18:26	6248	3.82	28.06
10-December-2023	18:26	5777	0.22	30.73

Table 11-15: AZNS-040 Pumping Post-Acid Bullhead Operation Report

Pumping ADR at Perforation Operation Report				
Date (DD-MM-YYYY)	Time (HH:MM)	Pumping pressure (PSI)	Pump Rate (BPM)	Volume total (BBL)
13-December-2023	10:01	3502	0.72	0.33
13-December-2023	10:02	3669	0.77	0.94
13-December-2023	10:03	3724	0.67	1.53
13-December-2023	10:04	3623	0.77	2.12
13-December-2023	10:05	3758	0.69	2.70
13-December-2023	10:06	3797	0.69	3.28
13-December-2023	10:06	3922	0.67	3.83
13-December-2023	10:07	3918	0.72	4.39
13-December-2023	10:08	3949	0.69	4.93
13-December-2023	10:09	4053	0.69	5.47
13-December-2023	10:10	4186	0.69	6.02
13-December-2023	10:11	4558	0.80	6.69
13-December-2023	10:11	4504	0.80	7.35
13-December-2023	10:12	4597	0.80	8.03
13-December-2023	10:13	4566	0.75	8.69
13-December-2023	10:14	4512	0.83	9.36
13-December-2023	10:15	4555	0.88	10.03
13-December-2023	10:16	4570	0.88	10.70
13-December-2023	10:16	4659	0.77	11.37
13-December-2023	10:17	4519	0.81	12.03
13-December-2023	10:18	4558	0.80	12.70
13-December-2023	10:19	4632	0.80	13.38
13-December-2023	10:20	4585	0.75	14.05
13-December-2023	10:21	4643	0.80	14.71
13-December-2023	10:21	4515	0.83	15.39
13-December-2023	10:22	4519	0.88	16.05
13-December-2023	10:23	4512	0.83	16.74
13-December-2023	10:24	4582	0.80	17.41
13-December-2023	10:25	4582	0.80	18.08
13-December-2023	10:26	4616	0.80	18.76
13-December-2023	10:26	4454	0.83	19.44
13-December-2023	10:27	2656	0.00	19.98

Table 11-16: AZNS-040 Pumping ADR at Perforation Operation Report

Post ADR Bullheading Operation Report				
Date (DD-MM-YYYY)	**Time (HH:MM)**	**Pumping pressure (PSI)**	**Pump Rate (BPM)**	**Volume total (BBL)**
13-December-2023	11:30	4671	1.2	0.65
13-December-2023	11:30	4605	0.59	0.98
13-December-2023	11:31	4865	2.96	2.14
13-December-2023	11:32	4981	2.22	4.26
13-December-2023	11:33	4834	2.16	6.07
13-December-2023	11:34	5044	2.19	7.91
13-December-2023	11:35	5086	2.22	9.7
13-December-2023	11:35	5141	2.16	11.53
13-December-2023	11:36	5184	2.19	13.37
13-December-2023	11:37	5117	2.22	15.17
13-December-2023	11:38	5211	2.16	16.99
13-December-2023	11:39	5242	2.14	18.82

Table 11-17: AZNS-040 Post ADR Bullheading 1

Post ADR Bullheading Operation Report				
Date (DD-MM-YYYY)	**Time (HH:MM)**	**Pumping pressure (PSI)**	**Pump Rate (BPM)**	**Volume total (BBL)**
13-December-2023	17:21	4531	1.28	0.83
13-December-2023	17:22	4512	1.01	1.75
13-December-2023	17:23	4582	1.74	2.93
13-December-2023	17:24	4536	1.98	4.62
13-December-2023	17:25	4900	1.92	6.25
13-December-2023	17:25	5156	3.10	8.75
13-December-2023	17:26	4981	3.20	11.35
13-December-2023	17:27	5470	3.18	13.98
13-December-2023	17:28	4788	3.15	16.64
13-December-2023	17:29	5040	3.18	19.2

Table 11-18: AZNS-040 Post ADR Bullheading 2

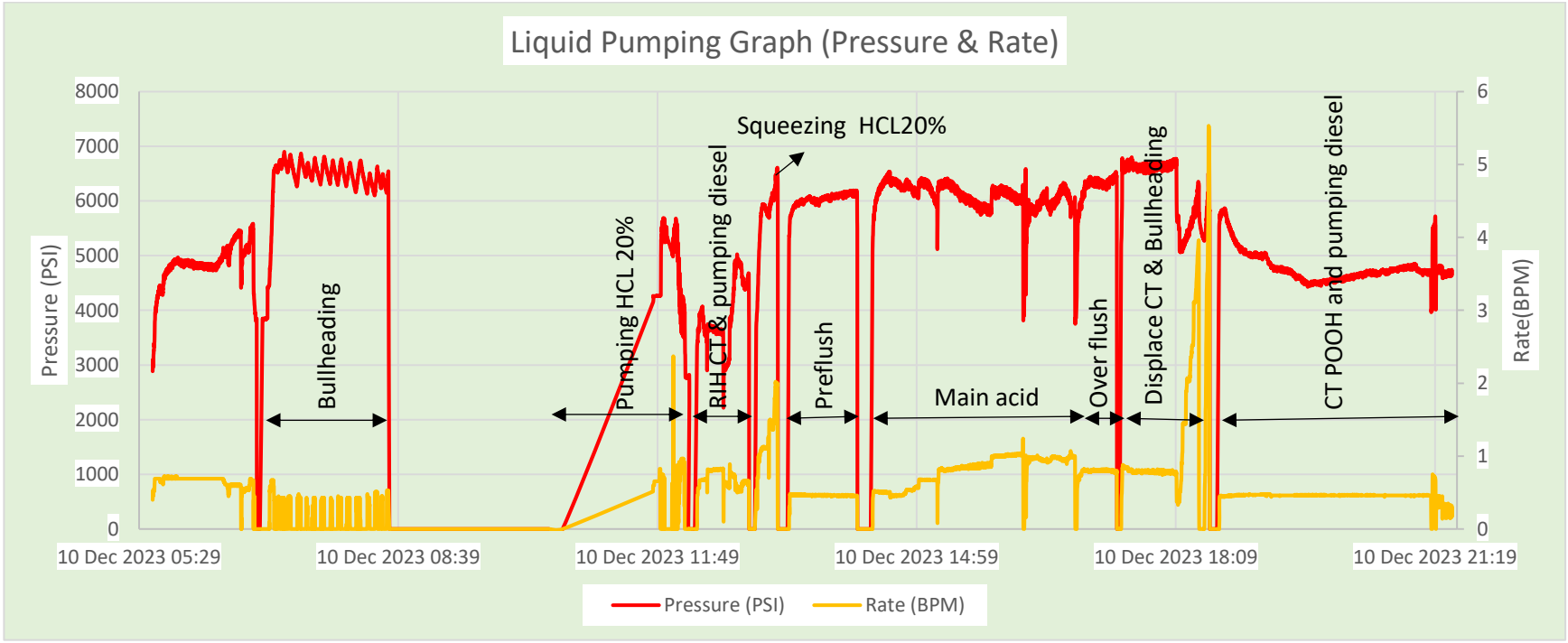


Figure 11-7: AZNS-40 Liquid Pumping Graph (Pr. & Rate)

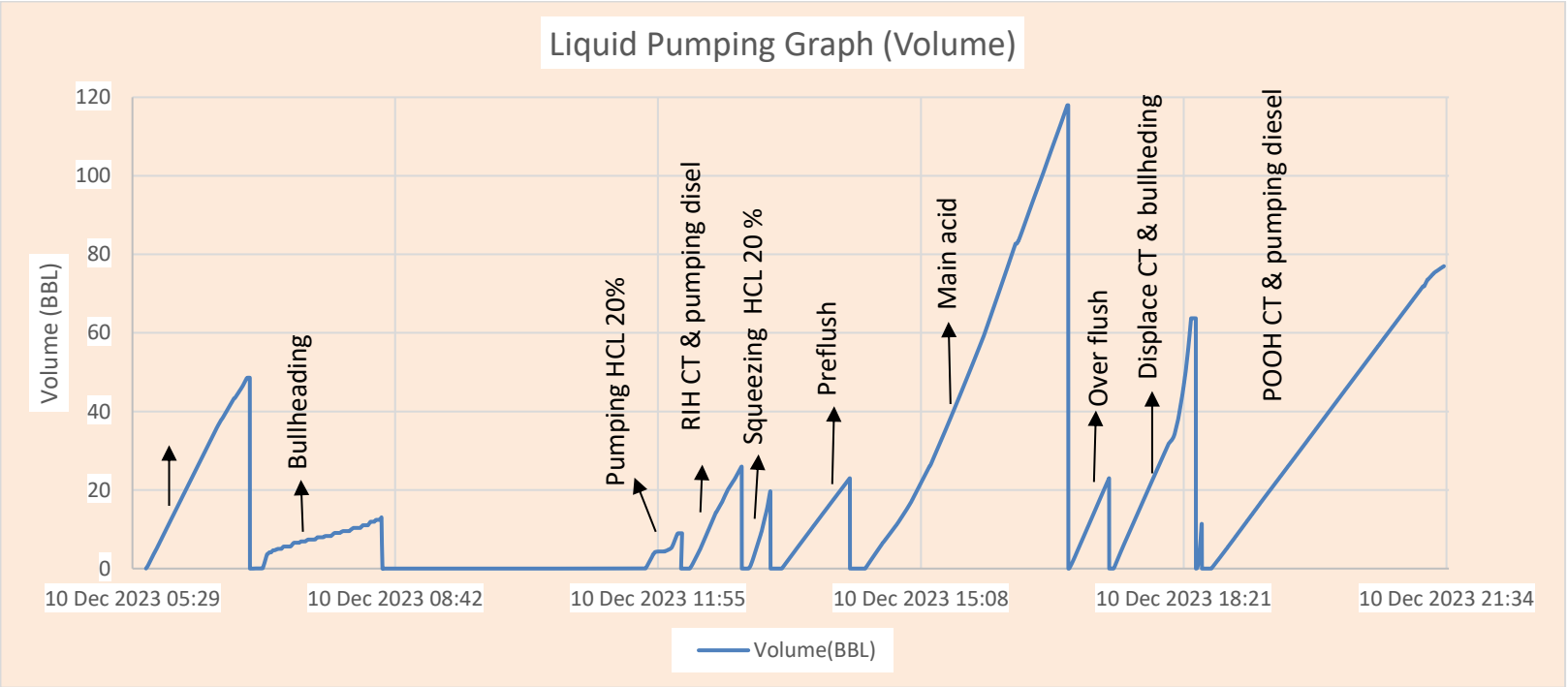


Figure 11-8: AZNS-40 Liquid Pumping Graph (Volume)

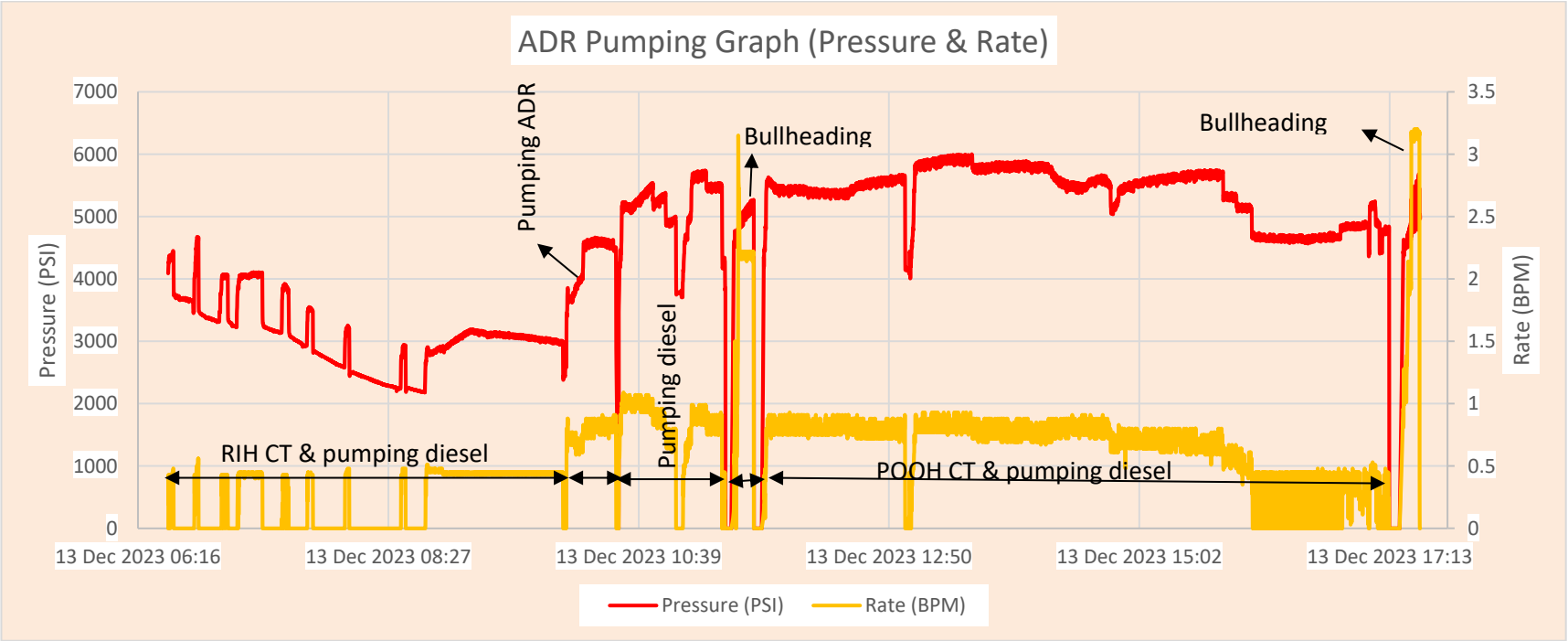


Figure 11-9: AZNS-40 ADR Pumping Graph (PR. & Rate)

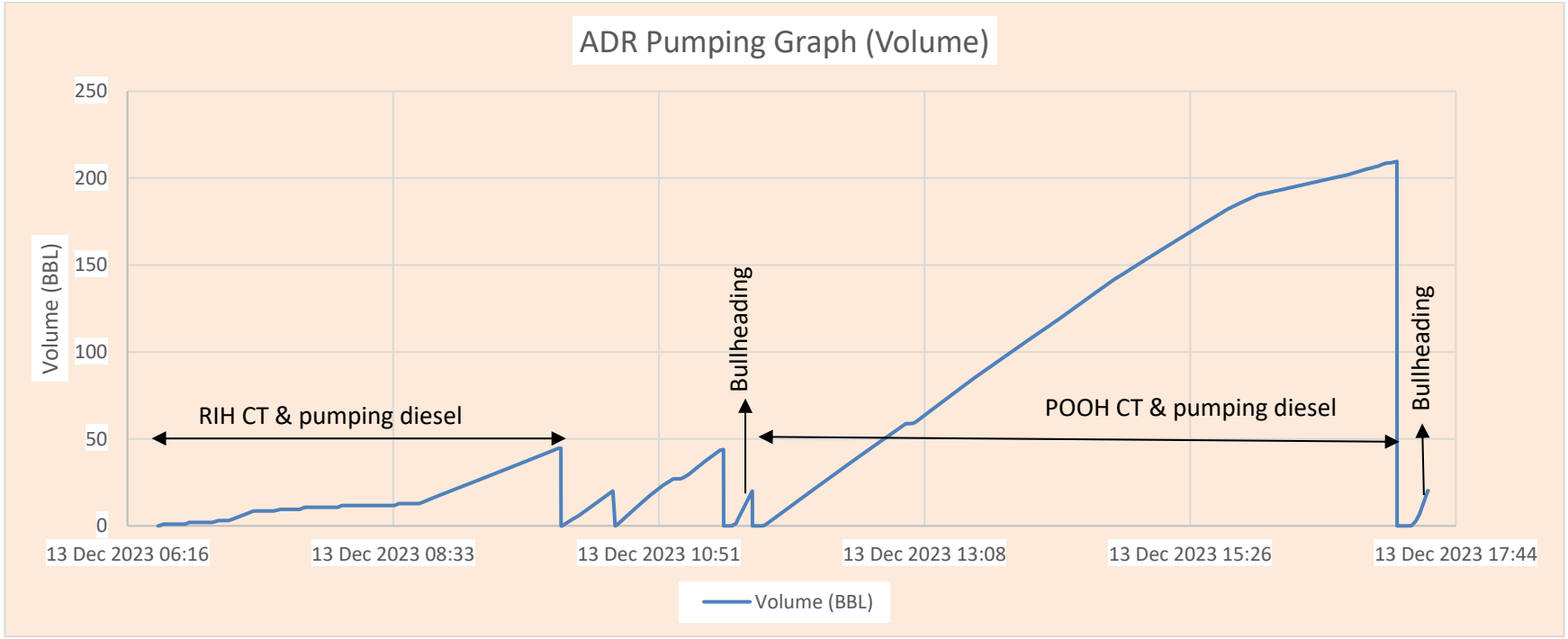


Figure 11-10: AZNS-40 ADR Pumping Graph (Volume)

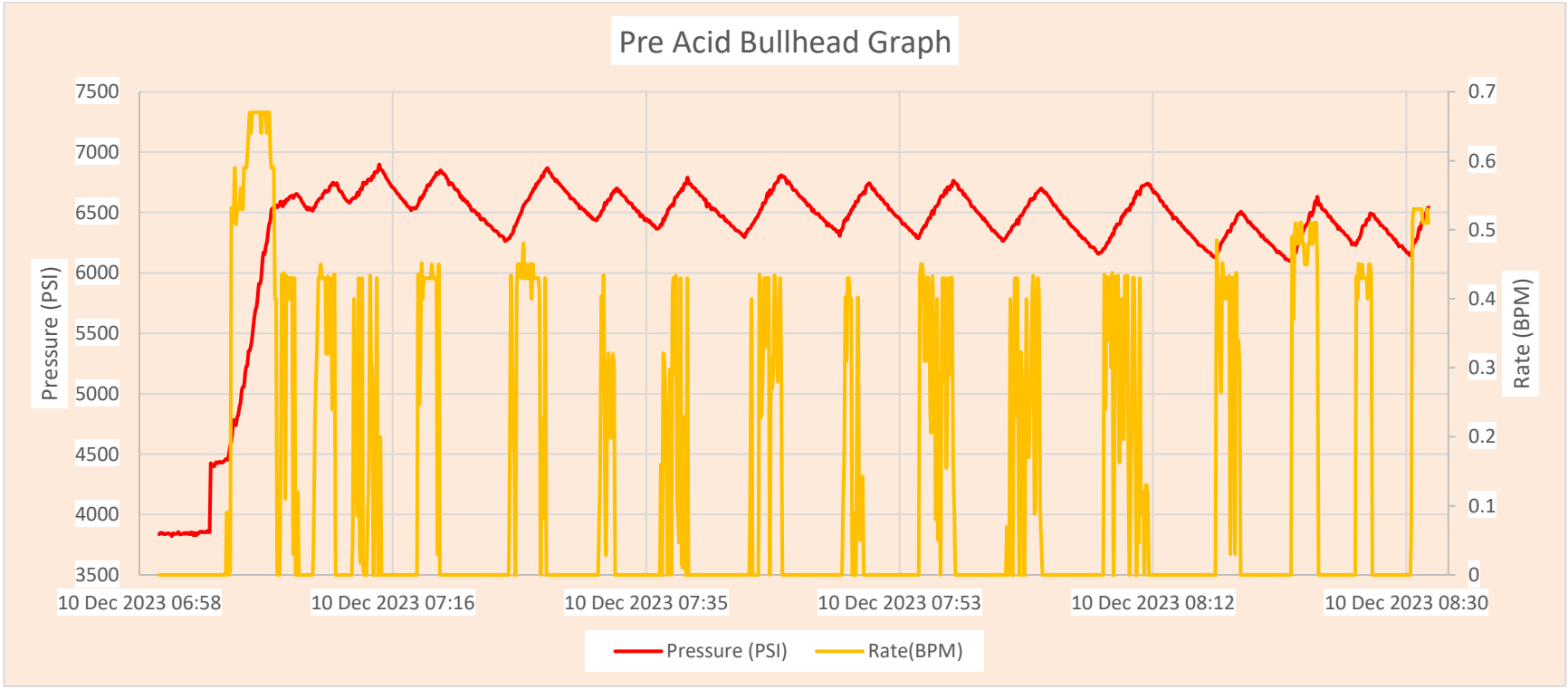


Figure 11-11: AZNS-40 Pre-Acid Bullhead Graph

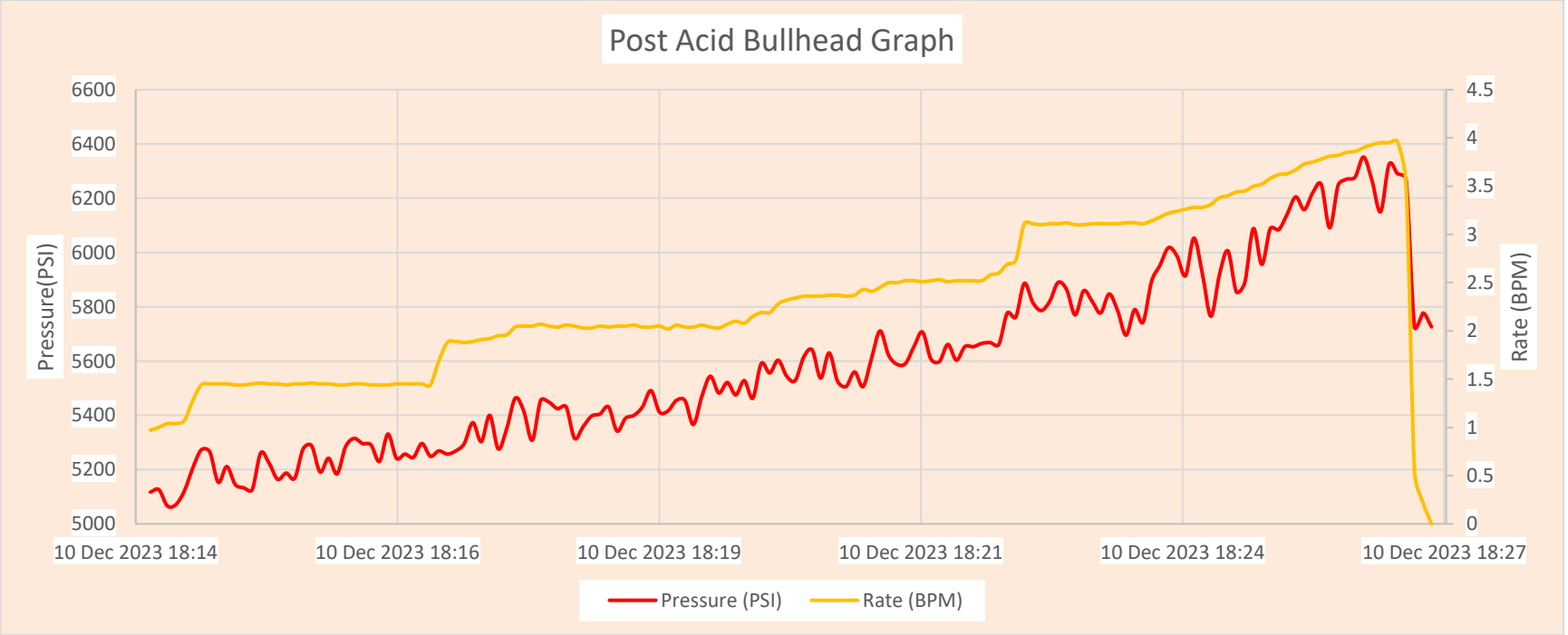


Figure 11-12: AZNS-40 Post-Acid Bullhead Graph

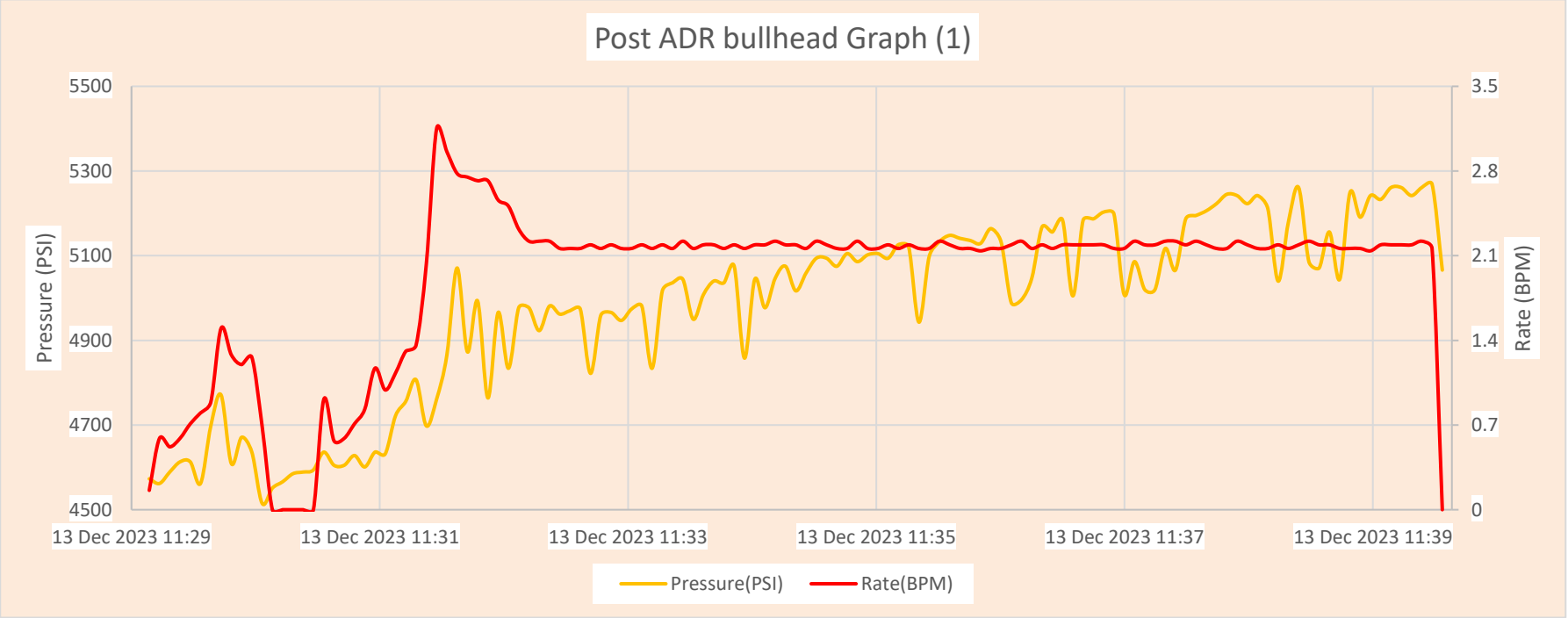


Figure 11-13: AZNS-40 Post ADR Bullhead Graph 1

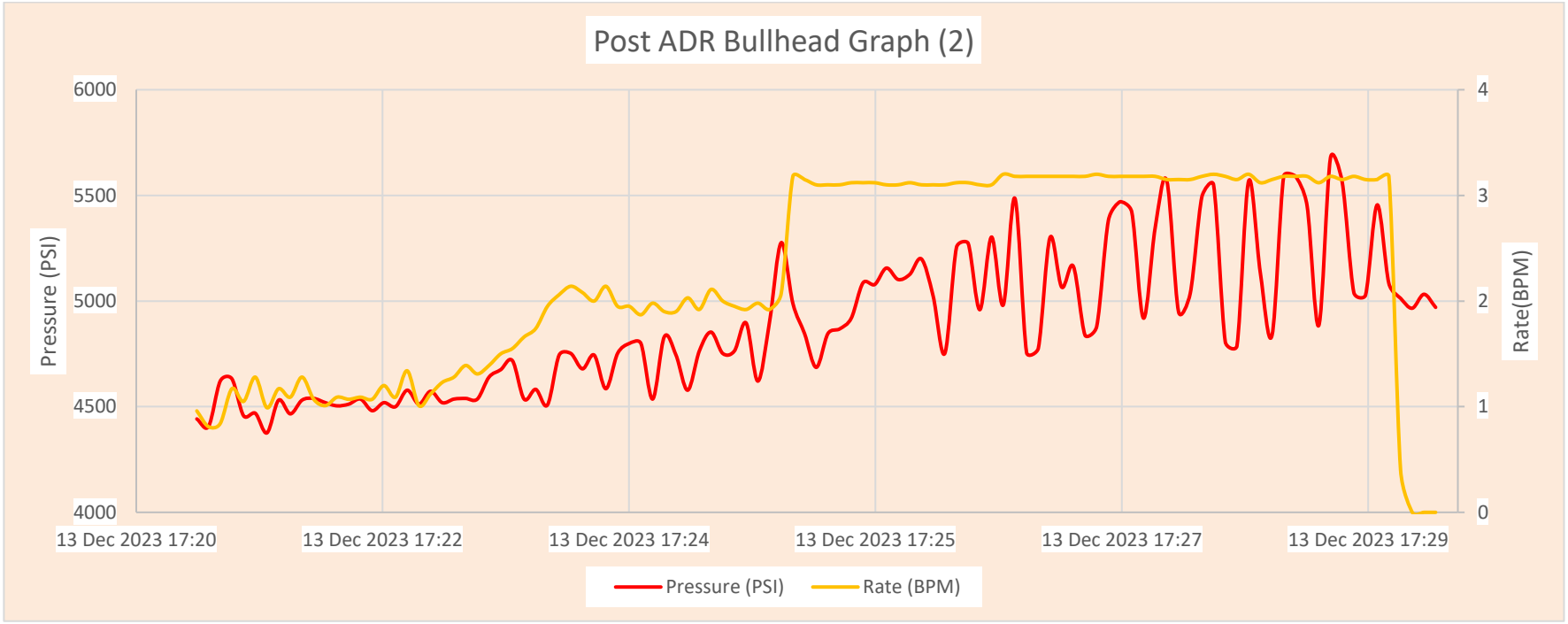


Figure 11-14: AZNS-40 Post ADR Bullhead Graph 2

Pumping Parameter in Pre-stimulation	
Pumping Pressure (PSI)	Pumping rate (bpm)
6899	0.67
Pumping Parameter in Post-stimulation	
Pumping Pressure (PSI)	Pumping rate (bpm)
6352	3.96
Pumping Parameter in Post-ADR	
5677	3.2

Table 11-19: AZNS-040 Pumping Parameter

11.4.4.6 Flowing/Static Survey Data & Interpretation

11.4.4.6.1 Pre – Stimulation Downhole Memory Gauge Data, Results & Plots

AZNS # 40 - Before Stimulation										
Row	Date	Time(hh:mm:ss)	Event	Stop POOH Duration (min)	Depth MD (FT) Dipping Valve	Depth TVD (FT) Dipping Valve	Pressure (PSIA) PPS 90506	Temperature (°F) PPS 90506	Pressure Gradient (PSIA/FT) PPS 90506	Temperature Gradient (°F/FT) PPS 90506
1	09-December-2023		Connect		0	0				
2			Open the well		0	0				
3			Start POOH		0	0				
4		11:28:15 - 11:38:15	POOH Step 01	10	13230	13230	3813	280	0.42500	0.00000
5		11:39:40 - 11:49:40	POOH Step 02	10	13190	13190	3796	280	0.33158	0.00526
6		11:54:13 - 11:59:13	POOH Step 03	05	13000	13000	3733	279	0.22000	0.01000
7		12:02:00 - 12:09:00	POOH Step 04	07	12900	12900	3711	278	0.29000	0.01000
8		12:11:30 - 12:16:30	POOH Step 05	05	12800	12800	3682	277	0.30333	0.00667
9		12:21:10 - 12:26:10	POOH Step 06	05	12500	12500	3591	275	0.24200	0.01000
10		12:31:12 - 12:36:12	POOH Step 07	05	12000	12000	3470	270	0.24900	0.01100
11		12:42:10 - 13:40:10	POOH Step 08	58	11000	11000	3221	259	0.25500	0.01200
12		13:45:45 - 13:50:45	POOH Step 09	05	10000	10000	2966	247	0.24400	0.01400
13		13:53:50 - 14:00:50	POOH Step 10	07	9500	9500	2844	240	0.23800	0.01400
14		14:03:10 - 14:08:10	POOH Step 11	05	9000	9000	2725	233	0.24600	0.01400

AZNS # 40 - Before Stimulation

Row	Date	Time(hh:mm:ss)	Event	Stop POOH Duration (min)	Depth MD (FT) Dipping Valve	Depth TVD (FT) Dipping Valve	Pressure (PSIA) PPS 90506	Temperature (°F) PPS 90506	Pressure Gradient (PSIA/FT) PPS 90506	Temperature Gradient (°F/FT) PPS 90506
15		14:11:00 - 14:16:00	POOH Step 12	05	8500	8500	2602	226	0.24000	0.01200
16		14:18:30 - 14:25:30	POOH Step 13	07	8000	8000	2482	220	0.26600	0.01600
17		14:28:10 - 14:33:10	POOH Step 14	05	7500	7500	2349	212	0.19600	0.01000
18		14:36:20 - 14:41:20	POOH Step 15	05	7000	7000	2251	207	0.22900	0.01100
19		14:46:10 - 14:53:10	POOH Step 16	07	6000	6000	2022	196	0.22400	0.01200
20		14:58:08 - 15:03:08	POOH Step 17	05	5000	5000	1798	184	0.20300	0.01500
21		15:07:20 - 15:12:20	POOH Step 18	05	4000	4000	1595	169	0.19400	0.01000
22		15:15:00 - 15:20:00	POOH Step 19	05	3000	3000	1401	159	0.18300	0.01200
23		15:25:00 - 15:30:00	POOH Step 20	05	2000	2000	1218	147	0.15600	0.01600
24		15:34:05 - 15:41:05	POOH Step 21	07	1500	1500	1140	139	0.16000	0.01400
25		15:44:30 - 15:49:30	POOH Step 22	05	1000	1000	1060	132	0.16000	0.01000
26		15:52:38 - 15:57:38	POOH Step 23	05	500	500	980	127	0.11667	0.01667
27		16:02:00 - 16:07:00	POOH Step 24	05	200	200	945	122	0.11039	0.01299
27		16:09:09 - 16:14:09	POOH Step 25	05	46	46	928	120		

Table 11-20: AZNS-040 Pre – Stimulation Flowing Survey Report – gauge No. 90506

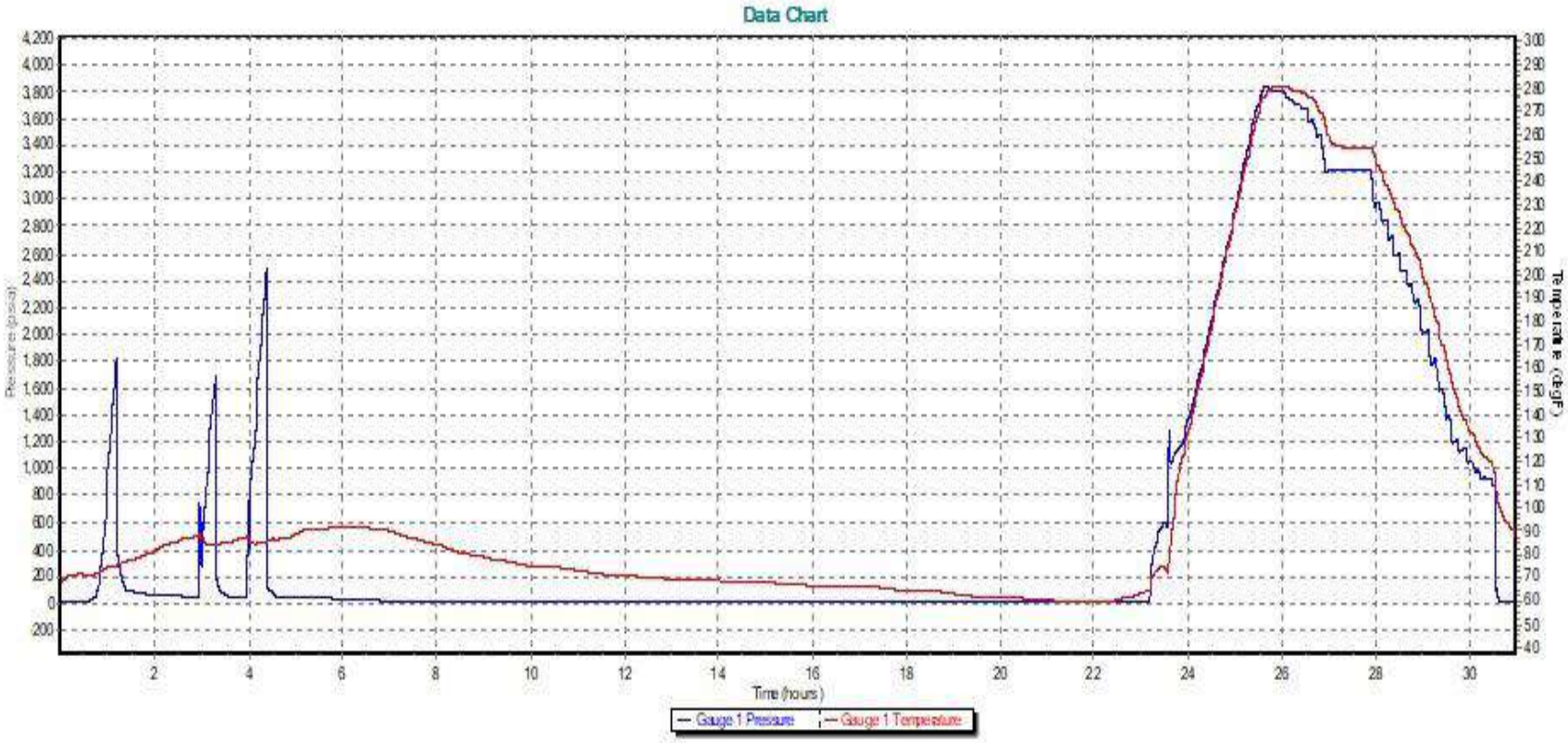


Figure 11-15: AZNS-40 Pre – Stimulation Downhole Memory Gauge (P&T) Plot – gauge No. 90506

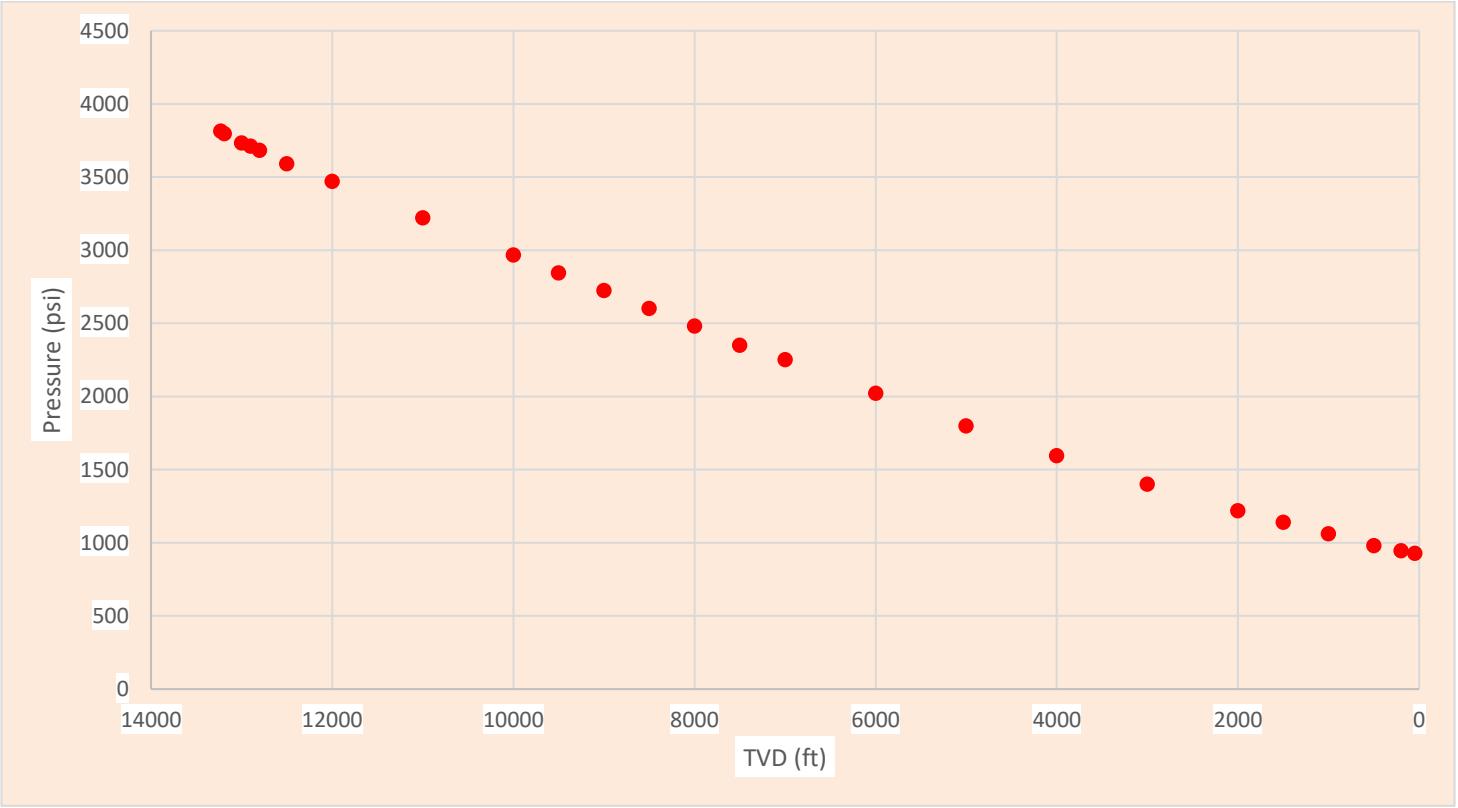


Figure 11-16: AZNS-40 Pre – Stimulation Pressure vs TVD Plot – gauge No. 90506

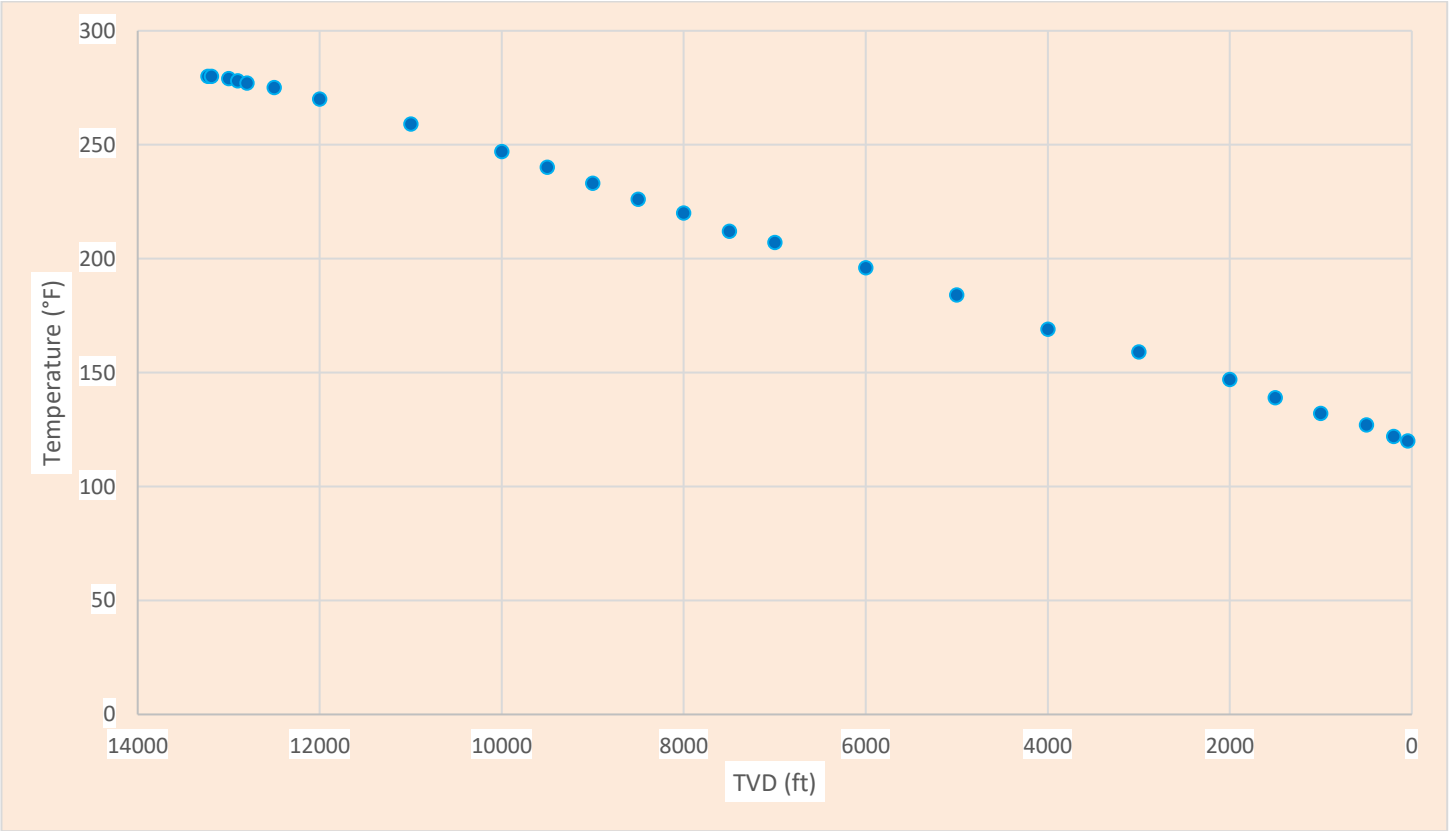


Figure 11-17: AZNS-40 Pre – Stimulation Temperature vs TVD Plot – gauge No. 90506

11.4.4.6.2 Post – Stimulation Downhole Memory Gauge Data, Results & Plots

AZNS # 40 - After Stimulation												
Row	Date	Time(hh:mm:ss)			Event	Stop POOH Duration (min)	Depth MD (FT) Dipping Valve	Depth TVD (FT) Dipping Valve	Pressure (PSIA) PPS 90506	Temperature (°F) PPS 90506	Pressure Gradient (PSIA/FT) PPS 90506	Temperature Gradient (°F/FT) PPS 90506
1	12-December-2023				Connect		0	0				
2					Open the well		0	0				
3					Start POOH		0	0				
4		10:43:05	10:43:05	10:43:05	POOH Step 01	10	13230	13230	6448	268	0.55000	-0.02500
5		10:54:40	10:54:40	10:54:40	POOH Step 02	10	13190	13190	6426	269	0.27895	0.00000
6		11:09:30	11:09:30	11:09:30	POOH Step 03	05	13000	13000	6373	269	0.32000	0.01000
7		11:16:55	11:16:55	11:16:55	POOH Step 04	07	12900	12900	6341	268	0.27000	0.00000
8		11:26:50	11:26:50	11:26:50	POOH Step 05	05	12800	12800	6314	268	0.28667	0.01000
9		11:35:50	11:35:50	11:35:50	POOH Step 06	05	12500	12500	6228	265	0.28600	0.00200
10		11:45:10	11:45:10	11:45:10	POOH Step 07	05	12000	12000	6085	264	0.28000	0.00200
11		11:58:40	11:58:40	11:58:40	POOH Step 08	36	11000	11000	5805	262	0.30300	0.00600
12		12:43:25	12:43:25	12:43:25	POOH Step 09	05	10000	10000	5502	256	0.26000	0.00800
13		12:55:10	12:55:10	12:55:10	POOH Step 10	07	9500	9500	5372	252	0.28200	0.00800
14		13:12:30	13:12:30	13:12:30	POOH Step 11	05	9000	9000	5231	248	0.30400	0.00800
15		13:26:15	13:26:15	13:26:15	POOH Step 12	05	8500	8500	5079	244	0.24600	0.00800

AZNS # 40 - After Stimulation

Row	Date	Time(hh:mm:ss)			Event	Stop POOH Duration (min)	Depth MD (FT) Dipping Valve	Depth TVD (FT) Dipping Valve	Pressure (PSIA) PPS 90506	Temperature (°F) PPS 90506	Pressure Gradient (PSIA/FT) PPS 90506	Temperature Gradient (°F/FT) PPS 90506
16		13:41:15	13:41:15	13:41:15	POOH Step 13	07	8000	8000	4956	240	0.32400	0.00800
17		13:56:15	13:56:15	13:56:15	POOH Step 14	05	7500	7500	4794	236	0.27600	0.00800
18		14:09:50	14:09:50	14:09:50	POOH Step 15	05	7000	7000	4656	232	0.00800	-0.01200
19		14:30:16	14:30:16	14:30:16	POOH Step 16	07	6000	6000	4648	244	0.27400	0.02900
20		14:49:50	14:49:50	14:49:50	POOH Step 17	05	5000	5000	4374	215	0.29600	0.01000
21		15:06:50	15:06:50	15:06:50	POOH Step 18	05	4000	4000	4078	205	0.27300	0.00800
22		15:20:25	15:20:25	15:20:25	POOH Step 19	05	3000	3000	3805	197	0.29900	0.01100
23		15:36:25	15:36:25	15:36:25	POOH Step 20	05	2000	2000	3506	186	0.58000	0.01000
24		15:45:40	15:45:40	15:45:40	POOH Step 21	07	1500	1500	3216	181	0.24800	0.01200
25		15:56:57	15:56:57	15:56:57	POOH Step 22	05	1000	1000	3092	175	0.27200	0.01000
26		16:07:20	16:07:20	16:07:20	POOH Step 23	05	500	500	2956	170	0.45000	0.01667
27		16:19:40	16:19:40	16:19:40	POOH Step 24	05	200	200	2821	165	0.29870	0.01299
28		16:29:15	16:29:15	16:29:15	POOH Step 25	05	46	46	2775	163		

Table 11-21: AZNS-040 Post – Stimulation Flowing Survey Report – gauge No. 90506

AZNS # 40 - After Stimulation

Row	Date	Time(hh:mm:ss)			Event	Stop POOH Duration (min)	Depth MD (FT) Dipping Valve	Depth TVD (FT) Dipping Valve	Pressure (PSIA) PPS 90504	Temperature (°F) PPS 90504	Pressure Gradient (PSIA/FT) PPS 90504	Temperature Gradient (°F/FT) PPS 90504
1	12-December-2023				Connect		0	0				
2					Open the well		0	0				
3					Start POOH		0	0				
4		10:43:05	10:43:05	10:43:05	POOH Step 01	10	13230	13230	6448	268	-0.11111	-0.02500
5		10:54:40	10:54:40	10:54:40	POOH Step 02	10	13190	13190	6426	269	0.00000	0.00000
6		11:09:30	11:09:30	11:09:30	POOH Step 03	05	13000	13000	6373	269	0.03333	0.01000
7		11:16:55	11:16:55	11:16:55	POOH Step 04	07	12900	12900	6341	268	0.00000	0.00000
8		11:26:50	11:26:50	11:26:50	POOH Step 05	05	12800	12800	6314	268	0.03297	0.01000
9		11:35:50	11:35:50	11:35:50	POOH Step 06	05	12500	12500	6228	265	0.00654	0.00200
10		11:45:10	11:45:10	11:45:10	POOH Step 07	05	12000	12000	6085	264	0.00658	0.00200
11		11:58:40	11:58:40	11:58:40	POOH Step 08	36	11000	11000	5805	262	0.01967	0.00600
12		12:43:25	12:43:25	12:43:25	POOH Step 09	05	10000	10000	5502	256	0.02614	0.00800
13		12:55:10	12:55:10	12:55:10	POOH Step 10	07	9500	9500	5372	252	0.02632	0.00800
14		13:12:30	13:12:30	13:12:30	POOH Step 11	05	9000	9000	5231	248	0.02632	0.00800
15		13:26:15	13:26:15	13:26:15	POOH Step 12	05	8500	8500	5079	244	0.02614	0.00800
16		13:41:15	13:41:15	13:41:15	POOH Step 13	07	8000	8000	4956	240	0.02632	0.00800
17		13:56:15	13:56:15	13:56:15	POOH Step 14	05	7500	7500	4794	236	0.02614	0.00800
18		14:09:50	14:09:50	14:09:50	POOH Step 15	05	7000	7000	4656	232	0.02961	0.00900

AZNS # 40 - After Stimulation

Row	Date	Time(hh:mm:ss)			Event	Stop POOH Duration (min)	Depth MD (FT) Dipping Valve	Depth TVD (FT) Dipping Valve	Pressure (PSIA) PPS 90504	Temperature (°F) PPS 90504	Pressure Gradient (PSIA/FT) PPS 90504	Temperature Gradient (°F/FT) PPS 90504
19		14:30:16	14:30:16	14:30:16	POOH Step 16	07	6000	6000	4648	244	0.02951	0.00900
20		14:49:50	14:49:50	14:49:50	POOH Step 17	05	5000	5000	4374	215	0.02927	0.00600
21		15:06:50	15:06:50	15:06:50	POOH Step 18	05	4000	4000	4078	205	0.02963	0.01200
22		15:20:25	15:20:25	15:20:25	POOH Step 19	05	3000	3000	3805	197	0.03289	0.01000
23		15:36:25	15:36:25	15:36:25	POOH Step 20	05	2000	2000	3506	186	0.03268	0.01000
24		15:45:40	15:45:40	15:45:40	POOH Step 21	07	1500	1500	3216	181	0.03947	0.01200
25		15:56:57	15:56:57	15:56:57	POOH Step 22	05	1000	1000	3092	175	0.03268	0.01000
26		16:07:20	16:07:20	16:07:20	POOH Step 23	05	500	500	2956	170	0.05495	0.01667
27		16:19:40	16:19:40	16:19:40	POOH Step 24	05	200	200	2821	165	0.04255	0.01299
28		16:29:15	16:29:15	16:29:15	POOH Step 25	05	46	46	2775	163		

Table 11-22: AZNS-040 Post – Stimulation Flowing Survey Report – gauge No. 90504

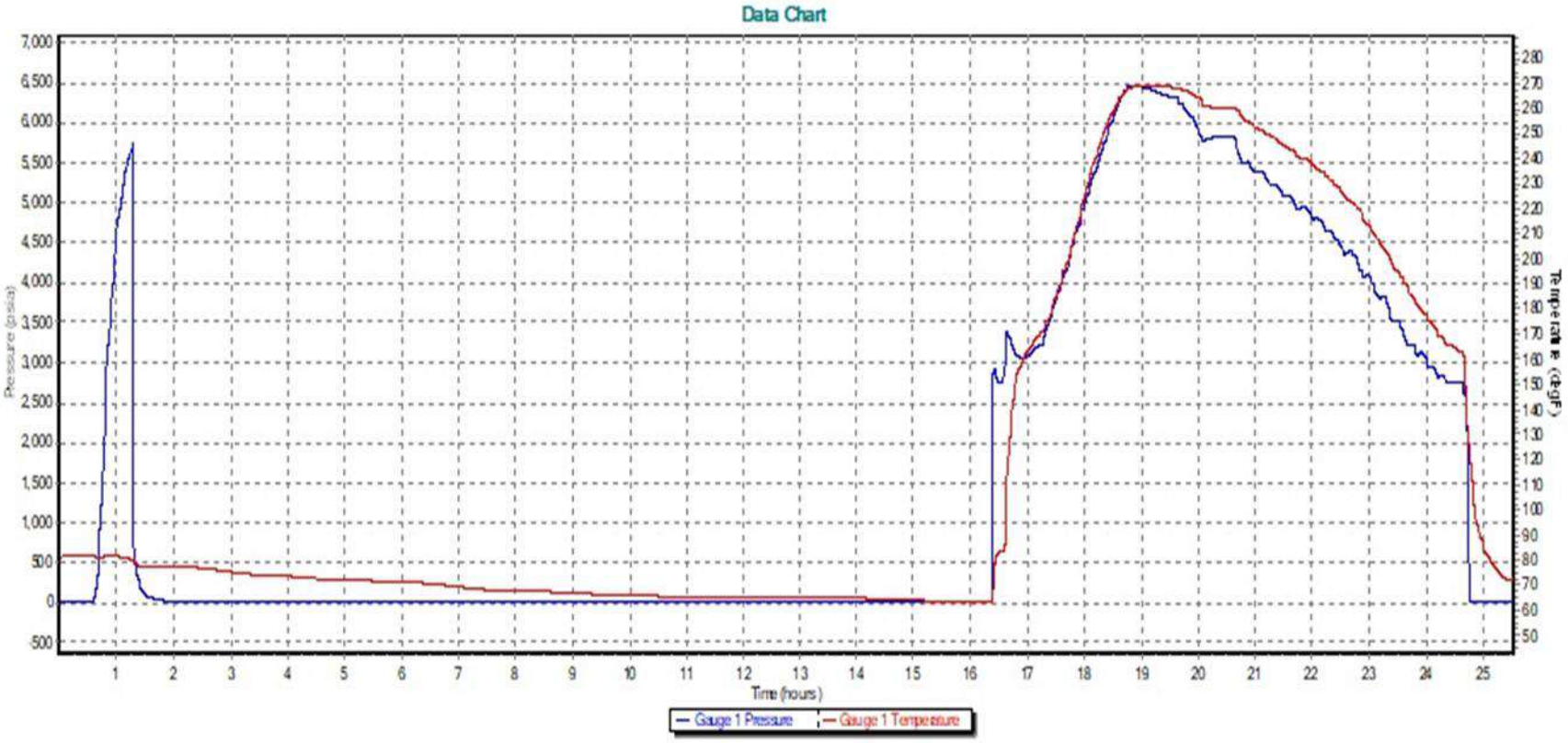


Figure 11-18: AZNS-40 Post – Stimulation Downhole Memory Gauge (P&T) Plot – gauge No. 90506

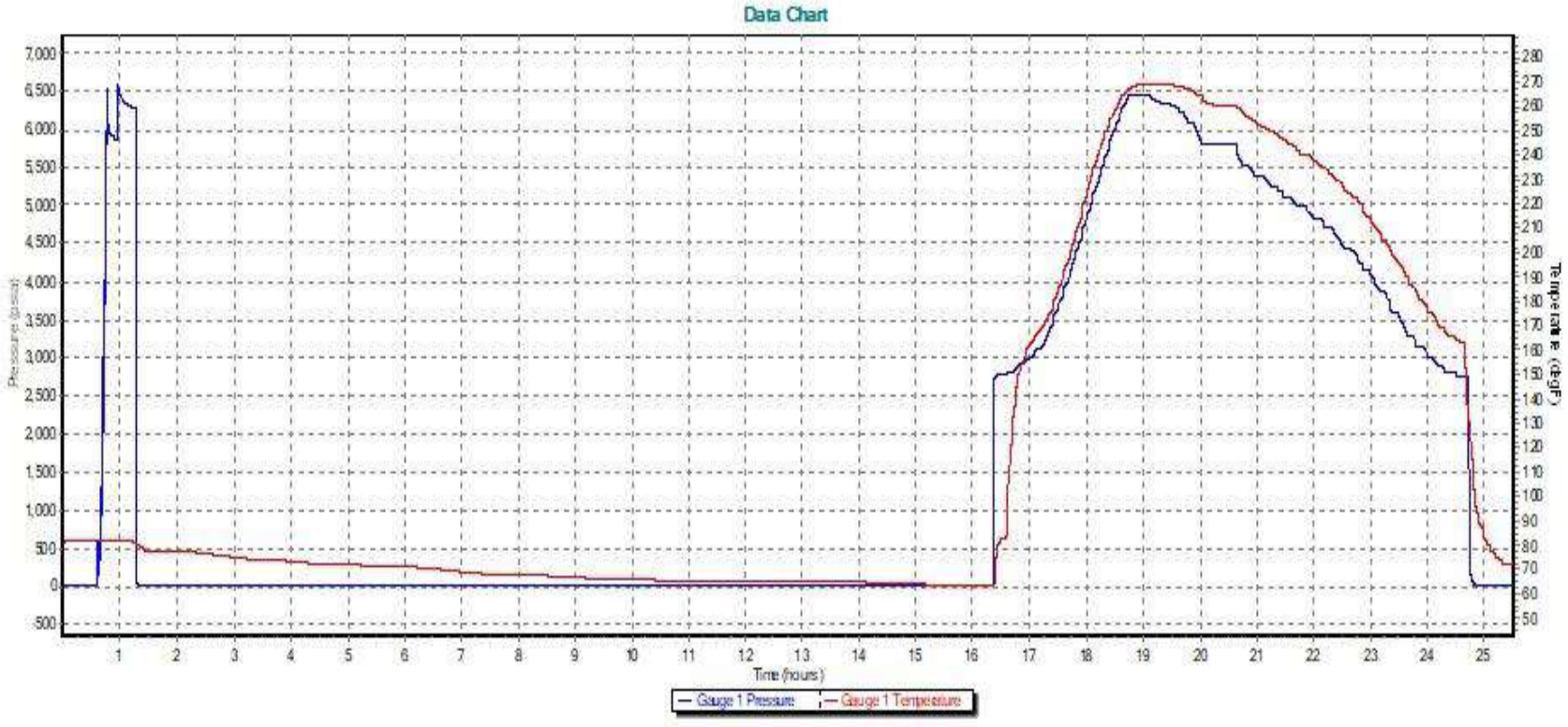


Figure 11-19: AZNS-40 Post – Stimulation Downhole Memory Gauge (P&T) Plot – gauge No. 90504

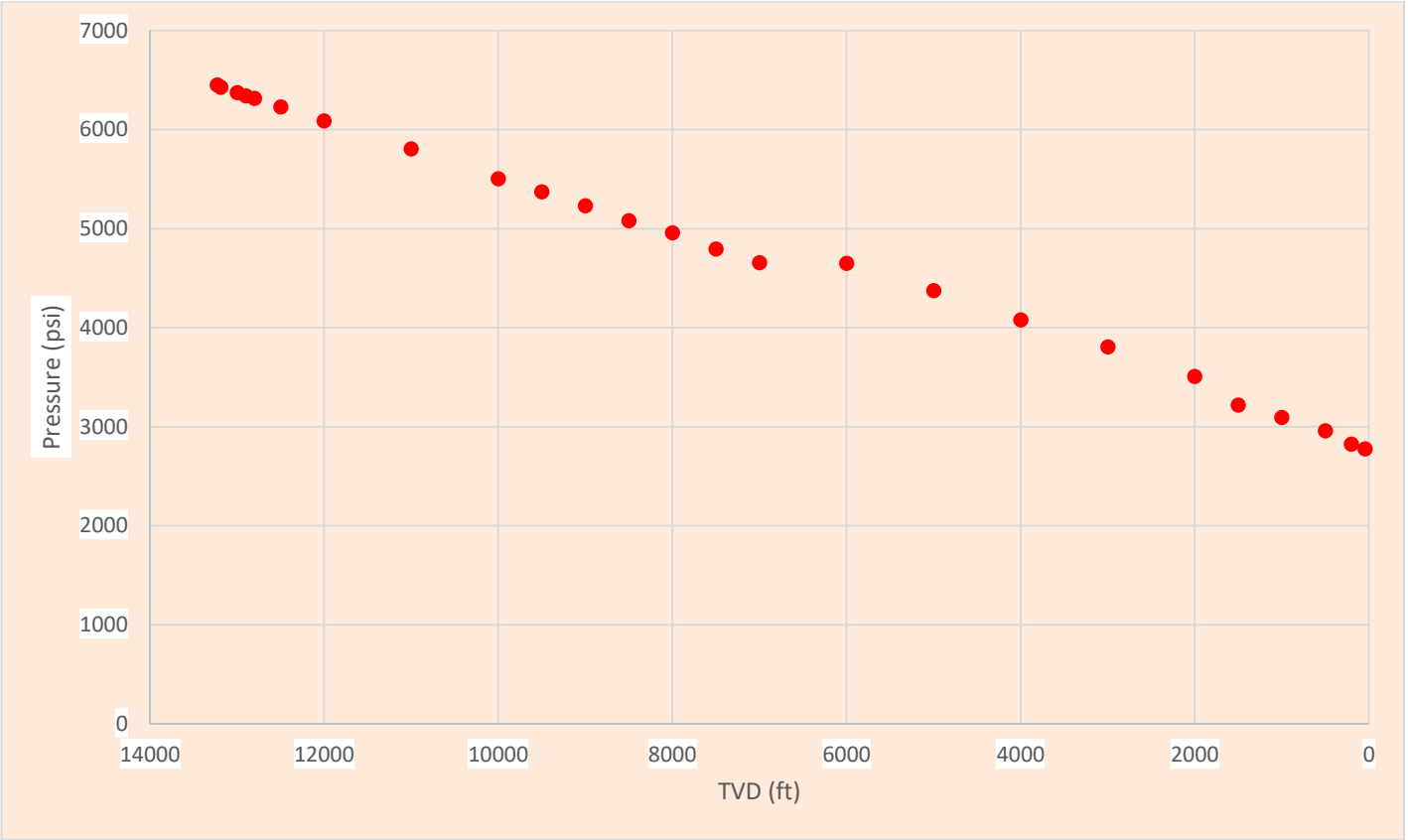


Figure 11-20: AZNS-40 Post – Stimulation Pressure vs TVD Plot – gauge No. 90506

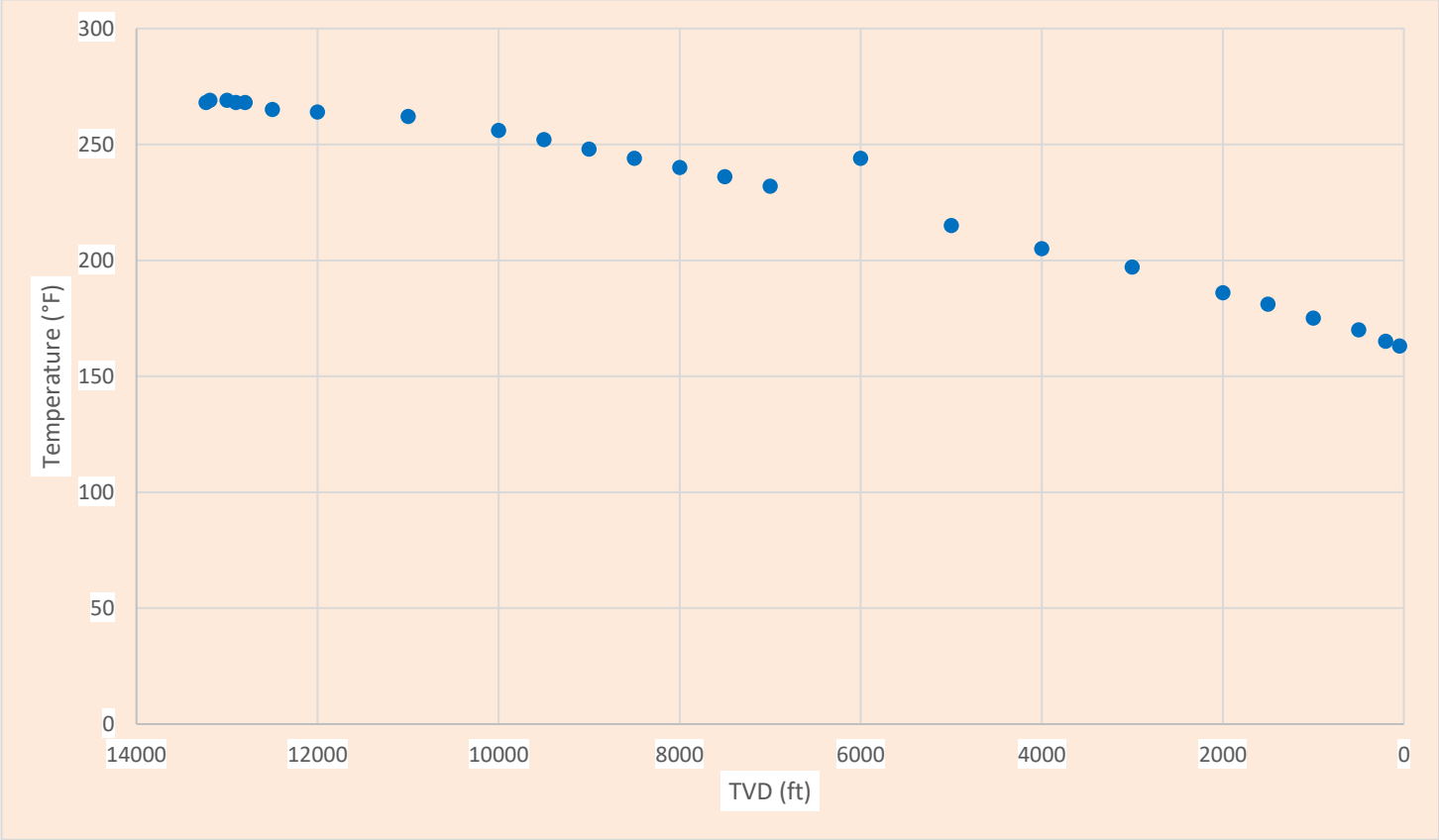


Figure 11-21: AZNS-40 Post – Stimulation Temperature vs TVD Plot – gauge No. 90506

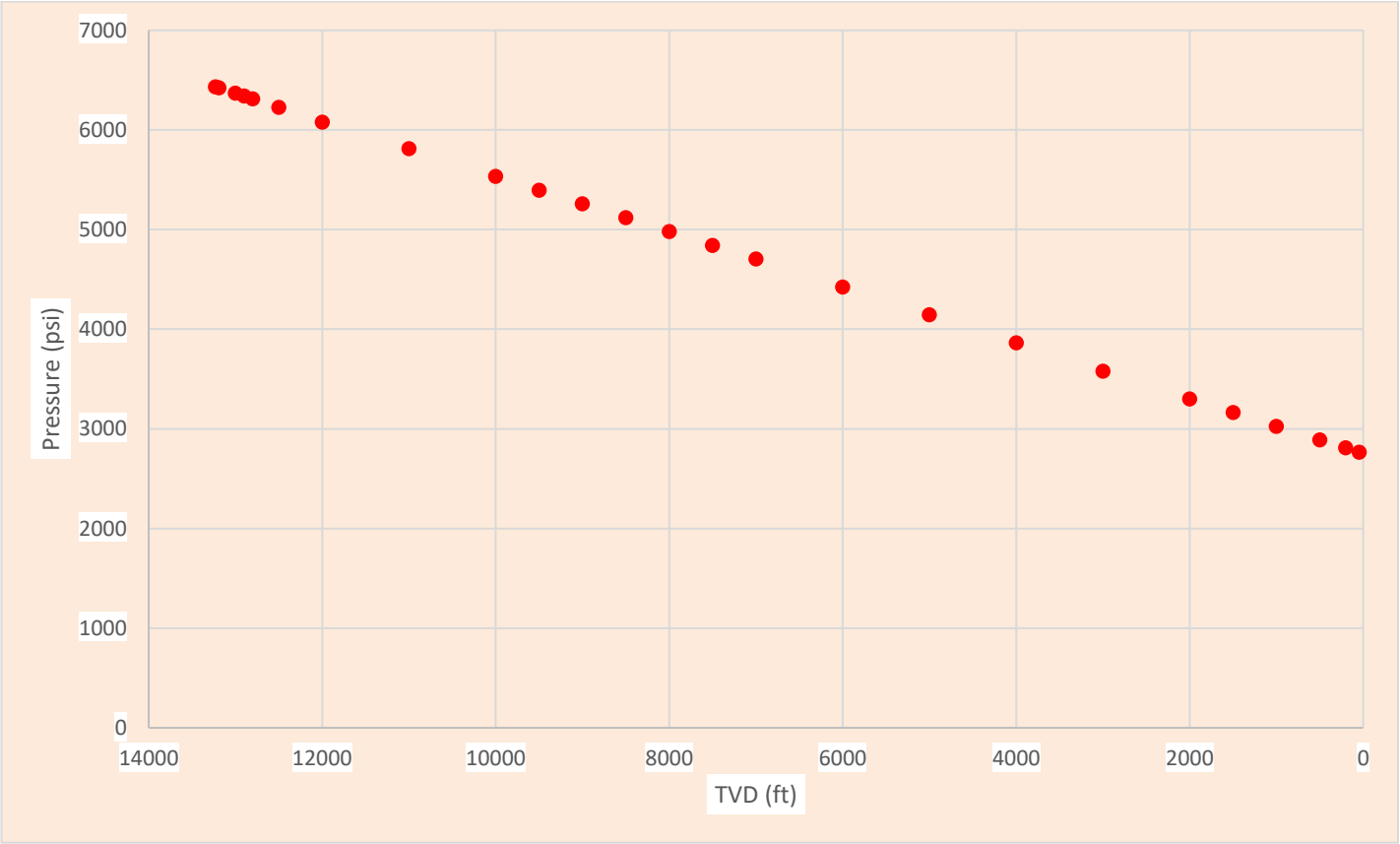


Figure 11-22: AZNS-40 Post – Stimulation Pressure vs TVD Plot – gauge No. 90504

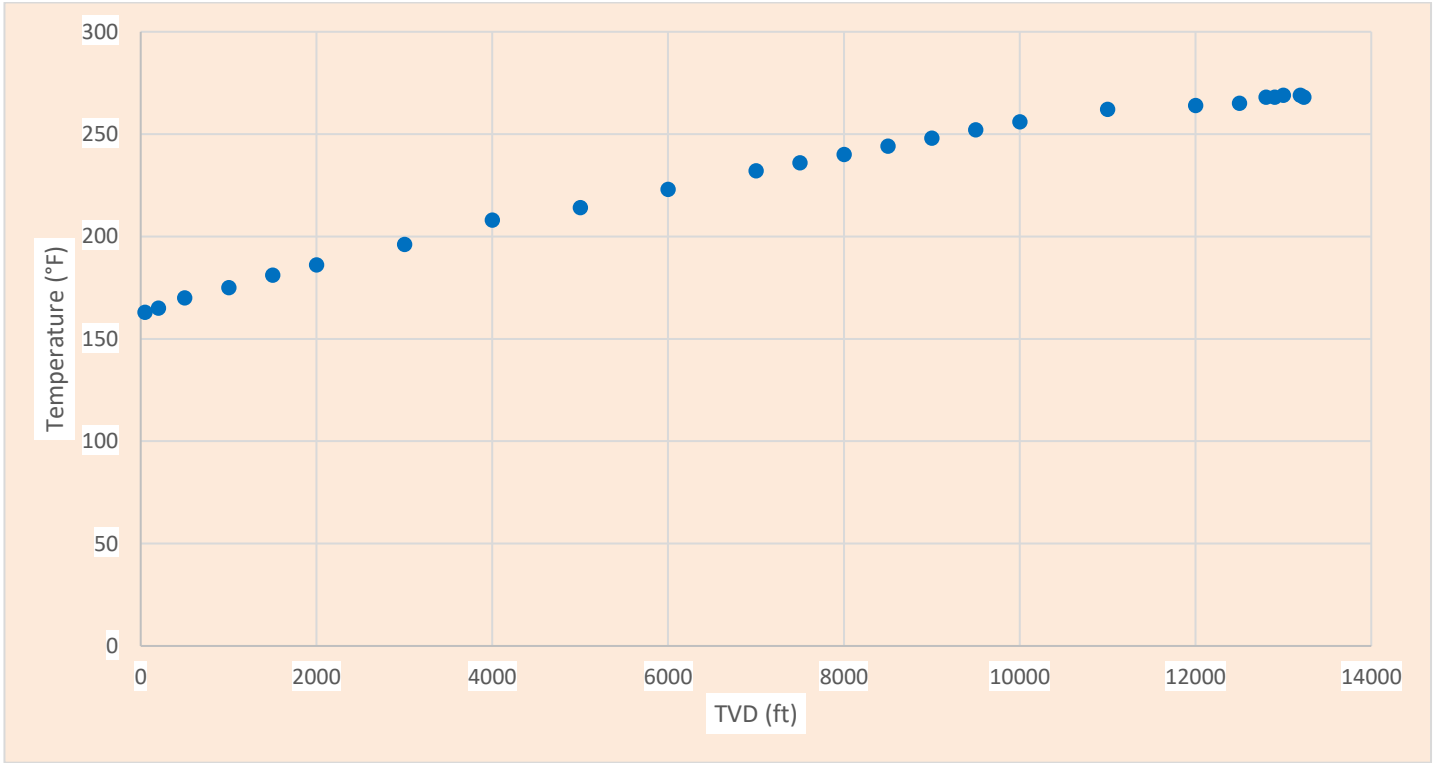


Figure 11-23: AZNS-40 Post – Stimulation Temperature vs TVD Plot – gauge No. 9050

The pre-stimulation PI of the well at the datum depth is calculated utilizing average reservoir pressure of the well extracted from the data provided by client, datum flowing pressure from flowing survey, and measured flow rate during flowing survey operation according to table below:

Pre-stimulation PI (@ 13124 FtTVDss): 0.18 (STB/D.PSI)	
Choke size (1/64" Fix)	24
Average Reservoir Pressure (PSIA)	9276
Bottom-hole Flowing Pressure (PSIA)	3788
Est. Rate (BBL/day)	995

Table 11-23: AZNS-040 Pre – Acid Productivity Index

The post-stimulation PI of the well at the datum depth is calculated utilizing average reservoir pressure of the well extracted from the data provided by client, datum flowing pressure from flowing survey, and measured flow rate during flowing survey operation according to table below:

Post-stimulation PI (@ 13124 ftTVDss): 0.88 (STB/D.PSI)	
Choke size (1/64" Fix)	24
Average Reservoir Pressure (PSIA)	9276
Bottom-hole Flowing Pressure (PSIA)	6426
Est. Rate (BBL/day)	2532

Table 11-24: AZNS-040 Post – Acid Productivity Index

11.5 Failure & problems

N/A

11.6 Lesson Learned

N/A

11.7 Appendix

Gauges Information

DHMG raw data is attached as memory gauges appendix.

Well Testing Raw Data

Well testing raw data is attached as well testing appendix.

Pumping Raw Data

Pumping raw data is attached as pumping appendix.

N_2 Lifting Raw Data

N_2 lifting raw data is attached as N_2 lifting appendix.

CT Raw Data

CT raw data is attached as CT appendix

Surface Equipment Layout

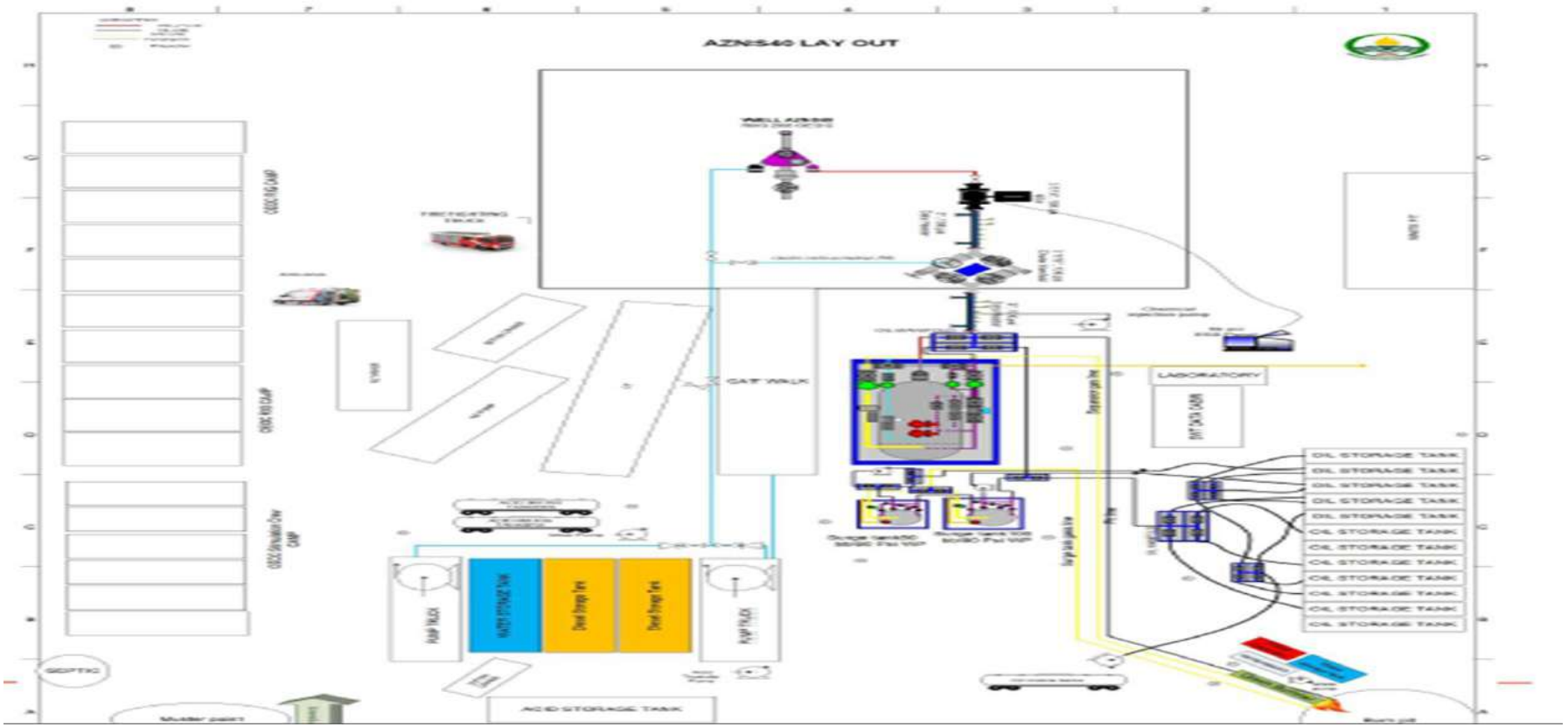


Figure 11-24: AZNS-40 Actual Surface Equipment Layout

CT BHA Diagram

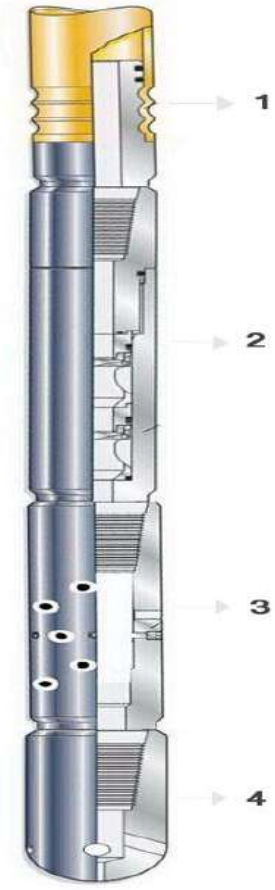
Row	BHA	Length (in)	OD size (in)	Schematic
1	Roll On Connector	3.3	1 3/4	
2	Two Check valves (Flapper Type)	8.3		
3	Perforation Sub (10 mm holes)	6.3		
4	Wash Nozzle (8 mm holes)	5		

Table 11-25: AZNS-040 Coiled Tubing BHA Design

S/L BHA Diagram

OD size (in)	Tools	Length (ft)	schematic
1 7/8	Rope socket	0.5	
1 7/8	Stem	10	
1 7/8	Mechanical Jar	4 (closed) & 6 (fully open)	
1 7/8	Hydraulic Jar	2	
3.450	Gauge Cutter or GS Pulling or Memory Gauge	1	 

Table 11-26: AZNS-040 Slickline BHA Design

11.8 Onsite QC Test Report of Stimulation Additives & Fluids

QC tests and checklist fill up process were performed by Client, OEOC, and Energy Semnan Co. representatives prior to AZNS-40 well's stimulation operation and its report is mentioned in detail in the AZNS-40 operation program. It should be noted that samples of fresh water, raw acid, diesel, and additives were taken from onsite materials under client's supervision, and it was approved by client's Rep.

12. Waste Management Report

12.1 Waste Management Package

The required equipments had been located on operattional area, according to the employer recall, which is tabled below.

1	Dewatering unit		One	Complete with related accessories
2	Solid separation unit	Automatic filteration unit	One	Capableto filter completion fluid
3	Buffer tanks and acceories	Buffer tanks and agitators	Two	Optional/ 25 HP
4	Pumps	Centrifugal pump	One	3”x4” ,3”x6” wirh related accessories
5		Submersible pump	Two	4”, 40-50 m3/hr, With related accessories
6		Air diaphragm pump	Two	3”, with related accessories
7		Wash down Gun	One	200 Br, with related accessories
8	Laboratory unit and waste water analysis equipment	Photo meter / ion detector	One	Complete with related accessories
9		Jar test / Centifuge test kit	One	
10		TDS/ pH /EC /Retort kit	One	
11		Caravan including office and lab.	One	

Table 12-1: AZNS-040 Well Waste Management Information

It also, worth to be mentioned that the service personnel had been working through the pre-spud process.

12.2 Waste Management End of Well Report

AZNS-40 well, was planned to be drilled as Horizontal well in Sarvak formation on Pad-50. The well was drilled by CNPCI to total depth 1493^{m} in Gachsaran formation as 17-1/2" hole section, 13-3/8" CSG ran and cemented to surface and well has been suspended by CMT plug.

As the Previous exploration/delineation drilling in Azadegan oilfield has proved presence of a very prolific hydrocarbon bearing sedimentary column, particularly within the Cretaceous sequences. The FahliyanFormation was divided into 6 zones, named as Fahliyan top, Fah-1, Fah-2, Fah-3, Fah-4 and Fah-5.

The objective of Re-Entry this well is to De-suspend the well, vertical drilling 12-1/4″ & 8-1/2″ hole section to new well target in Fahliyan reservoir (Well target has been changed from Sarvak horizontal well to Fahliyan vertical well), run 9-5/8" CSG and 7" Liner, perforate 7" Liner in Fahliyan formation and complete the well as Fahliyan vertical oil producer well.

12-1/4" Hole Section

Mud System	Salt Saturated KCL/Glycol/Polymer/$CaCO_3$ **WBM** 75-85^{pcf}
Rig	208 OEOC
Start Date	Jul. 13th 2023
Start Depth	1492^{m}
End Date	Sep. 9th 2023
End Depth	3939^{m}
Differential Depth	2447^{m}
Casing Depth	13 3/8” @ 1492^{m}
Duration	59 Days

Table 12-2: AZNS-040 12-1/4" Hole Waste Management Information

Total Cuttings (bbl)	5473
Total Fixed Cuttings (bbl)	4820
Total Disposed Cuttings (bbl)	4820
Total Used Cement (kg)	24000
Total Used Sodium silicate (kg)	385

Fixation									
Report No.	Date	Depth	Drilled	Total Cuttings		Fixed Cuttings		Sent to Disposal	
				Daily	Cum.	Daily	Cum.	Daily	Cum.
-	dd-M-yy	m	m	bbl.	bbl.	bbl.	bbl.	bbl.	bbl.
1	13-Jul-23	0	0	0	0	0	0	0	0
2	14-Jul-23	1,492	0	0	0	0	0	0	0
3	15-Jul-23	1,492	0	140	140	0	0	0	0
4	16-Jul-23	1,510	18	47	187	0	0	0	0
5	17-Jul-23	1,556	46	96	283	0	0	0	0
6	18-Jul-23	1,567	11	55	338	0	0	0	0
7	19-Jul-23	1,756	189	235	573	0	0	0	0
8	20-Jul-23	1,908	152	247	819	700	700	0	0
9	21-Jul-23	1,960	52	244	1,063	0	700	0	0
10	22-Jul-23	2,010	50	114	1,177	0	700	0	0
11	23-Jul-23	2,084	74	144	1,322	200	900	0	0
12	24-Jul-23	2,115	31	99	1,421	0	900	0	0
13	25-Jul-23	2,115	0	80	1,501	0	900	0	0
14	26-Jul-23	2,161	46	84	1,585	0	900	0	0
15	27-Jul-23	2,191	30	116	1,701	200	1,100	0	0
16	28-Jul-23	2,197	6	94	1,795	190	1,290	95	95
17	29-Jul-23	2,197	0	24	1,819	0	1,290	95	190
18	30-Jul-23	2,253	56	99	1,918	0	1,290	95	285
19	31-Jul-23	2,394	141	149	2,066	0	1,290	0	285
20	1-Aug-23	2,512	118	191	2,258	250	1,540	63	348
21	2-Aug-23	2,690	178	173	2,431	0	1,540	150	498
22	3-Aug-23	2,768	78	222	2,652	0	1,540	500	998
23	4-Aug-23	2,850	82	139	2,792	0	1,540	400	1,398
24	5-Aug-23	2,854	4	52	2,844	0	1,540	140	1,538
25	6-Aug-23	2,854	0	15	2,859	1,280	2,820	0	1,538
26	7-Aug-23	2,854	0	15	2,874	0	2,820	0	1,538
27	8-Aug-23	2,854	0	14	2,888	0	2,820	0	1,538
28	9-Aug-23	2,900	46	178	3,066	0	2,820	0	1,538
29	10-Aug-23	2,919	19	77	3,143	0	2,820	0	1,538
30	11-Aug-23	2,922	3	41	3,183	0	2,820	0	1,538
31	12-Aug-23	2,923	1	25	3,208	0	2,820	0	1,538
32	13-Aug-23	2,944	21	60	3,268	300	3,120	0	1,538
33	14-Aug-23	2,945	1	6	3,273	0	3,120	0	1,538
34	15-Aug-23	2,952	7	75	3,349	100	3,220	0	1,538
35	16-Aug-23	2,961	9	77	3,426	0	3,220	0	1,538
36	17-Aug-23	2,988	27	38	3,464	0	3,220	0	1,538
37	18-Aug-23	3,069	81	85	3,549	0	3,220	0	1,538
38	19-Aug-23	3,191	122	125	3,674	0	3,220	0	1,538
39	20-Aug-23	3,275	84	98	3,772	0	3,220	0	1,538
40	21-Aug-23	3,349	74	95	3,868	0	3,220	0	1,538

41	22-Aug-23	3,497	148	138	4,006	0	3,220	0	1,538
42	23-Aug-23	3,552	55	101	4,107	0	3,220	0	1,538
43	24-Aug-23	3,552	0	40	4,147	0	3,220	220	1,758
44	25-Aug-23	3,552	0	10	4,157	0	3,220	220	1,978
45	26-Aug-23	3,642	90	146	4,303	0	3,220	350	2,328
46	27-Aug-23	3,776	134	190	4,493	0	3,220	350	2,678
47	28-Aug-23	3,833	57	166	4,659	0	3,220	300	2,978
48	29-Aug-23	3,876	43	128	4,787	0	3,220	150	3,128
49	30-Aug-23	3,910	34	51	4,838	1,200	4,420	90	3,218
50	31-Aug-23	3,932	22	65	4,903	0	4,420	200	3,418
51	1-Sep-23	3,934	2	60	4,963	0	4,420	200	3,618
52	2-Sep-23	3,934	0	10	4,973	0	4,420	300	3,918
53	3-Sep-23	3,934	0	195	5,168	400	4,820	400	4,318
54	4-Sep-23	3,934	0	20	5,188	0	4,820	0	4,318
55	5-Sep-23	3,934	0	18	5,206	0	4,820	0	4,318
56	6-Sep-23	3,934	0	35	5,241	0	4,820	0	4,318
57	7-Sep-23	3,934	0	235	5,476	0	4,820	200	4,518
58	8-Sep-23	4,070	136	31	5,507	0	4,820	150	4,668
59	9-Sep-23	3,934	-136	-59	5,448	0	4,820	100	4,768
60	10-Sep-23	3,939	5	25	5,473	0	4,820	52	4,820

Figure 12-1: 12-1/4" Fixation Table

Fixation analysis								
Report No.	Date	Depth	Drilled	Can test	Cement/Waste		SodSil/Waste	
				Daily	Daily	Average	Daily	Average
				Pass/NA	kg/M3	kg/M3	kg/M3	kg/M3
1	13-Jul-23	0	0	NA	0.000	4.747	0.000	0.080
2	14-Jul-23	1,492	0	NA	0.000	4.555	0.000	0.080
3	15-Jul-23	1,492	0	Pass	0.000	4.747	0.000	0.080
4	16-Jul-23	1,510	18	Pass	0.000	4.555	0.000	0.080
5	17-Jul-23	1,556	46	Pass	0.000	4.747	0.000	0.080
6	18-Jul-23	1,567	11	Pass	0.000	4.555	0.000	0.080
7	19-Jul-23	1,756	189	Pass	0.000	4.747	0.000	0.080
8	20-Jul-23	1,908	152	Pass	6.429	4.555	0.007	0.080
9	21-Jul-23	1,960	52	Pass	0.000	4.579	0.000	0.092
10	22-Jul-23	2,010	50	Pass	0.000	4.347	0.000	0.092
11	23-Jul-23	2,084	74	Pass	2.000	4.579	0.000	0.092
12	24-Jul-23	2,115	31	Pass	0.000	4.640	0.000	0.092
13	25-Jul-23	2,115	0	Pass	0.000	4.866	0.000	0.092
14	26-Jul-23	2,161	46	Pass	0.000	4.640	0.000	0.092

15	27-Jul-23	2,191	30	Pass	2.250	4.866	0.000	0.092
16	28-Jul-23	2,197	6	Pass	2.632	4.982	0.008	0.092
17	29-Jul-23	2,197	0	Pass	0.000	5.558	0.000	0.109
18	30-Jul-23	2,253	56	Pass	0.000	5.374	0.000	0.109
19	31-Jul-23	2,394	141	Pass	0.000	5.558	0.000	0.109
20	1-Aug-23	2,512	118	Pass	4.000	5.374	0.011	0.109
21	2-Aug-23	2,690	178	Pass	0.000	5.818	0.000	0.133
22	3-Aug-23	2,768	78	Pass	0.000	5.648	0.000	0.133
23	4-Aug-23	2,850	82	Pass	0.000	5.818	0.000	0.133
24	5-Aug-23	2,854	4	Pass	0.000	5.648	0.000	0.133
25	6-Aug-23	2,854	0	Pass	3.242	5.818	0.000	0.133
26	7-Aug-23	2,854	0	Pass	0.000	6.250	0.000	0.133
27	8-Aug-23	2,854	0	Pass	0.000	6.333	0.000	0.133
28	9-Aug-23	2,900	46	Pass	0.000	6.250	0.000	0.133
29	10-Aug-23	2,919	19	Pass	0.000	6.333	0.000	0.133
30	11-Aug-23	2,922	3	Pass	0.000	6.250	0.000	0.133
31	12-Aug-23	2,923	1	Pass	0.000	6.333	0.000	0.133
32	13-Aug-23	2,944	21	Pass	6.667	6.250	0.012	0.133
33	14-Aug-23	2,945	1	Pass	0.000	6.250	0.000	0.174
34	15-Aug-23	2,952	7	Pass	10.000	6.111	0.014	0.174
35	16-Aug-23	2,961	9	Pass	0.000	5.000	0.000	0.253
36	17-Aug-23	2,988	27	Pass	0.000	4.167	0.000	0.253
37	18-Aug-23	3,069	81	Pass	0.000	5.000	0.000	0.253
38	19-Aug-23	3,191	122	Pass	0.000	4.167	0.000	0.253
39	20-Aug-23	3,275	84	Pass	0.000	5.000	0.000	0.253
40	21-Aug-23	3,349	74	Pass	0.000	4.167	0.000	0.253
41	22-Aug-23	3,497	148	Pass	0.000	5.000	0.000	0.253
42	23-Aug-23	3,552	55	Pass	0.000	4.167	0.000	0.253
43	24-Aug-23	3,552	0	Pass	0.000	5.000	0.000	0.253
44	25-Aug-23	3,552	0	Pass	0.000	4.167	0.000	0.253
45	26-Aug-23	3,642	90	Pass	0.000	5.000	0.000	0.253
46	27-Aug-23	3,776	134	Pass	0.000	4.167	0.000	0.253
47	28-Aug-23	3,833	57	Pass	0.000	5.000	0.000	0.253
48	29-Aug-23	3,876	43	Pass	0.000	4.167	0.000	0.253
49	30-Aug-23	3,910	34	Pass	3.333	5.000	0.444	0.253
50	31-Aug-23	3,932	22	Pass	0.000	5.000	0.000	0.063
51	1-Sep-23	3,934	2	Pass	0.000	5.833	0.000	0.063
52	2-Sep-23	3,934	0	Pass	0.000	5.000	0.000	0.063
53	3-Sep-23	3,934	0	Pass	5.000	5.833	0.063	0.063
54	4-Sep-23	3,934	0	Pass	0.000	0.000	0.000	0.000
55	5-Sep-23	3,934	0	Pass	0.000	6.667	0.000	0.000
56	6-Sep-23	3,934	0	Pass	0.000	0.000	0.000	0.000
57	7-Sep-23	3,934	0	Pass	0.000	6.667	0.000	0.000
58	8-Sep-23	4,070	136	Pass	0.000	0.000	0.000	0.000
59	9-Sep-23	3,934	-136	Pass	0.000	6.667	0.000	0.000
60	10-Sep-23	3,939	5	Pass	0.000	0.000	0.000	0.000

Figure 12-2: 12-1/4" Fixation Analysis

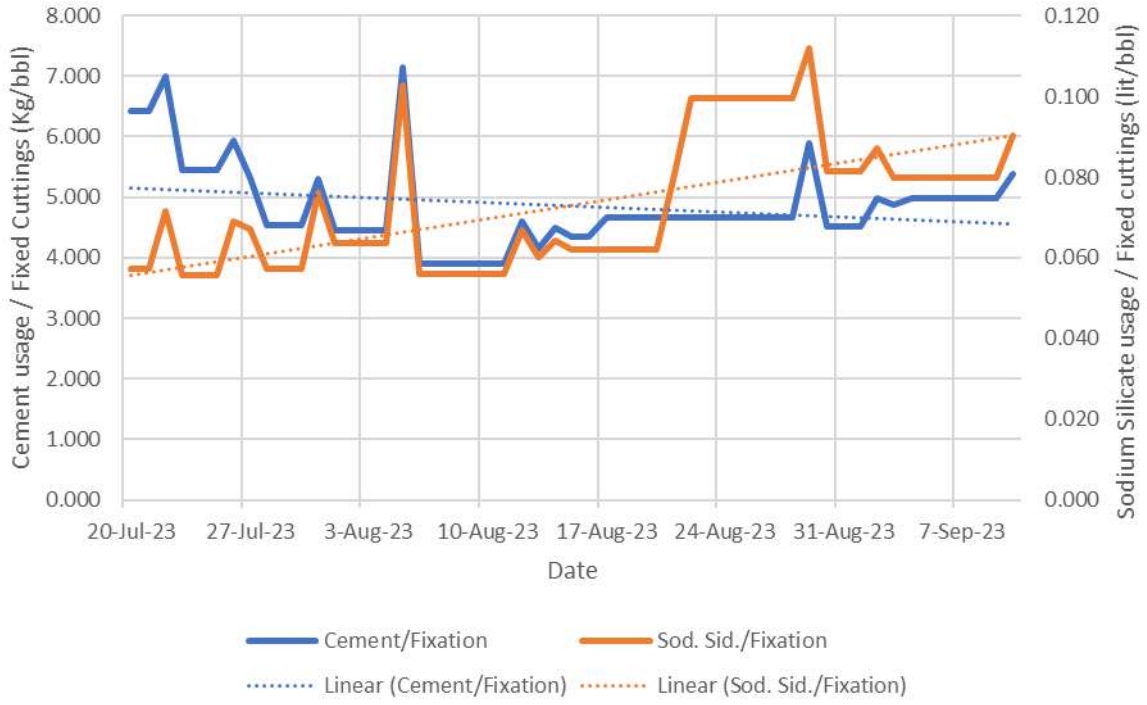


Figure 12-3: 12-1/4" Fixation Material Consumption Trend

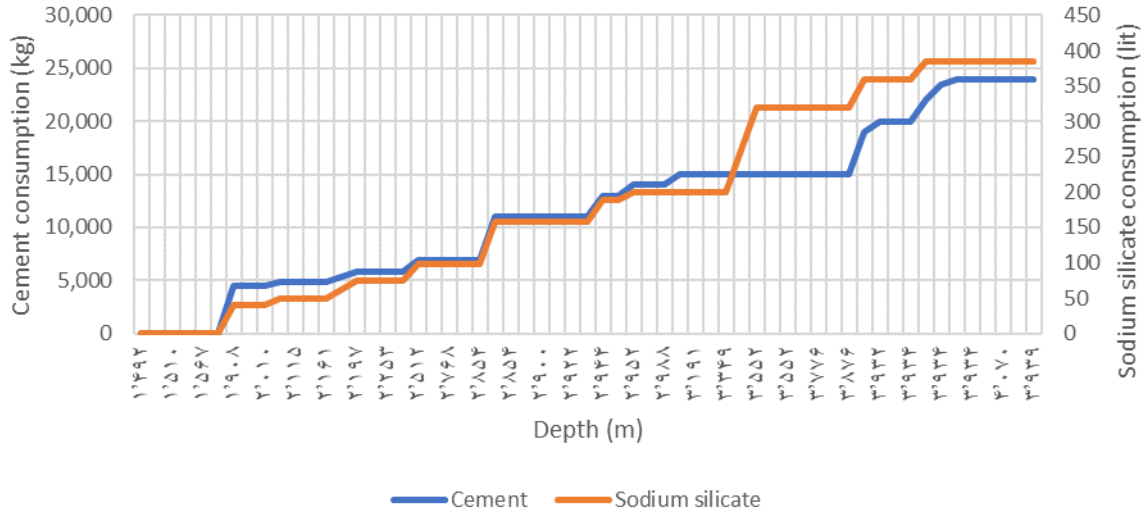


Figure 12-4: 12-1/4" Fixation Material Consumption vs. Depth

Table 12-3: AZNS-040 12-1/4" Used Material Volume

Total Waste-water (bbl)	6170
Total Dewater (bbl)	7400
Recycled (bbl)	5680
Disposed and Lost (bbl)	1392
Used PAC (kg)	225
Used Polymer (kg)	30
Used Lime (kg)	1675
Used Citric acid (kg)	25
Used Caustic (kg)	50
Used Soda ash (kg)	1845

Dewatering

Report No.	Date	Depth	Drilled	Total waste-water		Total Dewater		Recycled		Disposed and Lost	
				Daily	Cum.	Daily	Cum.	Daily	Cum.	Daily	Cum.
-	dd-M-yy	m	m	bbl.	bbl.	bbl.	bbl.	bbl.	bbl.	bbl.	bbl.
1	13-Jul-23	0	0	0	0	0	0	0	0	0	0
2	14-Jul-23	1,492	0	0	0	0	0	0	0	0	0
3	15-Jul-23	1,492	0	0	0	0	0	0	0	0	0
4	16-Jul-23	1,510	18	0	0	0	0	0	0	0	0
5	17-Jul-23	1,556	46	400	400	300	300	0	0	0	0
6	18-Jul-23	1,567	11	0	400	0	300	0	0	0	0
7	19-Jul-23	1,756	189	0	400	0	300	0	0	0	0
8	20-Jul-23	1,908	152	400	800	350	650	250	250	50	50
9	21-Jul-23	1,960	52	0	800	0	650	250	500	50	100
10	22-Jul-23	2,010	50	300	1,100	0	650	0	500	0	100
11	23-Jul-23	2,084	74	0	1,100	0	650	0	500	0	100
12	24-Jul-23	2,115	31	0	1,100	0	650	0	500	0	100
13	25-Jul-23	2,115	0	0	1,100	0	650	0	500	0	100
14	26-Jul-23	2,161	46	0	1,100	300	950	250	750	50	150
15	27-Jul-23	2,191	30	0	1,100	0	950	0	750	0	150
16	28-Jul-23	2,197	6	250	1,350	250	1,200	200	950	50	200
17	29-Jul-23	2,197	0	250	1,600	250	1,450	200	1,150	50	250
18	30-Jul-23	2,253	56	0	1,600	0	1,450	0	1,150	0	250
19	31-Jul-23	2,394	141	300	1,900	250	1,700	200	1,350	50	300
20	1-Aug-23	2,512	118	300	2,200	200	1,900	150	1,500	50	350
21	2-Aug-23	2,690	178	100	2,300	0	1,900	0	1,500	0	350
22	3-Aug-23	2,768	78	100	2,400	0	1,900	0	1,500	0	350
23	4-Aug-23	2,850	82	100	2,500	200	2,100	150	1,650	50	400
24	5-Aug-23	2,854	4	100	2,600	0	2,100	0	1,650	0	400
25	6-Aug-23	2,854	0	0	2,600	0	2,100	0	1,650	0	400

26	7-Aug-23	2,854	0	0	2,600	200	2,300	150	1,800	50	450
27	8-Aug-23	2,854	0	0	2,600	0	2,300	0	1,800	0	450
28	9-Aug-23	2,900	46	50	2,650	300	2,600	100	1,900	100	550
29	10-Aug-23	2,919	19	50	2,700	0	2,600	100	2,000	0	550
30	11-Aug-23	2,922	3	50	2,750	0	2,600	0	2,000	0	550
31	12-Aug-23	2,923	1	50	2,800	0	2,600	0	2,000	0	550
32	13-Aug-23	2,944	21	50	2,850	0	2,600	0	2,000	0	550
33	14-Aug-23	2,945	1	50	2,900	200	2,800	0	2,000	0	550
34	15-Aug-23	2,952	7	100	3,000	0	2,800	150	2,150	0	550
35	16-Aug-23	2,961	9	200	3,200	300	3,100	300	2,450	100	650
36	17-Aug-23	2,988	27	300	3,500	0	3,100	0	2,450	12	662
37	18-Aug-23	3,069	81	300	3,800	300	3,400	100	2,550	100	762
38	19-Aug-23	3,191	122	100	3,900	300	3,700	300	2,850	0	762
39	20-Aug-23	3,275	84	100	4,000	200	3,900	0	2,850	0	762
40	21-Aug-23	3,349	74	160	4,160	0	3,900	200	3,050	0	762
41	22-Aug-23	3,497	148	160	4,320	150	4,050	80	3,130	58	820
42	23-Aug-23	3,552	55	150	4,470	250	4,300	50	3,180	0	820
43	24-Aug-23	3,552	0	50	4,520	0	4,300	0	3,180	0	820
44	25-Aug-23	3,552	0	50	4,570	0	4,300	0	3,180	0	820
45	26-Aug-23	3,642	90	50	4,620	0	4,300	0	3,180	0	820
46	27-Aug-23	3,776	134	0	4,620	0	4,300	50	3,230	17	837
47	28-Aug-23	3,833	57	0	4,620	150	4,450	100	3,330	20	857
48	29-Aug-23	3,876	43	50	4,670	250	4,700	50	3,380	0	857
49	30-Aug-23	3,910	34	150	4,820	0	4,700	100	3,480	60	917
50	31-Aug-23	3,932	22	150	4,970	200	4,900	50	3,530	0	917
51	1-Sep-23	3,934	2	150	5,120	0	4,900	100	3,630	25	942
52	2-Sep-23	3,934	0	150	5,270	300	5,200	0	3,630	0	942
53	3-Sep-23	3,934	0	150	5,420	0	5,200	0	3,630	0	942
54	4-Sep-23	3,934	0	100	5,520	350	5,550	300	3,930	50	992
55	5-Sep-23	3,934	0	150	5,670	350	5,900	300	4,230	50	1,042
56	6-Sep-23	3,934	0	100	5,770	350	6,250	300	4,530	50	1,092
57	7-Sep-23	3,934	0	0	5,770	350	6,600	300	4,830	70	1,162
58	8-Sep-23	4,070	136	400	6,170	0	6,600	250	5,080	30	1,192
59	9-Sep-23	3,934	-136	0	6,170	400	7,000	300	5,380	100	1,292
60	10-Sep-23	3,939	5	0	6,170	400	7,400	300	5,680	100	1,392

Figure 12-5: 12-1/4" Dewatering Detail Report

Dewatering Analysis

Report No.	Date	Depth	Drilled	pH	Hardn.	Turb.	TSS	Nitrate	Al	Sulfate	Ca
				7-9	<800	<50	< 80	< 50	< 3	< 50	< 600
-	dd-M-yy	m	m	-	mg/L	NTU	mg/L	mg/L	mg/L	mg/L	mg/L
1	13-Jul-23	0	0	0	0	0	0	0	0	0	0
2	14-Jul-23	1,492	0	0	0	0	0	0	0	0	0
3	15-Jul-23	1,492	0	0	0	0	0	0	0	0	0
4	16-Jul-23	1,510	18	0	0	0	0	0	0	0	0
5	17-Jul-23	1,556	46	0	0	0	0	0	0	0	0
6	18-Jul-23	1,567	11	8	800	35	30	40	1	30	600
7	19-Jul-23	1,756	189	0	0	0	0	0	0	0	0

8	20-Jul-23	1,908	152	0	0	0	0	0	0	0	0
9	21-Jul-23	1,960	52	9	800	36	24	40	1	26	600
10	22-Jul-23	2,010	50	9	800	36	24	40	1	26	600
11	23-Jul-23	2,084	74	0	0	0	0	0	0	0	0
12	24-Jul-23	2,115	31	0	0	0	0	0	0	0	0
13	25-Jul-23	2,115	0	0	0	0	0	0	0	0	0
14	26-Jul-23	2,161	46	0	0	0	0	0	0	0	0
15	27-Jul-23	2,191	30	9	760	14	16	40	1	28	600
16	28-Jul-23	2,197	6	0	0	0	0	0	0	0	0
17	29-Jul-23	2,197	0	9	800	16	18	38	1	26	560
18	30-Jul-23	2,253	56	9	800	14	16	38	1	26	600
19	31-Jul-23	2,394	141	0	0	0	0	0	0	0	0
20	1-Aug-23	2,512	118	7	760	18	16	40	1	26	600
21	2-Aug-23	2,690	178	8	800	16	16	40	1	24	600
22	3-Aug-23	2,768	78	0	0	0	0	0	0	0	0
23	4-Aug-23	2,850	82	0	0	0	0	0	0	0	0
24	5-Aug-23	2,854	4	8	800	14	16	36	1	28	600
25	6-Aug-23	2,854	0	0	0	0	0	0	0	0	0
26	7-Aug-23	2,854	0	0	0	0	0	0	0	0	0
27	8-Aug-23	2,854	0	0	0	0	0	0	0	0	0
28	9-Aug-23	2,900	46	0	0	0	0	0	0	0	0
29	10-Aug-23	2,919	19	8	730	30	55	20	1	35	550
30	11-Aug-23	2,922	3	8	730	30	55	20	1	35	550
31	12-Aug-23	2,923	1	0	0	0	0	0	0	0	0
32	13-Aug-23	2,944	21	0	0	0	0	0	0	0	0
33	14-Aug-23	2,945	1	0	0	0	0	0	0	0	0
34	15-Aug-23	2,952	7	8	720	35	20	40	2	30	560
35	16-Aug-23	2,961	9	8	720	35	20	40	2	30	560
36	17-Aug-23	2,988	27	8	760	18	20	38	1	30	520
37	18-Aug-23	3,069	81	0	0	0	0	0	0	0	0
38	19-Aug-23	3,191	122	8	800	34	36	36	1	28	600
39	20-Aug-23	3,275	84	8	800	32	34	36	1	26	600
40	21-Aug-23	3,349	74	8	800	18	20	38	1	30	600
41	22-Aug-23	3,497	148	8	800	18	20	38	1	30	600
42	23-Aug-23	3,552	55	8	800	20	18	35	1	36	600
43	24-Aug-23	3,552	0	9	760	26	18	34	1	36	600
44	25-Aug-23	3,552	0	9	760	26	18	34	1	36	600
45	26-Aug-23	3,642	90	9	760	26	18	34	1	36	600
46	27-Aug-23	3,776	134	8	760	26	18	34	1	36	600
47	28-Aug-23	3,833	57	8	760	28	18	34	1	36	600
48	29-Aug-23	3,876	43	9	800	26	18	34	1	34	560
49	30-Aug-23	3,910	34	9	760	34	20	34	1	25	600
50	31-Aug-23	3,932	22	9	760	34	20	34	1	25	600
51	1-Sep-23	3,934	2	9	800	32	20	32	1	26	600
52	2-Sep-23	3,934	0	9	800	32	20	32	1	26	600
53	3-Sep-23	3,934	0	9	760	26	24	35	1	25	600
54	4-Sep-23	3,934	0	0	0	0	O	0	0	0	0
55	5-Sep-23	3,934	0	9	760	26	24	35	1	25	600
56	6-Sep-23	3,934	0	9	760	26	24	35	1	25	600
57	7-Sep-23	3,934	0	9	760	26	24	35	1	25	600
58	8-Sep-23	4,070	136	9	760	26	24	35	1	25	600
59	9-Sep-23	3,934	-136	9	760	26	24	35	1	25	600
60	10-Sep-23	3,939	5	9	760	26	24	35	1	25	600

Figure 12-6: 12-1/4" Dewatering Analysis

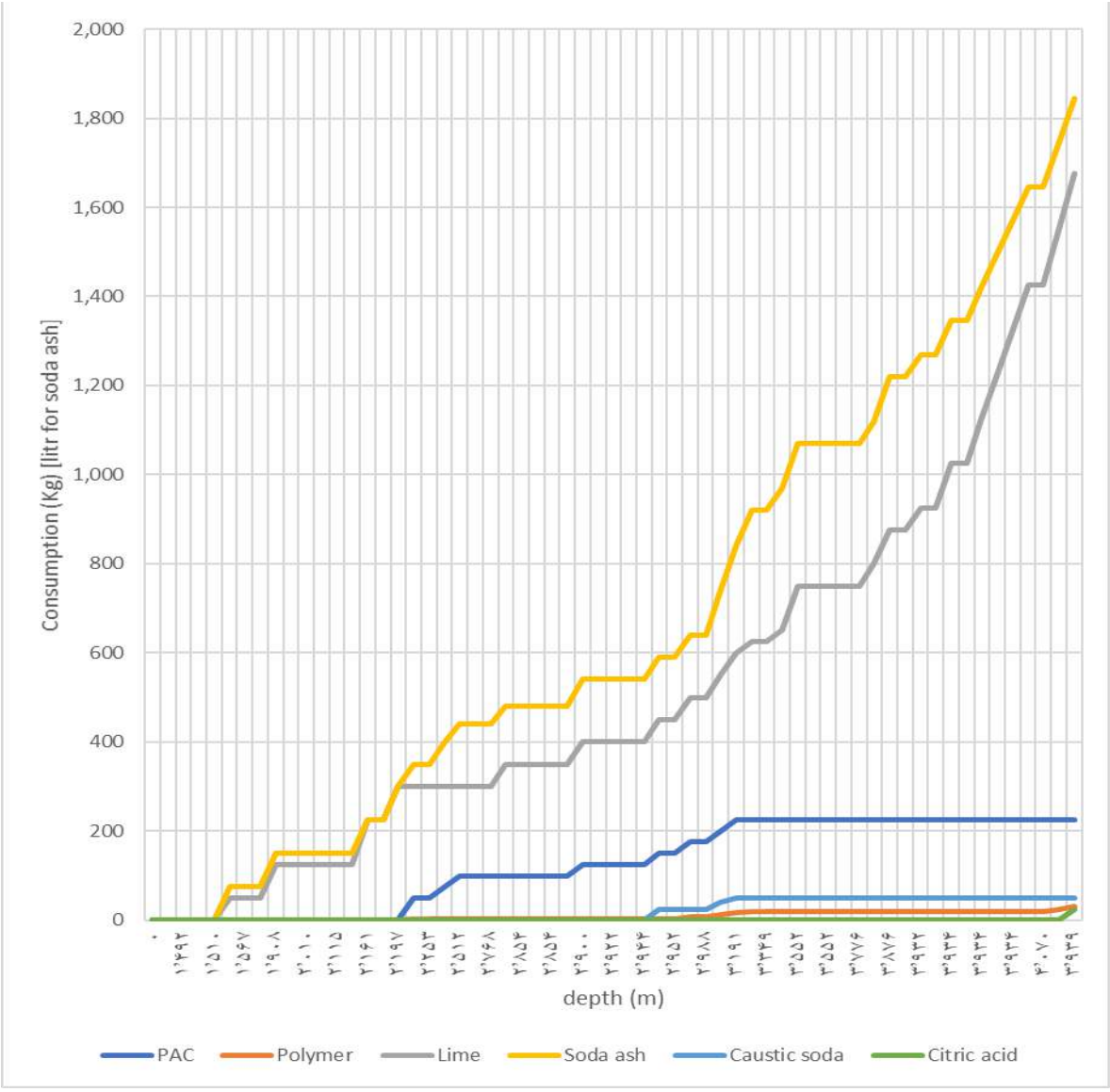


Figure 12-7: 12-1/4" Cumulative Consumption vs. Depth

8-1/2" Hole Section

Mud System	Salt Saturated KCL/Polymer/$CaCO_3$ WBM 100 – 106^{pcf}
Rig	208 OEOC
Start Date	Sep. 10th 2023
Start Depth	3939^{m}
End Date	Oct. 6th 2023
End Depth	4200^{m}
Differential Depth	261^{m}
Casing Depth	9 5/8” @ 3986^{m}
Duration	26 Days

Total Cuttings (bbl)	870
Total Fixed Cuttings (bbl)	1100
Total Disposed Cuttings (bbl)	1100
Total Used Cement (kg)	10100
Total Used Sodium silicate (kg)	235

Table 12-4: AZNS-040 8-1/2" Hole Waste Management Information

Fixation									
Report No.	Date	Depth	Drilled	Total Cuttings		Fixed Cuttings		Sent to Disposal	
				Daily	Cum.	Daily	Cum.	Daily	Cum.
-	dd-M-yy	m	m	bbl.	bbl.	bbl.	bbl.	bbl.	bbl.
61	11-Sep-23	3,939	0	20	5,493	300	5,120	0	4,820
62	12-Sep-23	3,939	0	0	5,493	100	5,220	0	4,820
63	13-Sep-23	3,956	17	108	5,601	0	5,220	0	4,820
64	14-Sep-23	4,013	57	53	5,654	0	5,220	0	4,820
65	15-Sep-23	4,070	57	71	5,725	0	5,220	0	4,820
66	16-Sep-23	4,129	59	149	5,874	0	5,220	130	4,950
67	17-Sep-23	4,186	57	80	5,954	0	5,220	130	5,080
68	18-Sep-23	4,200	14	11	5,965	0	5,220	70	5,150
69	19-Sep-23	4,200	0	50	6,015	0	5,220	50	5,200
70	20-Sep-23	4,200	0	0	6,015	400	5,620	20	5,220
71	21-Sep-23	4,200	0	0	6,015	0	5,620	20	5,240
72	22-Sep-23	4,200	0	15	6,030	300	5,920	0	5,240
73	23-Sep-23	4,200	0	30	6,060	0	5,920	0	5,240
74	24-Sep-23	4,200	0	0	6,060	0	5,920	0	5,240
75	25-Sep-23	4,200	0	70	6,130	0	5,920	0	5,240
76	26-Sep-23	4,200	0	0	6,130	0	5,920	0	5,240
77	27-Sep-23	4,200	0	97	6,227	0	5,920	0	5,240
78	28-Sep-23	4,200	0	33	6,260	0	5,920	200	5,440
79	29-Sep-23	4,200	0	50	6,310	0	5,920	200	5,640
80	30-Sep-23	4,200	0	0	6,310	0	5,920	150	5,790
81	1-Oct-23	4,200	0	0	6,310	0	5,920	130	5,920
82	2-Oct-23	4,200	0	0	6,310	300	6,220	0	5,920
83	3-Oct-23	4,200	0	0	6,310	0	6,220	0	5,920
84	4-Oct-23	4,200	0	0	6,310	0	6,220	0	5,920
85	5-Oct-23	4,200	0	48	6,358	0	6,220	0	5,920
86	6-Oct-23	4,200	0	5	6,363	0	6,220	100	6,020

Figure 12-8: 8-1/2" Fixation Table

Fixation analysis								
Report No.	Date	Depth	Drilled	Can test	Cement/Waste		SodSil/Waste	
				Daily	Daily	Average	Daily	Average
				Pass/NA	kg/M3	kg/M3	kg/M3	kg/M3
61	11-Sep-23	3,939	0	Pass	0.000	0.000	0.000	0.000
62	12-Sep-23	3,939	0	NA	0.000	0.000	0.000	0.000
63	13-Sep-23	3,956	17	Pass	0.000	0.000	0.000	0.000
64	14-Sep-23	4,013	57	Pass	0.000	0.000	0.000	0.000
65	15-Sep-23	4,070	57	Pass	0.000	0.000	0.000	0.000
66	16-Sep-23	4,129	59	Pass	0.000	0.000	0.000	0.000
67	17-Sep-23	4,186	57	Pass	0.000	0.000	0.000	0.000
68	18-Sep-23	4,200	14	Pass	0.000	0.000	0.000	0.000
69	19-Sep-23	4,200	0	Pass	0.000	0.000	0.000	0.000
70	20-Sep-23	4,200	0	NA	0.000	0.000	0.000	0.000
71	21-Sep-23	4,200	0	NA	0.000	0.000	0.500	0.500
72	22-Sep-23	4,200	0	Pass	0.000	0.000	0.000	0.500
73	23-Sep-23	4,200	0	Pass	0.000	0.000	0.000	0.500
74	24-Sep-23	4,200	0	NA	0.000	0.000	0.000	0.500
75	25-Sep-23	4,200	0	Pass	0.000	0.000	0.000	0.500
76	26-Sep-23	4,200	0	NA	0.000	0.000	0.000	0.500
77	27-Sep-23	4,200	0	Pass	0.000	0.000	0.000	0.500
78	28-Sep-23	4,200	0	Pass	0.000	0.000	0.000	0.500
79	29-Sep-23	4,200	0	Pass	0.000	0.000	0.000	0.500
80	30-Sep-23	4,200	0	NA	0.000	0.000	0.000	0.500
81	1-Oct-23	4,200	0	NA	0.000	0.000	0.000	0.500
82	2-Oct-23	4,200	0	NA	0.000	0.000	0.000	0.500
83	3-Oct-23	4,200	0	NA	0.000	0.000	0.000	0.500
84	4-Oct-23	4,200	0	NA	0.000	0.000	0.000	0.500
85	5-Oct-23	4,200	0	Pass	0.000	0.000	0.000	0.500
86	6-Oct-23	4,200	0	Pass	0.000	0.000	0.000	0.500

Figure 12-9: 8-1/2" Fixation Analysis



Figure 12-10: 8-1/2" Fixation material consumption trend

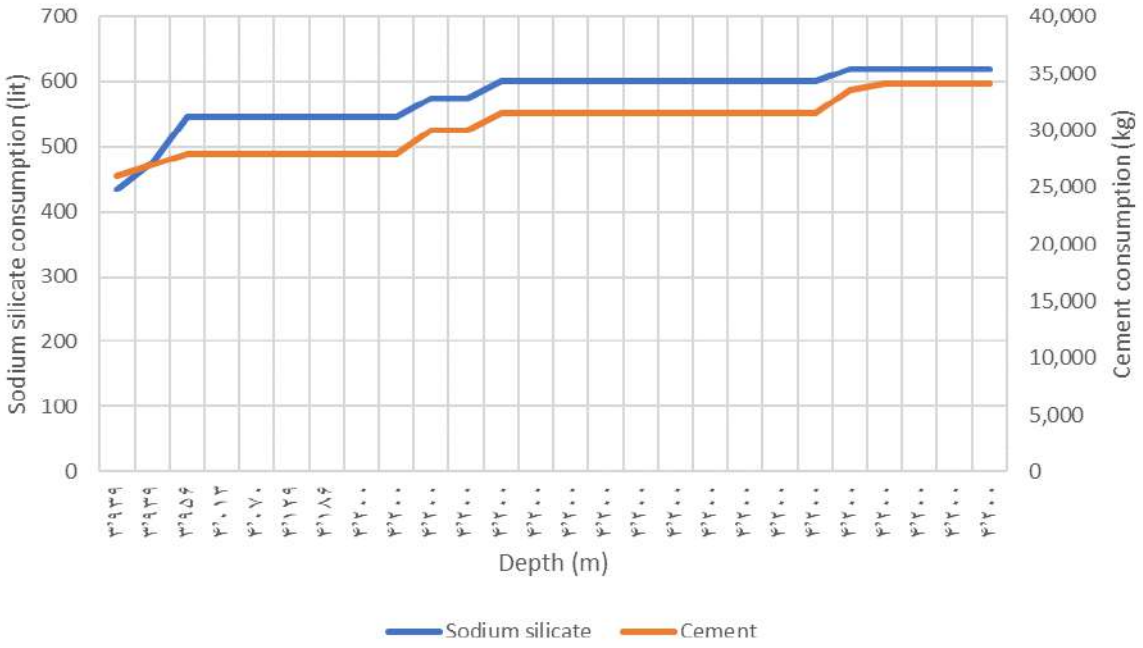


Figure 12-11: 8-1/2" Fixation material consumption vs. depth

Total Waste-water (bbl)	1650
Total Dewater (bbl)	1500
Recycled (bbl)	1350
Disposed and Lost (bbl)	460
Used PAC (kg)	25
Used Polymer (kg)	5
Used Lime (kg)	500
Used Citric acid (kg)	25
Used Caustic (kg)	0
Used Soda ash (kg)	400

Table 12-5: AZNS-040 8-1/2" Used Material Volume

Dewatering

Report No.	Date	Depth	Drilled	Total waste-water		Total Dewater		Recycled		Disposed and Lost	
				Daily	Cum.	Daily	Cum.	Daily	Cum.	Daily	Cum.
-	dd-M-yy	m	m	bbl.	bbl.	bbl.	bbl.	bbl.	bbl.	bbl.	bbl.
61	11-Sep-23	3,939	0	0	6,170	400	7,800	200	5,880	0	1,392
62	12-Sep-23	3,939	0	0	6,170	0	7,800	0	5,880	0	1,392
63	13-Sep-23	3,956	17	100	6,270	0	7,800	200	6,080	0	1,392
64	14-Sep-23	4,013	57	0	6,270	0	7,800	0	6,080	0	1,392
65	15-Sep-23	4,070	57	0	6,270	0	7,800	0	6,080	0	1,392
66	16-Sep-23	4,129	59	0	6,270	0	7,800	0	6,080	110	1,502
67	17-Sep-23	4,186	57	0	6,270	0	7,800	0	6,080	0	1,502
68	18-Sep-23	4,200	14	0	6,270	0	7,800	0	6,080	0	1,502
69	19-Sep-23	4,200	0	250	6,520	250	8,050	100	6,180	0	1,502
70	20-Sep-23	4,200	0	150	6,670	0	8,050	100	6,280	50	1,552
71	21-Sep-23	4,200	0	100	6,770	0	8,050	0	6,280	0	1,552
72	22-Sep-23	4,200	0	0	6,770	0	8,050	0	6,280	0	1,552
73	23-Sep-23	4,200	0	0	6,770	200	8,250	150	6,430	50	1,602
74	24-Sep-23	4,200	0	200	6,970	200	8,450	150	6,580	50	1,652
75	25-Sep-23	4,200	0	200	7,170	200	8,650	150	6,730	50	1,702
76	26-Sep-23	4,200	0	250	7,420	250	8,900	200	6,930	50	1,752
77	27-Sep-23	4,200	0	200	7,620	0	8,900	0	6,930	0	1,752
78	28-Sep-23	4,200	0	0	7,620	0	8,900	0	6,930	0	1,752
79	29-Sep-23	4,200	0	0	7,620	200	9,100	150	7,080	50	1,802
80	30-Sep-23	4,200	0	0	7,620	0	9,100	0	7,080	0	1,802
81	1-Oct-23	4,200	0	0	7,620	0	9,100	0	7,080	0	1,802
82	2-Oct-23	4,200	0	0	7,620	0	9,100	0	7,080	0	1,802
83	3-Oct-23	4,200	0	0	7,620	0	9,100	0	7,080	0	1,802
84	4-Oct-23	4,200	0	0	7,620	0	9,100	0	7,080	0	1,802
85	5-Oct-23	4,200	0	0	7,620	0	9,100	0	7,080	0	1,802
86	6-Oct-23	4,200	0	200	7,820	200	9,300	150	7,230	50	1,852

Figure 12-12: 8-1/2" Dewatering Detail Report

Dewatering Analysis											
Report No.	Date	Depth	Drilled	pH	Hardn.	Turb.	TSS	Nitrate	Al	Sulfate	Ca
				7-9	<800	<50	< 80	< 50	< 3	< 50	< 600
-	dd-M-yy	m	m	-	mg/L	NTU	mg/L	mg/L	mg/L	mg/L	mg/L
61	11-Sep-23	3,939	0	9	760	26	24	35	1	25	600
62	12-Sep-23	3,939	0	9	760	26	24	35	1	25	600
63	13-Sep-23	3,956	17	9	760	26	24	35	1	25	600
64	14-Sep-23	4,013	57	0	0	0	0	0	0	0	0
65	15-Sep-23	4,070	57	0	0	0	0	0	0	0	0
66	16-Sep-23	4,129	59	0	0	0	0	0	0	0	0
67	17-Sep-23	4,186	57	0	0	0	0	0	0	0	0
68	18-Sep-23	4,200	14	0	0	0	0	0	0	0	0
69	19-Sep-23	4,200	0	0	0	0	0	0	0	0	0
70	20-Sep-23	4,200	0	8	760	24	26	34	1	35	560
71	21-Sep-23	4,200	0	8	760	24	26	34	1	35	560
72	22-Sep-23	4,200	0	0	0	0	0	0	0	0	0
73	23-Sep-23	4,200	0	0	0	0	0	0	0	0	0
74	24-Sep-23	4,200	0	8	800	30	32	32	1	36	600
75	25-Sep-23	4,200	0	8	800	30	32	32	1	36	600
76	26-Sep-23	4,200	0	8	800	30	32	34	1	35	560
77	27-Sep-23	4,200	0	9	800	32	34	34	1	36	600
78	28-Sep-23	4,200	0	0	0	0	0	0	0	0	0
79	29-Sep-23	4,200	0	0	0	0	0	0	0	0	0
80	30-Sep-23	4,200	0	8	800	26	24	36	1	28	560
81	1-Oct-23	4,200	0	0	0	0	0	0	0	0	0
82	2-Oct-23	4,200	0	0	0	0	0	0	0	0	0
83	3-Oct-23	4,200	0	0	0	0	0	0	0	0	0
84	4-Oct-23	4,200	0	0	0	0	0	0	0	0	0
85	5-Oct-23	4,200	0	0	0	0	0	0	0	0	0
86	6-Oct-23	4,200	0	0	0	0	0	0	0	0	0

Figure 12-13: 8-1/2" Dewatering Analysis

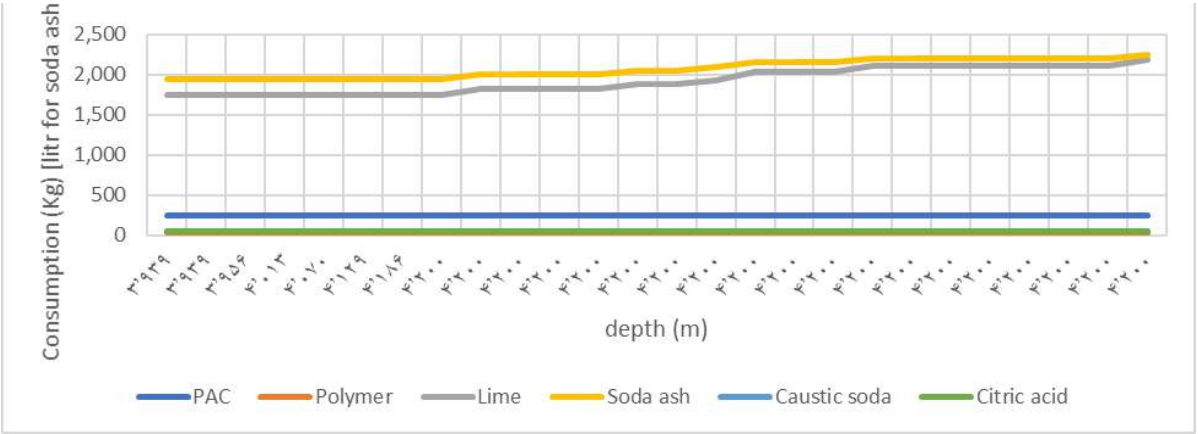


Figure 12-14: 8-1/2" Cumulative chemical consumption vs. depth

Completion

Mud System	-
Rig	208 OEOC
Start Date	Oct. 7th 2023
Start Depth	4200m
End Date	Dec. 26th 2023
End Depth	4200m
Differential Depth	0 m
Casing Depth	-
Duration	80 Days

Total Cuttings (bbl)	2140
Total Fixed Cuttings (bbl)	2103
Total Disposed Cuttings (bbl)	1983
Total Used Cement (kg)	18400
Total Used Sodium silicate (kg)	180

Table 12-6: AZNS-040 Completion Section Waste Management Information

Fixation									
Report No.	Date	Depth	Drilled	Total Cuttings		Fixed Cuttings		Sent to Disposal	
				Daily	Cum.	Daily	Cum.	Daily	Cum.
-	dd-M-yy	m	m	bbl.	bbl.	bbl.	bbl.	bbl.	bbl.
87	7-Oct-23	4,200	0	0	6,363	0	6,220	100	6,120
88	8-Oct-23	4,200	0	10	6,373	0	6,220	70	6,190
89	9-Oct-23	4,200	0	0	6,373	0	6,220	30	6,220
90	10-Oct-23	4,200	0	0	6,373	100	6,320	0	6,220
91	11-Oct-23	4,200	0	160	6,533	0	6,320	0	6,220
92	12-Oct-23	4,200	0	5	6,538	0	6,320	0	6,220
93	13-Oct-23	4,200	0	10	6,548	0	6,320	0	6,220
94	14-Oct-23	4,200	0	60	6,608	0	6,320	0	6,220
95	15-Oct-23	4,200	0	100	6,708	0	6,320	0	6,220
96	16-Oct-23	4,200	0	0	6,708	0	6,320	0	6,220
97	17-Oct-23	4,200	0	10	6,718	0	6,320	0	6,220
98	18-Oct-23	4,200	0	0	6,718	0	6,320	0	6,220
99	19-Oct-23	4,200	0	0	6,718	0	6,320	0	6,220
100	20-Oct-23	4,200	0	25	6,743	0	6,320	0	6,220

101	21-Oct-23	4,200	0	0	6,743	0	6,320	0	6,220
102	22-Oct-23	4,200	0	0	6,743	0	6,320	0	6,220
103	23-Oct-23	4,200	0	90	6,833	0	6,320	0	6,220
104	24-Oct-23	4,200	0	270	7,103	0	6,320	0	6,220
105	25-Oct-23	4,200	0	85	7,188	0	6,320	0	6,220
106	26-Oct-23	4,200	0	0	7,188	0	6,320	0	6,220
107	27-Oct-23	4,200	0	40	7,228	0	6,320	0	6,220
108	28-Oct-23	4,200	0	0	7,228	0	6,320	0	6,220
109	29-Oct-23	4,200	0	513	7,741	0	6,320	0	6,220
110	30-Oct-23	4,200	0	36	7,777	0	6,320	0	6,220
111	31-Oct-23	4,200	0	38	7,815	0	6,320	0	6,220
112	1-Nov-23	4,200	0	74	7,889	0	6,320	0	6,220
113	2-Nov-23	4,200	0	20	7,909	0	6,320	0	6,220
114	3-Nov-23	4,200	0	48	7,957	0	6,320	0	6,220
115	4-Nov-23	4,200	0	0	7,957	80	6,400	60	6,280
116	5-Nov-23	4,200	0	10	7,967	0	6,400	40	6,320
117	6-Nov-23	4,200	0	83	8,050	0	6,400	30	6,350
118	7-Nov-23	4,200	0	6	8,056	0	6,400	30	6,380
119	8-Nov-23	4,200	0	7	8,063	0	6,400	20	6,400
120	9-Nov-23	4,200	0	50	8,113	400	6,800	0	6,400
121	10-Nov-23	4,200	0	0	8,113	0	6,800	0	6,400
122	11-Nov-23	4,200	0	40	8,153	0	6,800	0	6,400
123	12-Nov-23	4,200	0	40	8,193	0	6,800	0	6,400
124	13-Nov-23	4,200	0	18	8,211	0	6,800	0	6,400
125	14-Nov-23	4,200	0	0	8,211	0	6,800	0	6,400
126	15-Nov-23	4,200	0	0	8,211	0	6,800	0	6,400
127	16-Nov-23	4,200	0	0	8,211	0	6,800	0	6,400
128	17-Nov-23	4,200	0	75	8,286	0	6,800	0	6,400
129	18-Nov-23	4,200	0	90	8,376	0	6,800	0	6,400
130	19-Nov-23	4,200	0	0	8,376	0	6,800	0	6,400
131	20-Nov-23	4,200	0	0	8,376	0	6,800	0	6,400
132	21-Nov-23	4,200	0	0	8,376	0	6,800	0	6,400
133	22-Nov-23	4,200	0	0	8,376	0	6,800	0	6,400
134	23-Nov-23	4,200	0	0	8,376	0	6,800	0	6,400
135	24-Nov-23	4,200	0	0	8,376	0	6,800	0	6,400
136	25-Nov-23	4,200	0	0	8,376	0	6,800	90	6,490
137	26-Nov-23	4,200	0	0	8,376	0	6,800	0	6,490
138	27-Nov-23	4,200	0	0	8,376	0	6,800	0	6,490
139	28-Nov-23	4,200	0	73	8,449	0	6,800	0	6,490
140	29-Nov-23	4,200	0	0	8,449	0	6,800	24	6,514
141	30-Nov-23	4,200	0	0	8,449	0	6,800	62	6,576
142	1-Dec-23	4,200	0	0	8,449	300	7,100	110	6,686
143	2-Dec-23	4,200	0	0	8,449	0	7,100	70	6,756
144	3-Dec-23	4,200	0	0	8,449	0	7,100	12	6,768
145	4-Dec-23	4,200	0	0	8,449	0	7,100	32	6,800
146	5-Dec-23	4,200	0	54	8,503	550	7,650	0	6,800

147	6-Dec-23	4,200	0	0	8,503	0	7,650	0	6,800
148	7-Dec-23	4,200	0	0	8,503	0	7,650	0	6,800
149	8-Dec-23	4,200	0	0	8,503	0	7,650	0	6,800
150	9-Dec-23	4,200	0	0	8,503	0	7,650	0	6,800
151	10-Dec-23	4,200	0	0	8,503	0	7,650	0	6,800
152	11-Dec-23	4,200	0	0	8,503	0	7,650	0	6,800
153	12-Dec-23	4,200	0	0	8,503	0	7,650	0	6,800
154	13-Dec-23	4,200	0	0	8,503	0	7,650	100	6,900
155	14-Dec-23	4,200	0	0	8,503	0	7,650	200	7,100
156	15-Dec-23	4,200	0	0	8,503	0	7,650	200	7,300
157	16-Dec-23	4,200	0	0	8,503	0	7,650	150	7,450
158	17-Dec-23	4,200	0	0	8,503	673	8,323	100	7,550
159	18-Dec-23	4,200	0	0	8,503	0	8,323	50	7,600
160	19-Dec-23	4,200	0	0	8,503	0	8,323	50	7,650
161	20-Dec-23	4,200	0	0	8,503	0	8,323	73	7,723
162	21-Dec-23	4,200	0	0	8,503	0	8,323	80	7,803
163	22-Dec-23	4,200	0	0	8,503	0	8,323	80	7,883
164	23-Dec-23	4,200	0	0	8,503	0	8,323	60	7,943
165	24-Dec-23	4,200	0	0	8,503	0	8,323	80	8,023
166	23-Dec-23	4,200	0	0	8,503	0	8,323	80	8,103

Figure 12-15: Completion Fixation Detail Table

Fixation analysis								
Report No.	Date	Depth	Drilled	Can test	Cement/Waste		SodSil/Waste	
				Daily	Daily	Average	Daily	Average
				Pass/NA	kg/M3	kg/M3	kg/M3	kg/M3
87	7-Oct-23	4,200	0	NA	0.000	6.733	0.000	0.019
88	8-Oct-23	4,200	0	Pass	0.000	6.733	0.000	0.019
89	9-Oct-23	4,200	0	NA	0.000	6.733	0.000	0.019
90	10-Oct-23	4,200	0	NA	0.000	6.733	0.000	0.019
91	11-Oct-23	4,200	0	Pass	0.000	6.733	0.000	0.019
92	12-Oct-23	4,200	0	Pass	0.000	6.733	0.000	0.019
93	13-Oct-23	4,200	0	Pass	0.000	6.733	0.000	0.019
94	14-Oct-23	4,200	0	Pass	0.000	6.733	0.000	0.019
95	15-Oct-23	4,200	0	Pass	0.000	6.733	0.000	0.019
96	16-Oct-23	4,200	0	NA	0.000	6.733	0.000	0.019
97	17-Oct-23	4,200	0	Pass	0.000	6.733	0.000	0.019
98	18-Oct-23	4,200	0	NA	0.000	6.733	0.000	0.019
99	19-Oct-23	4,200	0	NA	0.000	6.733	0.000	0.019
100	20-Oct-23	4,200	0	Pass	0.000	6.733	0.000	0.019
101	21-Oct-23	4,200	0	NA	0.000	6.733	0.000	0.019
102	22-Oct-23	4,200	0	NA	0.000	6.733	0.000	0.019
103	23-Oct-23	4,200	0	Pass	0.000	6.733	0.000	0.019
104	24-Oct-23	4,200	0	Pass	0.000	6.733	0.000	0.019
105	25-Oct-23	4,200	0	Pass	0.000	6.733	0.000	0.019

106	26-Oct-23	4,200	0	NA	0.000	6.733	0.000	0.019
107	27-Oct-23	4,200	0	Pass	0.000	6.733	0.000	0.019
108	28-Oct-23	4,200	0	NA	0.000	6.733	0.000	0.019
109	29-Oct-23	4,200	0	Pass	0.000	6.733	0.000	0.019
110	30-Oct-23	4,200	0	Pass	0.000	6.733	0.000	0.019
111	31-Oct-23	4,200	0	Pass	0.000	6.733	0.000	0.019
112	1-Nov-23	4,200	0	Pass	0.000	6.733	0.000	0.019
113	2-Nov-23	4,200	0	Pass	0.000	6.733	0.000	0.019
114	3-Nov-23	4,200	0	Pass	0.000	6.733	0.000	0.019
115	4-Nov-23	4,200	0	NA	0.000	6.733	0.000	0.019
116	5-Nov-23	4,200	0	Pass	0.000	6.733	0.000	0.019
117	6-Nov-23	4,200	0	Pass	0.000	6.733	0.000	0.019
118	7-Nov-23	4,200	0	Pass	0.000	6.733	0.000	0.019
119	8-Nov-23	4,200	0	Pass	0.000	6.733	0.000	0.019
120	9-Nov-23	4,200	0	Pass	0.000	6.733	0.000	0.019
121	10-Nov-23	4,200	0	NA	0.000	6.733	0.000	0.019
122	11-Nov-23	4,200	0	Pass	0.000	6.733	0.000	0.019
123	12-Nov-23	4,200	0	Pass	0.000	6.733	0.000	0.019
124	13-Nov-23	4,200	0	Pass	0.000	6.733	0.000	0.019
125	14-Nov-23	4,200	0	NA	0.000	6.733	0.000	0.019
126	15-Nov-23	4,200	0	NA	0.000	6.733	0.000	0.019
127	16-Nov-23	4,200	0	NA	0.000	6.733	0.000	0.019
128	17-Nov-23	4,200	0	Pass	0.000	6.733	0.000	0.019
129	18-Nov-23	4,200	0	Pass	0.000	6.733	0.000	0.019
130	19-Nov-23	4,200	0	NA	0.000	6.733	0.000	0.019
131	20-Nov-23	4,200	0	NA	0.000	6.733	0.000	0.019
132	21-Nov-23	4,200	0	NA	0.000	6.733	0.000	0.019
133	22-Nov-23	4,200	0	NA	0.000	6.733	0.000	0.019
134	23-Nov-23	4,200	0	NA	0.000	6.733	0.000	0.019
135	24-Nov-23	4,200	0	NA	0.000	6.733	0.000	0.019
136	25-Nov-23	4,200	0	NA	0.000	6.733	0.000	0.019
137	26-Nov-23	4,200	0	NA	0.000	6.733	0.000	0.019
138	27-Nov-23	4,200	0	NA	0.000	6.733	0.000	0.019
139	28-Nov-23	4,200	0	Pass	0.000	6.733	0.000	0.019
140	29-Nov-23	4,200	0	NA	0.000	6.733	0.000	0.019
141	30-Nov-23	4,200	0	NA	0.000	6.733	0.000	0.019
142	1-Dec-23	4,200	0	NA	0.000	6.733	0.000	0.019
143	2-Dec-23	4,200	0	NA	0.000	6.733	0.000	0.019
144	3-Dec-23	4,200	0	NA	0.000	6.733	0.000	0.019
145	4-Dec-23	4,200	0	NA	0.000	6.733	0.000	0.019
146	5-Dec-23	4,200	0	Pass	0.000	6.733	0.000	0.019
147	6-Dec-23	4,200	0	NA	0.000	6.733	0.000	0.019
148	7-Dec-23	4,200	0	NA	0.000	6.733	0.000	0.019
149	8-Dec-23	4,200	0	NA	0.000	6.733	0.000	0.019
150	9-Dec-23	4,200	0	NA	0.000	6.733	0.000	0.019

151	10-Dec-23	4,200	0	NA	0.000	6.733	0.000	0.019
152	11-Dec-23	4,200	0	NA	0.000	6.733	0.000	0.019
153	12-Dec-23	4,200	0	NA	0.000	6.733	0.000	0.019
154	13-Dec-23	4,200	0	NA	0.000	6.733	0.000	0.019
155	14-Dec-23	4,200	0	NA	0.000	6.733	0.000	0.019
156	15-Dec-23	4,200	0	NA	0.000	6.733	0.000	0.019
157	16-Dec-23	4,200	0	NA	0.000	6.733	0.000	0.019
158	17-Dec-23	4,200	0	NA	0.000	6.733	0.000	0.019
159	18-Dec-23	4,200	0	NA	0.000	6.733	0.000	0.019
160	19-Dec-23	4,200	0	NA	0.000	6.733	0.000	0.019
161	20-Dec-23	4,200	0	NA	0.000	6.733	0.000	0.019
162	21-Dec-23	4,200	0	NA	0.000	6.733	0.000	0.019
163	22-Dec-23	4,200	0	NA	0.000	6.733	0.000	0.019
164	23-Dec-23	4,200	0	NA	0.000	6.733	0.000	0.019
165	24-Dec-23	4,200	0	NA	0.000	6.733	0.000	0.019
166	23-Dec-23	4,200	0	NA	0.000	6.733	0.000	0.019

Figure 12-16: Completion Fixation Analysis

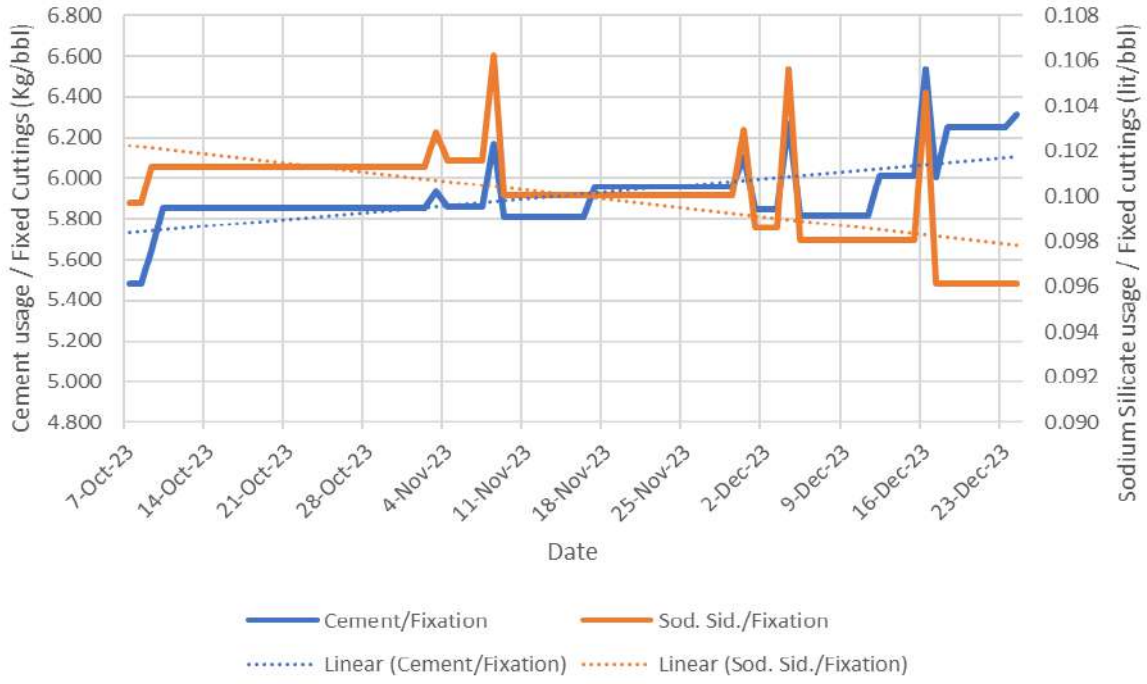


Figure 12-17: Completion Fixation Material Consumption Trend

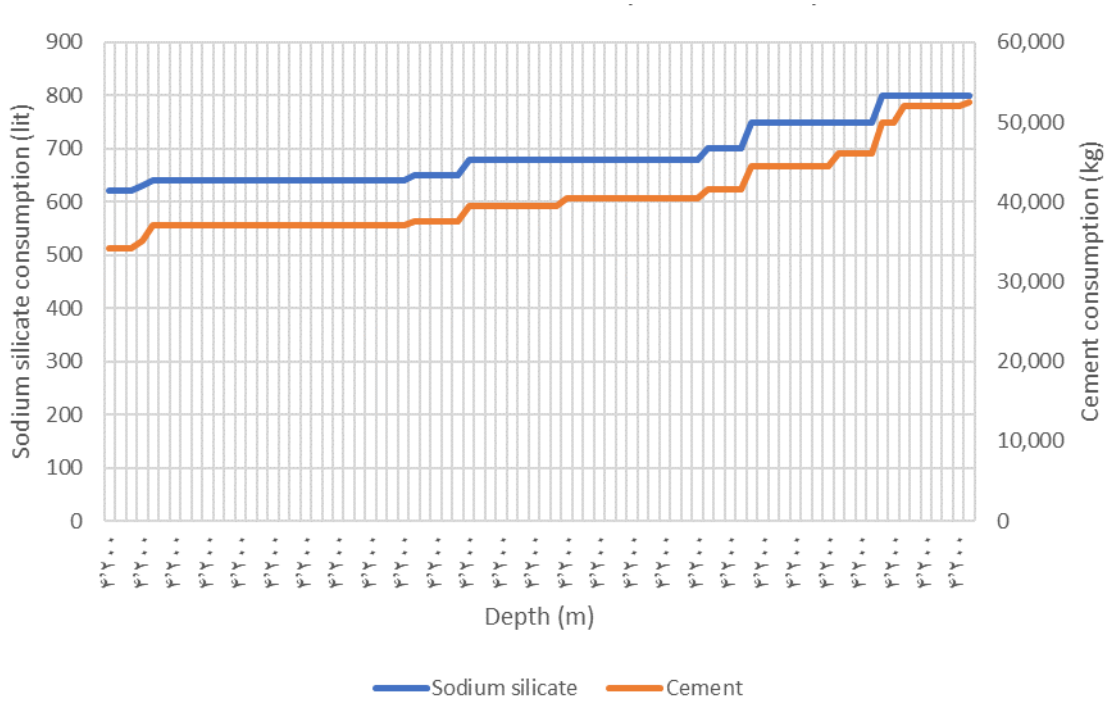


Figure 12-18: Completion Fixation Material Consumption vs. Depth.

Table 12-7: AZNS-040 Completion Used Material Volume

Total Waste-water (bbl)	2910
Total Dewater (bbl)	2410
Recycled (bbl)	2610
Disposed and Lost (bbl)	350
Used PAC (kg)	0
Used Polymer (kg)	20
Used Lime (kg)	550
Used Citric acid (kg)	25
Used Caustic (kg)	0
Used Soda ash (kg)	475

Dewatering											
Report No.	Date	Depth	Drilled	Total waste-water		Total Dewater		Recycled		Disposed and Lost	
				Daily	Cum.	Daily	Cum.	Daily	Cum.	Daily	Cum.
-	dd-M-yy	m	m	bbl.	bbl.	bbl.	bbl.	bbl.	bbl.	bbl.	bbl.
87	7-Oct-23	4,200	0	120	7,940	120	9,420	70	7,300	50	1,902
88	8-Oct-23	4,200	0	20	7,960	0	9,420	0	7,300	0	1,902
89	9-Oct-23	4,200	0	20	7,980	0	9,420	0	7,300	0	1,902
90	10-Oct-23	4,200	0	30	8,010	0	9,420	0	7,300	0	1,902
91	11-Oct-23	4,200	0	30	8,040	0	9,420	0	7,300	0	1,902
92	12-Oct-23	4,200	0	10	8,050	0	9,420	0	7,300	0	1,902
93	13-Oct-23	4,200	0	100	8,150	200	9,620	150	7,450	50	1,952
94	14-Oct-23	4,200	0	50	8,200	0	9,620	0	7,450	0	1,952
95	15-Oct-23	4,200	0	100	8,300	0	9,620	0	7,450	0	1,952
96	16-Oct-23	4,200	0	300	8,600	250	9,870	200	7,650	50	2,002
97	17-Oct-23	4,200	0	200	8,800	250	10,120	200	7,850	50	2,052
98	18-Oct-23	4,200	0	100	8,900	0	10,120	0	7,850	0	2,052
99	19-Oct-23	4,200	0	100	9,000	0	10,120	0	7,850	0	2,052
100	20-Oct-23	4,200	0	200	9,200	200	10,320	150	8,000	50	2,102
101	21-Oct-23	4,200	0	100	9,300	0	10,320	0	8,000	0	2,102
102	22-Oct-23	4,200	0	100	9,400	0	10,320	0	8,000	0	2,102
103	23-Oct-23	4,200	0	50	9,450	0	10,320	0	8,000	0	2,102
104	24-Oct-23	4,200	0	0	9,450	0	10,320	0	8,000	0	2,102
105	25-Oct-23	4,200	0	0	9,450	0	10,320	0	8,000	0	2,102
106	26-Oct-23	4,200	0	0	9,450	0	10,320	0	8,000	0	2,102
107	27-Oct-23	4,200	0	0	9,450	0	10,320	0	8,000	0	2,102
108	28-Oct-23	4,200	0	0	9,450	0	10,320	0	8,000	0	2,102
109	29-Oct-23	4,200	0	0	9,450	0	10,320	0	8,000	0	2,102
110	30-Oct-23	4,200	0	0	9,450	0	10,320	0	8,000	0	2,102
111	31-Oct-23	4,200	0	0	9,450	0	10,320	0	8,000	0	2,102
112	1-Nov-23	4,200	0	0	9,450	0	10,320	0	8,000	0	2,102
113	2-Nov-23	4,200	0	0	9,450	0	10,320	0	8,000	0	2,102
114	3-Nov-23	4,200	0	0	9,450	0	10,320	0	8,000	0	2,102
115	4-Nov-23	4,200	0	0	9,450	0	10,320	0	8,000	0	2,102
116	5-Nov-23	4,200	0	0	9,450	260	10,580	170	8,170	30	2,132
117	6-Nov-23	4,200	0	200	9,650	0	10,580	60	8,230	0	2,132
118	7-Nov-23	4,200	0	200	9,850	400	10,980	0	8,230	0	2,132
119	8-Nov-23	4,200	0	0	9,850	0	10,980	150	8,380	50	2,182
120	9-Nov-23	4,200	0	200	10,050	0	10,980	150	8,530	0	2,182
121	10-Nov-23	4,200	0	0	10,050	200	11,180	230	8,760	20	2,202
122	11-Nov-23	4,200	0	0	10,050	0	11,180	0	8,760	0	2,202
123	12-Nov-23	4,200	0	200	10,250	350	11,530	200	8,960	0	2,202
124	13-Nov-23	4,200	0	0	10,250	0	11,530	0	8,960	0	2,202
125	14-Nov-23	4,200	0	0	10,250	0	11,530	0	8,960	0	2,202
126	15-Nov-23	4,200	0	0	10,250	0	11,530	0	8,960	0	2,202
127	16-Nov-23	4,200	0	0	10,250	0	11,530	0	8,960	0	2,202
128	17-Nov-23	4,200	0	0	10,250	0	11,530	100	9,060	0	2,202
129	18-Nov-23	4,200	0	0	10,250	0	11,530	0	9,060	0	2,202
130	19-Nov-23	4,200	0	0	10,250	0	11,530	0	9,060	0	2,202
131	20-Nov-23	4,200	0	0	10,250	0	11,530	0	9,060	0	2,202

132	21-Nov-23	4,200	0	0	10,250	0	11,530	100	9,160	0	2,202
133	22-Nov-23	4,200	0	0	10,250	0	11,530	100	9,260	0	2,202
134	23-Nov-23	4,200	0	0	10,250	0	11,530	100	9,360	0	2,202
135	24-Nov-23	4,200	0	0	10,250	0	11,530	100	9,460	0	2,202
136	25-Nov-23	4,200	0	0	10,250	0	11,530	100	9,560	0	2,202
137	26-Nov-23	4,200	0	0	10,250	0	11,530	100	9,660	0	2,202
138	27-Nov-23	4,200	0	300	10,550	0	11,530	0	9,660	0	2,202
139	28-Nov-23	4,200	0	300	10,850	300	11,830	0	9,660	0	2,202
140	29-Nov-23	4,200	0	0	10,850	0	11,830	0	9,660	0	2,202
141	30-Nov-23	4,200	0	0	10,850	0	11,830	0	9,660	0	2,202
142	1-Dec-23	4,200	0	0	10,850	0	11,830	0	9,660	0	2,202
143	2-Dec-23	4,200	0	0	10,850	0	11,830	0	9,660	0	2,202
144	3-Dec-23	4,200	0	0	10,850	0	11,830	0	9,660	0	2,202
145	4-Dec-23	4,200	0	0	10,850	0	11,830	0	9,660	0	2,202
146	5-Dec-23	4,200	0	0	10,850	0	11,830	0	9,660	0	2,202
147	6-Dec-23	4,200	0	0	10,850	0	11,830	0	9,660	0	2,202
148	7-Dec-23	4,200	0	0	10,850	0	11,830	0	9,660	0	2,202
149	8-Dec-23	4,200	0	0	10,850	0	11,830	0	9,660	0	2,202
150	9-Dec-23	4,200	0	0	10,850	0	11,830	0	9,660	0	2,202
151	10-Dec-23	4,200	0	0	10,850	0	11,830	0	9,660	0	2,202
152	11-Dec-23	4,200	0	0	10,850	0	11,830	0	9,660	0	2,202
153	12-Dec-23	4,200	0	0	10,850	0	11,830	0	9,660	0	2,202
154	13-Dec-23	4,200	0	0	10,850	0	11,830	250	9,910	50	2,252
155	14-Dec-23	4,200	0	0	10,850	0	11,830	0	9,910	0	2,252
156	15-Dec-23	4,200	0	0	10,850	0	11,830	0	9,910	0	2,252
157	16-Dec-23	4,200	0	0	10,850	0	11,830	0	9,910	0	2,252
158	17-Dec-23	4,200	0	0	10,850	0	11,830	0	9,910	0	2,252
159	18-Dec-23	4,200	0	0	10,850	0	11,830	0	9,910	0	2,252
160	19-Dec-23	4,200	0	0	10,850	0	11,830	0	9,910	0	2,252
161	20-Dec-23	4,200	0	0	10,850	0	11,830	0	9,910	0	2,252
162	21-Dec-23	4,200	0	0	10,850	0	11,830	0	9,910	0	2,252
163	22-Dec-23	4,200	0	0	10,850	0	11,830	0	9,910	0	2,252
164	23-Dec-23	4,200	0	0	10,850	0	11,830	0	9,910	0	2,252
165	24-Dec-23	4,200	0	0	10,850	0	11,830	0	9,910	0	2,252
166	25-Dec-23	4,200	0	0	10,850	0	11,830	0	9,910	0	2,252

Figure 12-19: Completion Dewatering Detail Report

Dewatering Analysis											
Report No.	Date	Depth	Drilled	pH	Hardn.	Turb.	TSS	Nitrate	Al	Sulfate	Ca
				7-9	<800	<50	< 80	< 50	< 3	< 50	< 600
-	dd-M-yy	m	m	-	mg/L	NTU	mg/L	mg/L	mg/L	mg/L	mg/L
87	7-Oct-23	4,200	0	9	800	32	30	36	1	38	600
88	8-Oct-23	4,200	0	9	760	30	32	36	1	36	560
89	9-Oct-23	4,200	0	0	0	0	0	0	0	0	0
90	10-Oct-23	4,200	0	0	0	0	0	0	0	0	0
91	11-Oct-23	4,200	0	0	0	0	0	0	0	0	0
92	12-Oct-23	4,200	0	0	0	0	0	0	0	0	0
93	13-Oct-23	4,200	0	0	0	0	0	0	0	0	0
94	14-Oct-23	4,200	0	8	720	30	45	31	1	41	600
95	15-Oct-23	4,200	0	0	0	0	0	0	0	0	0
96	16-Oct-23	4,200	0	0	0	0	0	0	0	0	0

97	17-Oct-23	4,200	0	9	800	32	26	36	1	32	560
98	18-Oct-23	4,200	0	9	760	30	28	32	1	28	600
99	19-Oct-23	4,200	0	0	0	0	0	0	0	0	0
100	20-Oct-23	4,200	0	0	0	0	0	0	0	0	0
101	21-Oct-23	4,200	0	9	800	36	38	32	1	32	520
102	22-Oct-23	4,200	0	0	0	0	0	0	0	0	0
103	23-Oct-23	4,200	0	0	0	0	0	0	0	0	0
104	24-Oct-23	4,200	0	0	0	0	0	0	0	0	0
105	25-Oct-23	4,200	0	0	0	0	0	0	0	0	0
106	26-Oct-23	4,200	0	0	0	0	0	0	0	0	0
107	27-Oct-23	4,200	0	0	0	0	0	0	0	0	0
108	28-Oct-23	4,200	0	0	0	0	0	0	0	0	0
109	29-Oct-23	4,200	0	0	0	0	0	0	0	0	0
110	30-Oct-23	4,200	0	0	0	0	0	0	0	0	0
111	31-Oct-23	4,200	0	0	0	0	0	0	0	0	0
112	1-Nov-23	4,200	0	0	0	0	0	0	0	0	0
113	2-Nov-23	4,200	0	0	0	0	0	0	0	0	0
114	3-Nov-23	4,200	0	0	0	0	0	0	0	0	0
115	4-Nov-23	4,200	0	0	0	0	0	0	0	0	0
116	5-Nov-23	4,200	0	0	0	0	0	0	0	0	0
117	6-Nov-23	4,200	0	8	720	30	45	31	1	41	600
118	7-Nov-23	4,200	0	8	720	30	45	31	1	41	600
119	8-Nov-23	4,200	0	8	720	30	45	31	1	41	600
120	9-Nov-23	4,200	0	8	720	30	45	31	1	41	600
121	10-Nov-23	4,200	0	8	720	30	45	31	1	41	600
122	11-Nov-23	4,200	0	8	720	30	45	30	1	40	600
123	12-Nov-23	4,200	0	8	720	30	45	30	1	40	600
124	13-Nov-23	4,200	0	8	720	30	45	30	1	40	600
125	14-Nov-23	4,200	0	8	720	30	45	30	1	40	600
126	15-Nov-23	4,200	0	8	720	30	45	30	1	40	600
127	16-Nov-23	4,200	0	8	720	30	45	30	1	40	600
128	17-Nov-23	4,200	0	8	720	30	45	30	1	40	600
129	18-Nov-23	4,200	0	8	800	40	42	36	1	34	560
130	19-Nov-23	4,200	0	0	0	0	0	0	0	0	0
131	20-Nov-23	4,200	0	0	0	0	0	0	0	0	0
132	21-Nov-23	4,200	0	0	0	0	0	0	0	0	0
133	22-Nov-23	4,200	0	8	800	40	42	36	1	34	560
134	23-Nov-23	4,200	0	8	800	40	42	36	1	34	560
135	24-Nov-23	4,200	0	8	800	40	42	36	1	34	560
136	25-Nov-23	4,200	0	0	0	0	0	0	0	0	0
137	26-Nov-23	4,200	0	0	0	0	0	0	0	0	0
138	27-Nov-23	4,200	0	0	0	0	0	0	0	0	0
139	28-Nov-23	4,200	0	0	0	0	0	0	0	0	0
140	29-Nov-23	4,200	0	8	720	30	45	30	2	40	600
141	30-Nov-23	4,200	0	8	720	30	45	30	2	40	600
142	1-Dec-23	4,200	0	8	720	30	45	30	2	40	600
143	2-Dec-23	4,200	0	8	720	30	45	30	2	40	600
144	3-Dec-23	4,200	0	8	720	30	45	30	2	40	600
145	4-Dec-23	4,200	0	8	720	30	45	30	2	40	600
146	5-Dec-23	4,200	0	8	720	30	45	30	2	40	600

147	6-Dec-23	4,200	0	8	720	30	45	30	2	40	600
148	7-Dec-23	4,200	0	8	720	30	45	30	2	40	600
149	8-Dec-23	4,200	0	8	720	30	45	30	2	40	600
150	9-Dec-23	4,200	0	8	720	30	45	30	2	40	600
151	10-Dec-23	4,200	0	8	720	30	45	30	2	40	600
152	11-Dec-23	4,200	0	8	720	30	45	30	2	40	600
153	12-Dec-23	4,200	0	8	720	30	45	30	2	40	600
154	13-Dec-23	4,200	0	0	0	0	0	0	0	0	0
155	14-Dec-23	4,200	0	8	760	30	45	30	1	40	600
156	15-Dec-23	4,200	0	8	760	30	45	30	1	40	600
157	16-Dec-23	4,200	0	0	0	0	0	0	0	0	0
158	17-Dec-23	4,200	0	0	0	0	0	0	0	0	0
159	18-Dec-23	4,200	0	0	0	0	0	0	0	0	0
160	19-Dec-23	4,200	0	0	0	0	0	0	0	0	0
161	20-Dec-23	4,200	0	0	0	0	0	0	0	0	0
162	21-Dec-23	4,200	0	0	0	0	0	0	0	0	0
163	22-Dec-23	4,200	0	0	0	0	0	0	0	0	0
164	23-Dec-23	4,200	0	0	0	0	0	0	0	0	0
165	24-Dec-23	4,200	0	0	0	0	0	0	0	0	0
166	25-Dec-23	4,200	0	0	0	0	0	0	0	0	0

Figure 12-20: Completion Dewatering Analysis

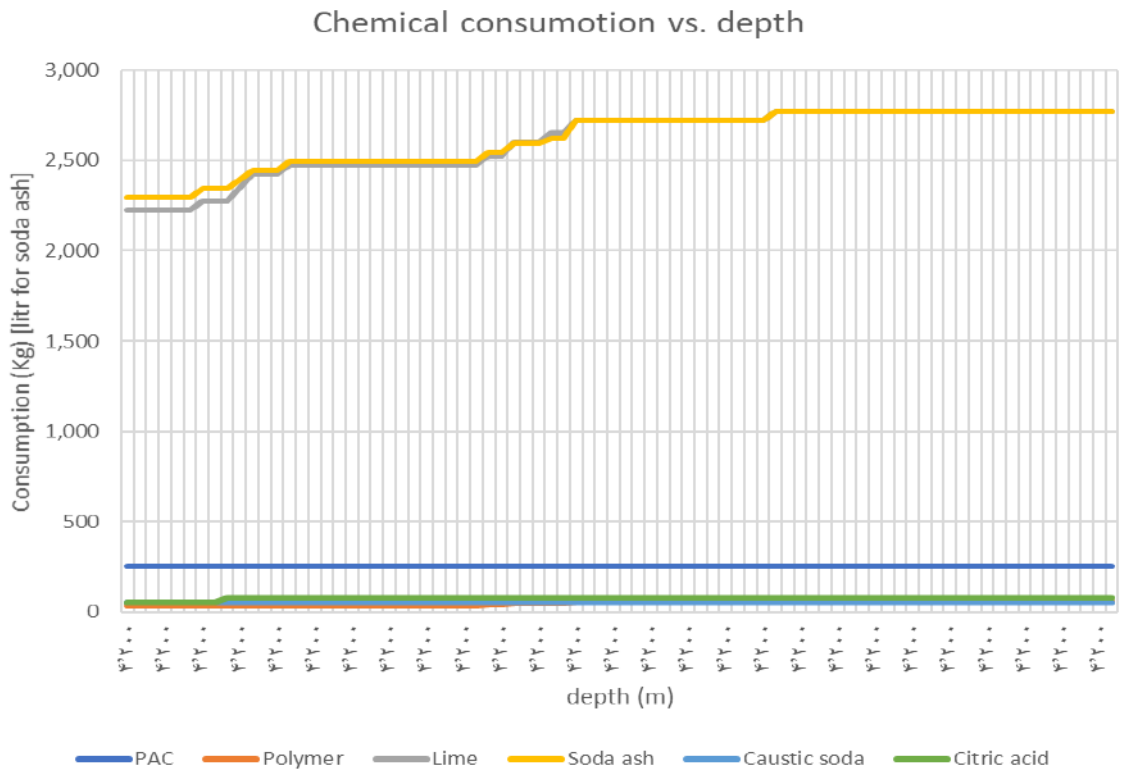


Figure 12-21: Completion Cumulative Chemical Consumption vs. Depth

Overall Report

Section	Unit	12 ¼"	8 ½"	Completion	Total
Total Cuttings	bbl	5473	870	2140	**8483**
Total Fixed Cuttings	bbl	4820	1100	2103	**8023**
Total Disposed Cuttings	bbl	4820	1100	1983	**7903**
Total Used Cement	Kg	24000	10100	18400	**52500**
Avg. Cement Fixation ratio	Kg/bbl	4.979	9.182	8.749	**6.544**
Total Used Sodium silicate	lit	385	235	180	**800**
Avg. Sodium Silicate Fixation ratio	Kg/bbl	0.080	0.214	0.086	**0.100**
Total Waste-water	bbl	6170	1650	2910	**10730**
Total Dewater	bbl	7400	1500	2410	**11310**
Recycled	bbl	5680	1350	2610	**9640**
Disposed and Lost	bbl	1392	460	350	**2202**
Used PAC	Kg	225	25	0	**250**
Avg. PAC treatment ratio	Kg/bbl	0.030	0.017	0	**0.022**
Used Polymer	Kg	30	5	20	**55**
Avg. polymer treatment ratio	Kg/bbl	0.004	0.003	0.008	**0.005**
Used Lime	Kg	1675	500	550	**2725**
Used Citric acid	Kg	25	25	25	**75**
Used Caustic	Kg	50	0	0	**50**
Used Soda ash	Kg	1845	400	475	**2720**

Table 12.8. Waste Management End of Well Report Summary.

13. List of Contractors

Item	Service Item	Contractor
1	Drilling Rig, Mobilization and Demobilization the Rig, Rig Move Between Well Pads	OEOC
2	Drilling Fluids and Completion Fluids	VIRA
3	Waste Management	Pars Haffaran Arvand
4	Electrical Logging and Perforation	PANIZ
5	Mud Logging	MAPSA KISH
6	Directional Drilling (MWD)	VIRA
7	H2S Monitoring and Control	MAPSA KISH BONYAN IMEN
8	Casing and Tubing Running	OEOC
9	Cementing	SPAD ENERGY
10	Dry Test, DST and RTTS	OEOC
11	Slick Line	OEOC
12	Stimulation (Coiled Tubing and Acidizing and Nitrogen Lifting)	OEOC
13	Surface Well Testing	OEOC
14	Liner Hanger	PARS POOLAD
15	Well Head	OEOC

14. Appendix

14.1 DDR Summary

Hole Section 12-1/4"

Item	Date	Remark
1	13 – July	Spud the rig @ 00:00, 13-Jul-2023, HPJSM, Record WH Press, Perform Press test 20-3/4"-3K*13-5/8"-5K CHS P-Seal & G.Valve, N/U 13 5/8" - 10K BOP Stack & P/T same till M.N.
2	14 – July	PRESS TEST BURNING LINE, FUNCTION TEST ACCUMULATOR SYSTEM, PERFORM WH INTEGRITY TEST AGAINST CMT PLUG, RIH W/ 12-1/4" R.BIT & D/O BHA TO 47 M, D/O CMT TO 160M.
3	15 - July	CONT. D/O HARD CMT F/ 160 TO 171 M, RIH W/ 12-1/4" MT BIT & D/O BHATO 1468 M, D/O TOP, BTM & F. COLLAR TO 1472 M. D/O CMT TO 1490 M.DISPLACE THE WELL W/ 73 PCF S.S.KCL POLY MUD.
4	16 - July	Cont. Circ. & Displace the well w/ 73 pcf S.KCl.Poly mud. Perform CIT. D/O hard CMT to 1493 m. Commence Drlg 12-1/4" H.S f/ 1493 to 1498 m. Perform FIT. Resume Drlg 12-1/4" H.S f/ 1498 to 1505.5 m. CBU. POOH 12-1/4" D/O BHA to surface. M/U & R/B 10 Stds 5" Dps
5	17 - July	RIH w/ 12-1/4" Drlg BHA to BTM. Resume Drlg 12-1/4” H.S to 1556.5m. Stop Drlg due to Obs. jumping string, drop 390 psi SPP & 5 klbs drop string Wt. POOH to surface, Obs. Twist off string from box of 1st Jt 8-1/2" DC, RIH w/ 10-1/8" Over shot & Fishing BHA to 1490m.
6	18 - July	Cont. RIH w/ 10-1/8" over shot to TOF (1547.2 m). Tag fish, Catch fish. POOH to surface, Recover all components of fish. Release Over Shot, L/D 12-1/4" PDC Bit & 1 Jt 8-1/2" DC, RIH w/ 12-1/4" new PDC bit & Drlg BHA (baffle & f.valve in B.sub) to 230m, Slip & Cut 97 ft Drlg line, RIH to 13-3/8" CSG Shoe) while F/U String, W/R to 1556.65m (BTM), Drlg 12-1/4" HS to 1571m, w/ 1-6 BPH loss.

7	19 - July	CONT DRLG 12-1/4" HS F/ 1561.65 to 1767m .
8	20 - July	CONT DRLG 12-1/4" HS F/ 1767 to 1908m W/4-8BPH LOSSES.
9	21 - July	Cont. Drlg 12-1/4" HS f/ 1910 to 1930m, 4-8 BPH loss, Circ. out 30 bbl Hi-vis pill, POOH on elevator as condition trip to 13-3/8" Csg shoe w/ good hole condition, Perform flow check; 2-3 BPH loss, Run back in hole to 1866m, Run totco survey; Inc.: 0.75 Deg, RIH to BTM), Cont. Drlg to 1960m w/ 2-8 BPH loss while Inc MW f/ 73 to 74 pcf. Due to Obs. 30 % chert in last geologist sample, stop Drlg, Circ. out 30 bbl Hi-vis pill, POOH as change Bit f/ BTM to 13-3/8" Csg shoe w/ good hole condition, POOH to 800m while F/U hole w/ 2-3 BPH loss.
10	22 - July	POOH W/12-1/4" PDC Bit to Surface . L/D Bit . RIH W / 12-1/4" New B.Bit & Drlg BHA TO BTM (1960 m) W/ good hole conection , 2-3 BPH loss . Cont , Drig 12-1/4" HS f/ 1960 m to 2010 m w/5-7 BPH loss.
11	23 - July	Cont , Drig 12-1/4" HS f/ 2010 m to 2084 m w/3-8 BPH loss.
12	24 - July	Cont. Drlg 12-1/4" HS f/ 2084 to 2115m w/ 3-8 BPH loss. Twist off string. POOH to Surf. String was twisted off f/ mandrel of 8" D.Jar. RIH w/ 10-1/8" O.Shot & Fishing BHA to 1800m. 2-3 BPHL.
13	25 - July	Cont. RIH w/ 10-1/8" O.Shot to TOF @ 1997.5m. Tag fish, Try to Catch fish; Succ. POOH to Surf. Recover fish. RIH w/ same 12-1/4" B.Bit & Drlg BHA to 200m, try to make Circ, Obs string plug. POOH to Surf. Observe string was plugged f/ first DC above B.Sub, unplug same. RIH back w/same B.Bit & BHA to 1100m, 2-3 BPHL.
14	26 - July	Cont. RIH w/ 12-1/4" B.Bit & Drlg BHA to BTM. Cont. Drlg 12-1/4" HS.
15	27 - July	Cont. Drlg 12-1/4" HS f/ 2161 to 2191m w/ 2-6 BPHL. take totco survey @ 2169m; Inc.: 0.75 Deg. Due to change Bit; Commence POOH f/ BTM to surface, 1-2 BPHL. L/D 12-1/4" B.Bit, (Dull: 2-2-WT-A-E-IG-CT-BHA).

16	28 - July	RIH w/ 12-1/4" PDC.Bit & Drlg BHA to BTM (2191m). Cont. Drlg 12-1/4" HS 2197m while add KCl solution in active mud Sys. Twist off string; Suddenly lost 30 Klbs of S.W. POOH to surface. string was twisted off f/ pin of 8 1/2" DC. RIH 10-1/8" O.Shot & Fishing BHA to 2100m, 1-2 BPHL.
17	29 - July	Cont. RIH w/ 10-1/8" O.Shot to TOF @ 2124m. Tag fish, Try to Catch fish; Succ. POOH to Surf. Recover all components of fish, L/D Bit & all 8-1/2" DC in derrick & BHA (15 Jts), P/U & M/U 12-1/4" new PDC.Bit & RIH w/ new BHA to 38m.
18	30 - July	Cont. RIH w/ 12-1/4" PDC.Bit & Drlg BHA to BTM (2197m). Cont. Drlg 12-1/4" HS to 2253m while Inc MW f/ 75 to 76 Pcf & add KCl solution in active mud Sys f/ 2.5% to 3%, 2-6 BPHL.
19	31 - July	CONT. DRILLING FORMATION F/ 2253m TO 2394m.
20	01 - August	Cont. Drlg 12-1/4" HS f/ 2392 to 2491, CBU, POOH to 2160m w/ good hole condition as wiper trip, Run back in hole to 2480m (11m off BTM), Circ. out 20 bbl Hi-vis pill, Run totco survey @ 2480m; Inc.: 1 Deg, RIH to BTM, Cont. Drlg 12-1/4" HS to 2512m, 1-8 BPHL.
21	02 - August	Cont. Drlg 12-1/4" HS f/ 2512 to 2690 m, 6-10 BPHL.
22	03 - August	Cont. Drlg 12-1/4" HS f/ 2690 to 2768 m, 6-10 BPHL.
23	04 - August	Cont. Drlg 12-1/4" HS f/ 2768 to 2850m, 8-10 BPHL.
24	05 - August	Cont. Drlg 12-1/4" HS f/ 2850 to 2854m, CBU, POOH to 2800m w/ good hole condition, driller didn't care about weight indicator. P/U string on elevator to 2795 m w/ 170 Klbs overpull, Suddenly lost 120 Klbs string Wt, due to jump TDS; Burst TDS hydraulic hose, Work on TDS, POOH to surfce, String was twisted off , RIH w/ 10-1/8" O.Shot & Fishing BHA & Catch fish, Pump out w/ 10-1/8" O.Shot & Fishing BHA to 2600 m (Bit Depth).
25	06 - August	Cont. POOH w/ 10-1/8" Over shot & Fishing BHA to 2100 m, Release over shot, Cont. POOH to surface, while L/D buckled tubular.

26	07 - August	Perform BOP test as NIOC policy, L/D 60 Jts 5" DP,RIH with 12-1/4" PDC Bit and drlg BHA to 234m,Slip & Cut.
27	08 - August	RIH w/ 12-1/4" PDC Bit & Drlg BHA to 2765m.
28	09 - August	Wash & R/D to 2854m, CBU, Take Survey, Resume Drlg 12-1/4" HS to 2900m.
29	10 - August	Cont. Drlg 12-1/4□□□.S to 2919m, Stop Drlg due to low ROP, POOH w/ 12-1/4" PDC Bit to 400 m.
30	11 - August	Cont. POOH w/ 12-1/4" PDC Bit to surface, RIH w/ 12-1/4" TCI Bit (include J.Sub) to BTM, While clear tight spot, Circ. Out 30 bbl Hi-Vis Pill, Work on J.Sub, Resume Drlg 12-1/4" HS f/ 2919 to 2922m.
31	12 - August	Cont. Drlg 12-1/4" HS f/ 2922 to 2923m, Work on J.Sub, POOH w/ 12-1/4" TCI Bit & J Sub f/ 2923 to surface, Static flow check, RIH W/ 12-1/4" PDC Bit & Drlg BHA to 1720m
32	13 - August	Cont. RIH W/ 12-1/4" PDC Bit & Drlg BHA f/ 1720 to 2923m, Resume Drlg 12-1/4” H.S to 2944m.
33	14 - August	Cont. Drlg 12-1/4” H.S f/ 2944 to 2945 m, Stop Drlg due to low progress, CBU , POOH w/ 12-1/4" PDC Bit & Drlg BHA f/ 2945m to surface, RIH w/ 12-1/4" TCI Bit, 2 No J Sub & Pendulum BHA to 1250m.
34	15 - August	Cont. RIH w/ 12-1/4" TCI Bit & 2 No J.Sub f/ 1250 to BTM (2945m). Circ out 30 bbl Hi-vis Pill. Work on J.Sub. Drlg 12-1/4" HS f/ 2945 to 2949.5m. Stop Drlg due to work on Weght indicator & fix same, while Circ out 90 bbl Hi-vis LCM Pill. Cont Drlg 12-1/4" HS to 2952m w/ 2-4 BPHL.
35	16 - August	Cont Drlg 12-1/4" HS f/ 2952 to 2961.5m. Due to low ROP; Stop Drlg, CBU to shaker clean. Work on J.Sub. POOH f/ BTM to 13-3/8" Csg shoe. Spot 30 bbl Hi-Vis LCM Pill. POOH to 1350m. Stop operation due to change Sencator. POOH to surface. L/D Bit & 2 No. J.Sub, 2-7 BPHL.
36	17 - August	P/U & M/U 12-1/4" PDC.Bit & RIH w/ Drlg BHA to 230m, change hook load Sensator, Cont RIH to BTM (2961.5m).

		Commence Drlg 12-1/4" HS f/ 2961.5m to 2988m, 1-5 BPHL.
37	18 - August	Cont Drlg 12-1/4" HS f/ 2988 to 3069m, 3-13 BPHL.
38	19 - August	Cont Drlg 12-1/4" HS f/3069m to 3191m, 5-15 BPHL.
39	20 - August	Cont Drlg 12-1/4" HS f/3191 to 3275m, 3-8 BPHL.
40	21 - August	CBU,POOH as Condition Trip f/3275 to 2960m RIH to BTM. Cont Drlg 12-1/4" HS f/3275 to 3349m, 4-10 BPHL.
41	22 - August	Cont Drlg 12-1/4" HS f/3350 to 3497m, 5-10 BPHL.While increased Mud weight from 82 to 83 pcf.
42	23 - August	Cont Drlg 12-1/4" HS f/3350 to3497m,Twist off string,suddenly lost 40 Klbs string Wt. POOH to 1100m.
43	24 - August	Cont. RIH w/ 10-1/8" O.Shot to TOF @ 3565m. Tag fish, Try to Catch fish; Succ. POOH to 300m, 5-10 BPHL.
44	25 - August	POOH w/ 10-1/8" O.Shot & Fishing BHA to Surf. Recover all components of fish, Release O.Shot. RIH back w/ same PDC Bit & Drlg BHA to 600m. Slip & Cut 110 ft of Drlg line. RIH to 1000m. Due to bad weather; stop operation. RIH to 3400m w/ good hole condition, 4-8 BPHL.
45	26 - August	Cont. RIH f/ 3400 to 3552m (BTM). CBU till shaker clean, POOH to 3546m, Run totco survey @ 3546m; Inc: 1 Deg. RIH to 3552m, Commence Drlg 12-1/4" HS w/ Hi TQ 3552 to 3656m while Inc MW f/ 83 to 84 Pcf, 4-10 BPH loss.
46	27 - August	Cont. Drlg 12-1/4" HS w/ Hi TQ f/ 3656 to 3776m, 5-10 BPH loss.
47	28 - August	Cont. Drlg 12-1/4" HS w/ Hi TQ f/ 3776 to 3833m, 5-10 BPH loss.
48	29 - August	Cont. Drlg 12-1/4" HS w/ Hi TQ & fluctuation TQ f/ 3833 to 3876m, 5-10 BPH loss.
49	30 - August	Cont. Drlg 12-1/4" HS f/ 3876 to 3910m.
50	31 - August	Cont. Drlg 12-1/4" HS f/ 3910 to 3932m. While CBU 2 time in

		order take last sample.
51	01 - Sep	Cont. Drlg 12-1/4" HS to 3934 m, CBU, POOH to 3500, Run back to BTM, CBU, Spot 200 bbl Hi-Vis Pill, POOH to 1150m while L/D 10 Stds 5" DP.
52	02 - Sep	L/D 17 Stds 5" DP, POOH w/ 12-1/4" PDC Bit & Drlg BHA f/ 1150m to surface, Retrieve 13-5/8" Wear Bushing, L/D mouse hole, Set shooting nipple. Install logging BOP & Test same. R/U Logging equipment. RIH w/ W/L logging tools (Full Suite) as Run #1 to 3934m & POOH to 300m.
53	03 - Sep	POOH W/L tools Run #1 to surface, Perform Neutron/Density Log as per program, R/D W/L Logging tools & Equip, Change top rams, R/U 9-5/8"CSG Running Equip, Run 9-5/8"CSG to 270m as per program.
54	04 - Sep	Cont. Run 9-5/8" CSG f/ 270 to 2586m.
55	05 - Sep	Cont. Run 9-5/8" CSG f/ 2586m to 3928m.Stuck 3925m @ this point.
56	06 - Sep	Wait on SPAD Energy Co. due not approved CMT Program & CMT Add. While wash down to 3933.8m. Try to wash down by apply 30 klb Wt; No-Succ. Confirmed BTM. Prepare 450 bbl CMT Solution.
57	07 - Sep	Prepare CMT Solution as per program. Mix & Pump 1154 bbl CMT G Delijan as lead slurry, 116 bbl CMT G Delijan blended w/ Silica Flour as tail slurry. Dispalce w/ 908 bbl 85 pcf mud. Set CSG slips, WOC, Rough cut CSG.
58	08 - Sep	Final cut 9-5/8"CSG neck. Install 13-5/8” 5k x 9-5/8" 10k THS & Test same. N/U BOP Stack. Perform BOP test as NIOC Policy. L/D 11 Jt 8-1/2" DC. P/U & M/U 8-1/2" TCI Bit, 2 Jts 6-1/2" DC. L/D 15 Jts 5" DP (Buckle).
59	09 - Sep	L/D 39 Jts 5" DP, RIH w/ 8-1/2" TCI Bit & D/O BHA to 3905.5m, D/O Hard CMT & F.Collar to 3912m.

Table 14-1: AZNS-040 12-14" Daily Drilling Report Summary

Hole Section 8-1/2"

Item	Date	Remark
1	10 - Sep	Cont. D/O Hard CMT f/ 3912 to 3932.5m, D/O 9-5/8" CSG Shoe f/ 3932.5 to 3933m. D/O 1m Hard CMT below CSG Shoe. While CBU & Perform CIT @ 3926m. Commence Drlg 8-1/2" HS f/ 3934 to 3939m, NL-NG. CBU. POOH w/ 8-1/2" TCI Bit & D/O BHA to surface. RIH w/ 8-1/2" TCI Bit & Scraper BHA (Include J.Sub) & Followed by 5" DPs to 500m
2	11 - Sep	Cont. RIH w/ 8-1/2" B.Bit & 9-5/8" Scraper BHA to 3939m (BTM), Work on Junk Sub, Scrape f/ 3885 to 3932m in several times. While work on Drawworks 2 Hrs @ 3470m. POOH w/ 8-1/2" B.Bit & 9-5/8" scraper BHA to surface. Retrieve 11" Wear Bushing, L/D mouse hole. Set shooting nipple. Install logging BOP.
3	12 - Sep	P/T logging BOP to 1000 Psi; OK. RIH w/ logging tools (CCL/GR/SBT/VDL) f/ surface to 1392m (100m above 13-3/8" Csg Shoe), Log Dn f/ 1392 to 3928m (5m Off-BTM). SBT tools Power off during Log Dn. Power on SBT tools, Log up f/ 3928 to 1385 m (107m above 13-3/8" CSG Shoe), Take repeat log 100m below 13-3/8"Csg shoe. POOH W/L to Surf. RIH w/ 8-1/2" PDC Bit & Drlg BHA to 2600m, NL-NG.
4	13 - Sep	Cont. RIH w/ 8-1/2" PDC Bit to 3923m (10m above 9-5/8" CSG Shoe), Perform FIT w/ mud 85 Pcf by Rig pump to 2100 Psi (EMW: 108 Pcf), Constant Press. Displace the well w/ Limestone mud 100 Pcf. RIH to BTM (3939m), Drlg 8-1/2" HS to 3942m. Perform FIT w/ mud 100 Pcf by Pump truck to 720 Psi (EMW: 108 Pcf), Constant Press. Drlg 8-1/2" HS to 3956m while Inc MW f/ 100 to 101 Pcf, NL-NG
5	14 - Sep	Cont Drlg 8-1/2" HS f/ 3956 to 3969m while Inc MW f/ 101 to 103 Pcf, Cont Drlg to 3982m. Due to failure of wash pipe; stop Drlg. B/R to 9-5/8" Csg shoe (3930m), change & test same. Run back in hole to Btm. Resume Drlg 8-1/2" HS to 4013m while work on mud rheology, 0-2 BPHL.
6	15 - Sep	Cont Drlg 8-1/2" HS f/ 4013 to 4070m while Inc MW f/ 103 to 104 Pcf due to Inc TQ & T.Gas to 92%, B/R & R/D each

		drilled Std, 1-3 BPHL.
7	16 – Sep	Cont Drlg 8-1/2" HS f/ 4070 to 4129m 1-4 BPHL.
8	17 – Sep	Cont Drlg 8-1/2" HS f/ 4129 to 4186m 1-4 BPHL.
9	18 – Sep	Cont Drlg 8-1/2" HS f/ 4186 to 4200m. CBU. confirm TD w/ Client. Confirm TD @ 4200m. POOH to 9-5/8" Csg shoe as condition trip. Run back in hole to TD. Spot Hi-Vis pill. POOH w/ 8-1/2" Drlg BHA to 250m (@ BHA depth). Perform Static flow check; 2 BPHL. POOH to surface. Retrieve 11" Wear Bushing. L/D mouse hole. Set shooting nipple, 1-3 BPHL.
10	19 – Sep	Perform W/L logging job as per program, 1-3 BPHL.
11	20 – Sep	Perform W/L logging job & run 7" liner as per program, 1-3 BPHL.
12	21 – Sep	Run 7" LNR to 3909m 1-3 BPHL.
13	22 – Sep	Cont. RIH w/ 7" LNR to 4190m, Took Wt, Try to pass, No Succ. Set Hanger @ 4189.5m, Perform 7" LNR CMT Job (SV: 53 bbl, SW: 118 Pcf), Obs Shearing/bumping, NL-NG. POOH 11 Stds to 3492m, CBU w/NL-NG.
14	23 – Sep	POOH w/ 7" LNR setting tools to surface. Install 11" wear bushing. RIH w/ 8-1/2" D/O BHA 3655m. W/D to 3802m without drag & W/D to 3812.2m w/ 3-5 Kips drag, Confirm 7" L.Lap @ 3812.2m. CBU & dump 30 bbl contaminated mud w/ slurry. POOH to 1620m, NL-NG.
15	24 – Sep	Cont. POOH w/ 8-1/2" D/O BHA to surface, while L/D 24 Jts 5" DP. L/D Bit. Change top ram f/ 4-1/2" x 7" VBR to 2-7/8" x 5" VBR. Perform BOP test. RIH w/ 5-7/8" Clean out BHA to 2500m, NL-NG.
16	25 – Sep	RIH w/ 5-7/8" Clean out BHA D/O POB, POOH to Surf, NL/NG.
17	26 – Sep	Cont. RIH w/ 7" X 9-5/8" tandem scraper to 4154m, slip & cut Drlg line, Wait on Rec. D.Water, Spot 40 bbl CaCl2 Brine as balance, POOH w/ 5-7/8" MT Bit & 7" X 9-5/8" tandem

		scraper BHA f/ 4154 to 3900m.
18	27 – Sep	Cont. POOH w/ 5-7/8" MT Bit & 7" X 9-5/8" tandem scraper BHA f/ 3900m to surface, RIH w/ OEOC 9-5/8" Dry test tools to 1160m.
19	28 – Sep	Cont. RIH w/ OEOC 9-5/8" Dry test tools to 3795m. F/U string as per program. Set 9-5/8" Dry test PKR, Monitor Open period for 3Hrs till MN.
20	29 – Sep	Close DCIP valve. Record close period, Drop Bar. Establish reverse circ, unset PKR, flow check, POOH w/ 9-5/8" Dry test assy to surface.
21	30 – Sep	Retrieve Memory gauge. RIH w/ 5" OEDPs to 1200m, Circ. Out Hi-Vis Pill & Displace contaminated mud, POOH w/ OEDP to surface. Perform CCL/GR/SBT/VDL Logging inside 7" LNR as per program. Due to not acceptable results of SBT according to client order repeat W/L Logging w/ SBT back up. RIH w/ W/L Logging tools to 3770m. Due to failur of cable head, POOH to surface.
22	01 – Oct	R/D W/L Logging run #2, Perform Logging w/ 3rd SBT as per program, RIH w/ 7-5/16" Polish Mill to 3812, Clean inside 7" TBS, Try to take LG mark.
23	02 – Oct	Uniform MW, POOH w/ 7-5/16" Polish Mill to surface, L/D 7-5/16" Polish mill; Obs. Mark of LG on same, Cont. Run SN PKR to 3400m.

Table 14-2: AZNS-040 8-1/2" Daily Drilling Report Summary

Completion Section

Item	Date	Remark
1	03 – Oct	Set SN PKR on 7" L.Lap as per PPA Co Procedure, CBU, POOH w/ SN PKR running tool to surface, RIH w/ OEOC 9-5/8" Dry test tools to 50m, meanwhile wait on Rec. Spare part of DCIP valve.
2	04 – Oct	RIH w/ OEOC 9-5/8" Dry test tools to 3280m.
3	05 – Oct	Cont. RIH w/ OEOC 9-5/8" Dry test tools to 3790m. F/U string as per program. Set 9-5/8" Dry test PKR, Monitor Open period for 3Hrs, Record Close period for 6 Hrs, Drop Bar. Establish Reverse Circ., POOH 9-5/8" Dry Test Assy to 1900m.
4	06 – Oct	Cont. POOH 9-5/8" Dry Test Assy f/ 1900m to surface, Retrieve two Memory gauge, Change top rams, Set 7" shooting nipple, RIH W/L w/ 4.5" Gun (Without bullets) & CCL tools as dummy run to 4140, POOH to surface, RIH W/L w/ CCL/GR/Wt Bar toolsas client request to 2500m.
5	07 – Oct	Perform 2 Run of perforation operation as per program.
6	08 – Oct	Perform 2 Run of perforation operation as per program.
7	09 – Oct	Perform 2 Run of perforation operation as per program.
8	10 – Oct	Perform perforation operation as Run #7 (Perforation Interval: 4039-4043m). Change BOP Top rams f/ 7" Fix to 2-7/8" * 5" VBR & test same to 7500 Psi; Ok. RIH w/ 5-7/8" M.M J.Mill & Clean out BHA to 2200m, 0-1 BPHL
9	11 – Oct	Cont RIH w/ 5-7/8" M.M J.Mill & Clean out BHA f/ 2200 to 4154m (7" LDC depth) W/O any drag. CBU. POOH to surface. L/D Mill.
10	12 – Oct	RIH W/L with CCL/GR Log to Perforated intervals & confirm same, POOH & R/D W/L tools & Equip. RIH w/ 5-7/8" M.M J.Mill & Clean out BHA to 1500m while F/U string every 15 Stds & break Circ, 0-1 BPHL

11	13 – Oct	Cont RIH w/ 5-7/8" M.M J.Mill & Clean out BHA to 4154m. CBU w/ 220 GPM, 0-1 BPHL. Cont CBU w/ 50 GPM while decrease MW to 102.5 Pcf, 0-0.5 BPHL.
12	14 – Oct	wait on receive 3-1/2" * 4-1/2" Tubing. POOH w/ 5-7/8" M.M J.Mill & Clean out BHA f/ BTM (4154m) to surface while Slip & cut 97 ft Drlg line & L/D 4 Stds 5" DP & 1 Std 4-3/4" DC. L/D Mill. L/D 3 Stds 5" HWDP Jt by Jt, 0-0.5 BPHL.
13	15 – Oct	L/D 5 Stds 6-1/2" DC. Flush BOP Stack. RIH w/ OEDP & 7" scraper to 4150m while scrape packer depth several times. Due to shortage of Tubing; wait on receive 3-1/2" * 4-1/2" Tubing.
14	16 – Oct	Due to shortage of Tubing; wait on receive 3-1/2" * 4-1/2" Tubing.
15	17 – Oct	Due to shortage of Tubing; wait on receive 3-1/2" * 4-1/2" Tubing.
16	18 – Oct	Due to shortage of Tubing; wait on receive 3-1/2" * 4-1/2" Tubing.
17	19 – Oct	Due to shortage of Tubing; wait on receive 3-1/2" * 4-1/2" Tubing.
18	20 – Oct	Due to shortage of Tubing; wait on receive 3-1/2" * 4-1/2" Tubing.
19	21 – Oct	Due to shortage of Tubing; wait on receive 3-1/2" * 4-1/2" Tubing.
20	22 – Oct	Due to shortage of Tubing; wait on receive 3-1/2" * 4-1/2" Tubing.
21	23 – Oct	Due to shortage of Tubing; wait on receive 3-1/2" * 4-1/2" Tubing.Prepared 930bbl completion fluid.
22	24 – Oct	While wait on receive 3-1/2" X 4-1/2" Tubing, Displace the well, Perform flow check, POOH w/ OEDPs & 7" Scraper f/ 4147 to 3720m while L/D 31 Jts 5" DP.
23	25 – Oct	While wait on receive 3-1/2" X 4-1/2" Tubing. Cont. POOH

		f/ 3720 to 1700 m while L/D totally 263 Jts 5" DP. Circ. CWH; NL-NG.
24	26 – Oct	While Wait on receive remained 4-1/2" TBGs as required, Stop POOH ,Circ. CWH. string already dropped in well. Perform Press test top rams. RIH w/ 7-5/8" O.Shot & Fishing BHA to 2080m, NL-NG.
25	27 – Oct	Catch fish, POOH 7-5/8" Over shot & Fishing BHA to 3800m, Perform flow check; Obs. Bubble. Close Top rams, Shut the well in, Monitor SICP, kill the well in progress till MN.
26	28 – Oct	Open the well and perform direct Circ, NL-NG. Try to RIH and uniform completion fluid @ BTM, unable to pass 7" L.Lap. POOH to 3797m, spot 60 bbl inhibited CaBr2, POOH to 3510m, Monitor the well. Decide to Displace well w/ 102pcf mud. Prepare 102 pcf mud.
27	29 – Oct	Displace the well thru C/M with 102 pcf mud, POOH w/ 7-5/8" Over shot & Fishing BHA f/ 3500 to 2600m.
28	30 – Oct	POOH w/ 7-5/8" Over shot & Fishing BHA to 1700m, Resume POOH w/ OEDP & 7" Scraper to surface, RIH W/ 5-7/8" M.M J.Mill to 360m.
29	31 – Oct	Cont. RIH w/ 5-7/8" M.M J.Mill & Clean out BHA to 3830m, While CBU & Flow check the well @ 2430, 2900, 3830m; NL-NG.
30	01 – Nov	Cont. RIH w/ 5-7/8" M.M J.Mill f/ 3830 to 4154m, Inc. MW to 103pcf, Uniform MW, Flow check, POOH to 1300m.
31	02 – Nov	Cont. POOH w/ 5-7/8" M.M J.Mill f/ 1300m to surface, Circ Across Wellhead; NL-NG. L/D J.Mill. Perform BOP Test. Install 11" wear bushing. RIH w/ 7-5/16" Polish mill to 3300m while F/U string & break Circ every 15 Stds, NL-NG.
32	03 – Nov	RIH w/ 7-5/16" Polish mill to 3811.4m, tag LG of 7" S.N Packer, Dress & Smooth Tie back sleeve several times, check W/O rotation. POOH to surface. RIH w/ 8-1/4" Dressing mill to 1800m, NL-NG.

33	04 – Nov	RIH w/ 8-1/4" Dressing mill 1800 to 3809.6m (7" L.Lap), Dress & smooth 7" L.Lap, NL-NG. Slip & Cut 97 ft Drlg line. POOH w/ 7-5/16" Dressing mill to 100m while F/U hole, NL-NG. Perform Static flow check; NL-NG
34	05 – Nov	POOH w/ 8-1/4" Dressing mill to surface. RIH w/ 5-7/8" M.M.J Mill to 2100m. As 7" L.Lap dry test operation decide to POOH to 770m. Stop POOH Due to Client requst & change program. Run back in hole to 1350m, NL-NG
35	06 – Nov	Cont RIH w/ 5-7/8" M.M.J Mill f/ 1350 to 3985m while Pass gently 7" L.Lap depth (3809m). Circ mud w/ 100 GPM @ 3985m (54m above perforation interval), NL-NG.
36	07 – Nov	Cont Circ mud w/ 100 GPM @ 3985m (54m above perforated interval), NL-NG.
37	08 – Nov	Cont Circ mud w/ 100 GPM @ 3985m (54m above perforated interval). POOH to 3400m, NL-NG.
38	09 – Nov	Cont POOH w/ 5-7/8" M.M.J.Mill f/ 3400m to surface. RIH w/ same 5-7/8" M.M.J.Mill & 7" Scraper to 3300m while F/U string & break Circ every 15 Stds, NL-NG.
39	10 – Nov	CBU, POOH to 3985m, Circ mud w/ 100 GPM @ 3985m (54m above perforated interval), NL-NG. CBU, POOH to 3985m, Circ mud w/ 100 GPM @ 3985m, NL-NG.
40	11 – Nov	Cont Circ mud w/ 100 GPM @ 3985m (54m above perforated interval), RIH to 4020m, Spot 50 bbl Hi-Vis Pill. POOH to surface, L/D J.Mill. RIH w/ 7" RTTS PKR to 500m, NL-NG.
41	12 – Nov	Cont RIH w/ 7" RTTS PKR f/ 500 to 3860m (51m below 7" L.Lap). Set 7" RTTS PKR @ 3860m, Perform 7" L.Lap positive test stepwise to 2000 Psi & hold for 40 min; Constant. Unset PKR. POOH to 2300, NL-NG.
42	13 – Nov	Cont POOH w/ 7" RTTS PKR f/ 2300m to surface. L/D same. RIH w/ 4-3/4" OEDP & 7" Scraper to 3000m while F/U string & break Circ every 15 Stds, NL-NG
43	14 – Nov	Cont RIH w/ 4-3/4" OEDP & 7" Scraper f/ 3000 to 3780m (29m above 7" L.Lap). CBU & uniform MW on 103 Pcf.

		Perform Static flow check; Static. POOH to 3670m while L/D 15 Jts 5" DP & 6 Jts 5" HWDP, Run back in hole to 3780m. Circ mud w/ 110 Gpm @ 3780m, NL-NG.
44	15 – Nov	Cont Circ mud w/ 110 Gpm @ 3780m (29m above 7" L.Lap), NL-NG.
45	16 – Nov	Cont Circ mud w/ 110 Gpm @ 3780m (29m above 7" L.Lap), NL-NG.
46	17 – Nov	Cont Circ mud w/ 110 Gpm @ 3780m (29m above 7" L.Lap), NL-NG.
47	18 – Nov	Cont Circ mud w/ 110 Gpm @ 3780m (29m above 7" L.Lap), NL-NG.
48	19 – Nov	Cont Circ mud w/ 110 Gpm @ 3780m (29m above 7" L.Lap), NL-NG. RIH w/ OEDPs & 7" Scraper f/ 3780 to 4013m while scrape f/ 3870 to 4013m. Held PJSM. L/U CMT to the string. Press test same. Pump 60 bbl 103 pcf Hi-Vis Pill. Displace the well w/ 104 pcf completion fluid in progress till M.N.
49	20 – Nov	Displace the well w/ 104 pcf completion fluid. Flow check the well; Static. POOH w/ OEDPs & 7" Scraper to 2650m.
50	21 – Nov	Cont. POOH w/ OEDPs & 7" Scraper to surface & L/D tubular.
51	22 – Nov	POOH to surface & L/D all tubular. Perform BOP test as NIOC Policy.Run 3-1/2" * 4-1/2" Completion string as per program to 84m. Perform CIV related test.
52	23 – Nov	Cont. RIH completion w/ totally 23 Jts 3-1/2" TBG & 71 Jts 4-1/2" TBG f/ 84 to 740m; NL-NG. While Press test sting to 7000 psi @ 262 & 740m; OK.
53	24 – Nov	Cont. RIH completion w/23 Jts 3-1/2" & 201 Jts 4-1/2" TBG f/ 740 to 2166m; NL-NG. While Press test sting to 7000 psi @ 1210, 1690 & 2166 m; OK.
54	25 – Nov	Cont. RIH completion w/23 Jts 3-1/2" & 316 Jts 4-1/2" TBG to 3270m; NL-NG. While Press test sting to 7000 psi @ 2166, 2641 & 3120 m; OK.

55	26 – Nov	Cont. RIH completion w/ totally 23 Jts 3-1/2" & 387 Jts 4-1/2" TBG & TR3V Assy to 3955m; NL-NG. While Press test sting to 7000 psi @ 3620 & 3940 m; OK & Perform TR3SV inflow test; Ok & Rupture CIVs. Check & Land TBG HGR in THS; OK.
56	27 – Nov	Press test string; Ok. Install 3.437" Protection Sleeve. Retrieve 2.75 WFB-2 check valve from SSD. Run and set 2.75 ” WRB-2 Check valve in No-Go L/N. Set PKR. Land TBG HGR in THS. Press test TBG HGR and Tie Dn screws. Final Press test string. Press test Ann. Ret. 2.750" check valve from No-Go. Ret. 3.437" Protection sleeve. Back off L/J. Install 4" TWCV in TBG HGR. SBR Press test; No-Succ.
57	28 – Nov	P/T TWCV against Top Rams to 2000 psi; OK. Change ram block f/ SBR to Blind Rams. P/T 4" TWCV to 7000 psi; OK. N/D 13-5/8"-10K BOP stack. N/U BMV & XMT upper section. Perform all P/T;OK. Ret. 4" TWCV.
58	29 – Nov	R/U Well test Equip. in progress While Wait on OEOC due to lack of well test choke manifold & Surge tank meanwhile wait on Rec. CTU.
59	30 – Nov	R/U Well test Equip. in progress
60	01 – Dec	R/U Well test Equip. in progress
61	02 – Dec	R/U Well test Equip. in progress
63	03 – Dec	R/U Well test Equip. in progress
64	04 – Dec	R/U Well test Equip. in progress
65	05 – Dec	R/U Well test Equip. R/U Slickline equipment.R/U CT service & Press test in progress.
66	06 – Dec	R/U CT service. Pump 5 bbl Acid as pickling thru CT reel, displace reel w/ 26 bbl (CT reel Cap) CaBr2. RIH 1-1/2" CT to 4000m (39m above perforated interval), displace well w/ NaCl brine.POOH CT to 1000m. Return on surface NaCl brine 75 Pcf. clean up the well.

67	07 – Dec	Cont clean up the well on choke size 20/64" to 52/64".
68	08 – Dec	Cont clean up the well on choke size 56/64" to 44/64". Stop clean up the well. R/D CT injector head. R/U Slickline service. Cont clean up the well on Fix choke size 24/64" till Constant Up stream Press for run memory gauge.
69	09 – Dec	Cont clean up the well on Fix choke size 24/64" till Constant Up stream Press for run memory gauge, constant same. Run memory gauge to 4029m, take flowing survey. R/U CT service, perform Eqp Press test w/ 6500 Psi, Ok.
70	10 – Dec	RIH 1-1/2" CT to 4117m (5m below perforated interval). Perform Acidizing job as per program. Clean up the well while POOH CT to surface.
71	11 – Dec	clean up the well on Fix choke size 48/64". Stop clean up the well. R/D CT injector head. R/U Slickline service. clean up the well. Stop clean up the well due to failure of well test separator & C/M, work on sames & rectify. Cont clean up the well on Fix choke size 24/64".
72	12 – Dec	Cont clean up the well on Fix choke size 24/64". Run 1-1/2" memory gauge w/ Slickline to 4029m, take flowing survey, POOH to surface. R/D Slickline Eqp & N/D lubricator. N/U CT riser & R/U Eqp, Perform CT BOP Press test in progress.
73	13 – Dec	RIH w/ 1-1/2" CT to 4030m, spot & bullhead 20 ADR, displace the well w/ 274 bbl diesel while POOH CT to surface, bullhead 20 diesel by kill line. Retrieve protection sleeve. Perform X-mass tree Valves inflow test in progress
74	14 – Dec	Perform Top master Vlv & 2SV & Swab Vlv & Production Vlv & Kill Vlv inflow test; Ok. Perform TR3SV inflow test; Ok. Set BPV in T.Hanger. secure well head, Moving the rig.

Table 14-3: AZNS-040 Completion Daily Drilling Report Summary

15. Comments & Revision

Drilling							
1	**7" Liner:** • مقدار spacer و lead cement در گزارش سیمان ۷ اینچ و اطلاعات درج شده در صفحه ۱۳۳ مغایرت دارد، اصلاح شود	132-133	CM	• اصلاح شد.			
2	**Wellhead Report:** • میزان فشار اعمال شده در تست سکشن C با گزارشات روزانه مغایرت دارد، بررسی و اصلاح شود.	187	CM	• در زمان نصب THS سکشن ”13-5/8 با فشار 4500^{psi} تست گردیده و در زمان تکمیل چاه سکشن ”11 با فشار 8000^{psi} تست صورت گرفته است.			
Geology							
3	در صفحه ۱۴ گزارش با Wellhead Coordinates گرافیک ول لاگ ارائه شده اختلاف دارد		CM	• پس از یررسی‌های صورت گرفته، با گرافیک ول‌لاگ مطابقت دارد.			
4	سرسازندهای گدوان، فهلیان، داریان، کژدمی در شکل با گرافیک ول AZNS-040 Well Sketchصفحه ۳۳ لاگ اصلاح گردد		CM	• اصلاح گردید.			

5	سرسازندها در صفحه ۳۳ و در صفحه ۳۲ (well Montage) با هم اختلاف دارد		CM	• اصلاح گردید.			